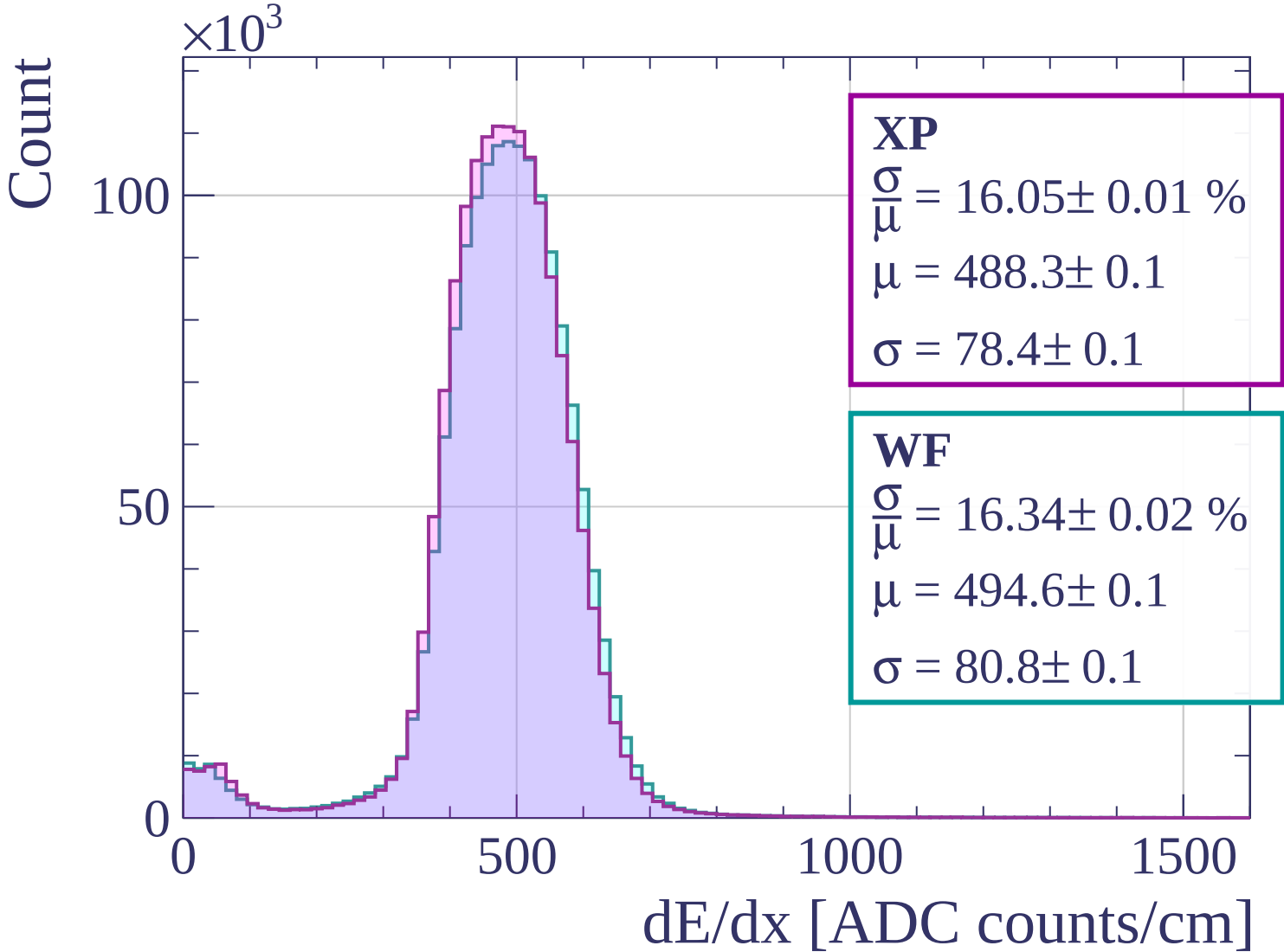
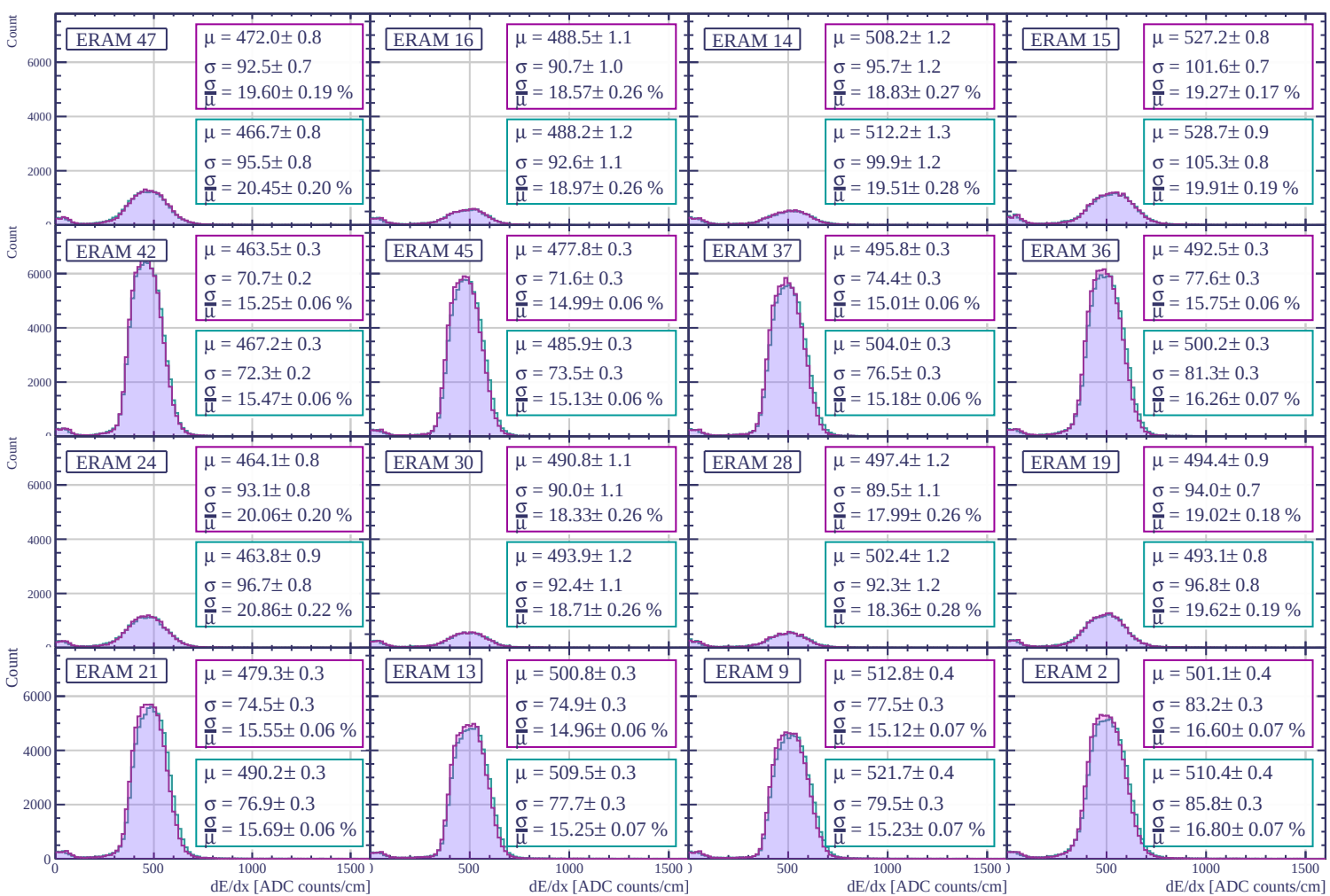
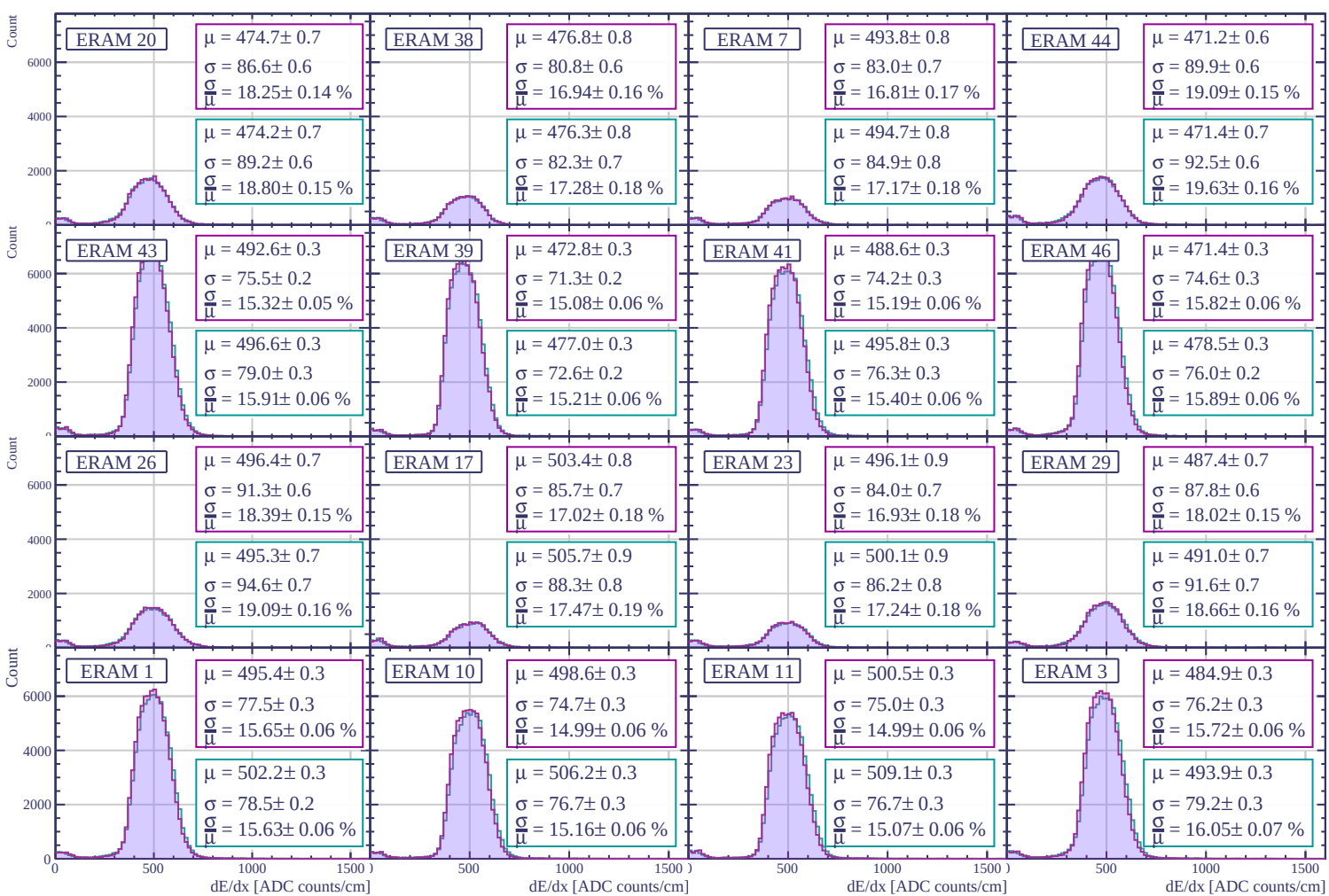
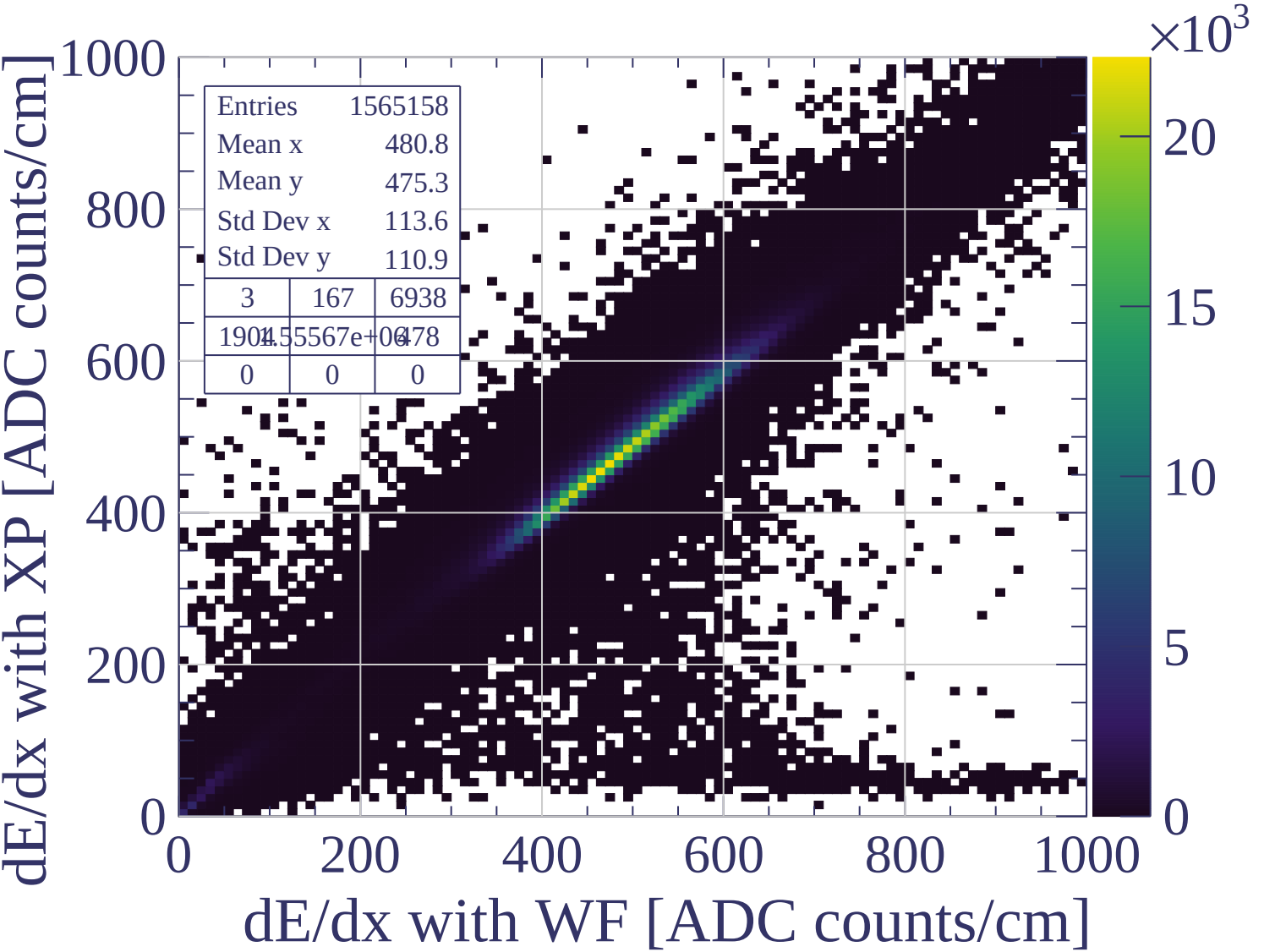
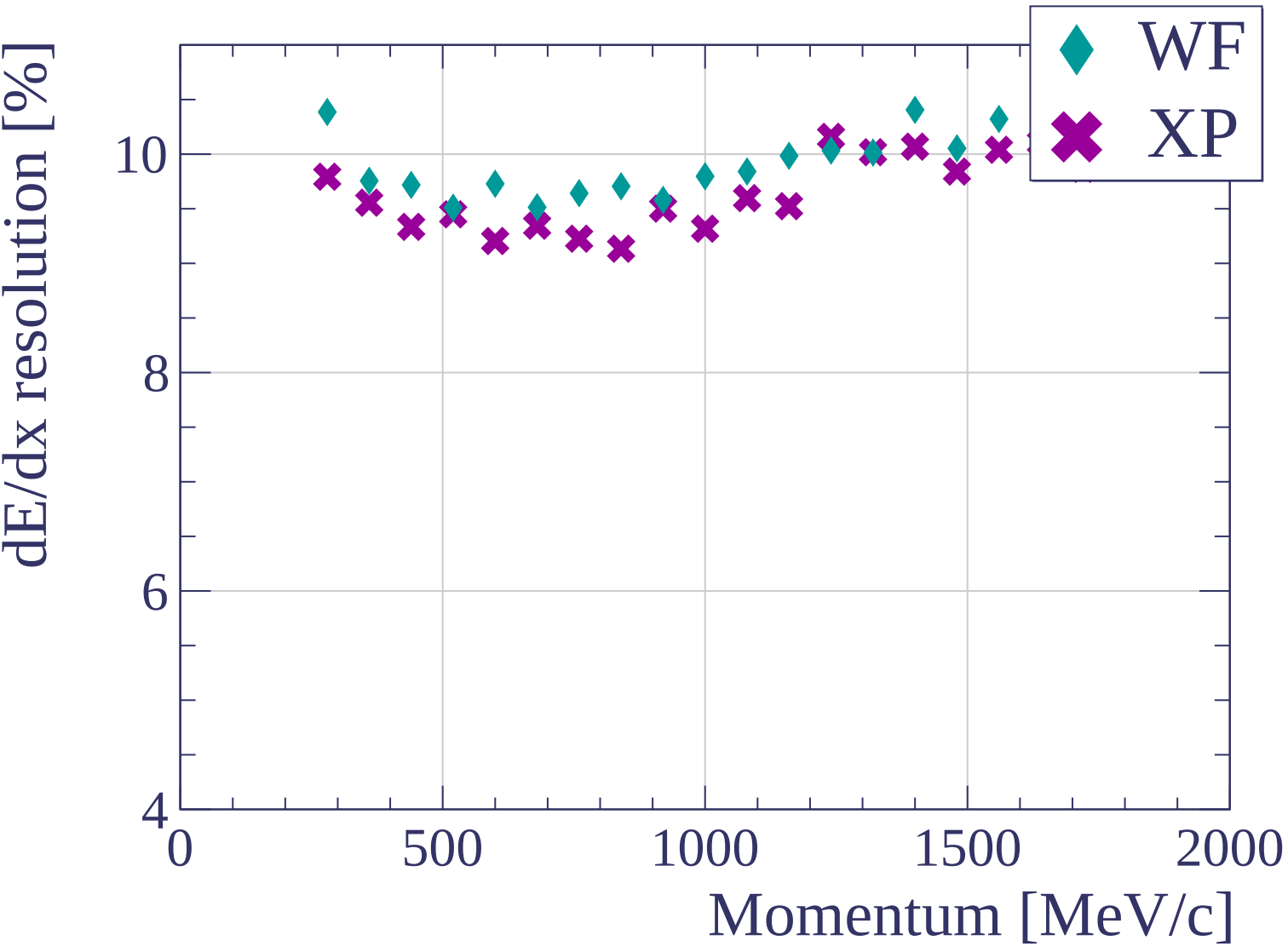


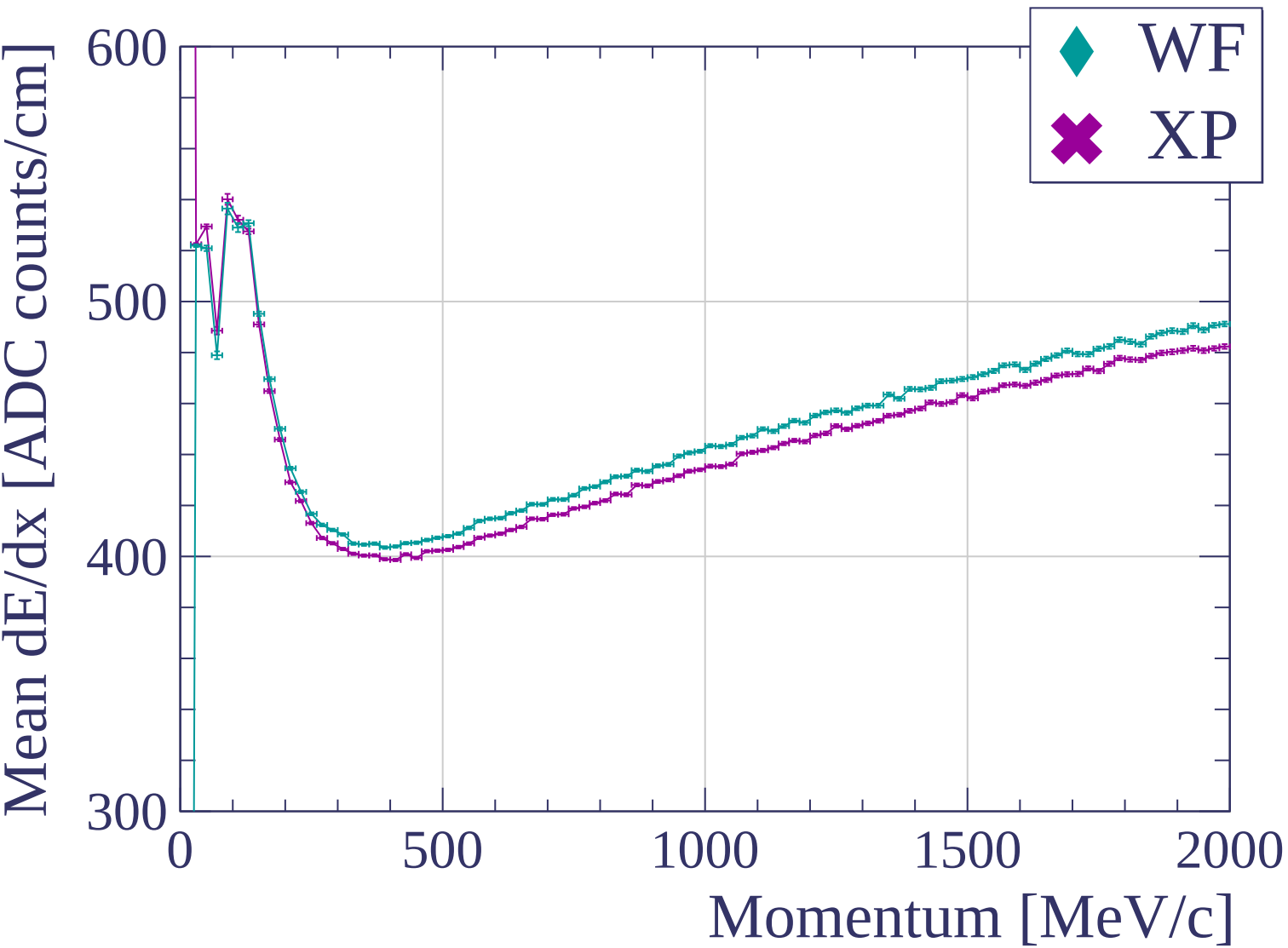


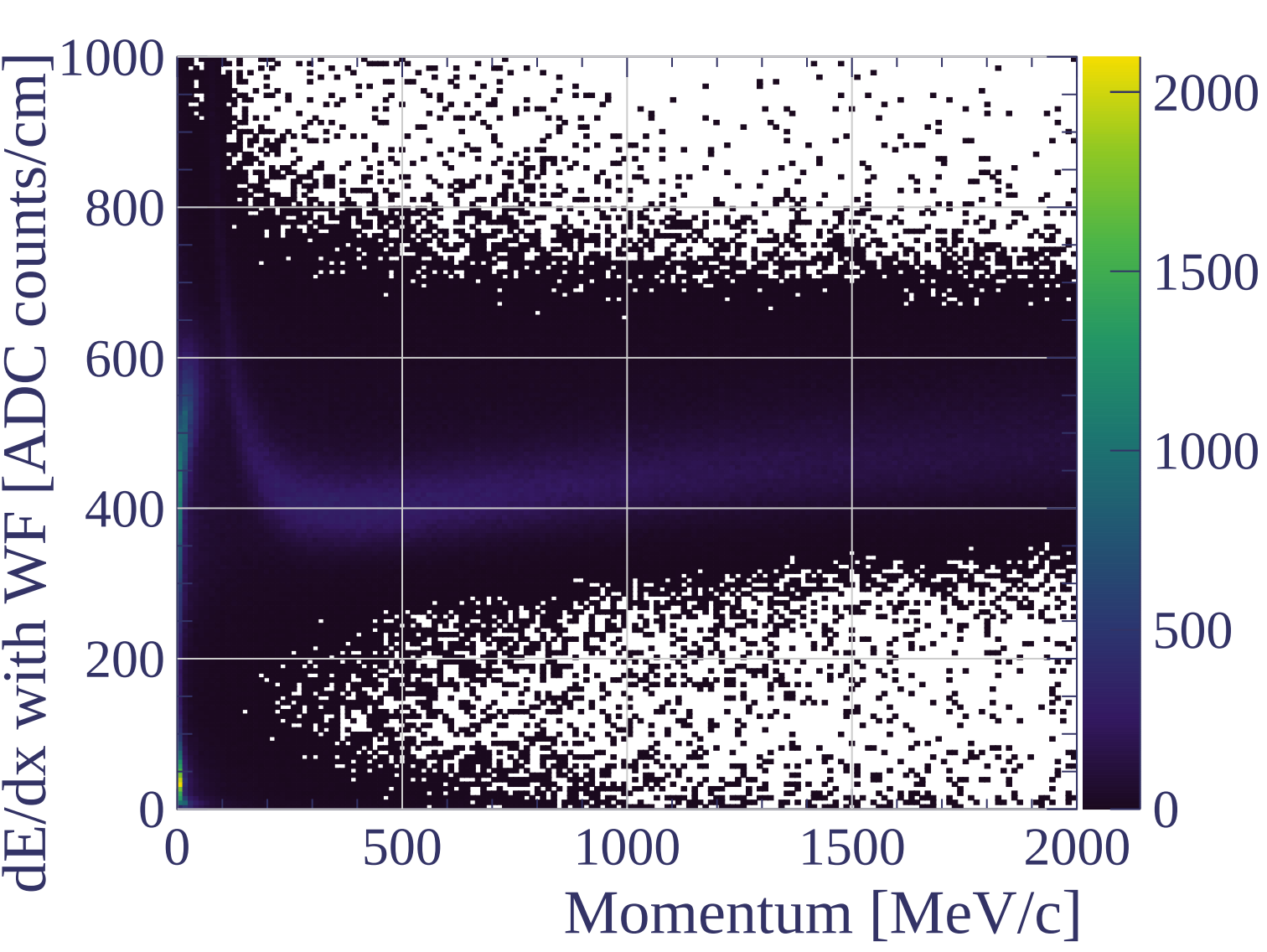


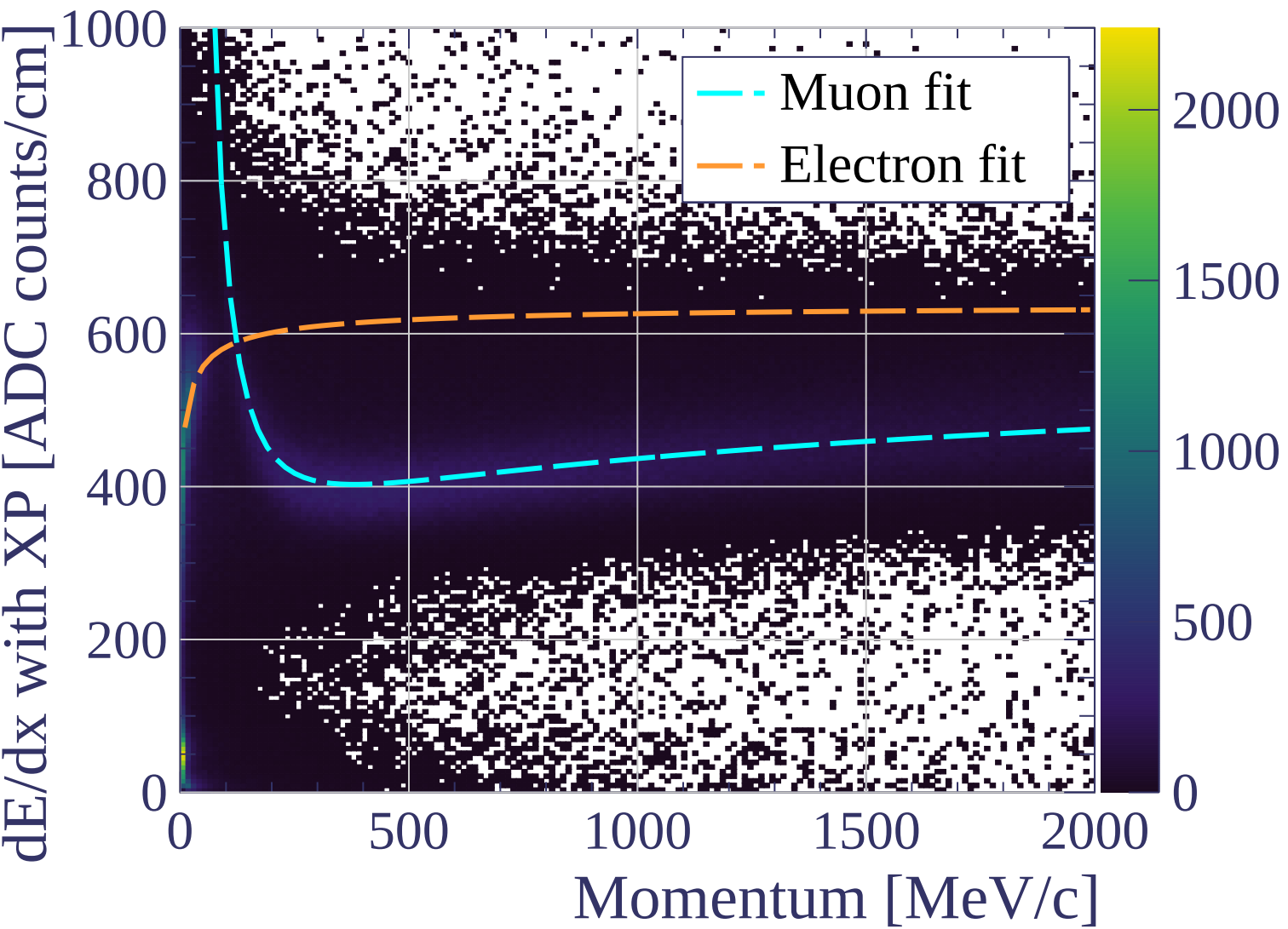




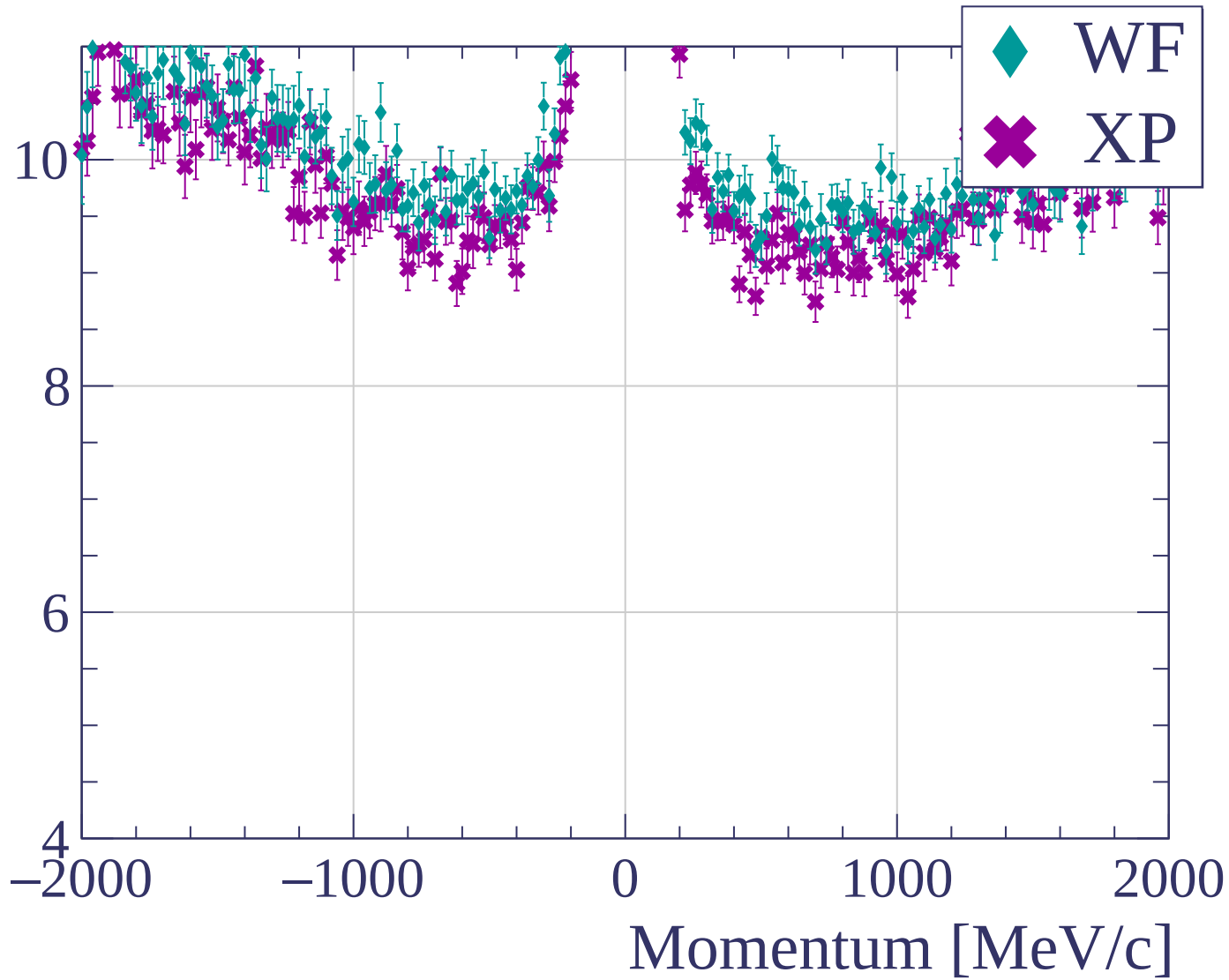


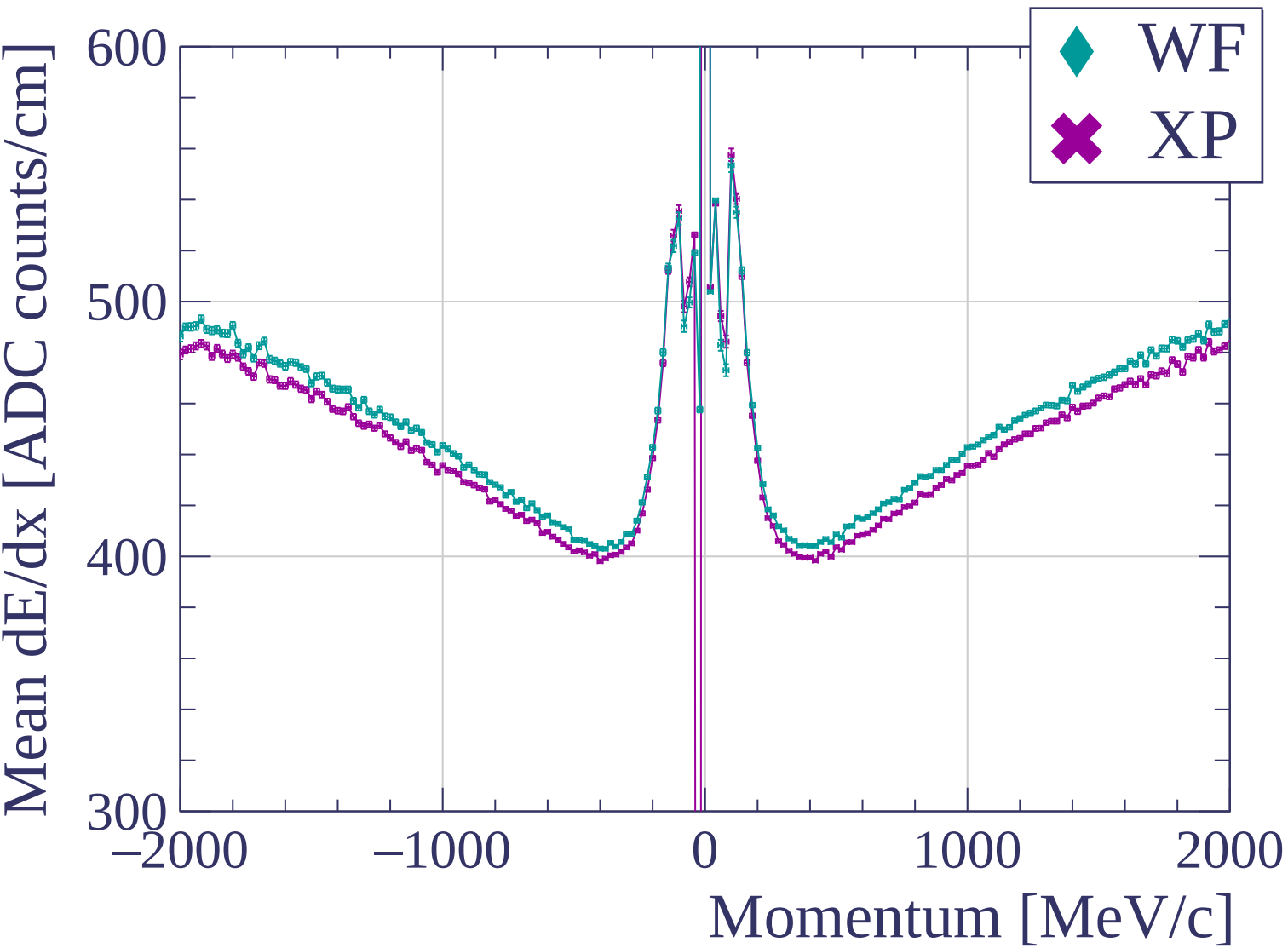


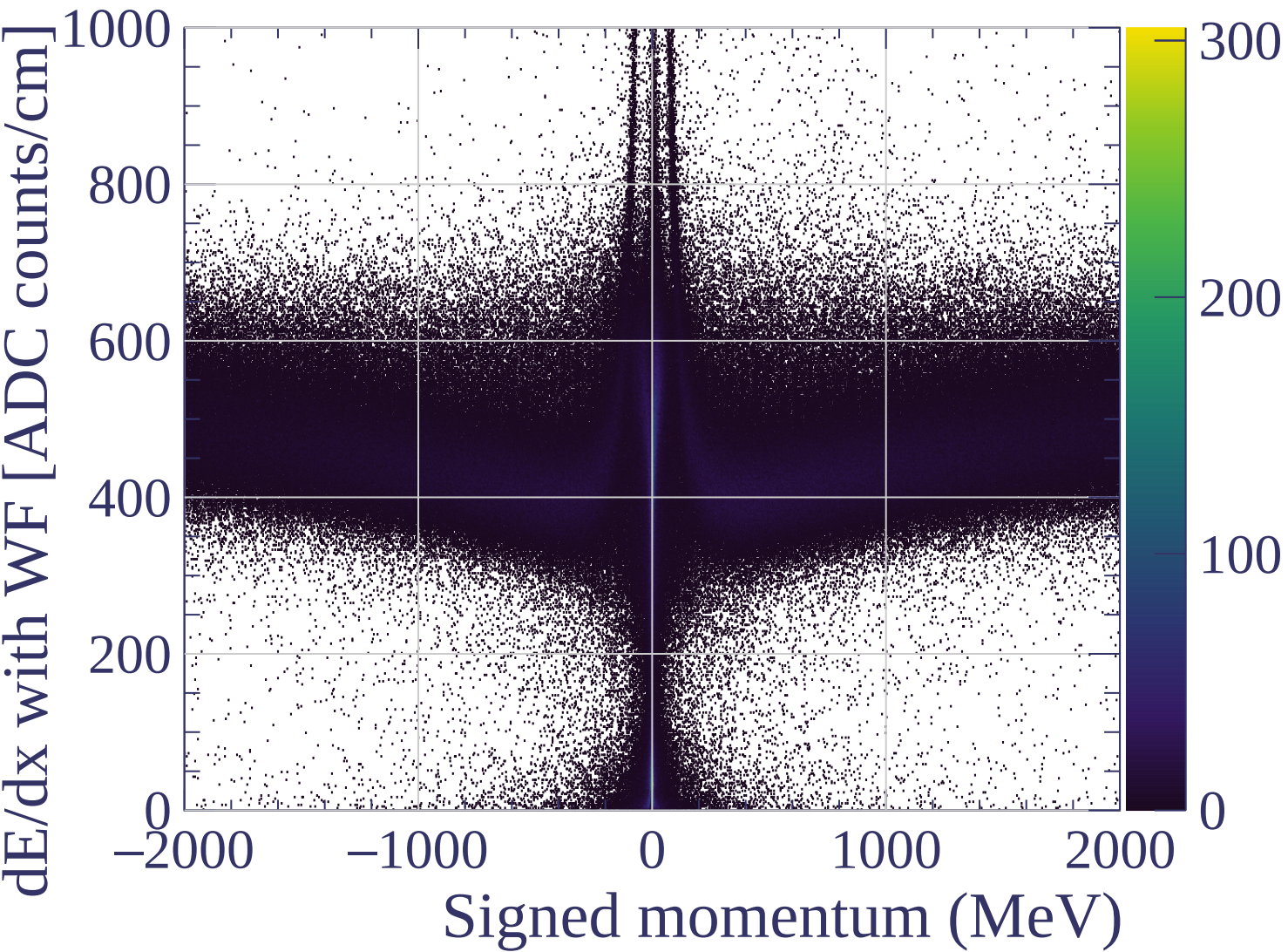


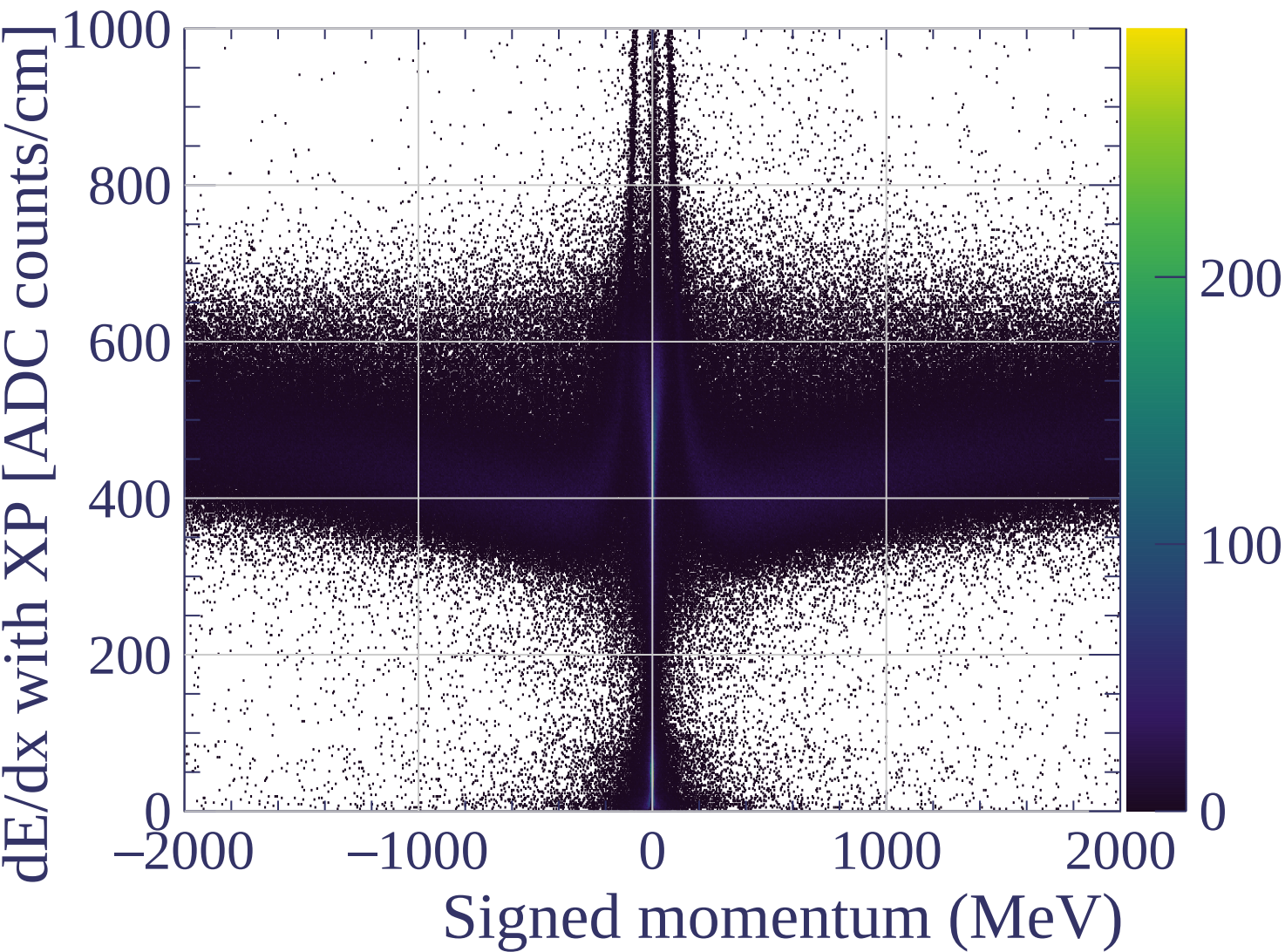


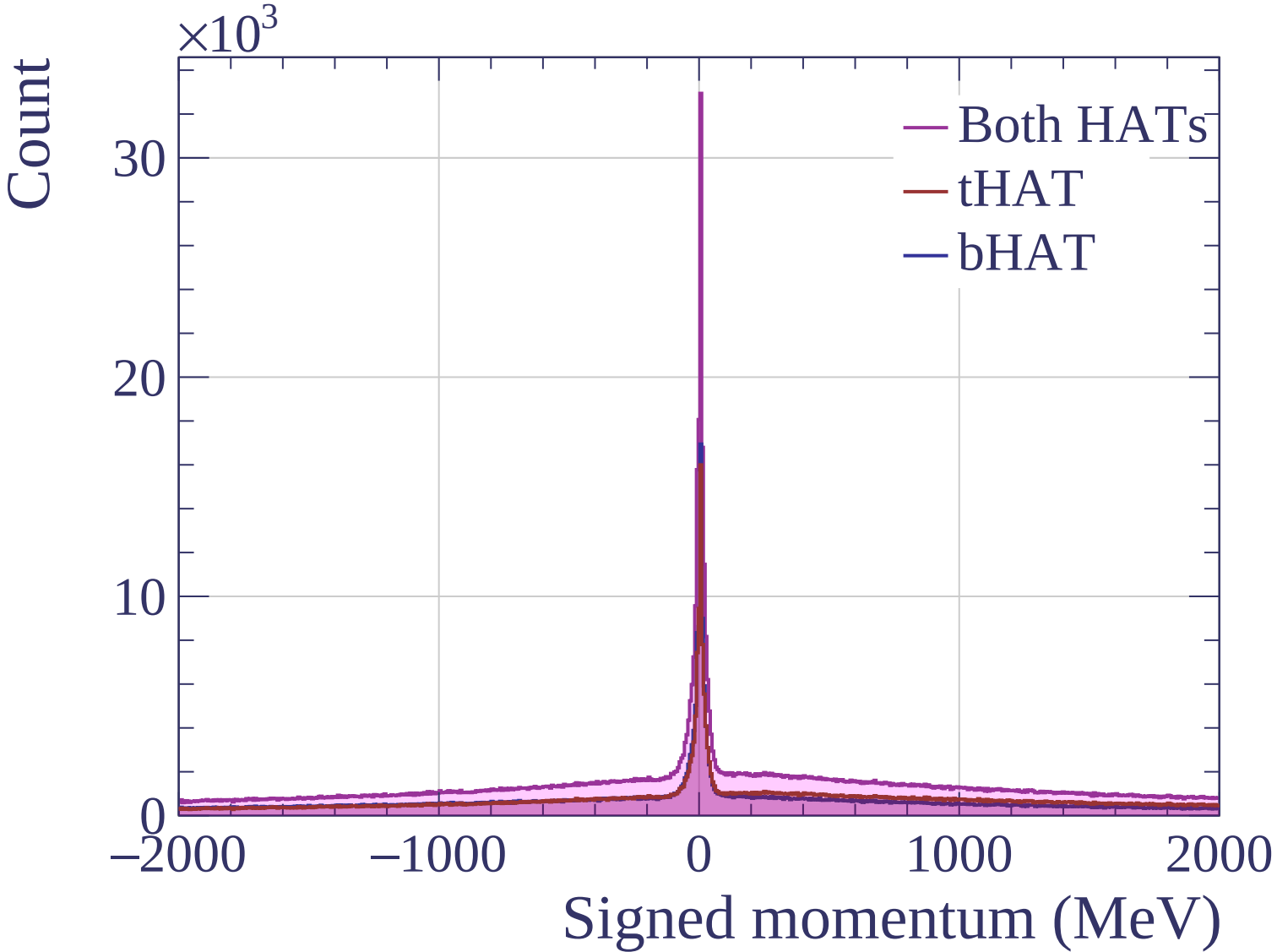
dE/dx resolution [%]

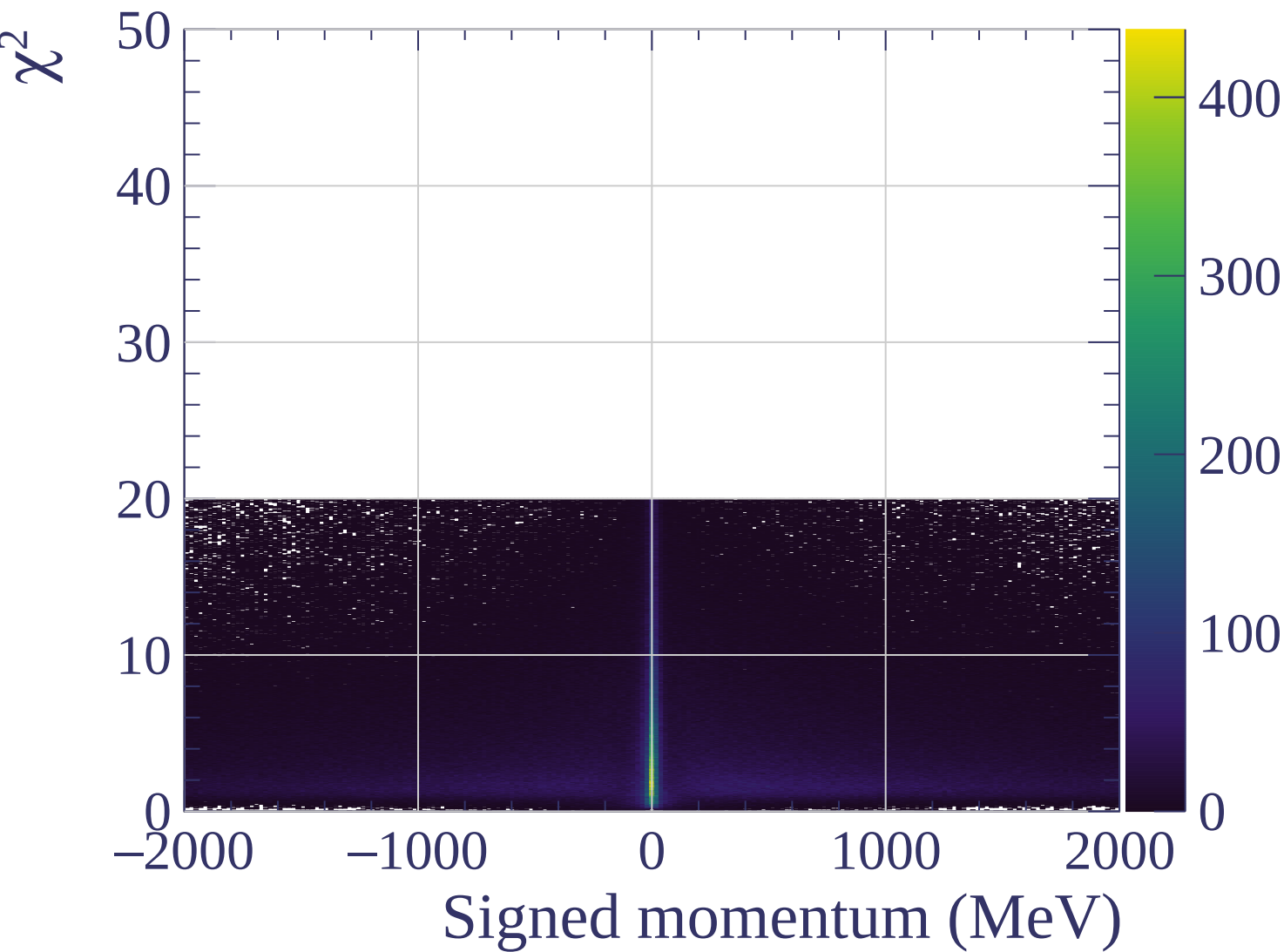


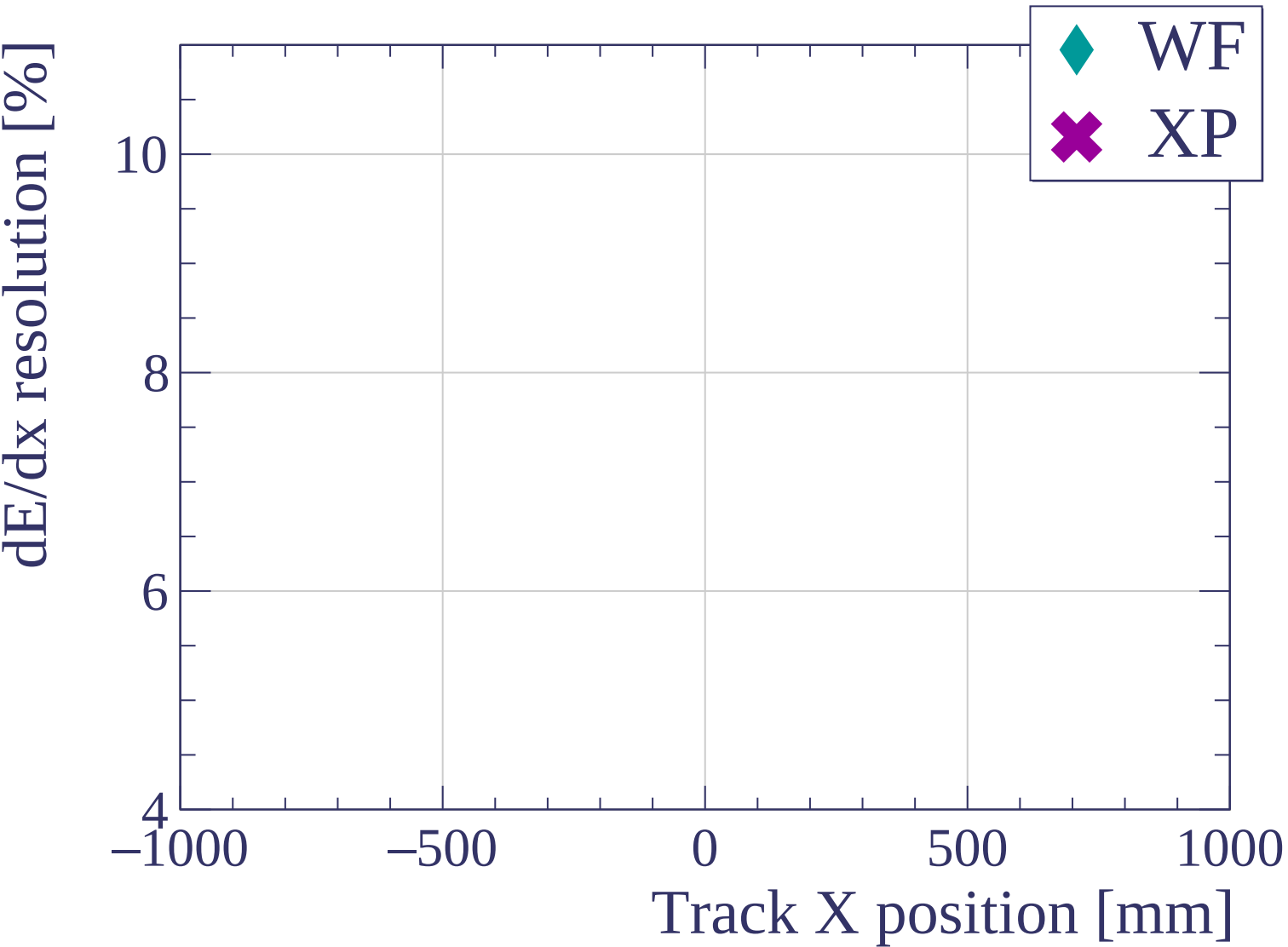


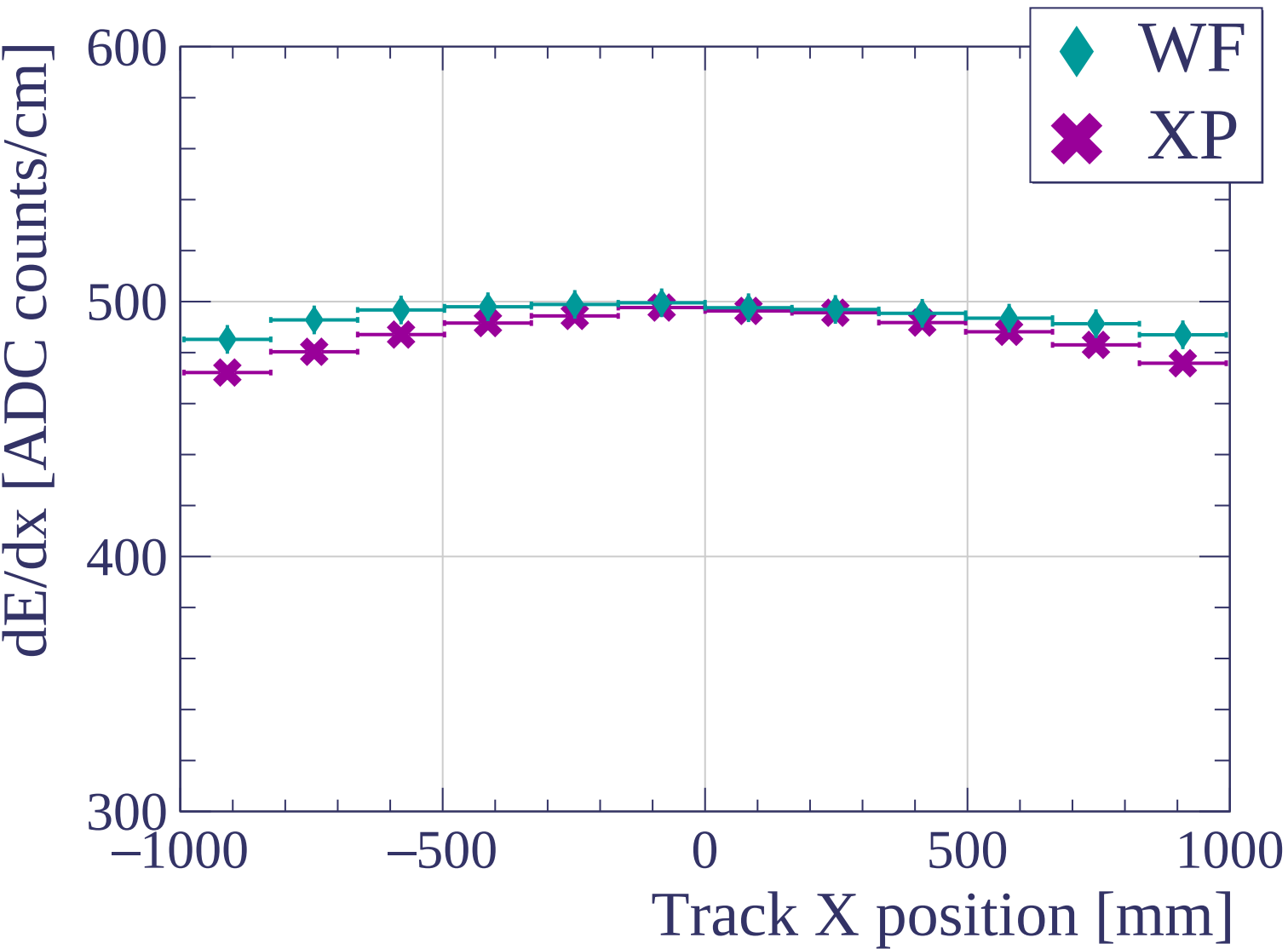


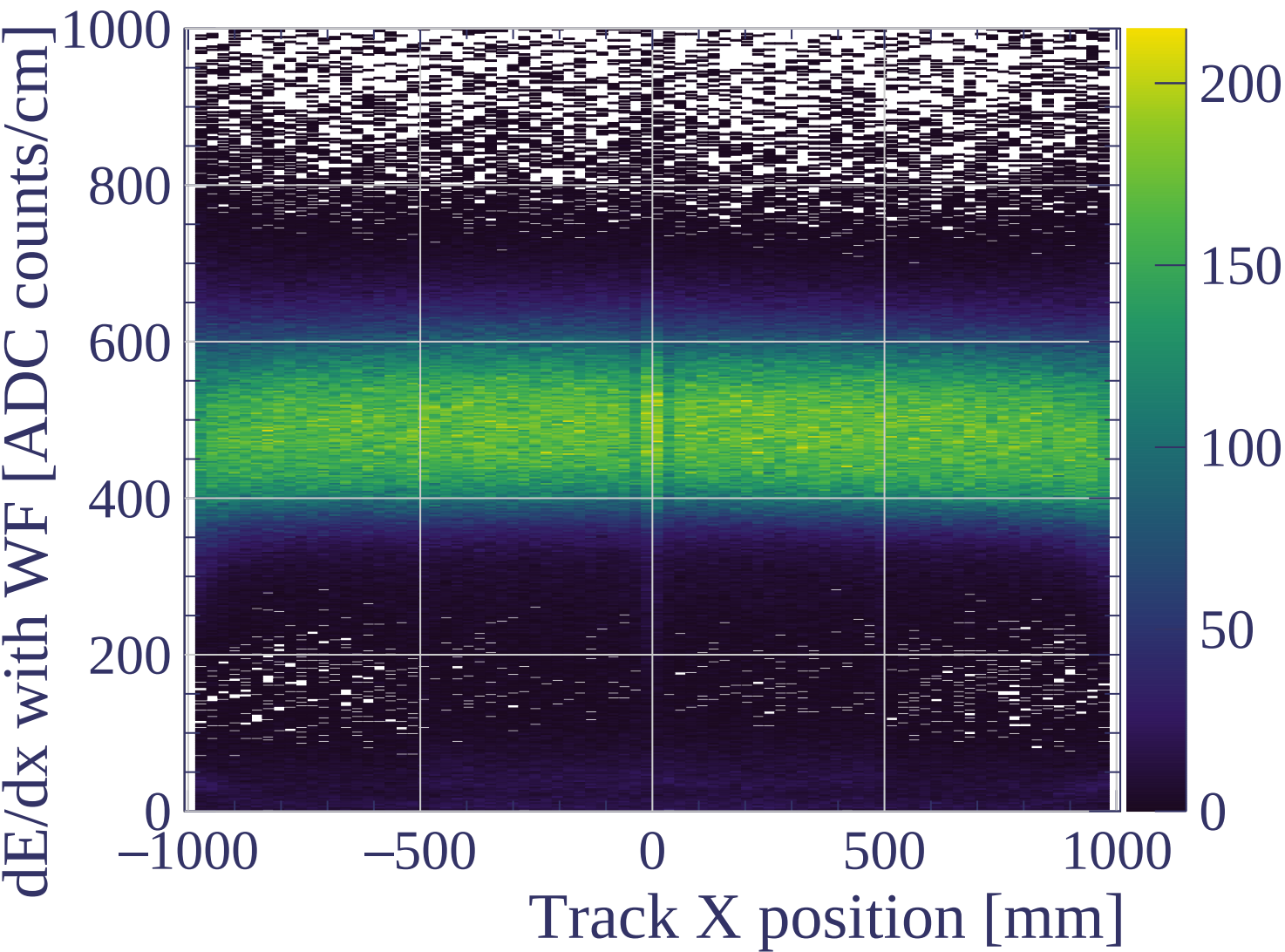


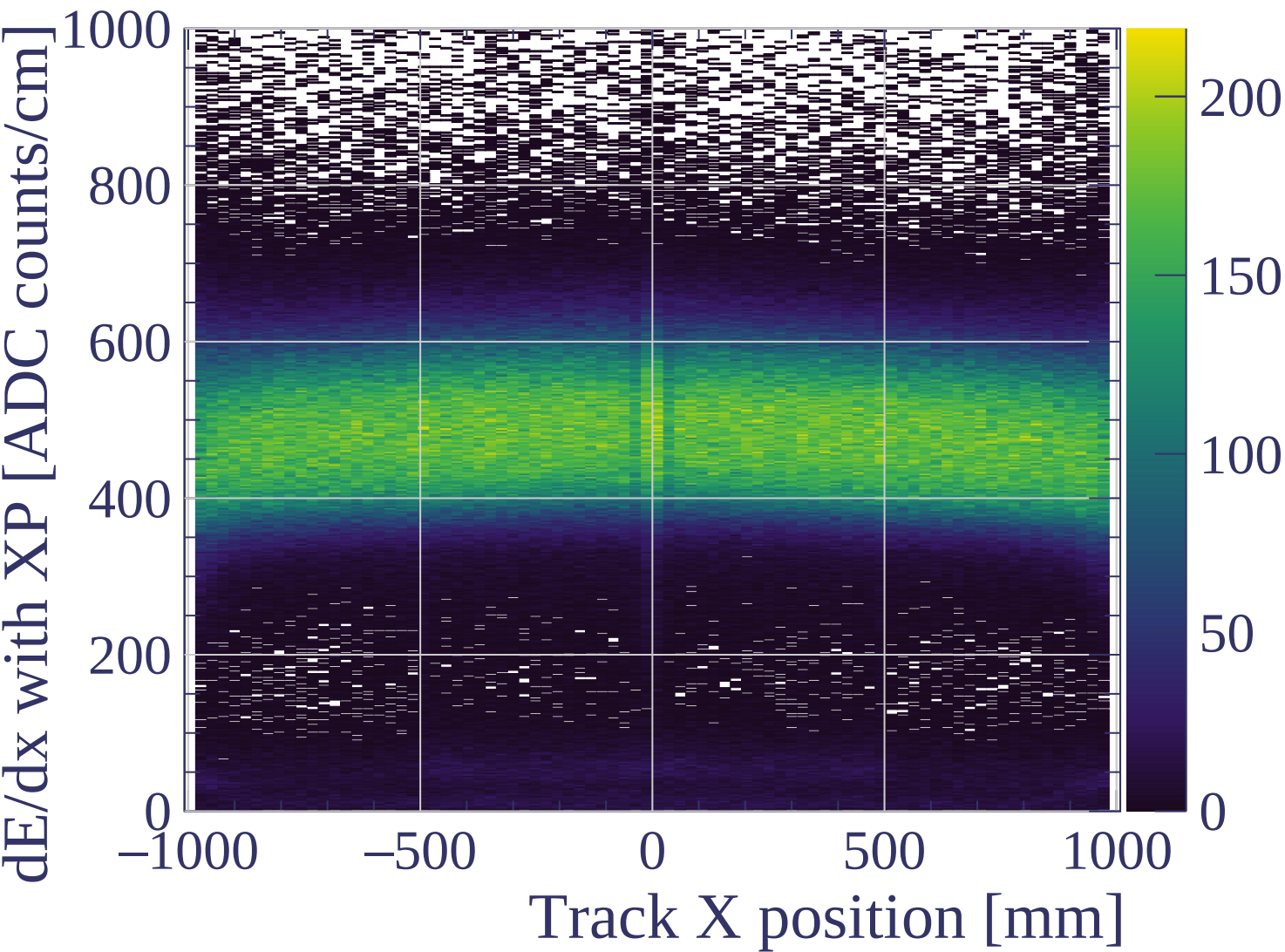




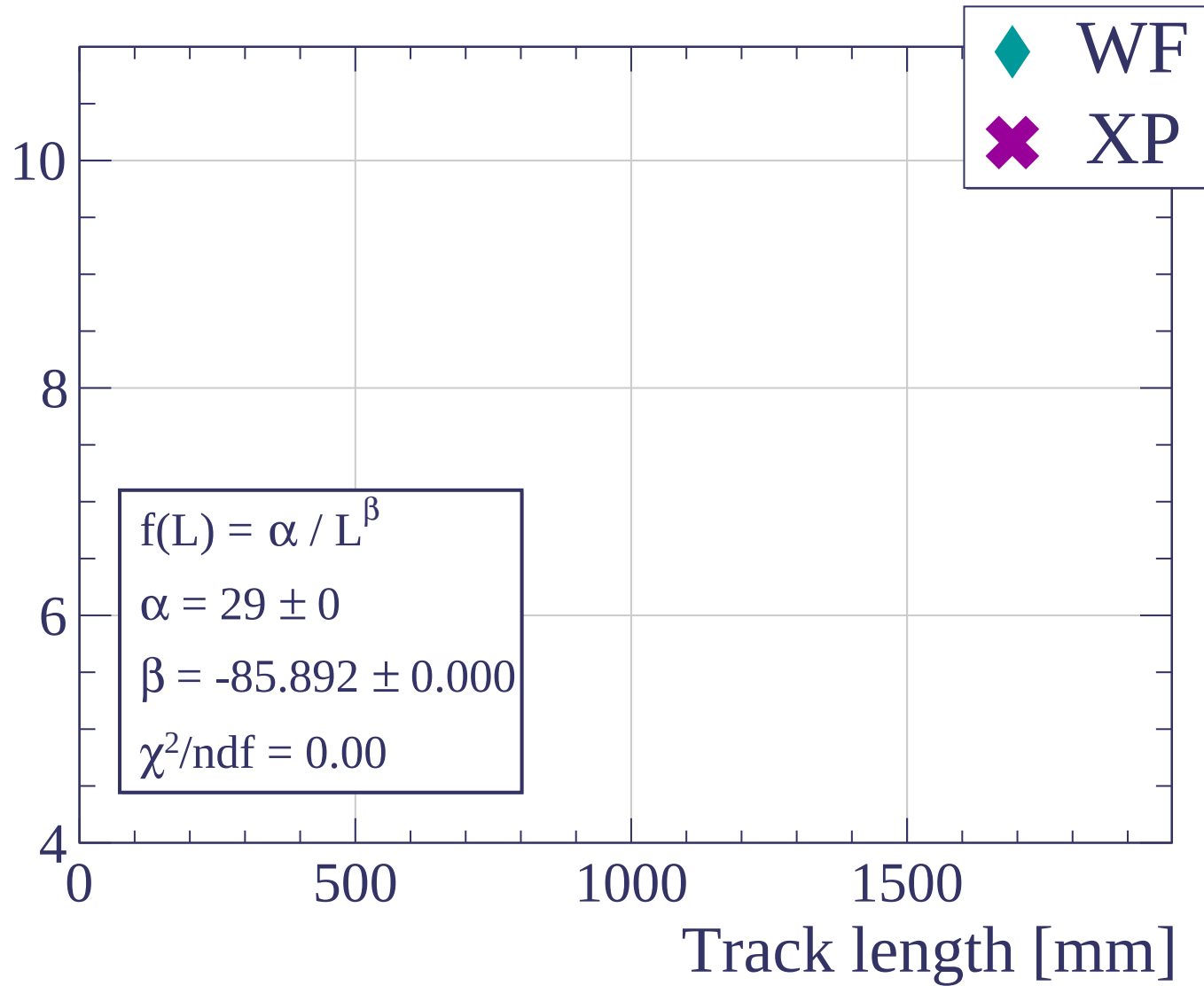


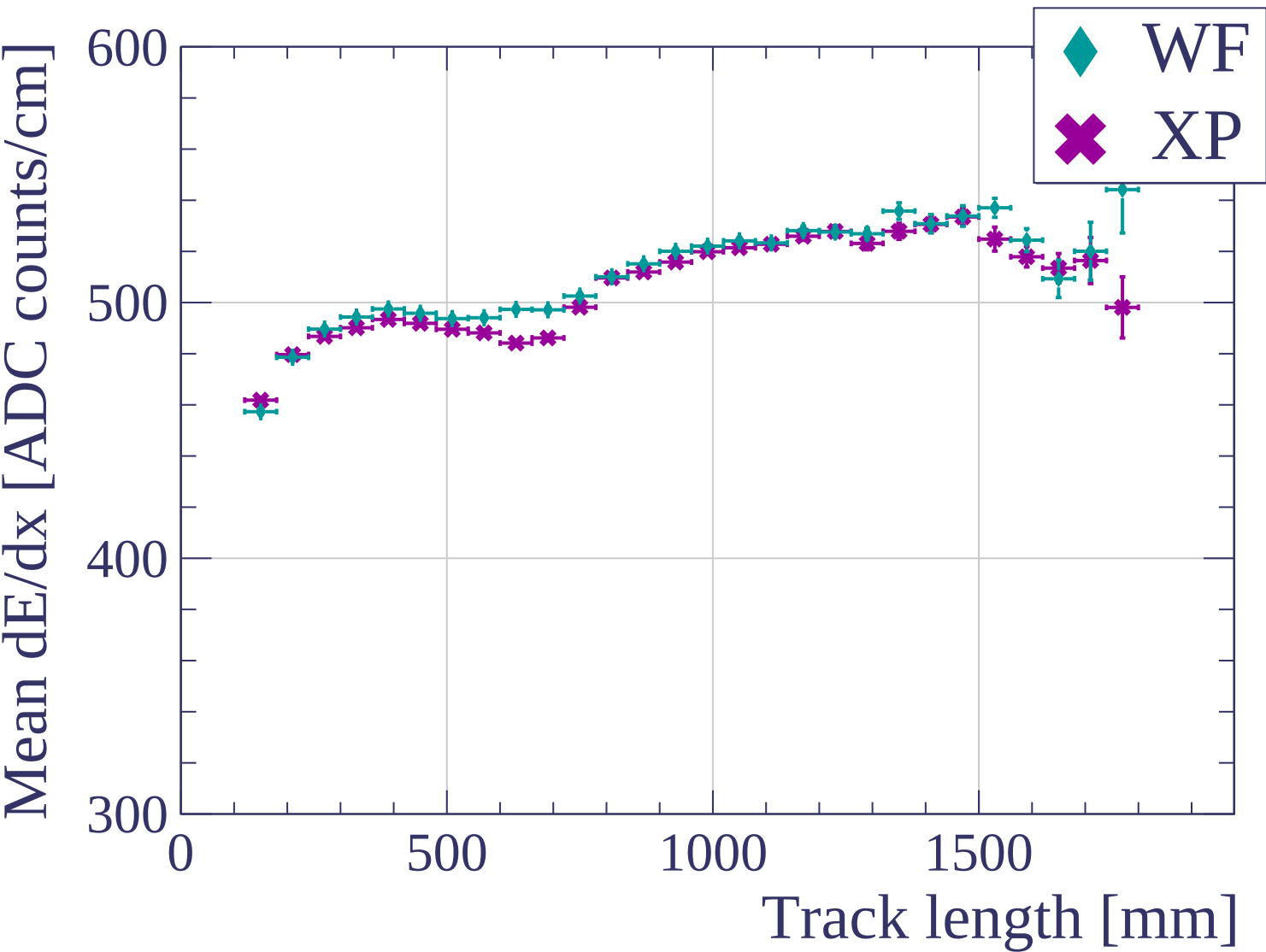


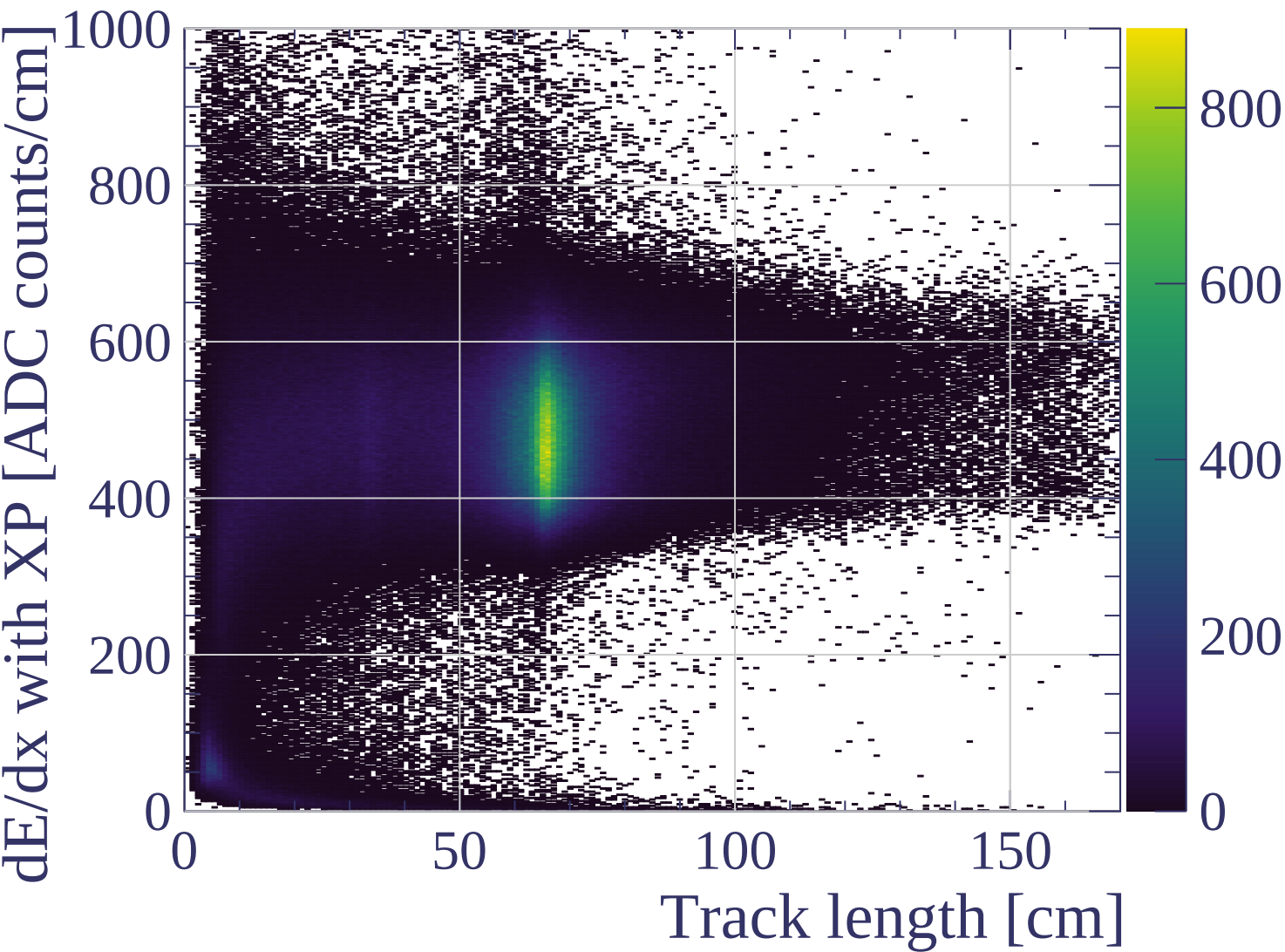


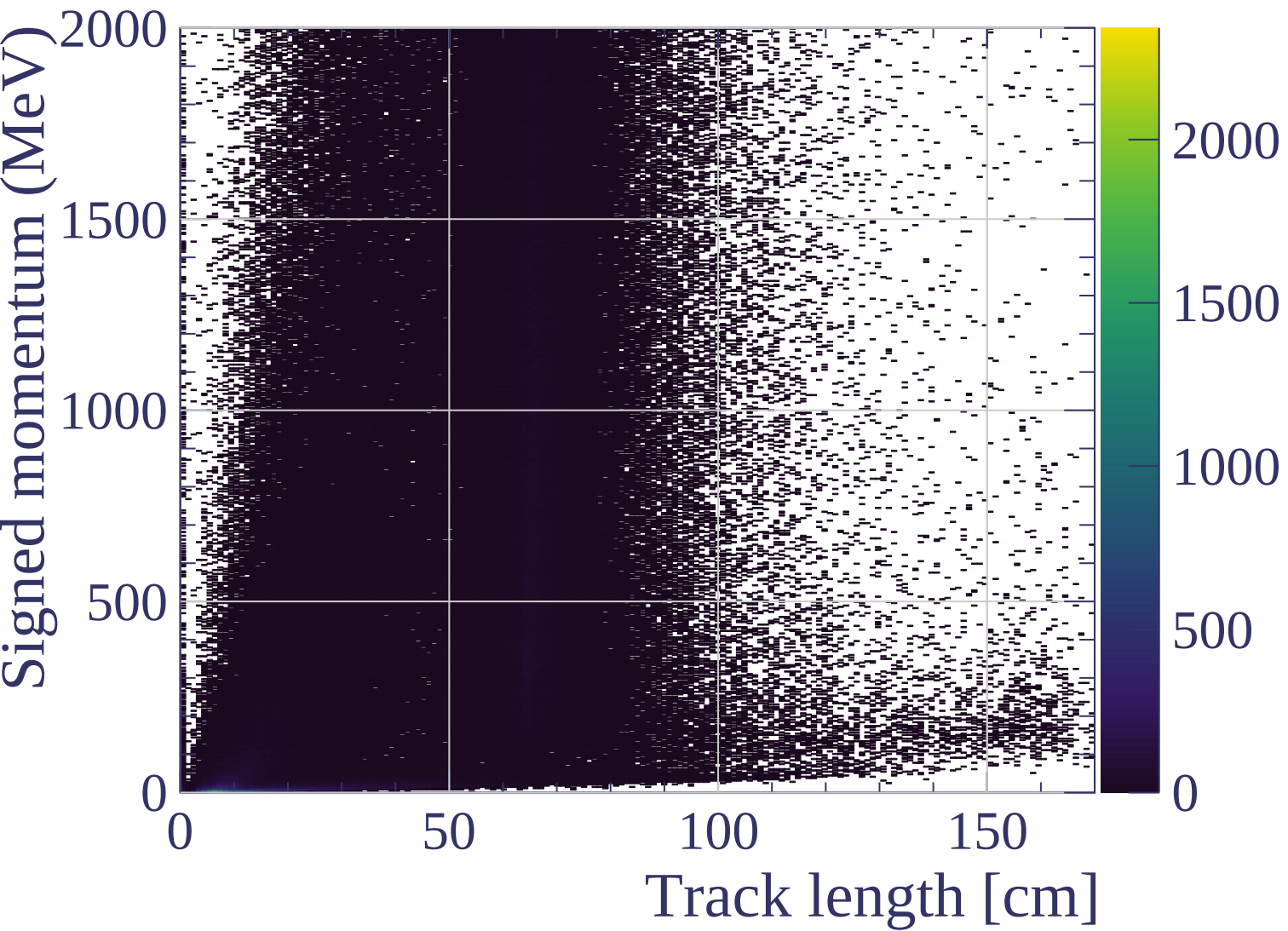


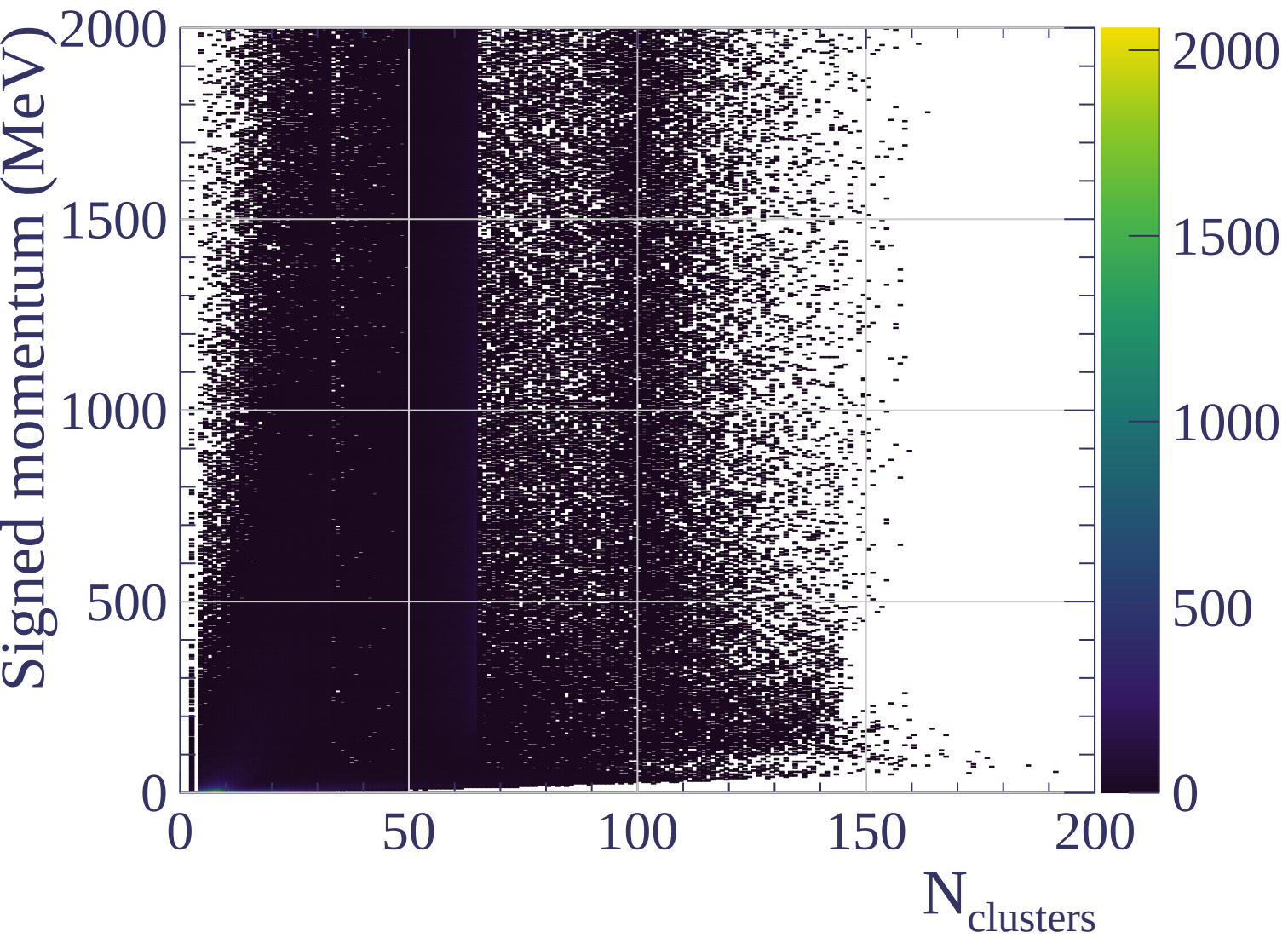
dE/dx resolution [%]

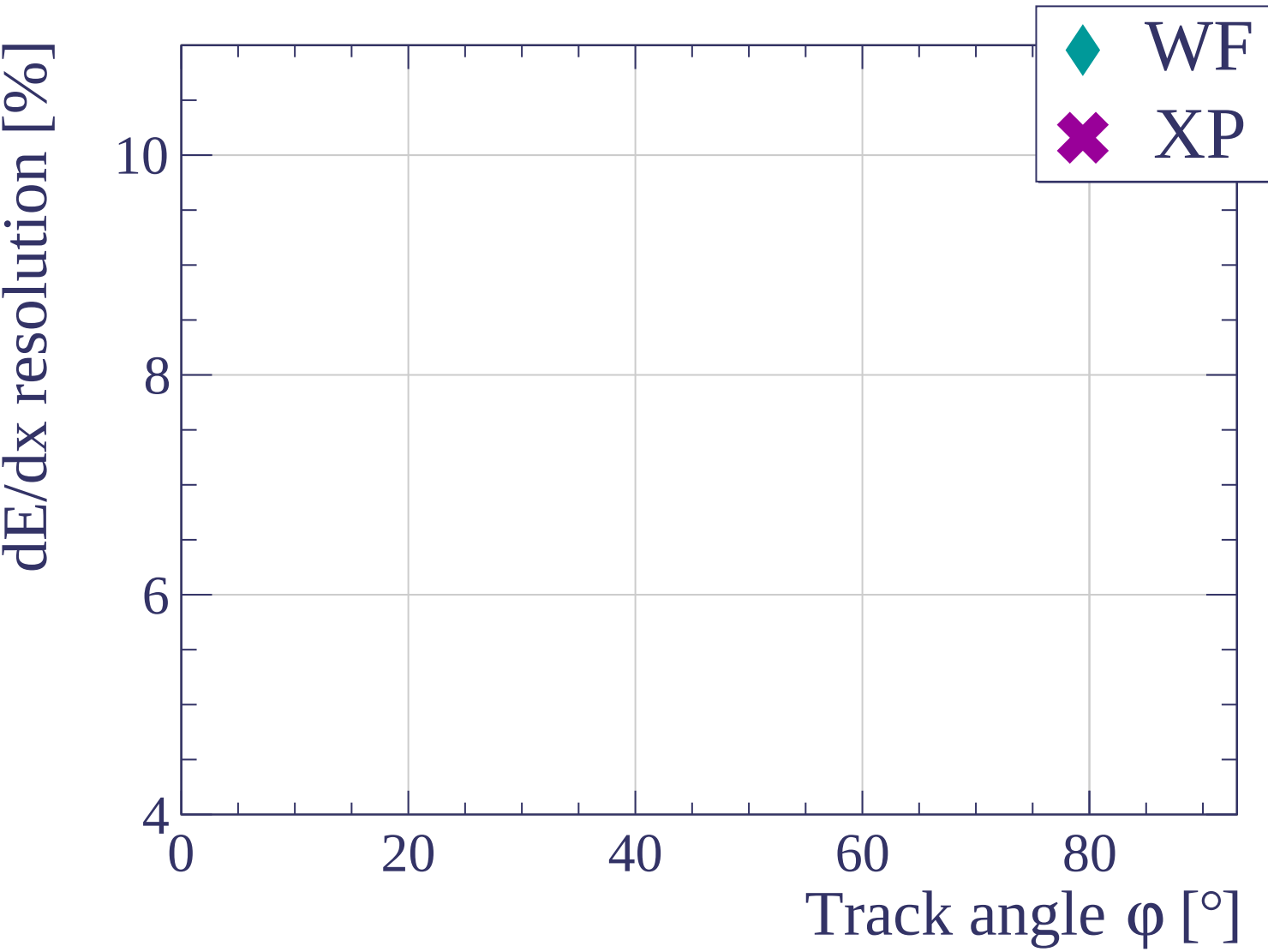


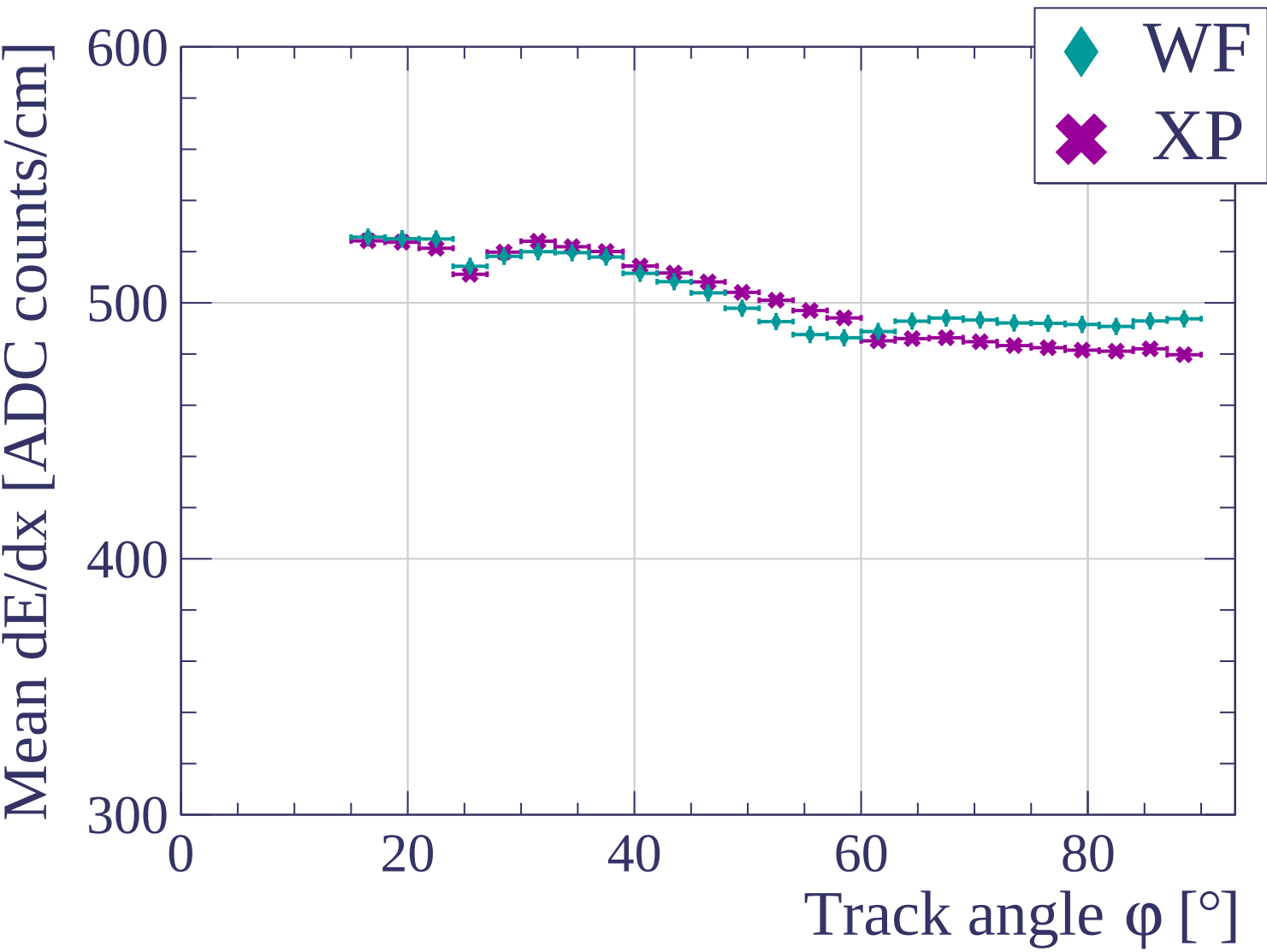


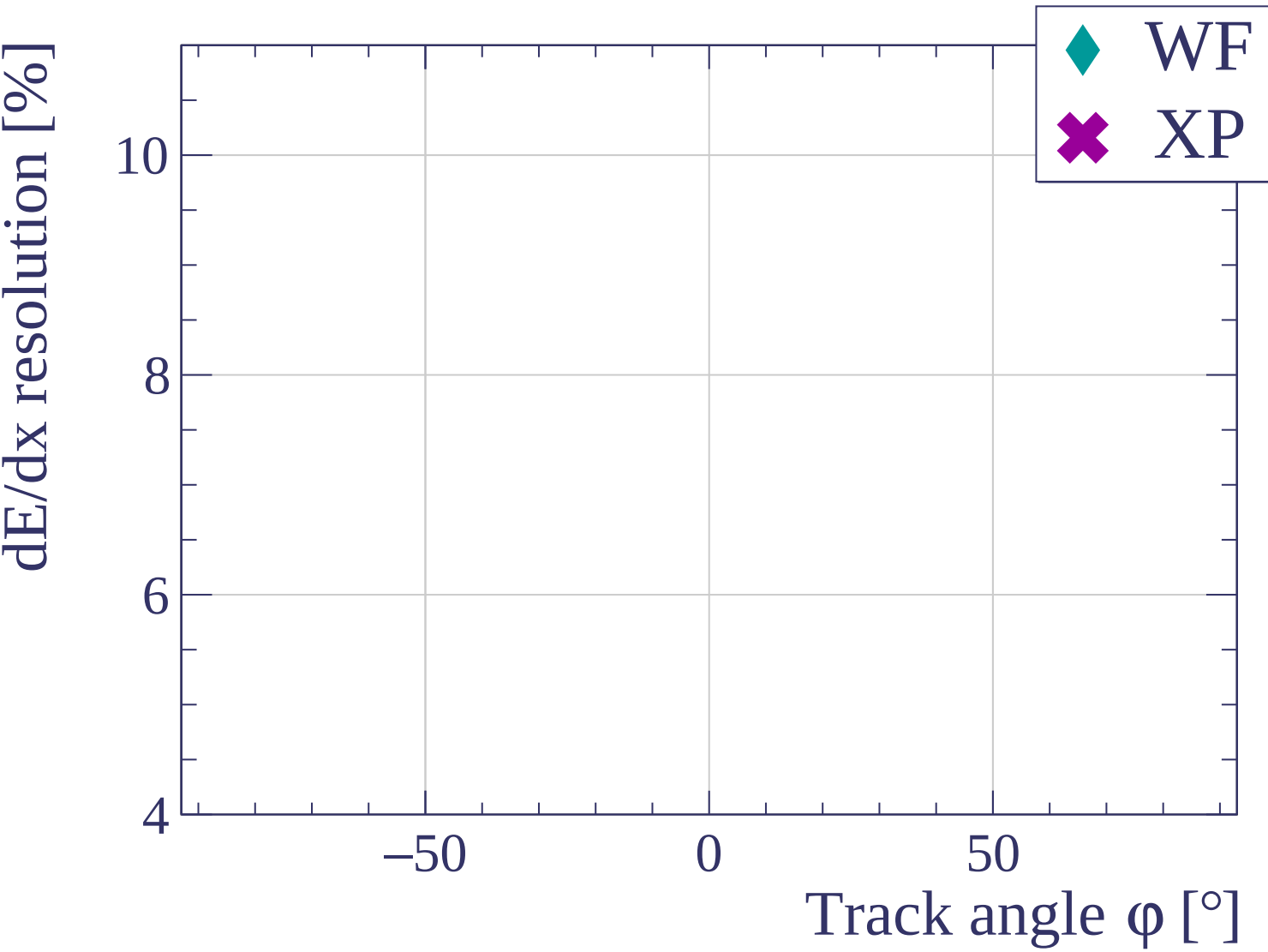


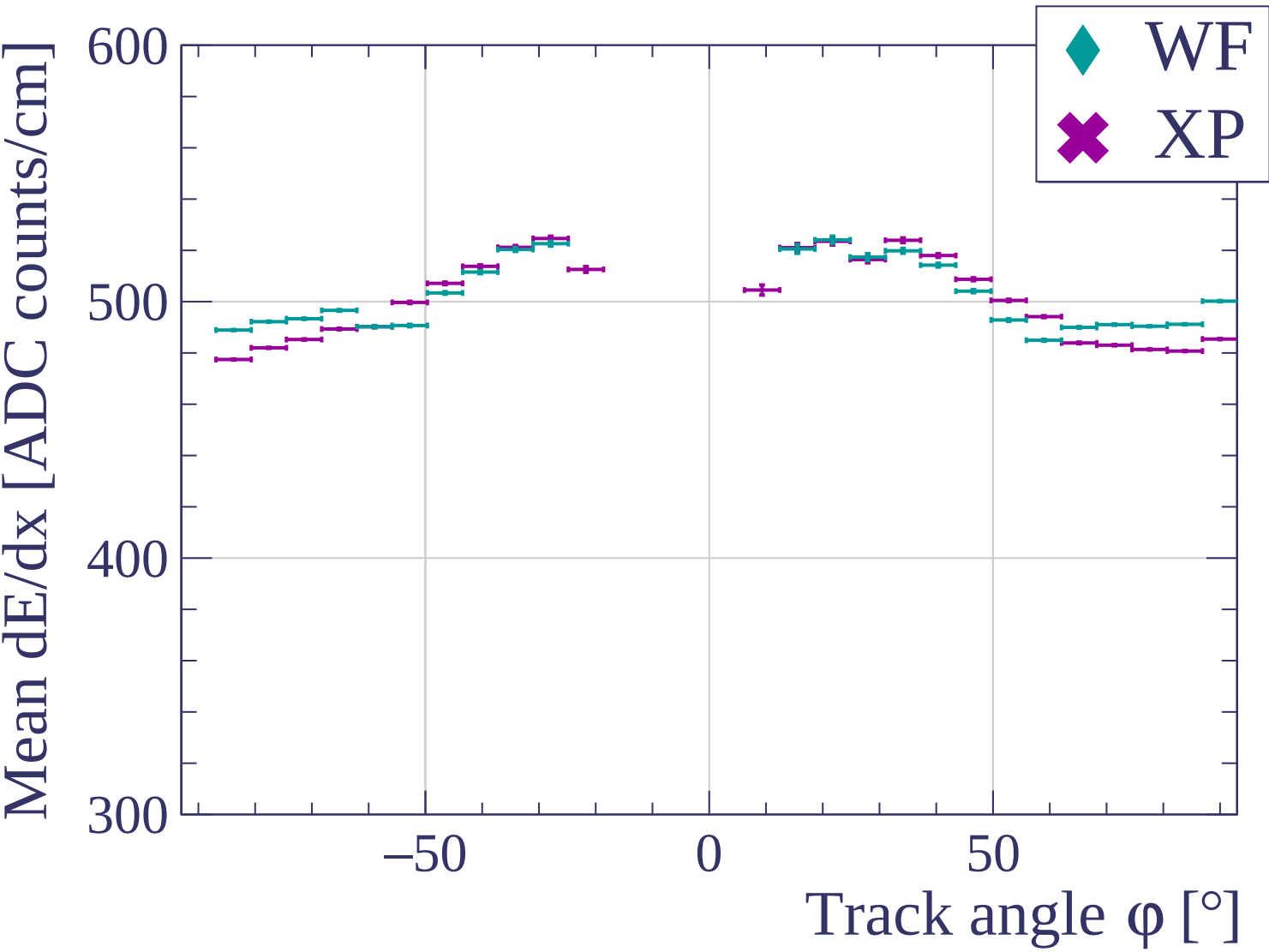


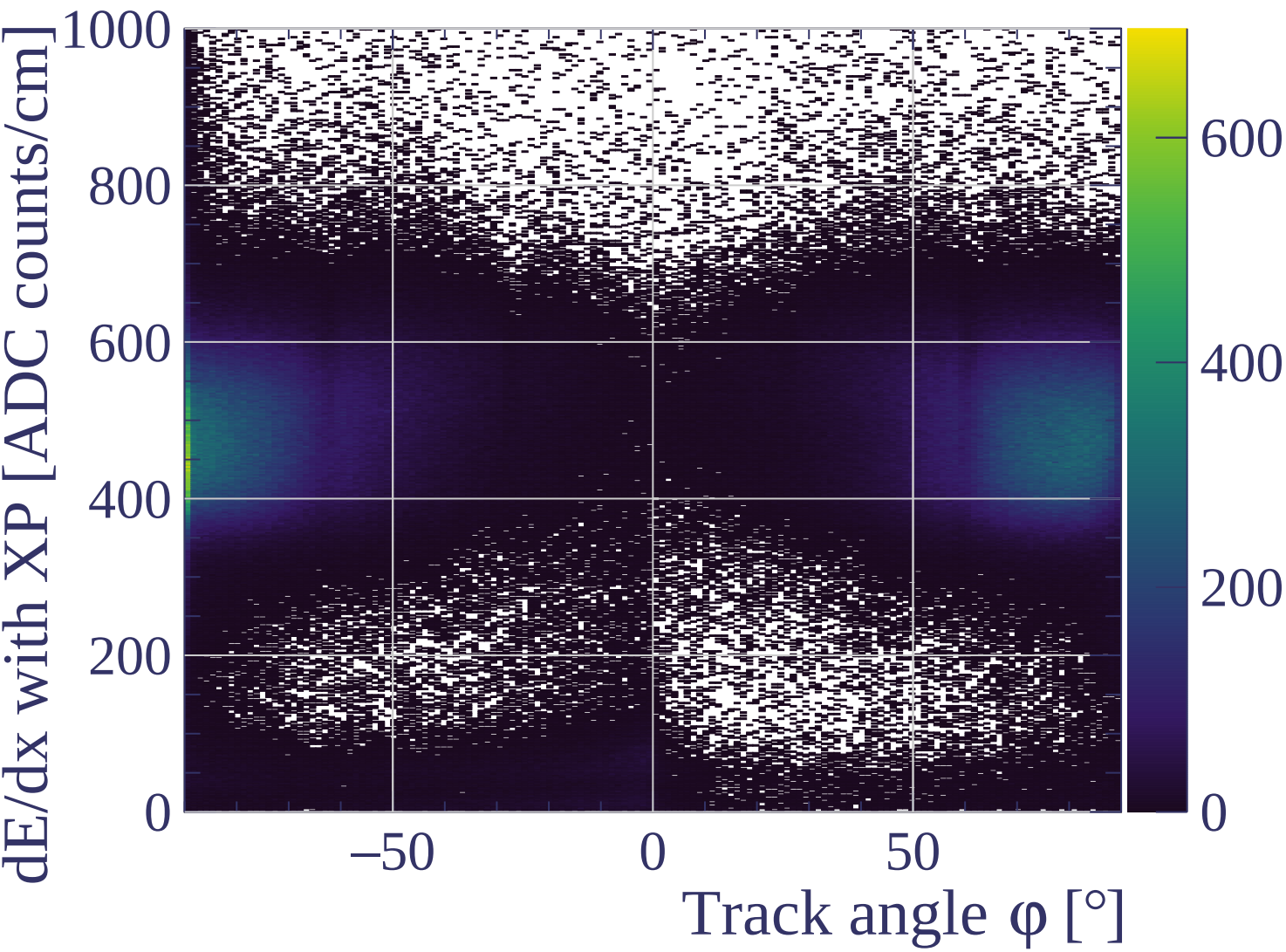




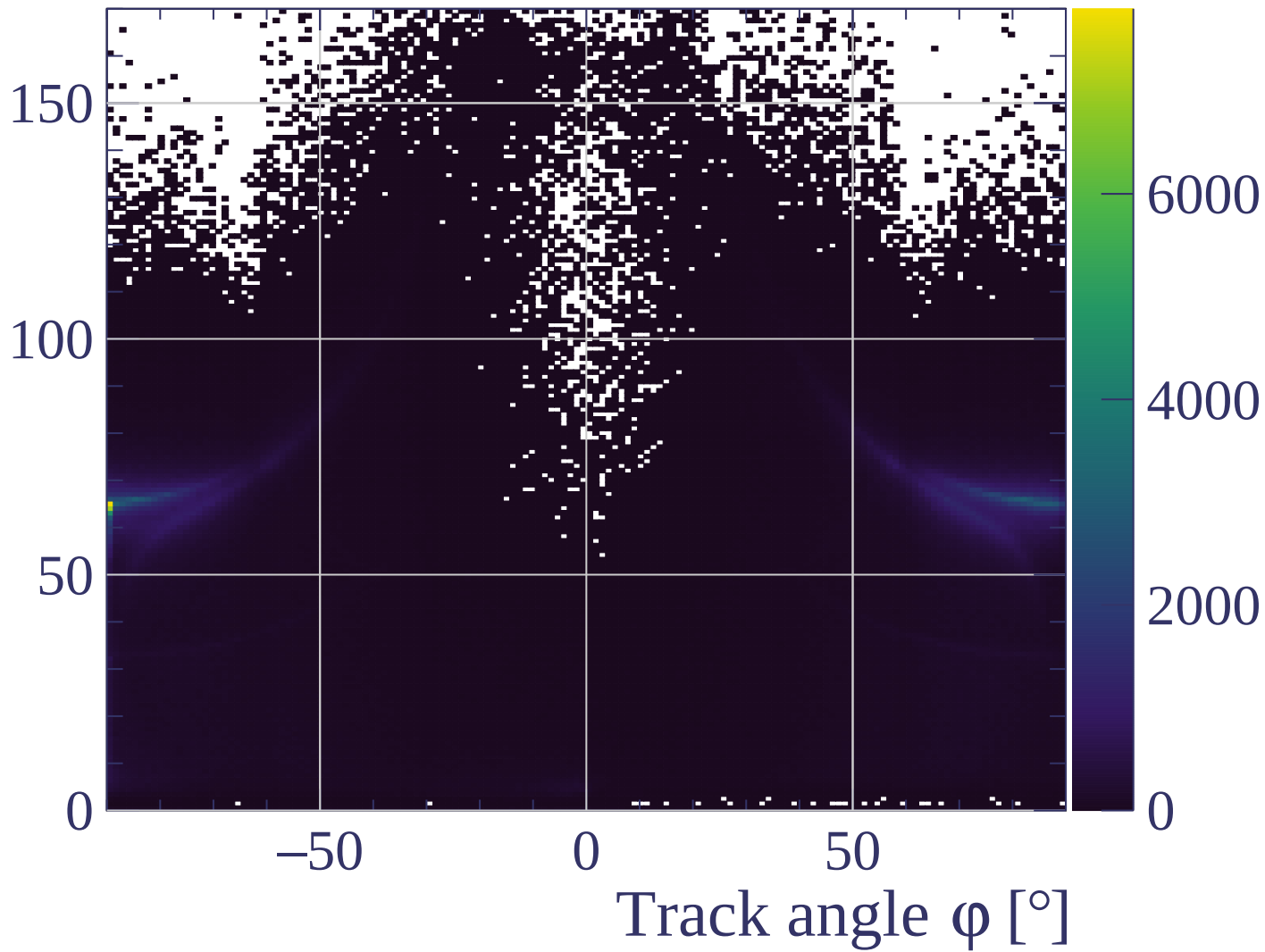


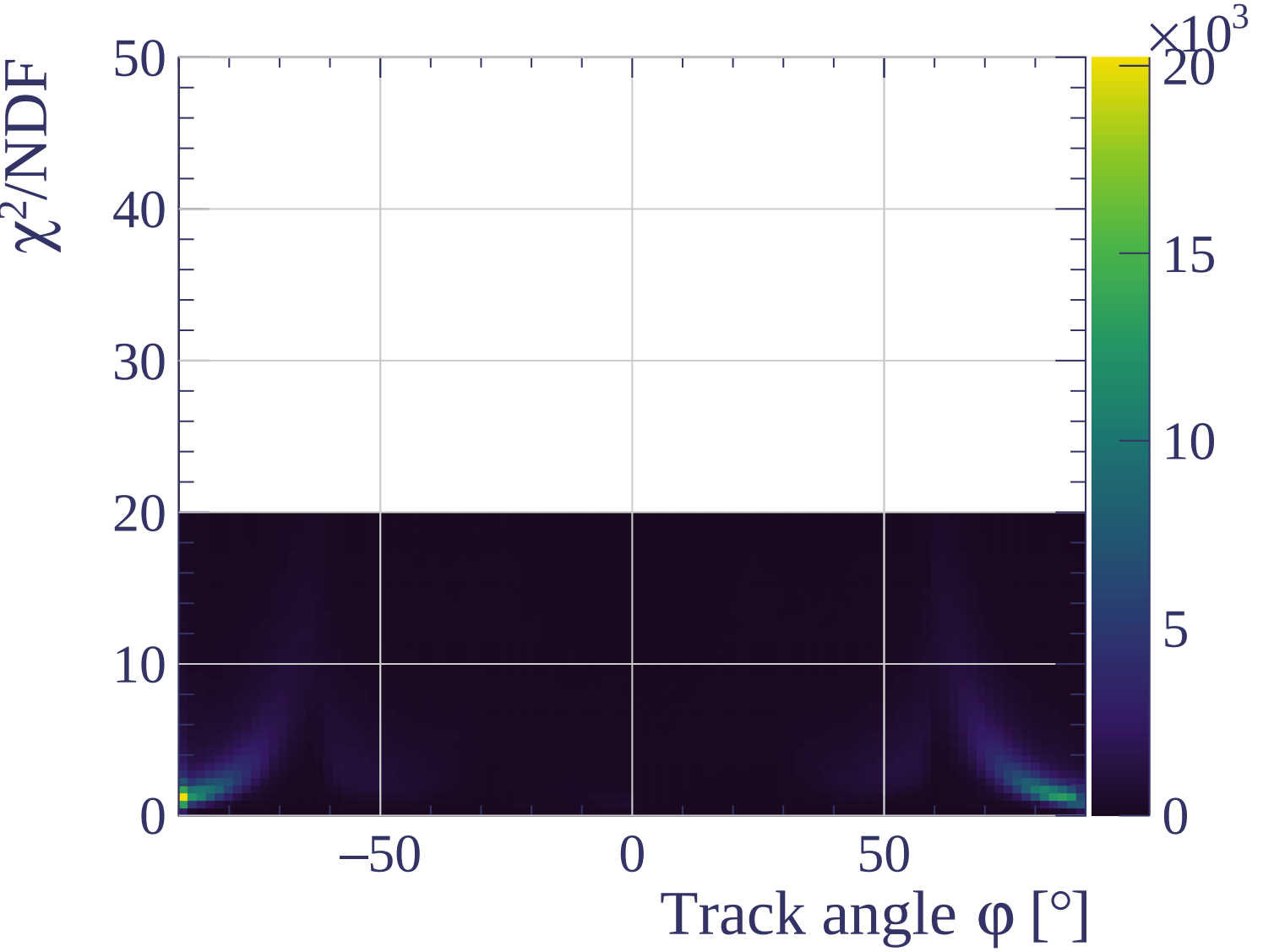


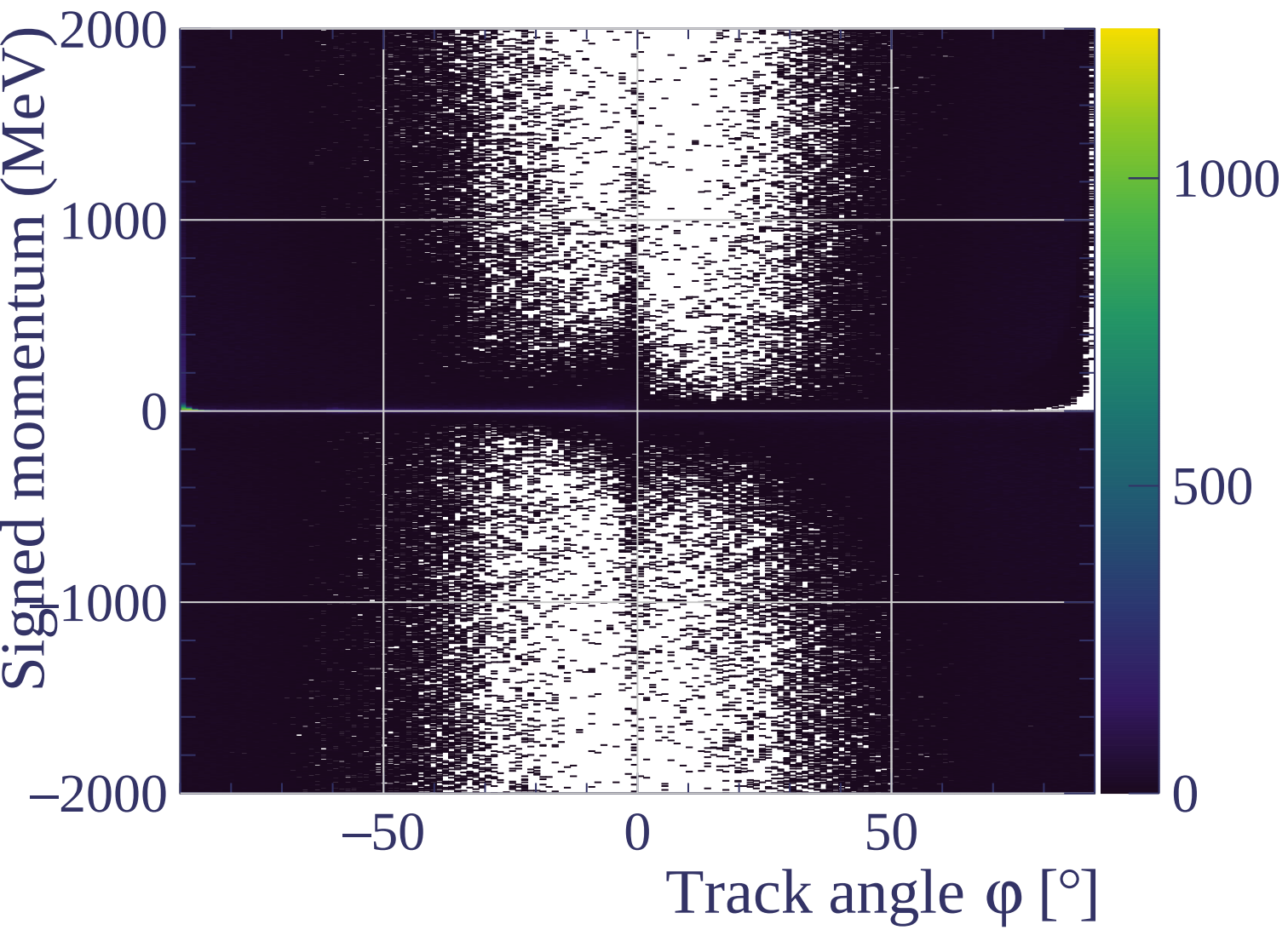


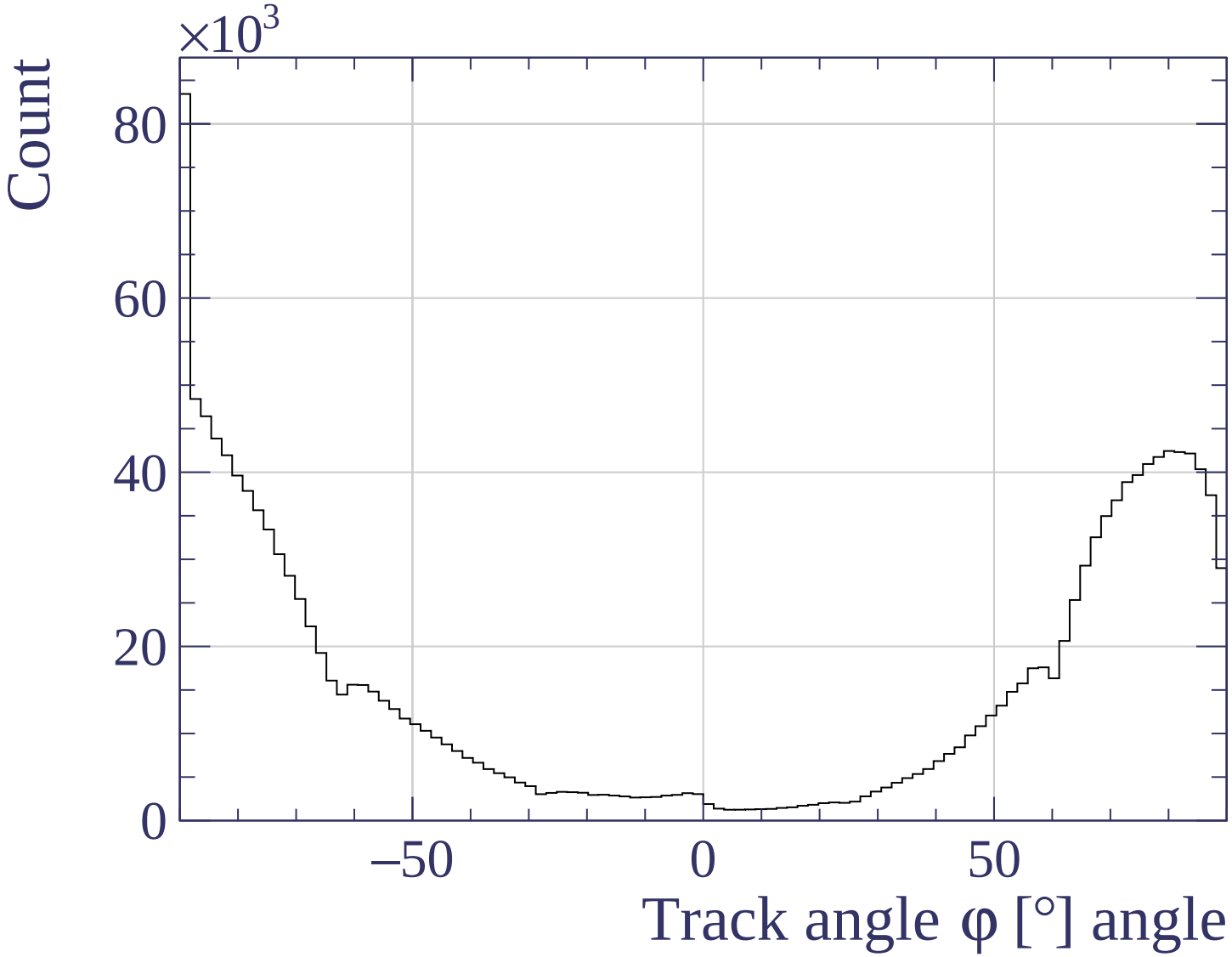


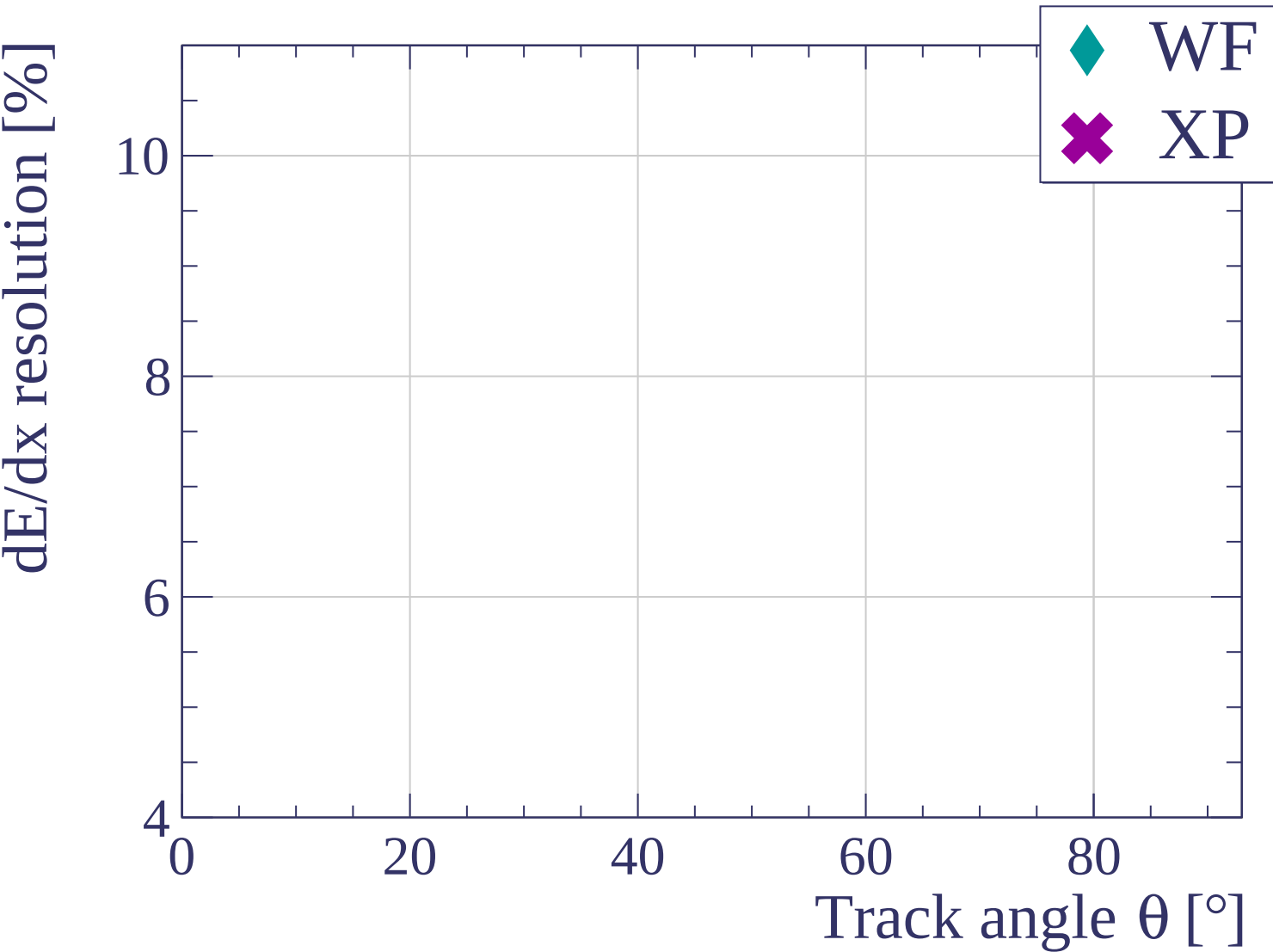
Track length [cm]

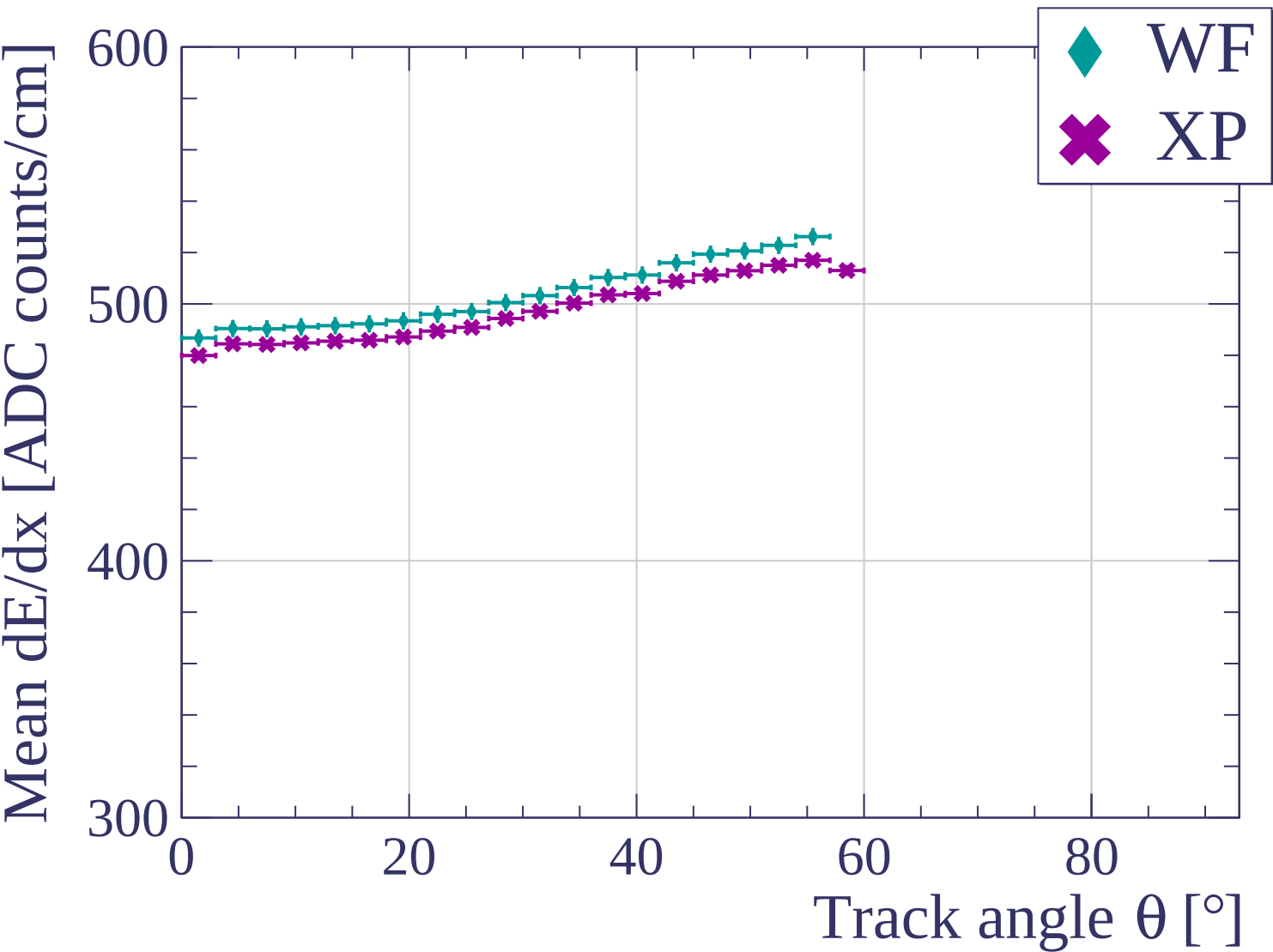


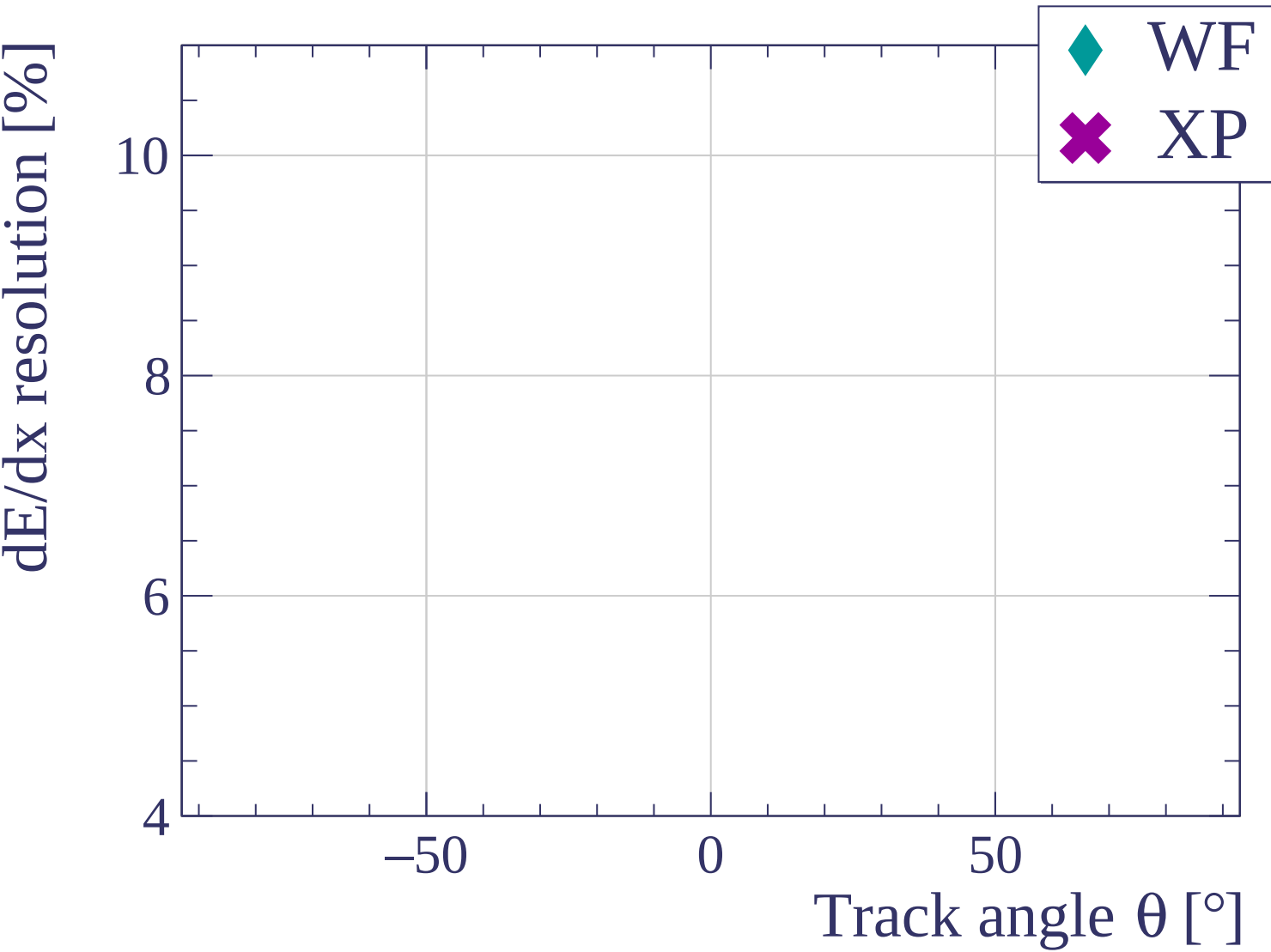


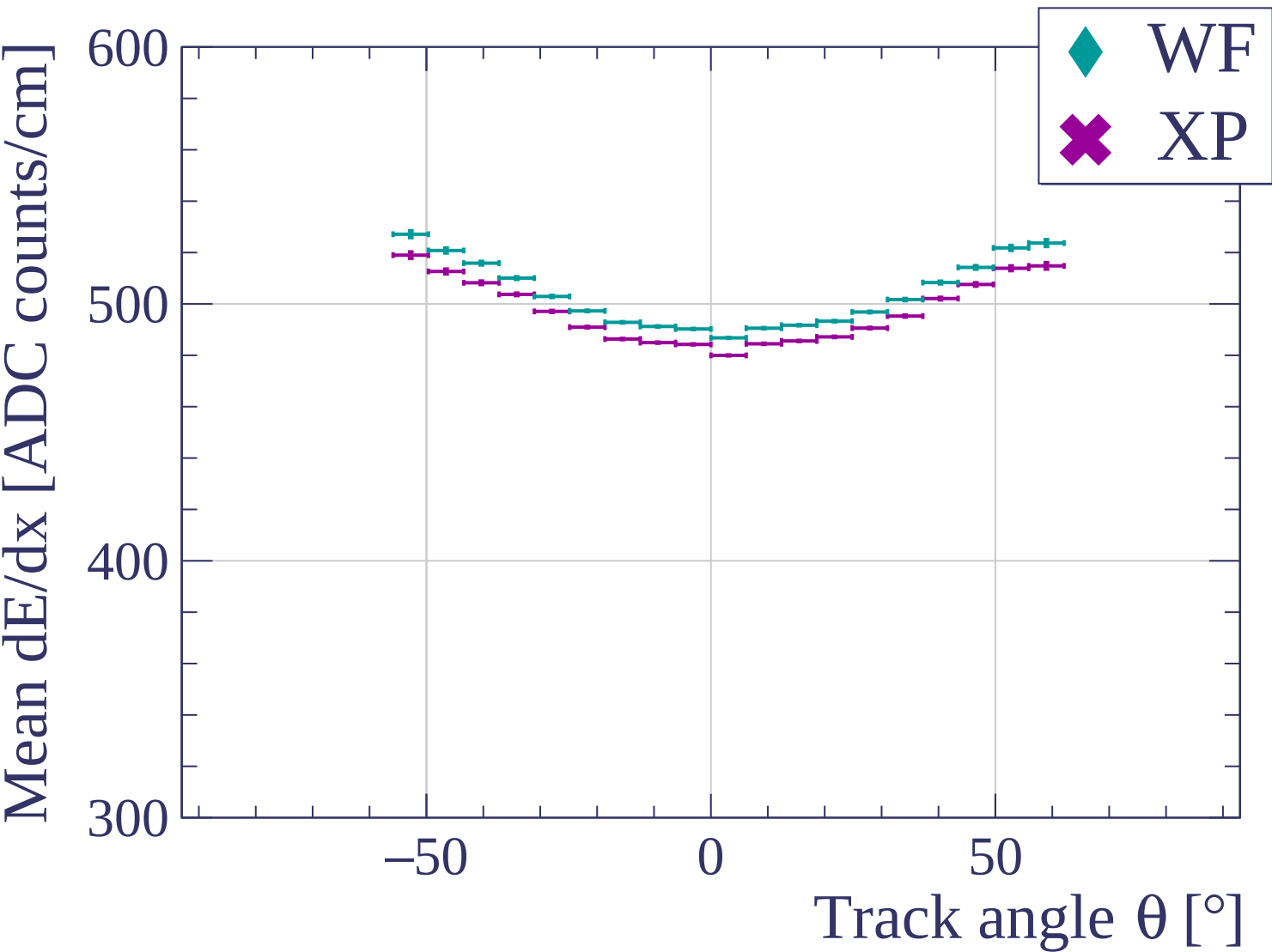


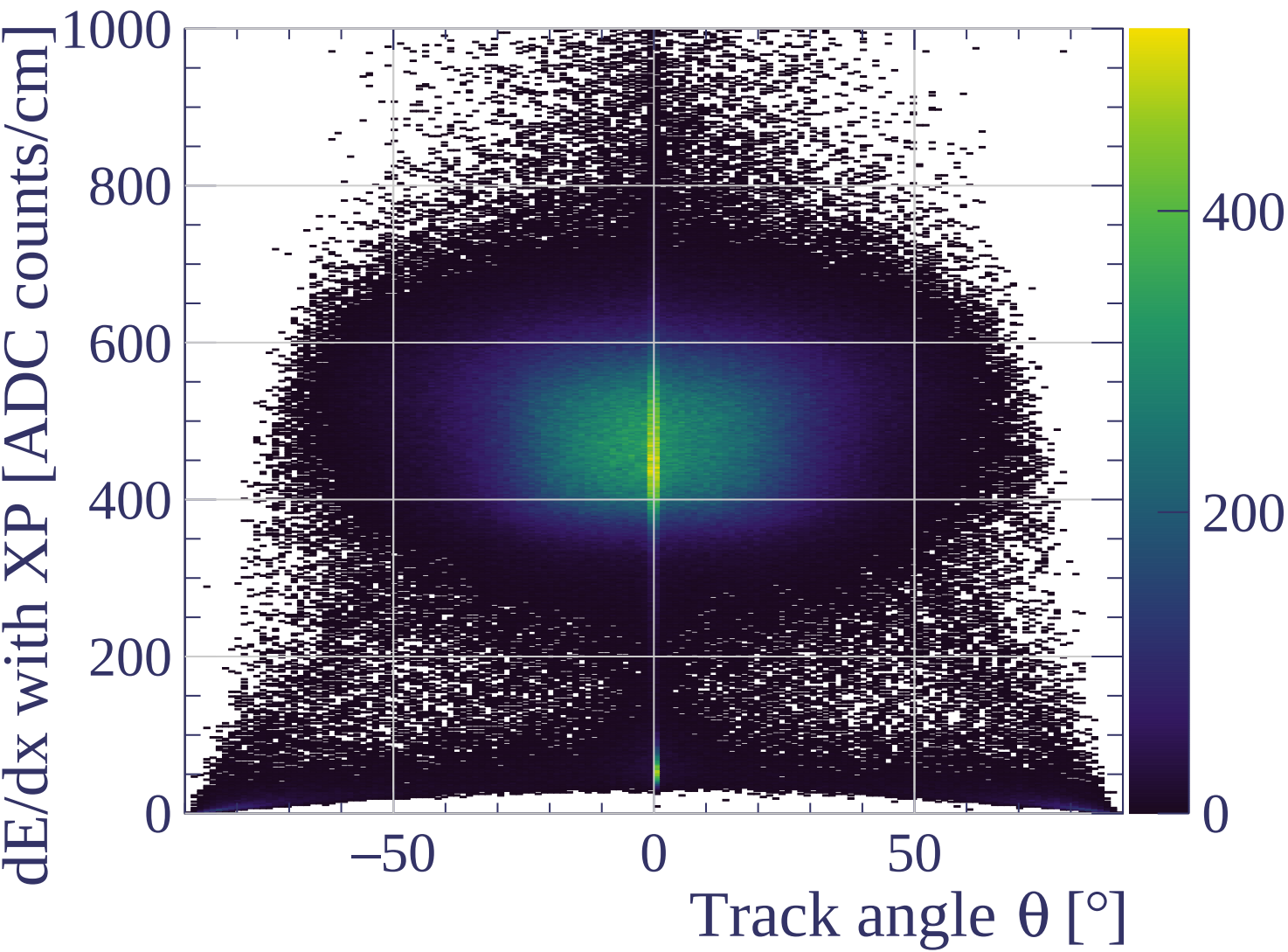




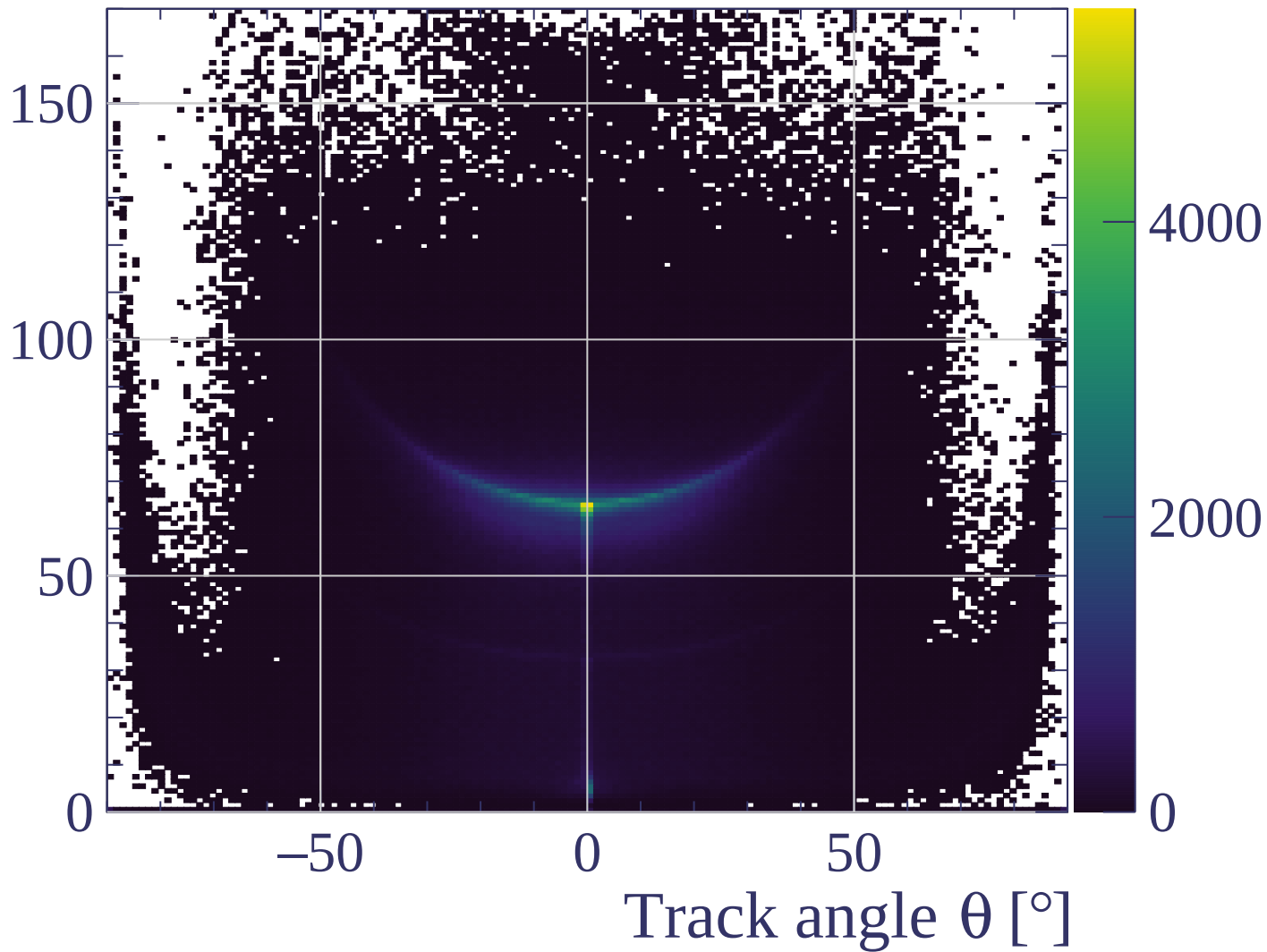


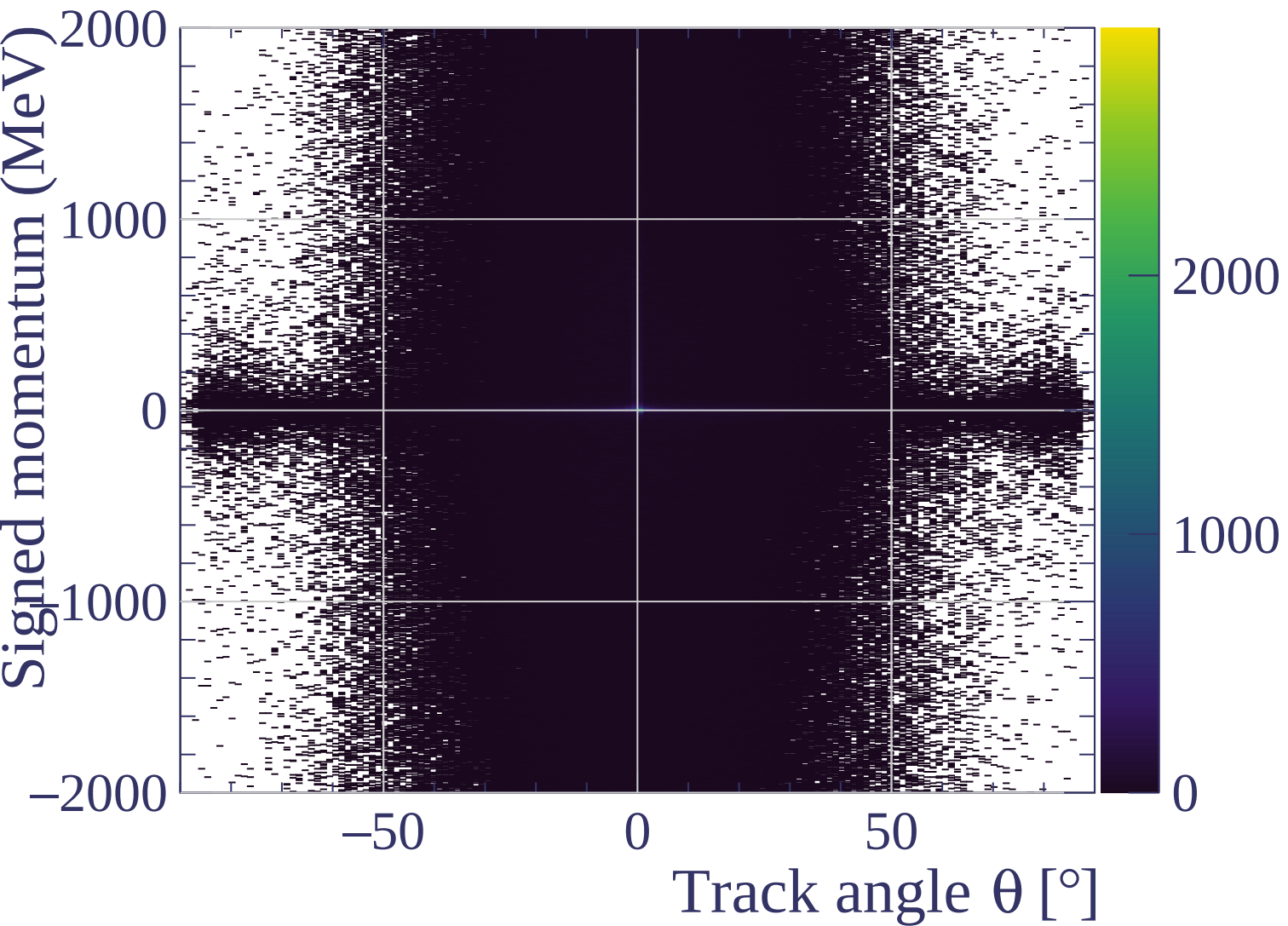


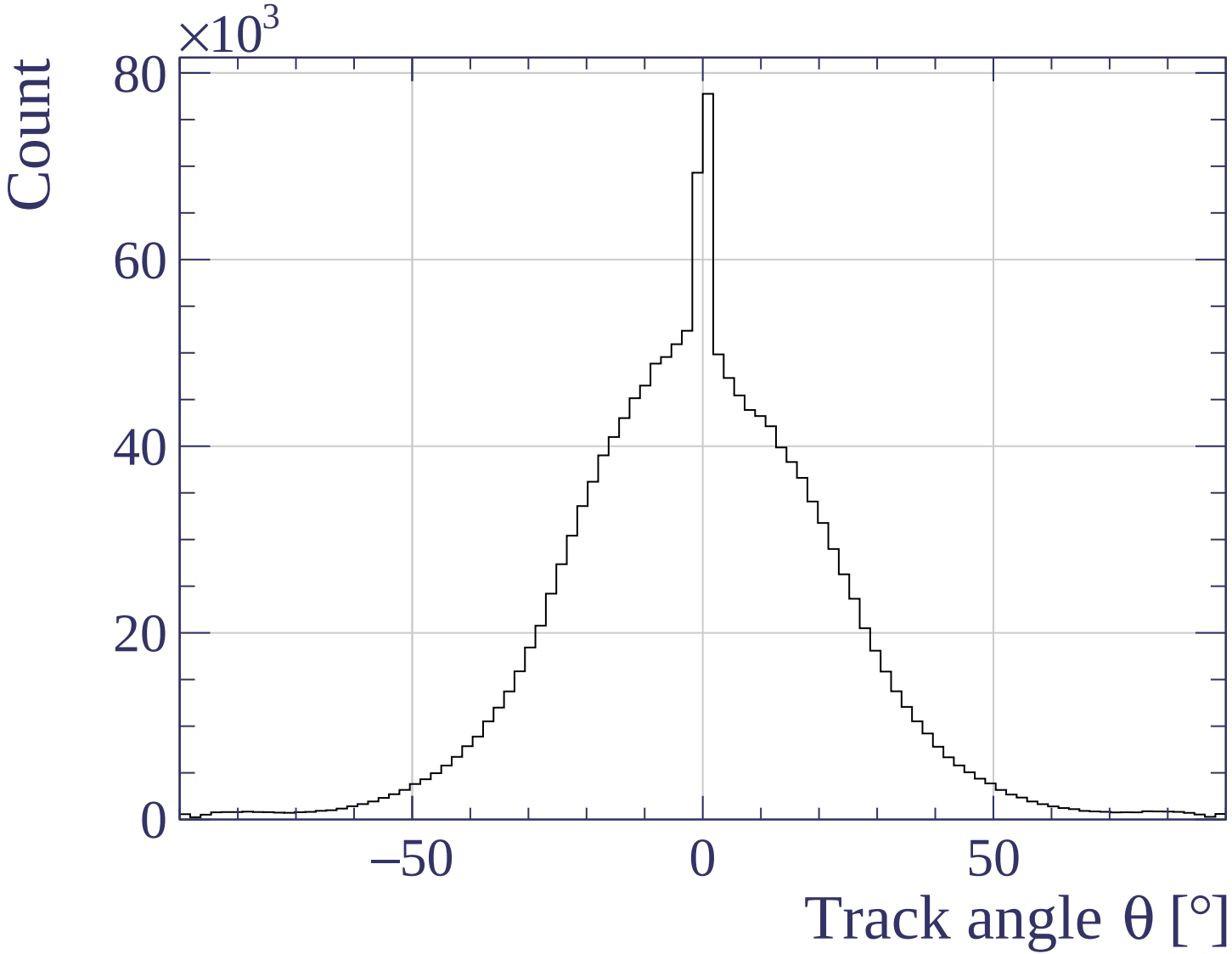


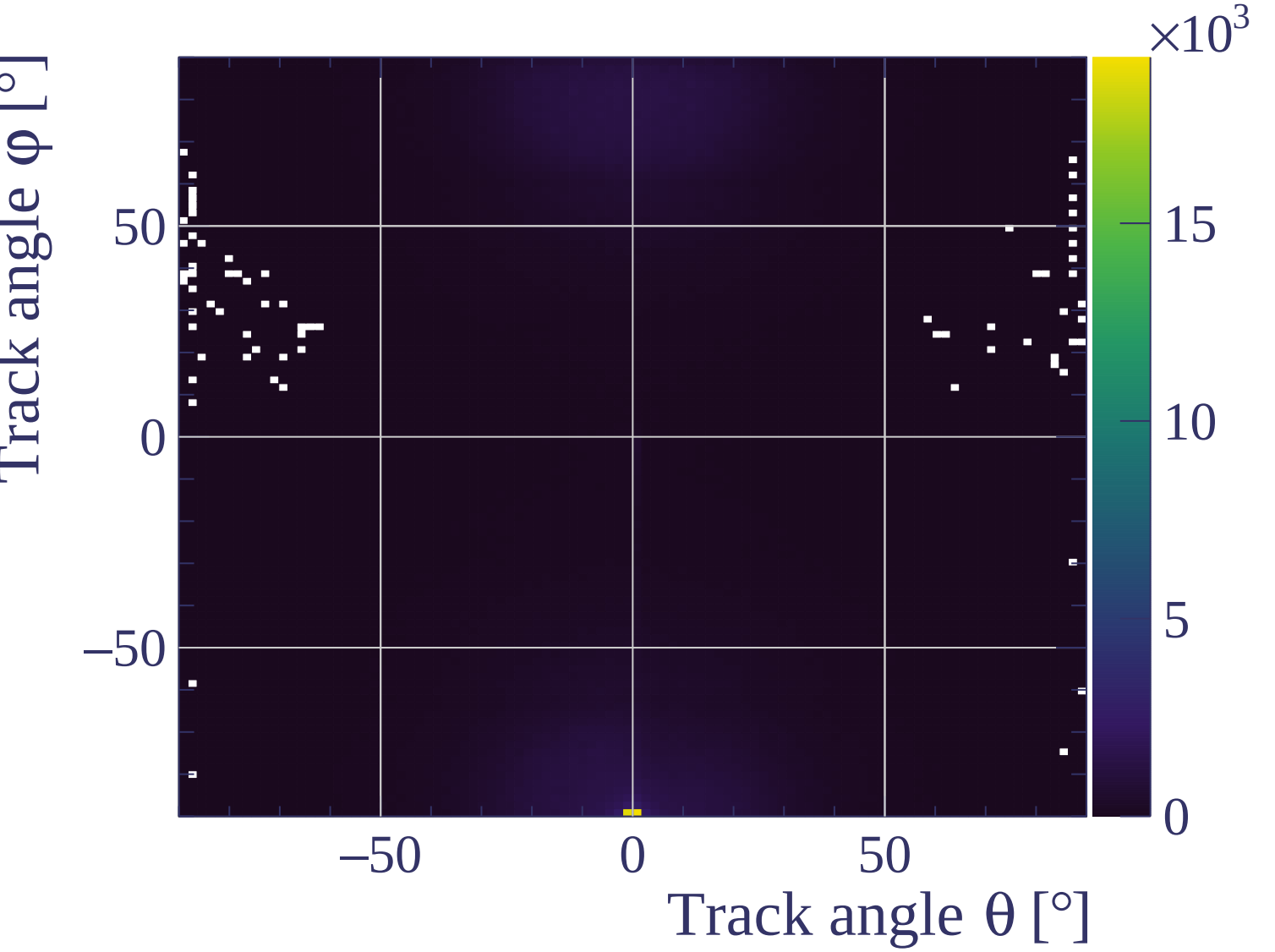


Track length [cm]

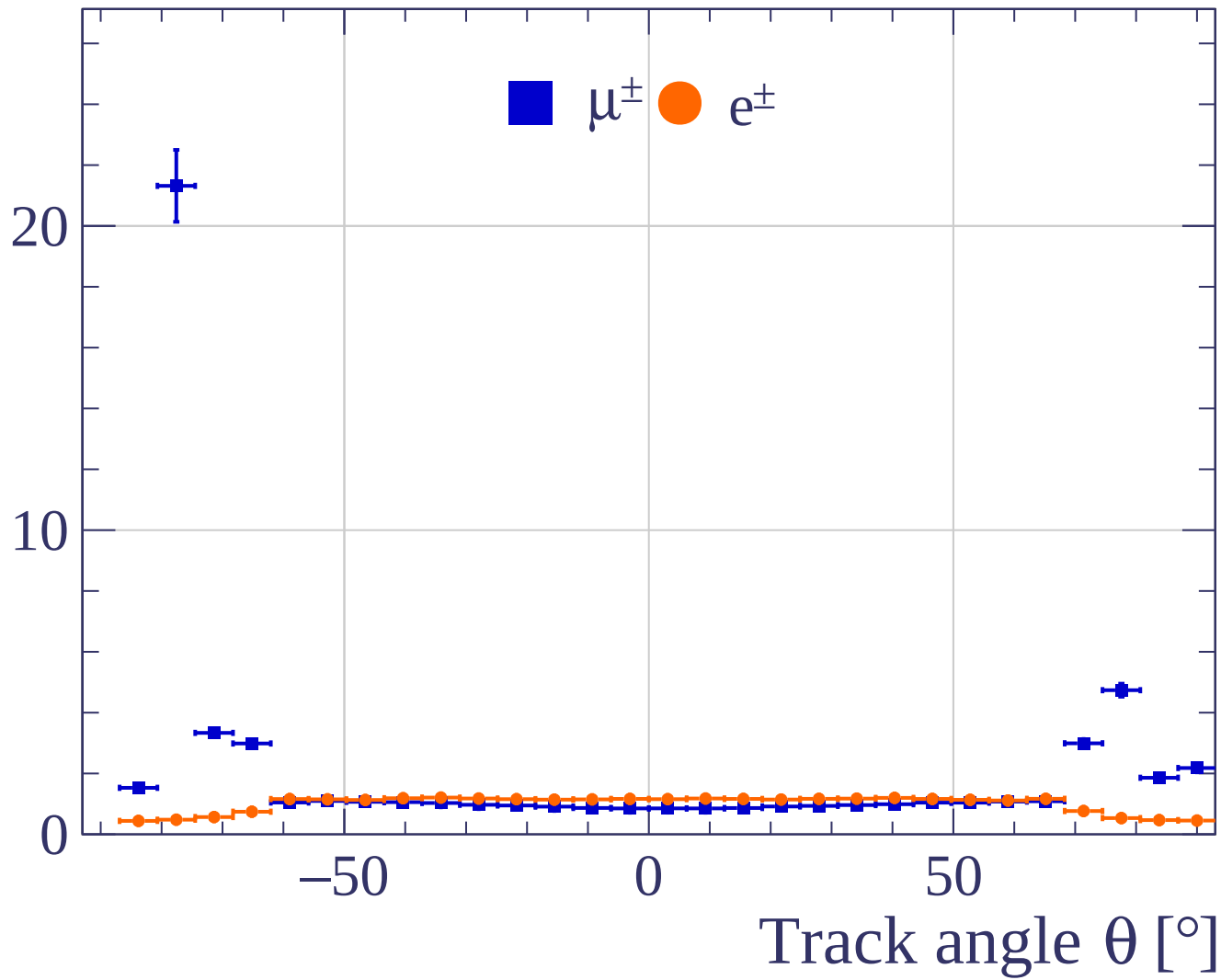




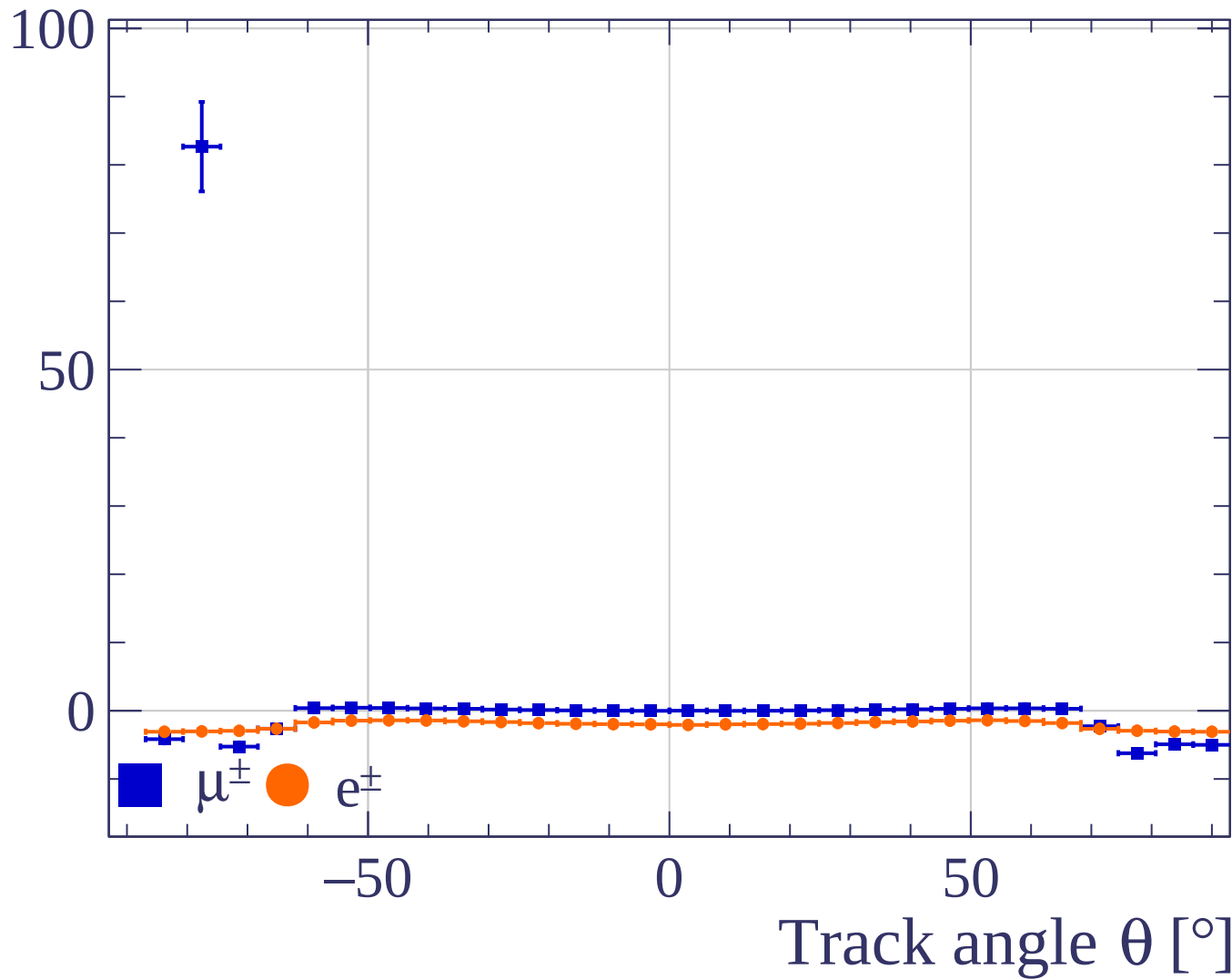




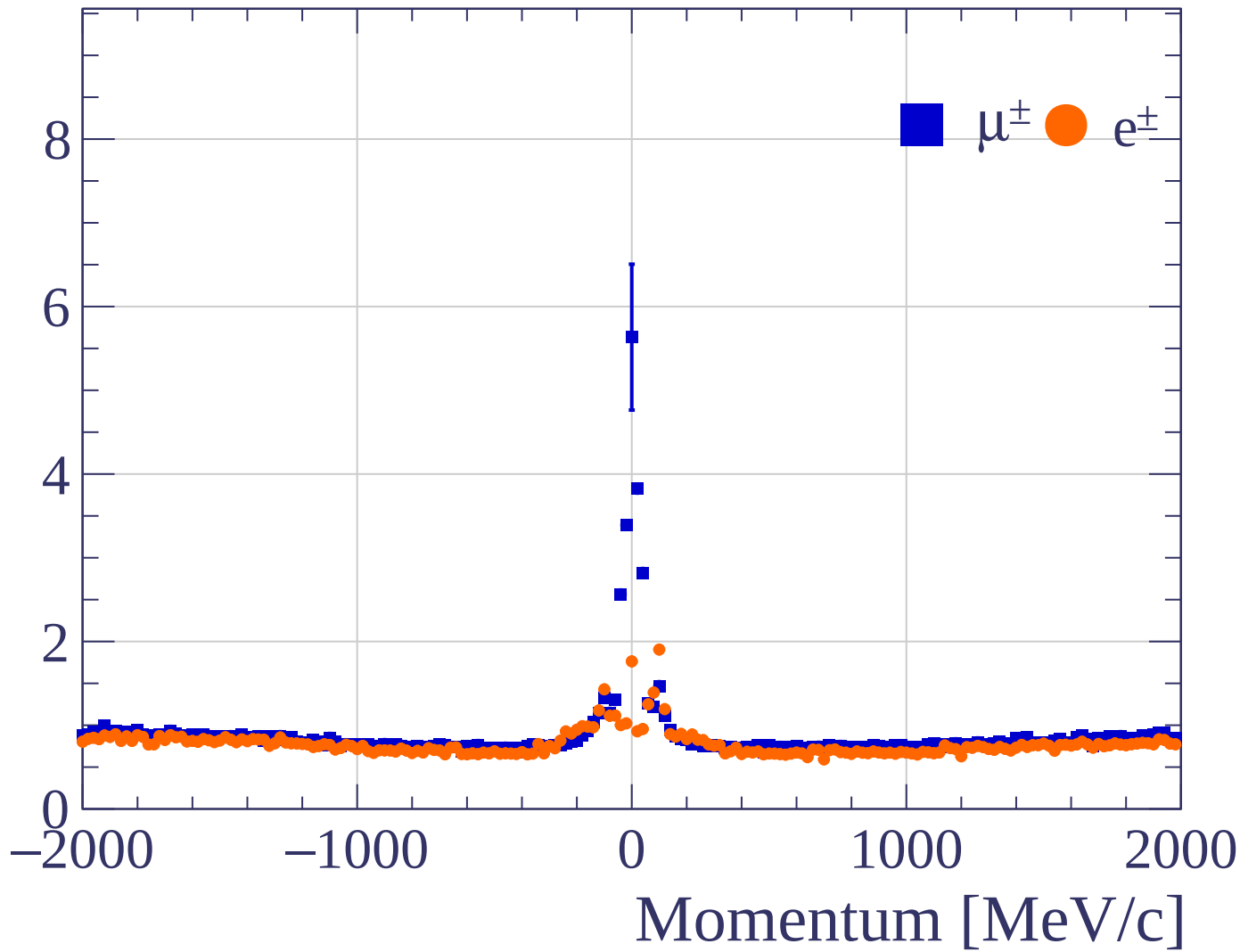
Pulls standard deviation



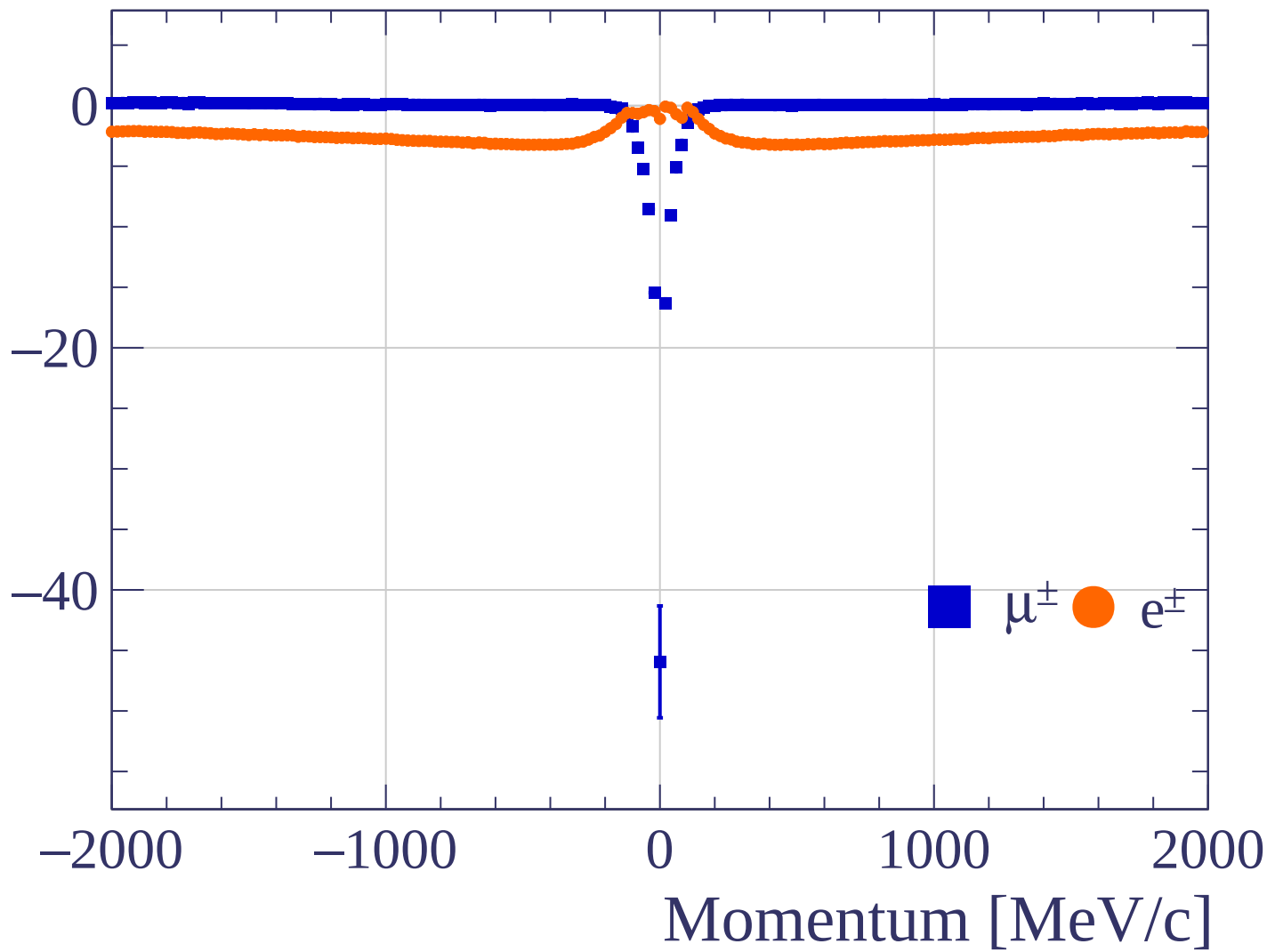
Pulls mean

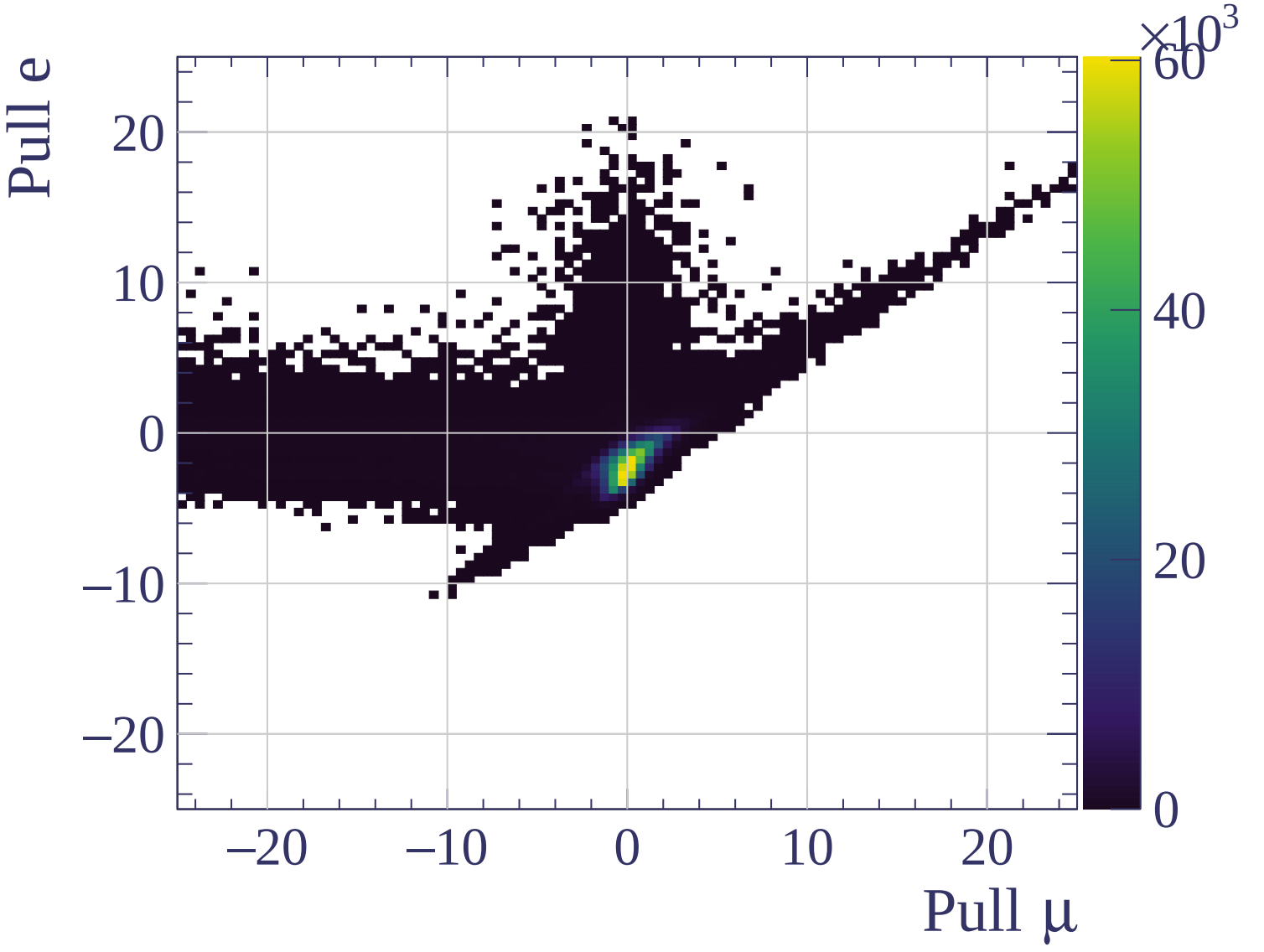


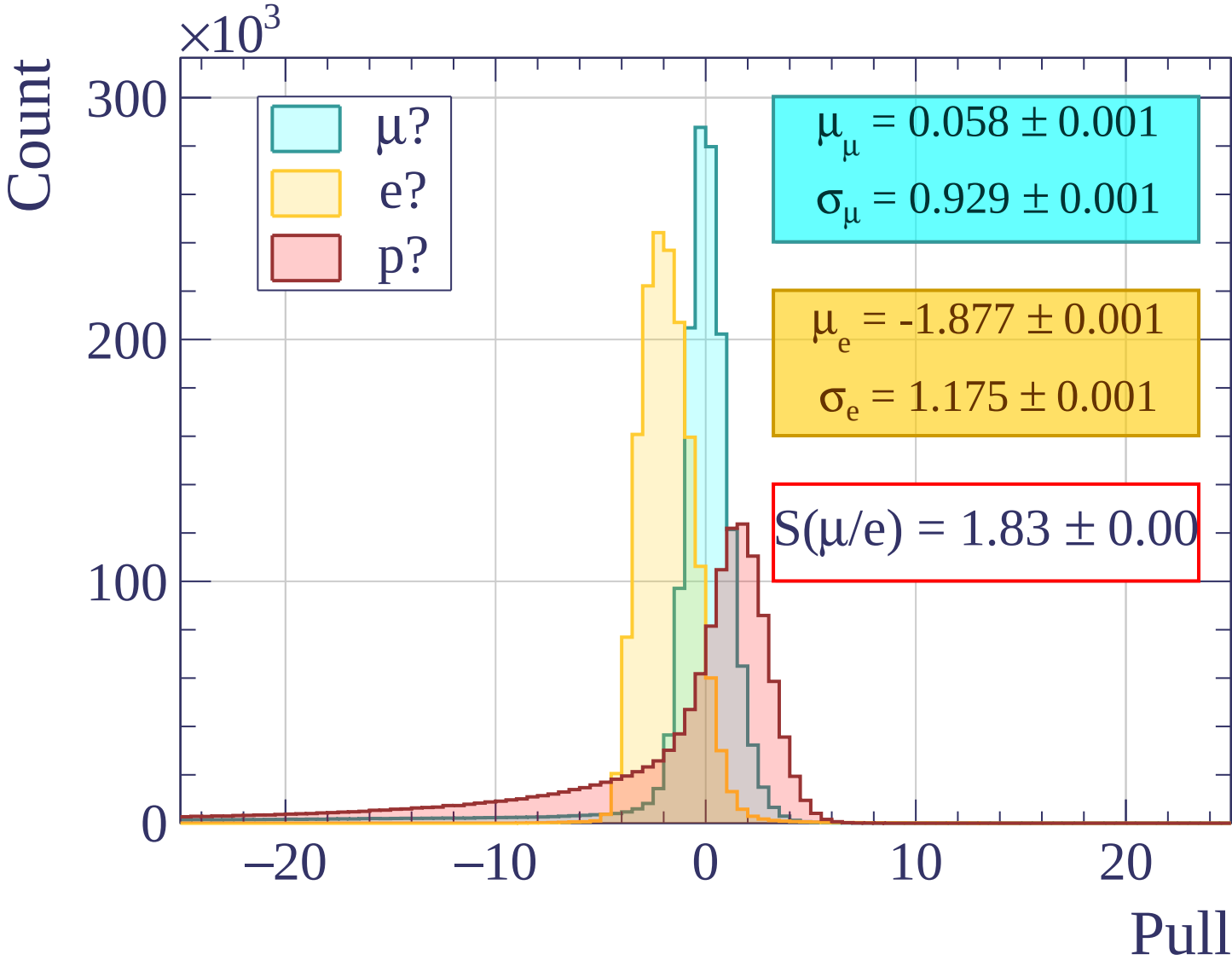
Pulls standard deviation

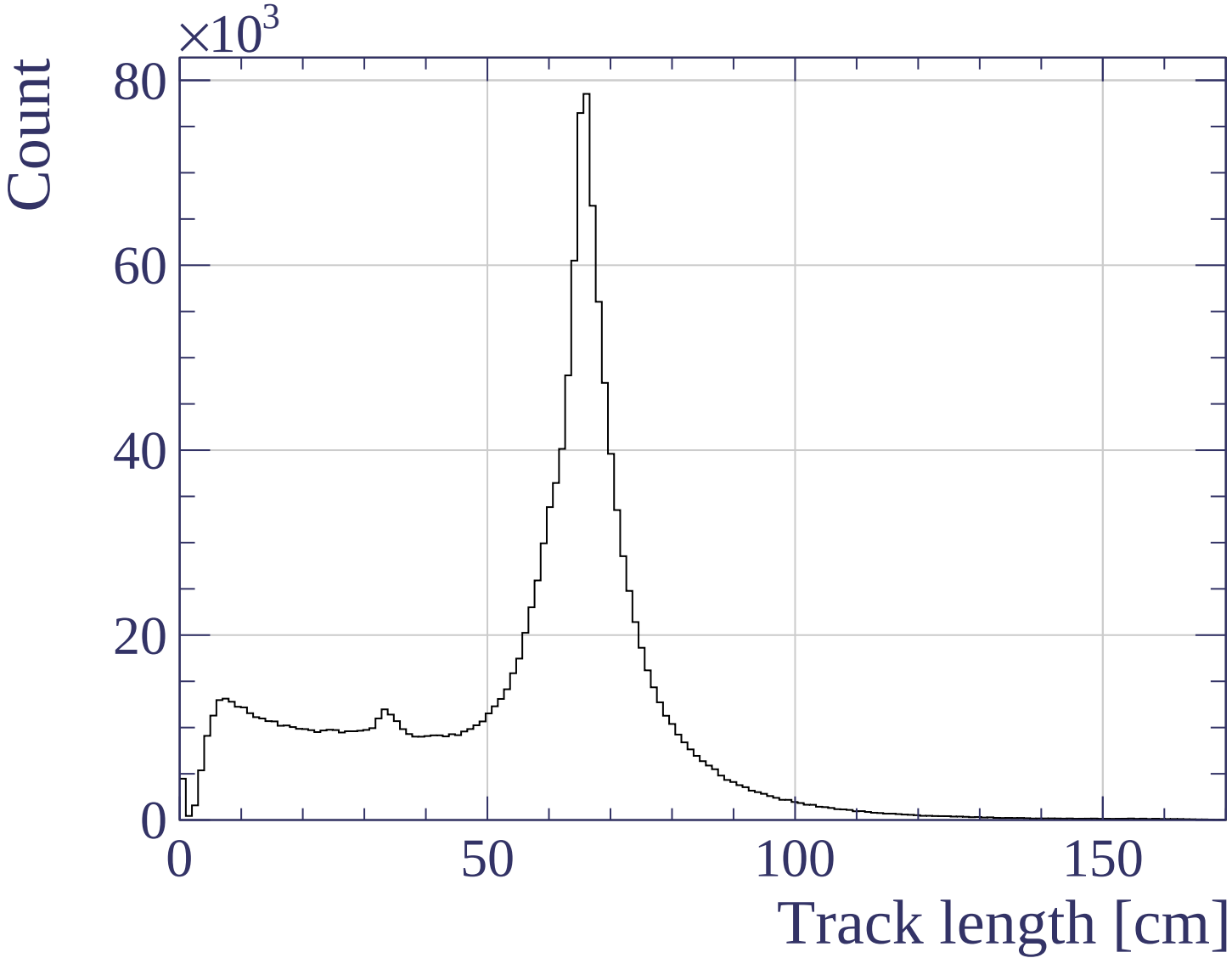


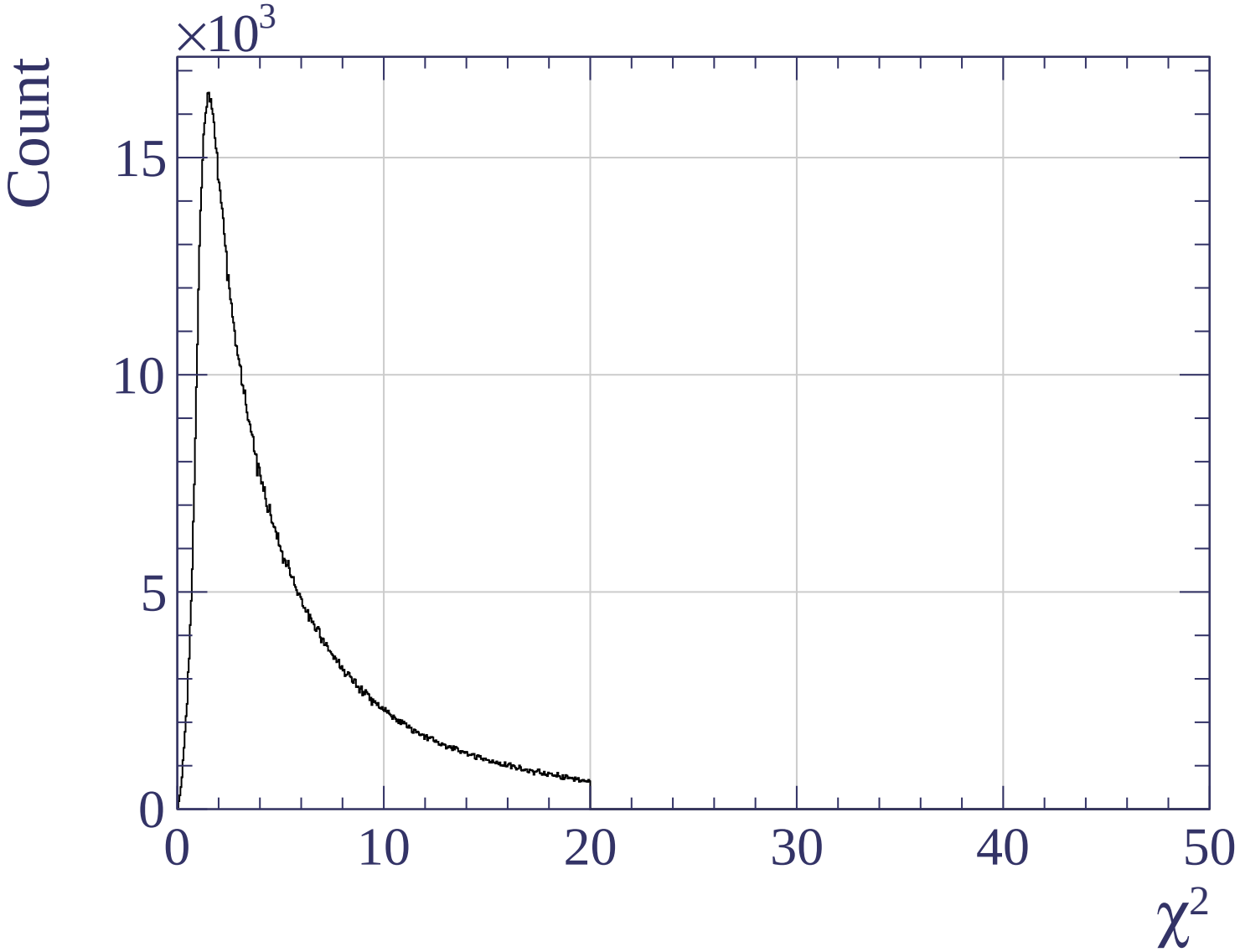
Pulls mean

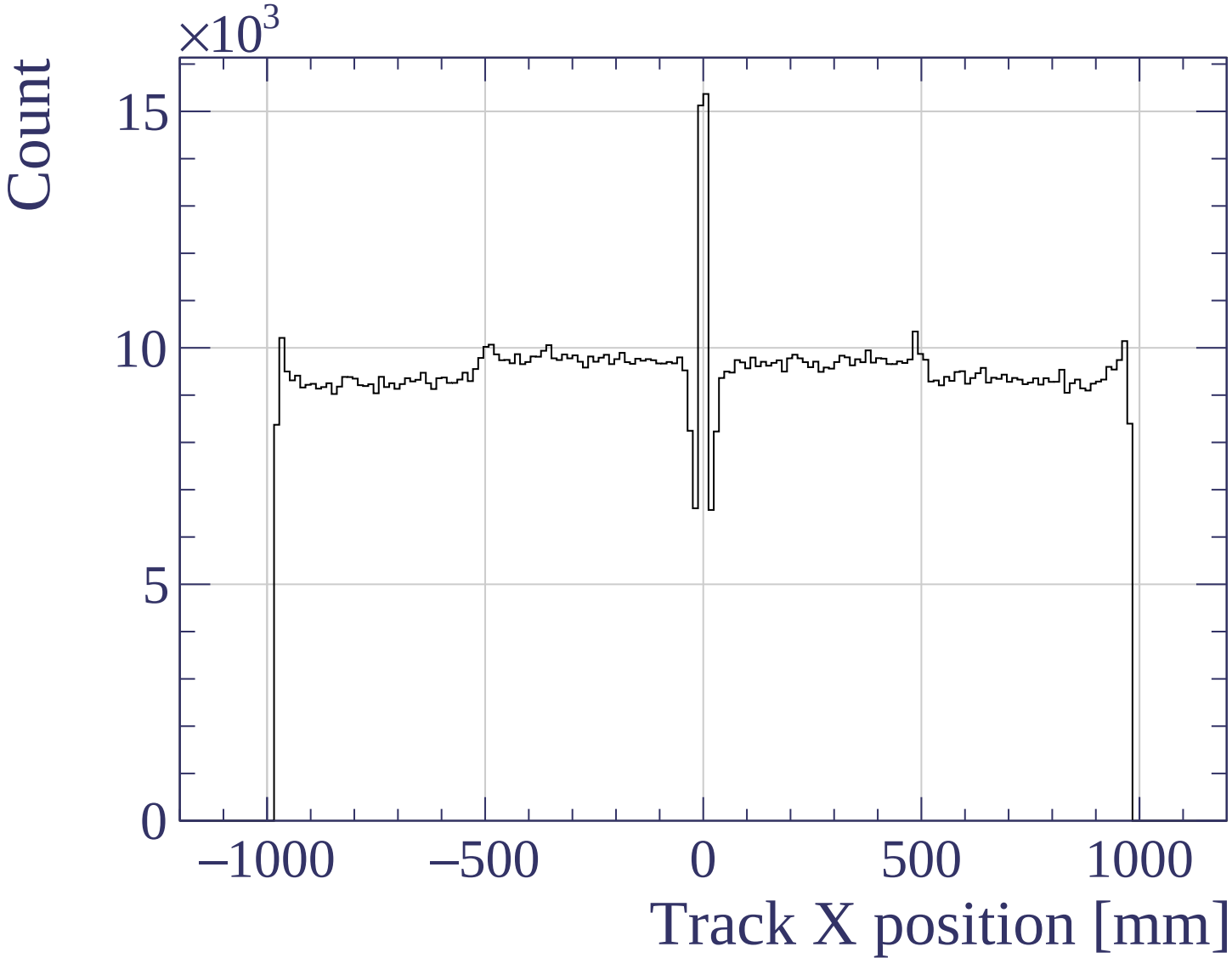


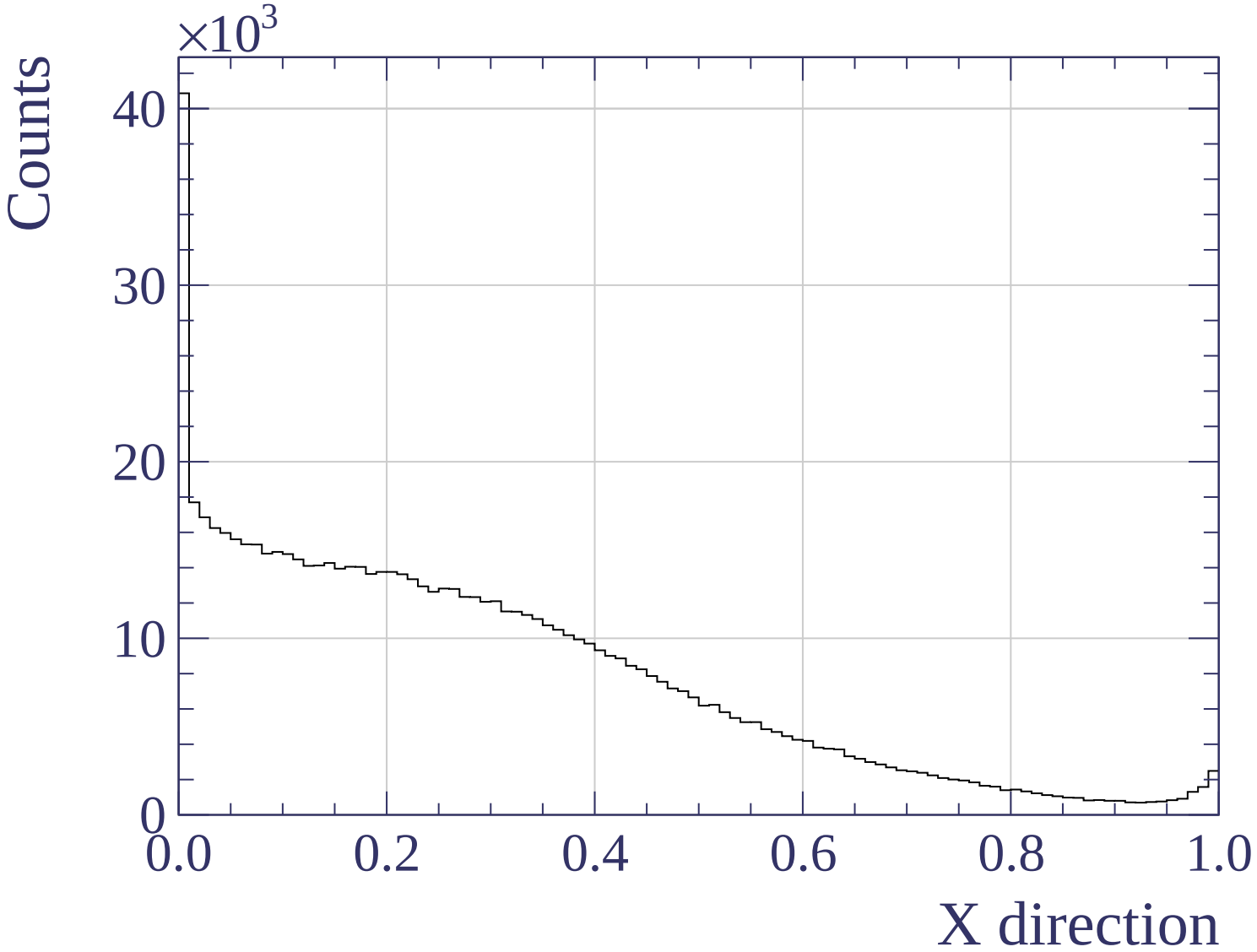


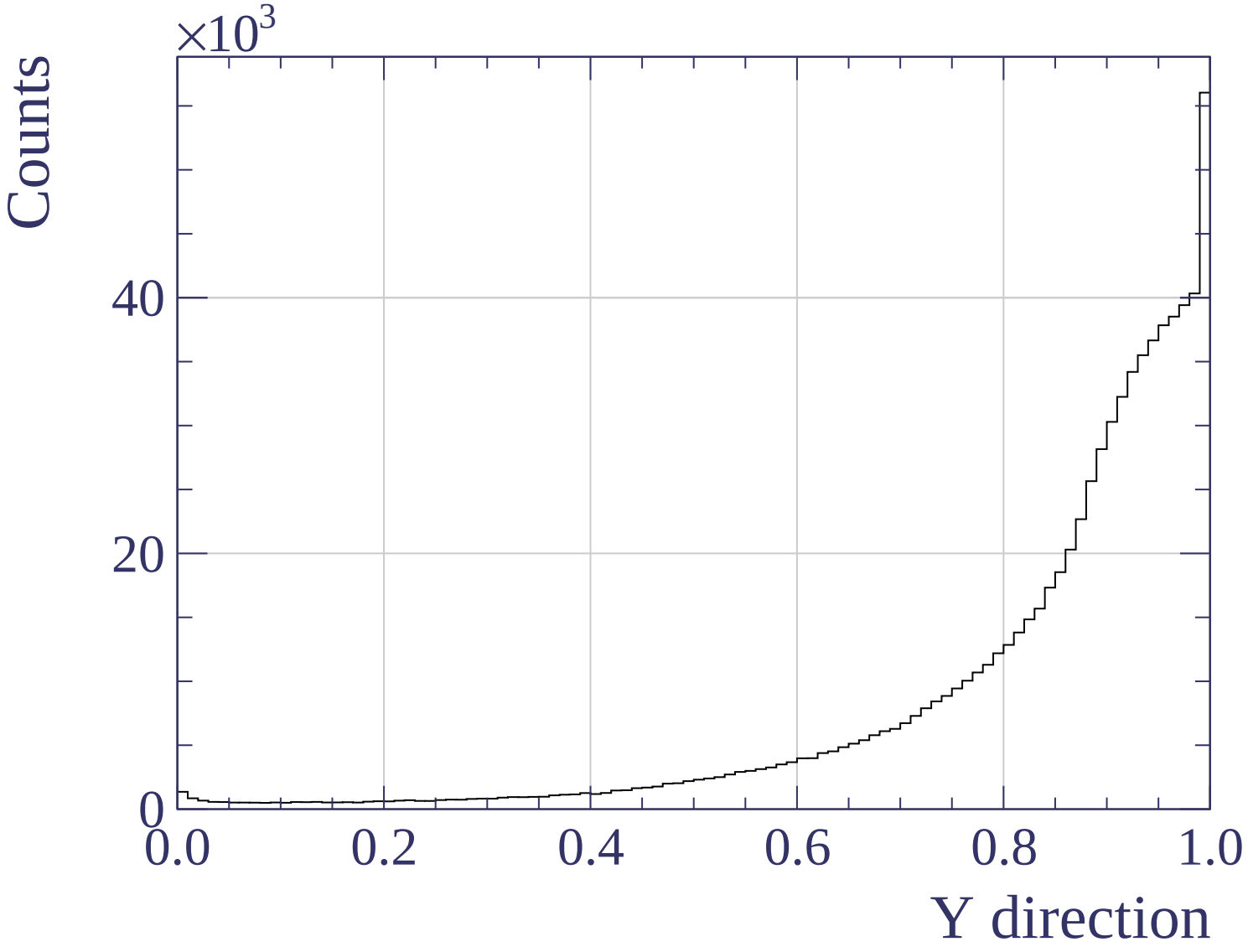


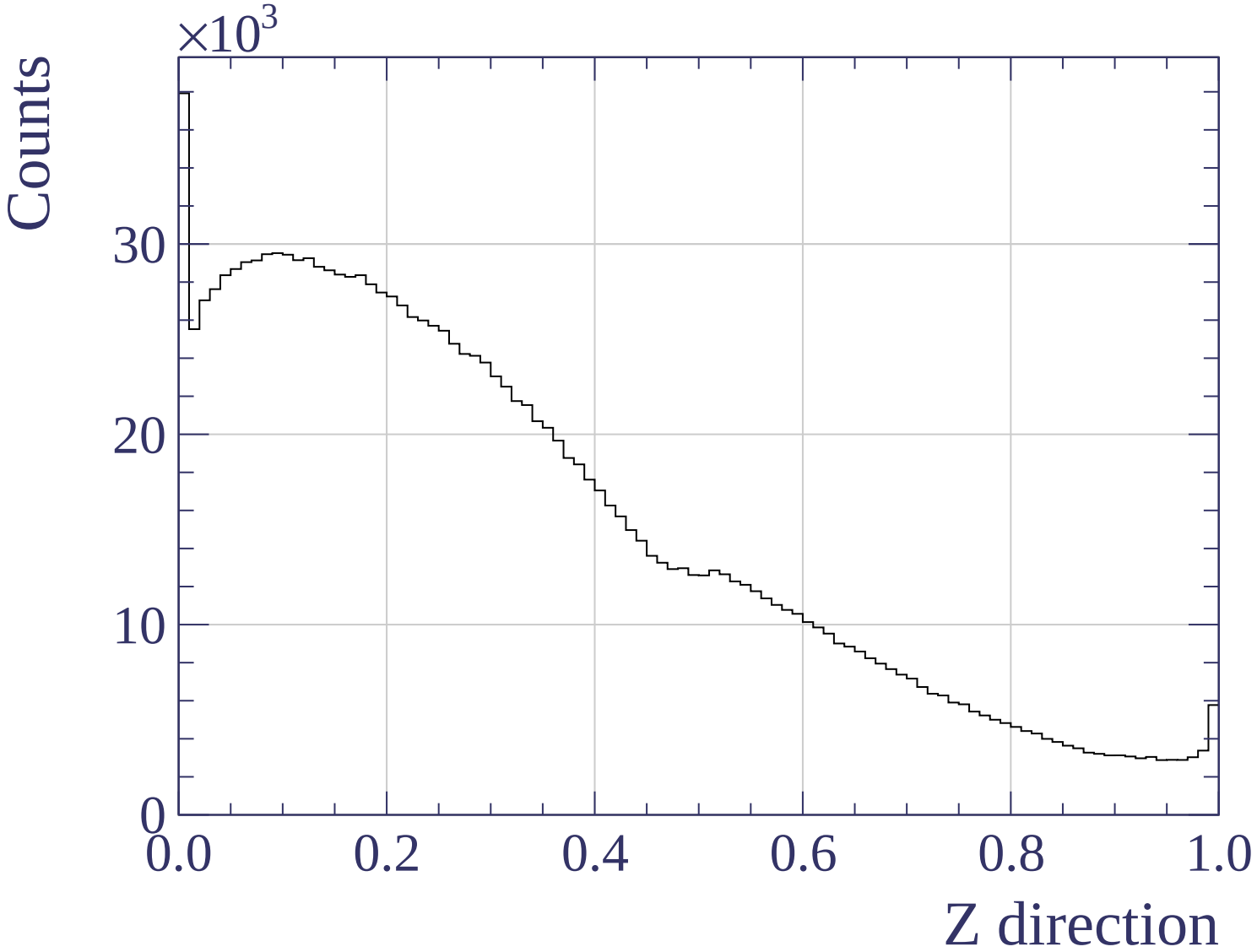


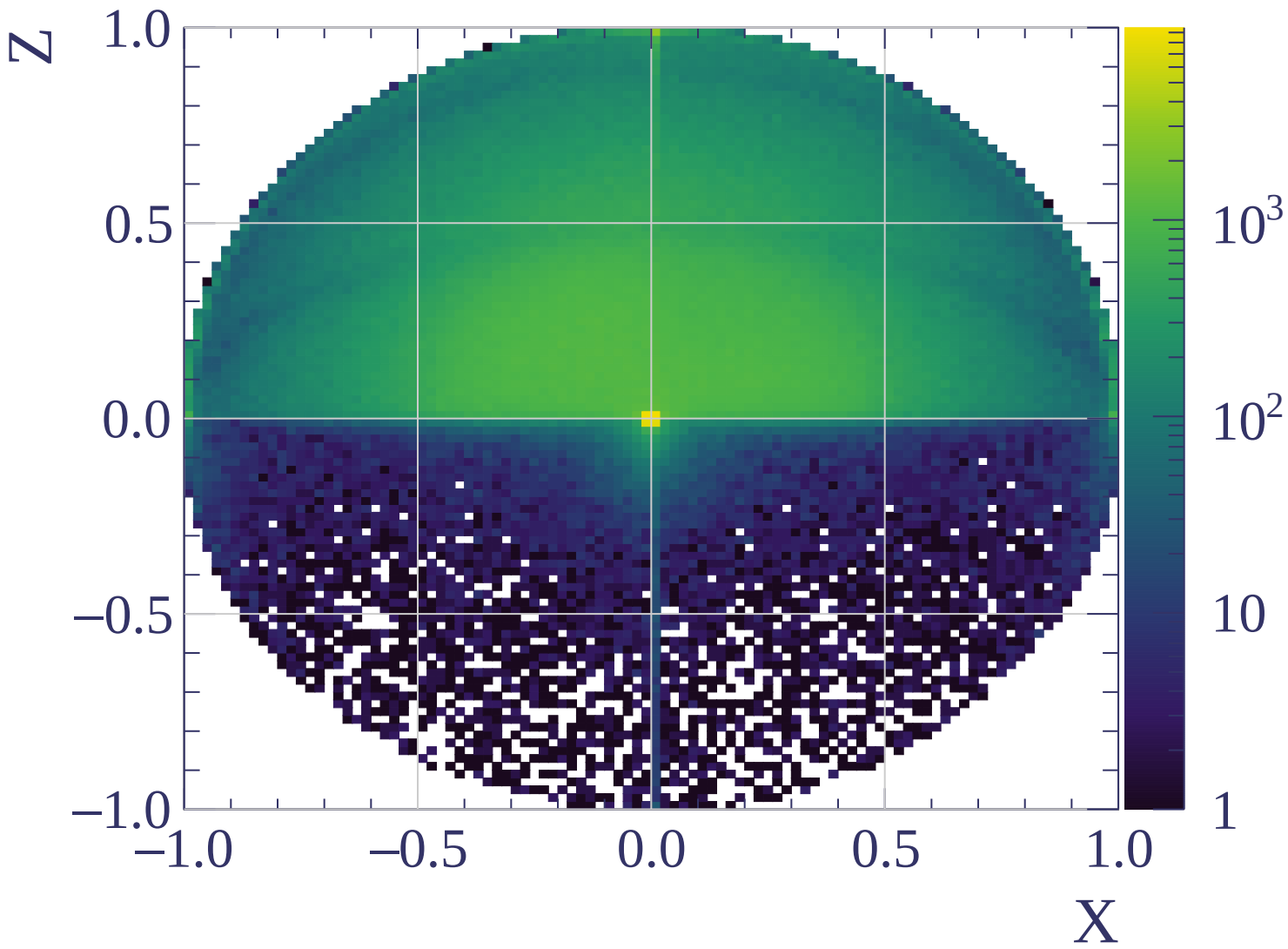


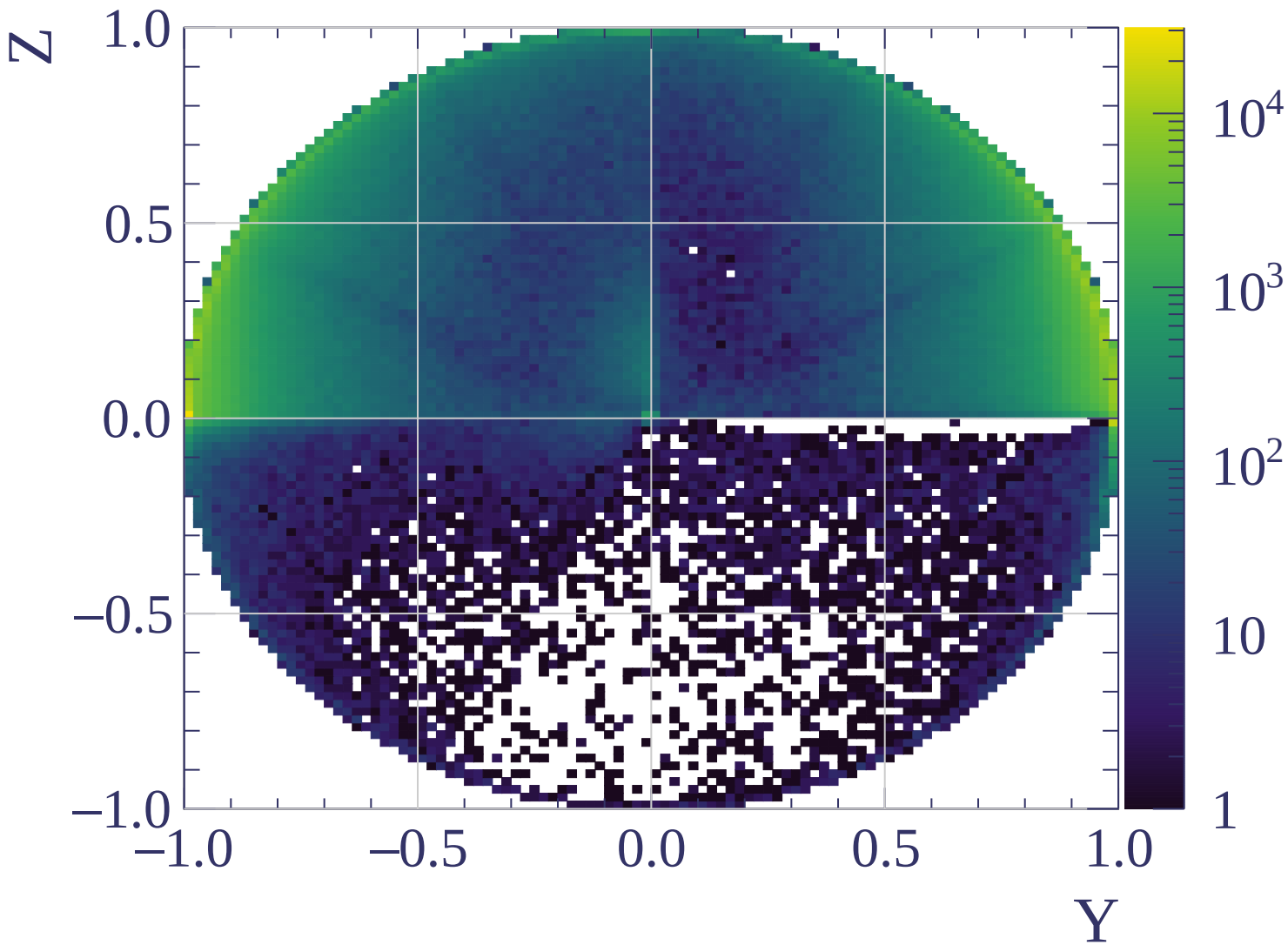


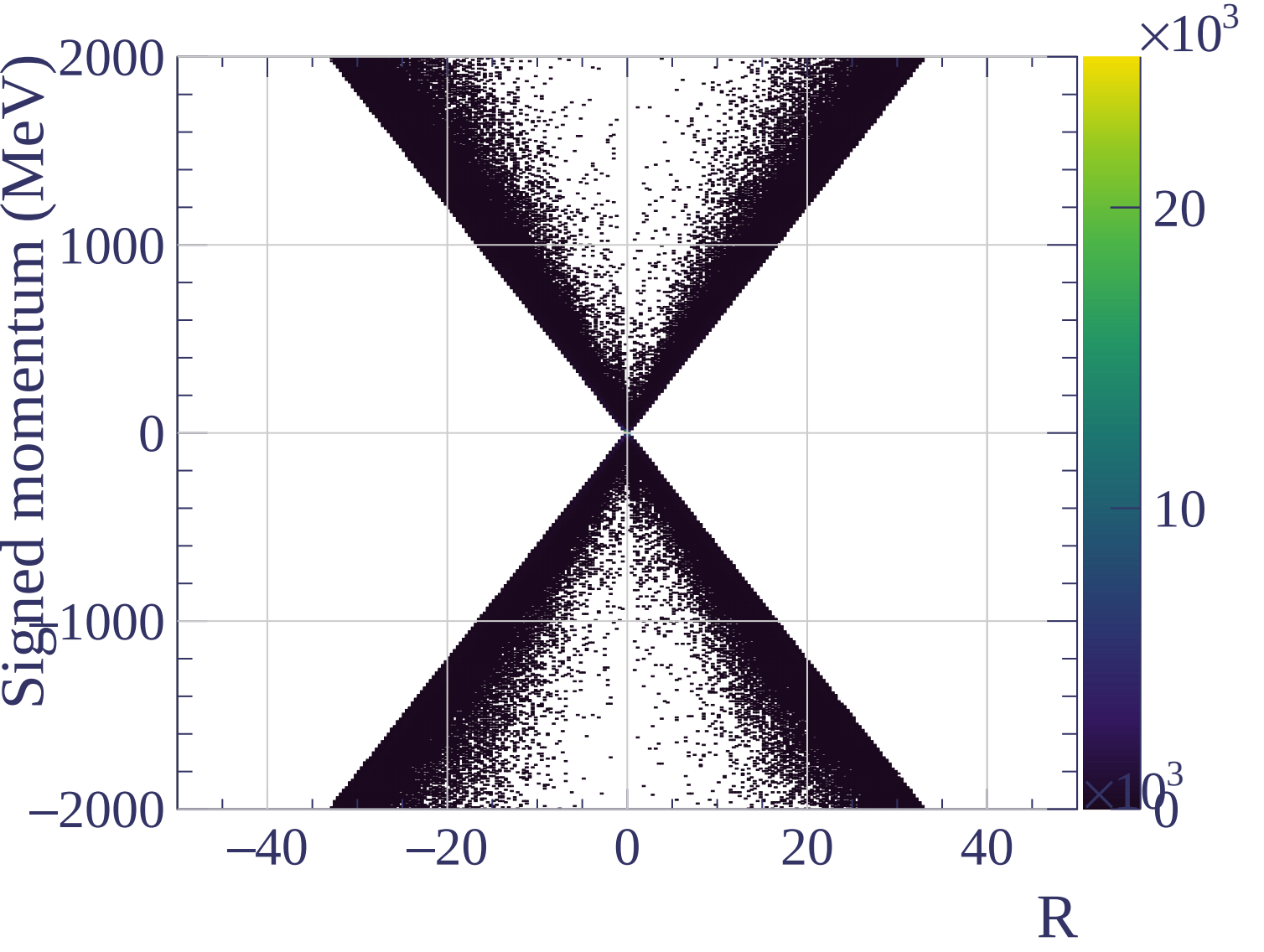


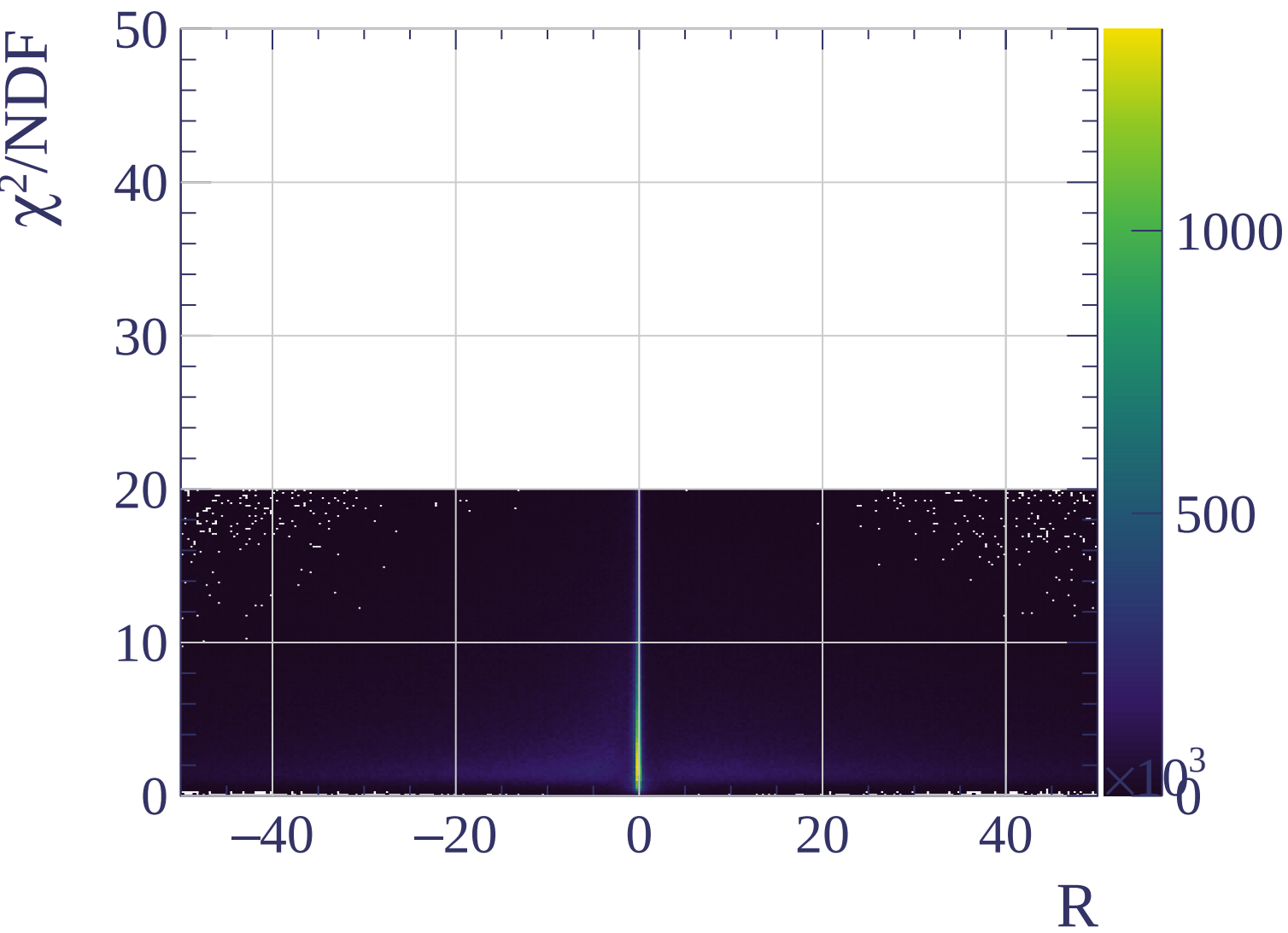


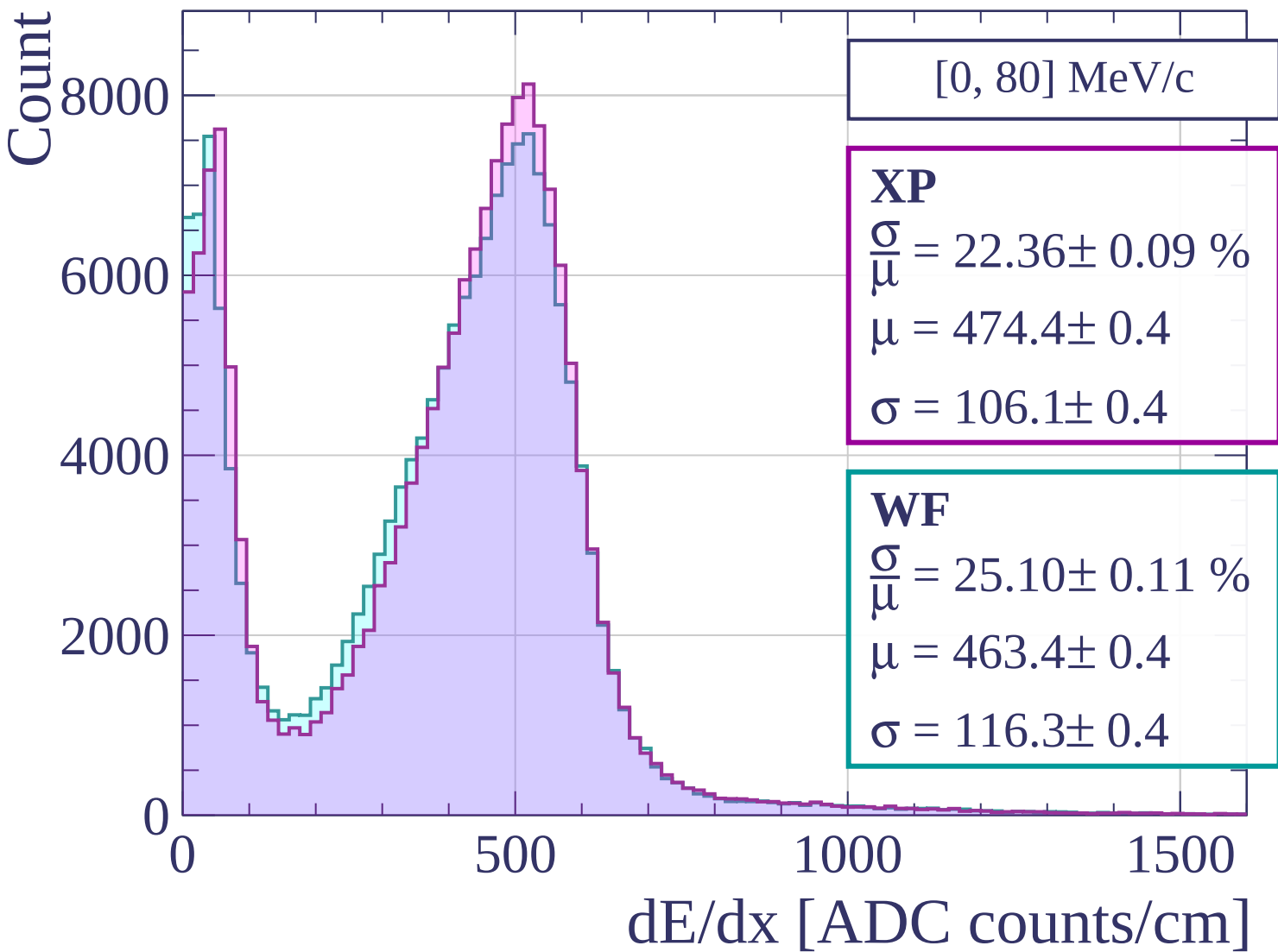


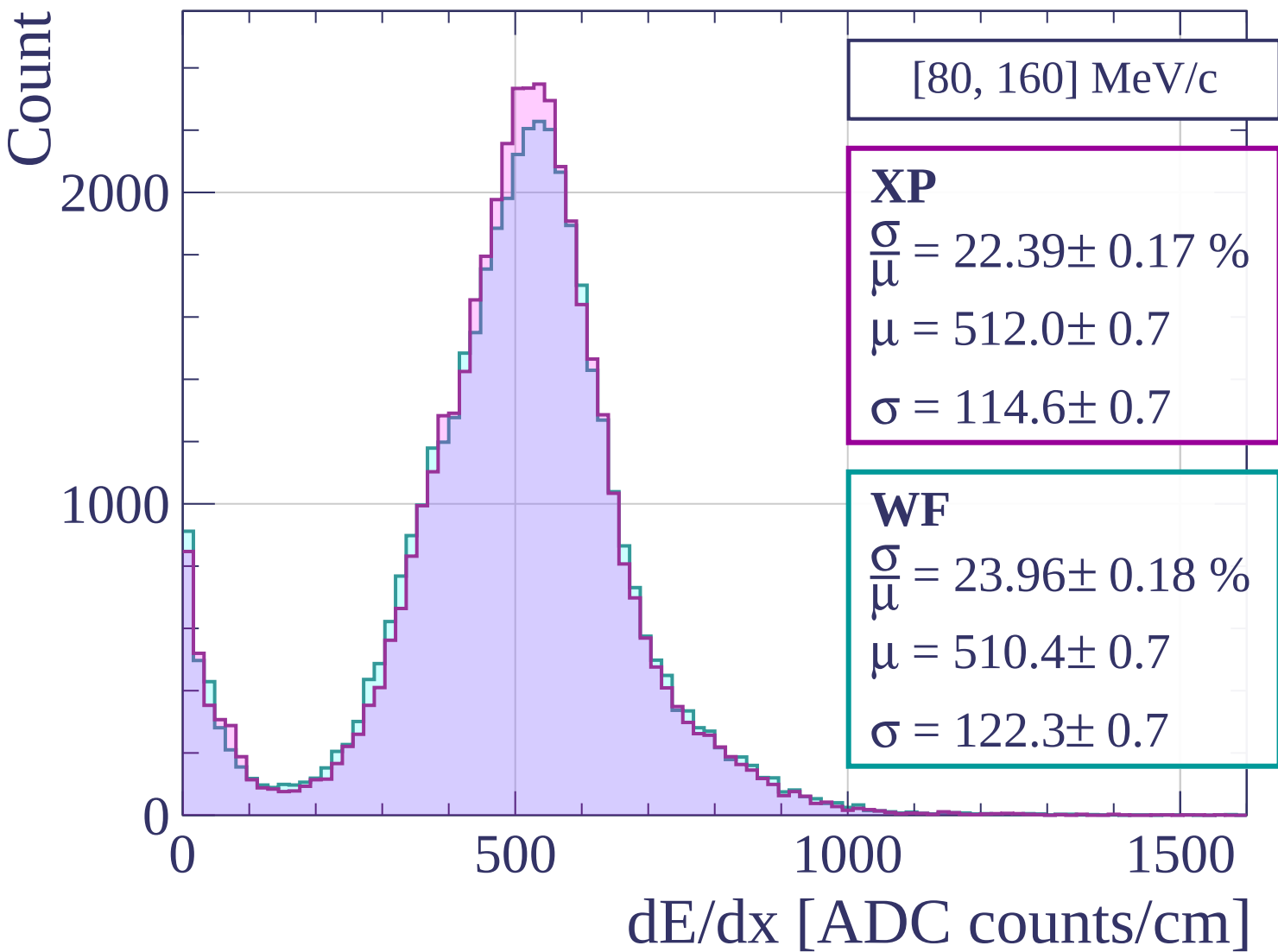


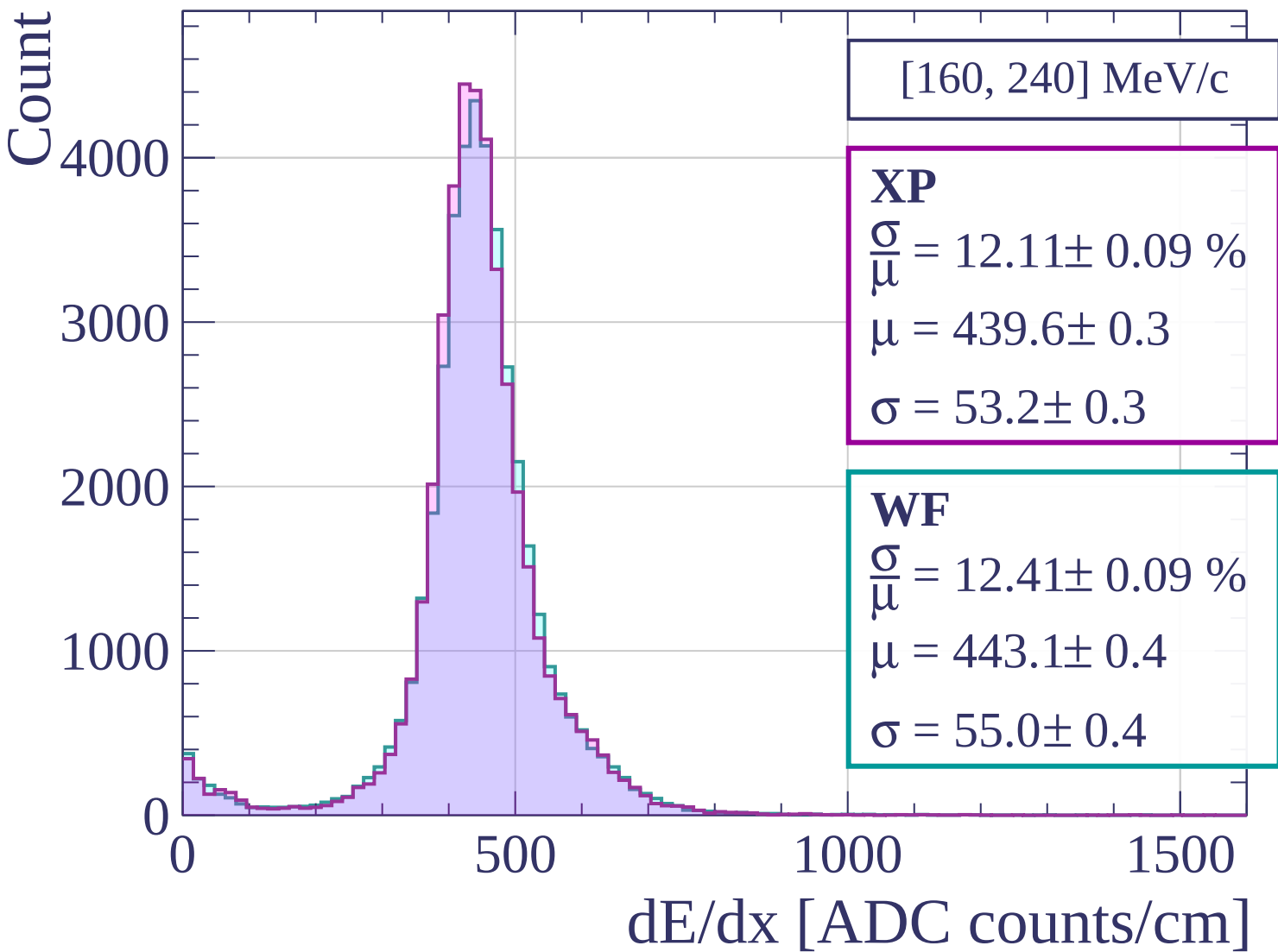


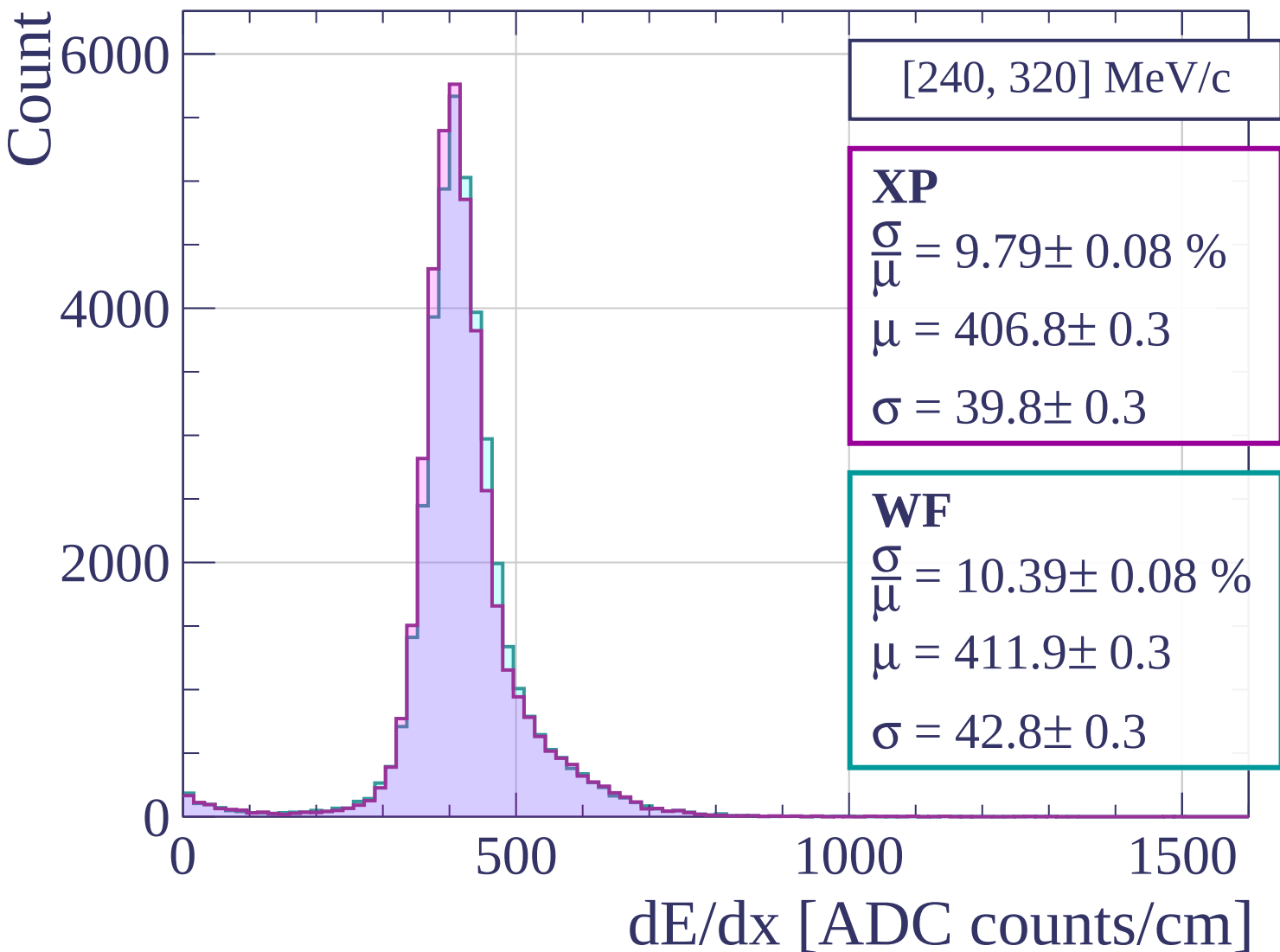


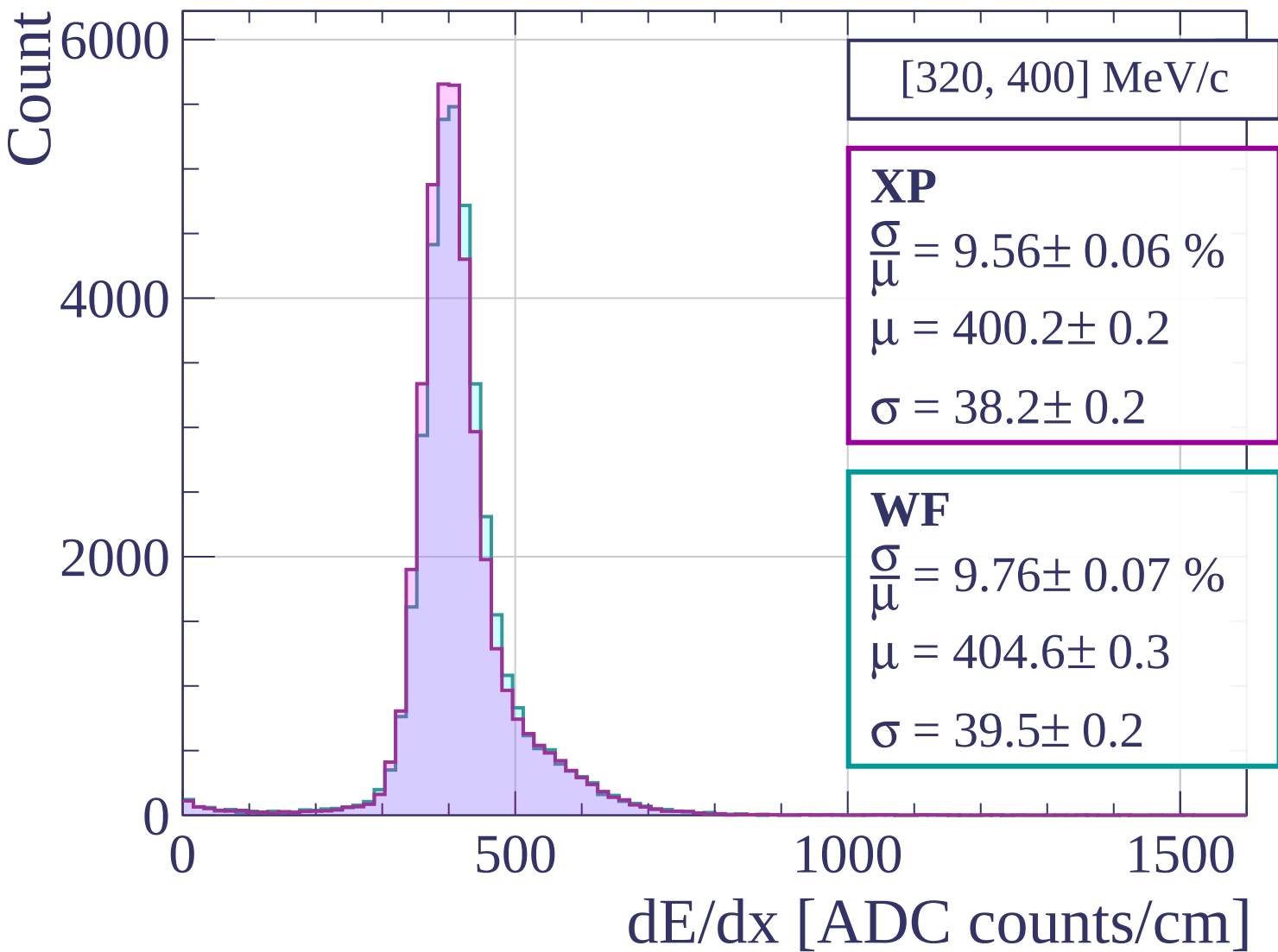


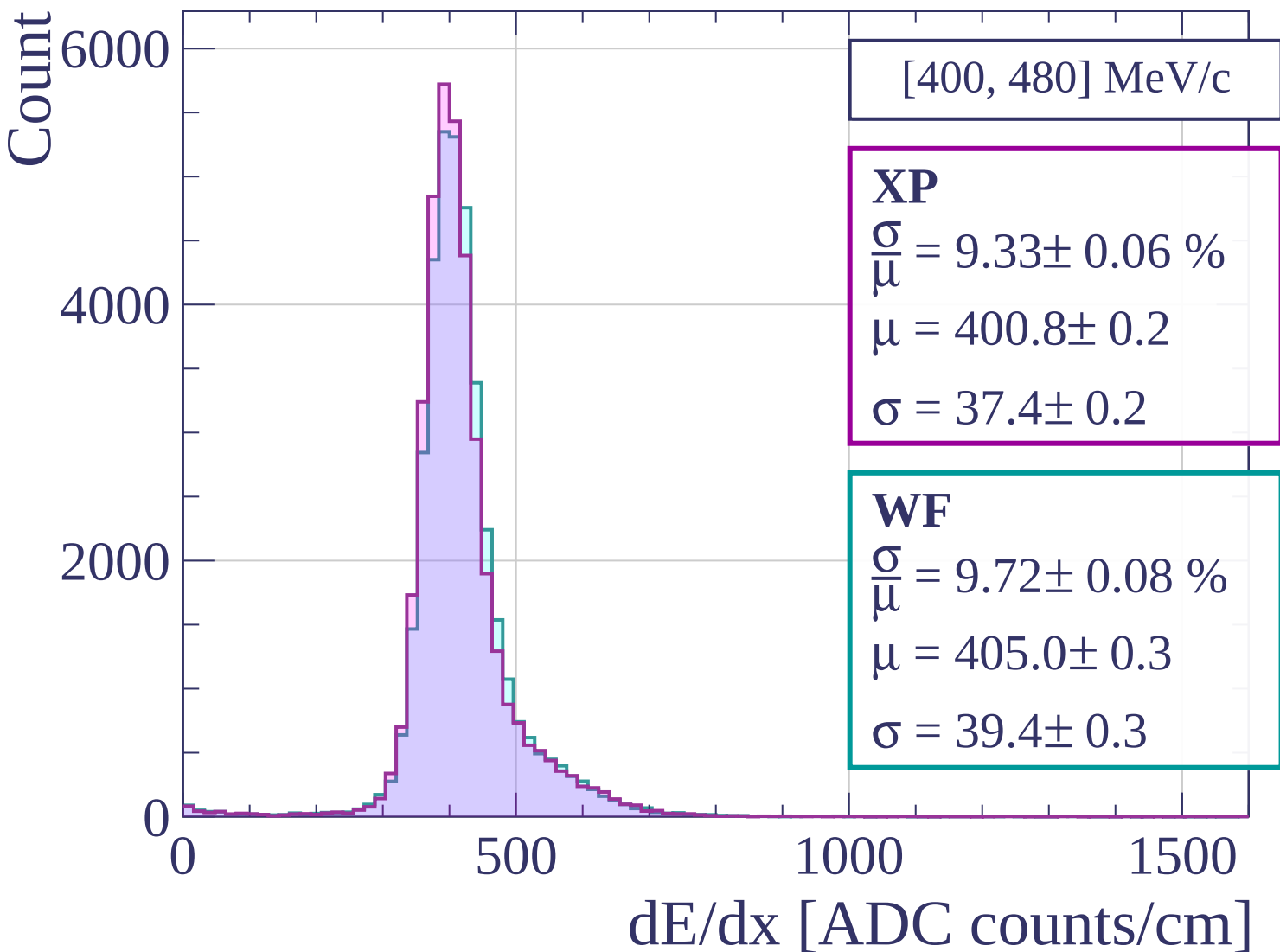




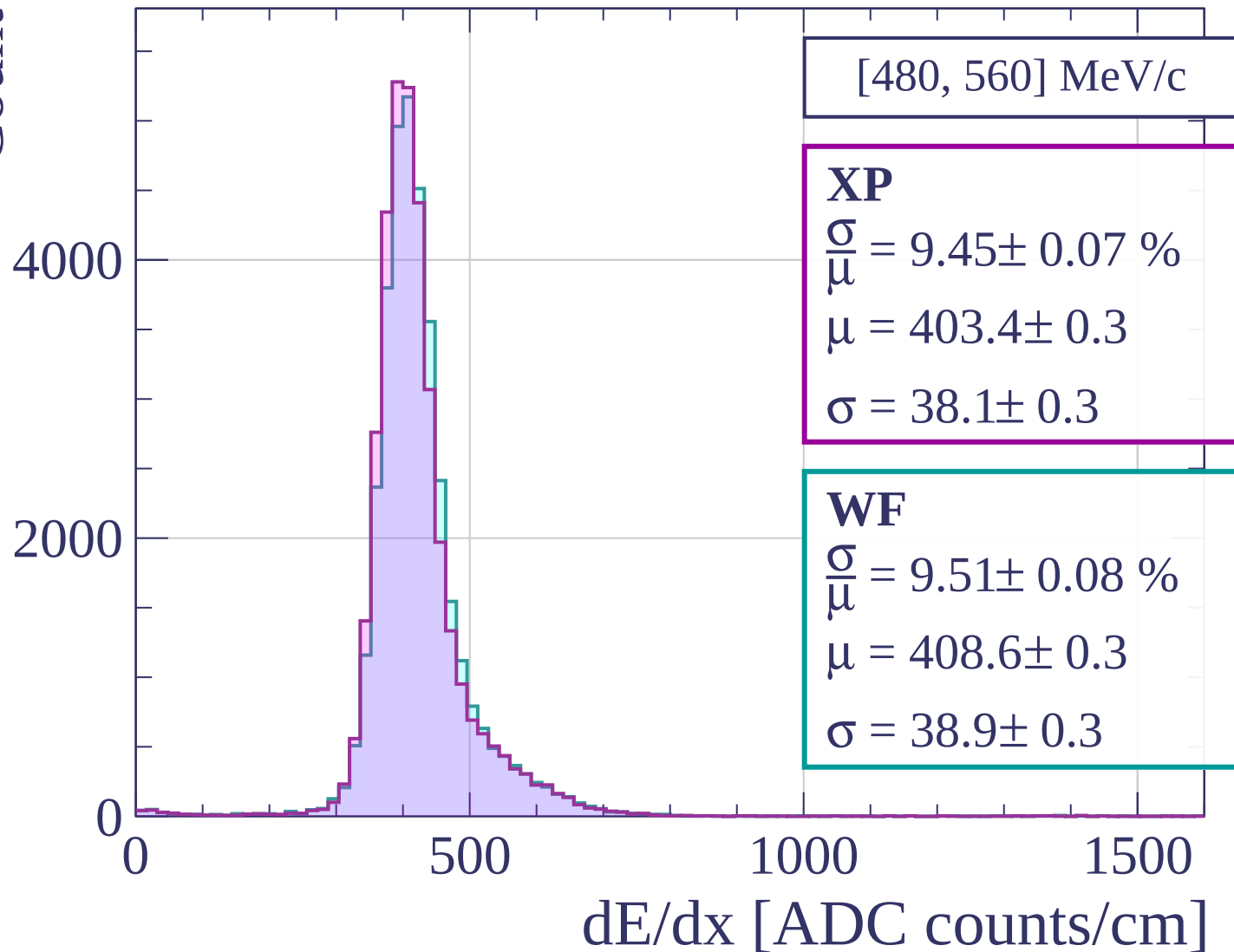








Count



[480, 560] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.45 \pm 0.07 \%$$

$$\mu = 403.4 \pm 0.3$$

$$\sigma = 38.1 \pm 0.3$$

WF

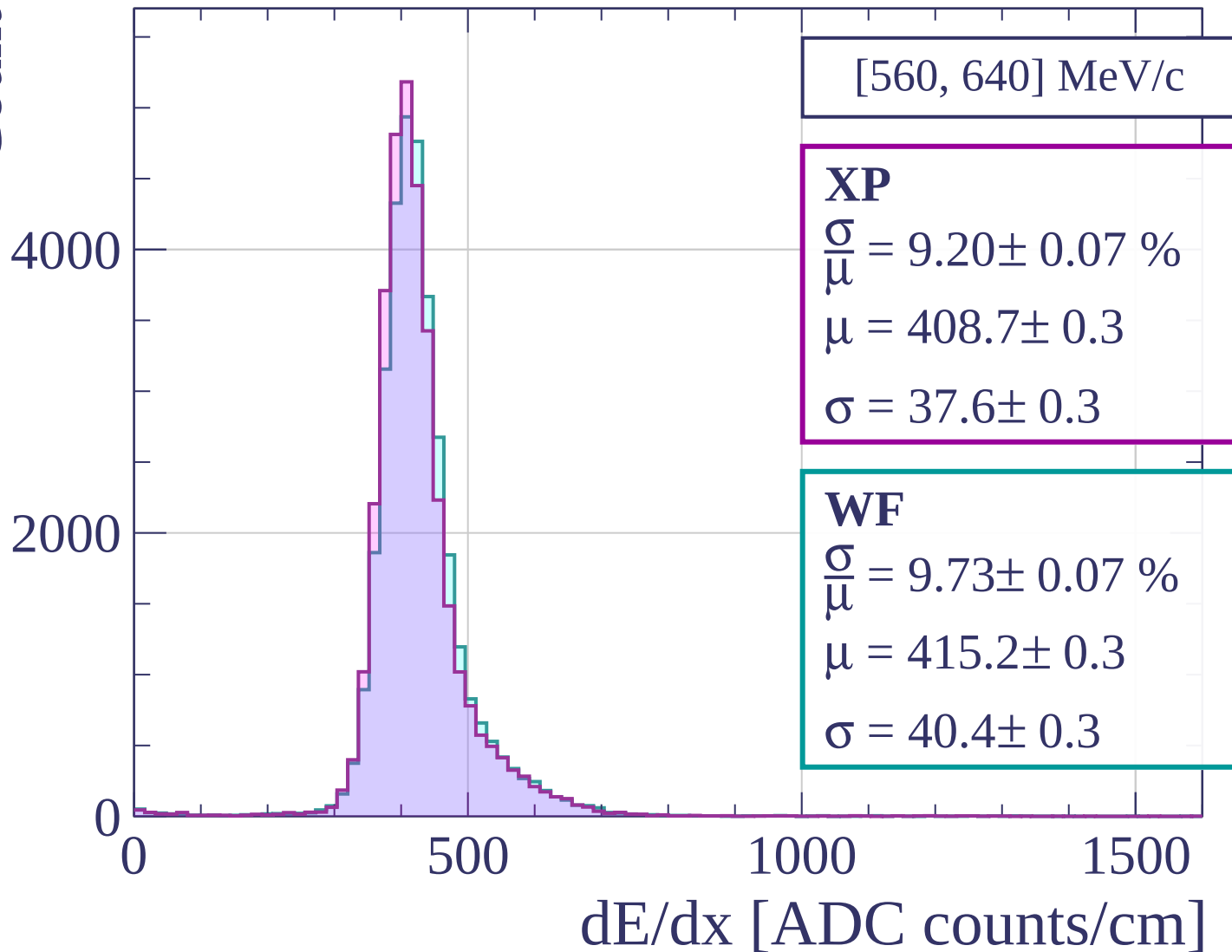
$$\frac{\sigma}{\mu} = 9.51 \pm 0.08 \%$$

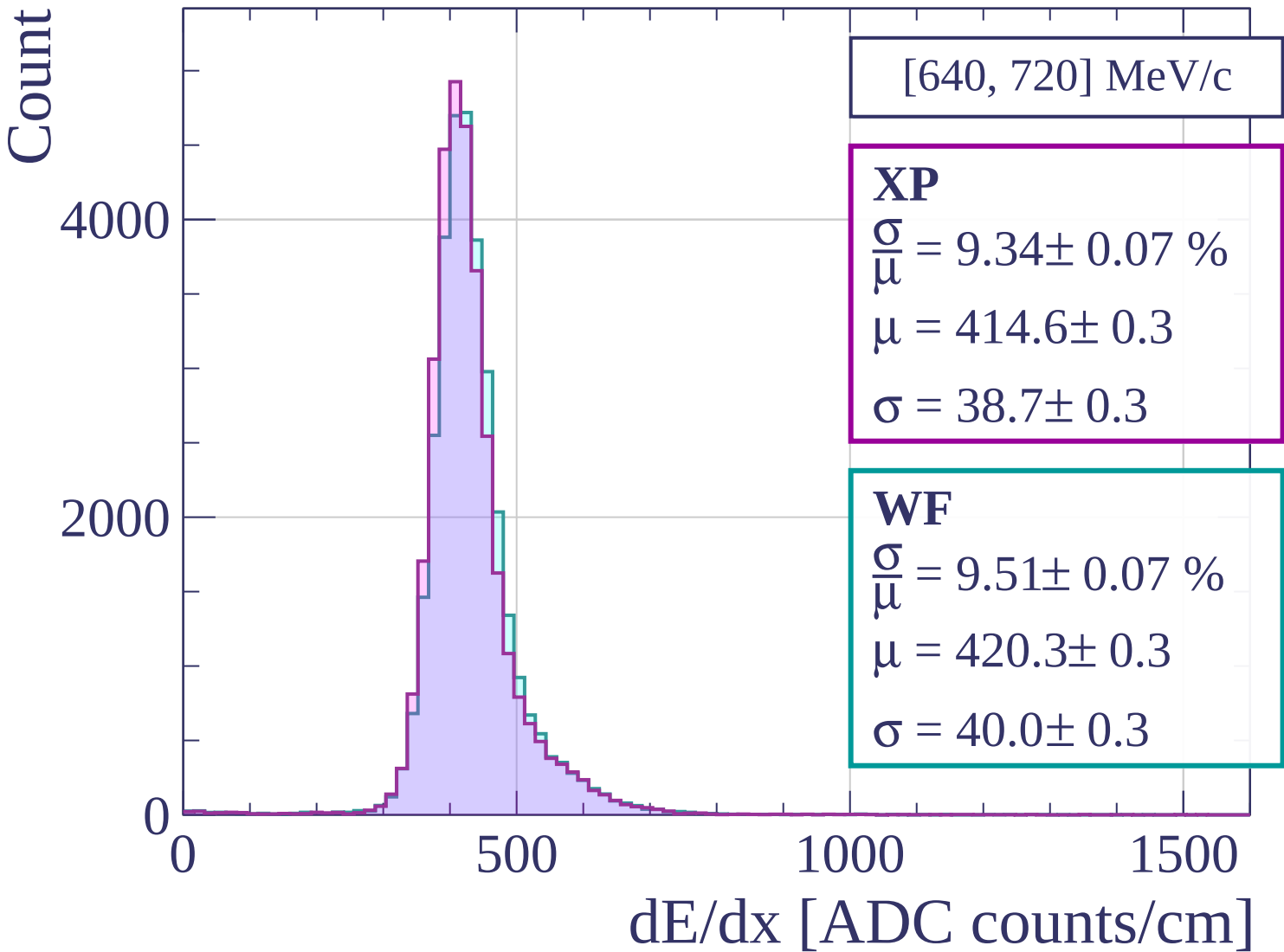
$$\mu = 408.6 \pm 0.3$$

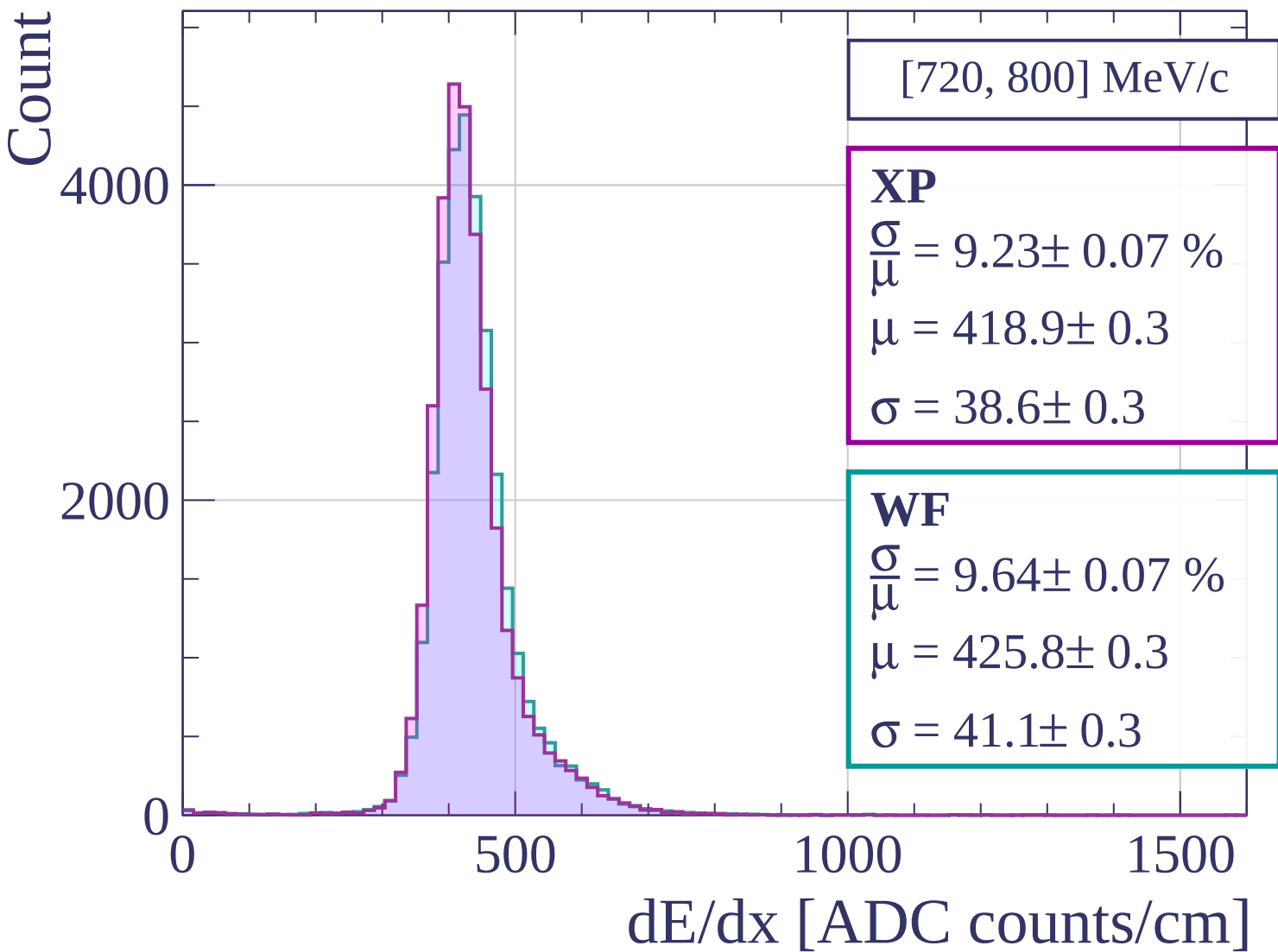
$$\sigma = 38.9 \pm 0.3$$

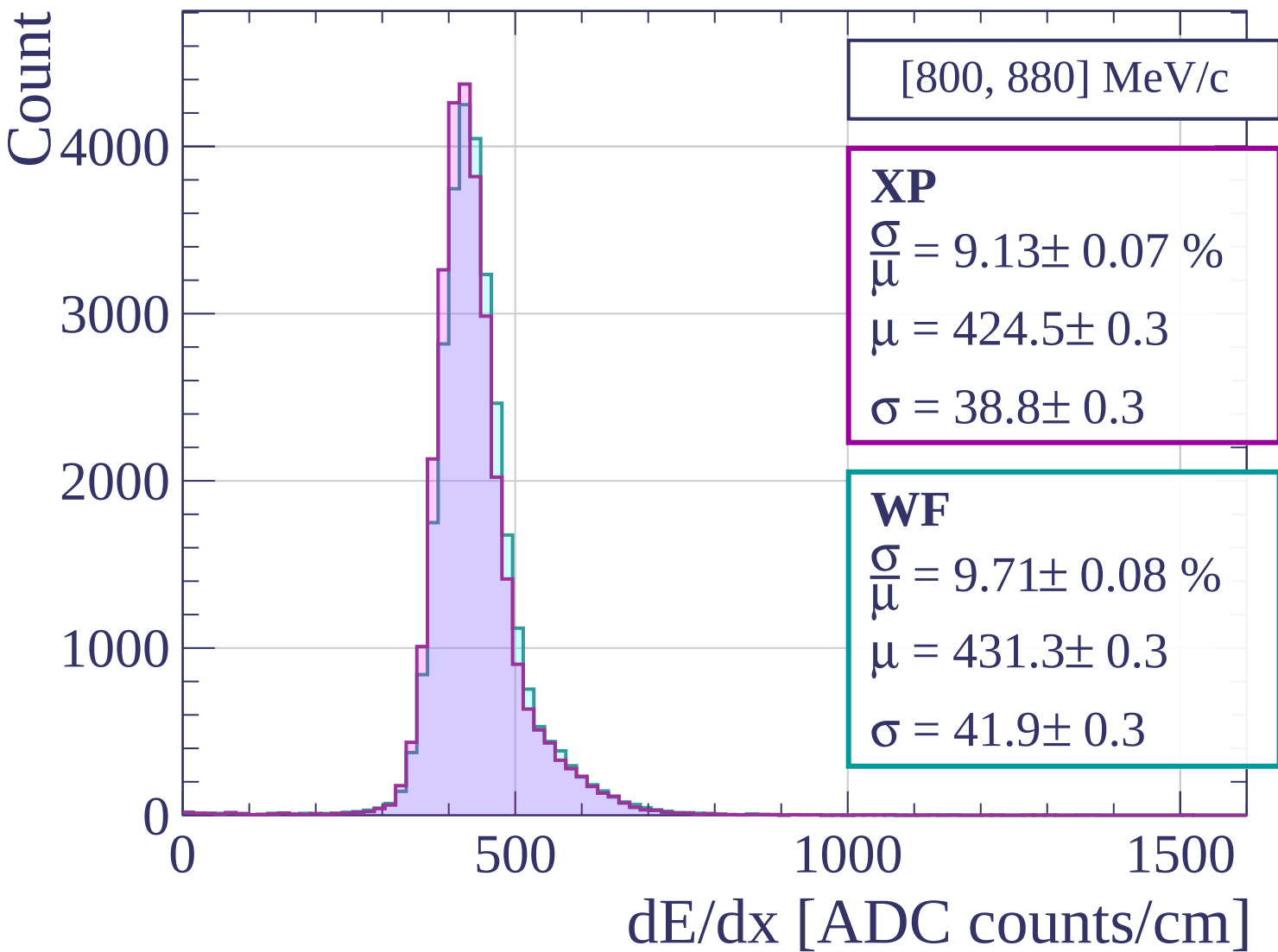
dE/dx [ADC counts/cm]

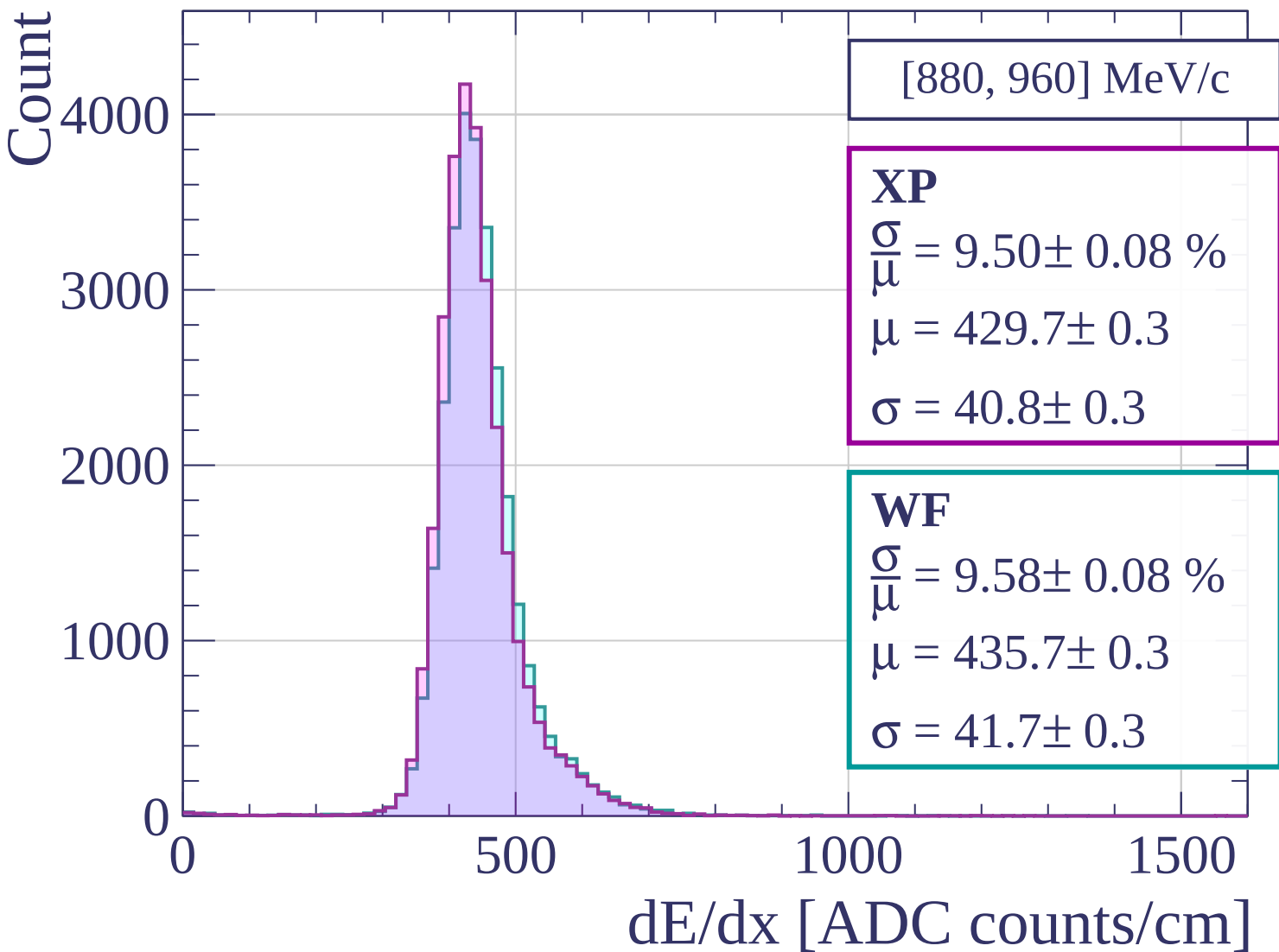
Count

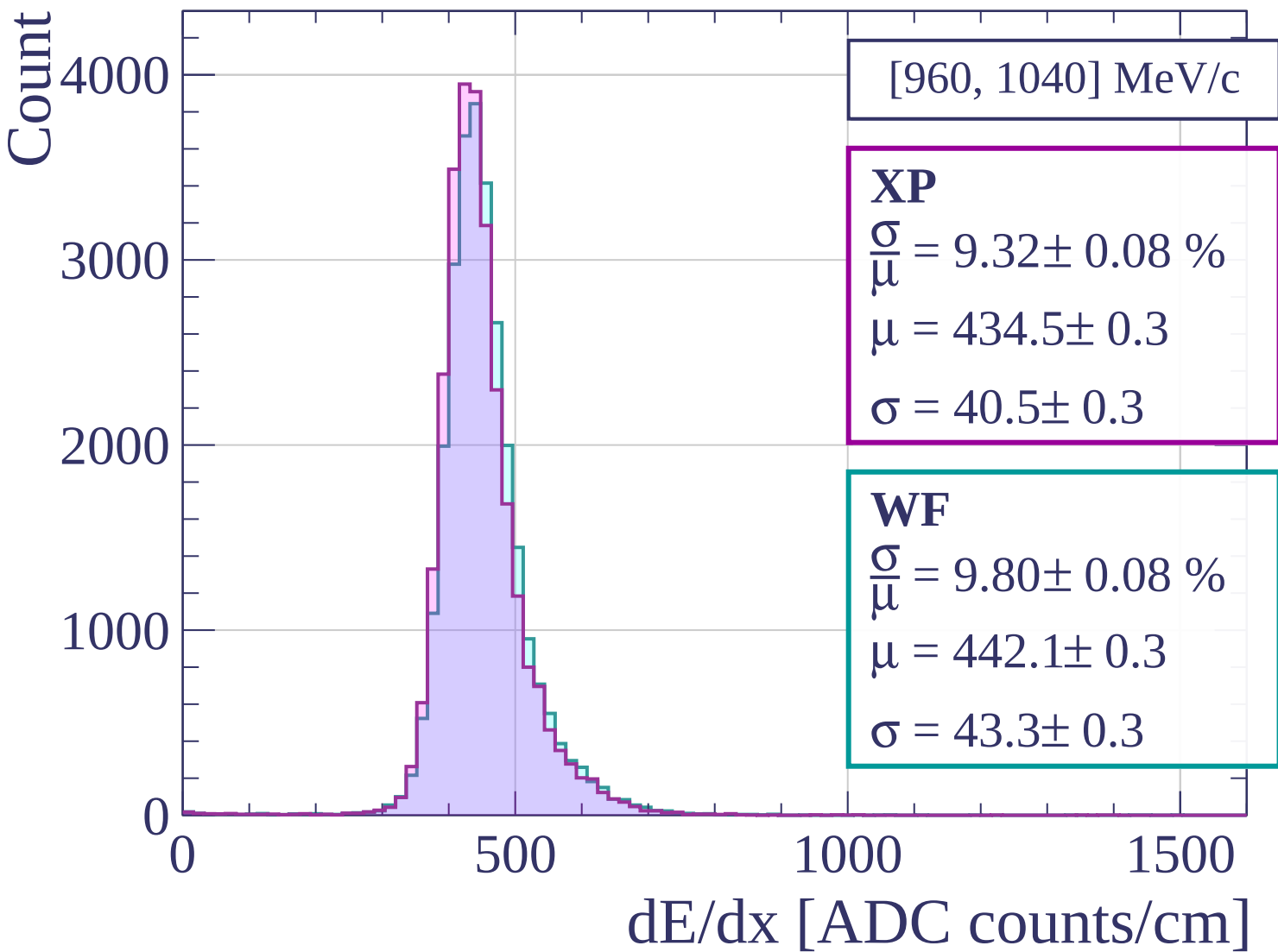


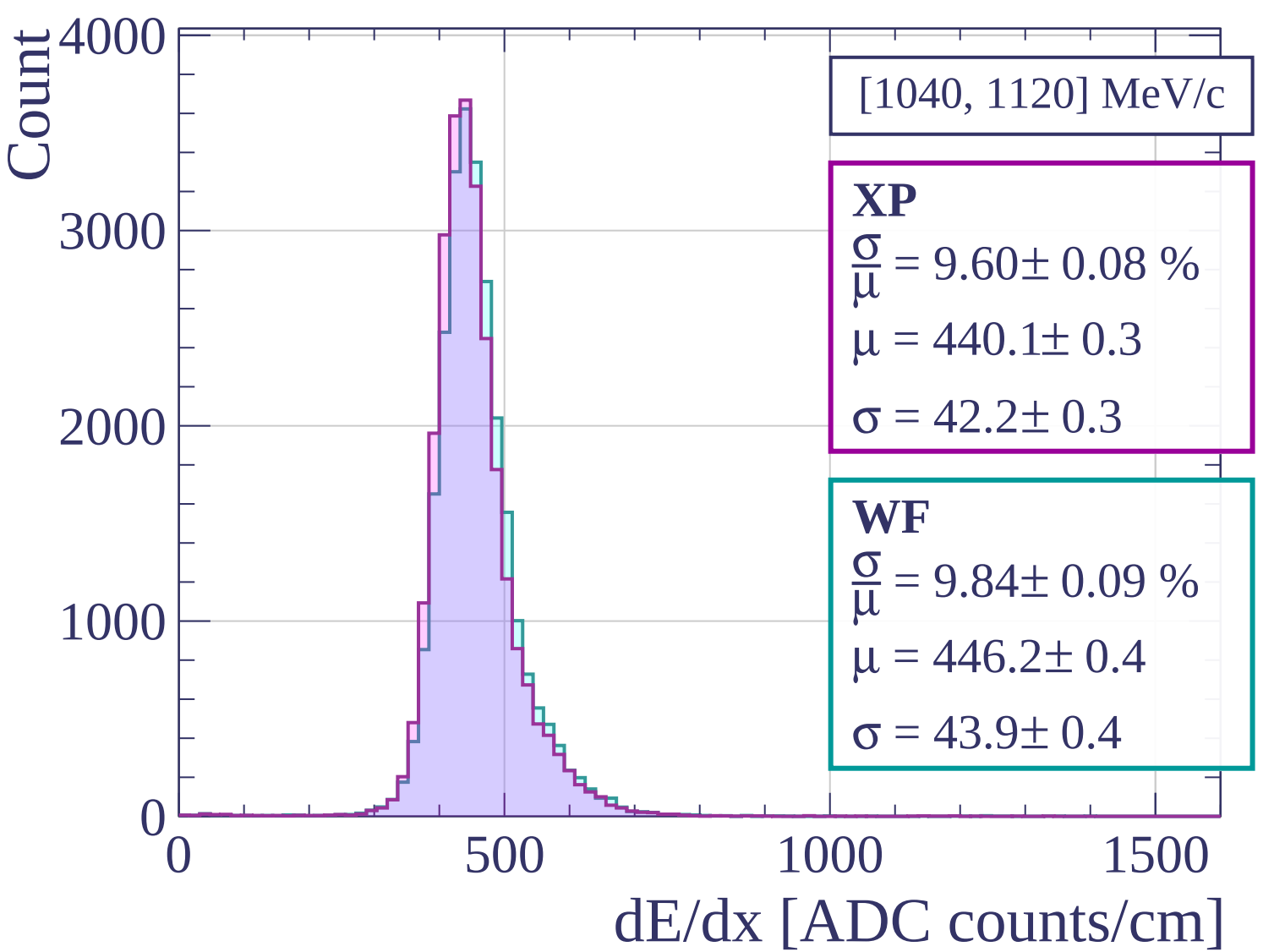


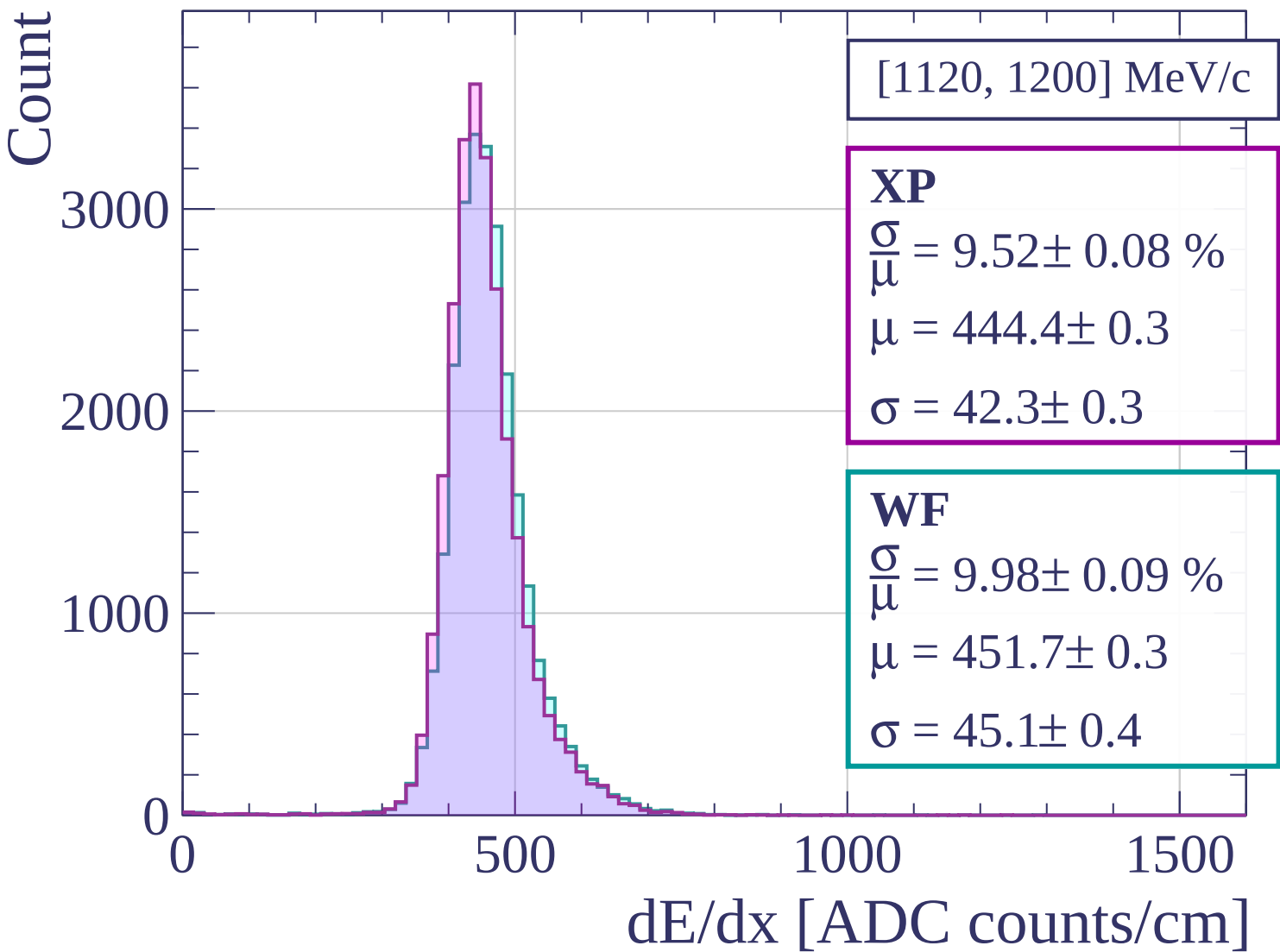


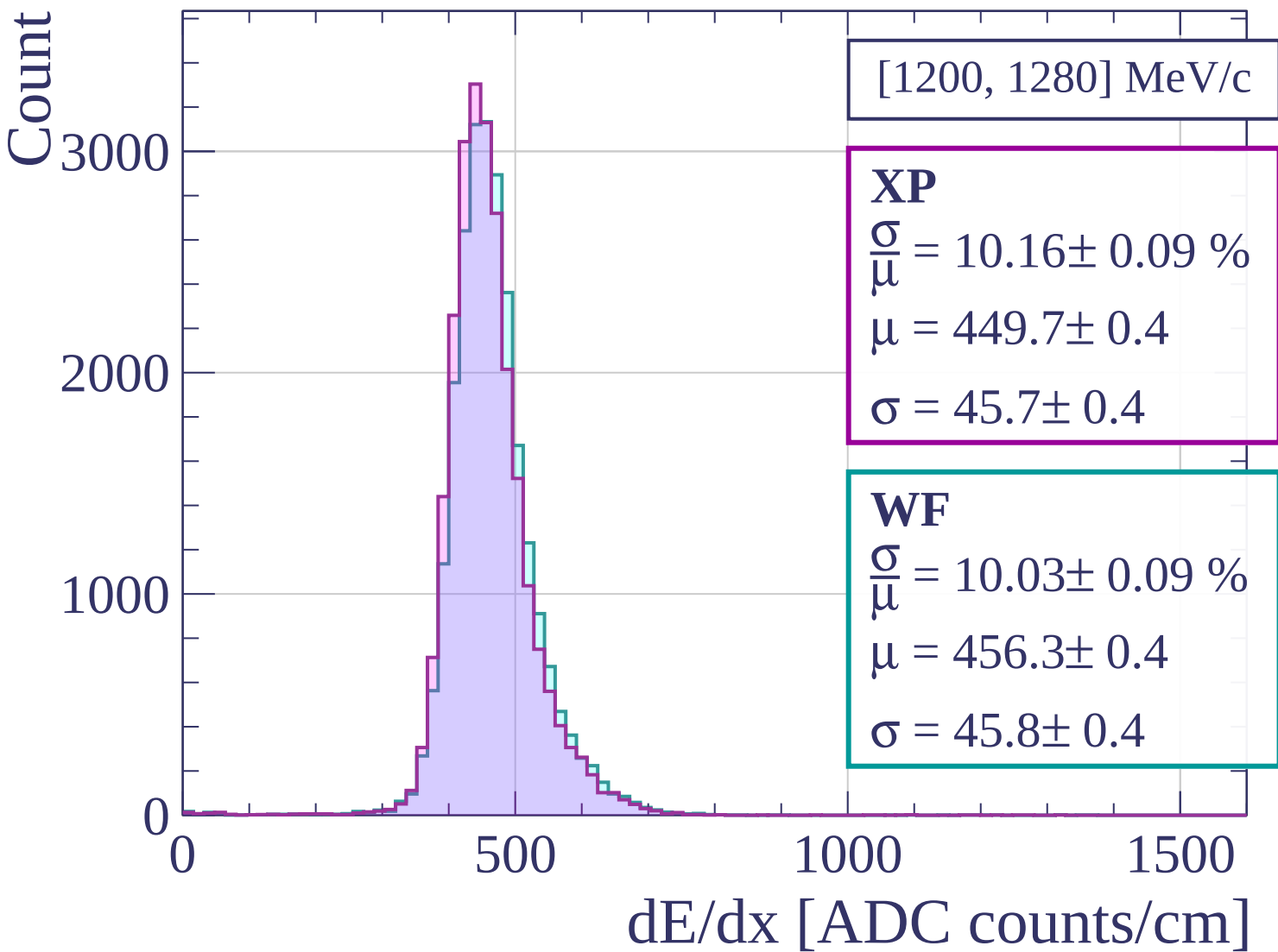


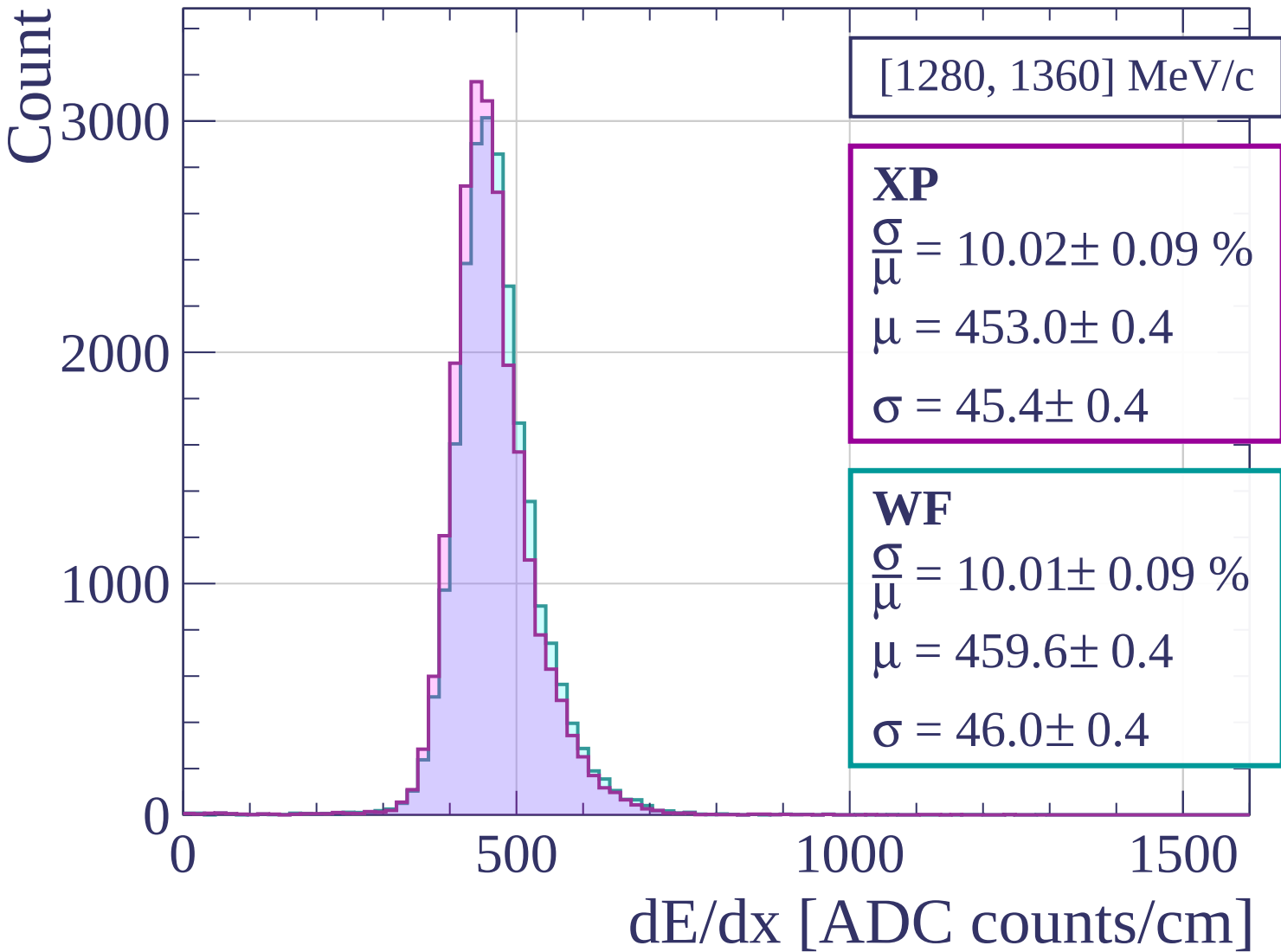


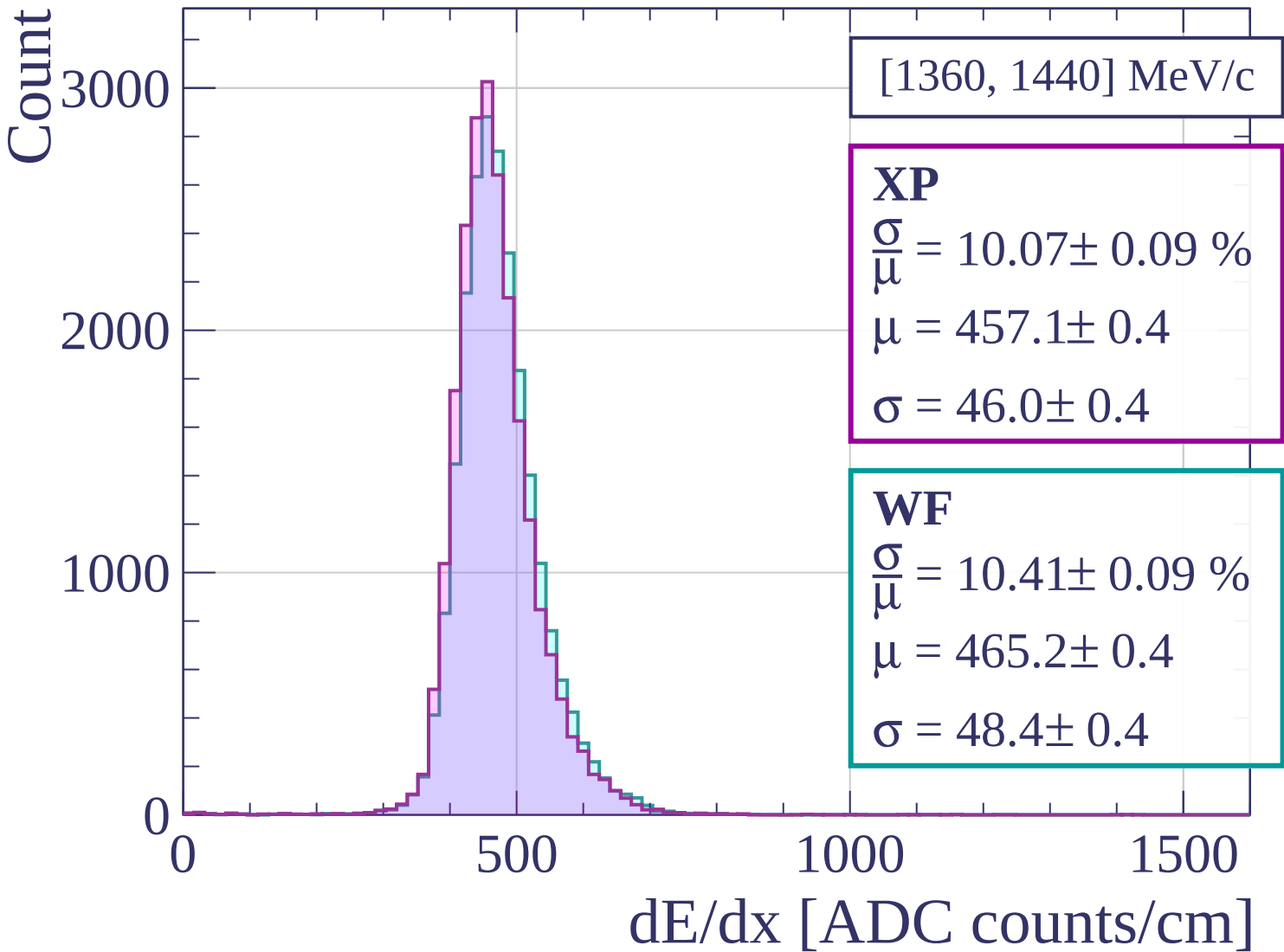


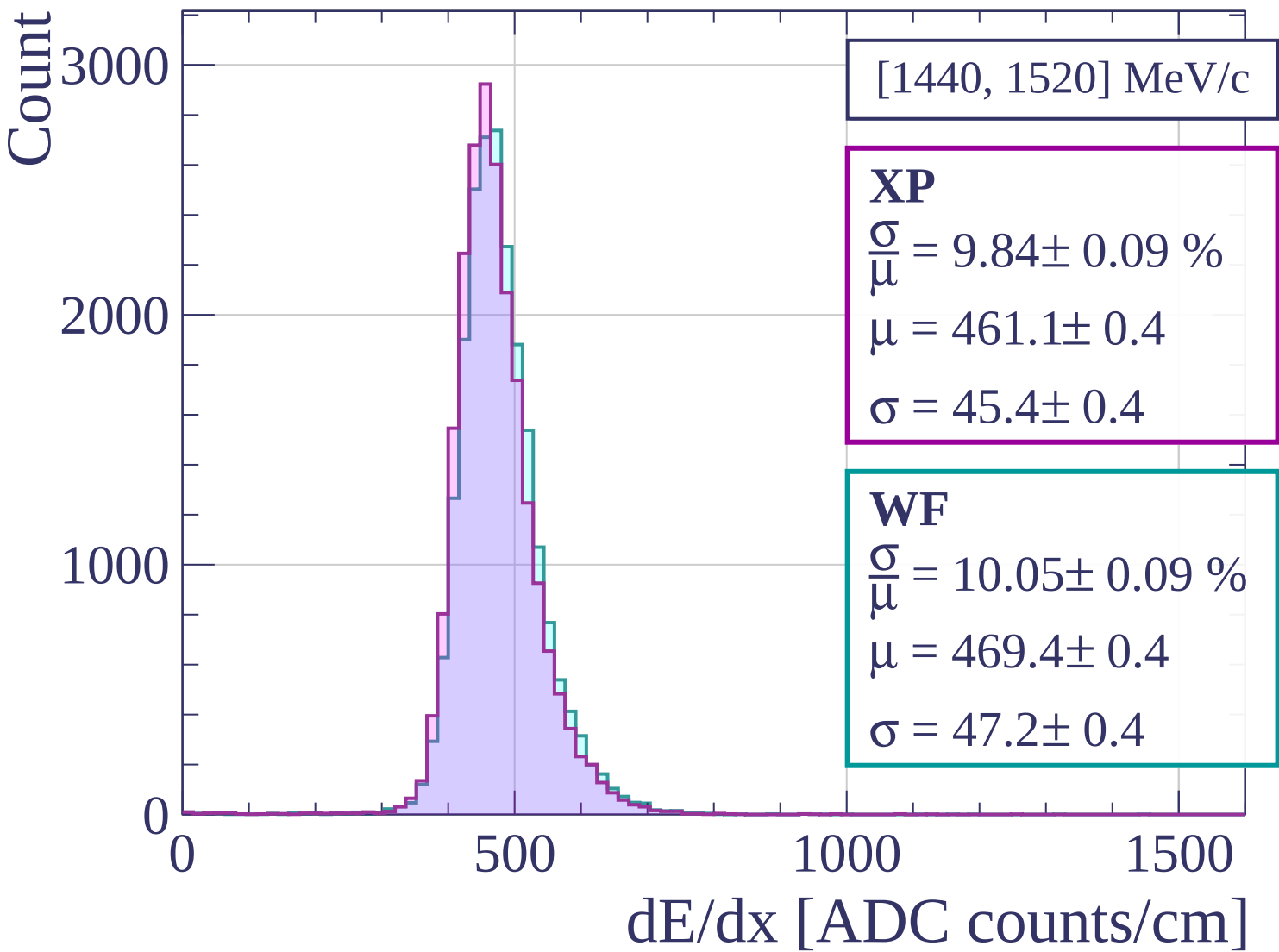


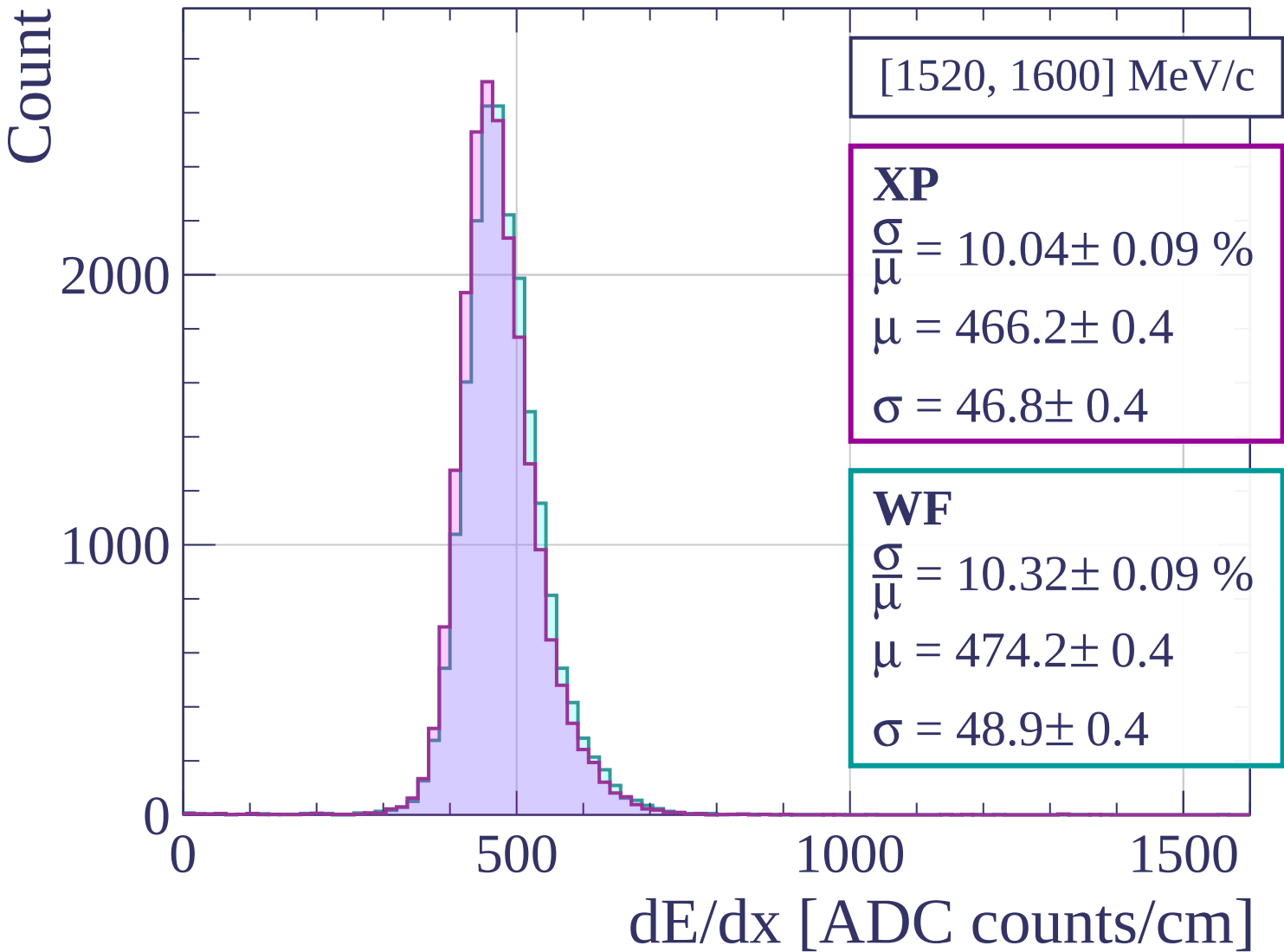


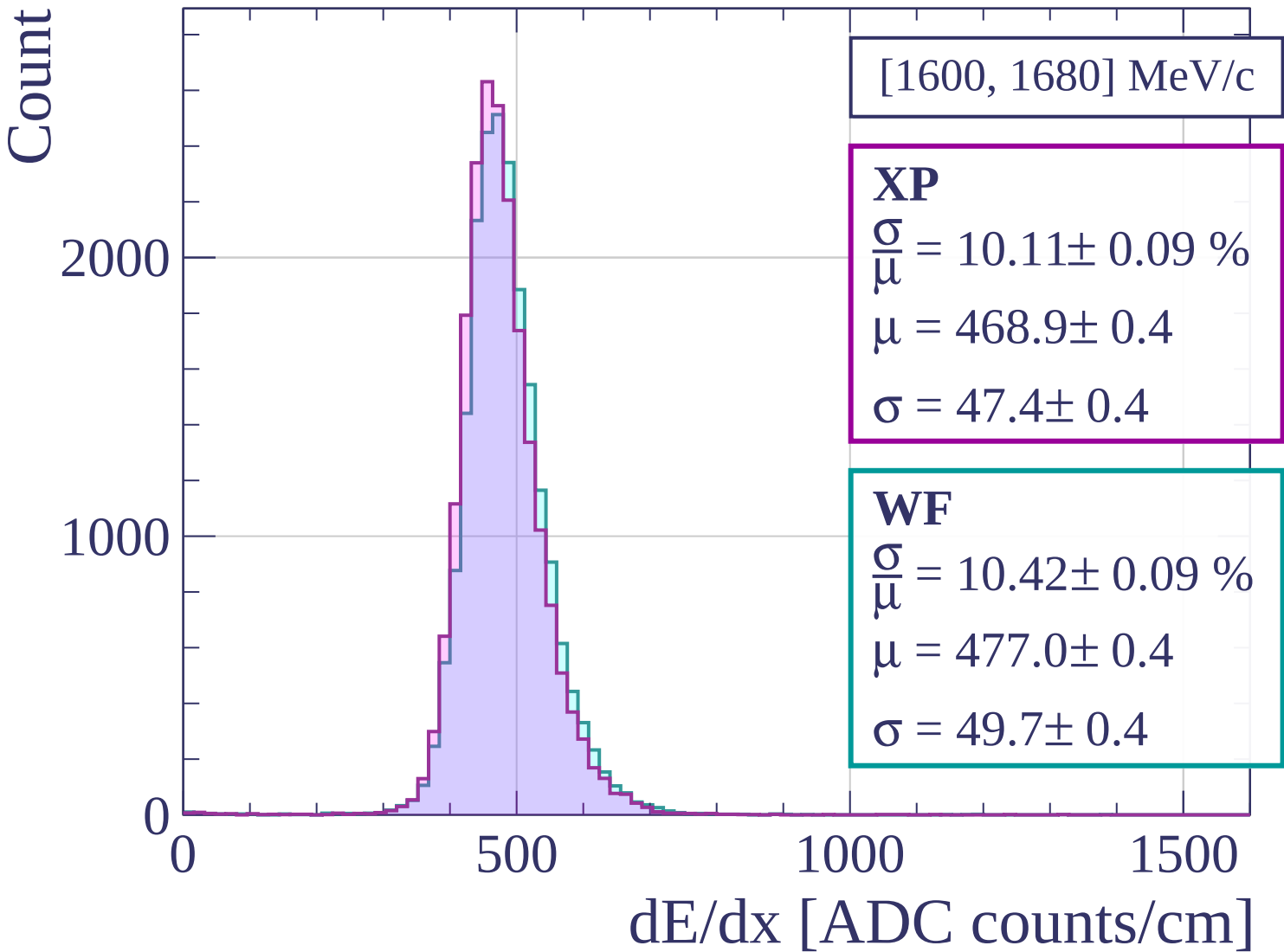


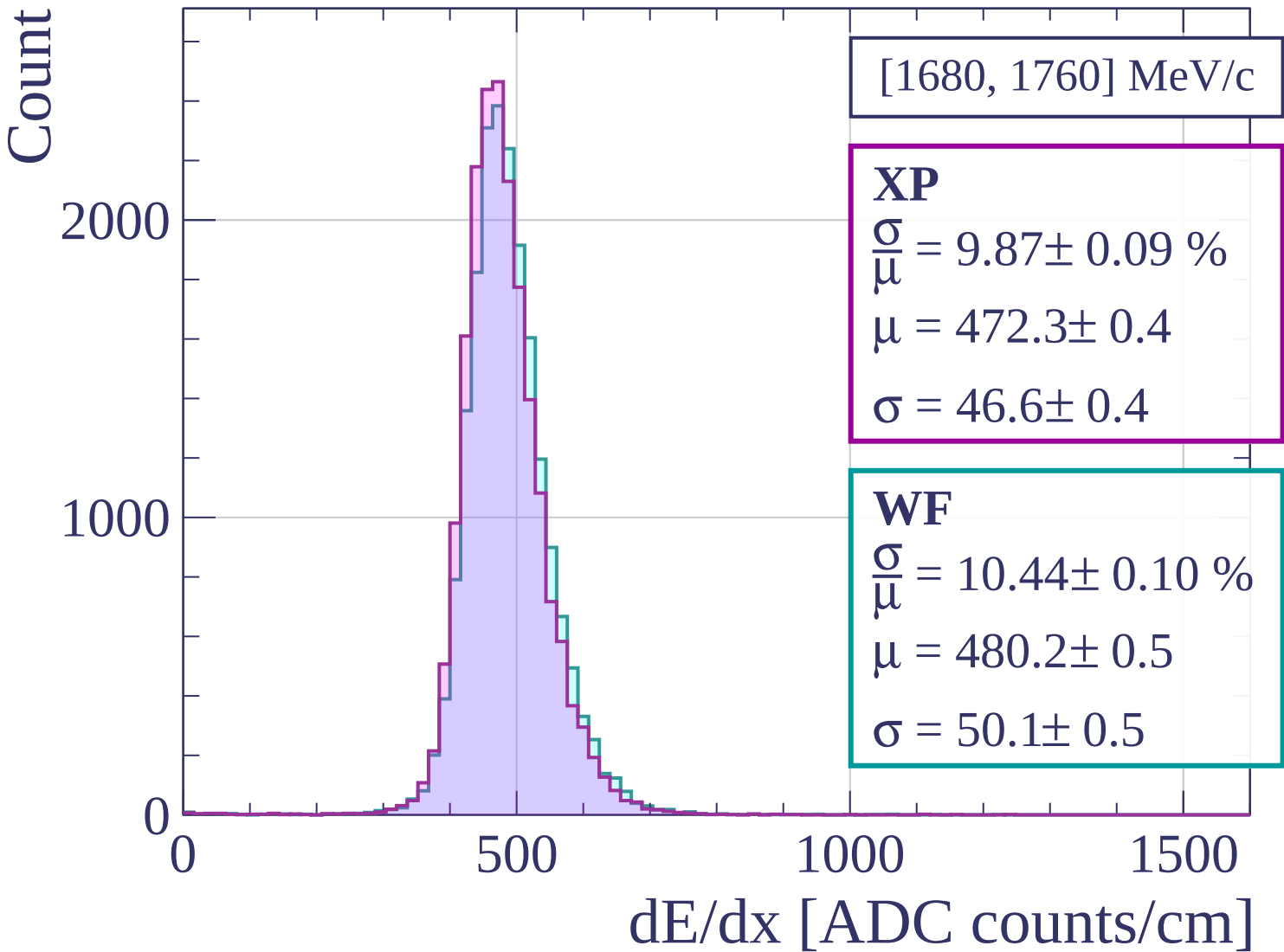


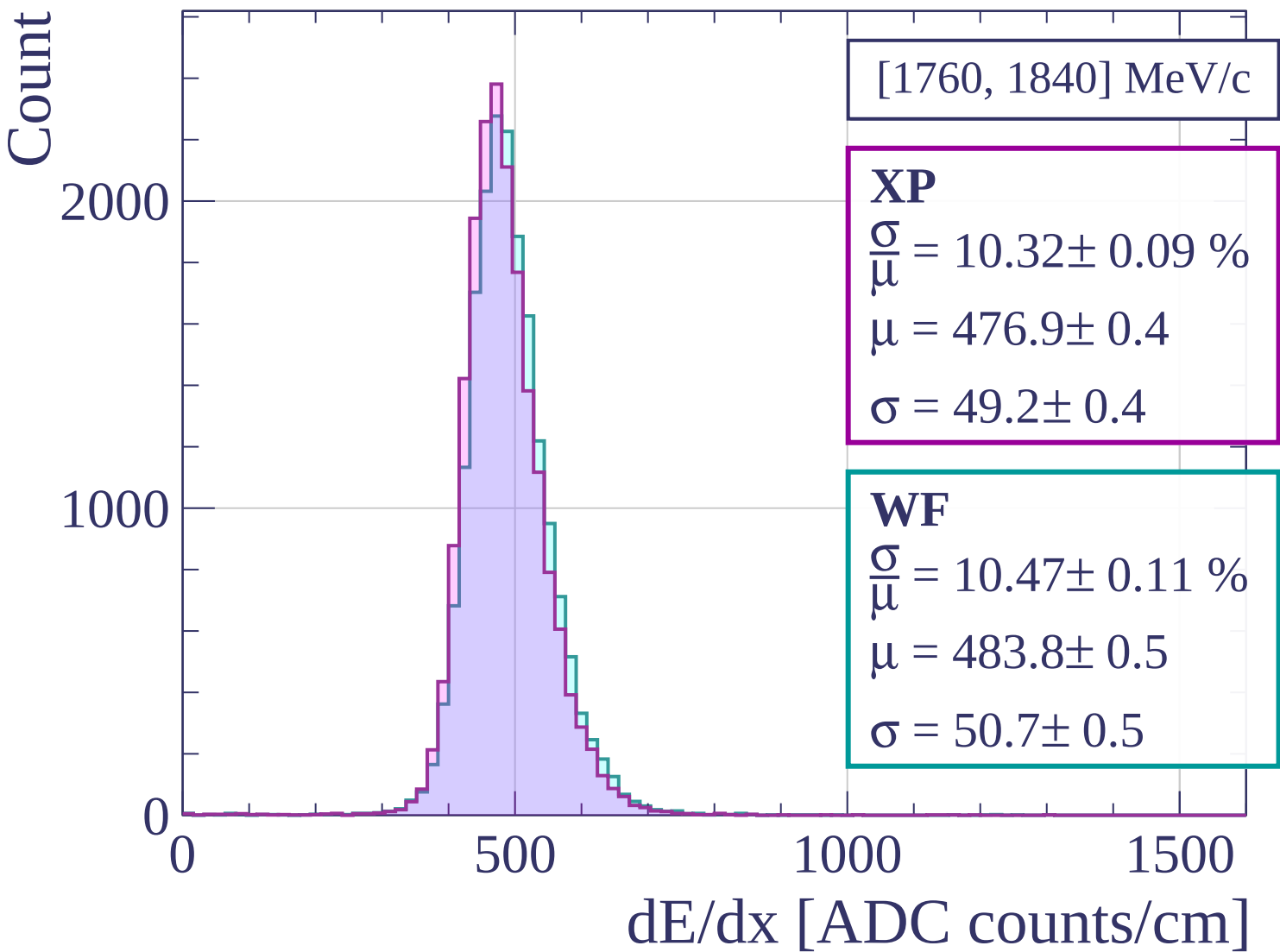


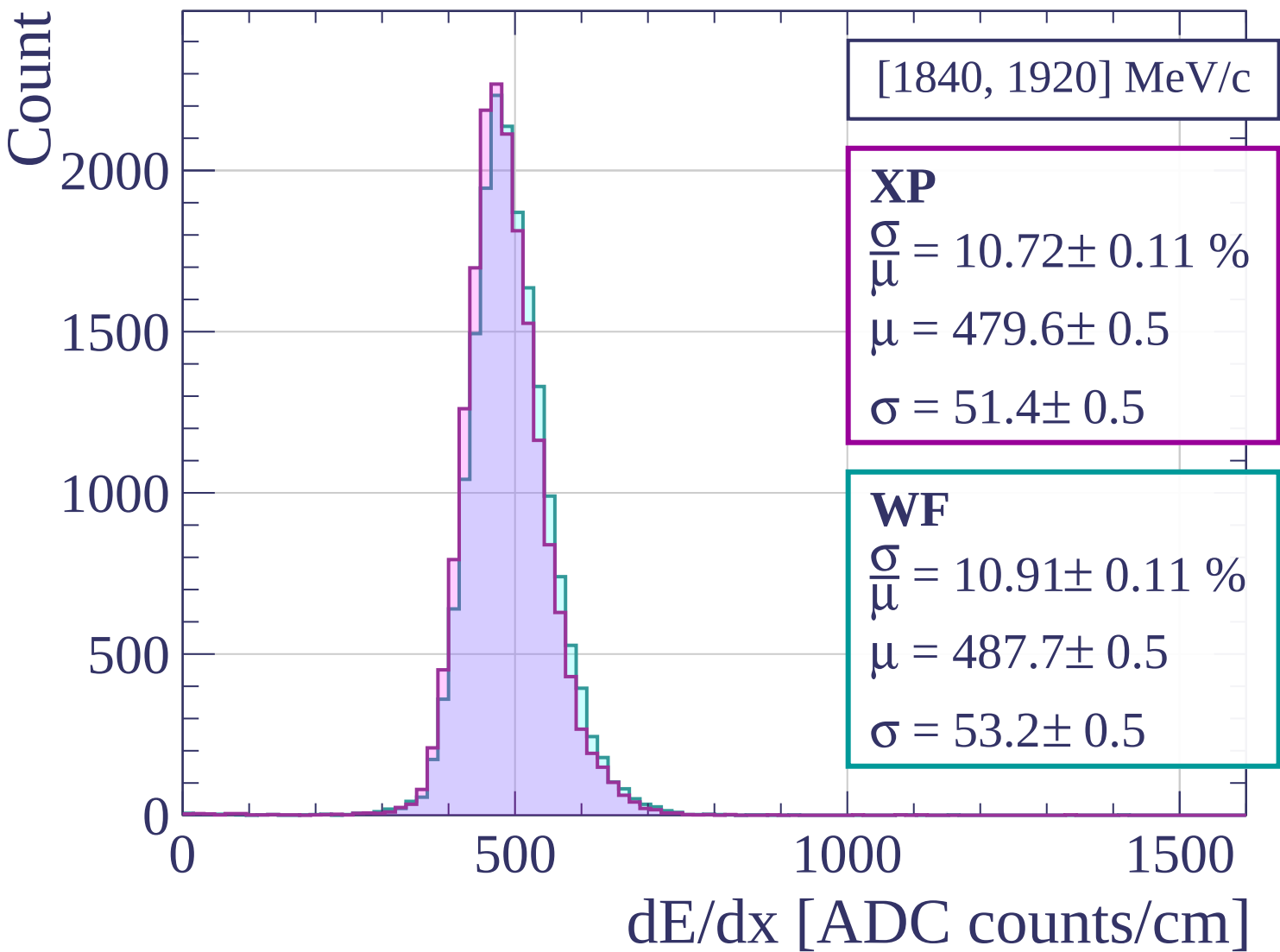


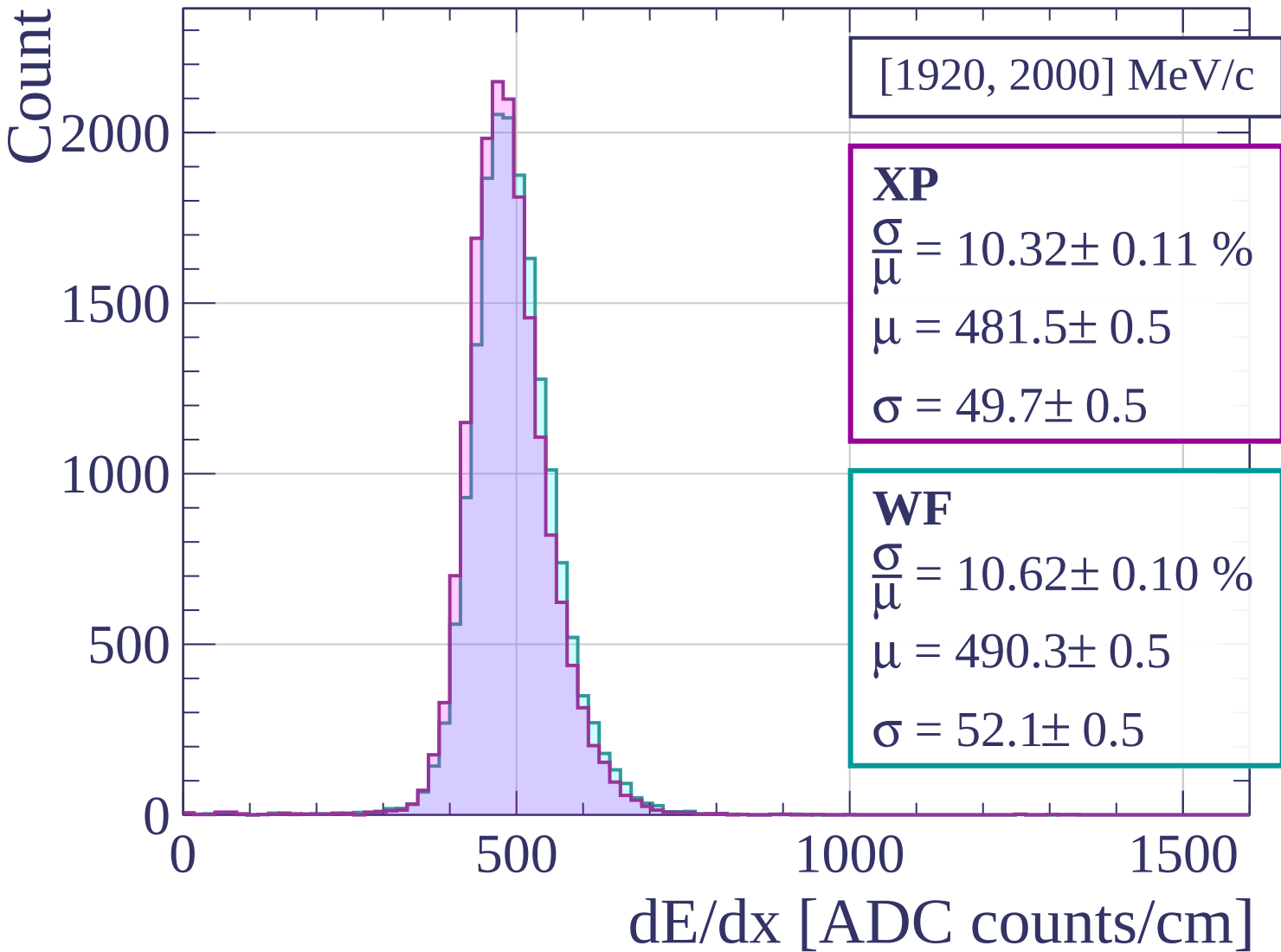












Count

[-2000, -1980] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.09 \pm 0.42 \%$$

$$\mu = 479.2 \pm 1.9$$

$$\sigma = 48.4 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 10.04 \pm 0.43 \%$$

$$\mu = 486.3 \pm 2.0$$

$$\sigma = 48.8 \pm 1.9$$

100

50

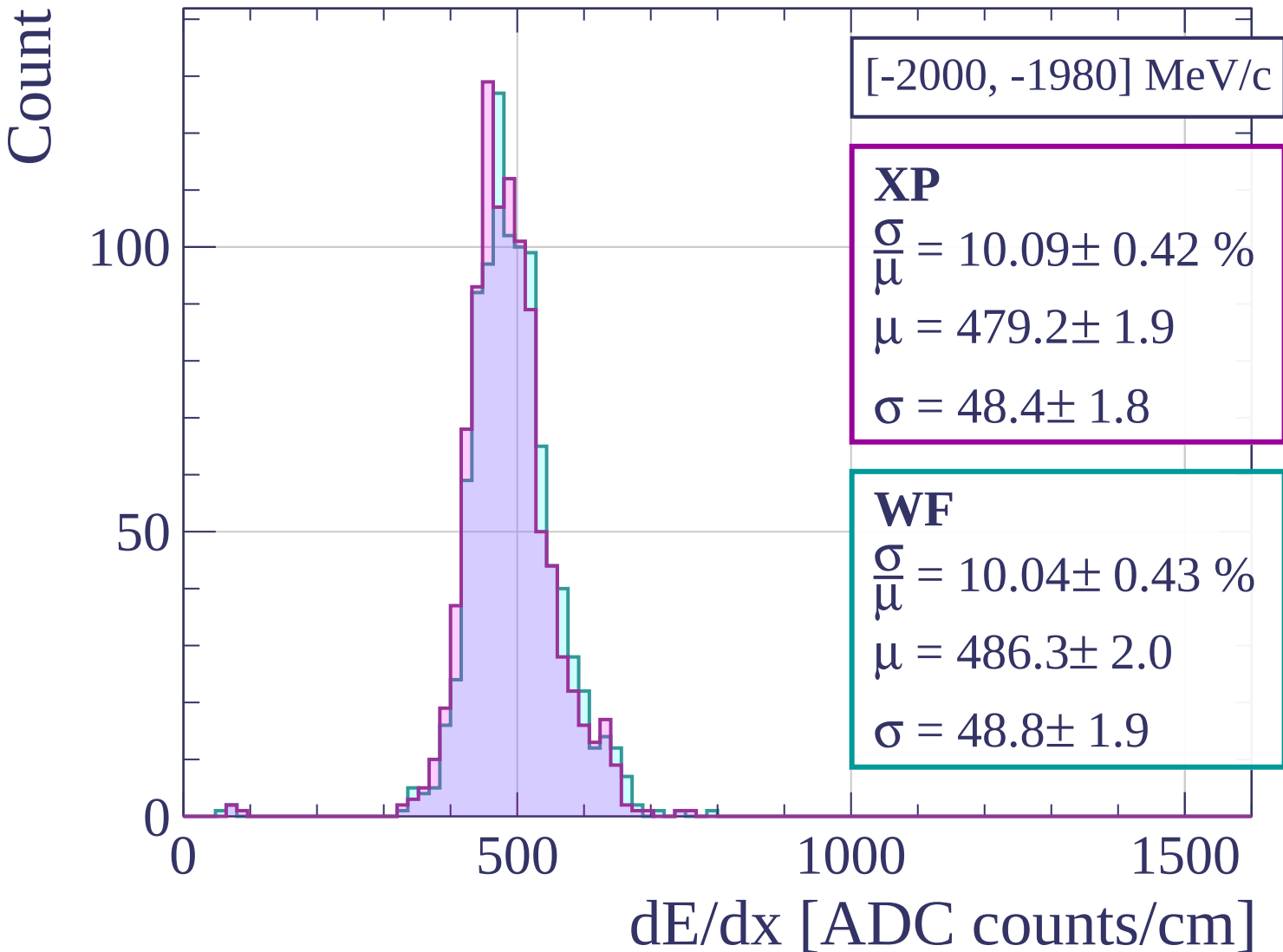
0

500

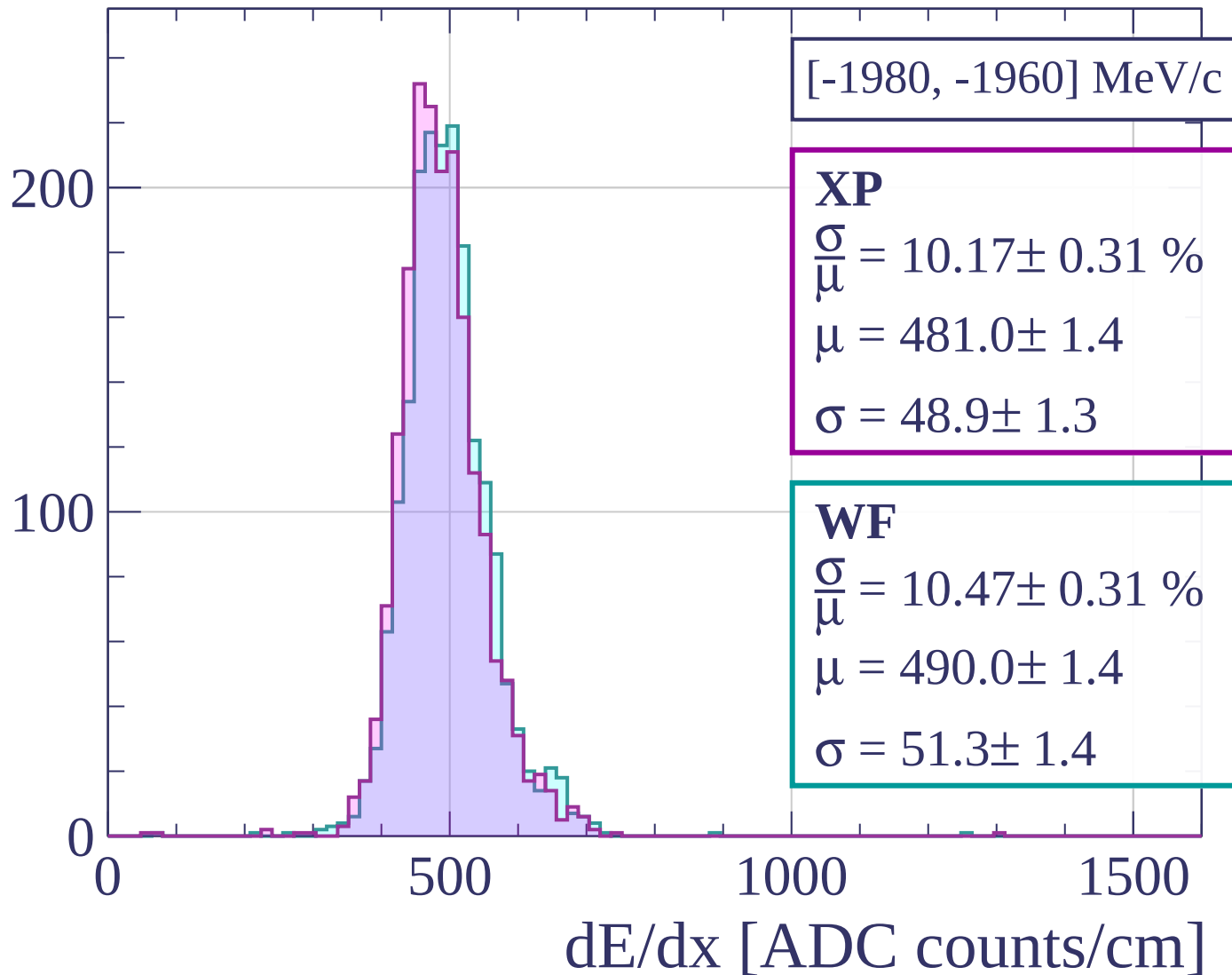
1000

1500

dE/dx [ADC counts/cm]



Count



Count

[-1960, -1940] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.55 \pm 0.35 \%$$

$$\mu = 481.4 \pm 1.5$$

$$\sigma = 50.8 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 10.99 \pm 0.35 \%$$

$$\mu = 490.1 \pm 1.5$$

$$\sigma = 53.8 \pm 1.5$$

200

100

0

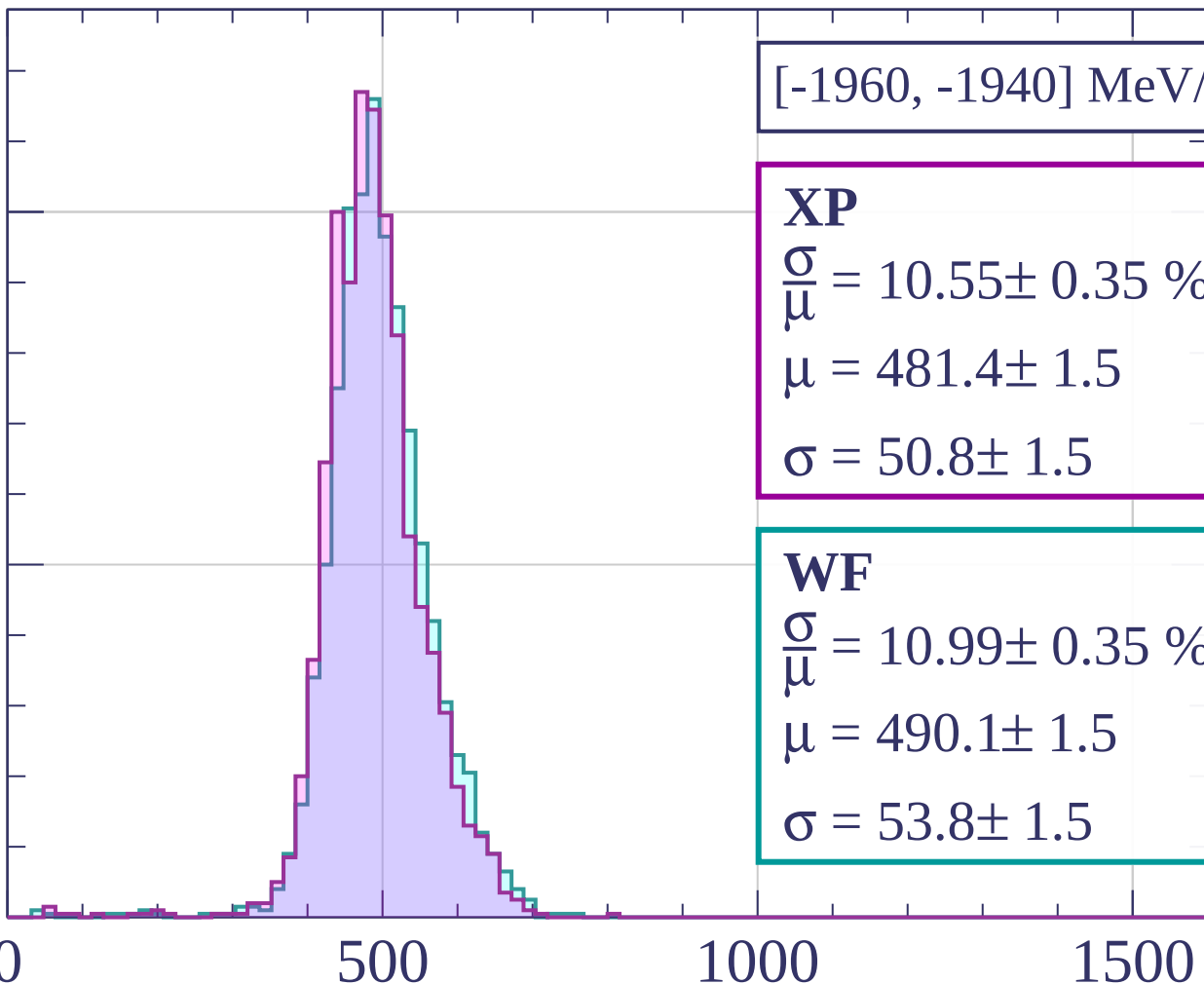
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1940, -1920] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.95 \pm 0.30 \%$$

$$\mu = 482.6 \pm 1.4$$

$$\sigma = 52.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 11.04 \pm 0.34 \%$$

$$\mu = 490.4 \pm 1.6$$

$$\sigma = 54.1 \pm 1.5$$

200

150

100

50

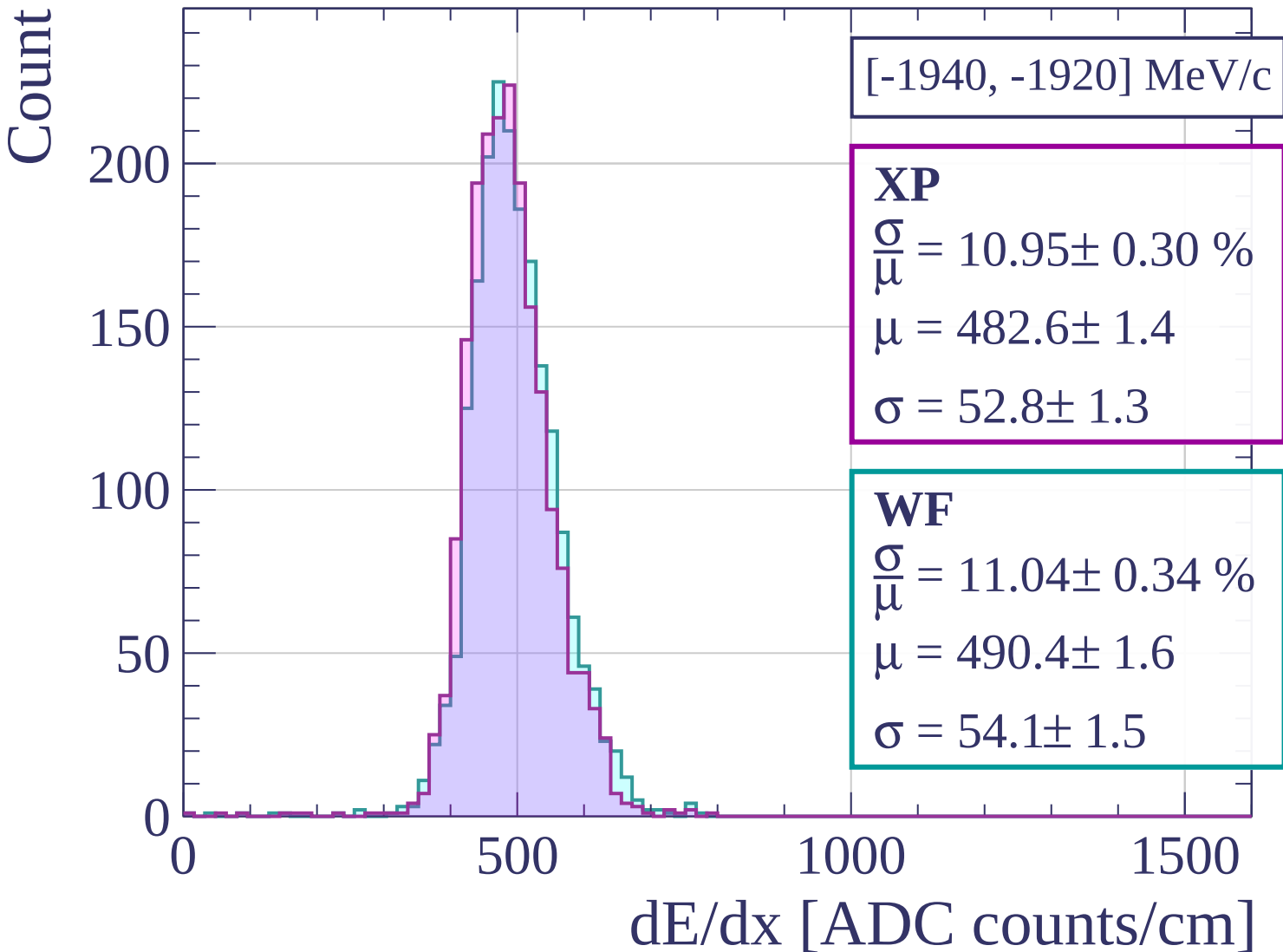
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1920, -1900] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.26 \pm 0.32 \%$$

$$\mu = 483.5 \pm 1.5$$

$$\sigma = 54.4 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.27 \pm 0.30 \%$$

$$\mu = 493.1 \pm 1.5$$

$$\sigma = 55.6 \pm 1.3$$

200

100

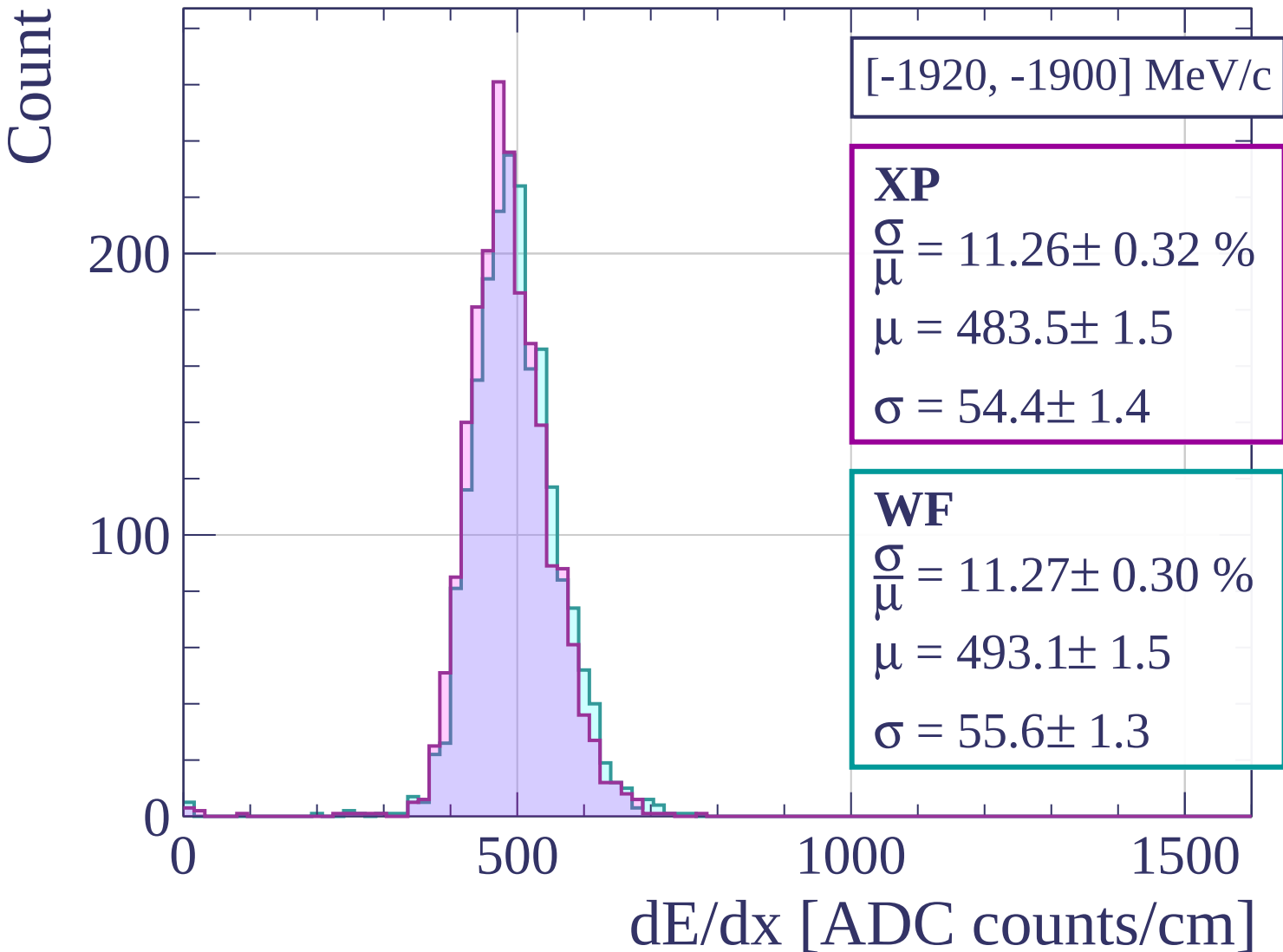
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1900, -1880] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.12 \pm 0.31 \%$$

$$\mu = 482.6 \pm 1.5$$

$$\sigma = 53.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 11.06 \pm 0.31 \%$$

$$\mu = 489.2 \pm 1.5$$

$$\sigma = 54.1 \pm 1.3$$

200

100

0

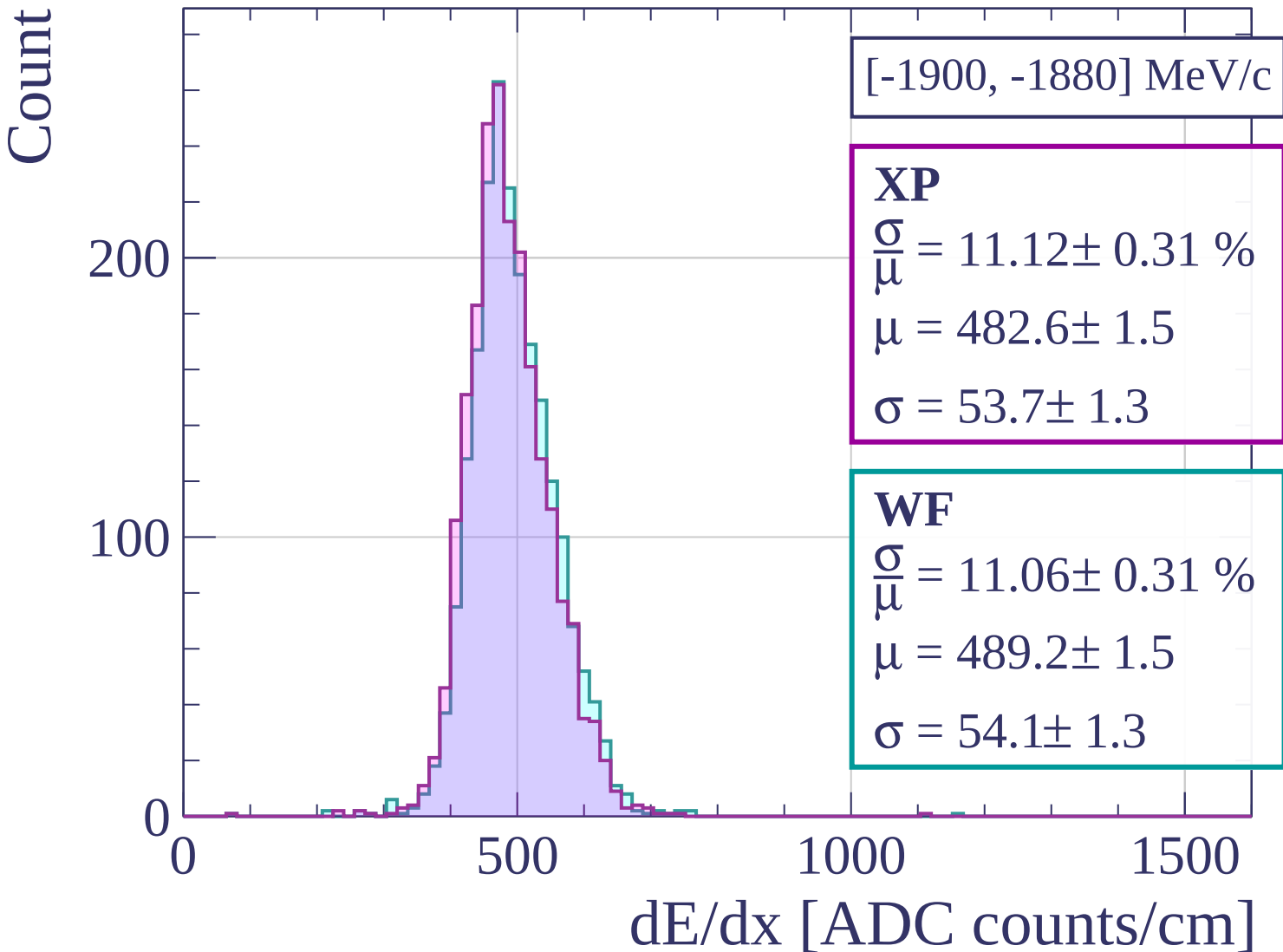
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1880, -1860] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.97 \pm 0.33 \%$$

$$\mu = 478.5 \pm 1.5$$

$$\sigma = 52.5 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.51 \pm 0.31 \%$$

$$\mu = 488.6 \pm 1.5$$

$$\sigma = 56.2 \pm 1.3$$

200

100

0

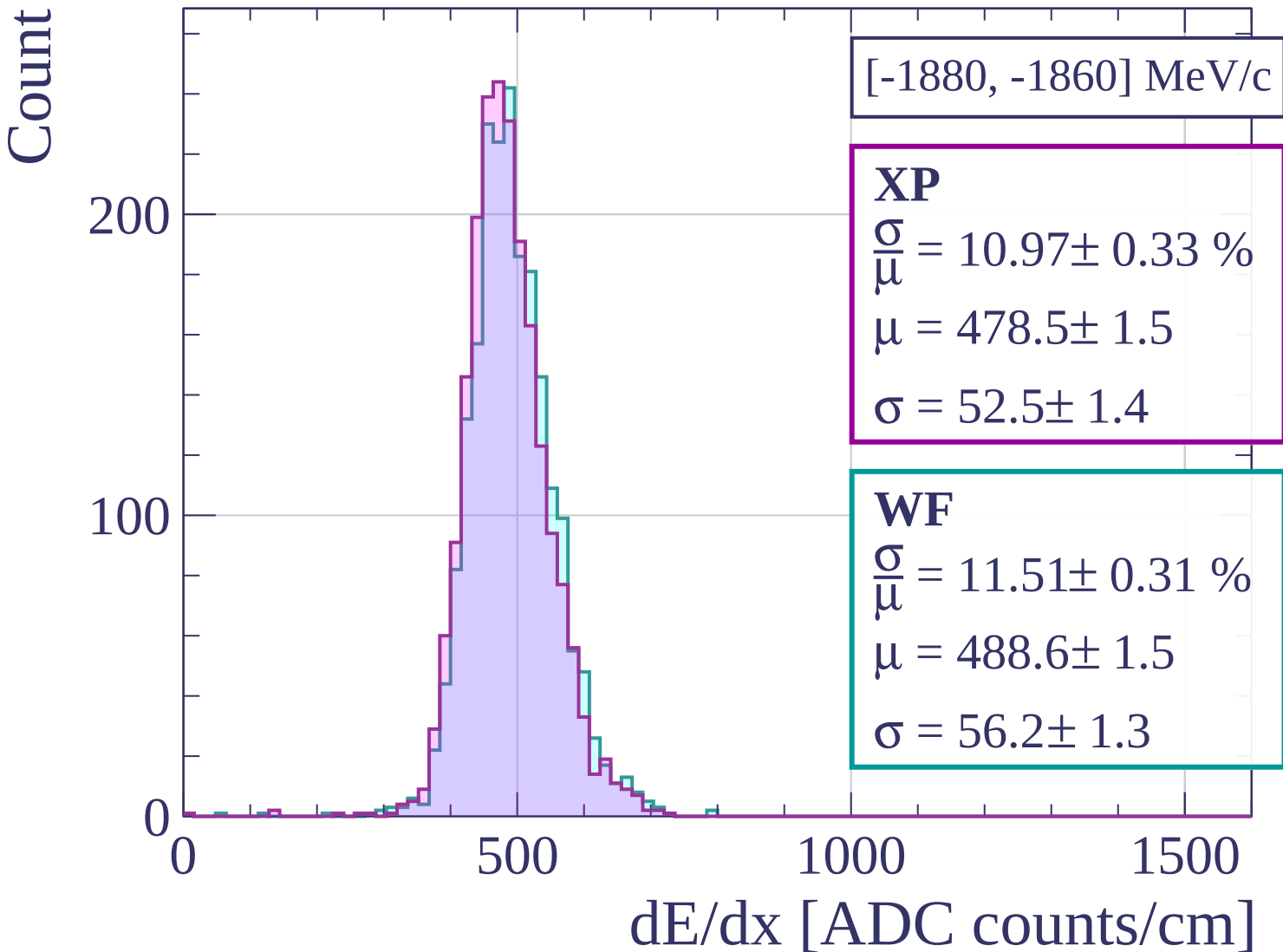
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1860, -1840] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.58 \pm 0.30 \%$$

$$\mu = 481.6 \pm 1.4$$

$$\sigma = 51.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 11.10 \pm 0.33 \%$$

$$\mu = 488.9 \pm 1.4$$

$$\sigma = 54.3 \pm 1.5$$

200

100

0

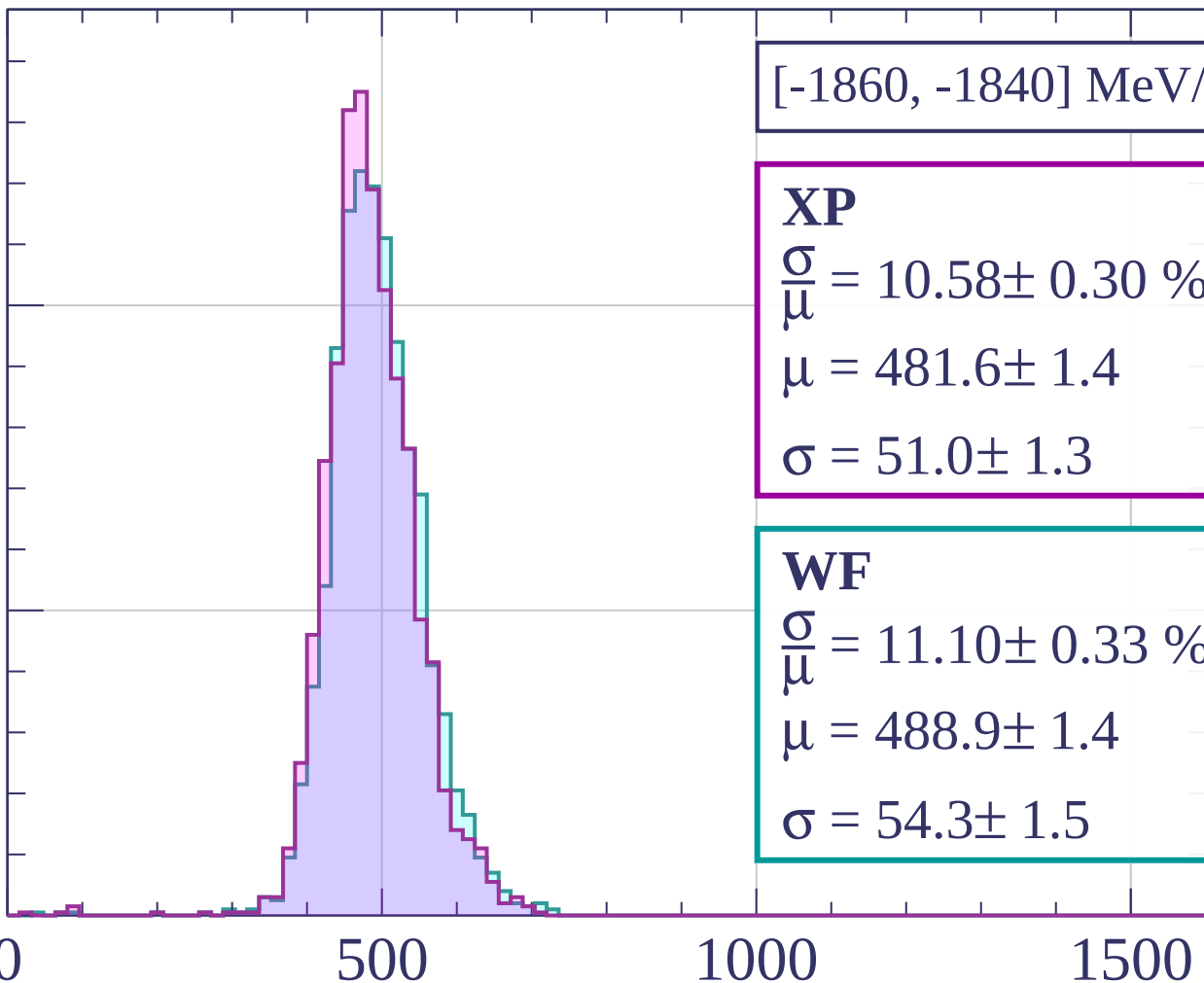
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1840, -1820] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.20 \pm 0.29 \%$$

$$\mu = 479.5 \pm 1.4$$

$$\sigma = 53.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.86 \pm 0.28 \%$$

$$\mu = 487.6 \pm 1.4$$

$$\sigma = 53.0 \pm 1.2$$

200

100

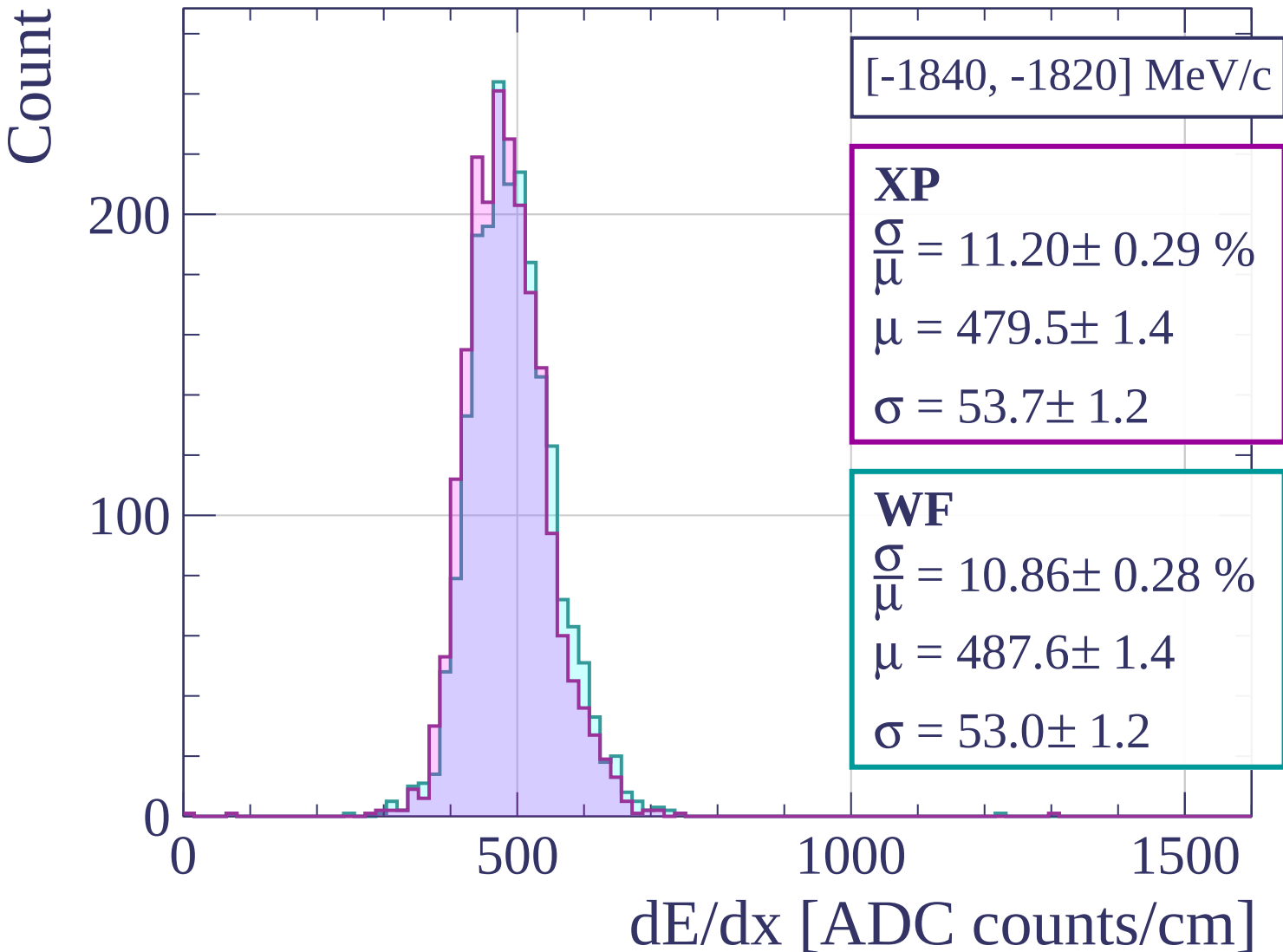
0

500

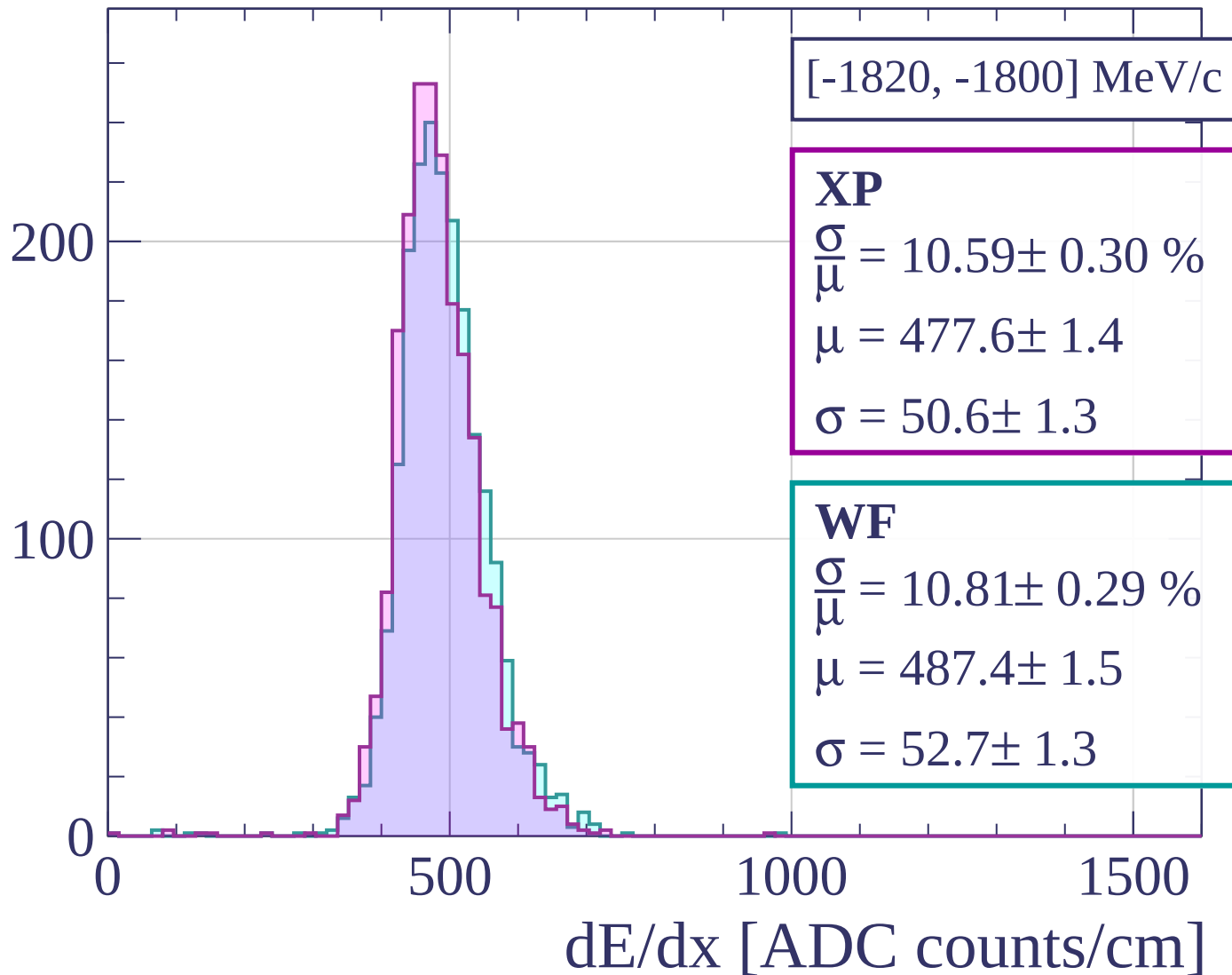
1000

1500

dE/dx [ADC counts/cm]



Count



Count

[-1800, -1780] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.71 \pm 0.30 \%$$

$$\mu = 479.3 \pm 1.5$$

$$\sigma = 51.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.59 \pm 0.25 \%$$

$$\mu = 490.6 \pm 1.4$$

$$\sigma = 52.0 \pm 1.1$$

200

100

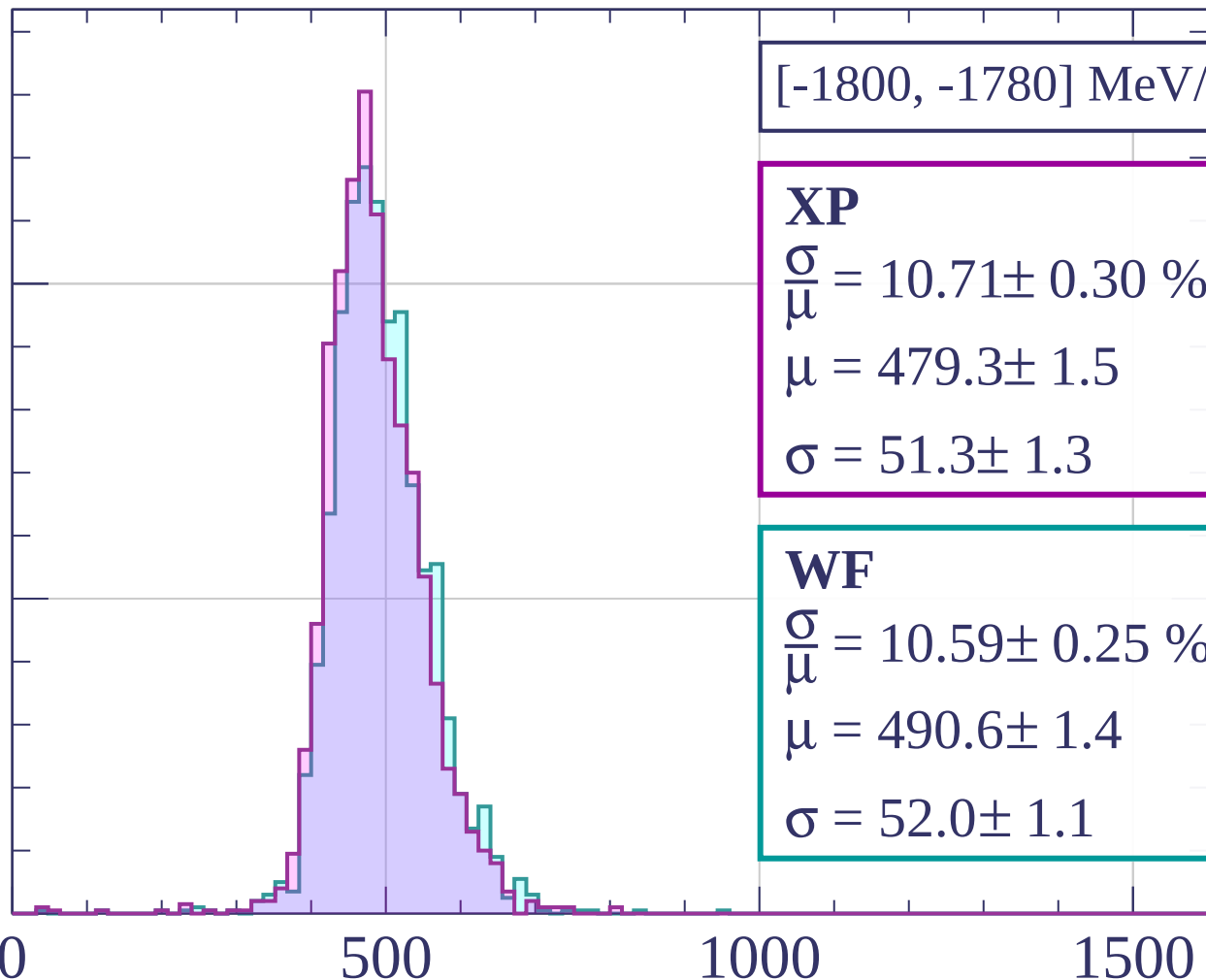
0

500

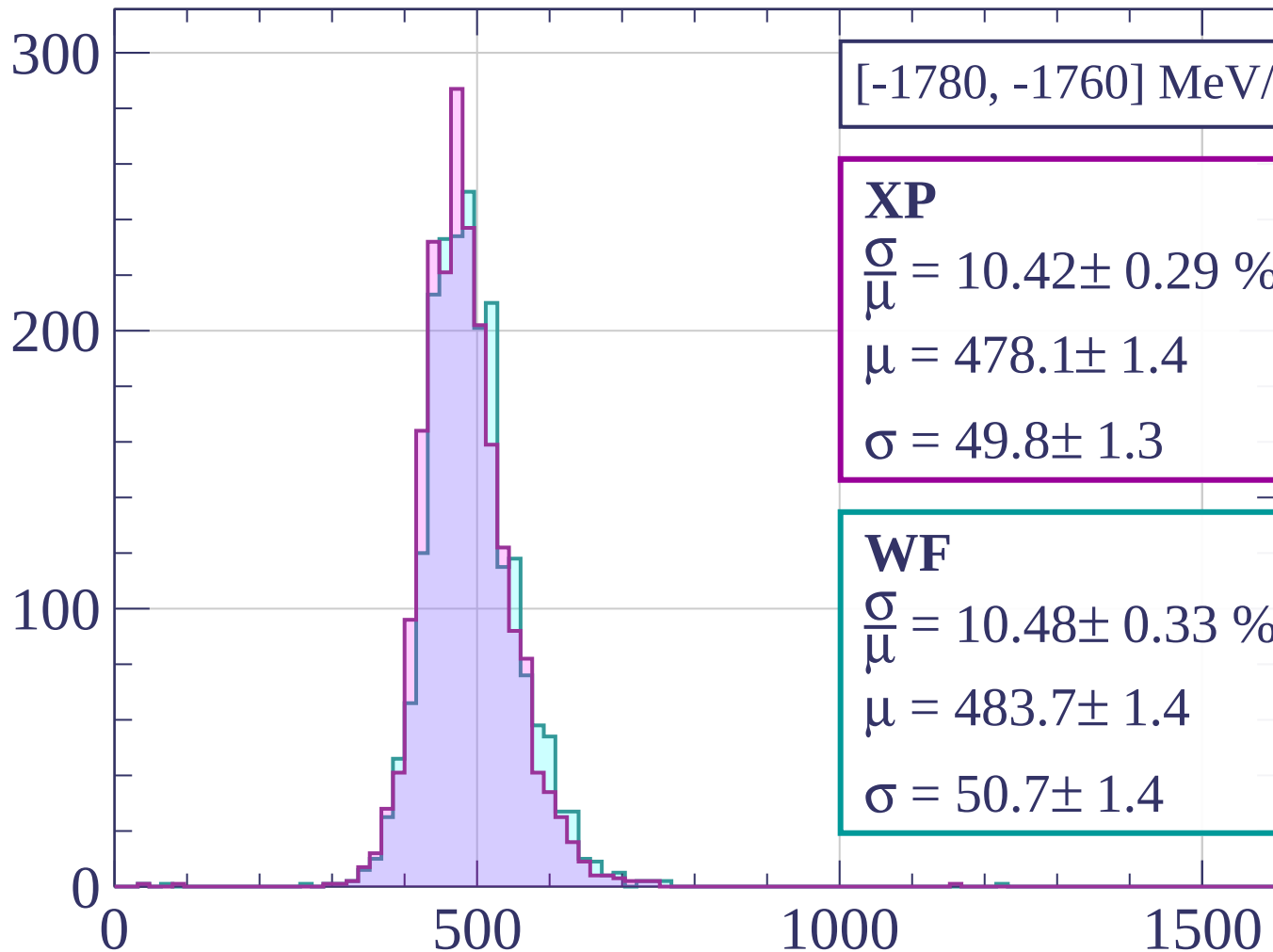
1000

1500

dE/dx [ADC counts/cm]



Count



[-1780, -1760] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.42 \pm 0.29 \%$$

$$\mu = 478.1 \pm 1.4$$

$$\sigma = 49.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.48 \pm 0.33 \%$$

$$\mu = 483.7 \pm 1.4$$

$$\sigma = 50.7 \pm 1.4$$

dE/dx [ADC counts/cm]

Count

[-1760, -1740] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.49 \pm 0.28 \%$$

$$\mu = 474.4 \pm 1.3$$

$$\sigma = 49.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.72 \pm 0.35 \%$$

$$\mu = 479.5 \pm 1.4$$

$$\sigma = 51.4 \pm 1.5$$

200

100

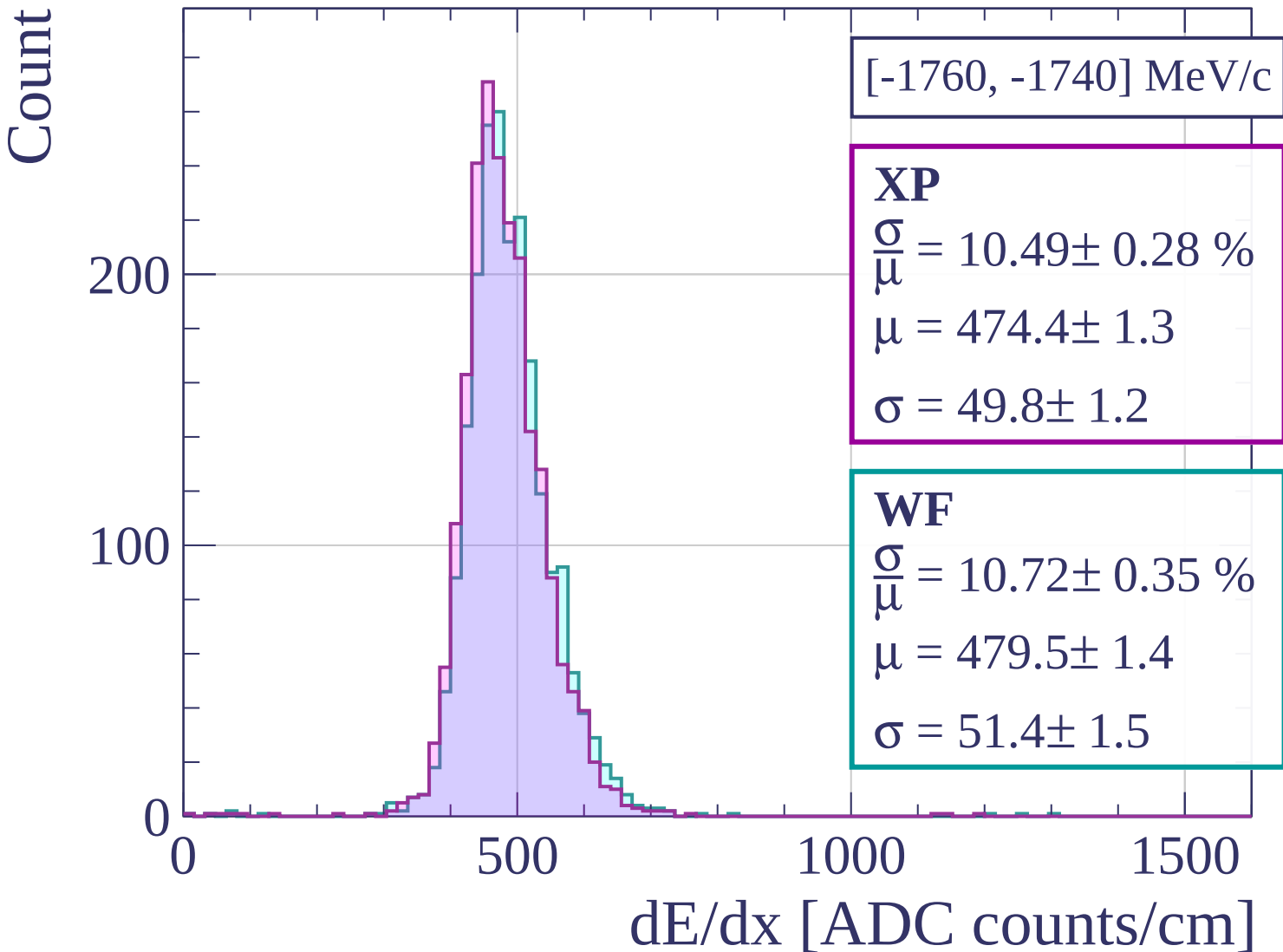
0

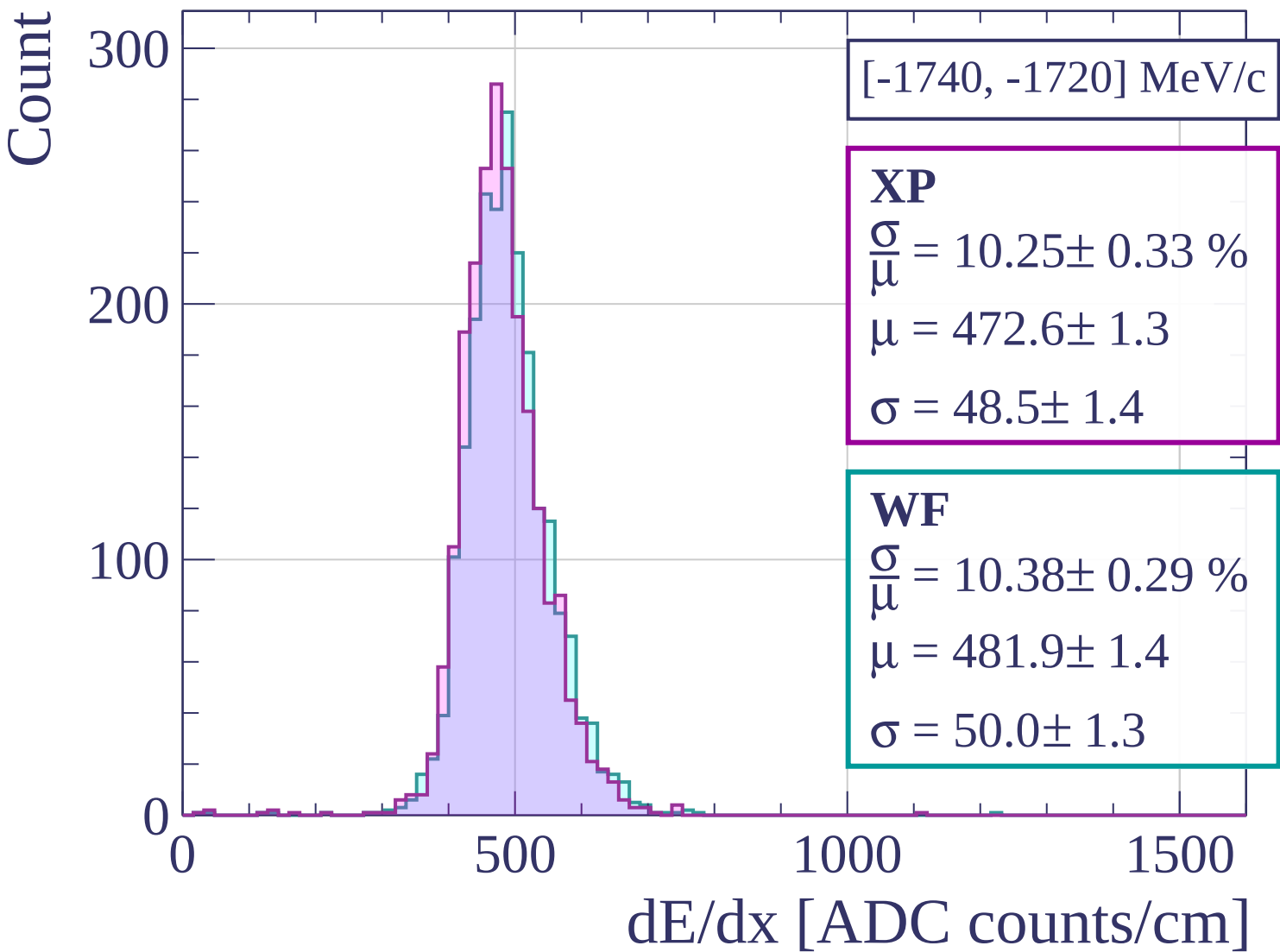
500

1000

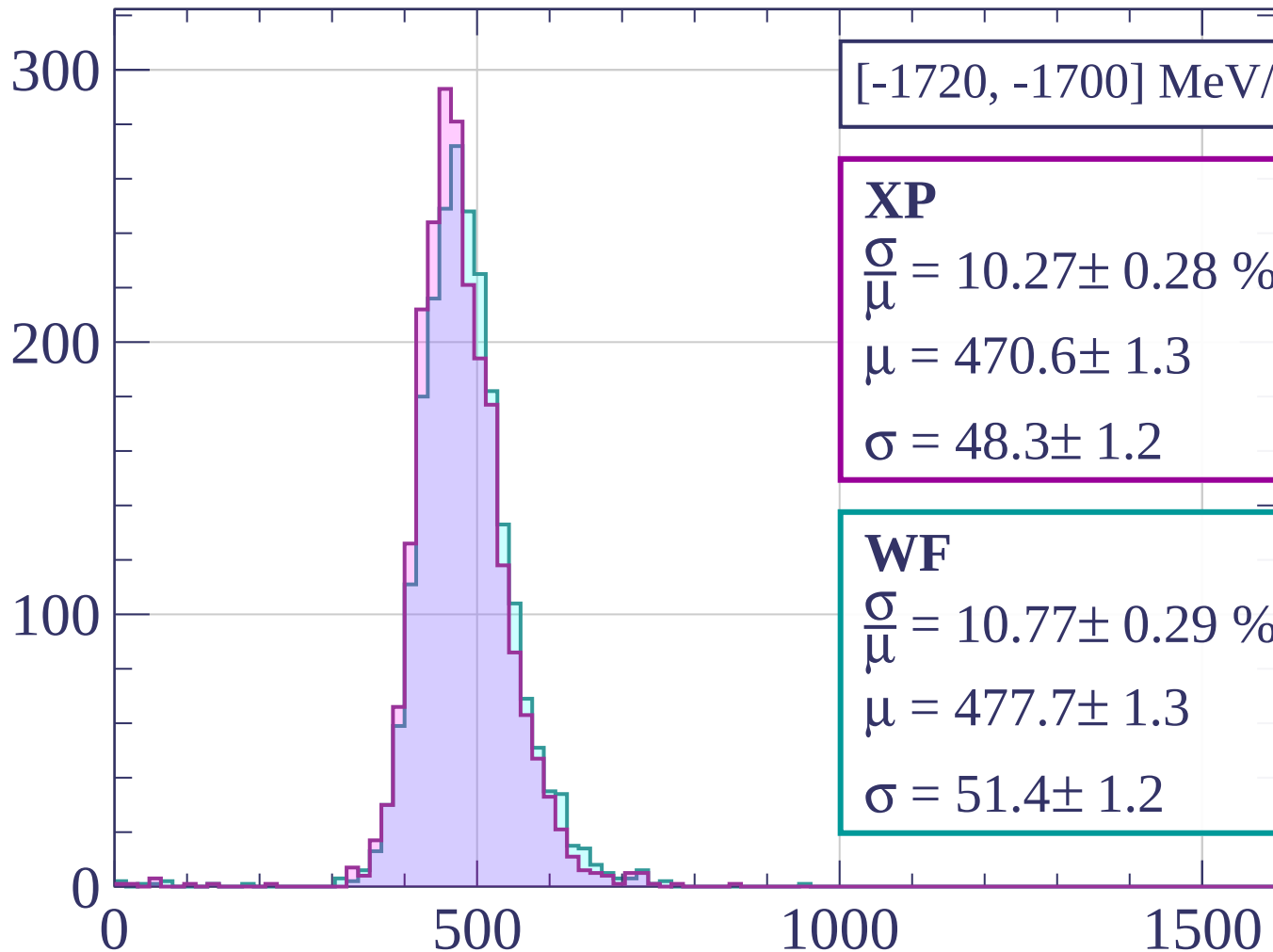
1500

dE/dx [ADC counts/cm]





Count



[-1720, -1700] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.27 \pm 0.28 \%$$

$$\mu = 470.6 \pm 1.3$$

$$\sigma = 48.3 \pm 1.2$$

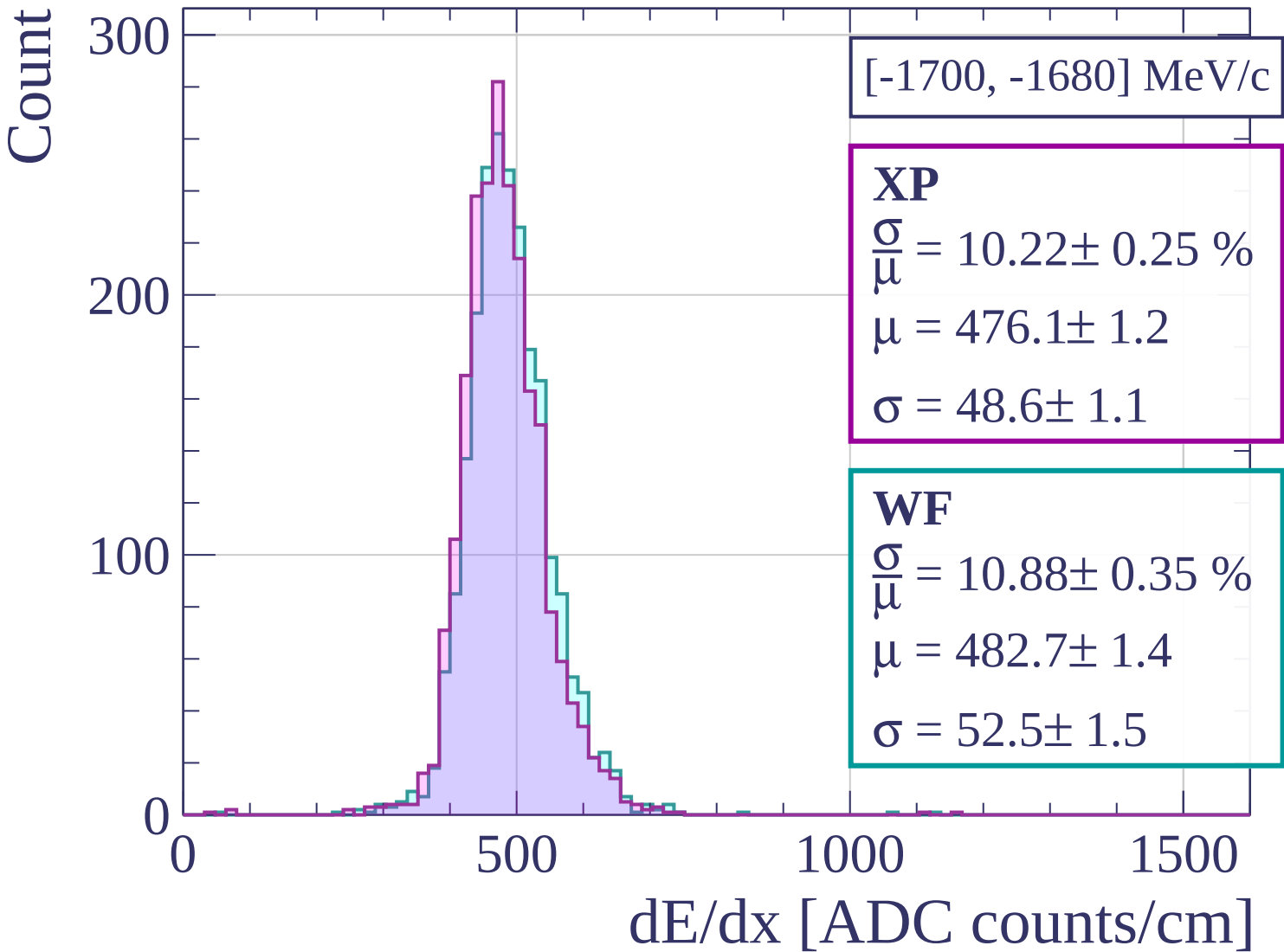
WF

$$\frac{\sigma}{\mu} = 10.77 \pm 0.29 \%$$

$$\mu = 477.7 \pm 1.3$$

$$\sigma = 51.4 \pm 1.2$$

dE/dx [ADC counts/cm]



Count

[-1680, -1660] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.07 \pm 0.30 \%$$

$$\mu = 475.7 \pm 1.4$$

$$\sigma = 52.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 11.16 \pm 0.28 \%$$

$$\mu = 484.5 \pm 1.4$$

$$\sigma = 54.1 \pm 1.2$$

200

100

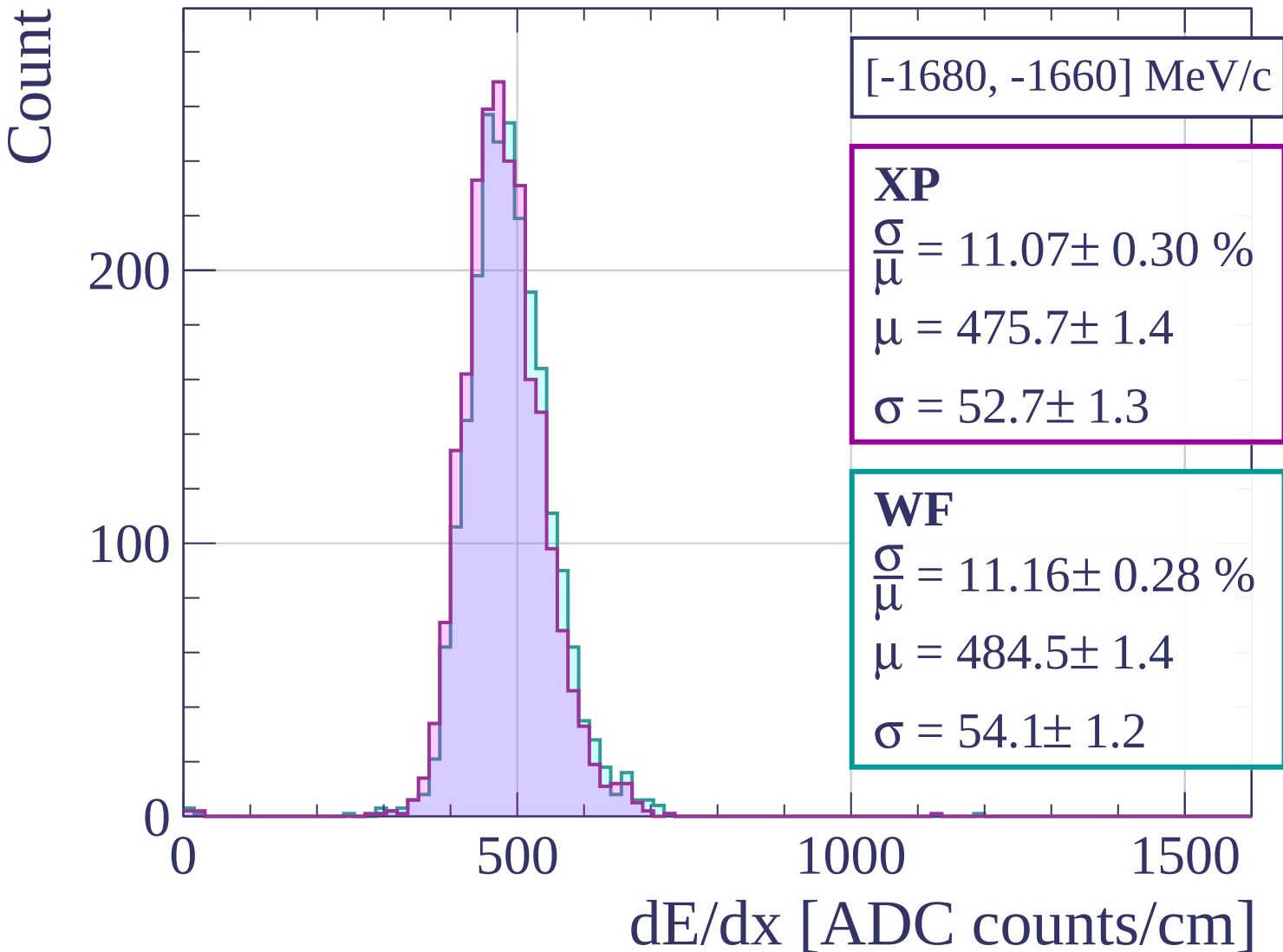
0

500

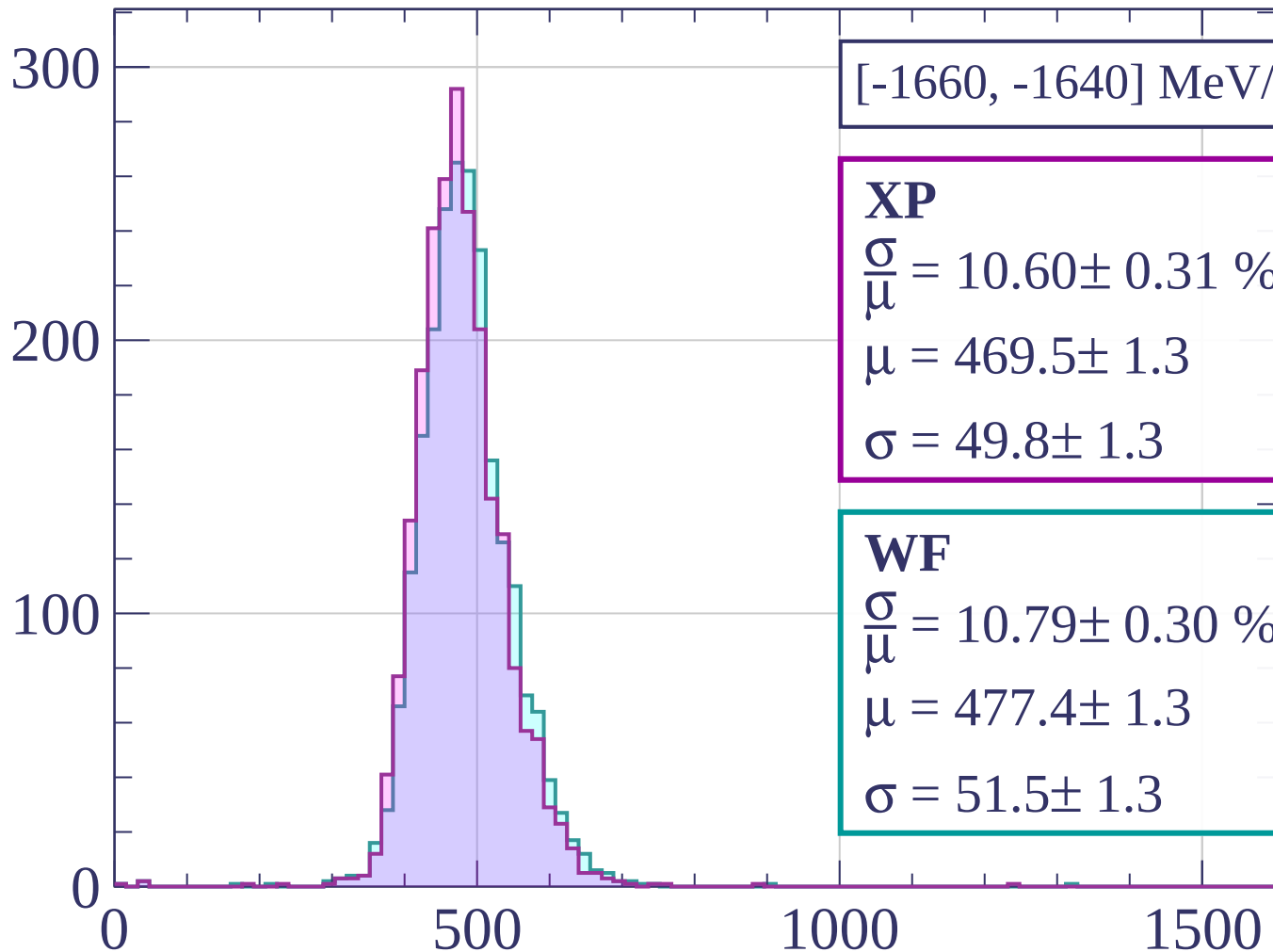
1000

1500

dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]

Count

[-1640, -1620] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.32 \pm 0.29 \%$$

$$\mu = 469.2 \pm 1.3$$

$$\sigma = 48.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.71 \pm 0.29 \%$$

$$\mu = 476.7 \pm 1.3$$

$$\sigma = 51.1 \pm 1.3$$

300

200

100

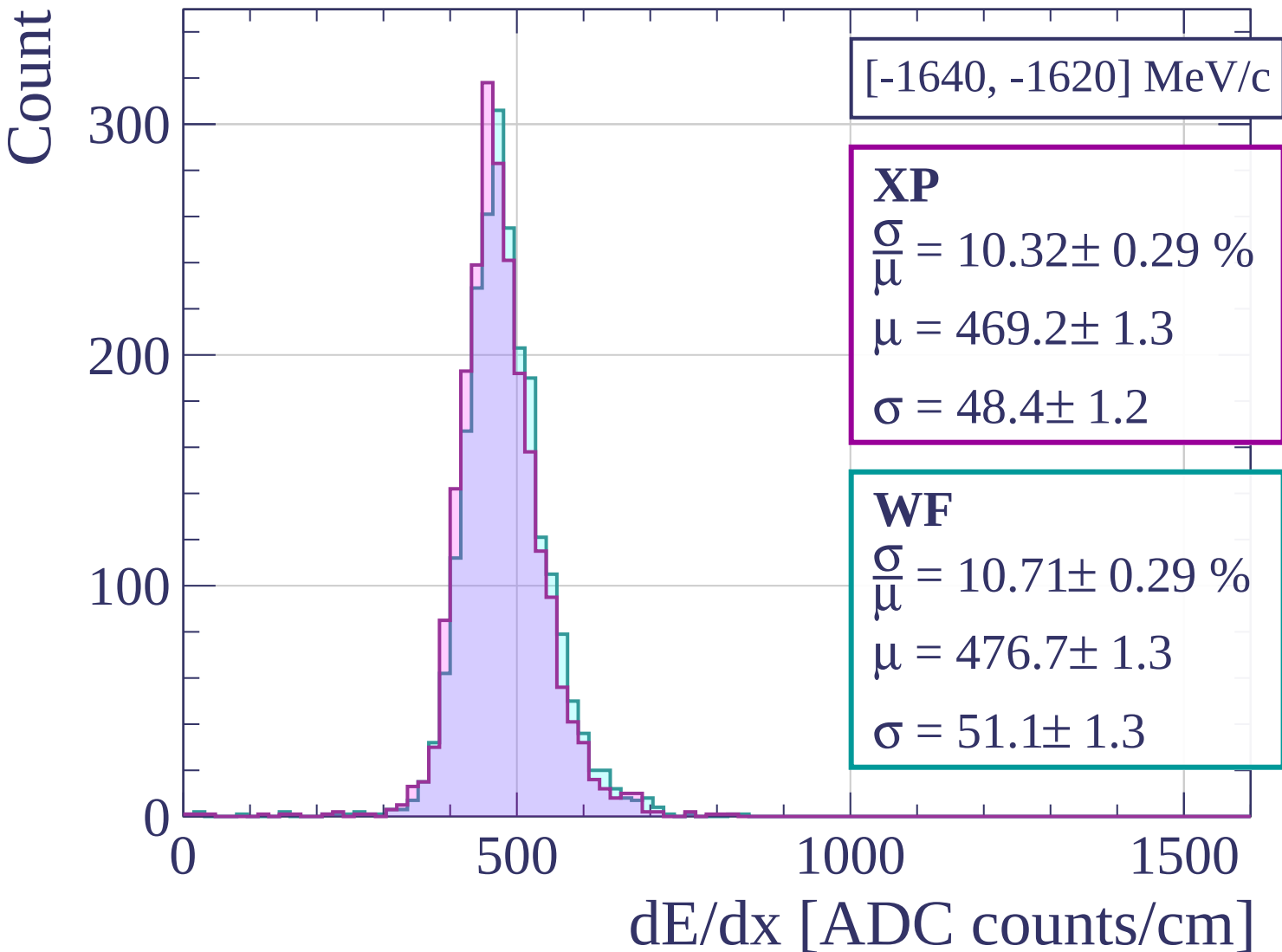
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1620, -1600] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.94 \pm 0.28 \%$$

$$\mu = 467.0 \pm 1.3$$

$$\sigma = 46.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.32 \pm 0.28 \%$$

$$\mu = 475.7 \pm 1.3$$

$$\sigma = 49.1 \pm 1.2$$

300

200

100

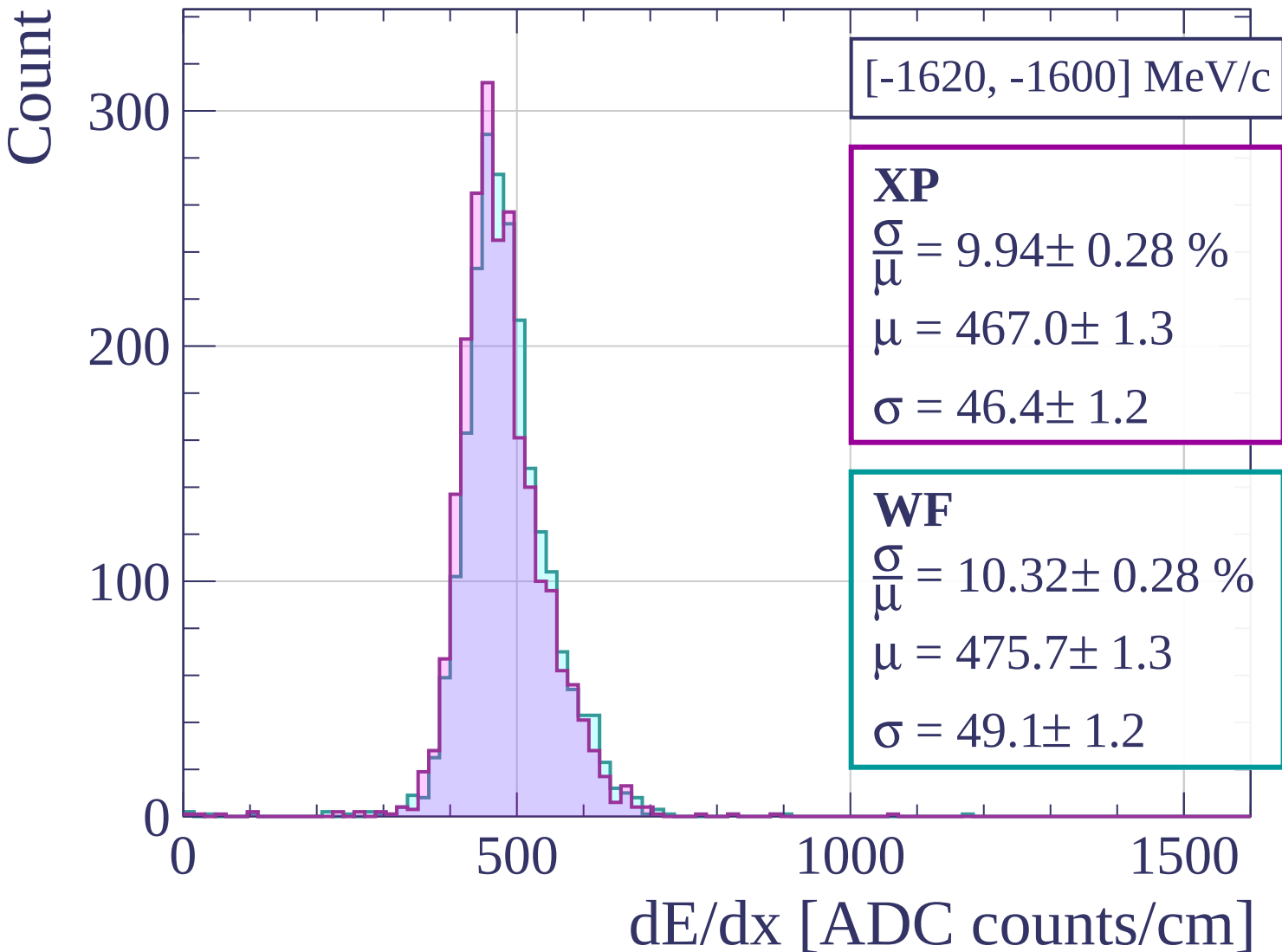
0

500

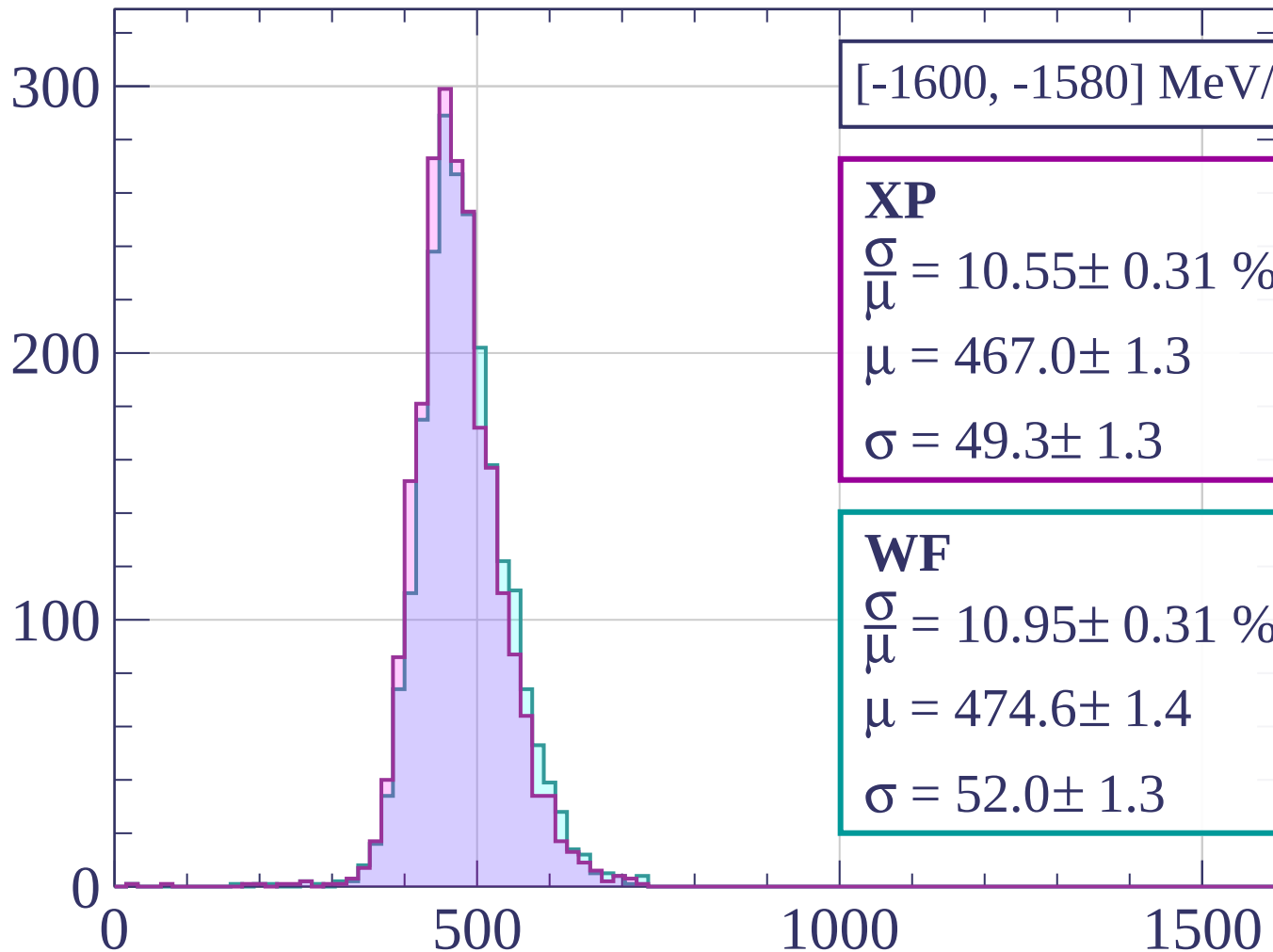
1000

1500

dE/dx [ADC counts/cm]



Count



[-1600, -1580] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.55 \pm 0.31 \%$$

$$\mu = 467.0 \pm 1.3$$

$$\sigma = 49.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.95 \pm 0.31 \%$$

$$\mu = 474.6 \pm 1.4$$

$$\sigma = 52.0 \pm 1.3$$

dE/dx [ADC counts/cm]

Count

[-1580, -1560] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.09 \pm 0.26 \%$$

$$\mu = 468.8 \pm 1.3$$

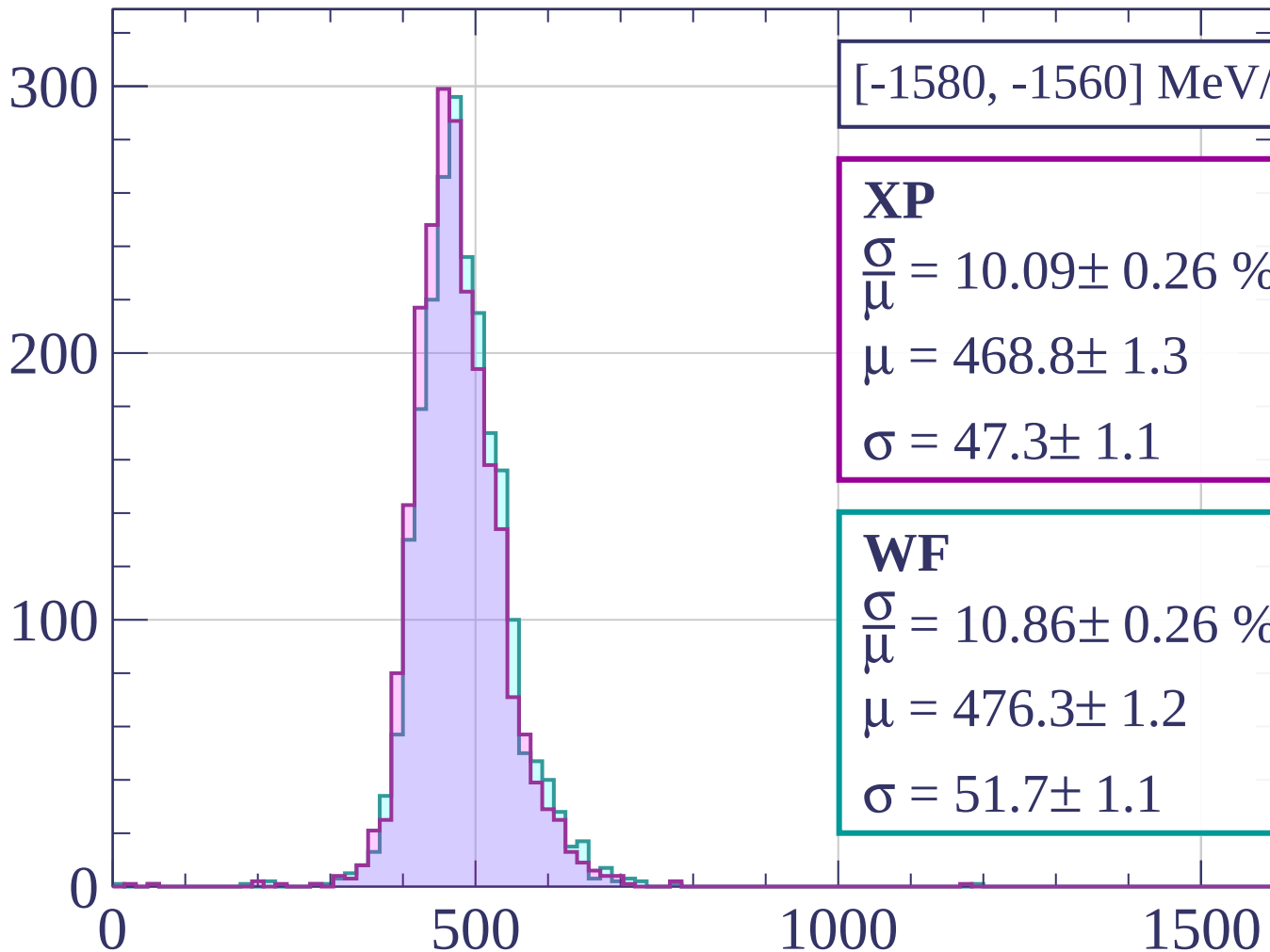
$$\sigma = 47.3 \pm 1.1$$

WF

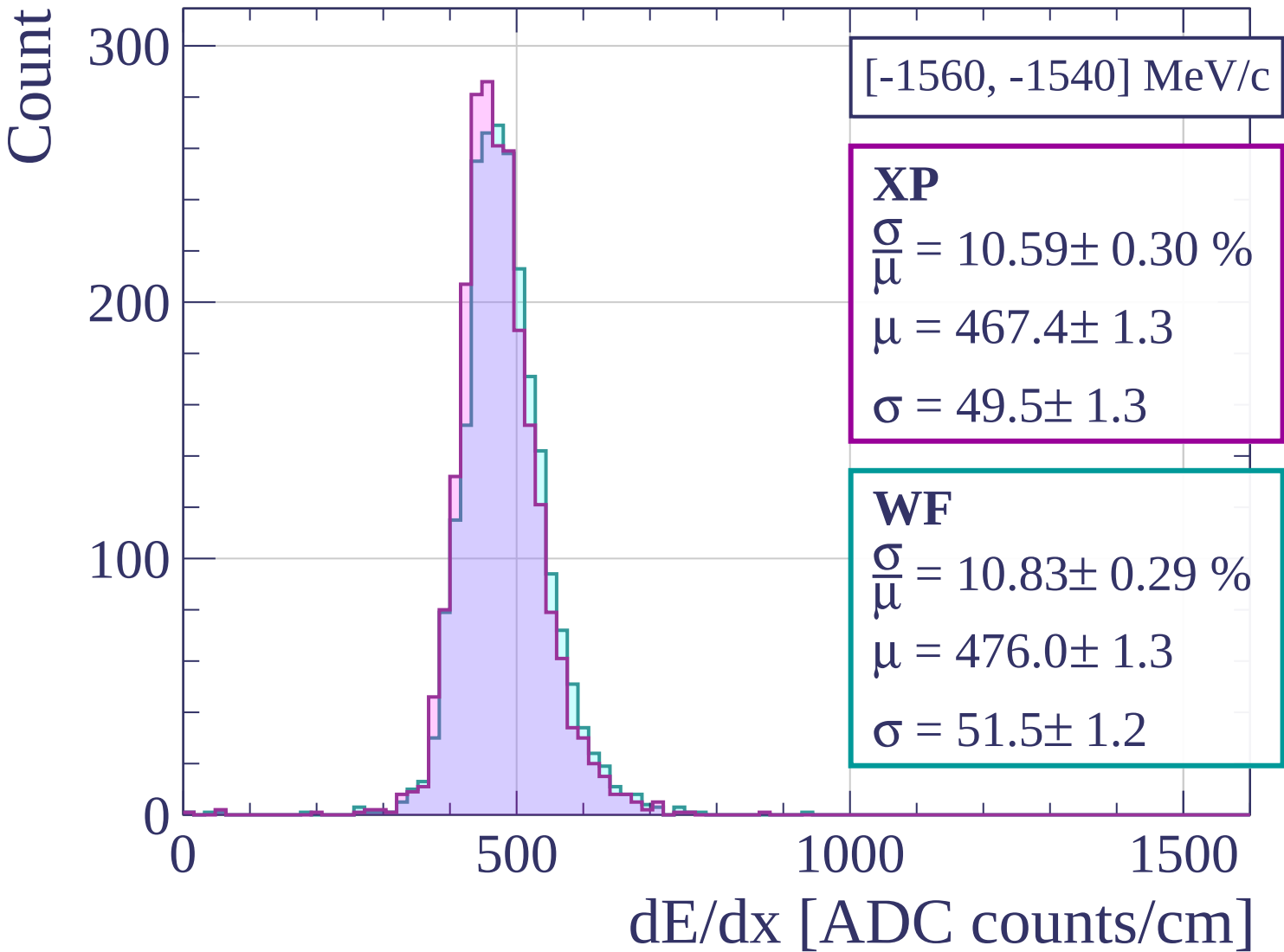
$$\frac{\sigma}{\mu} = 10.86 \pm 0.26 \%$$

$$\mu = 476.3 \pm 1.2$$

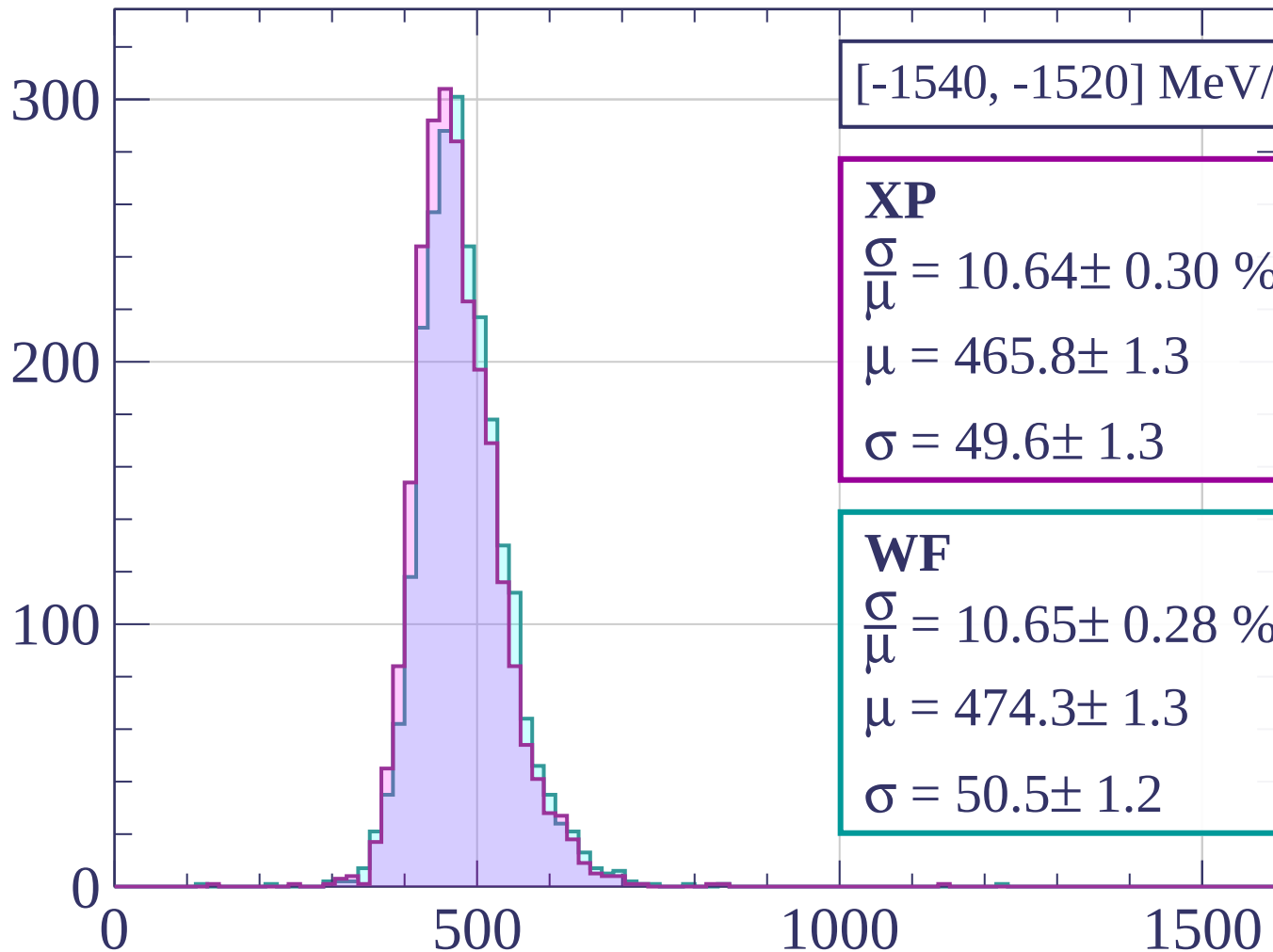
$$\sigma = 51.7 \pm 1.1$$



dE/dx [ADC counts/cm]



Count



[-1540, -1520] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.64 \pm 0.30 \%$$

$$\mu = 465.8 \pm 1.3$$

$$\sigma = 49.6 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.65 \pm 0.28 \%$$

$$\mu = 474.3 \pm 1.3$$

$$\sigma = 50.5 \pm 1.2$$

dE/dx [ADC counts/cm]

Count

[-1520, -1500] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.27 \pm 0.27 \%$$

$$\mu = 465.3 \pm 1.2$$

$$\sigma = 47.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.56 \pm 0.27 \%$$

$$\mu = 473.6 \pm 1.2$$

$$\sigma = 50.0 \pm 1.2$$

300

200

100

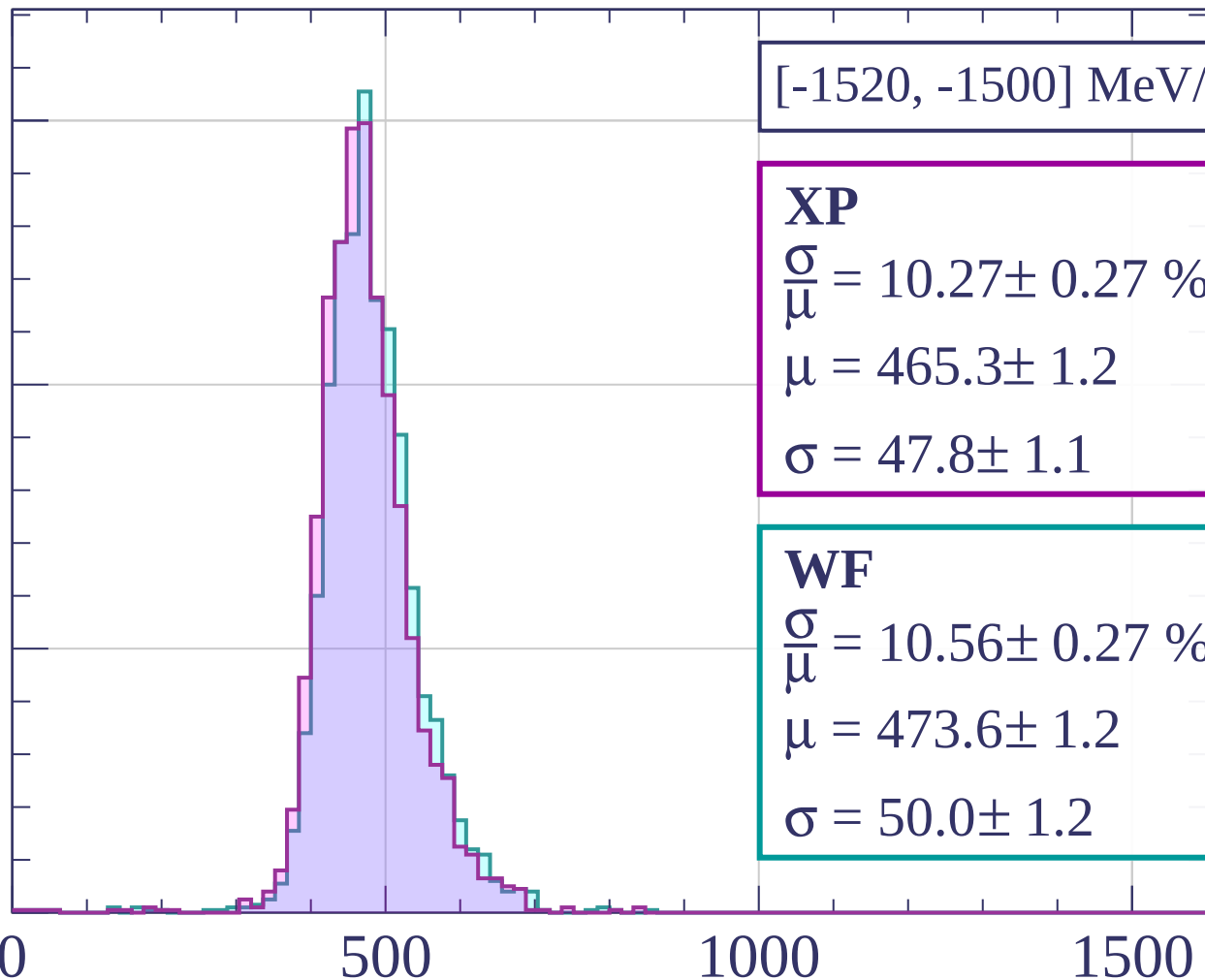
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1500, -1480] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.46 \pm 0.30 \%$$

$$\mu = 461.7 \pm 1.2$$

$$\sigma = 48.3 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.29 \pm 0.29 \%$$

$$\mu = 468.0 \pm 1.3$$

$$\sigma = 48.2 \pm 1.2$$

300

200

100

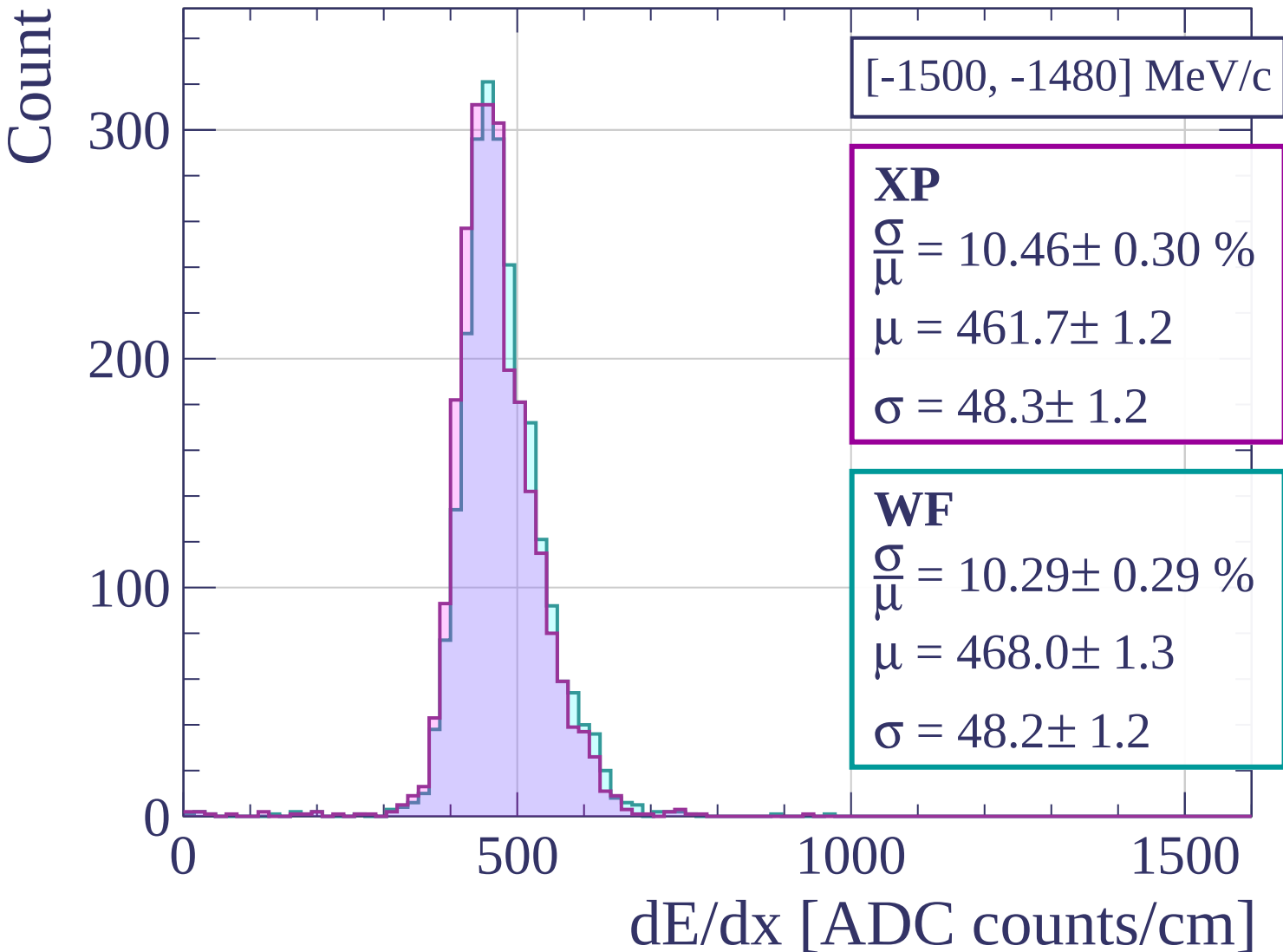
0

500

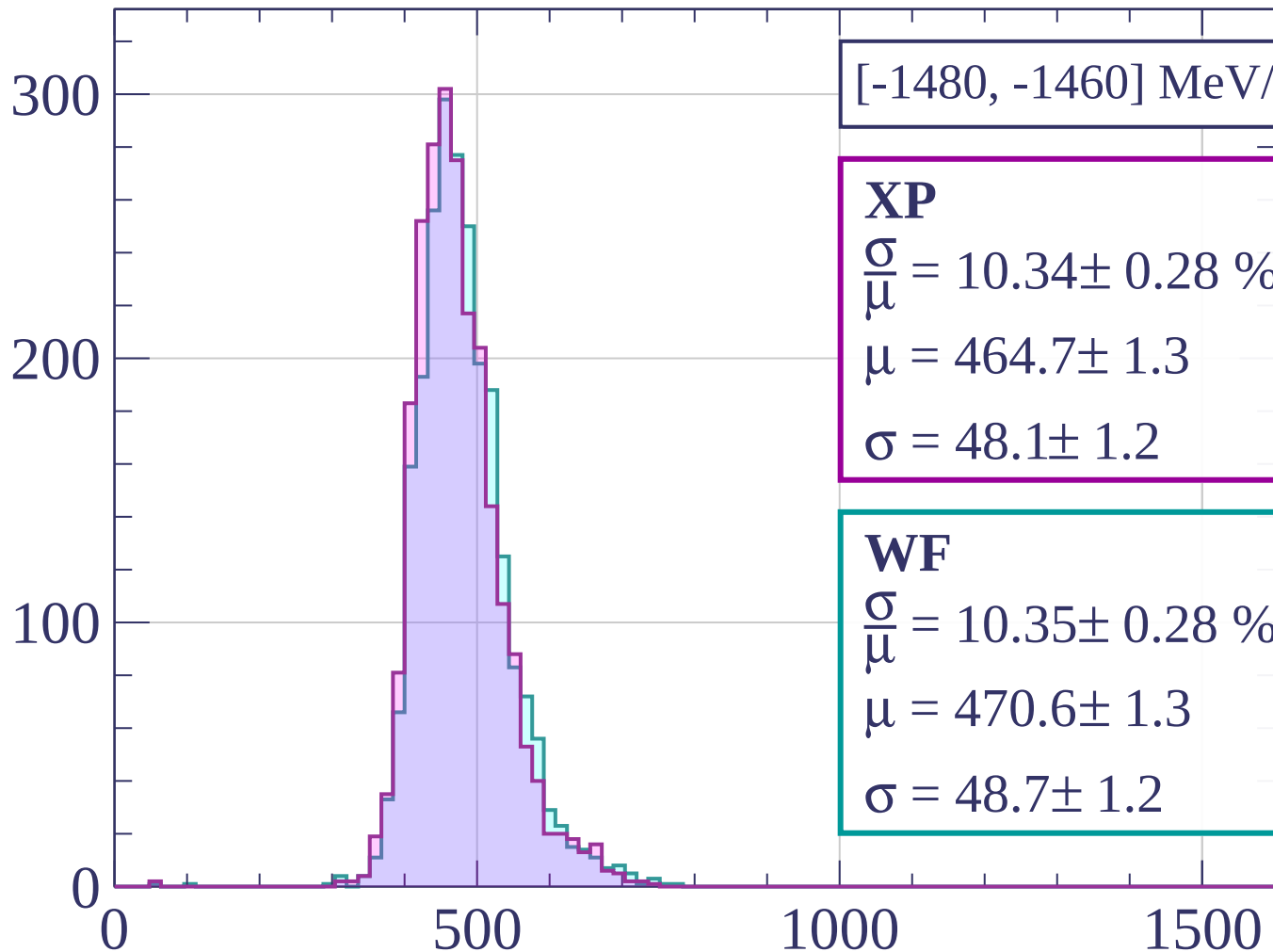
1000

1500

dE/dx [ADC counts/cm]



Count



[-1480, -1460] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.34 \pm 0.28 \%$$

$$\mu = 464.7 \pm 1.3$$

$$\sigma = 48.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.35 \pm 0.28 \%$$

$$\mu = 470.6 \pm 1.3$$

$$\sigma = 48.7 \pm 1.2$$

dE/dx [ADC counts/cm]

Count

[-1460, -1440] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.18 \pm 0.23 \%$$

$$\mu = 463.5 \pm 1.2$$

$$\sigma = 47.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.85 \pm 0.26 \%$$

$$\mu = 470.9 \pm 1.2$$

$$\sigma = 51.1 \pm 1.1$$

300

200

100

0

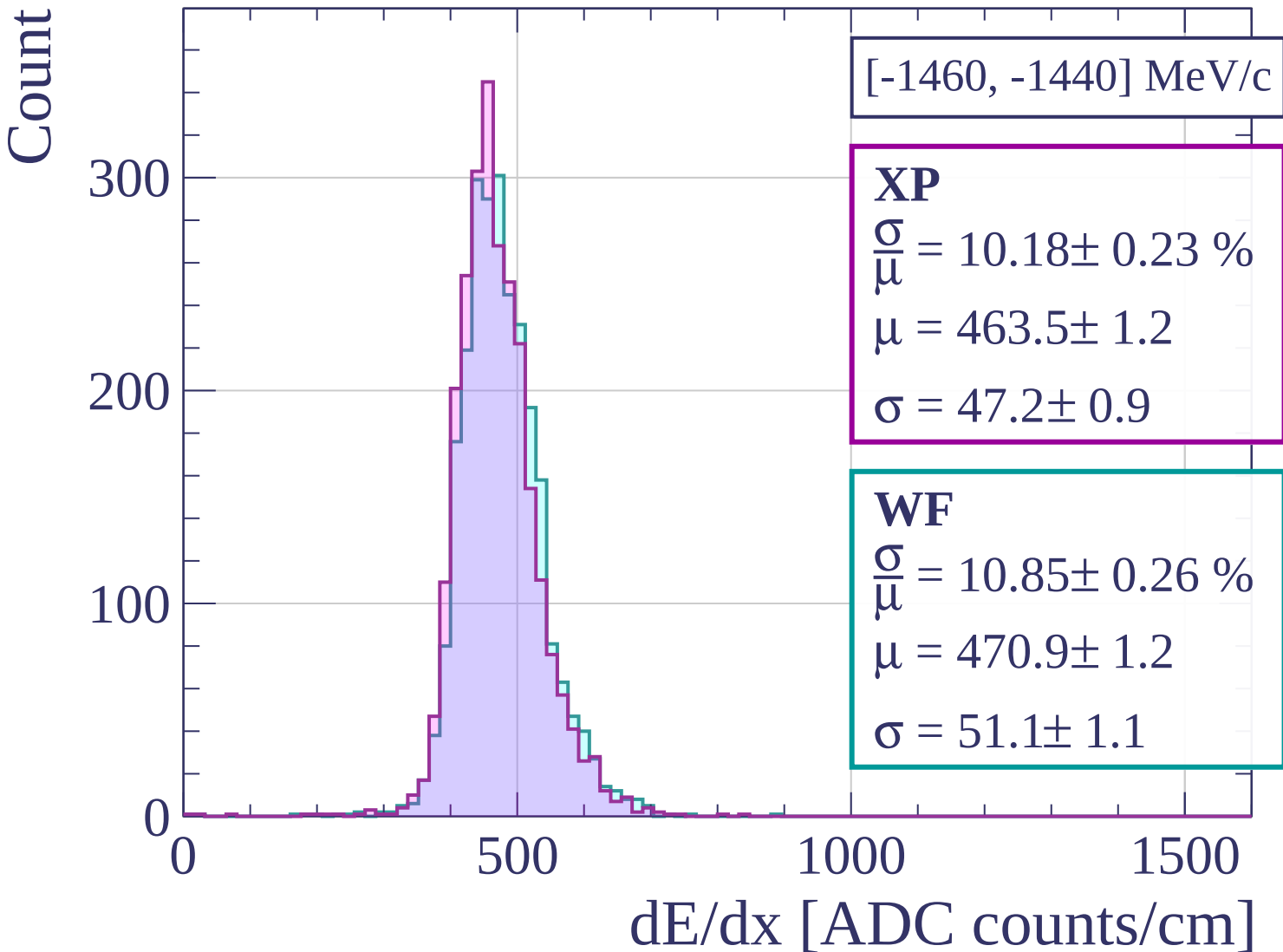
0

500

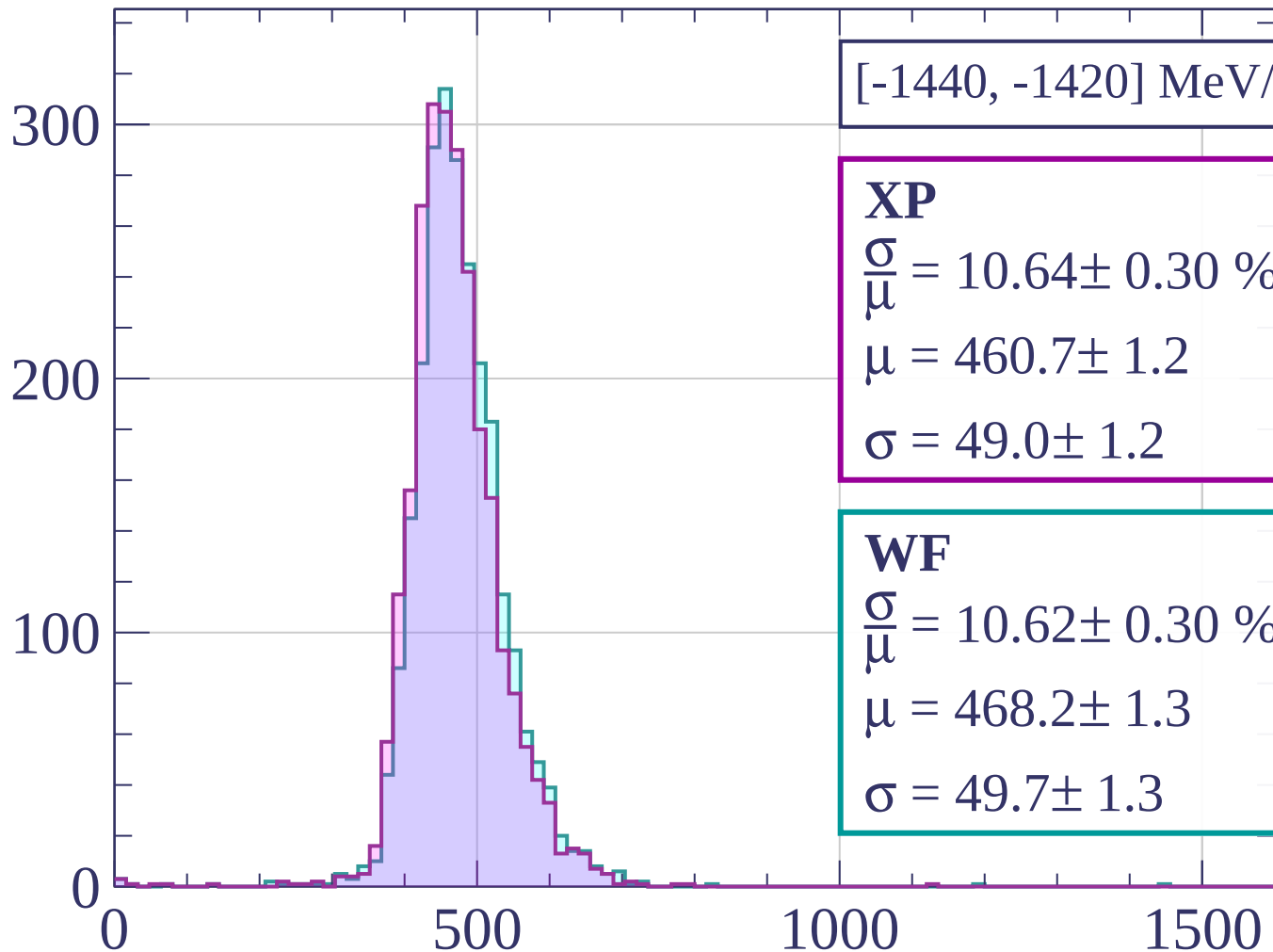
1000

1500

dE/dx [ADC counts/cm]



Count



[-1440, -1420] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.64 \pm 0.30 \%$$

$$\mu = 460.7 \pm 1.2$$

$$\sigma = 49.0 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.62 \pm 0.30 \%$$

$$\mu = 468.2 \pm 1.3$$

$$\sigma = 49.7 \pm 1.3$$

dE/dx [ADC counts/cm]

Count

[-1420, -1400] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.37 \pm 0.30 \%$$

$$\mu = 457.9 \pm 1.2$$

$$\sigma = 47.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.61 \pm 0.29 \%$$

$$\mu = 465.8 \pm 1.2$$

$$\sigma = 49.4 \pm 1.2$$

300

200

100

0

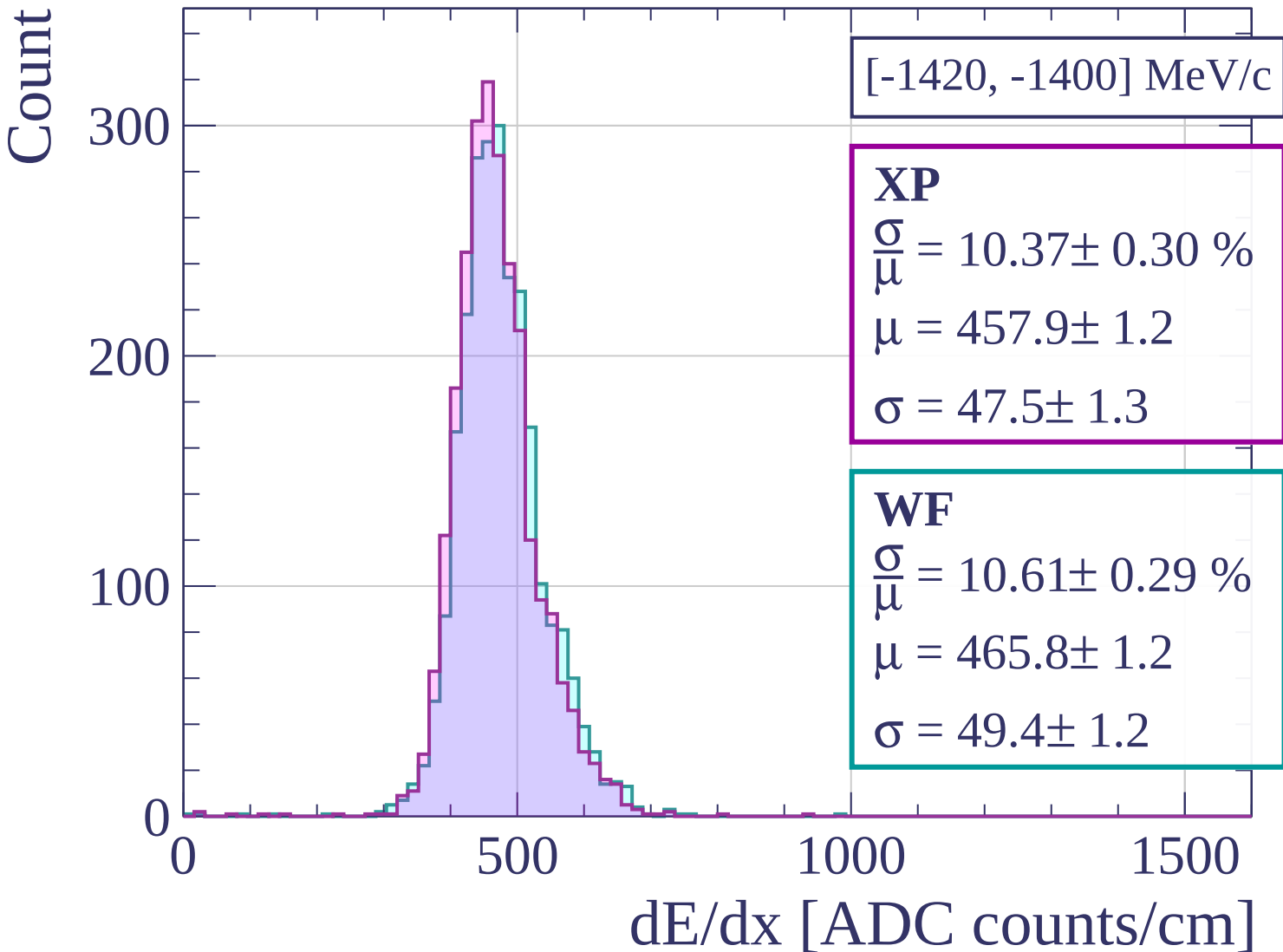
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1400, -1380] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.07 \pm 0.29 \%$$

$$\mu = 457.2 \pm 1.2$$

$$\sigma = 46.0 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.93 \pm 0.32 \%$$

$$\mu = 465.5 \pm 1.3$$

$$\sigma = 50.9 \pm 1.3$$

300

200

100

0

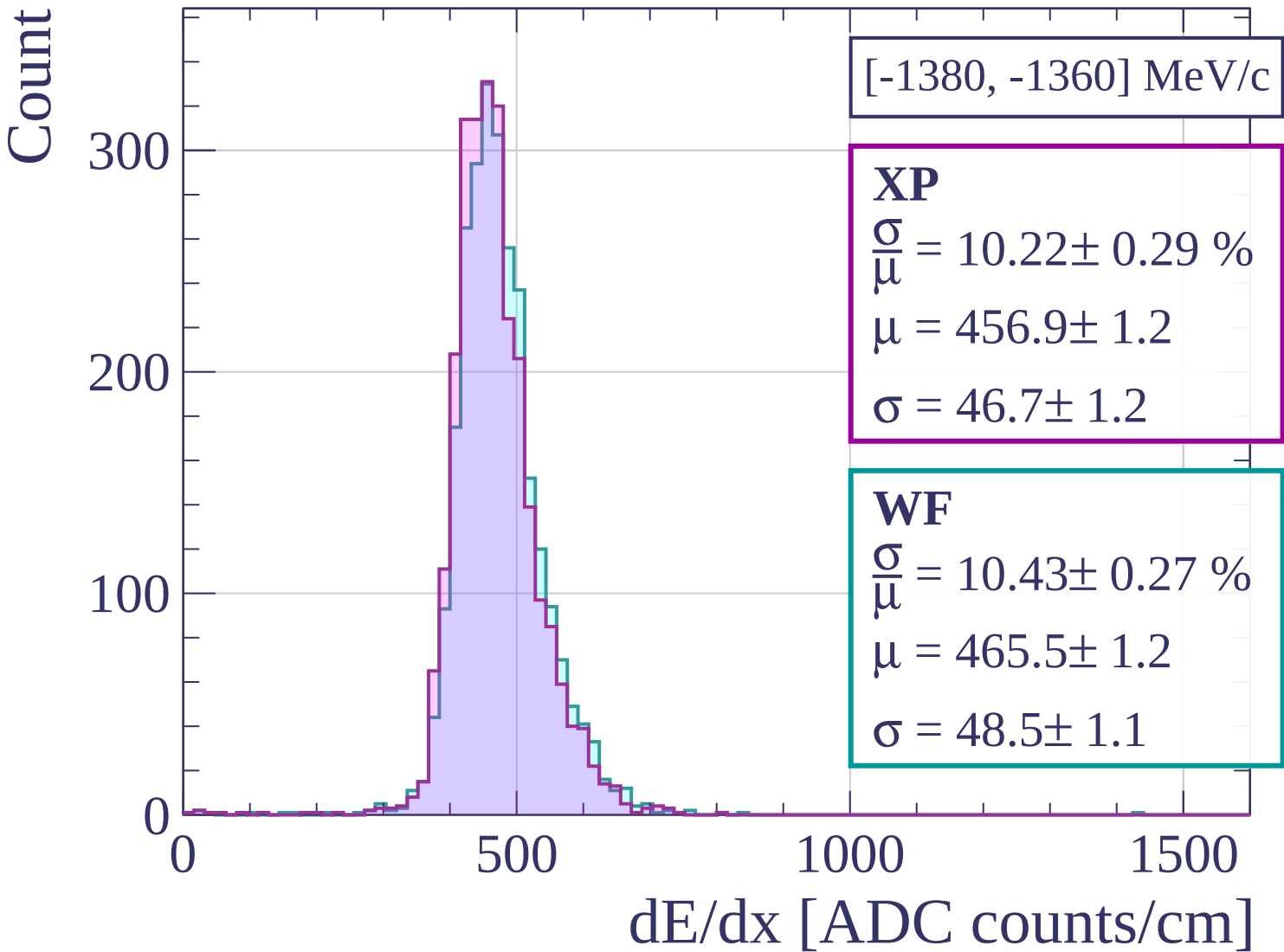
0

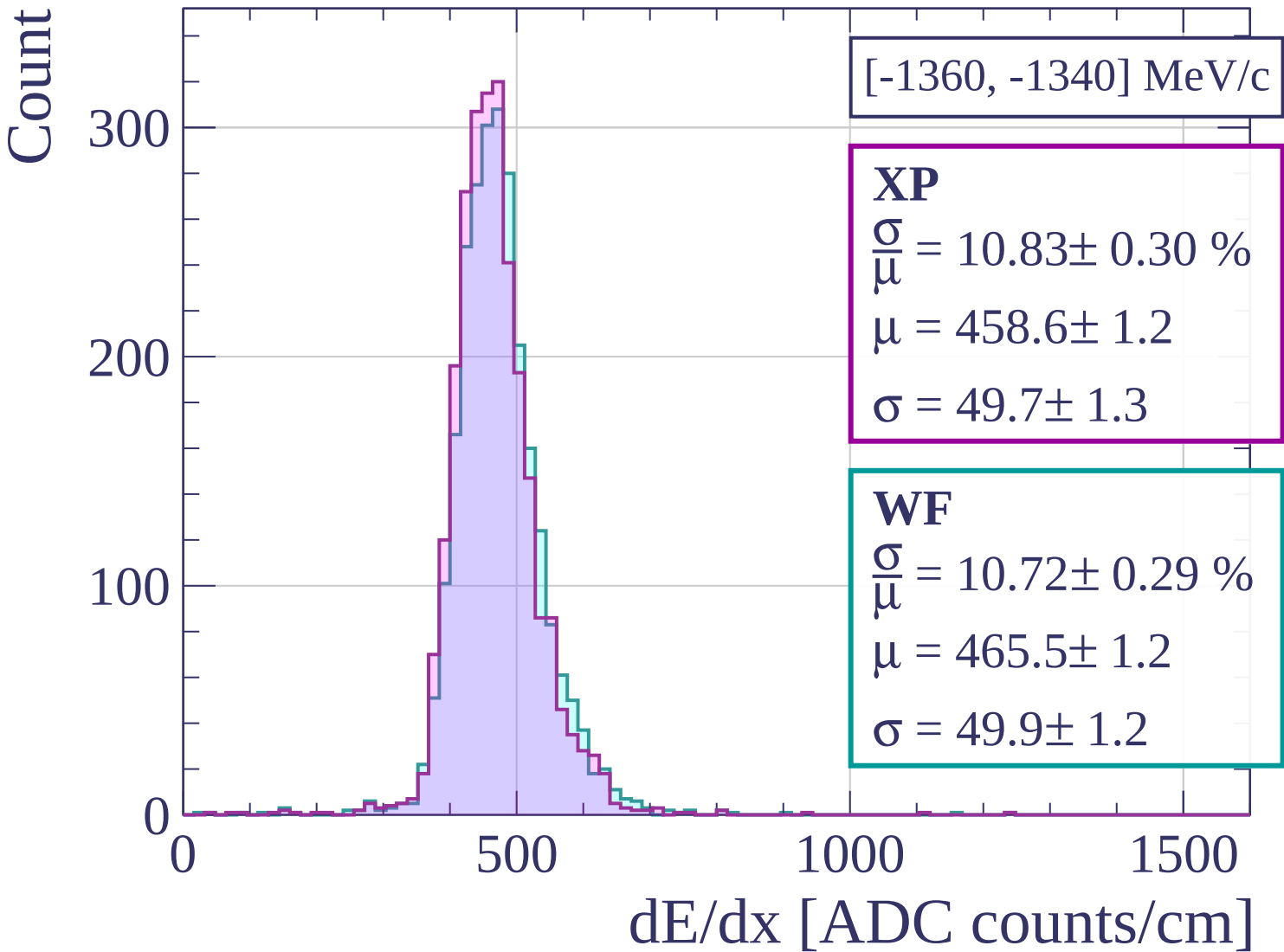
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-1340, -1320] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.01 \pm 0.28 \%$$

$$\mu = 454.8 \pm 1.1$$

$$\sigma = 45.5 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.13 \pm 0.29 \%$$

$$\mu = 461.1 \pm 1.2$$

$$\sigma = 46.7 \pm 1.2$$

300

200

100

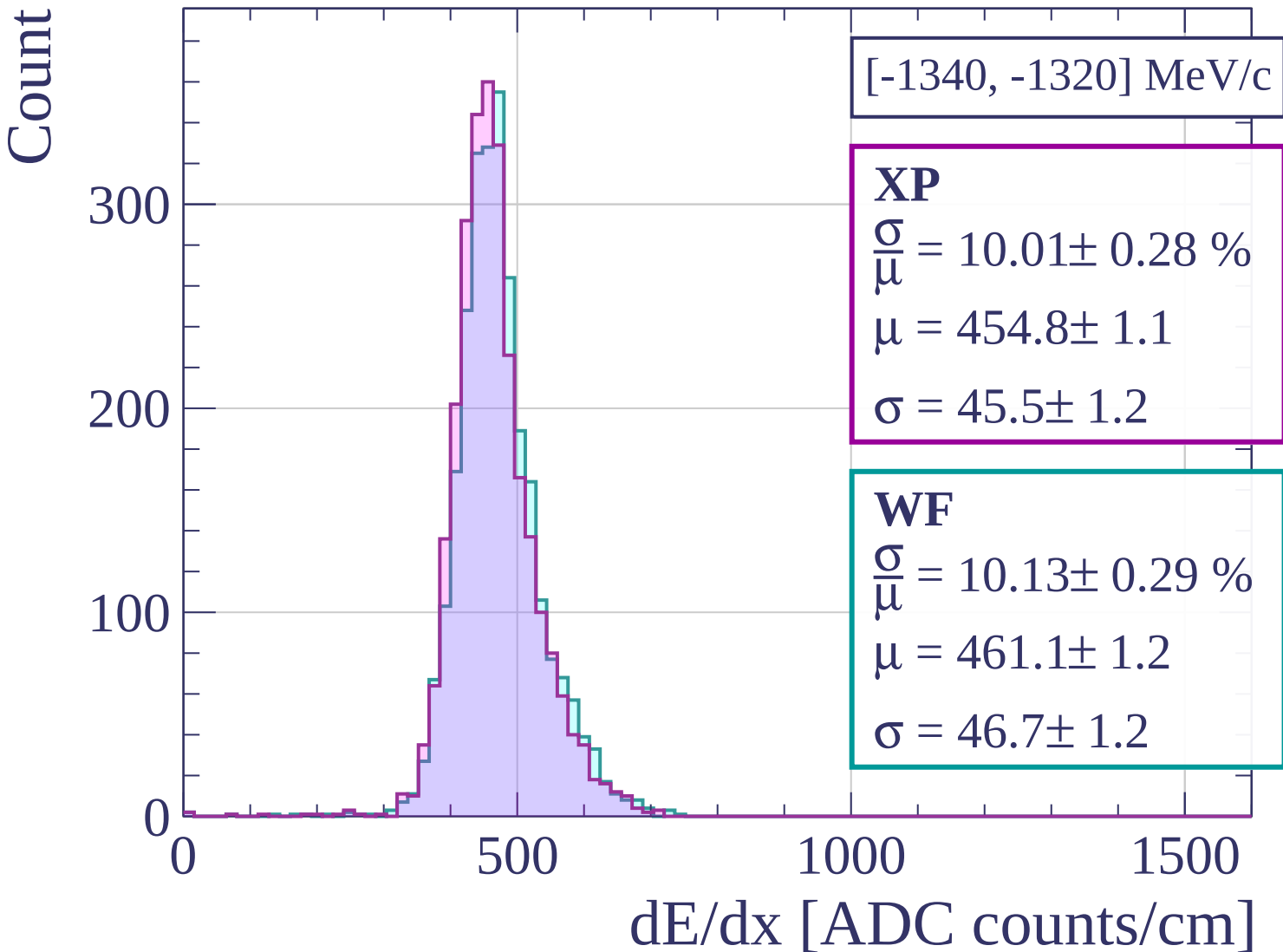
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1320, -1300] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.28 \pm 0.30 \%$$

$$\mu = 452.3 \pm 1.1$$

$$\sigma = 46.5 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.01 \pm 0.29 \%$$

$$\mu = 458.3 \pm 1.1$$

$$\sigma = 45.9 \pm 1.2$$

300

200

100

0

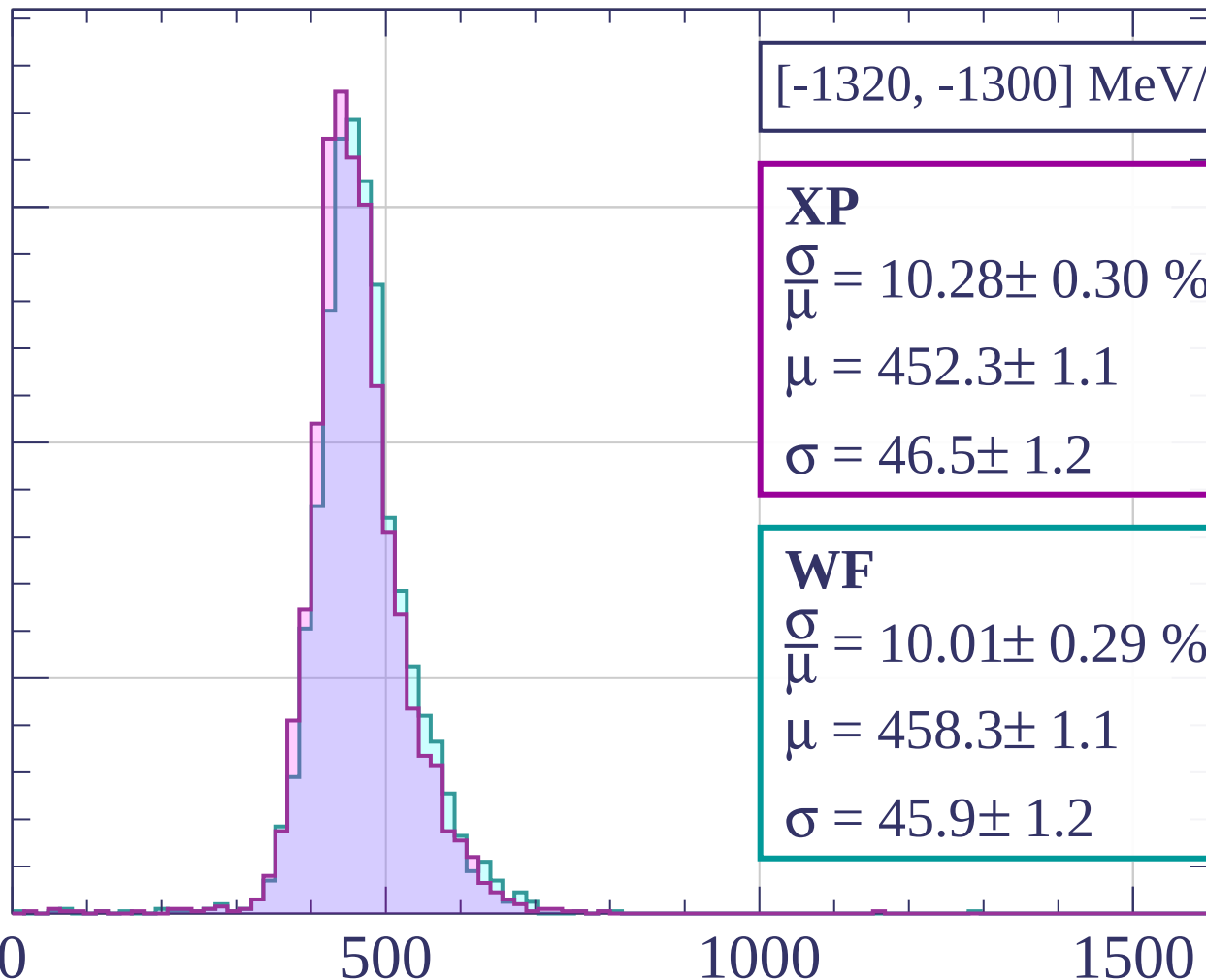
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1300, -1280] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.18 \pm 0.26 \%$$

$$\mu = 451.1 \pm 1.1$$

$$\sigma = 45.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.55 \pm 0.25 \%$$

$$\mu = 461.4 \pm 1.2$$

$$\sigma = 48.7 \pm 1.0$$

300

200

100

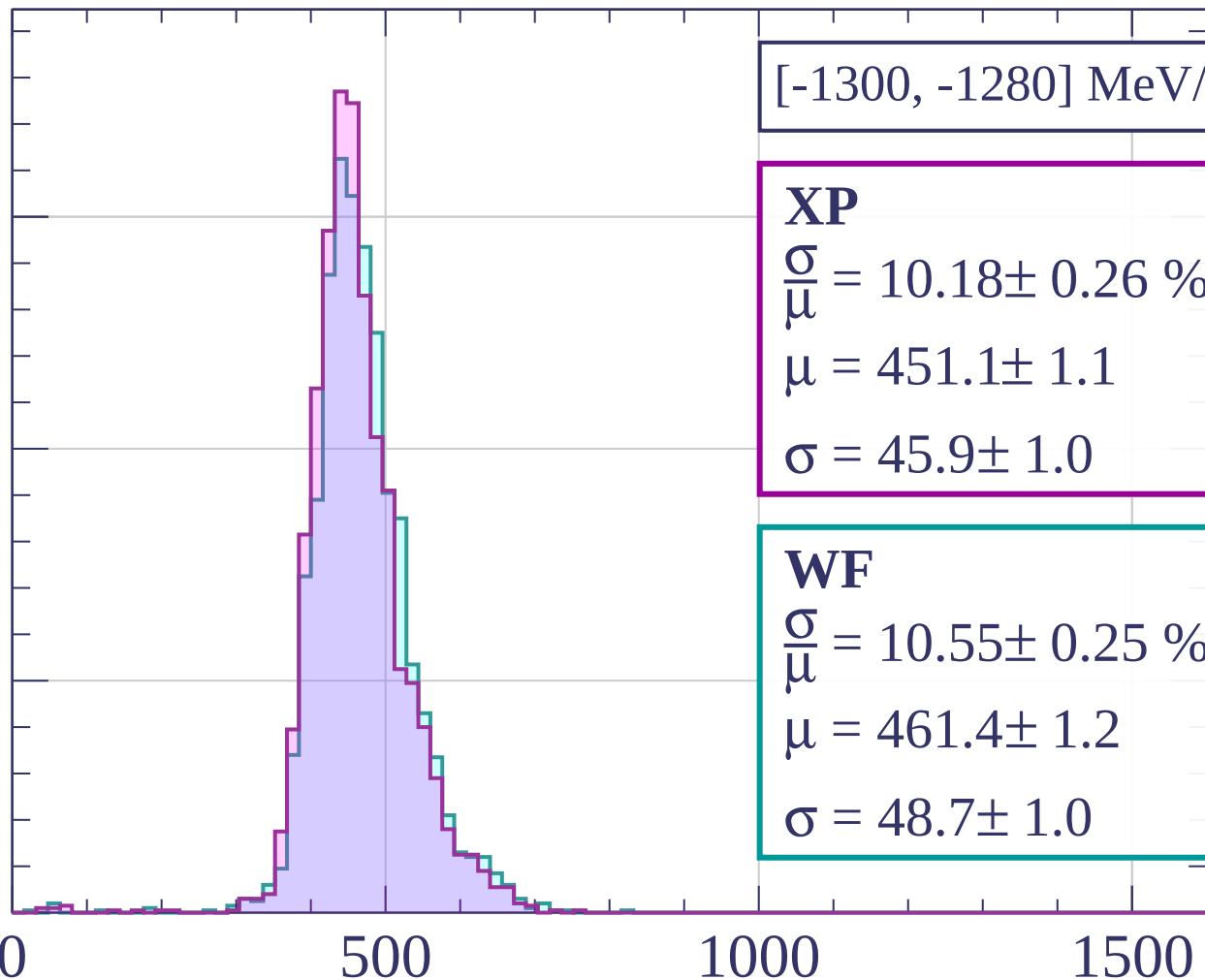
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1280, -1260] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.28 \pm 0.25 \%$$

$$\mu = 451.9 \pm 1.1$$

$$\sigma = 46.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.37 \pm 0.30 \%$$

$$\mu = 456.9 \pm 1.2$$

$$\sigma = 47.4 \pm 1.2$$

300

200

100

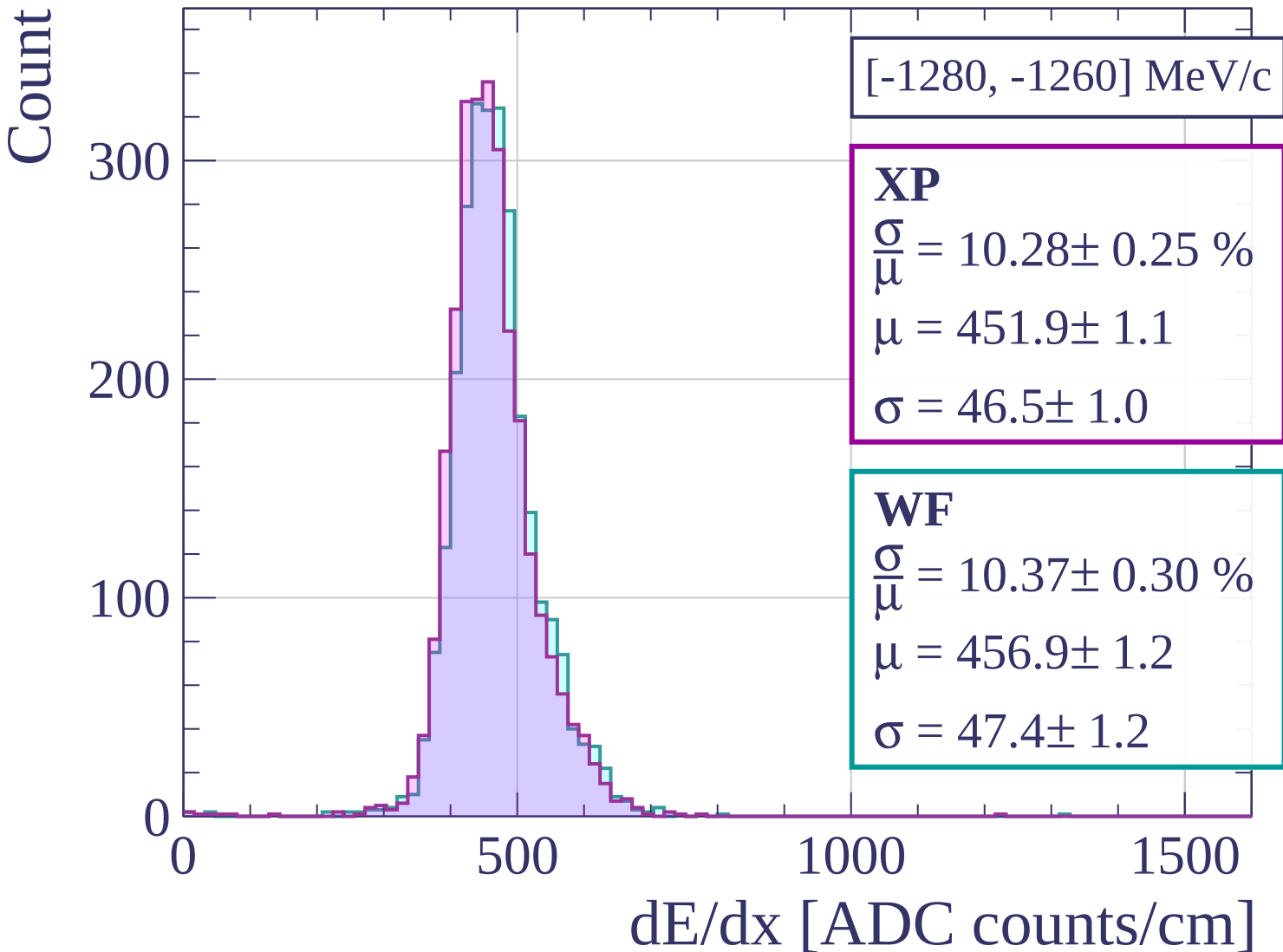
0

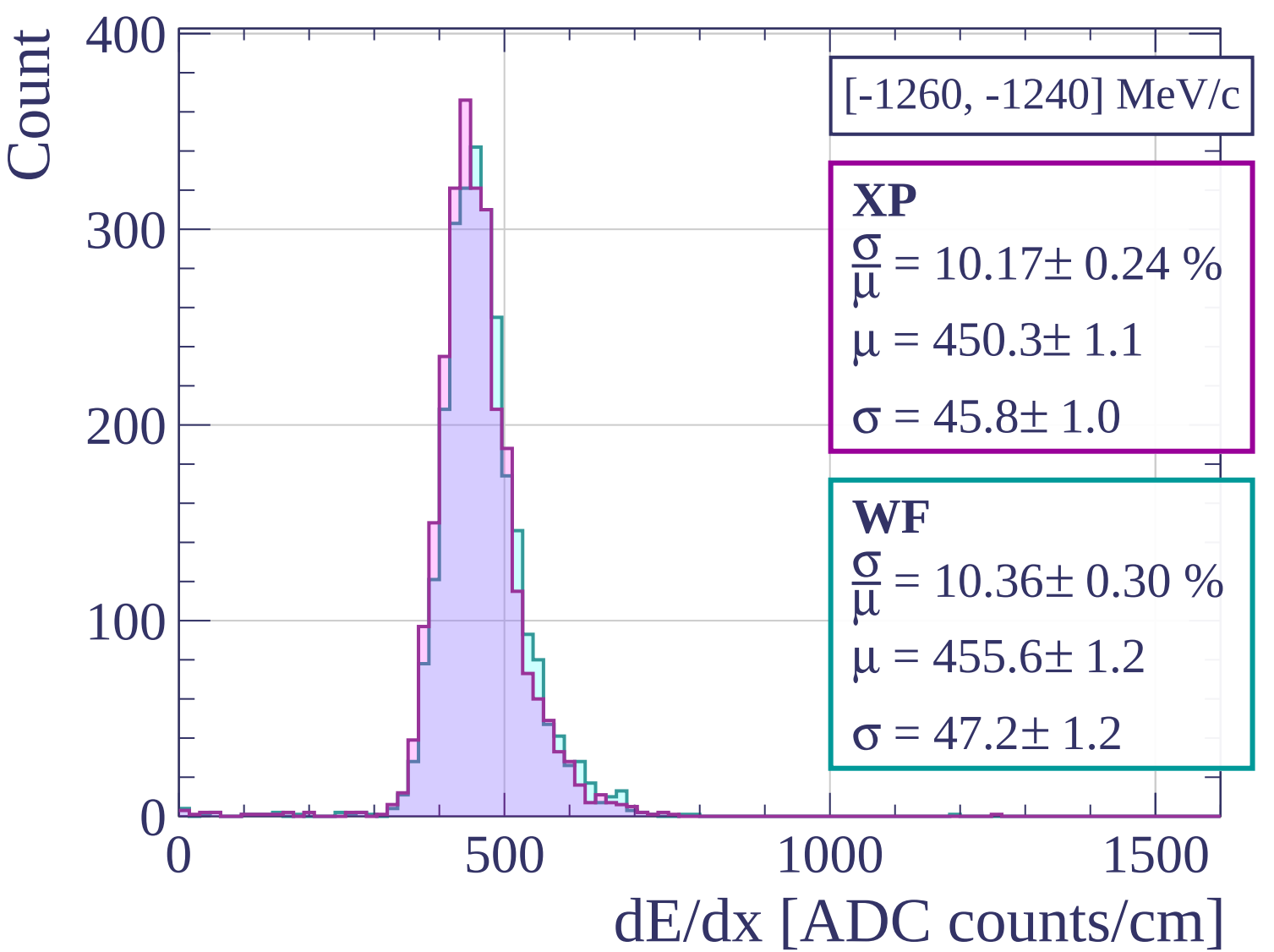
500

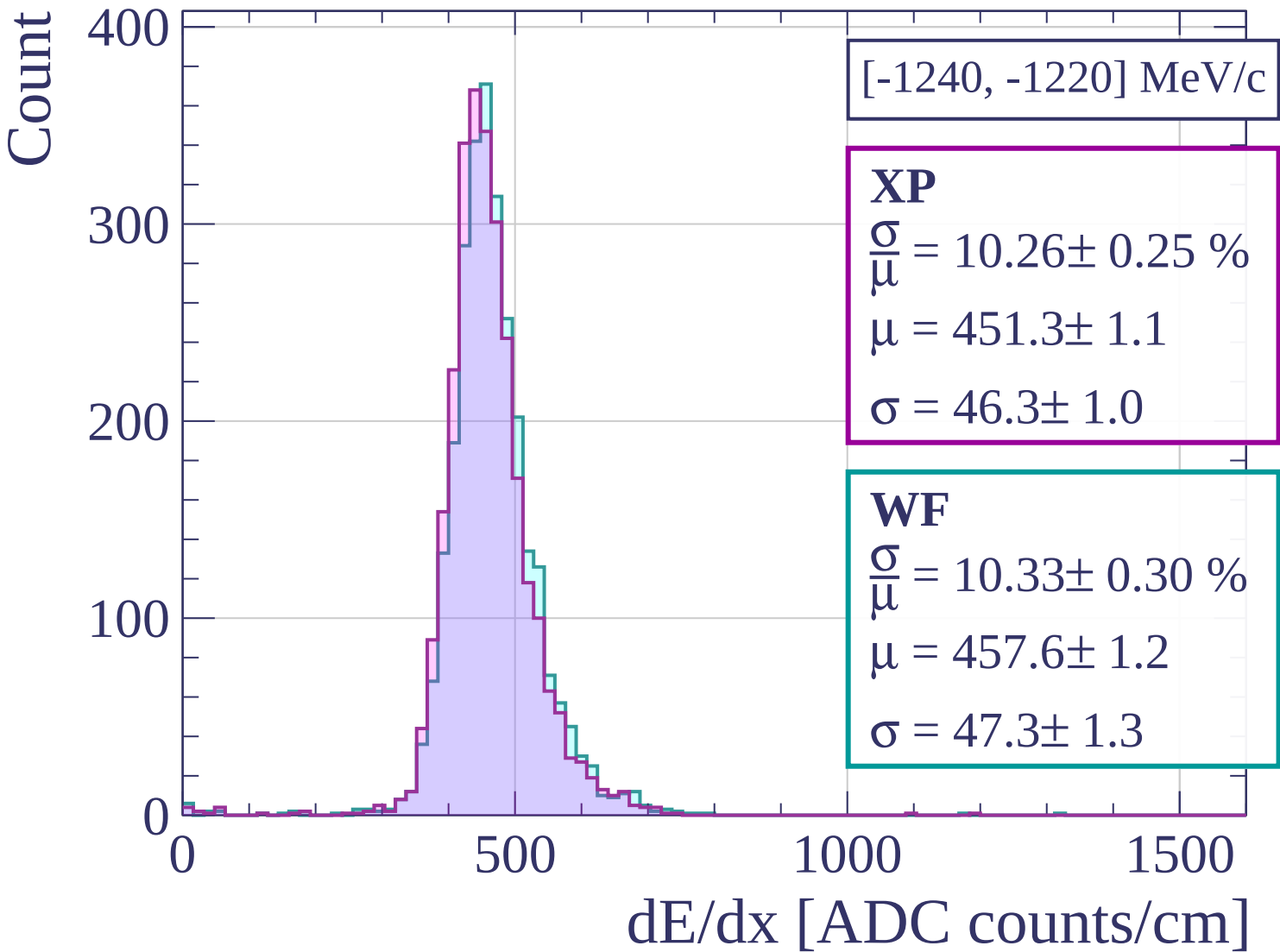
1000

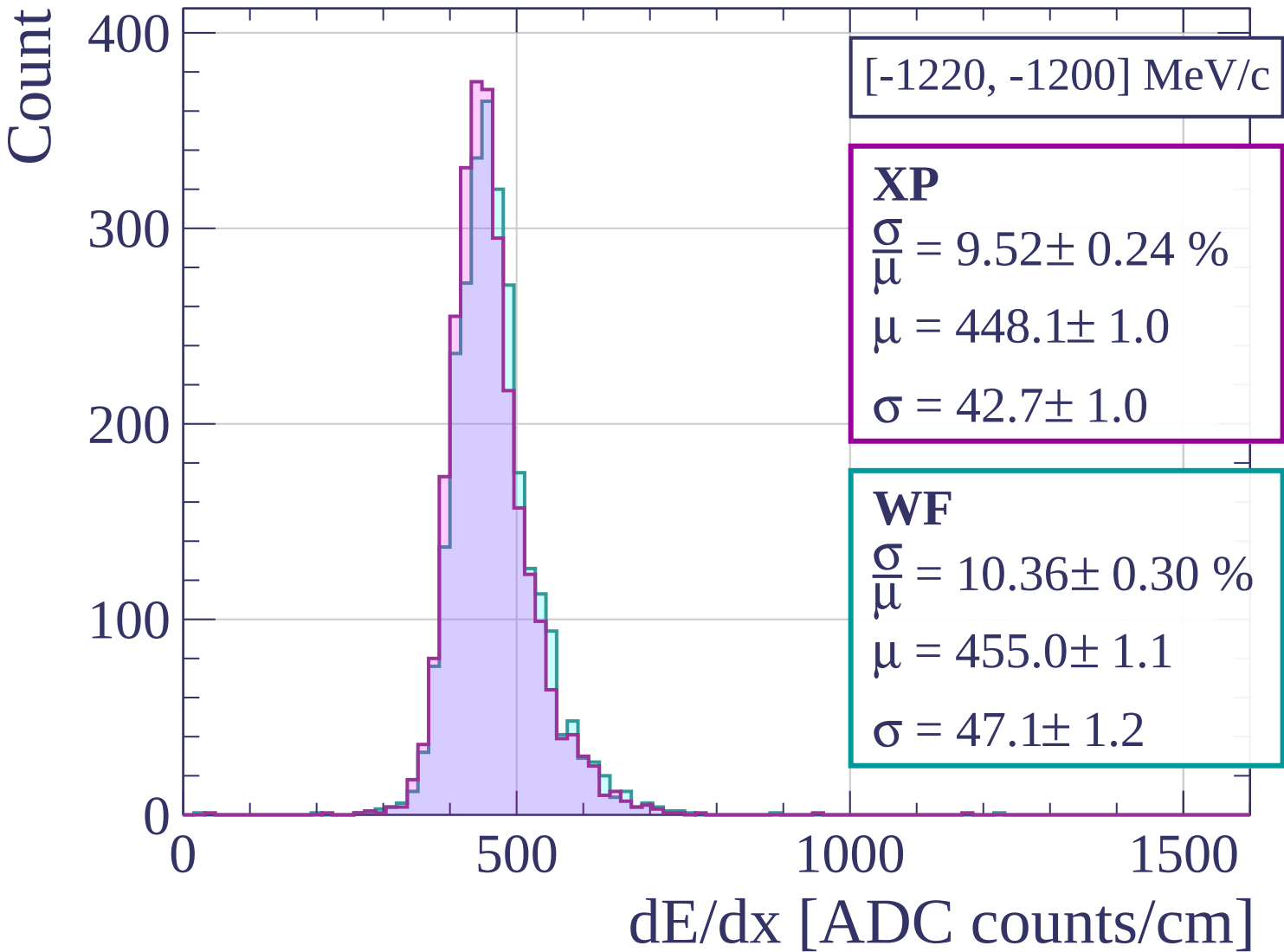
1500

dE/dx [ADC counts/cm]









Count

[-1200, -1180] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.85 \pm 0.25 \%$$

$$\mu = 446.5 \pm 1.1$$

$$\sigma = 44.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.48 \pm 0.29 \%$$

$$\mu = 454.6 \pm 1.2$$

$$\sigma = 47.6 \pm 1.2$$

300

200

100

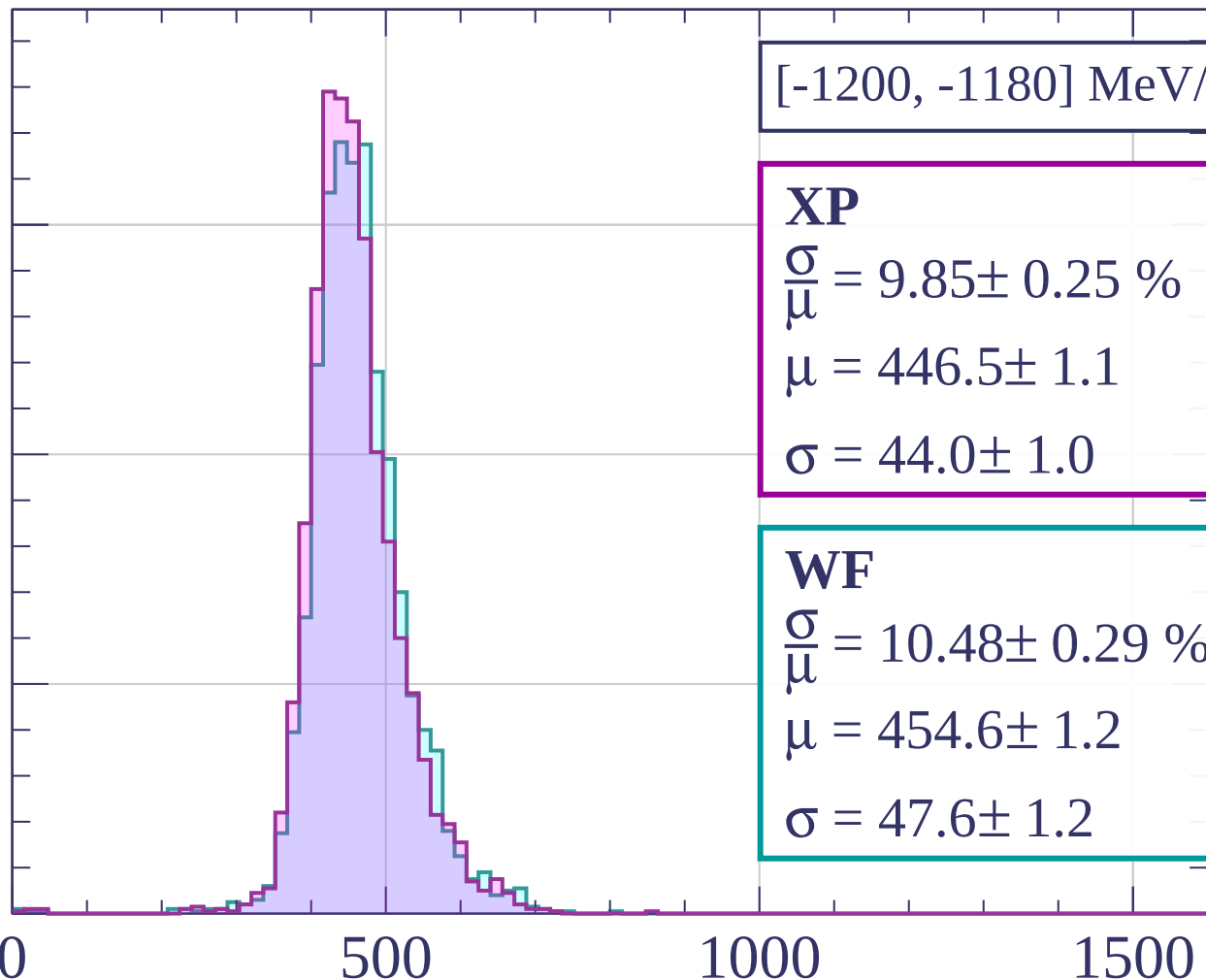
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1180, -1160] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.23 \%$$

$$\mu = 444.8 \pm 1.0$$

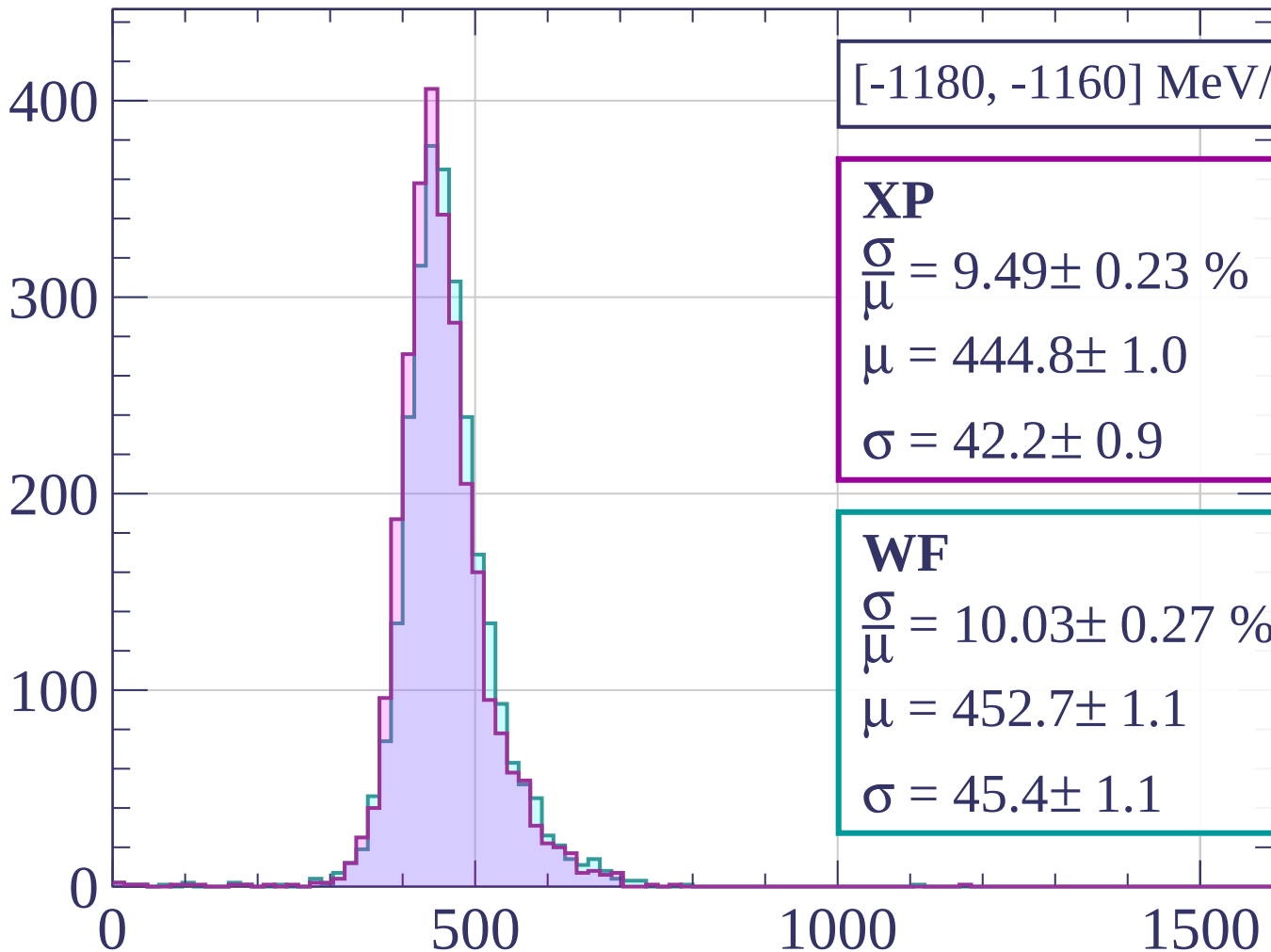
$$\sigma = 42.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.03 \pm 0.27 \%$$

$$\mu = 452.7 \pm 1.1$$

$$\sigma = 45.4 \pm 1.1$$



dE/dx [ADC counts/cm]

Count

[-1160, -1140] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.33 \pm 0.29 \%$$

$$\mu = 443.2 \pm 1.1$$

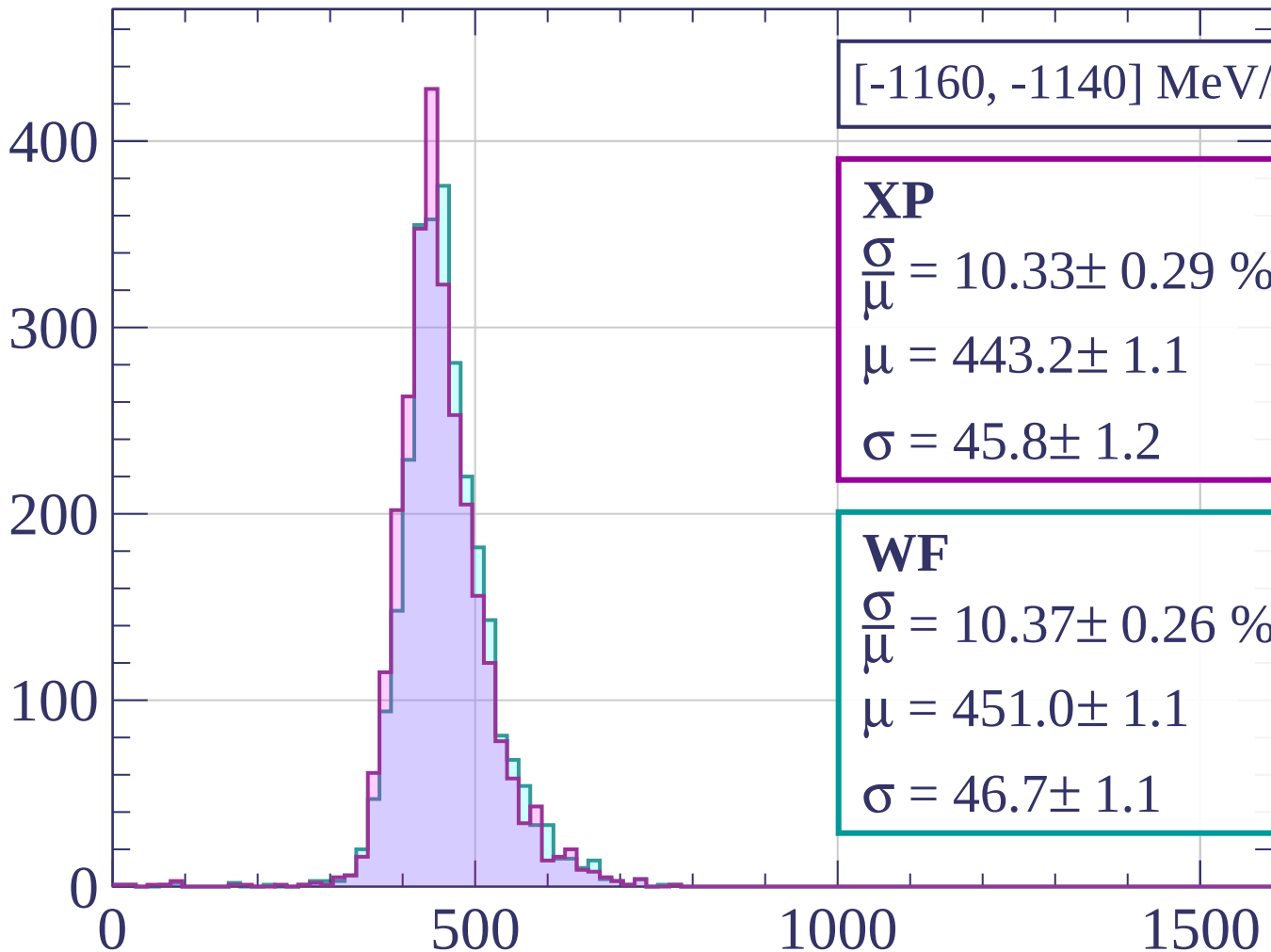
$$\sigma = 45.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.37 \pm 0.26 \%$$

$$\mu = 451.0 \pm 1.1$$

$$\sigma = 46.7 \pm 1.1$$



dE/dx [ADC counts/cm]

Count

[-1140, -1120] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.95 \pm 0.25 \%$$

$$\mu = 445.0 \pm 1.0$$

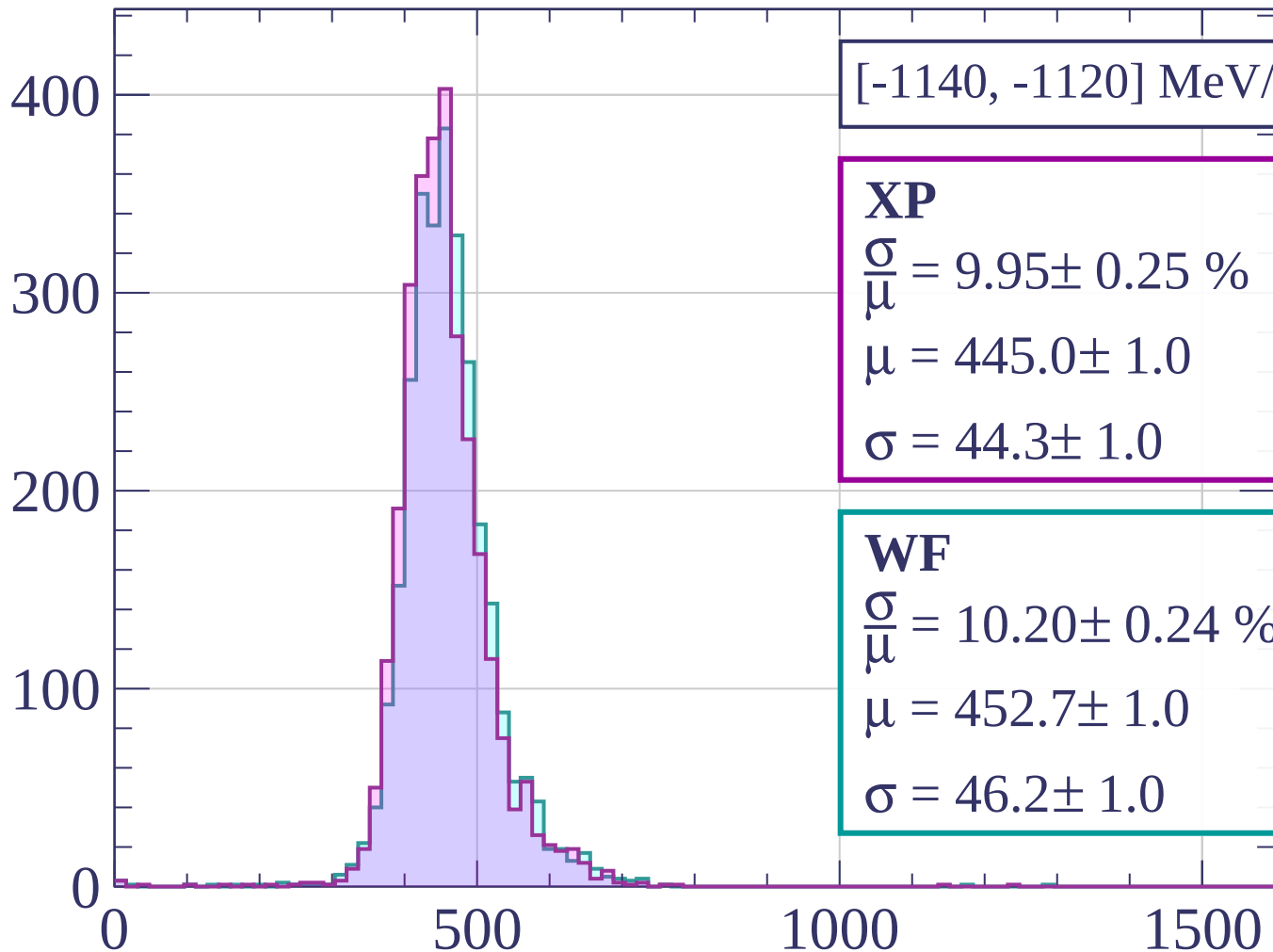
$$\sigma = 44.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.20 \pm 0.24 \%$$

$$\mu = 452.7 \pm 1.0$$

$$\sigma = 46.2 \pm 1.0$$



dE/dx [ADC counts/cm]

Count

[-1120, -1100] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.53 \pm 0.22 \%$$

$$\mu = 441.5 \pm 1.0$$

$$\sigma = 42.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.25 \pm 0.25 \%$$

$$\mu = 449.5 \pm 1.0$$

$$\sigma = 46.1 \pm 1.0$$

400

300

200

100

0

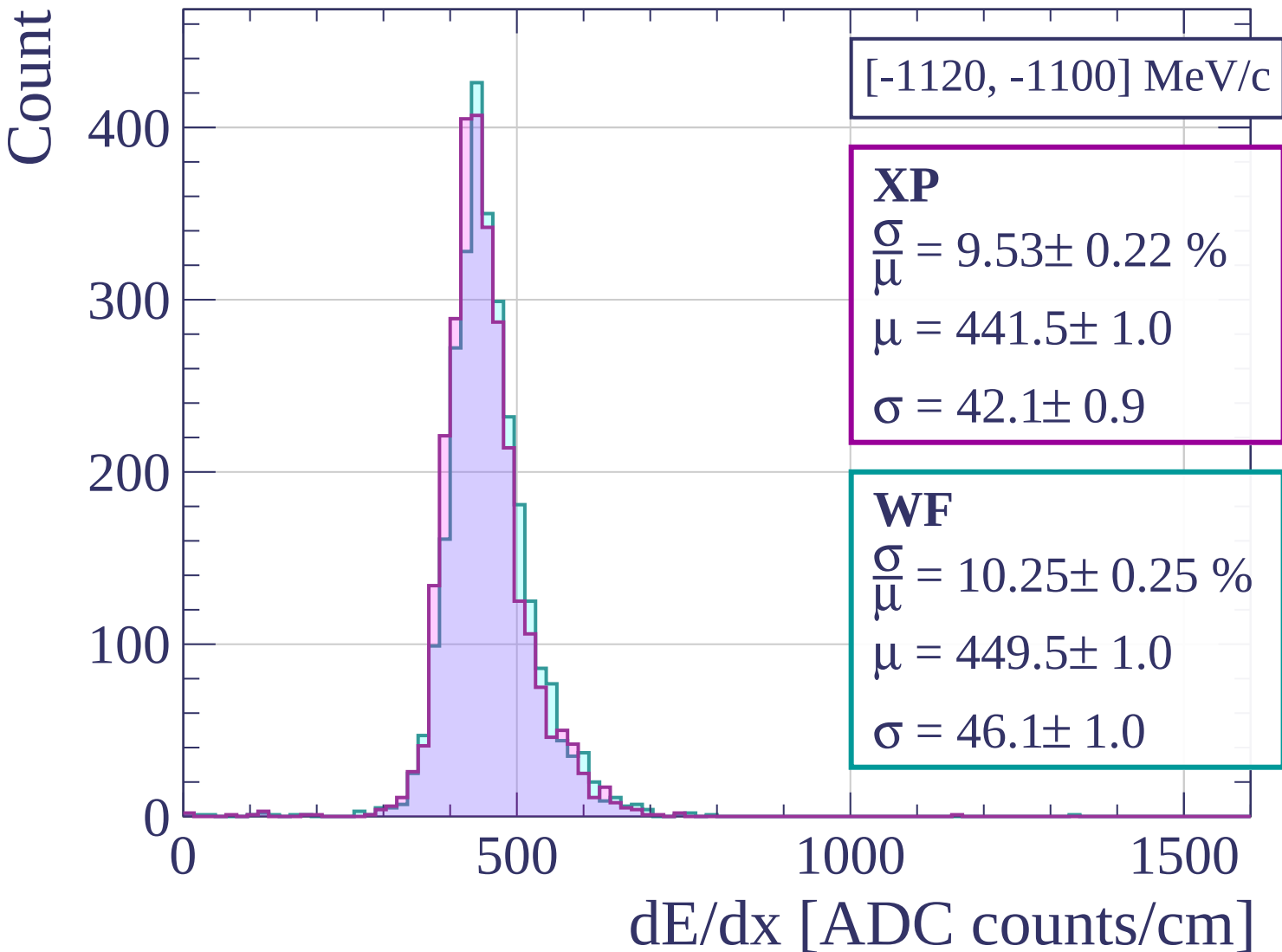
0

500

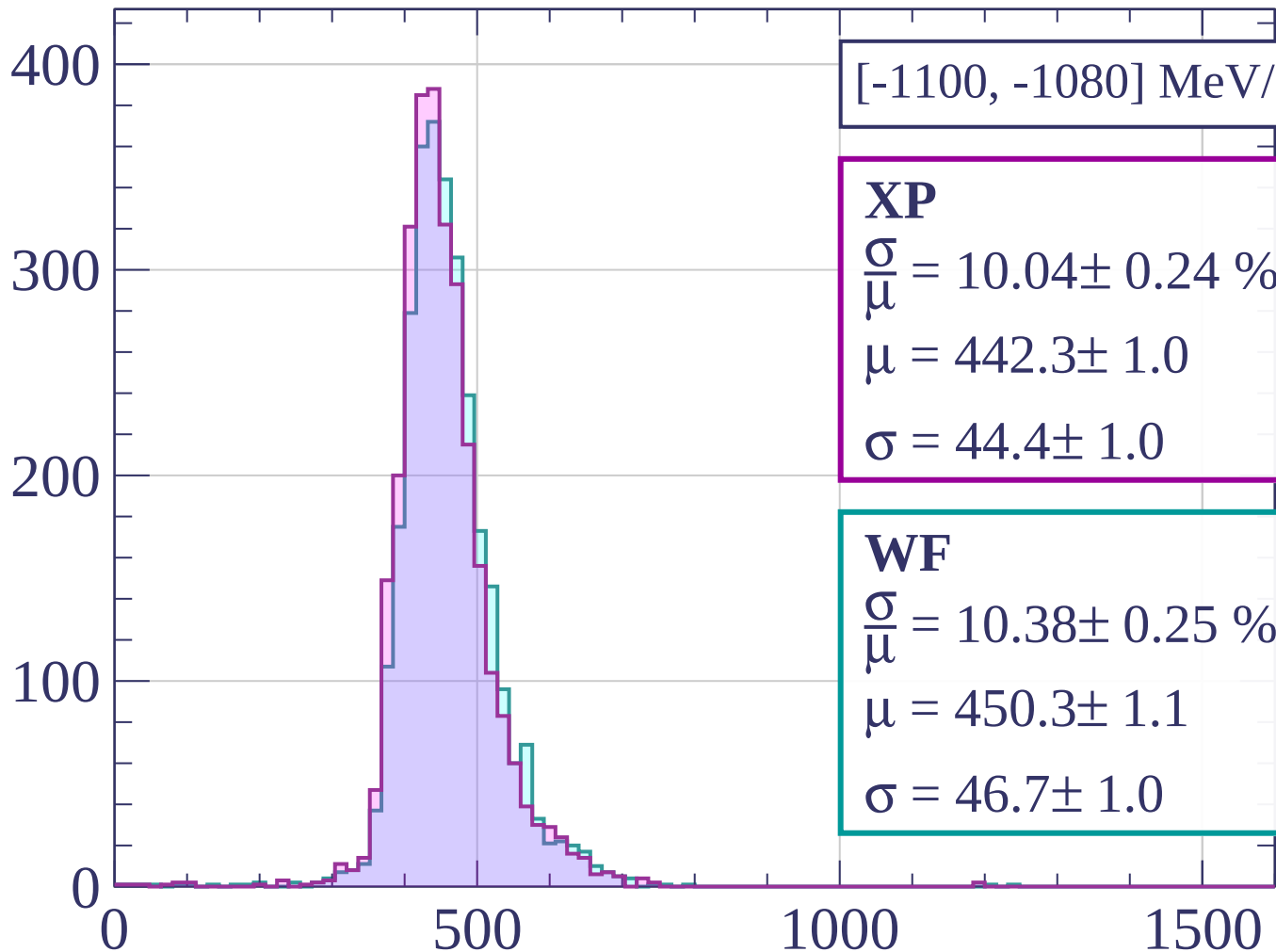
1000

1500

dE/dx [ADC counts/cm]



Count



[-1100, -1080] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.04 \pm 0.24 \%$$

$$\mu = 442.3 \pm 1.0$$

$$\sigma = 44.4 \pm 1.0$$

WF

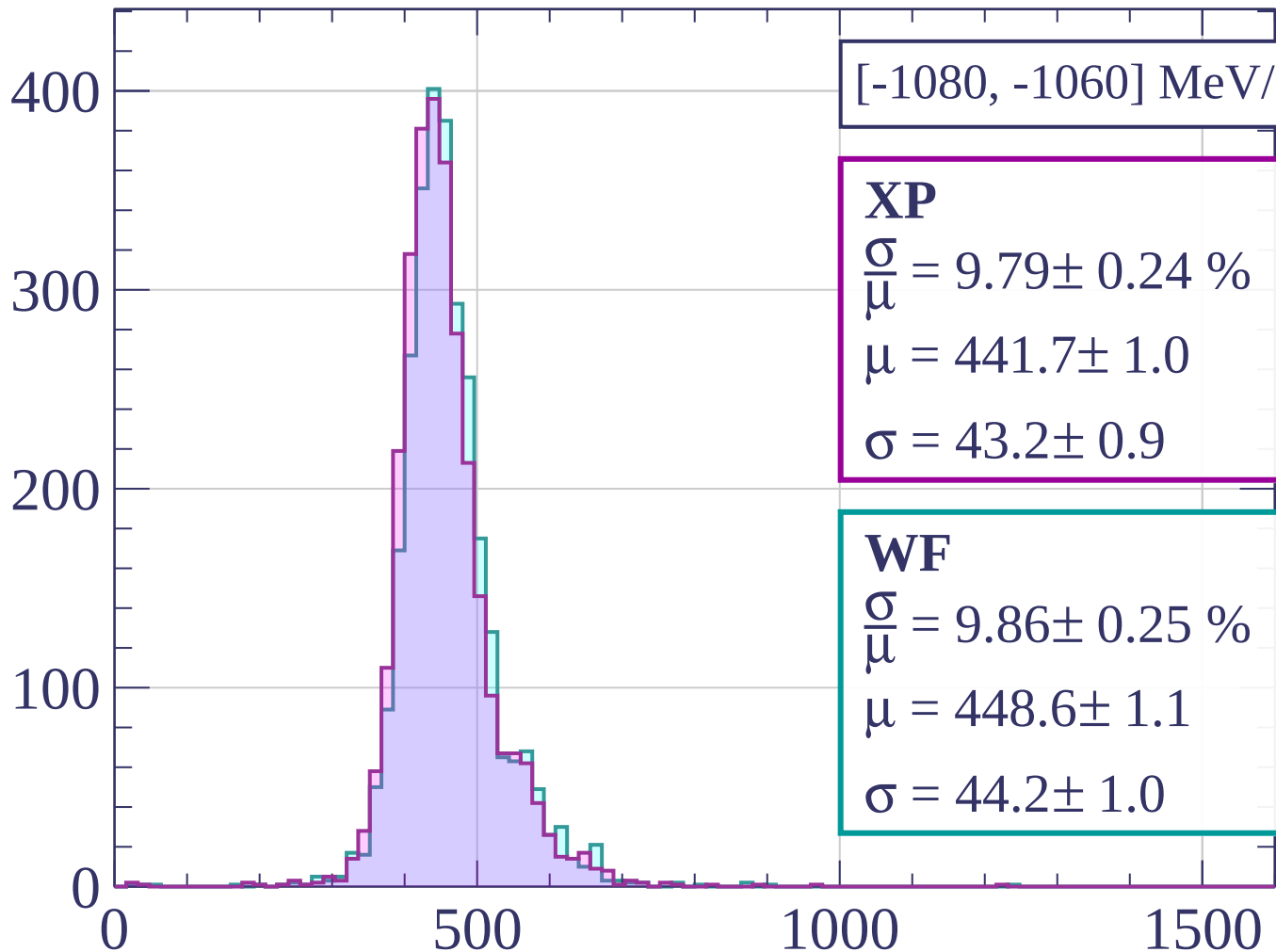
$$\frac{\sigma}{\mu} = 10.38 \pm 0.25 \%$$

$$\mu = 450.3 \pm 1.1$$

$$\sigma = 46.7 \pm 1.0$$

dE/dx [ADC counts/cm]

Count



[-1080, -1060] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.79 \pm 0.24 \%$$

$$\mu = 441.7 \pm 1.0$$

$$\sigma = 43.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.86 \pm 0.25 \%$$

$$\mu = 448.6 \pm 1.1$$

$$\sigma = 44.2 \pm 1.0$$

dE/dx [ADC counts/cm]

Count

[-1060, -1040] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.16 \pm 0.22 \%$$

$$\mu = 437.0 \pm 0.9$$

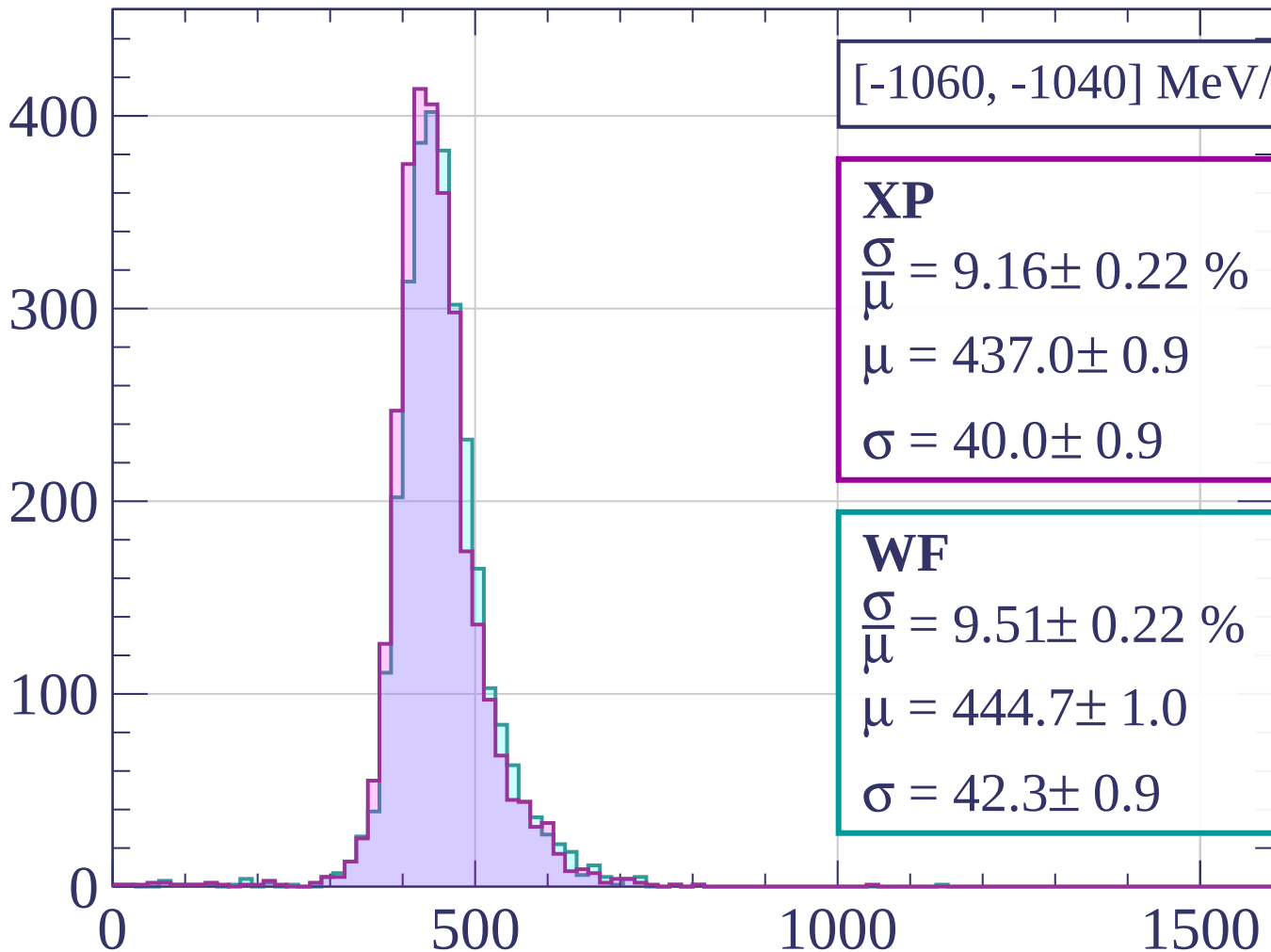
$$\sigma = 40.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.51 \pm 0.22 \%$$

$$\mu = 444.7 \pm 1.0$$

$$\sigma = 42.3 \pm 0.9$$



dE/dx [ADC counts/cm]

Count

[-1040, -1020] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.54 \pm 0.25 \%$$

$$\mu = 435.9 \pm 1.0$$

$$\sigma = 41.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.96 \pm 0.24 \%$$

$$\mu = 443.9 \pm 1.0$$

$$\sigma = 44.2 \pm 1.0$$

400

200

0

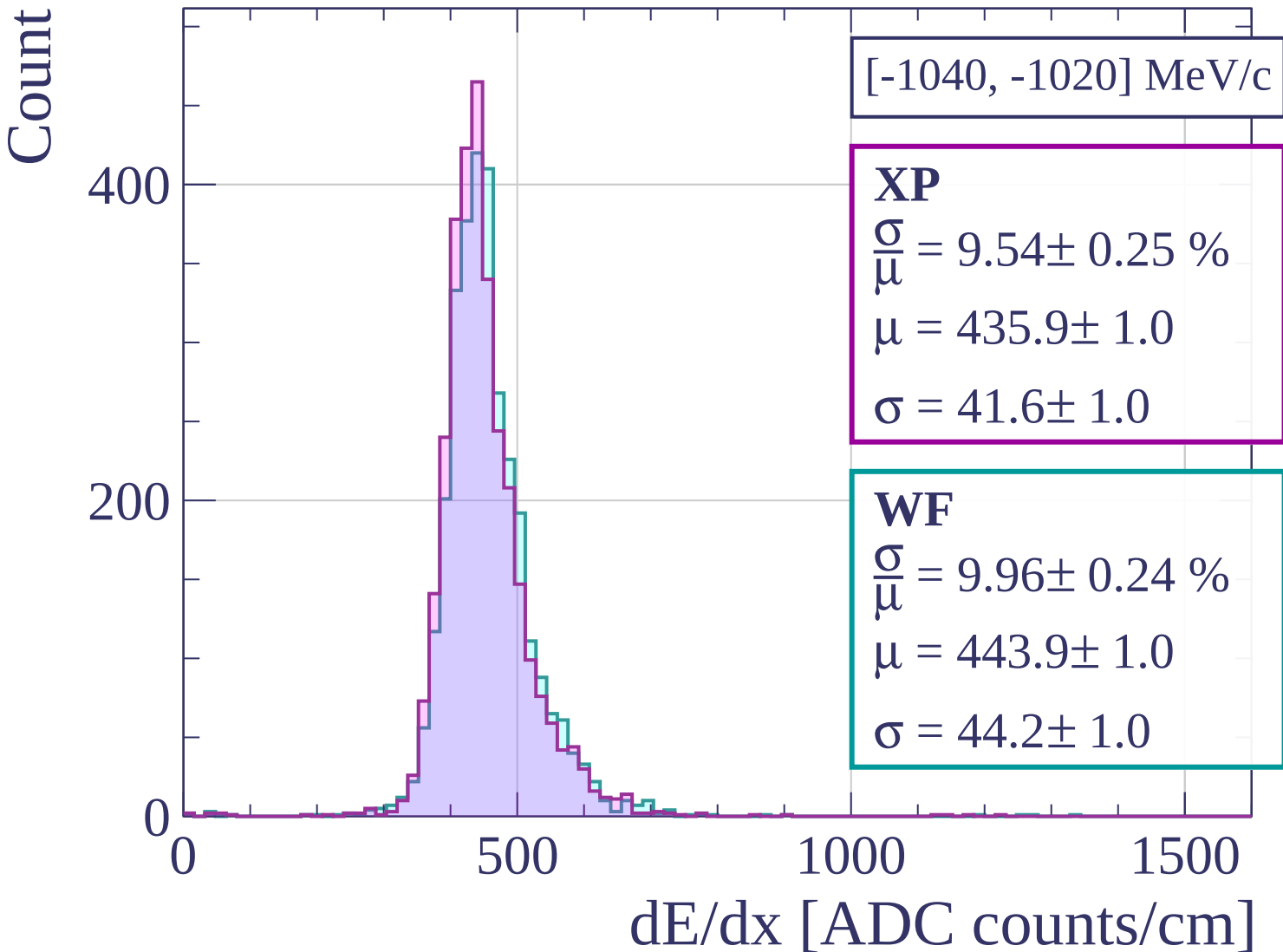
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1020, -1000] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.25 \%$$

$$\mu = 433.0 \pm 0.9$$

$$\sigma = 41.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.01 \pm 0.26 \%$$

$$\mu = 441.0 \pm 1.0$$

$$\sigma = 44.1 \pm 1.0$$

400

200

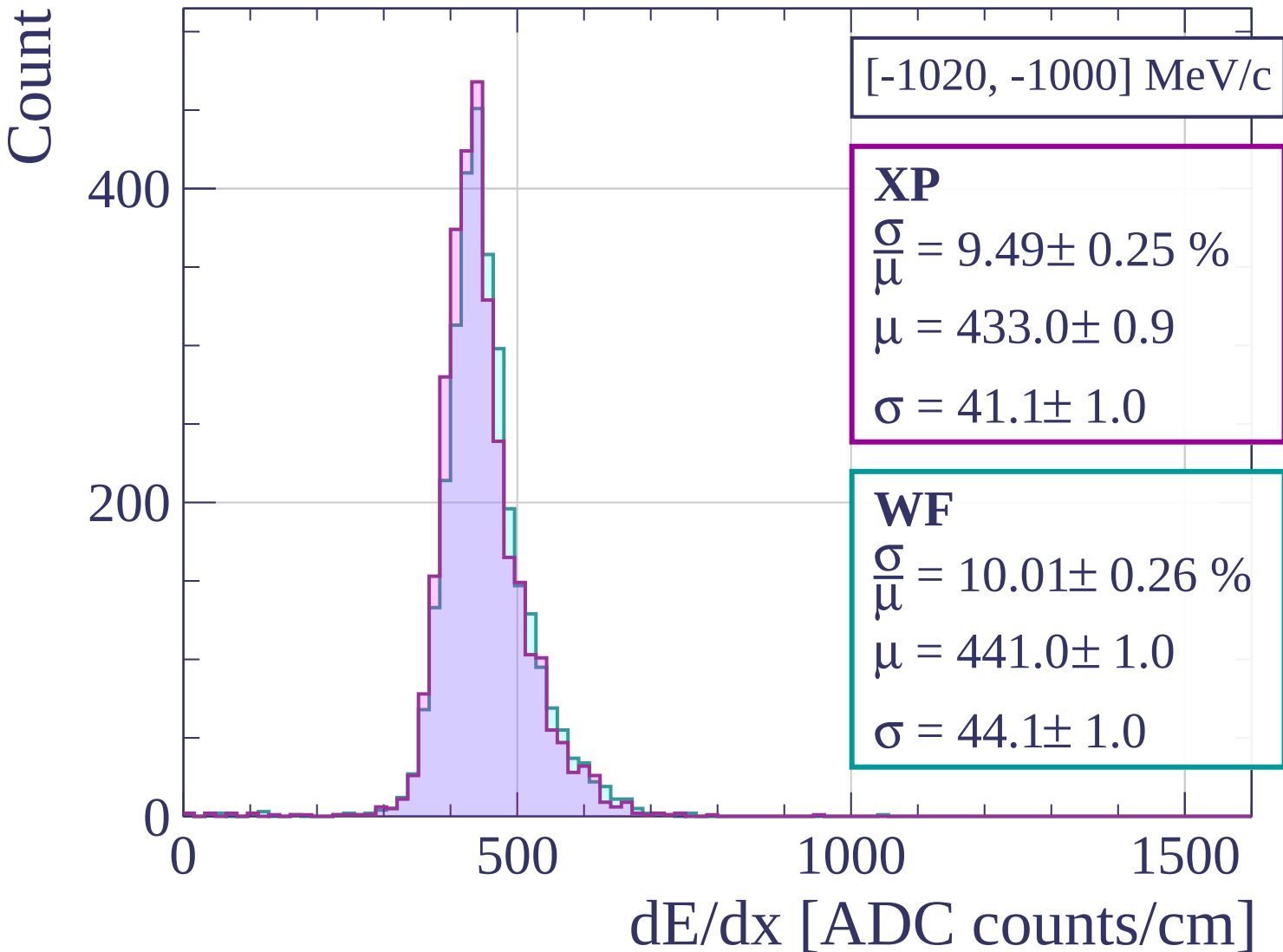
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1000, -980] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.40 \pm 0.23 \%$$

$$\mu = 435.7 \pm 1.0$$

$$\sigma = 41.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.63 \pm 0.21 \%$$

$$\mu = 443.5 \pm 1.0$$

$$\sigma = 42.7 \pm 0.9$$

400

200

0

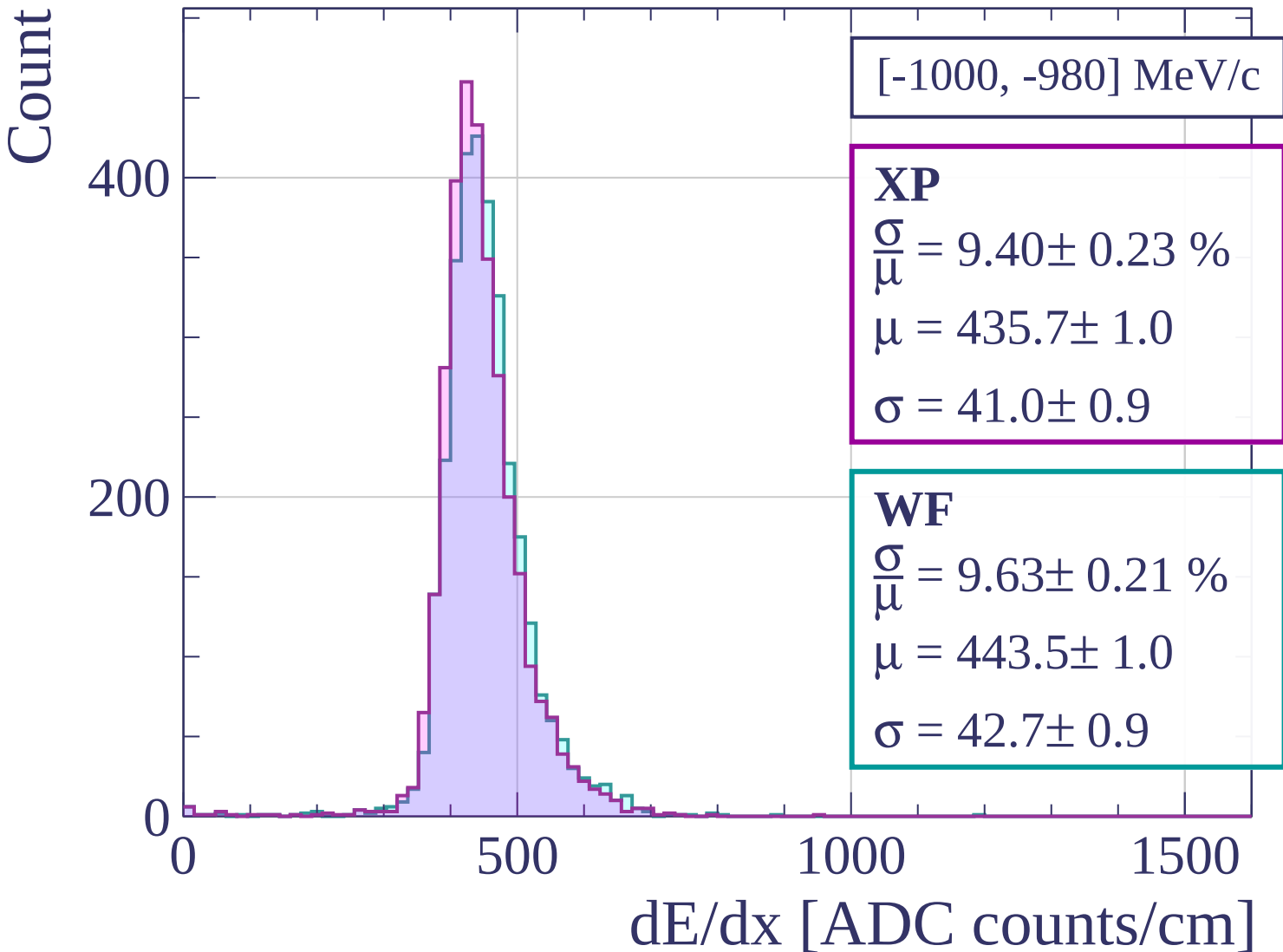
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-980, -960] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.57 \pm 0.24 \%$$

$$\mu = 433.9 \pm 0.9$$

$$\sigma = 41.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.14 \pm 0.25 \%$$

$$\mu = 442.1 \pm 1.0$$

$$\sigma = 44.8 \pm 1.0$$

400

300

200

100

0

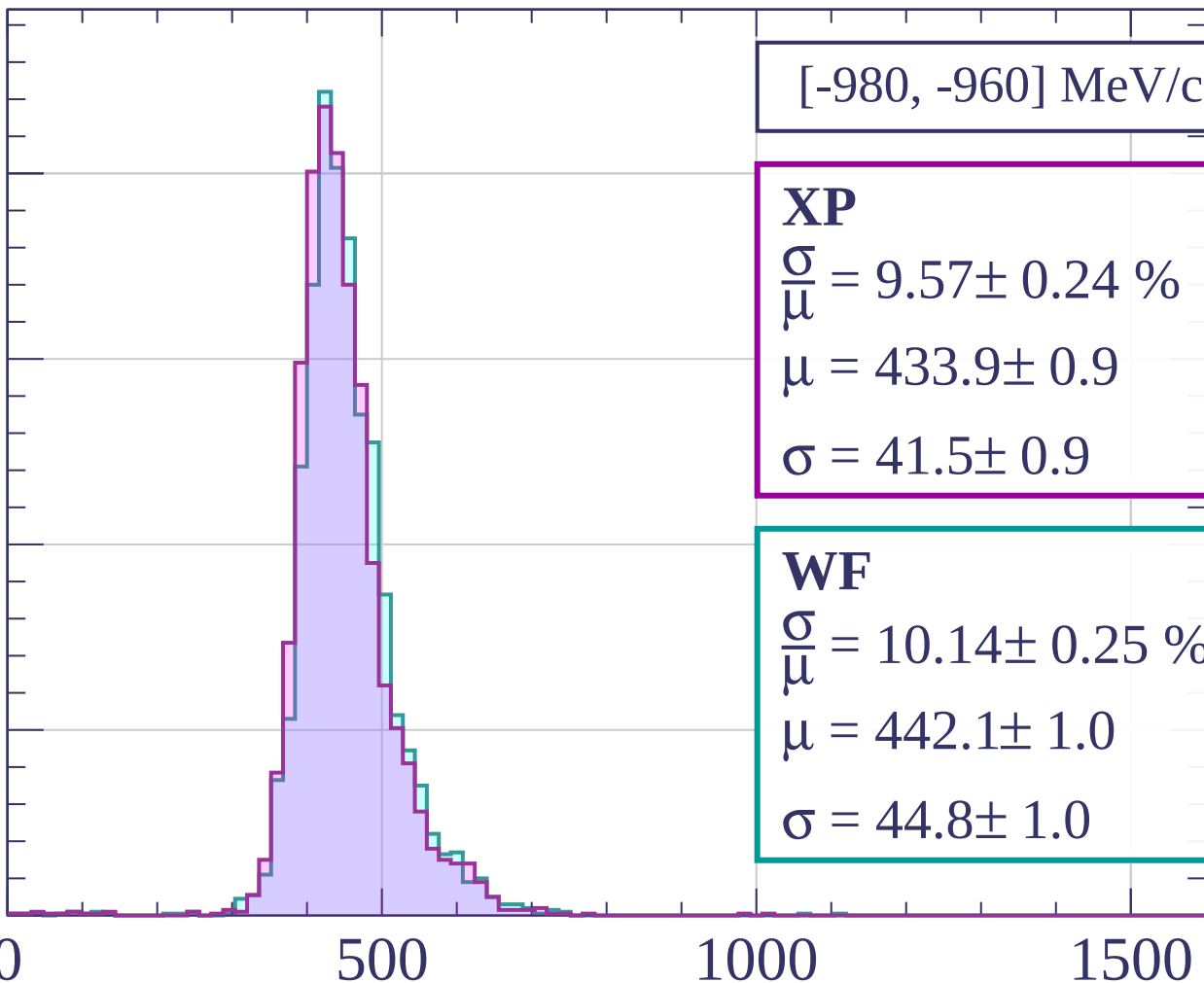
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-960, -940] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.46 \pm 0.22 \%$$

$$\mu = 433.6 \pm 0.9$$

$$\sigma = 41.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.11 \pm 0.23 \%$$

$$\mu = 440.4 \pm 1.0$$

$$\sigma = 44.5 \pm 0.9$$

400

200

0

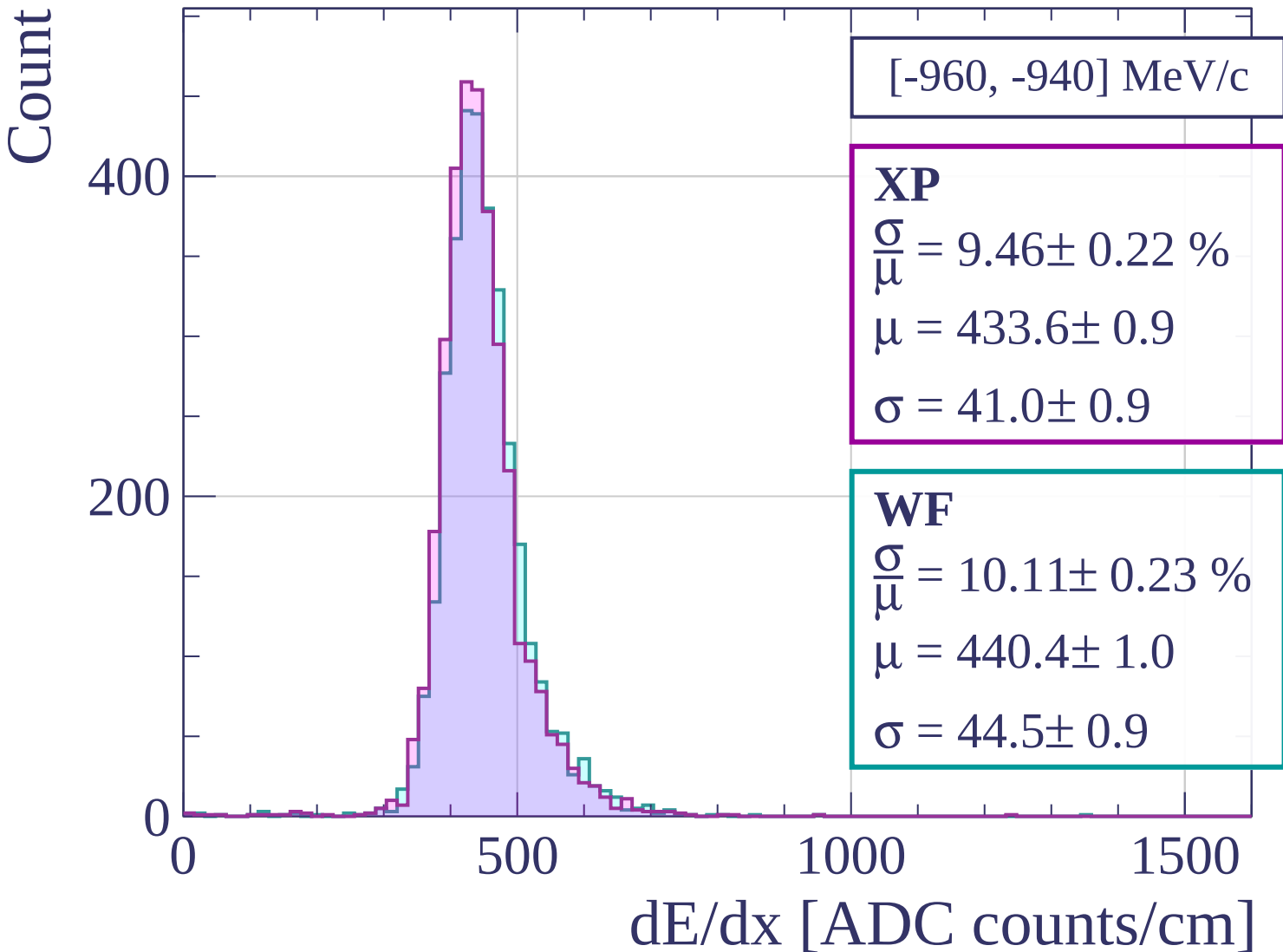
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-940, -920] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.54 \pm 0.23 \%$$

$$\mu = 432.3 \pm 0.9$$

$$\sigma = 41.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.75 \pm 0.22 \%$$

$$\mu = 439.3 \pm 0.9$$

$$\sigma = 42.8 \pm 0.9$$

400

300

200

100

0

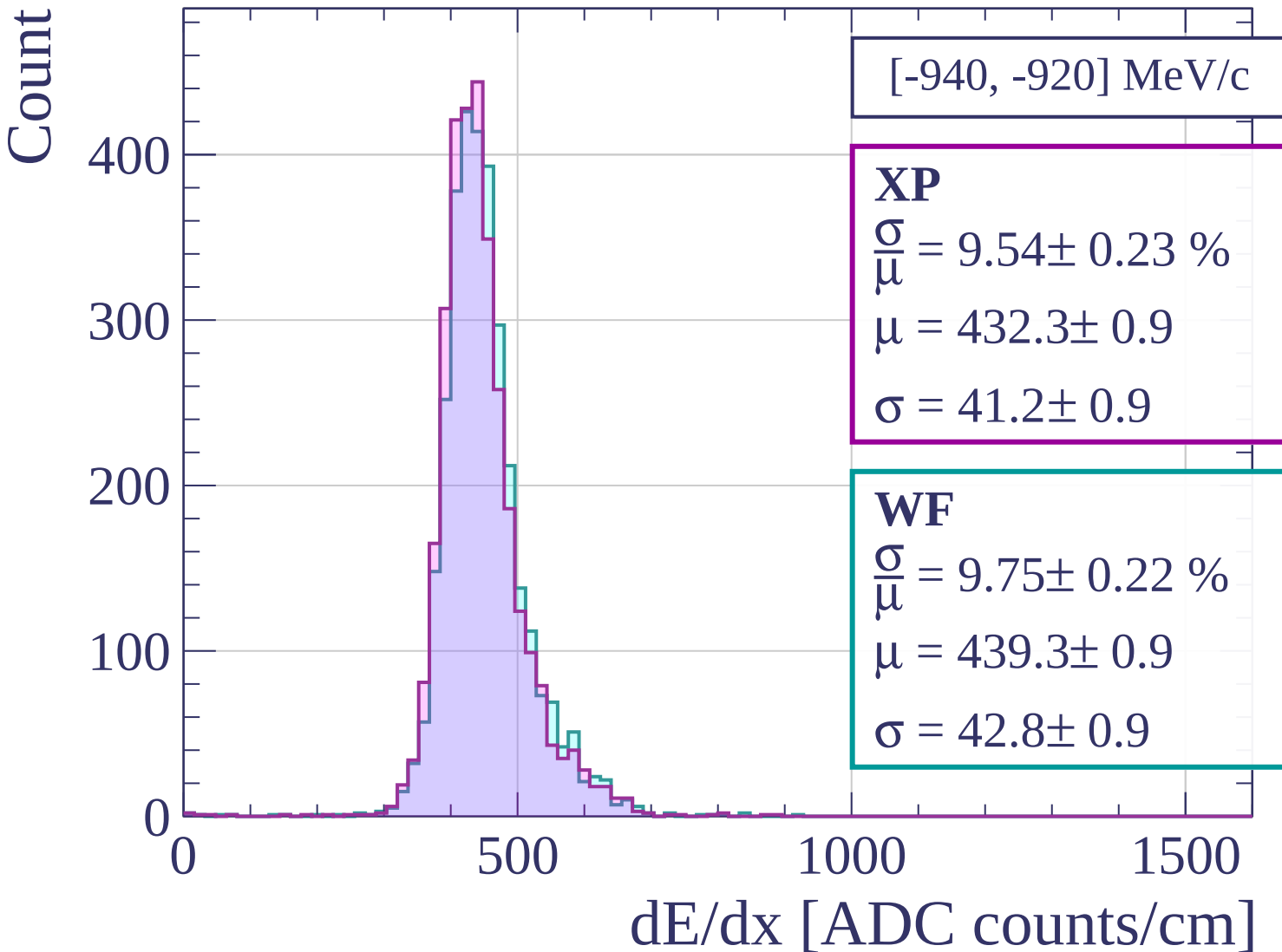
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-920, -900] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.61 \pm 0.24 \%$$

$$\mu = 429.1 \pm 0.9$$

$$\sigma = 41.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.79 \pm 0.25 \%$$

$$\mu = 434.9 \pm 1.0$$

$$\sigma = 42.6 \pm 1.0$$

400

200

0

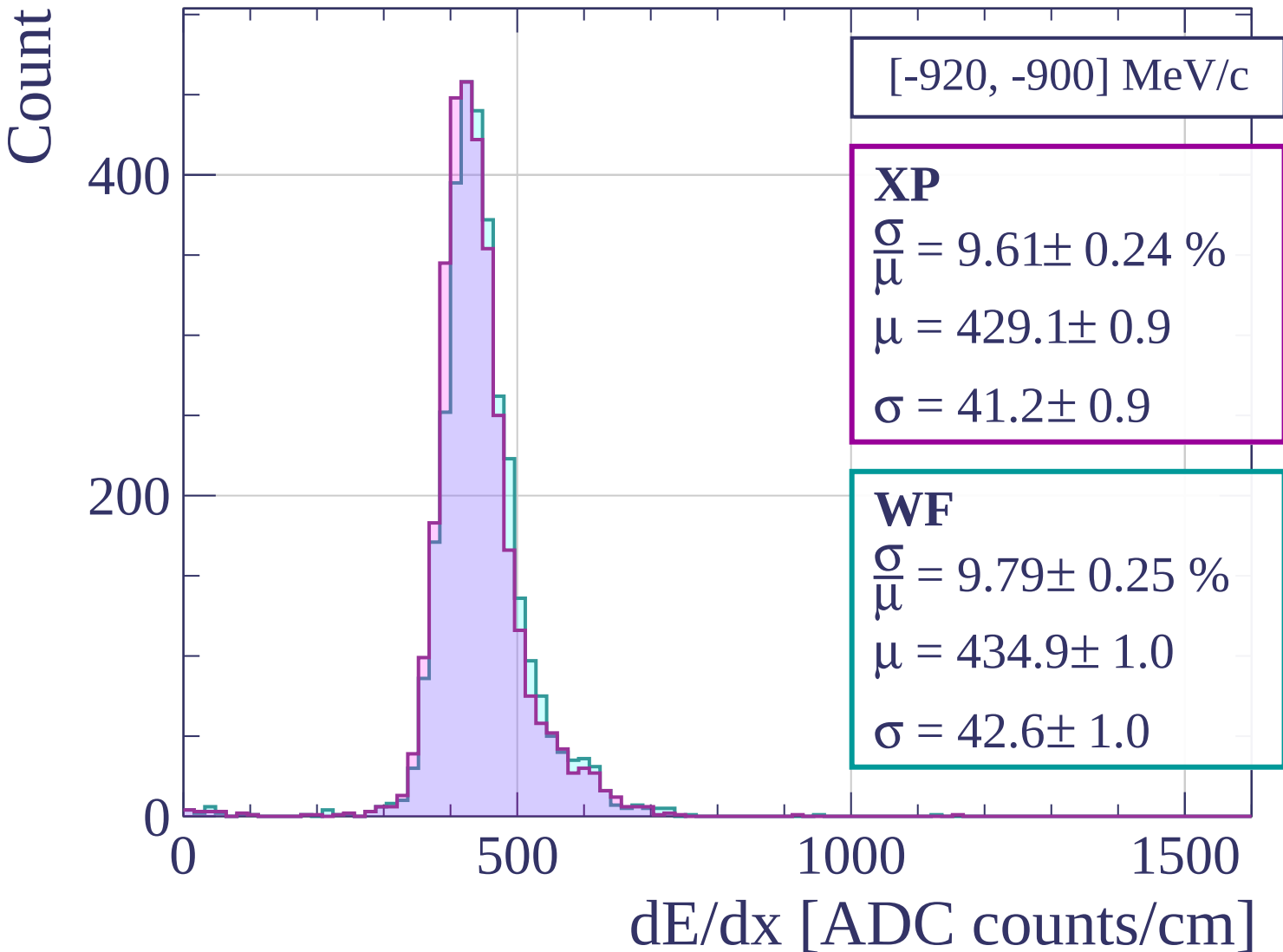
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-900, -880] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.62 \pm 0.25 \%$$

$$\mu = 428.7 \pm 0.9$$

$$\sigma = 41.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.42 \pm 0.26 \%$$

$$\mu = 435.9 \pm 1.0$$

$$\sigma = 45.4 \pm 1.0$$

400

300

200

100

0

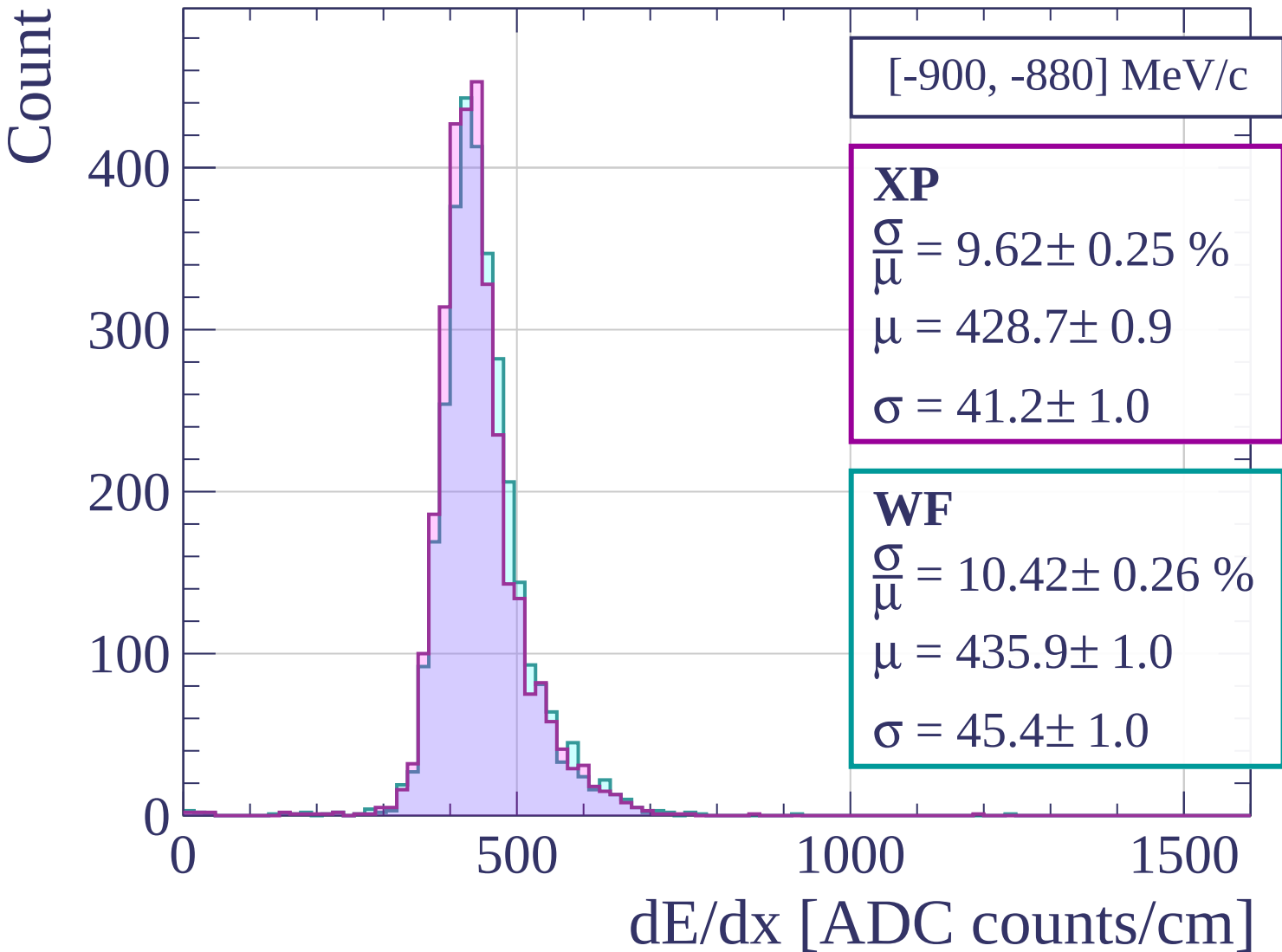
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-880, -860] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.87 \pm 0.25 \%$$

$$\mu = 428.0 \pm 0.9$$

$$\sigma = 42.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.73 \pm 0.25 \%$$

$$\mu = 433.9 \pm 1.0$$

$$\sigma = 42.2 \pm 1.0$$

400

200

0

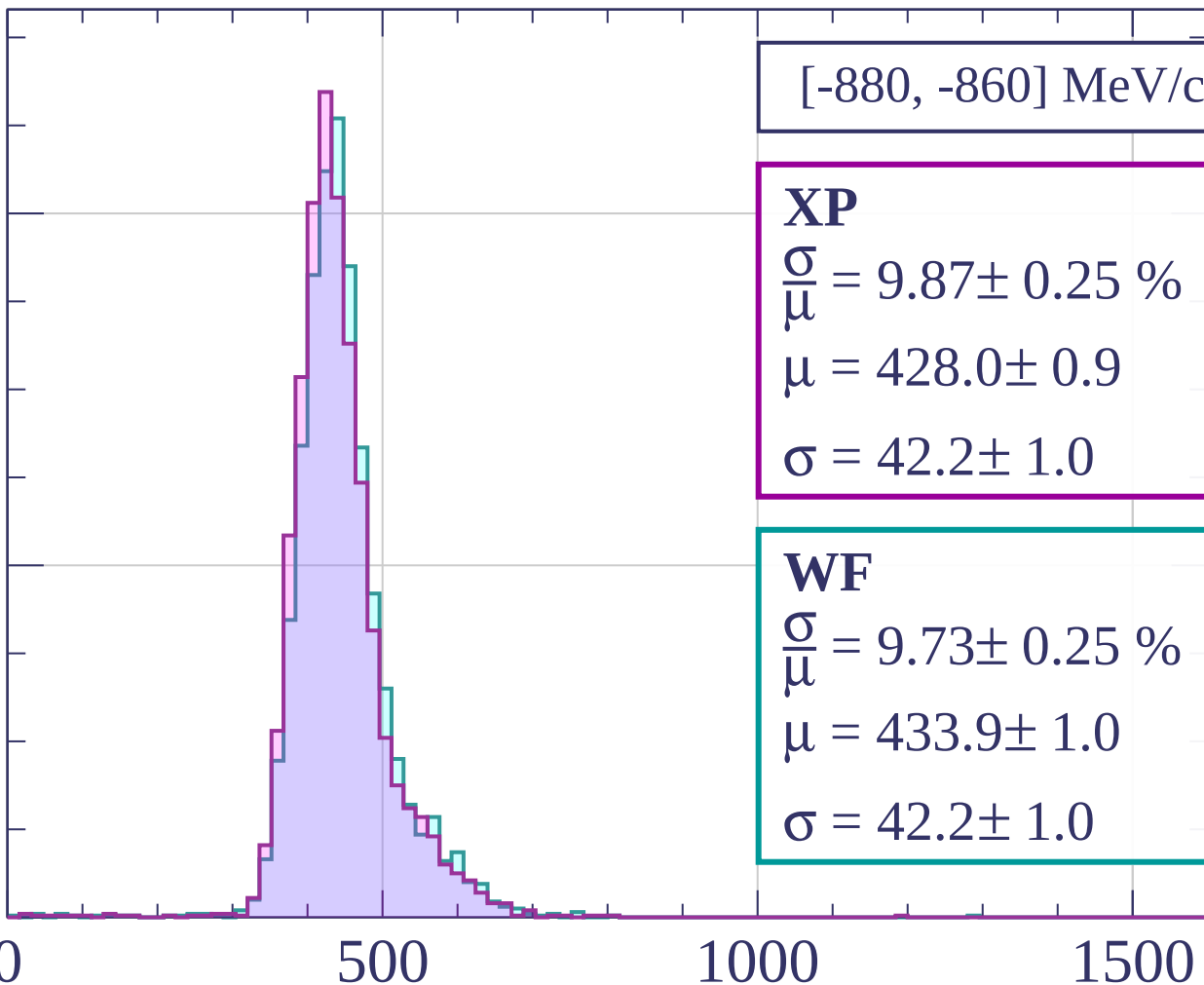
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-860, -840] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.61 \pm 0.20 \%$$

$$\mu = 427.0 \pm 0.9$$

$$\sigma = 41.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.78 \pm 0.25 \%$$

$$\mu = 432.2 \pm 0.9$$

$$\sigma = 42.3 \pm 1.0$$

400

200

0

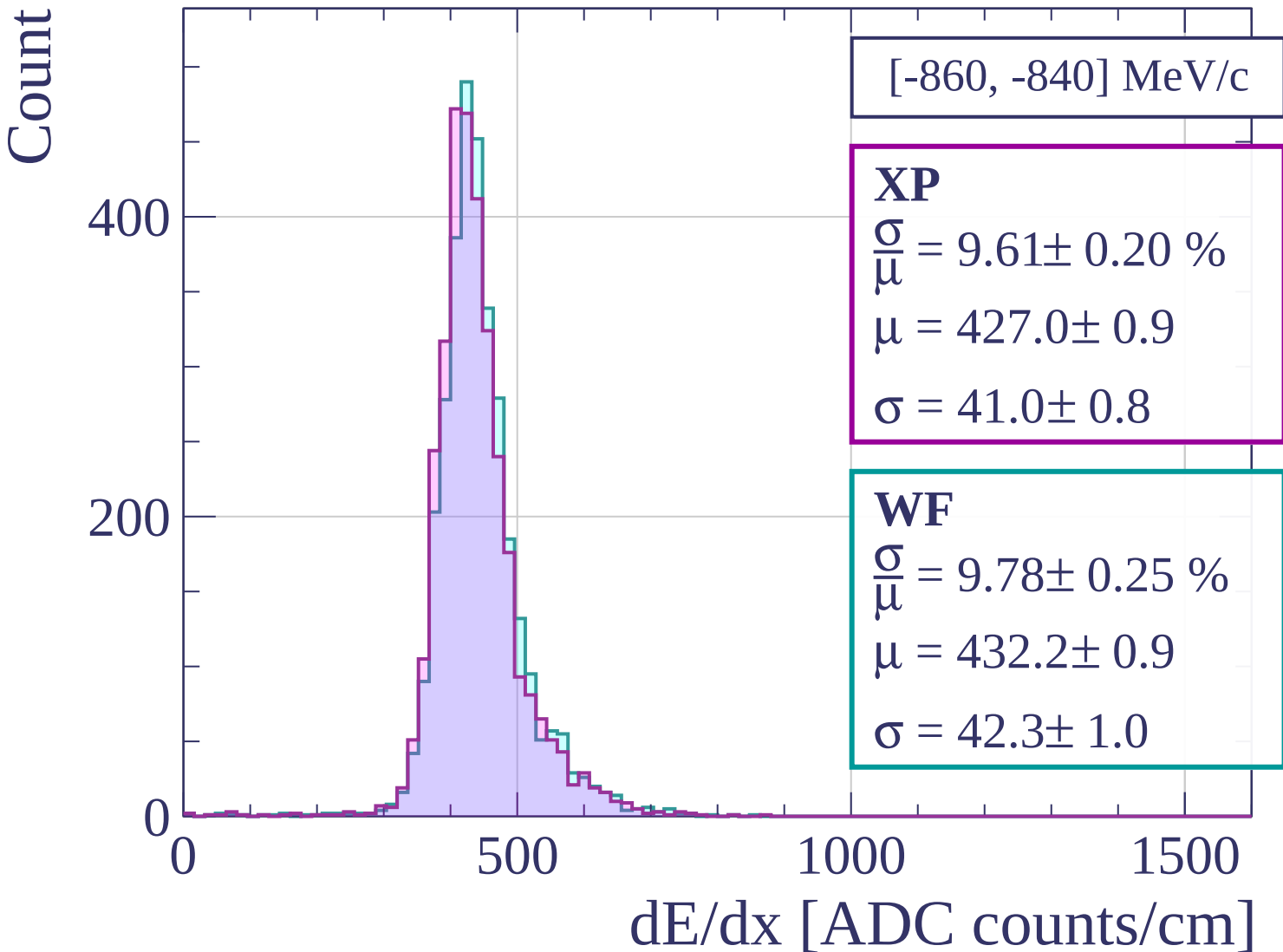
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-840, -820] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.74 \pm 0.21 \%$$

$$\mu = 426.3 \pm 0.9$$

$$\sigma = 41.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.08 \pm 0.23 \%$$

$$\mu = 432.1 \pm 1.0$$

$$\sigma = 43.5 \pm 0.9$$

400

200

0

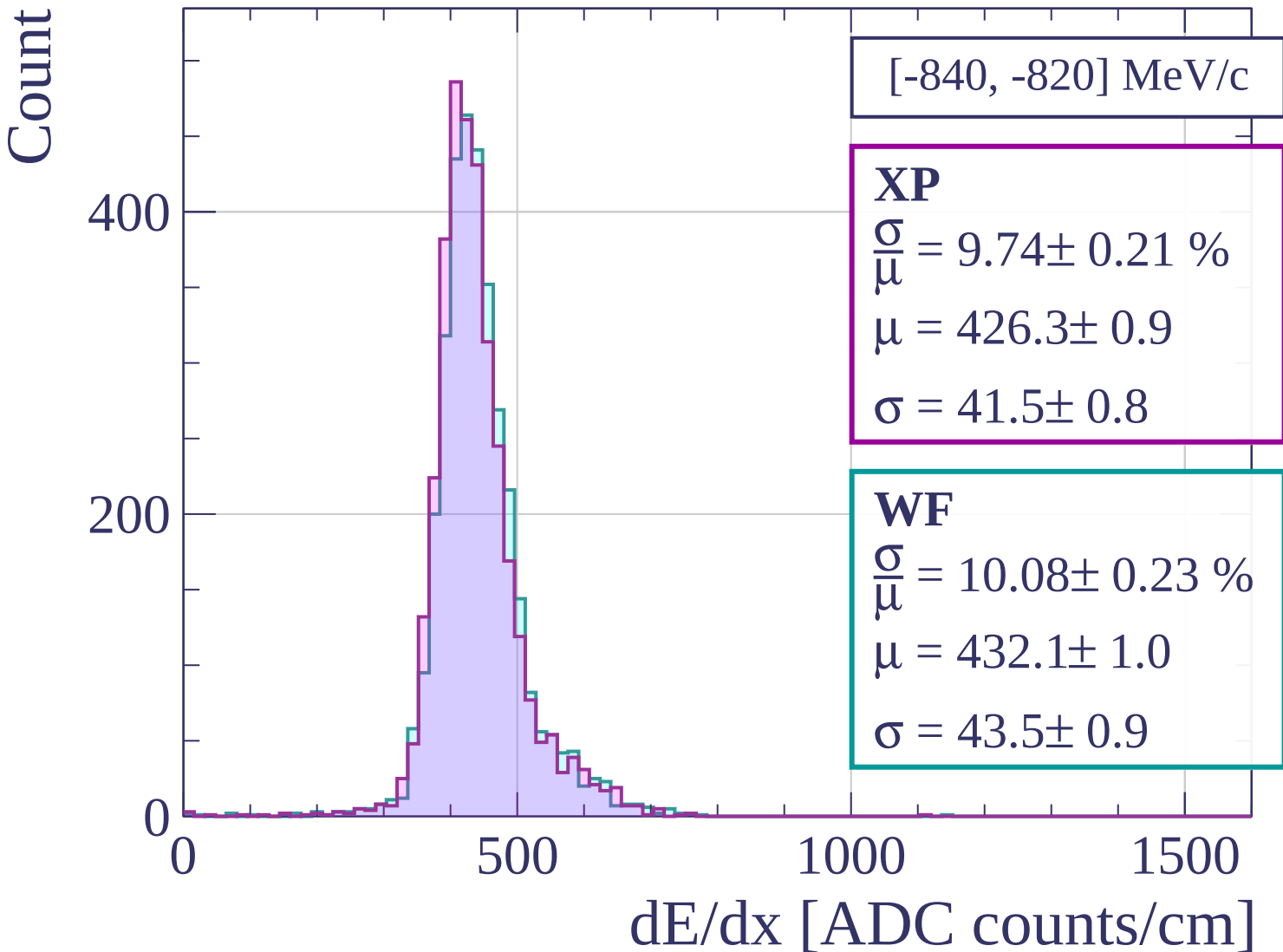
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-820, -800] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.36 \pm 0.24 \%$$

$$\mu = 421.6 \pm 0.9$$

$$\sigma = 39.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.56 \pm 0.19 \%$$

$$\mu = 429.1 \pm 0.9$$

$$\sigma = 41.0 \pm 0.8$$

400

200

0

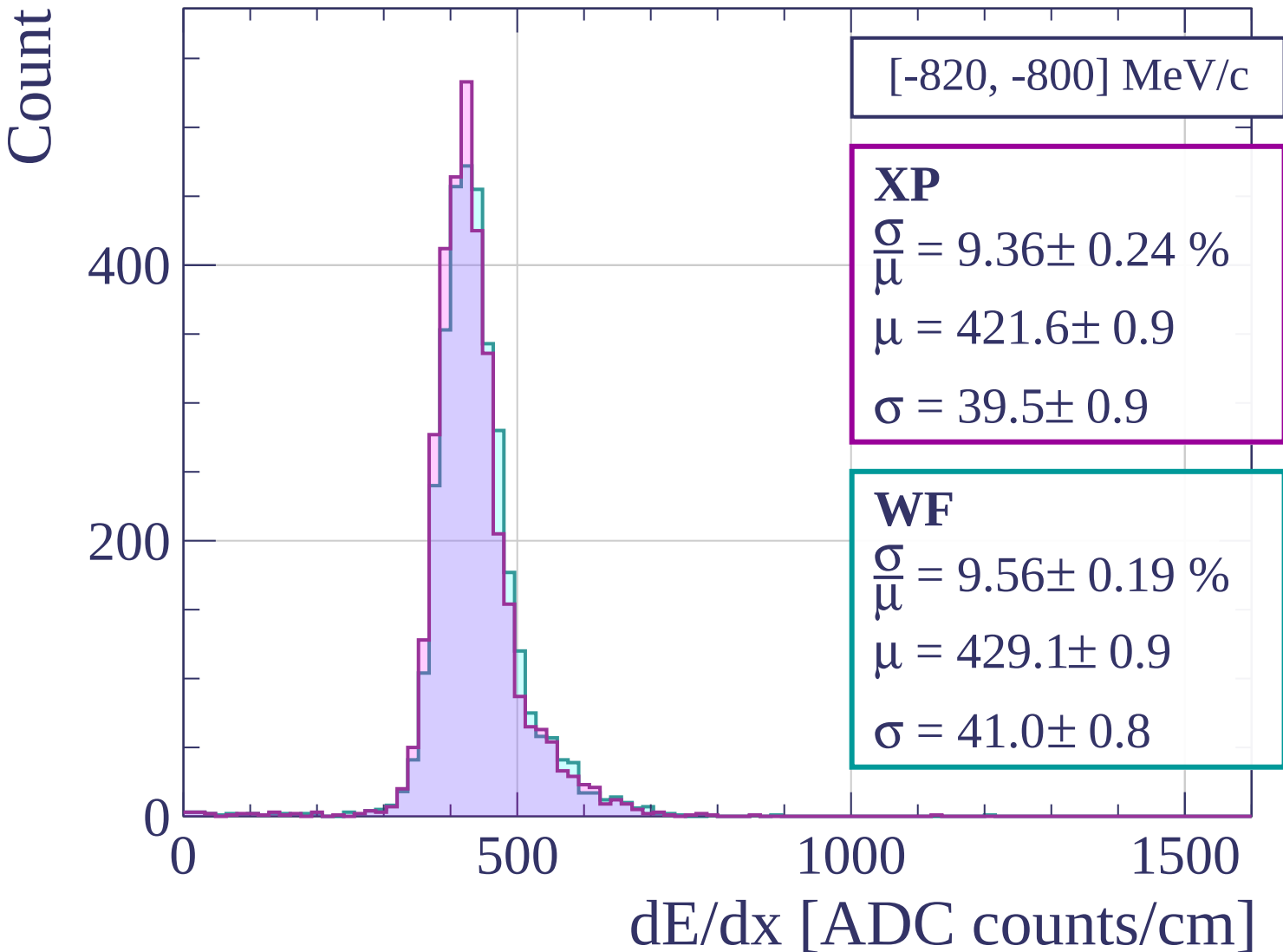
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-800, -780] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.04 \pm 0.19 \%$$

$$\mu = 422.0 \pm 0.8$$

$$\sigma = 38.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.59 \pm 0.23 \%$$

$$\mu = 428.1 \pm 0.9$$

$$\sigma = 41.1 \pm 0.9$$

400

200

0

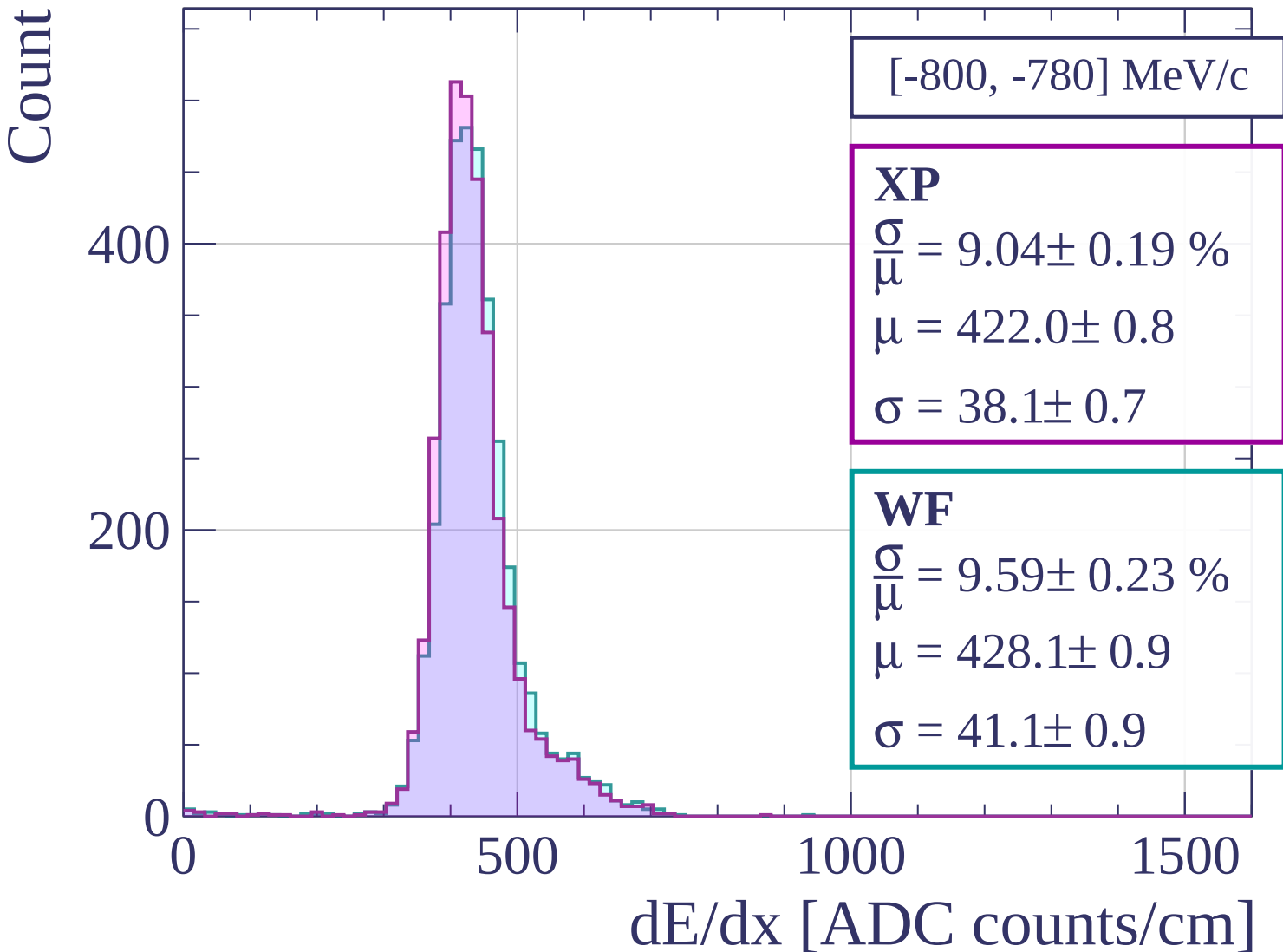
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-780, -760] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.23 \pm 0.20 \%$$

$$\mu = 420.5 \pm 0.8$$

$$\sigma = 38.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.71 \pm 0.21 \%$$

$$\mu = 427.1 \pm 0.9$$

$$\sigma = 41.5 \pm 0.8$$

400

200

0

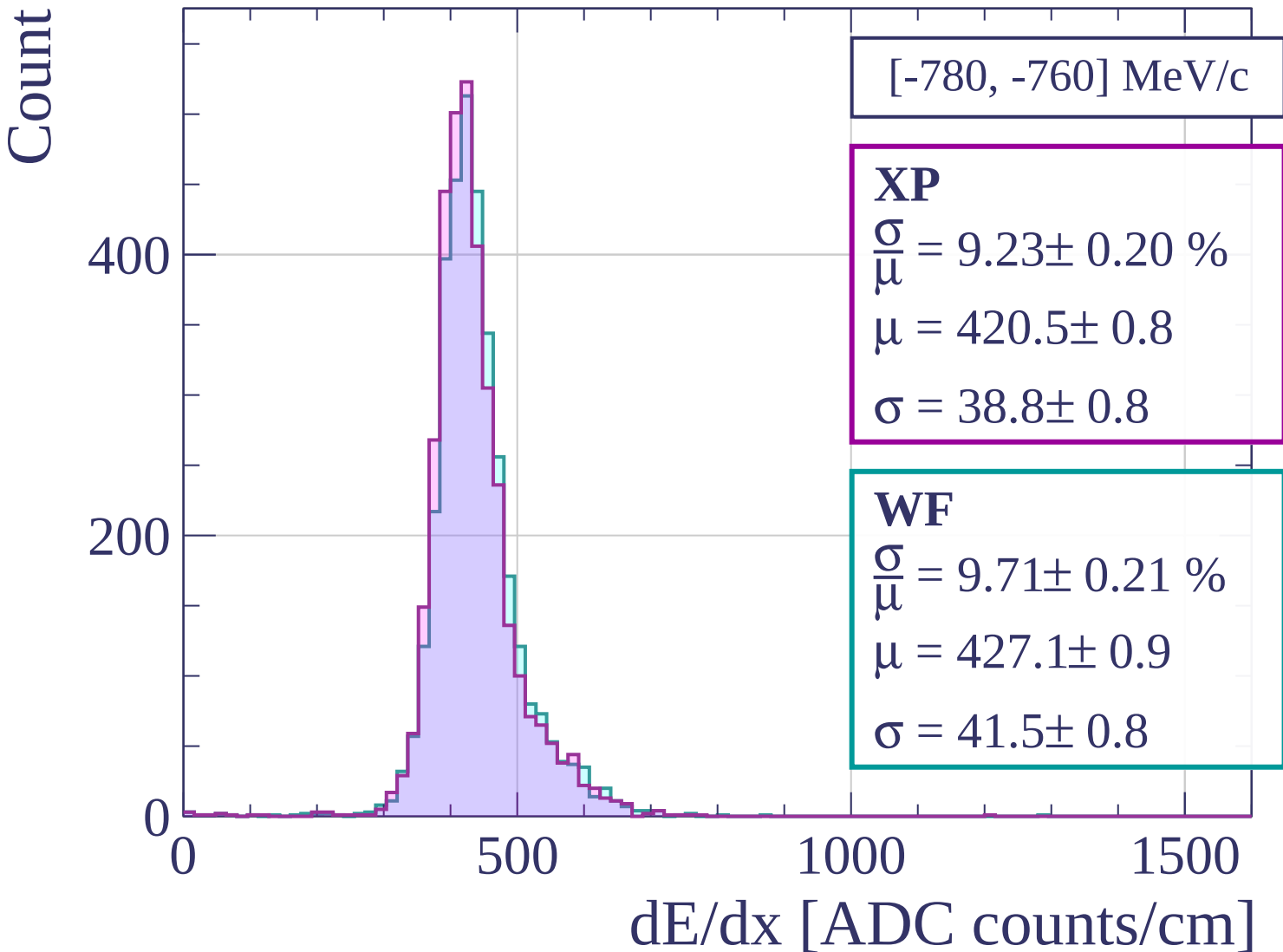
0

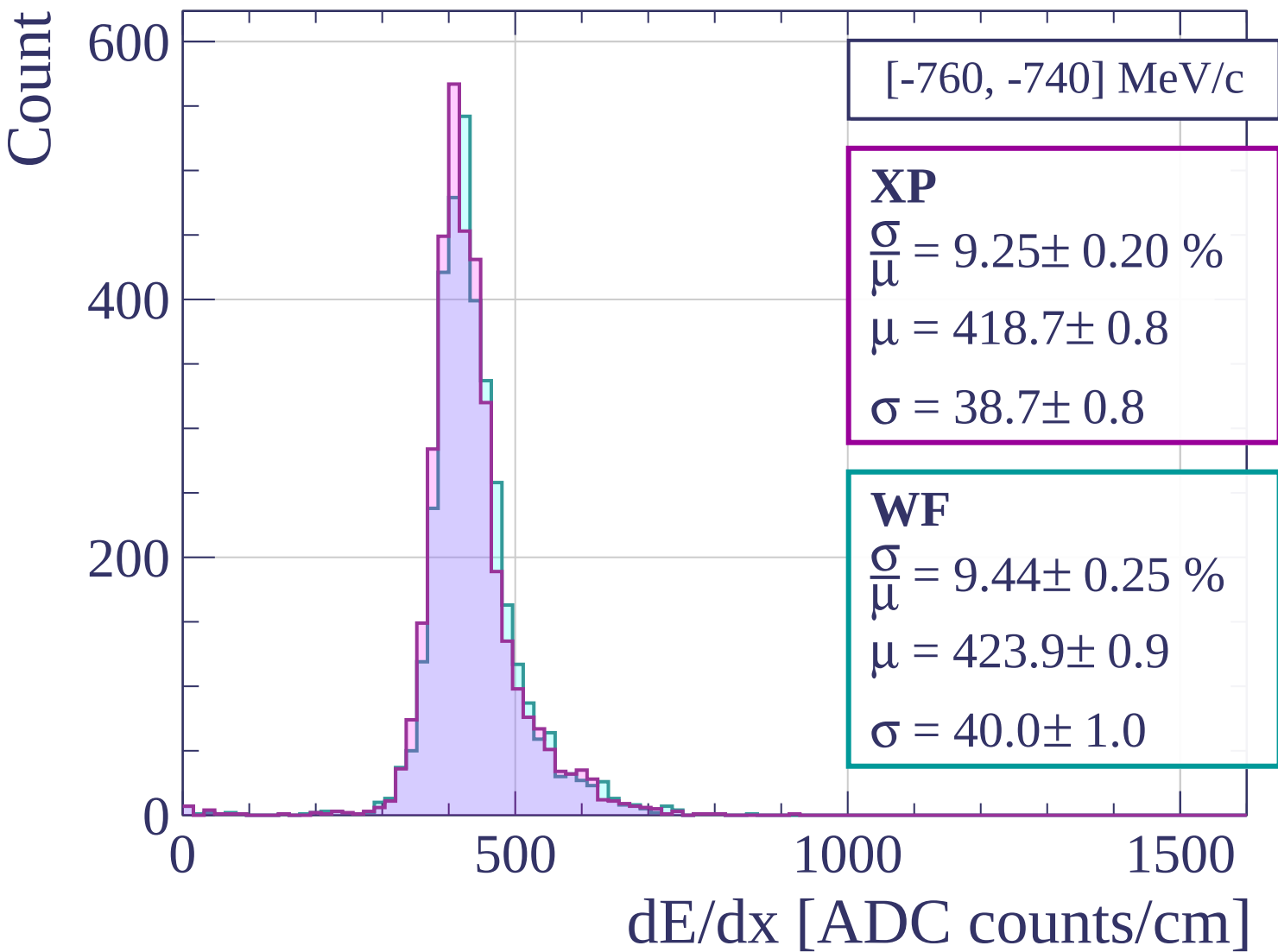
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-740, -720] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.29 \pm 0.20 \%$$

$$\mu = 418.0 \pm 0.8$$

$$\sigma = 38.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.77 \pm 0.20 \%$$

$$\mu = 425.3 \pm 0.8$$

$$\sigma = 41.6 \pm 0.8$$

400

200

0

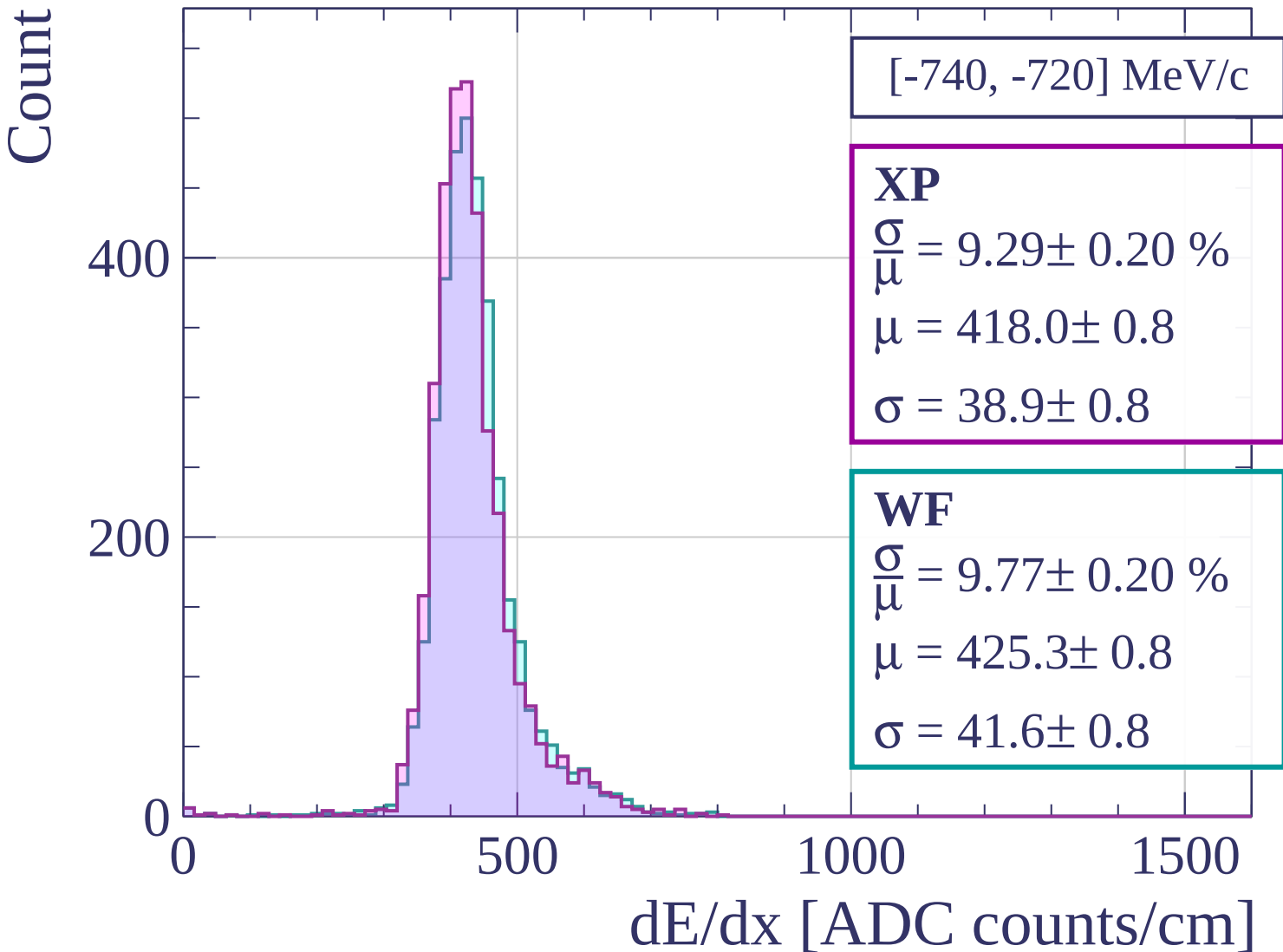
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-720, -700] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.55 \pm 0.22 \%$$

$$\mu = 415.9 \pm 0.8$$

$$\sigma = 39.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.61 \pm 0.22 \%$$

$$\mu = 421.5 \pm 0.9$$

$$\sigma = 40.5 \pm 0.9$$

400

200

0

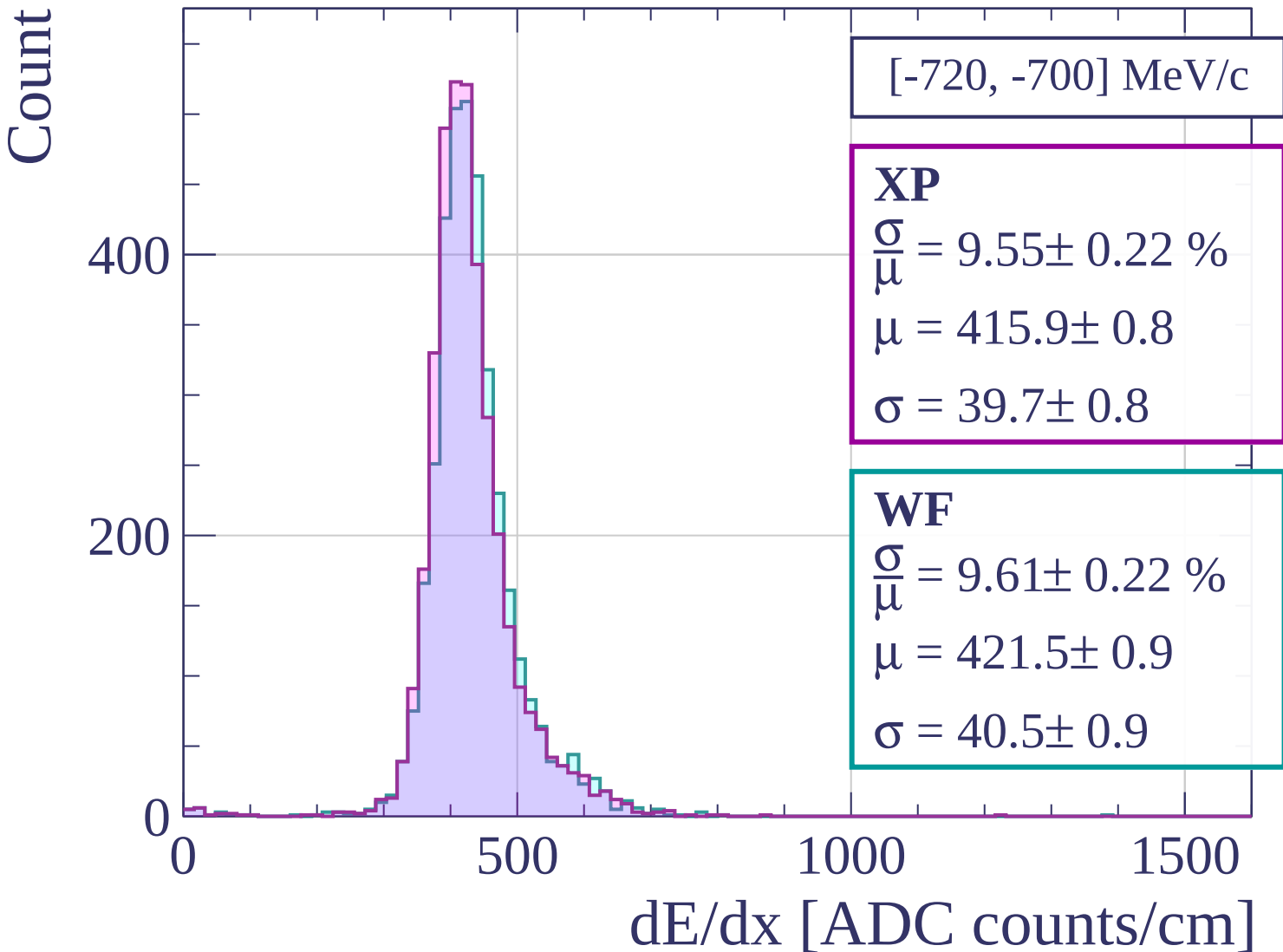
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-700, -680] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.12 \pm 0.19 \%$$

$$\mu = 416.4 \pm 0.8$$

$$\sigma = 38.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.46 \pm 0.21 \%$$

$$\mu = 422.3 \pm 0.9$$

$$\sigma = 40.0 \pm 0.8$$

400

200

0

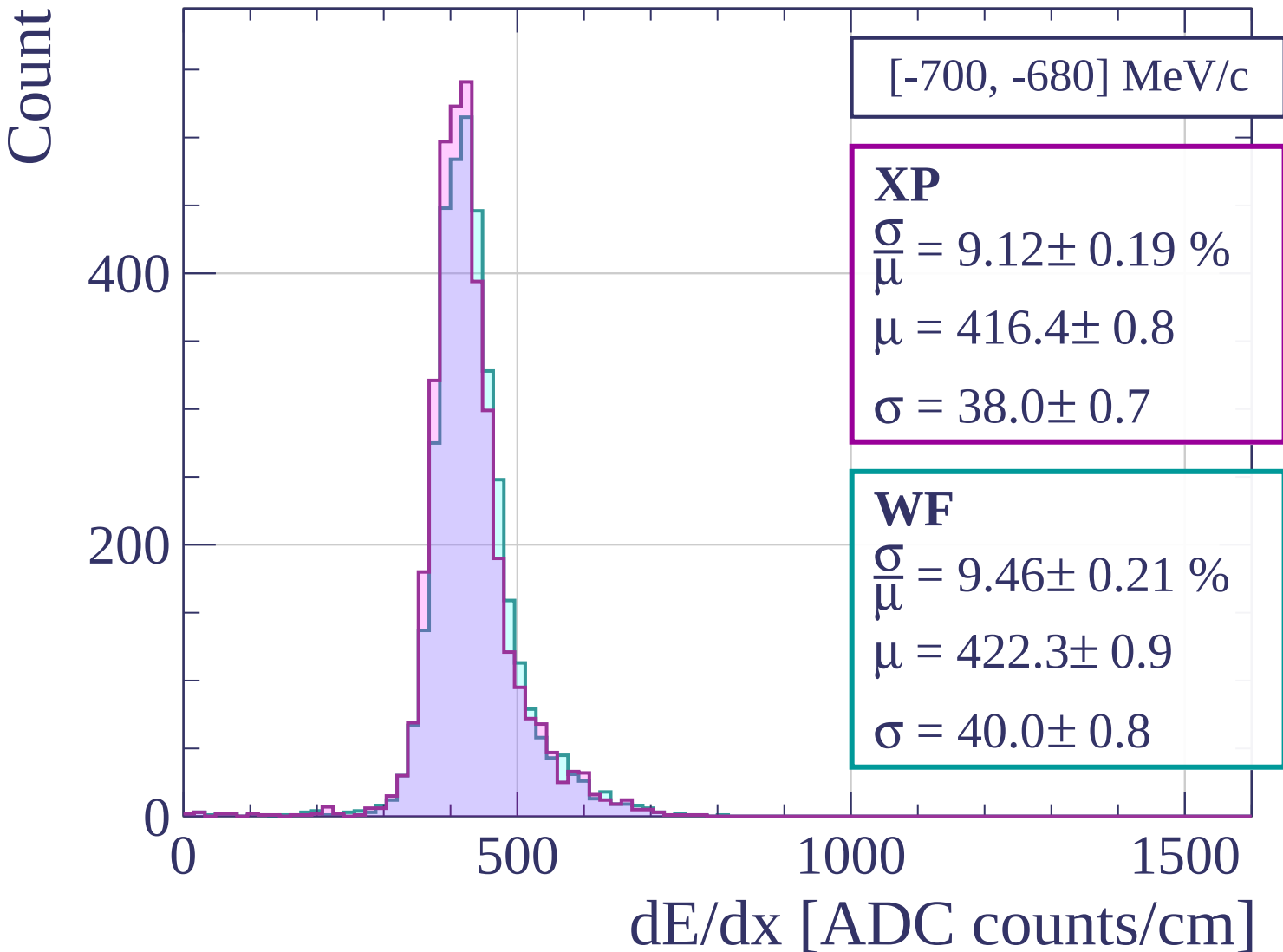
0

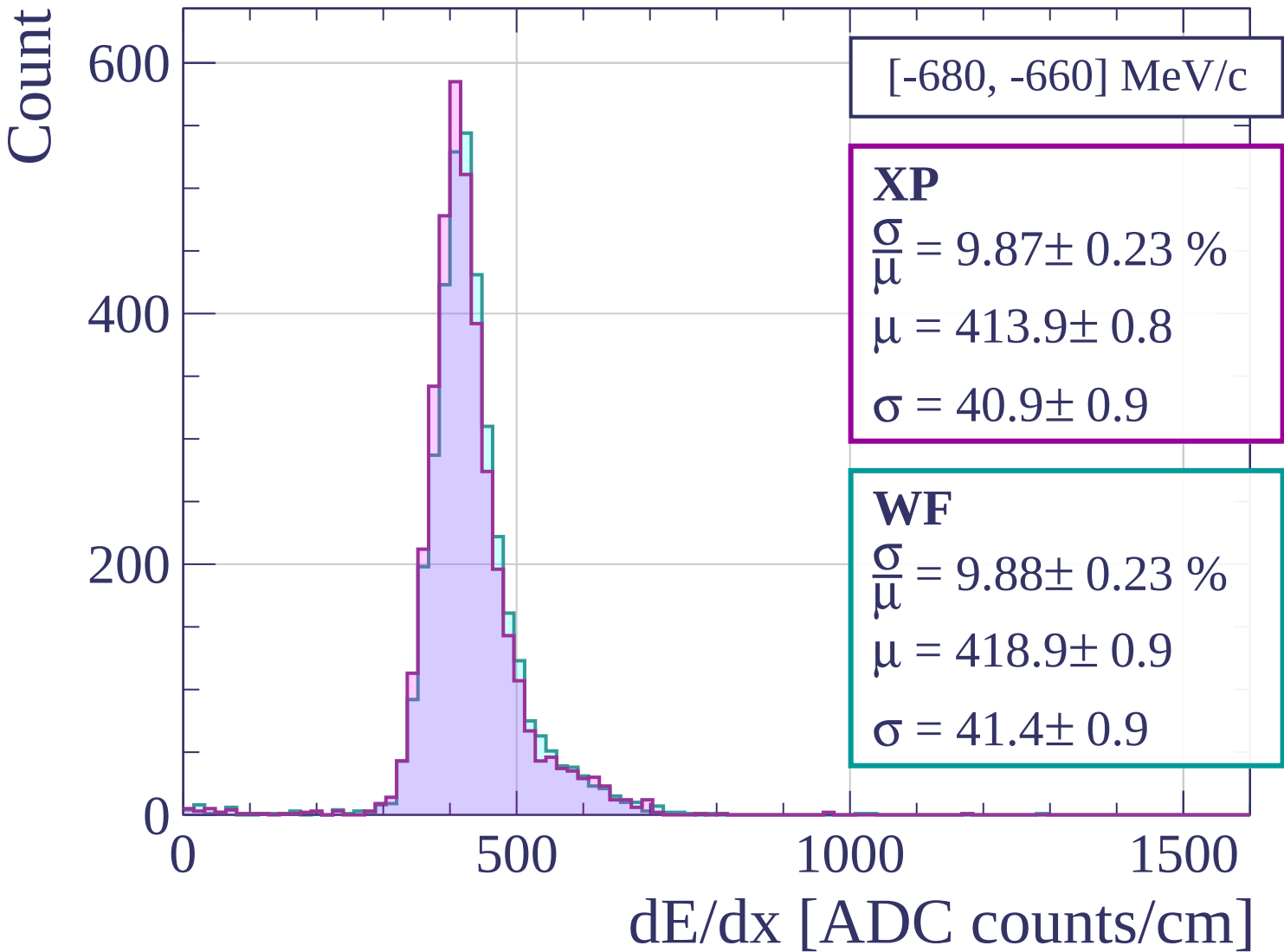
500

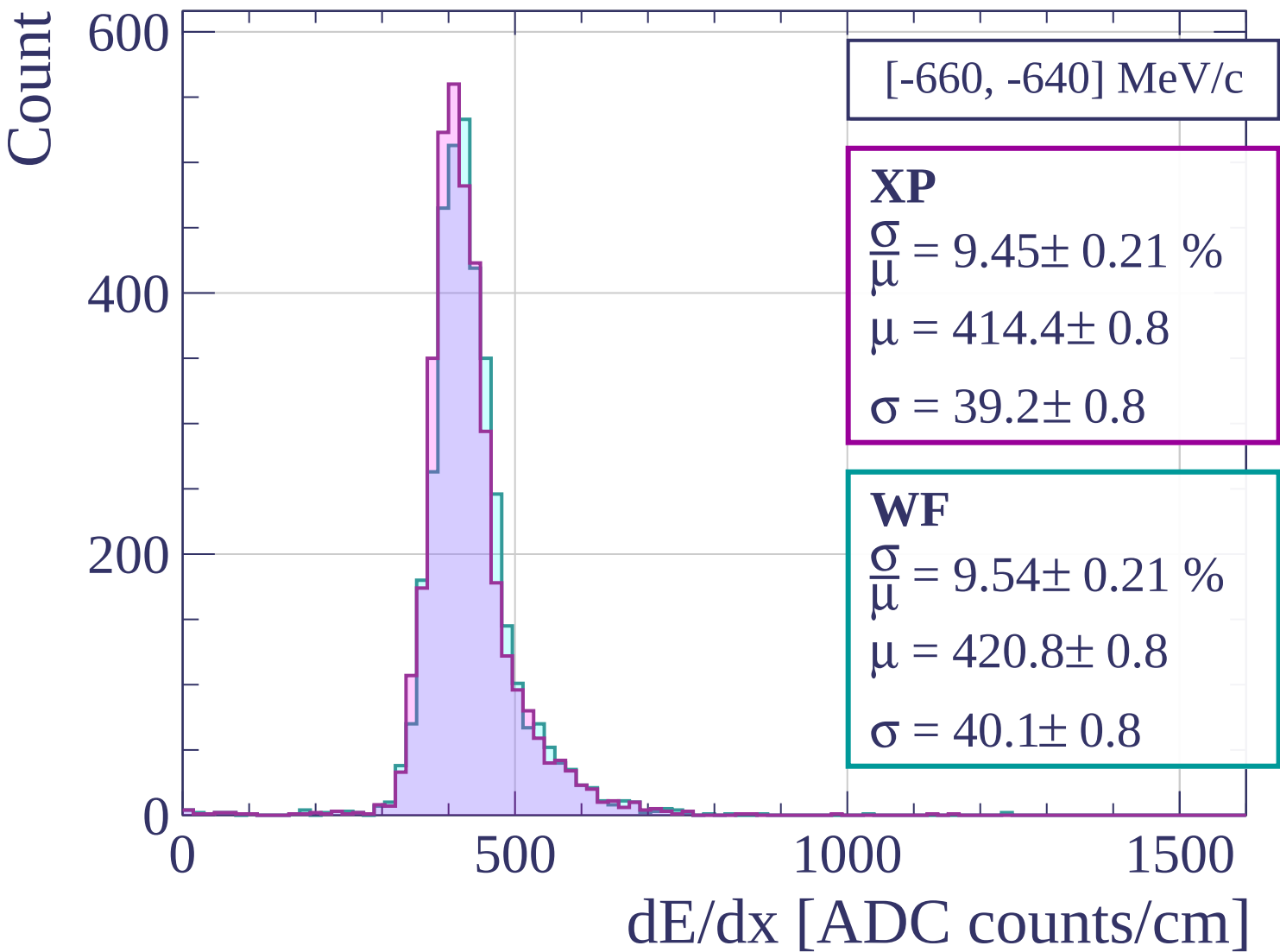
1000

1500

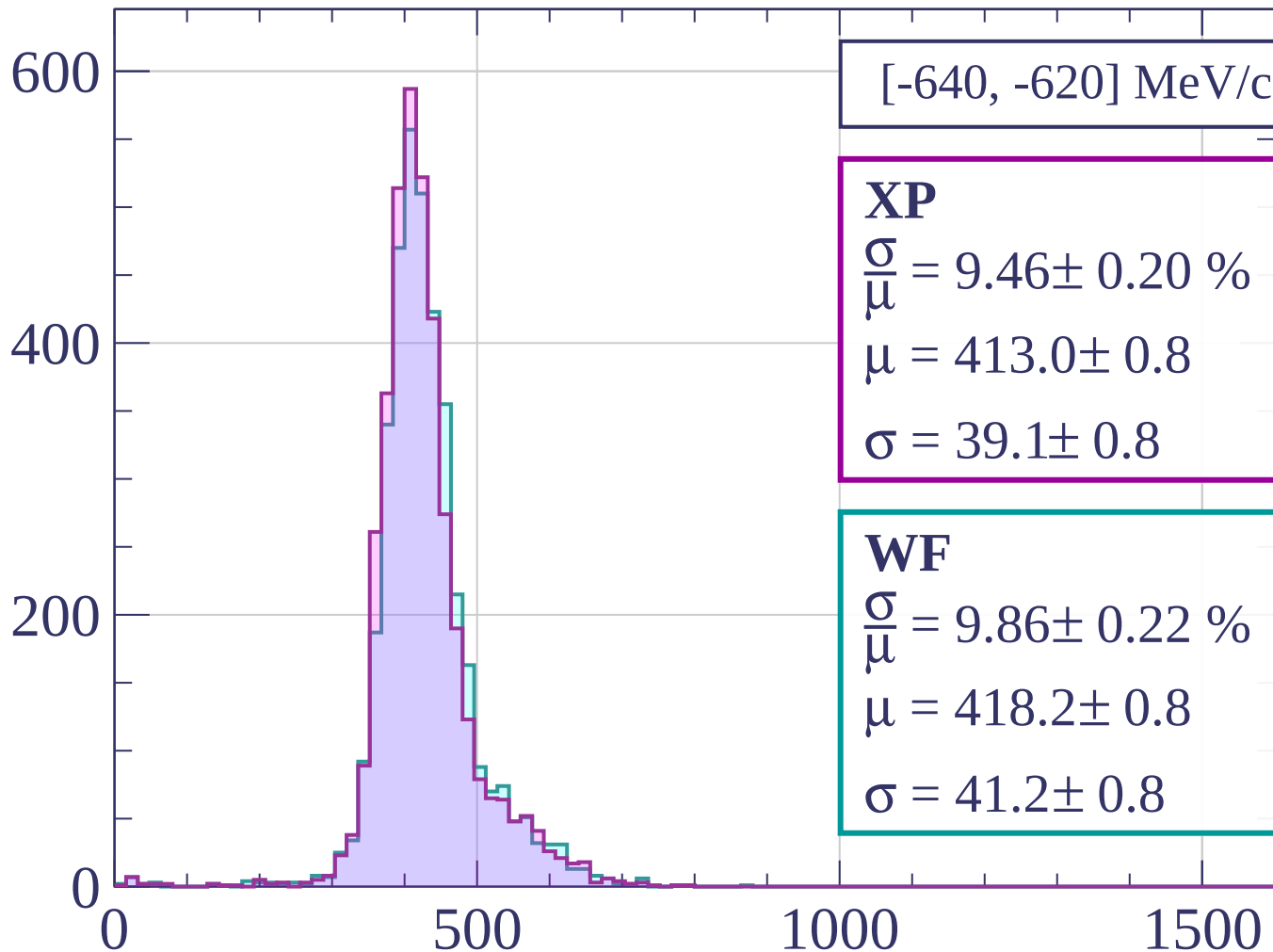
dE/dx [ADC counts/cm]







Count



[-640, -620] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.46 \pm 0.20 \%$$

$$\mu = 413.0 \pm 0.8$$

$$\sigma = 39.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.86 \pm 0.22 \%$$

$$\mu = 418.2 \pm 0.8$$

$$\sigma = 41.2 \pm 0.8$$

dE/dx [ADC counts/cm]

Count

[-620, -600] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.91 \pm 0.20 \%$$

$$\mu = 409.2 \pm 0.7$$

$$\sigma = 36.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.64 \pm 0.21 \%$$

$$\mu = 415.4 \pm 0.8$$

$$\sigma = 40.0 \pm 0.8$$

600

400

200

0

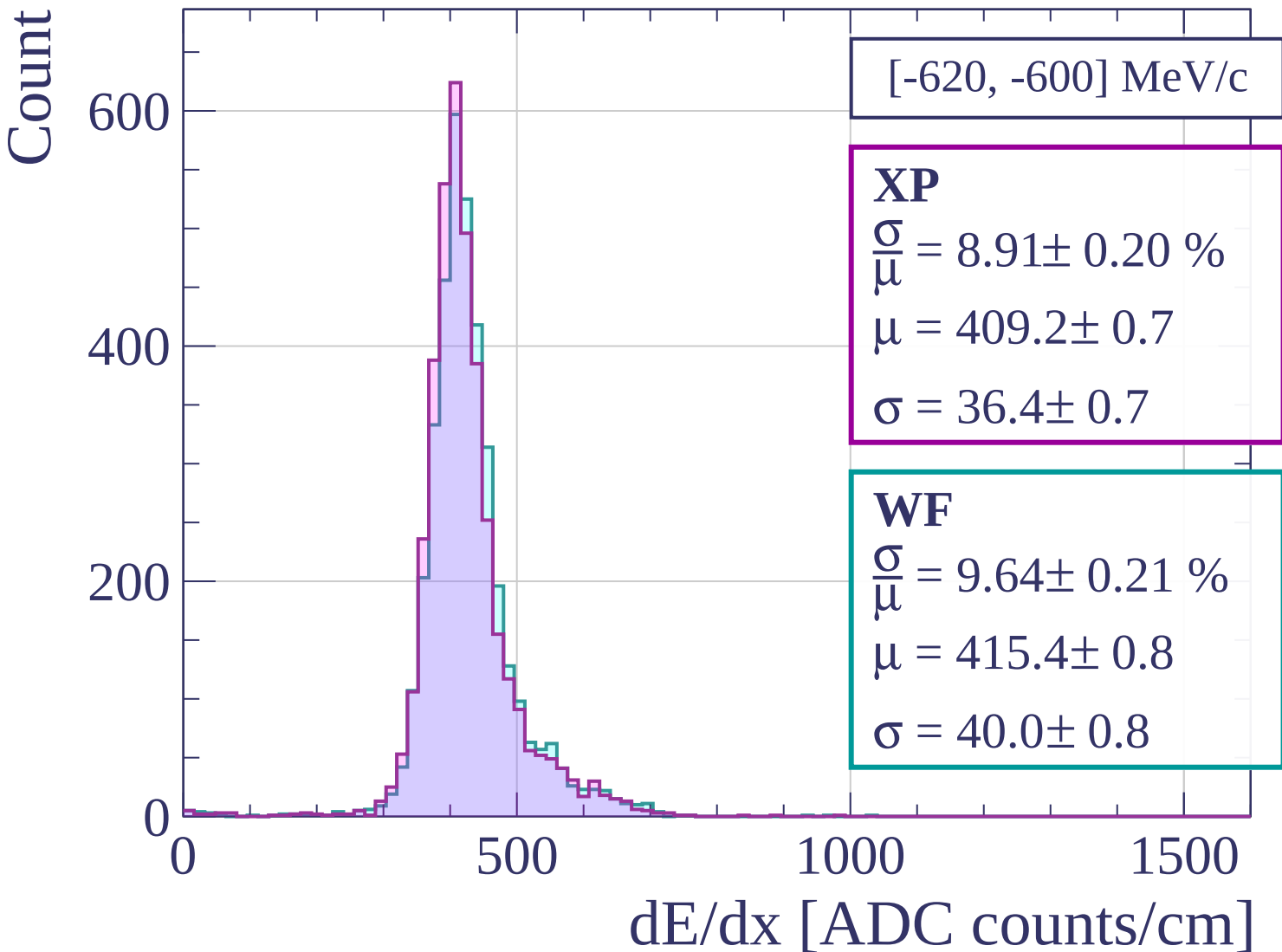
0

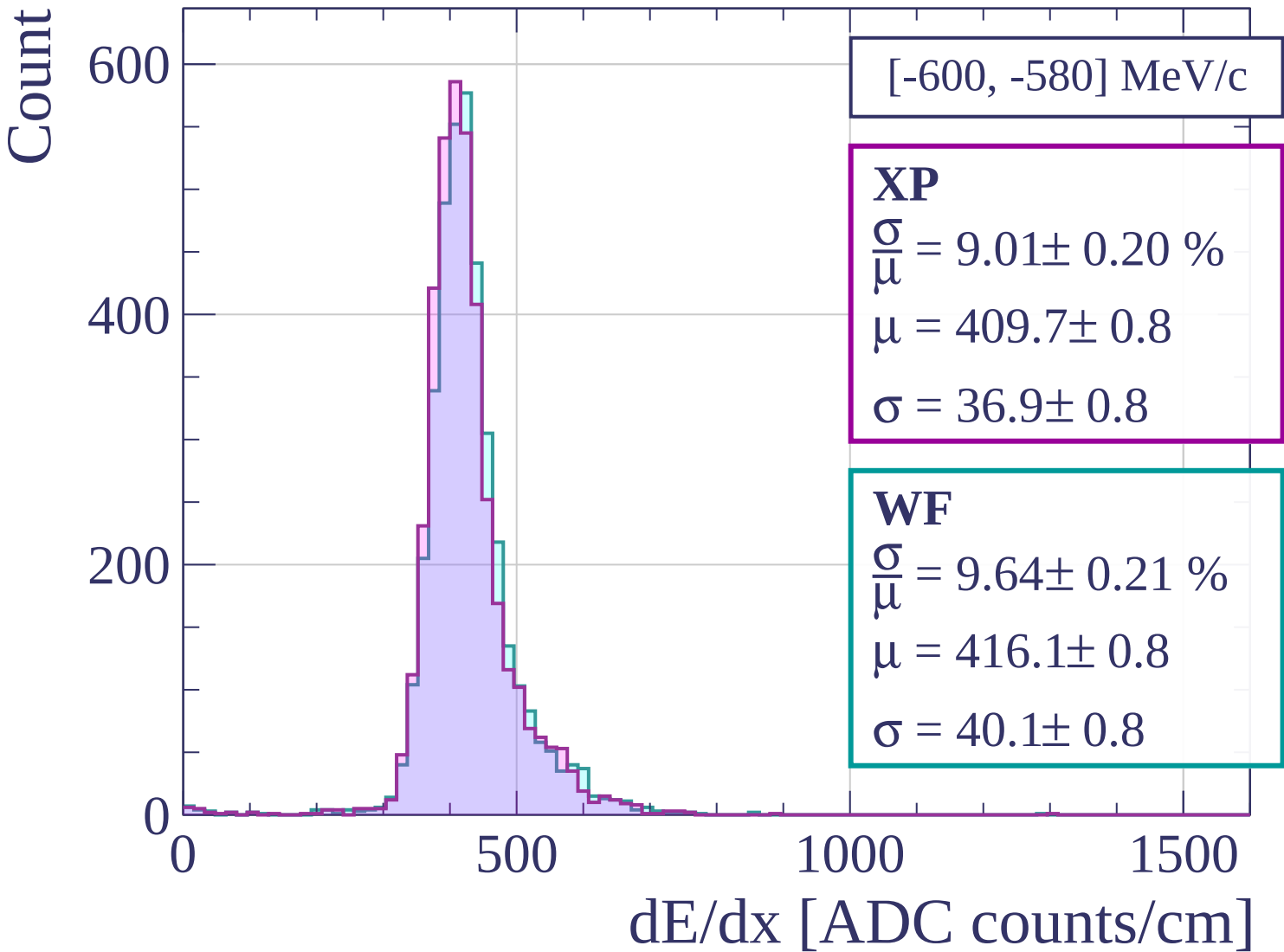
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-580, -560] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.29 \pm 0.21 \%$$

$$\mu = 407.8 \pm 0.8$$

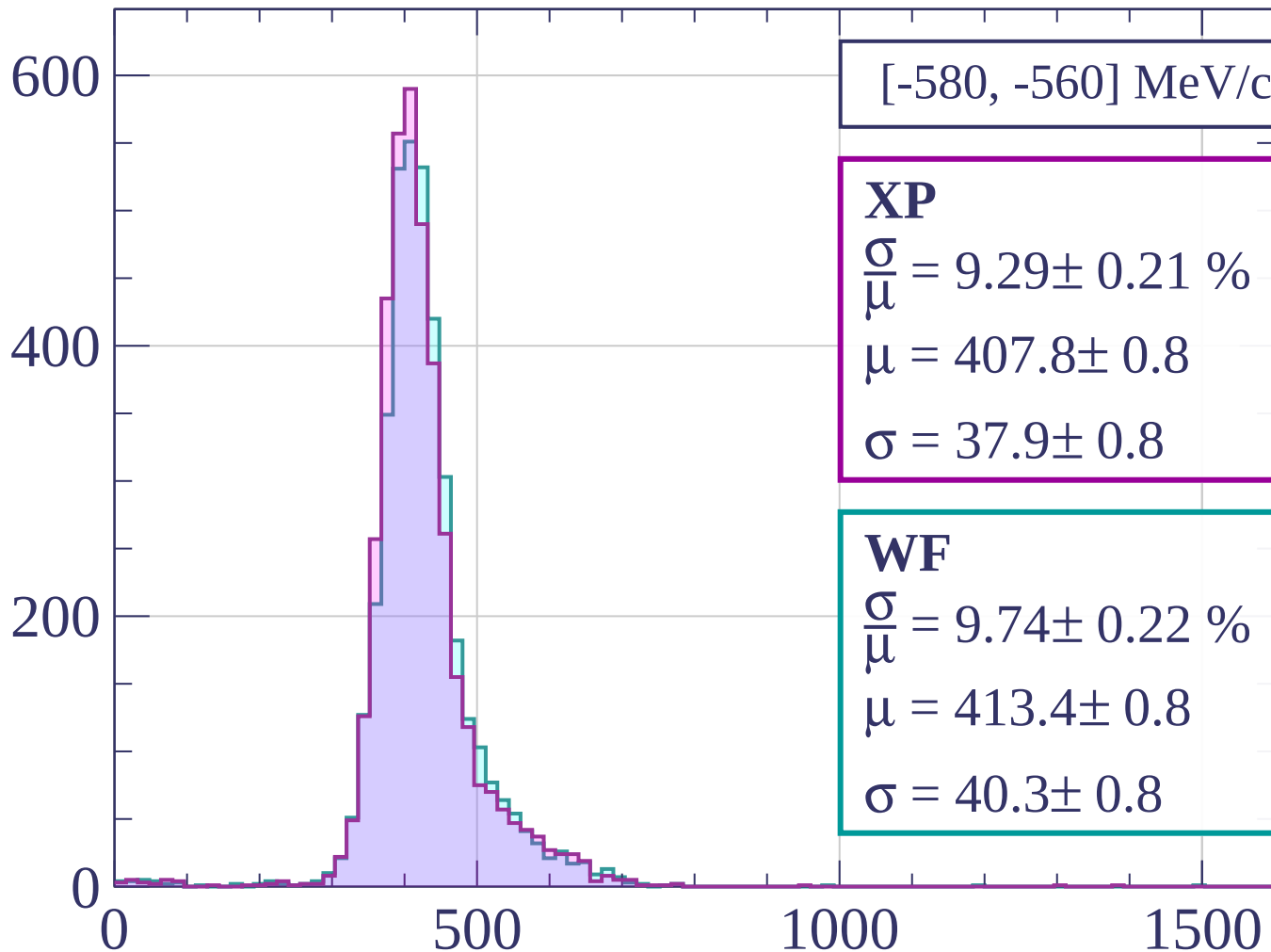
$$\sigma = 37.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.74 \pm 0.22 \%$$

$$\mu = 413.4 \pm 0.8$$

$$\sigma = 40.3 \pm 0.8$$



dE/dx [ADC counts/cm]

Count

[-560, -540] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.26 \pm 0.22 \%$$

$$\mu = 406.3 \pm 0.8$$

$$\sigma = 37.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.79 \pm 0.22 \%$$

$$\mu = 412.7 \pm 0.8$$

$$\sigma = 40.4 \pm 0.8$$

600

400

200

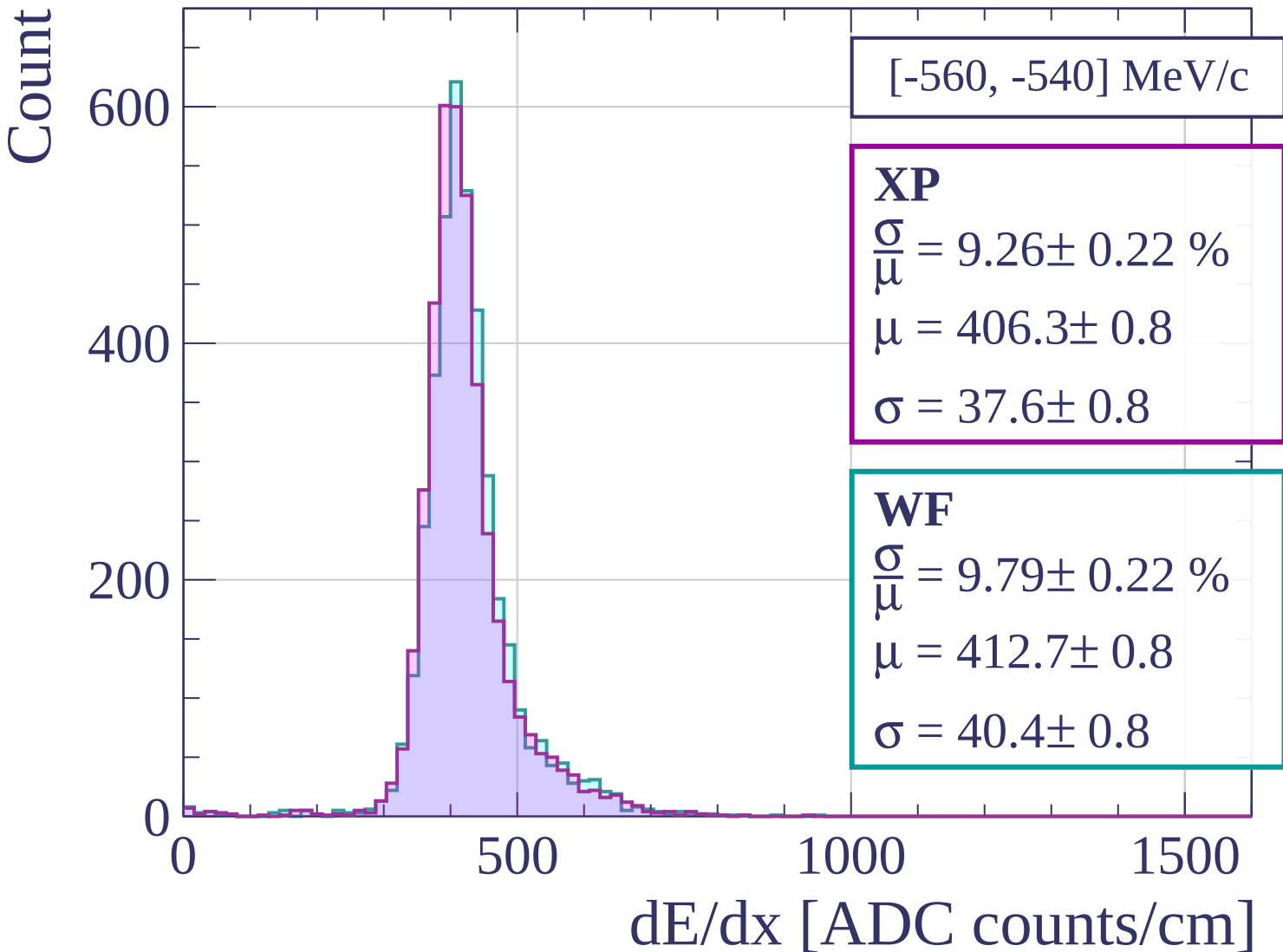
0

500

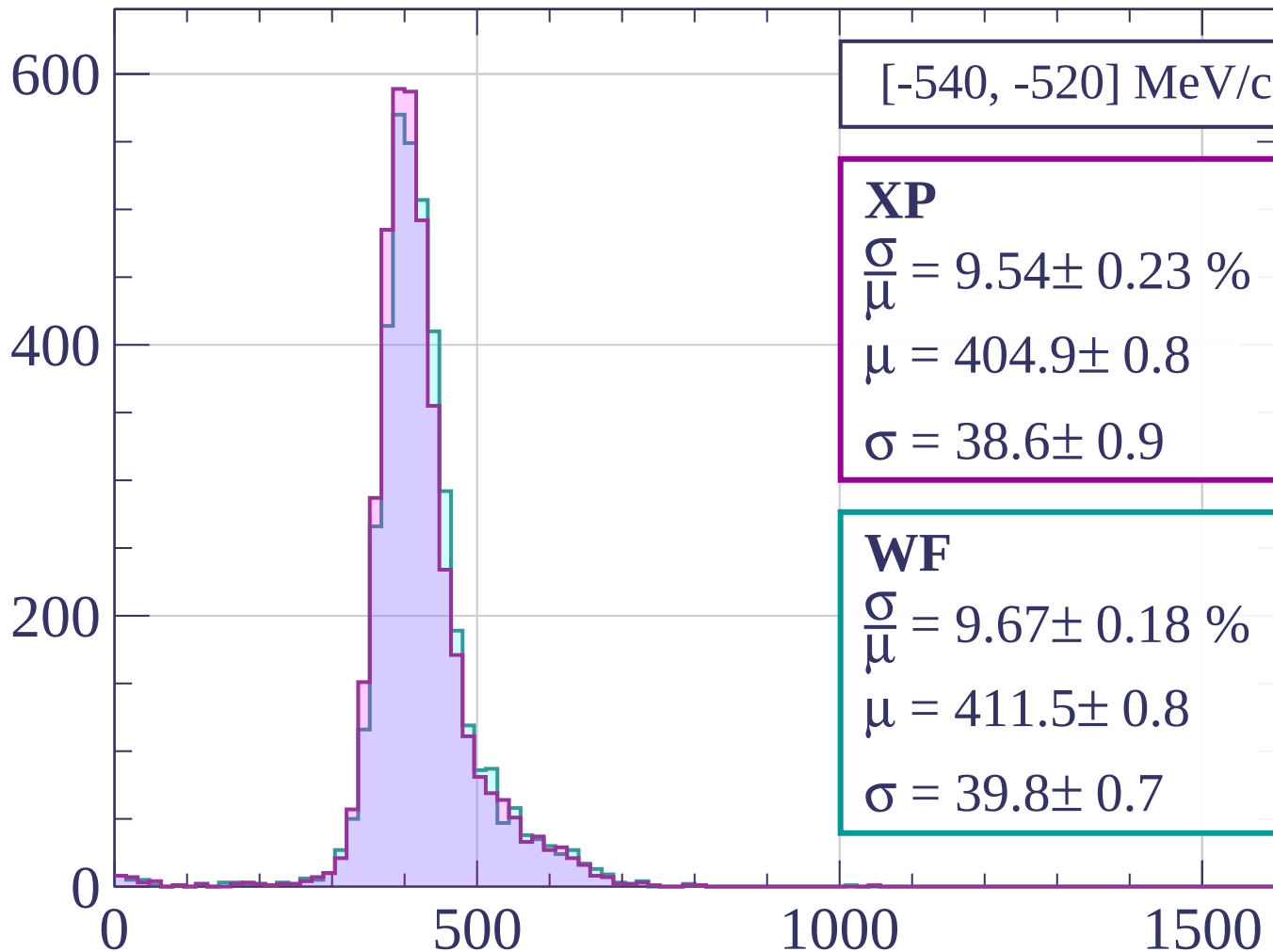
1000

1500

dE/dx [ADC counts/cm]



Count



$[-540, -520] \text{ MeV/c}$

XP

$$\frac{\sigma}{\mu} = 9.54 \pm 0.23 \%$$

$$\mu = 404.9 \pm 0.8$$

$$\sigma = 38.6 \pm 0.9$$

WF

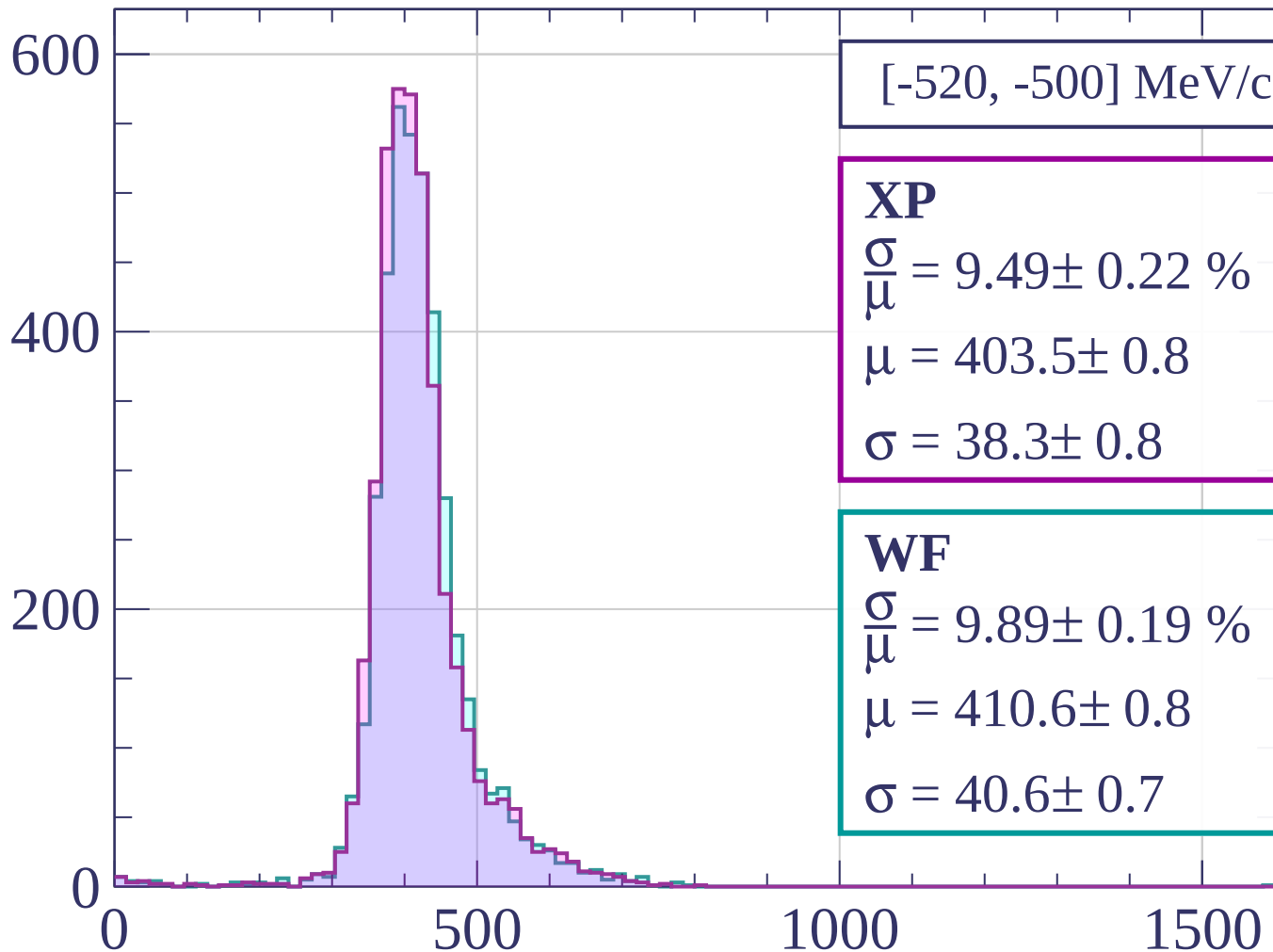
$$\frac{\sigma}{\mu} = 9.67 \pm 0.18 \%$$

$$\mu = 411.5 \pm 0.8$$

$$\sigma = 39.8 \pm 0.7$$

dE/dx [ADC counts/cm]

Count



[-520, -500] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.22 \%$$

$$\mu = 403.5 \pm 0.8$$

$$\sigma = 38.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.89 \pm 0.19 \%$$

$$\mu = 410.6 \pm 0.8$$

$$\sigma = 40.6 \pm 0.7$$

dE/dx [ADC counts/cm]

Count

[-500, -480] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.25 \pm 0.17 \%$$

$$\mu = 402.0 \pm 0.7$$

$$\sigma = 37.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.31 \pm 0.18 \%$$

$$\mu = 406.5 \pm 0.7$$

$$\sigma = 37.9 \pm 0.7$$

600

400

200

0

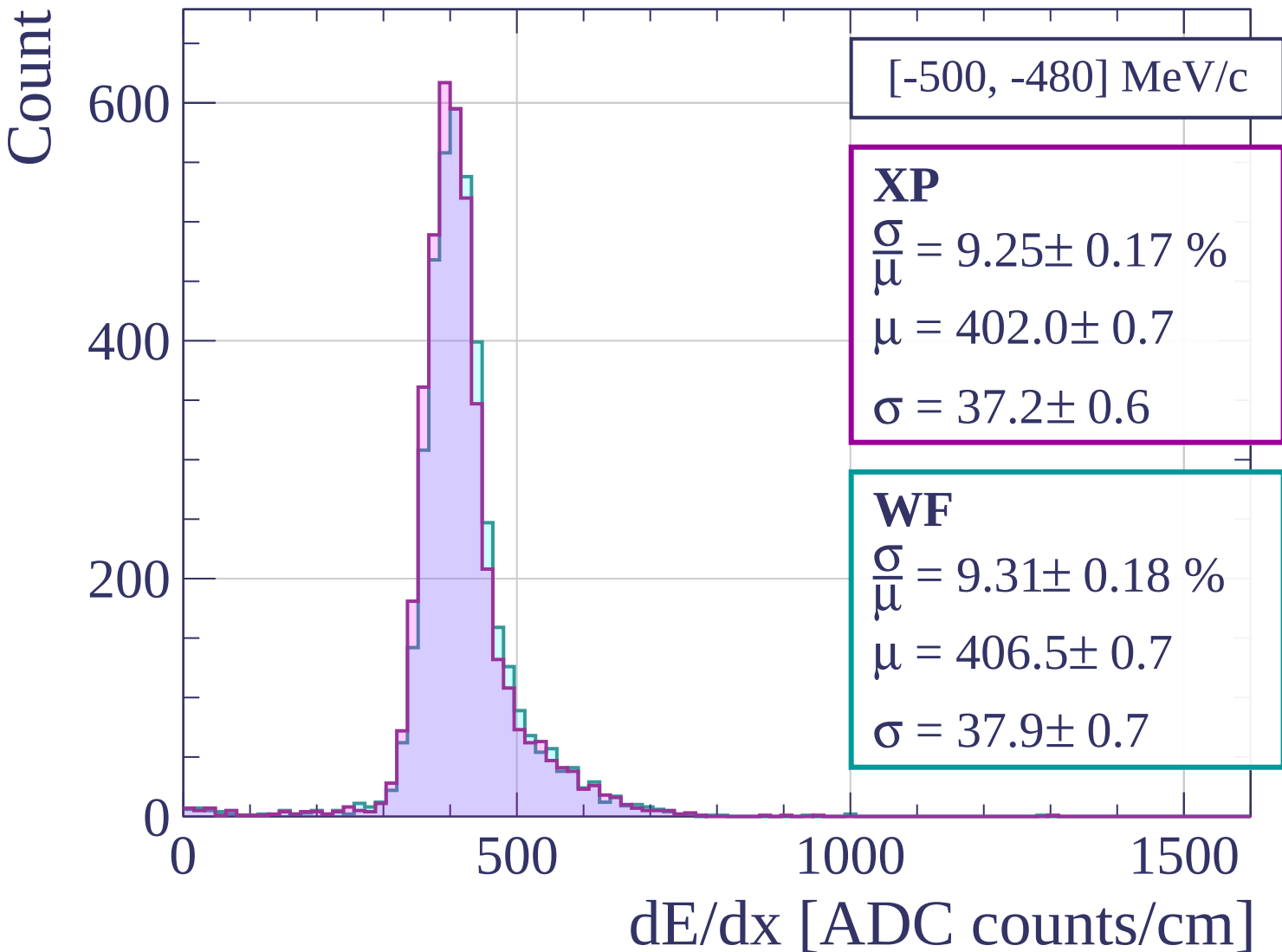
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-480, -460] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.42 \pm 0.18 \%$$

$$\mu = 402.4 \pm 0.7$$

$$\sigma = 37.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.74 \pm 0.24 \%$$

$$\mu = 406.5 \pm 0.8$$

$$\sigma = 39.6 \pm 0.9$$

600

400

200

0

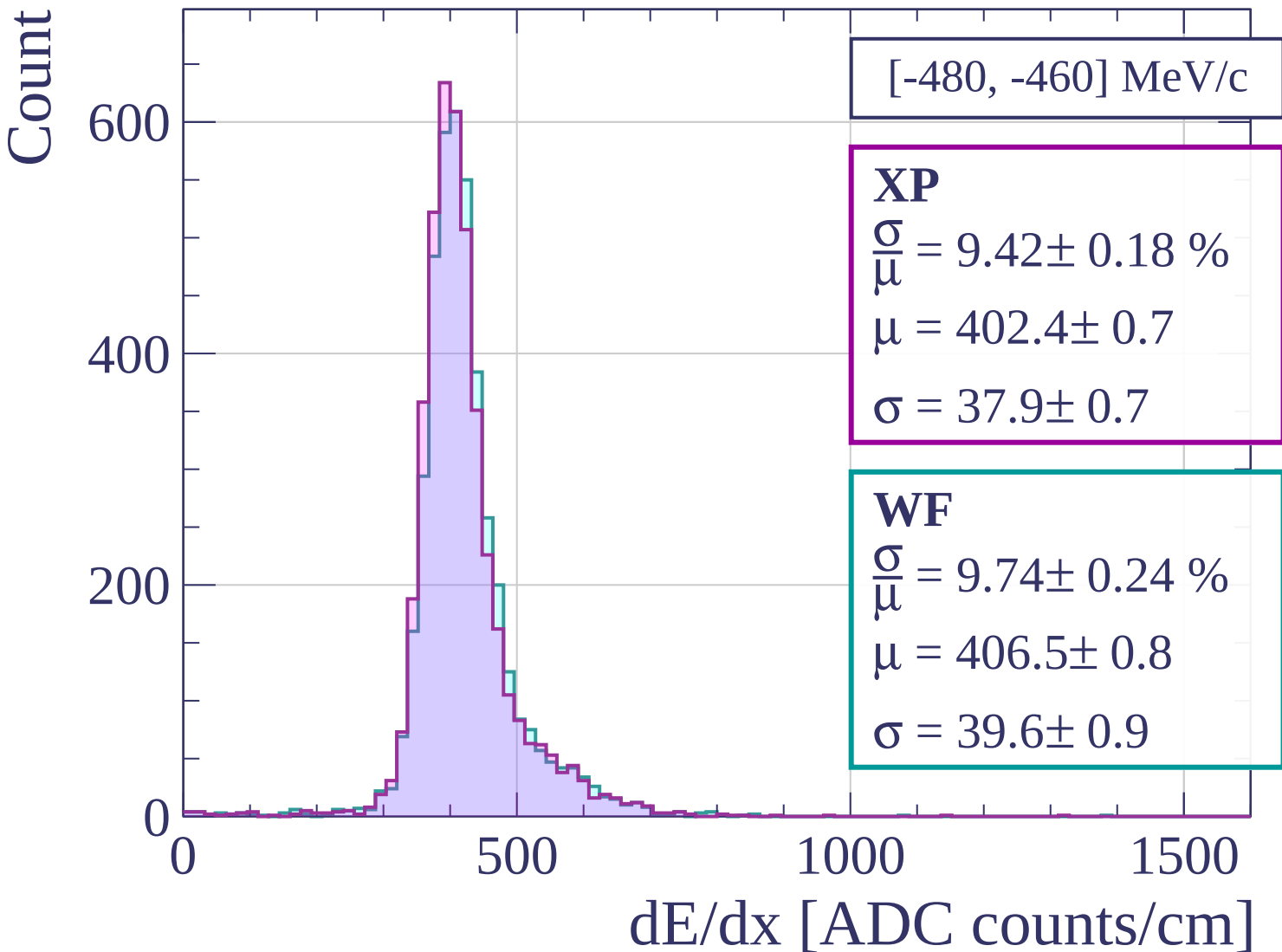
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-460, -440] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.41 \pm 0.19 \%$$

$$\mu = 401.6 \pm 0.7$$

$$\sigma = 37.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.55 \pm 0.20 \%$$

$$\mu = 406.1 \pm 0.8$$

$$\sigma = 38.8 \pm 0.7$$

600

400

200

0

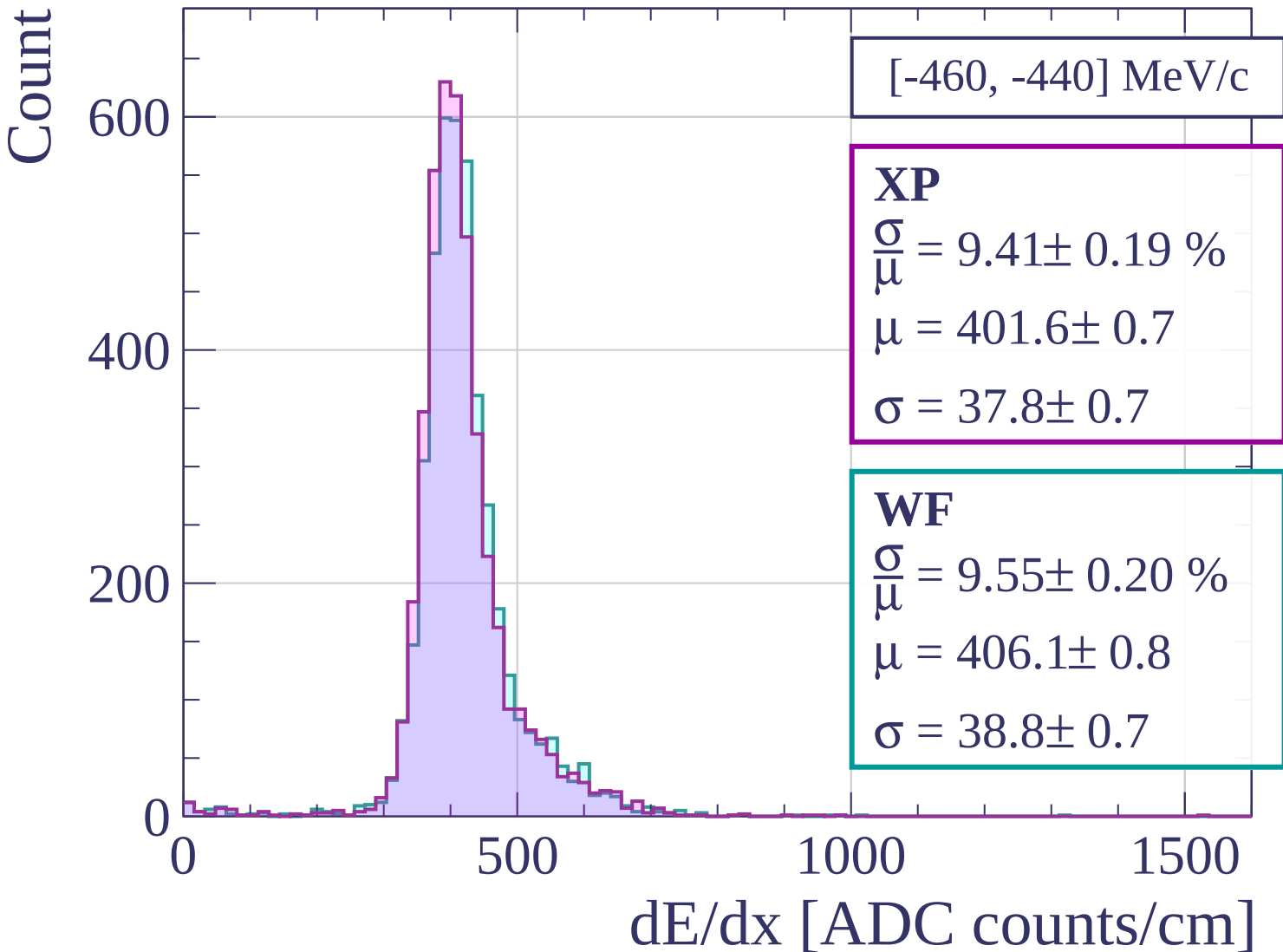
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-440, -420] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.19 \%$$

$$\mu = 400.3 \pm 0.7$$

$$\sigma = 38.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.66 \pm 0.19 \%$$

$$\mu = 404.9 \pm 0.8$$

$$\sigma = 39.1 \pm 0.7$$

600

400

200

0

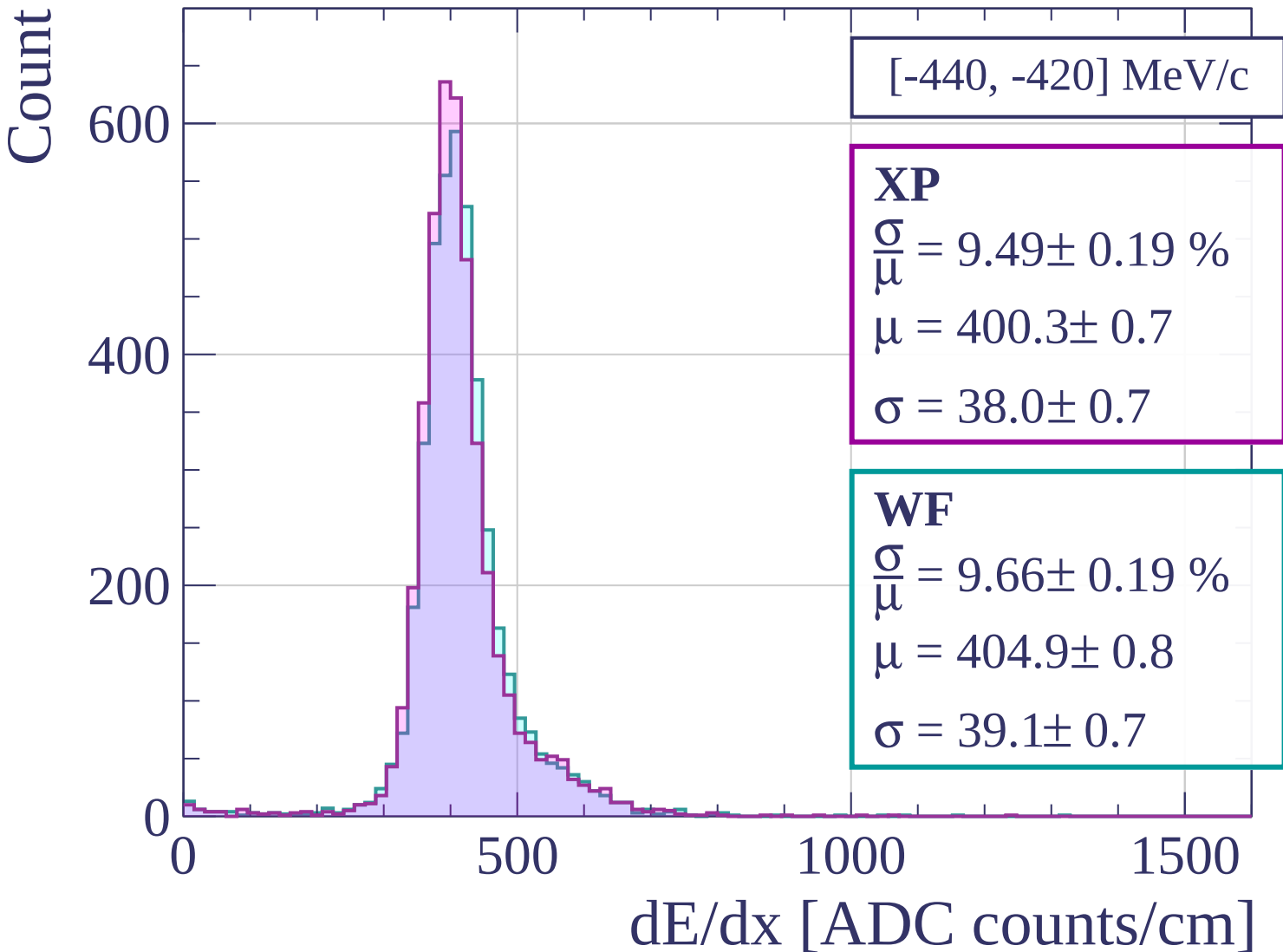
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-420, -400] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.29 \pm 0.18 \%$$

$$\mu = 401.0 \pm 0.7$$

$$\sigma = 37.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.56 \pm 0.21 \%$$

$$\mu = 404.2 \pm 0.8$$

$$\sigma = 38.6 \pm 0.8$$

600

400

200

0

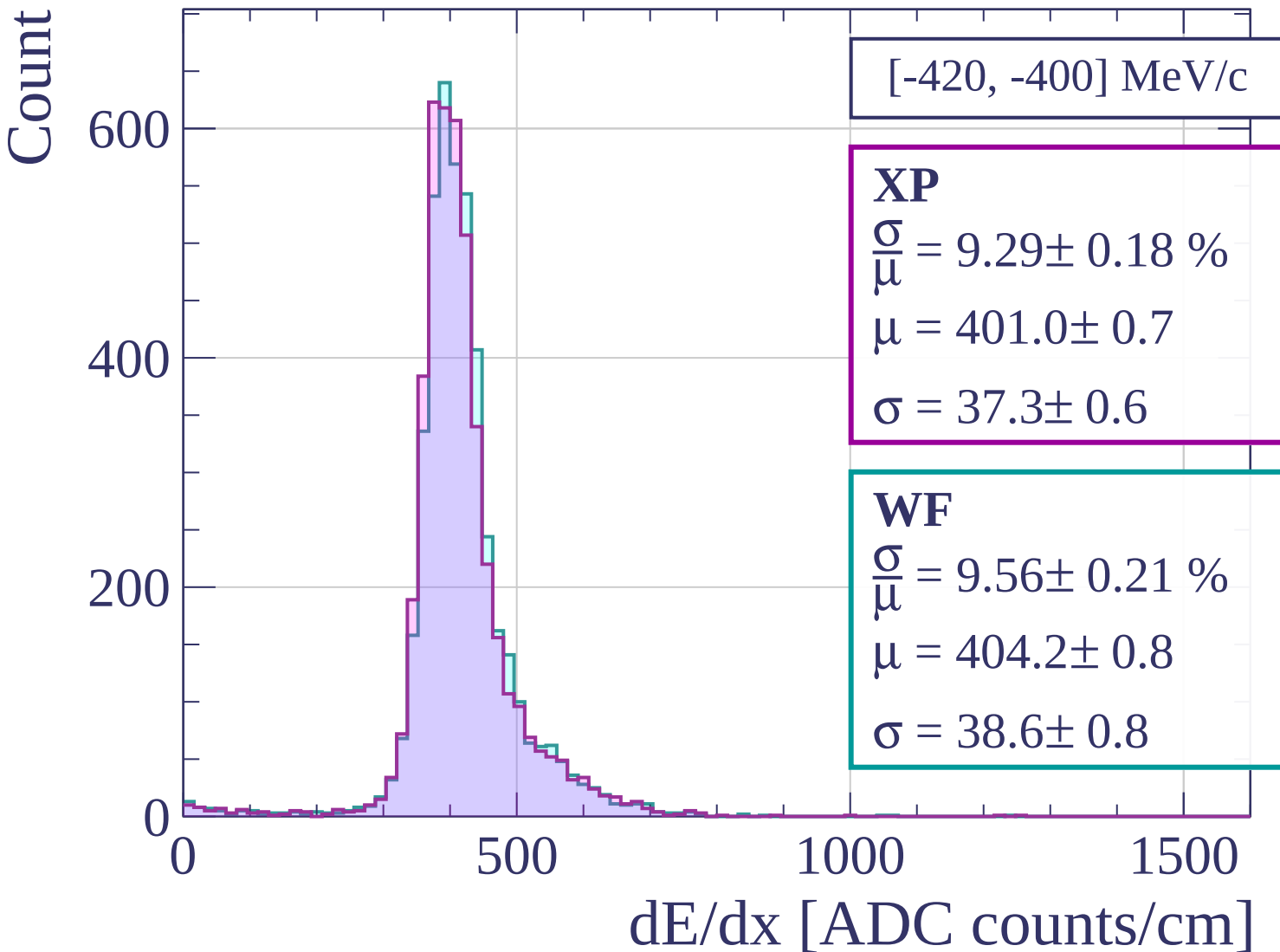
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-400, -380] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.03 \pm 0.18 \%$$

$$\mu = 398.1 \pm 0.7$$

$$\sigma = 35.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.72 \pm 0.20 \%$$

$$\mu = 403.1 \pm 0.7$$

$$\sigma = 39.2 \pm 0.7$$

600

400

200

0

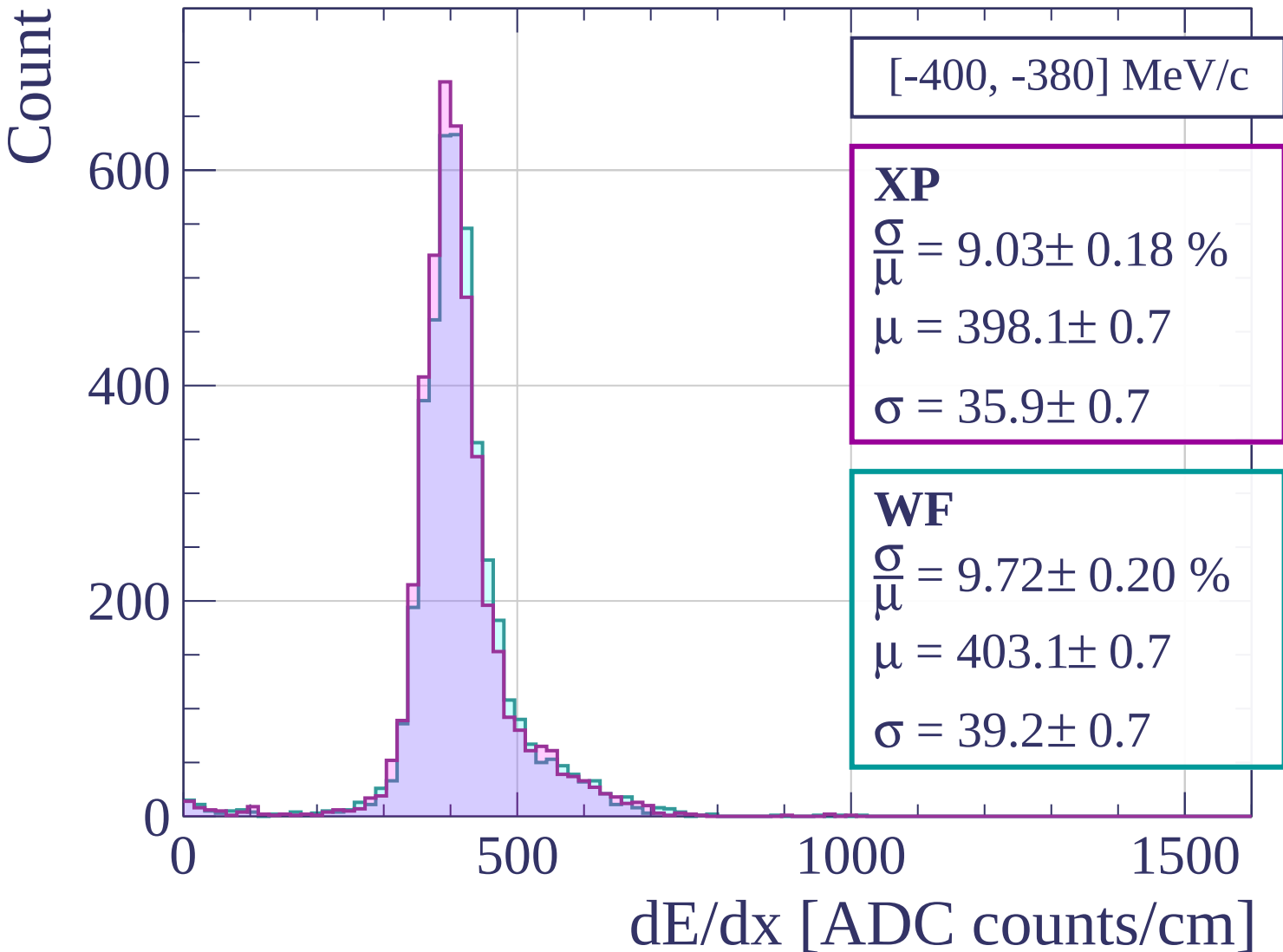
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-380, -360] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.44 \pm 0.19 \%$$

$$\mu = 399.2 \pm 0.7$$

$$\sigma = 37.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.60 \pm 0.20 \%$$

$$\mu = 403.0 \pm 0.7$$

$$\sigma = 38.7 \pm 0.7$$

600

400

200

0

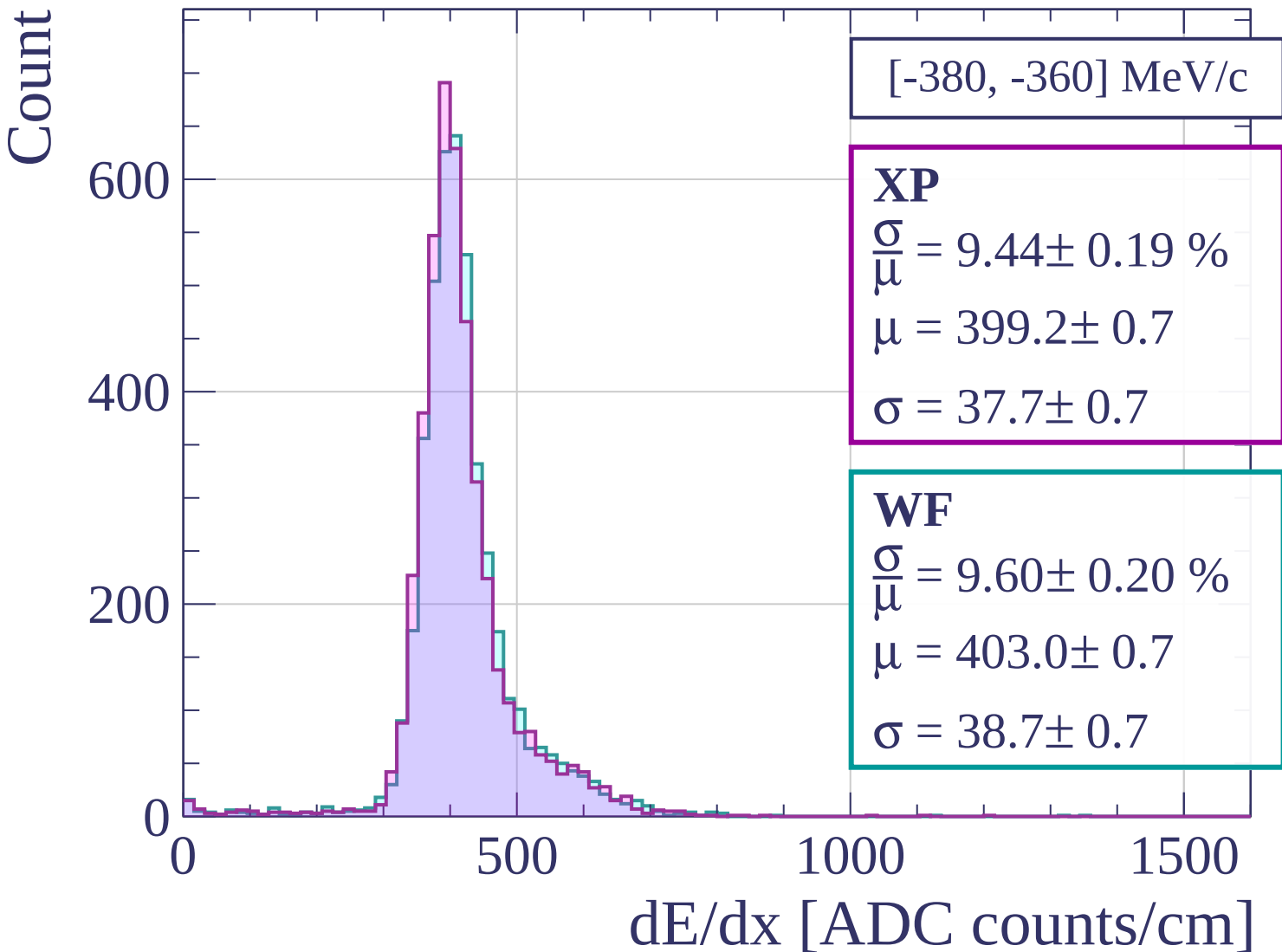
0

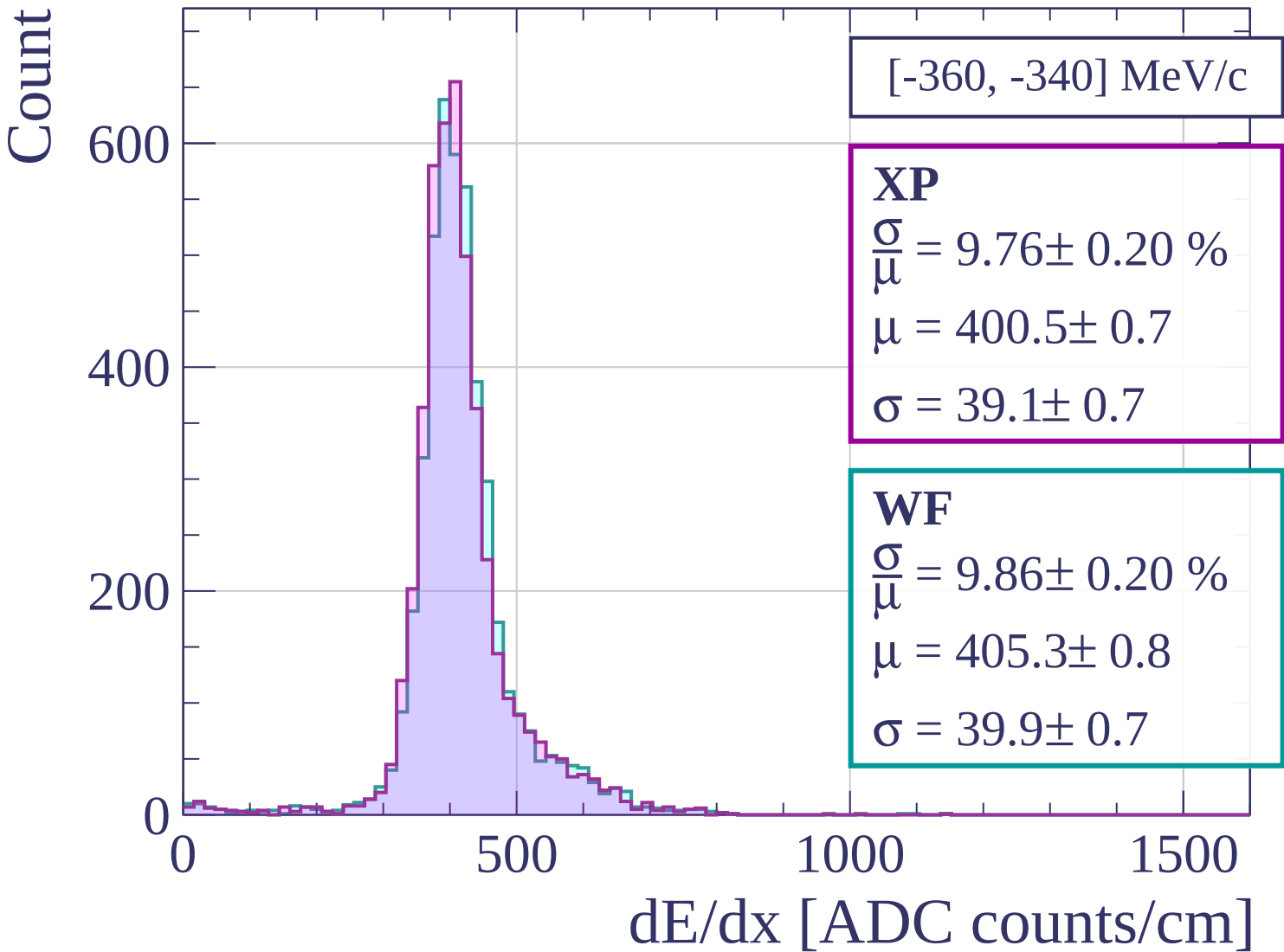
500

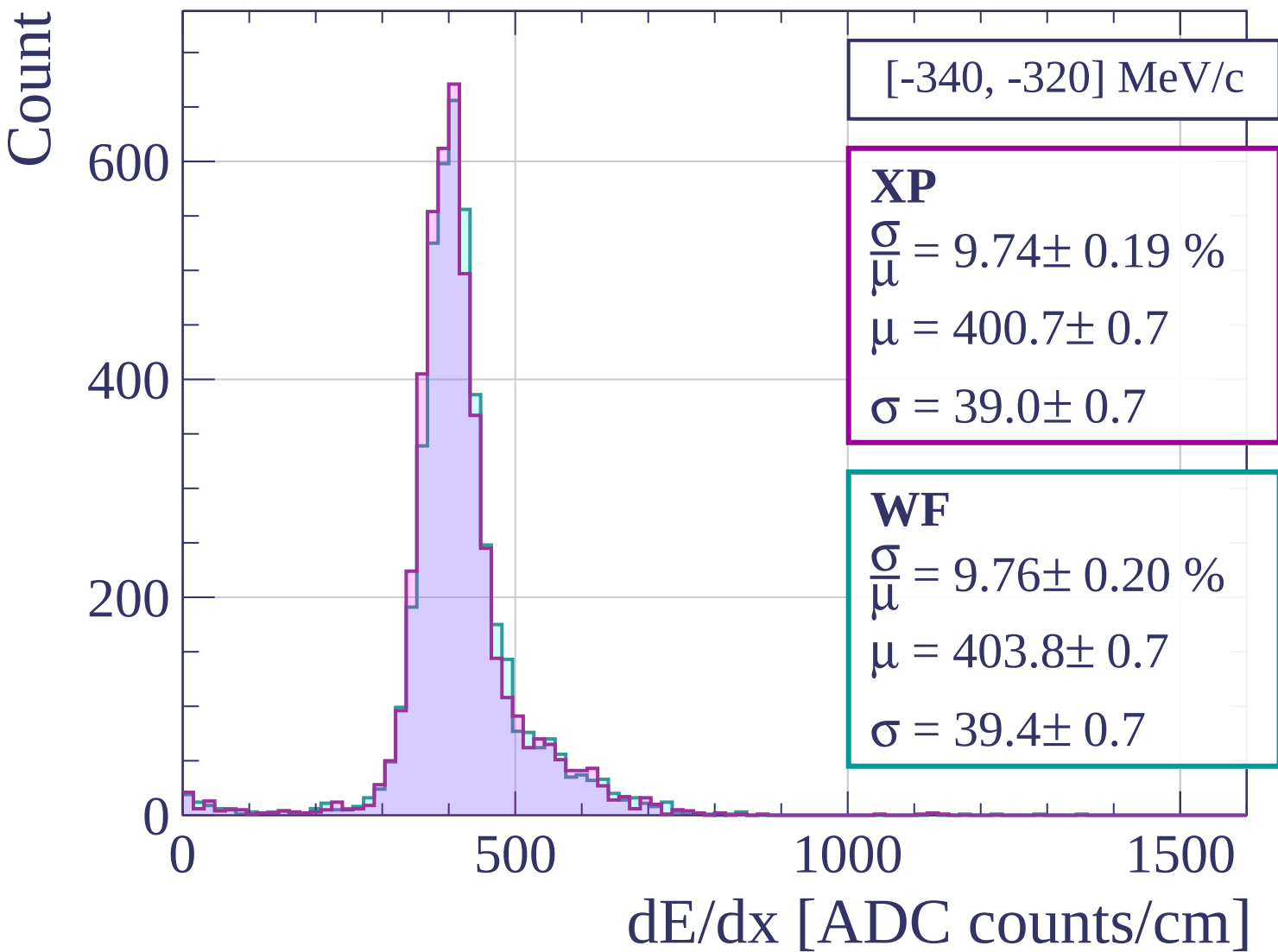
1000

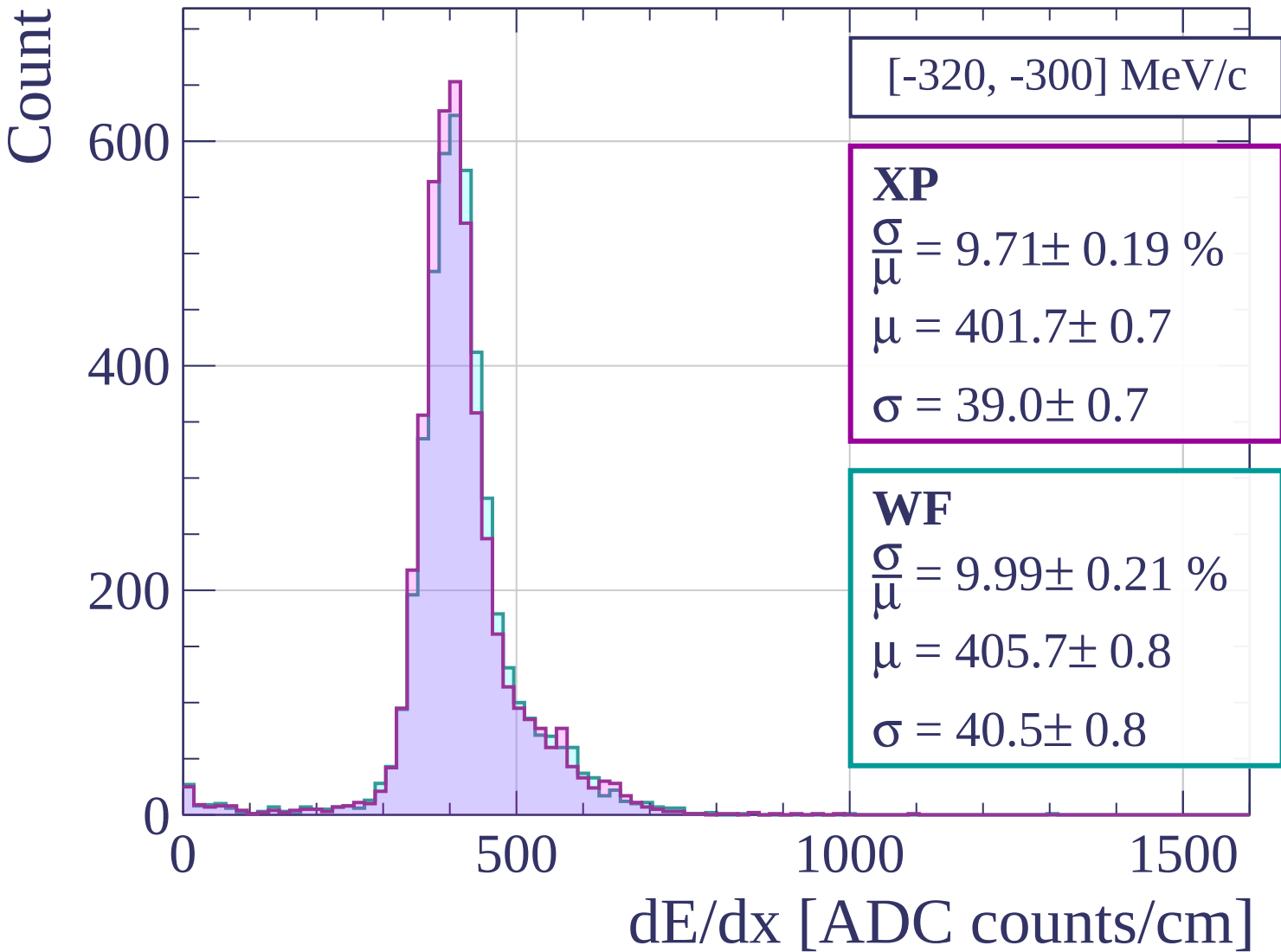
1500

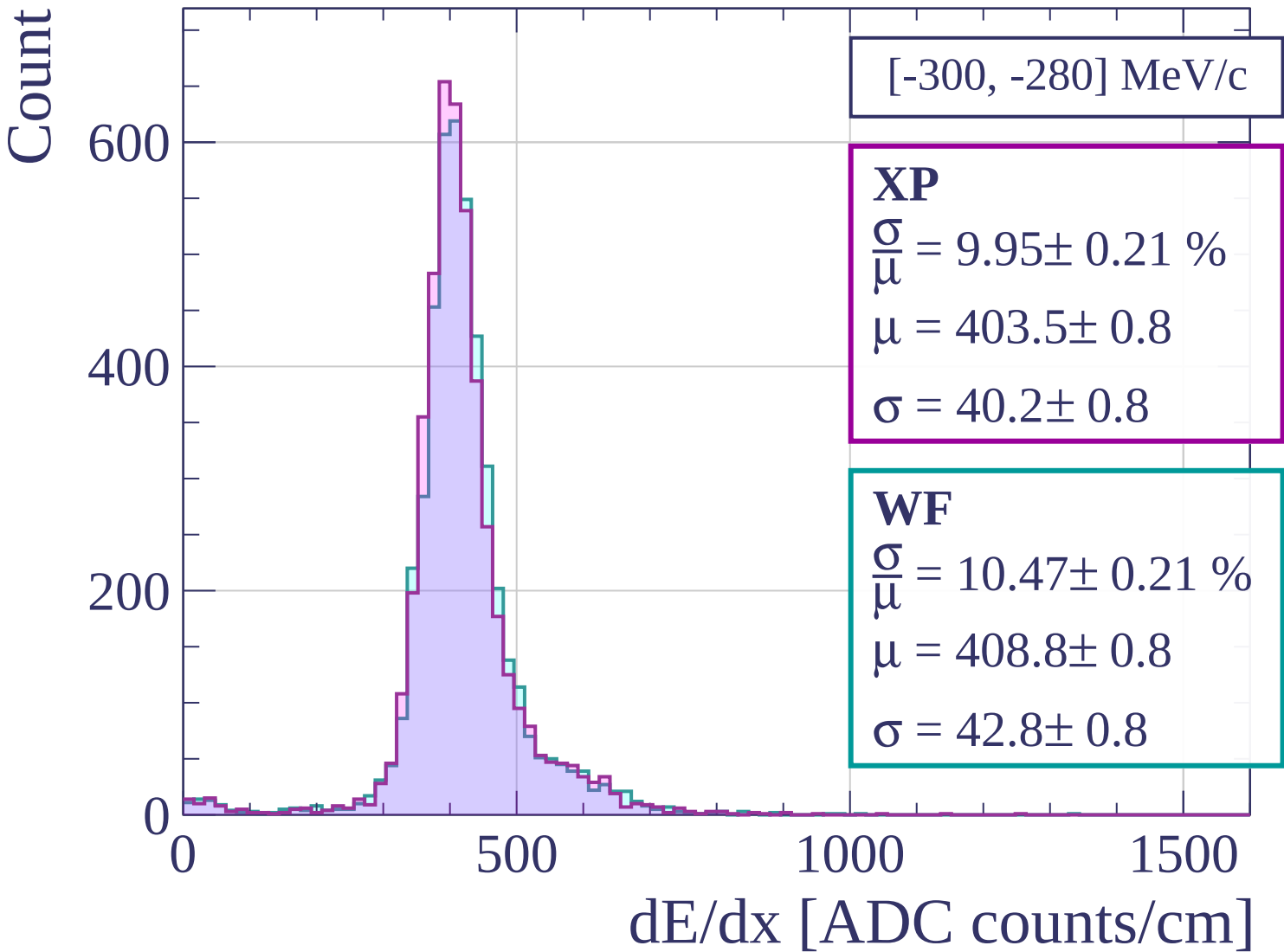
dE/dx [ADC counts/cm]











Count

[-280, -260] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.59 \pm 0.22 \%$$

$$\mu = 405.1 \pm 0.8$$

$$\sigma = 38.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.68 \pm 0.23 \%$$

$$\mu = 408.8 \pm 0.8$$

$$\sigma = 39.6 \pm 0.9$$

600

400

200

0

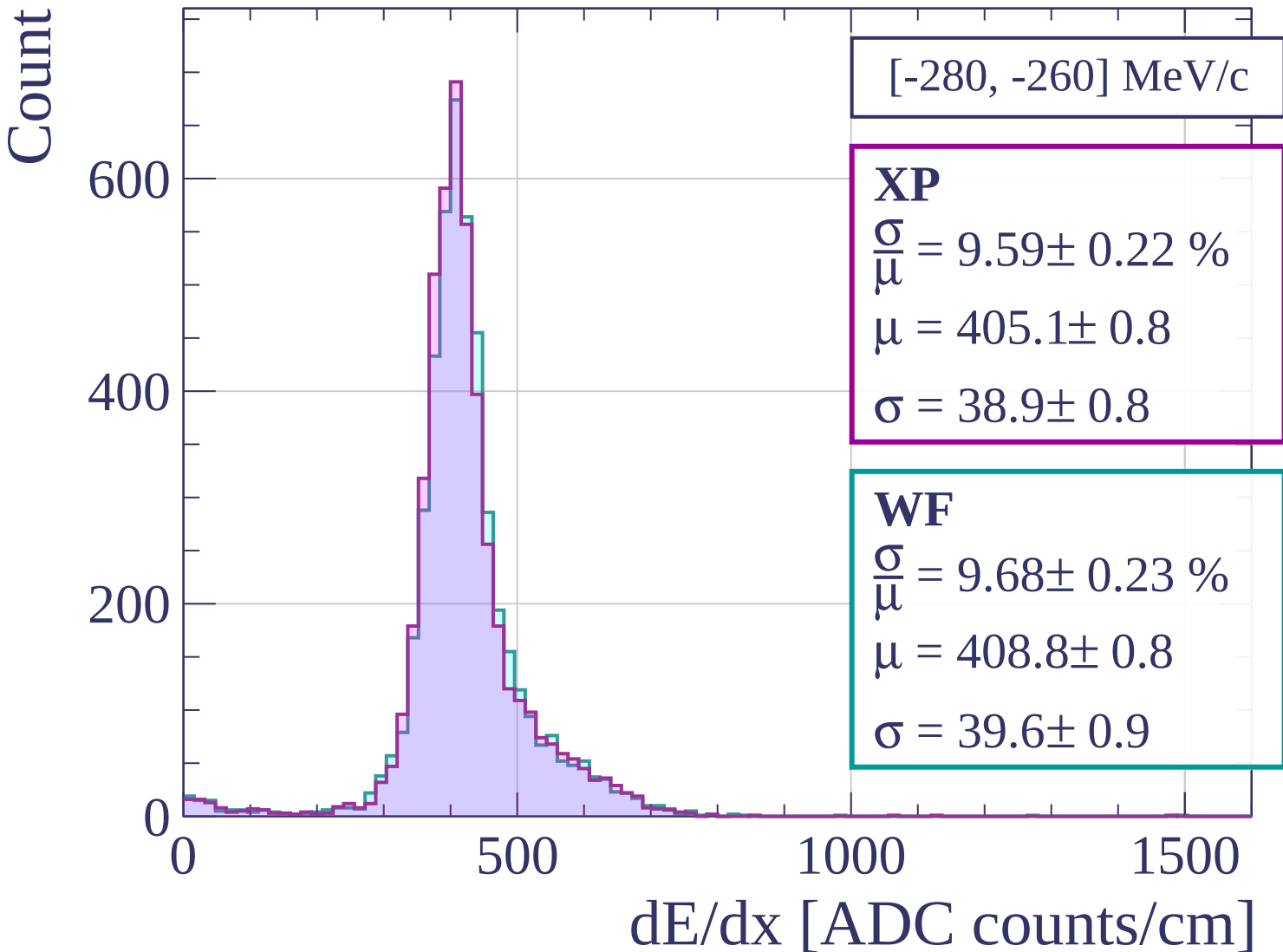
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-260, -240] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.98 \pm 0.18 \%$$

$$\mu = 410.1 \pm 0.7$$

$$\sigma = 40.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.23 \pm 0.23 \%$$

$$\mu = 414.0 \pm 0.8$$

$$\sigma = 42.3 \pm 0.9$$

600

400

200

0

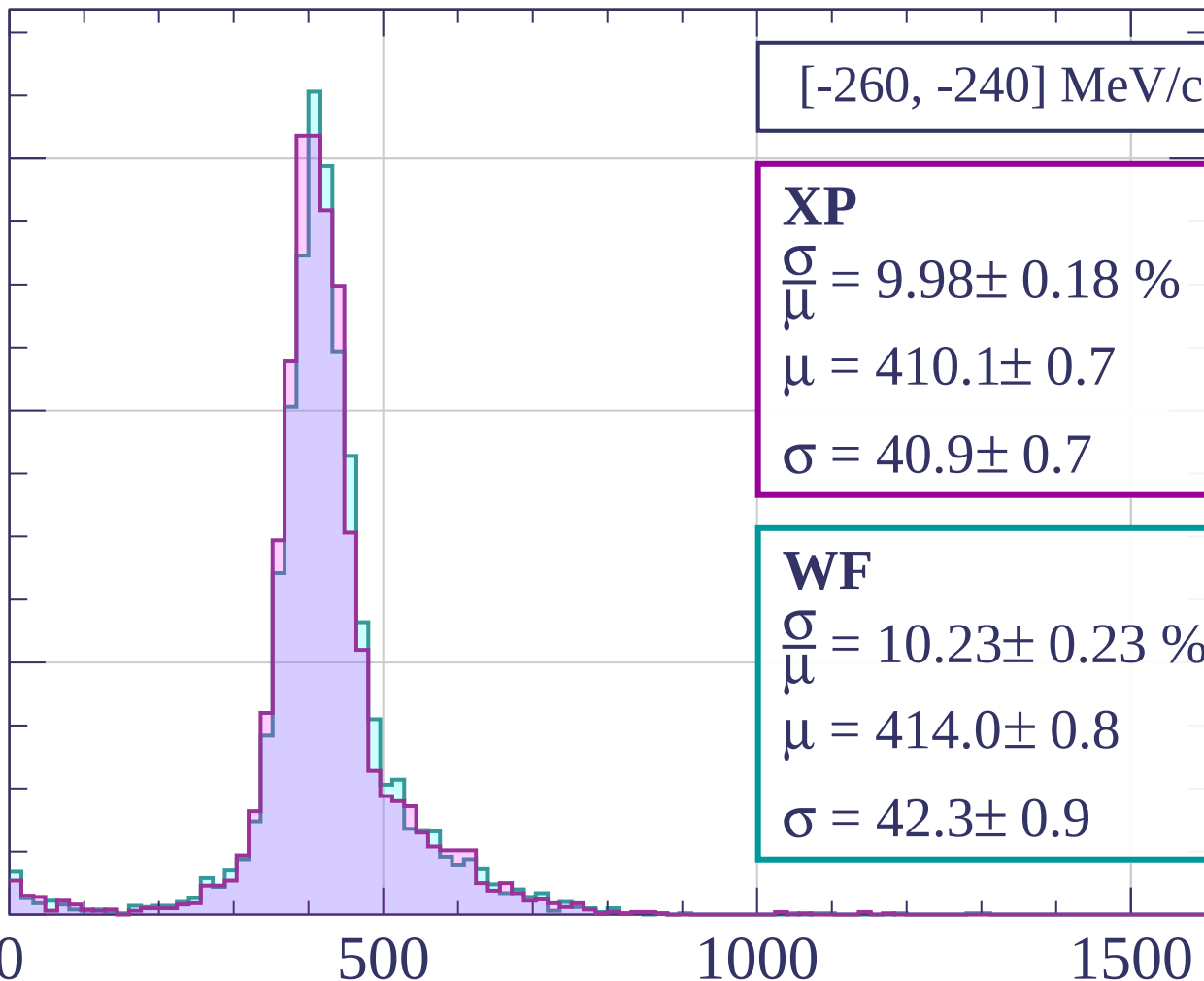
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-240, -220] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.21 \pm 0.23 \%$$

$$\mu = 416.9 \pm 0.8$$

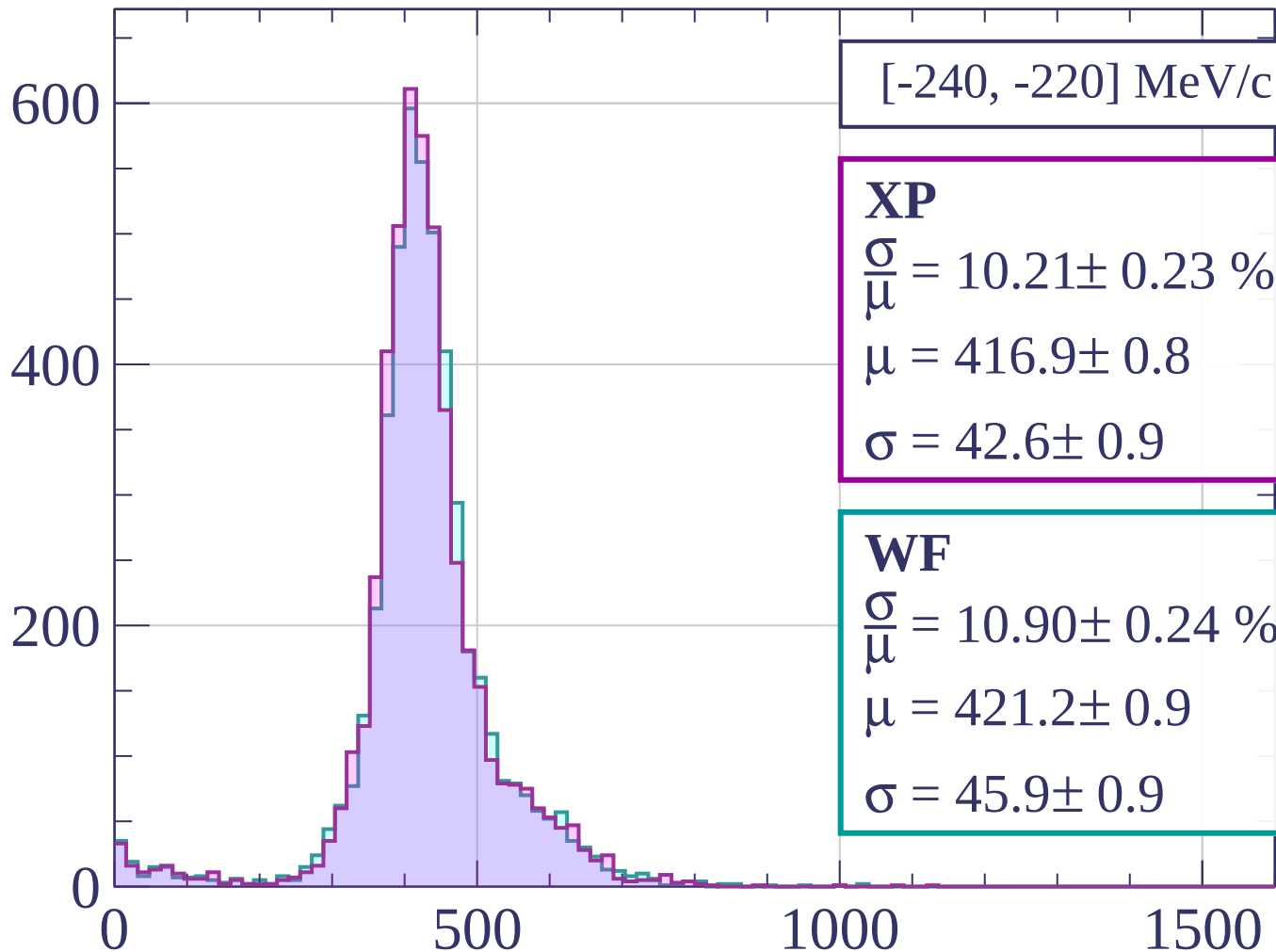
$$\sigma = 42.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.90 \pm 0.24 \%$$

$$\mu = 421.2 \pm 0.9$$

$$\sigma = 45.9 \pm 0.9$$



dE/dx [ADC counts/cm]

Count

[-220, -200] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.47 \pm 0.22 \%$$

$$\mu = 426.2 \pm 0.8$$

$$\sigma = 44.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.95 \pm 0.22 \%$$

$$\mu = 431.3 \pm 0.8$$

$$\sigma = 47.2 \pm 0.9$$

600

400

200

0

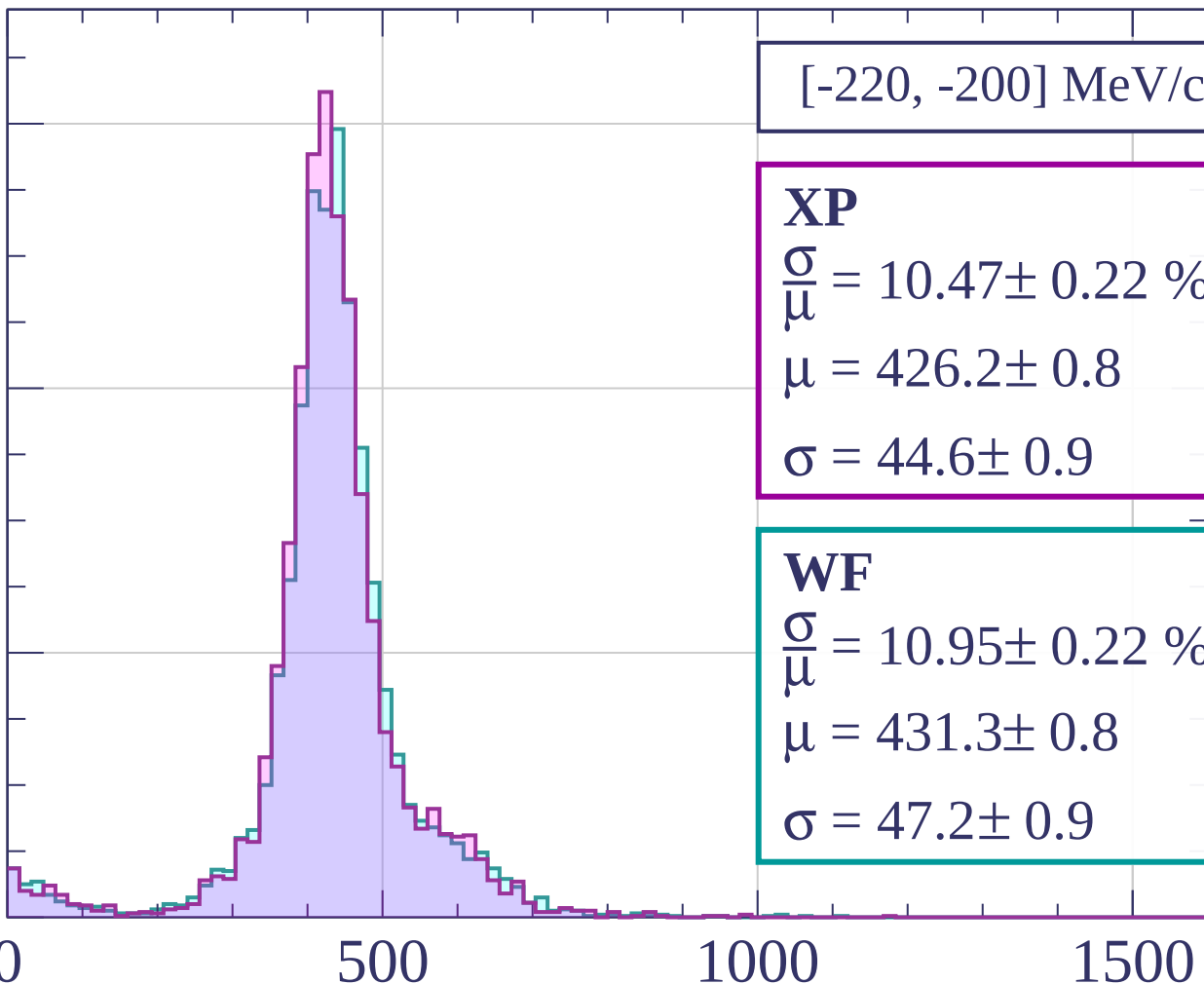
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-200, -180] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.71 \pm 0.25 \%$$

$$\mu = 438.6 \pm 0.9$$

$$\sigma = 47.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.39 \pm 0.26 \%$$

$$\mu = 442.8 \pm 1.0$$

$$\sigma = 50.4 \pm 1.0$$

600

400

200

0

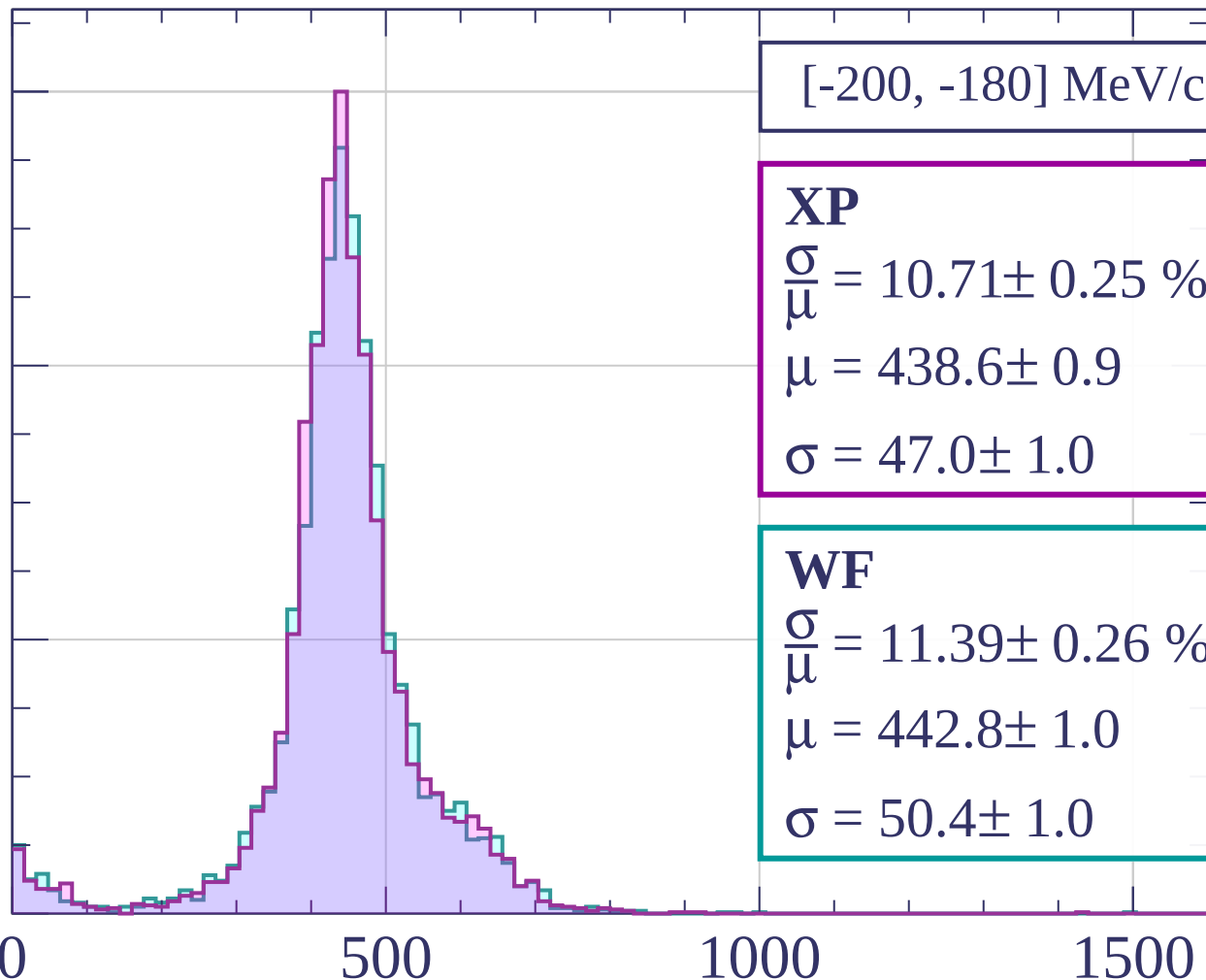
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-180, -160] MeV/c

XP

$$\frac{\sigma}{\mu} = 12.34 \pm 0.30 \%$$

$$\mu = 453.5 \pm 1.1$$

$$\sigma = 56.0 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 13.62 \pm 0.32 \%$$

$$\mu = 457.2 \pm 1.2$$

$$\sigma = 62.3 \pm 1.3$$

400

200

0

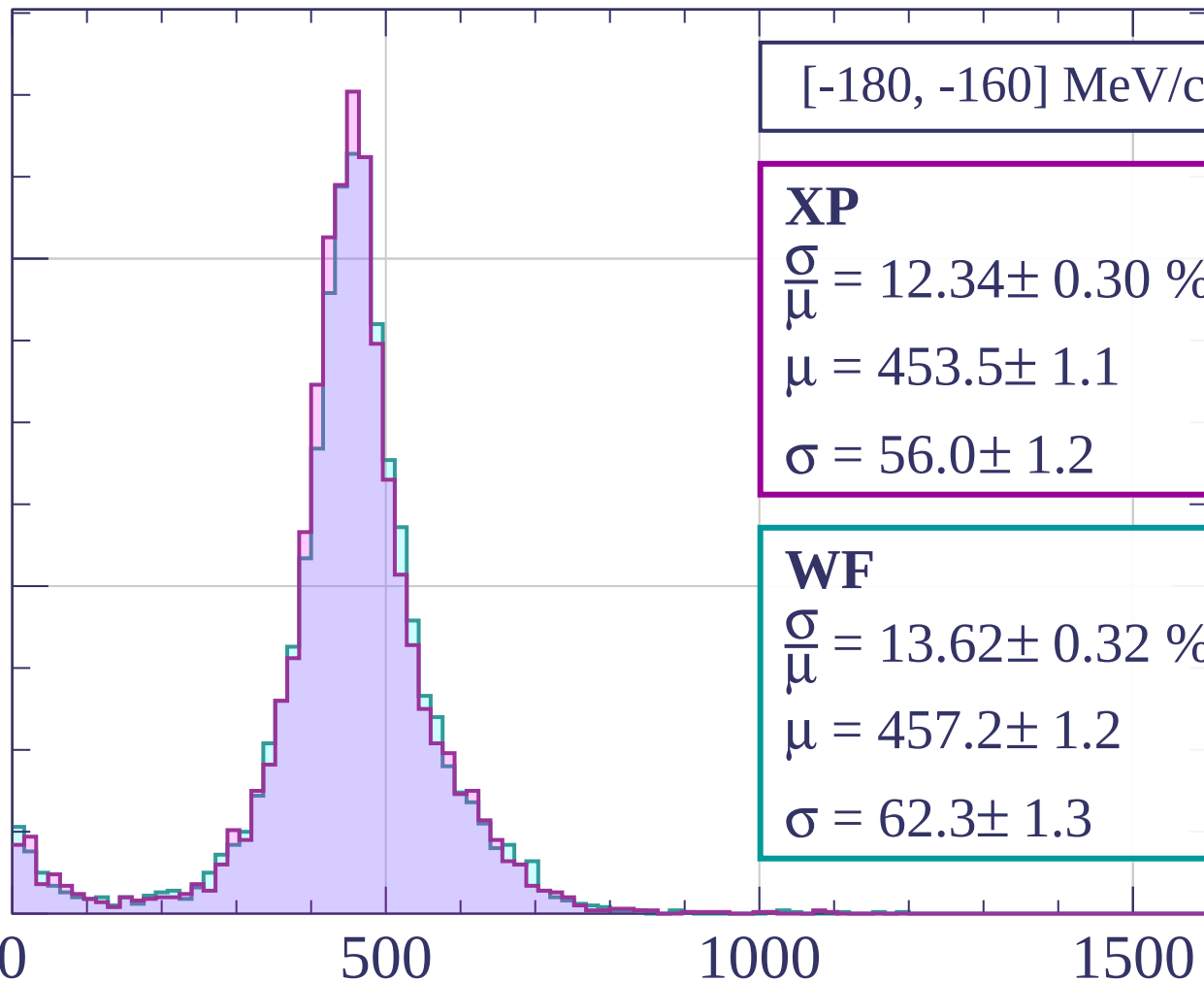
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-160, -140] MeV/c

XP

$$\frac{\sigma}{\mu} = 14.71 \pm 0.33 \%$$

$$\mu = 475.9 \pm 1.3$$

$$\sigma = 70.0 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 14.37 \pm 0.36 \%$$

$$\mu = 480.1 \pm 1.3$$

$$\sigma = 69.0 \pm 1.5$$

400

300

200

100

0

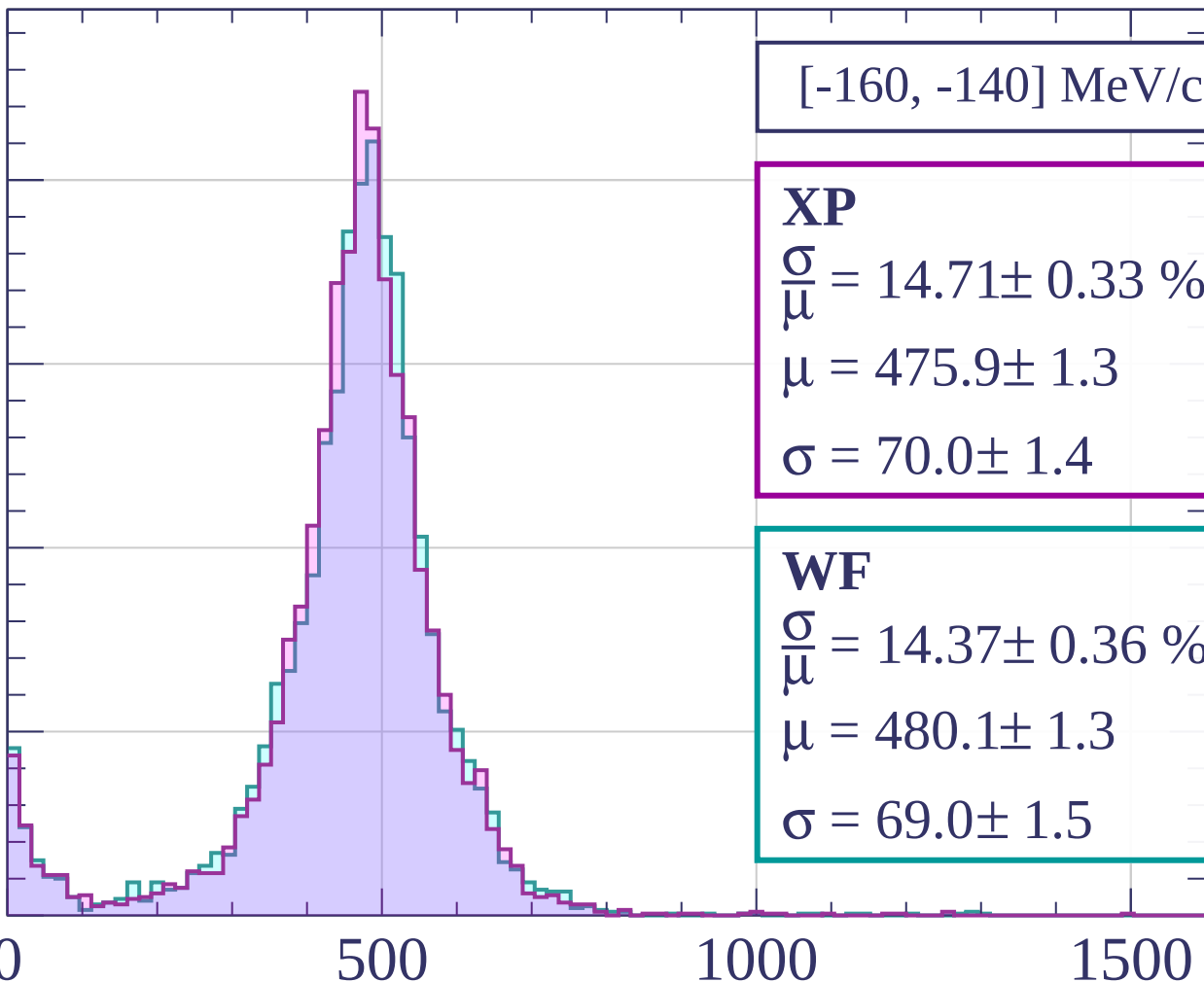
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-140, -120] MeV/c

XP

$$\frac{\sigma}{\mu} = 15.75 \pm 0.38 \%$$

$$\mu = 512.4 \pm 1.5$$

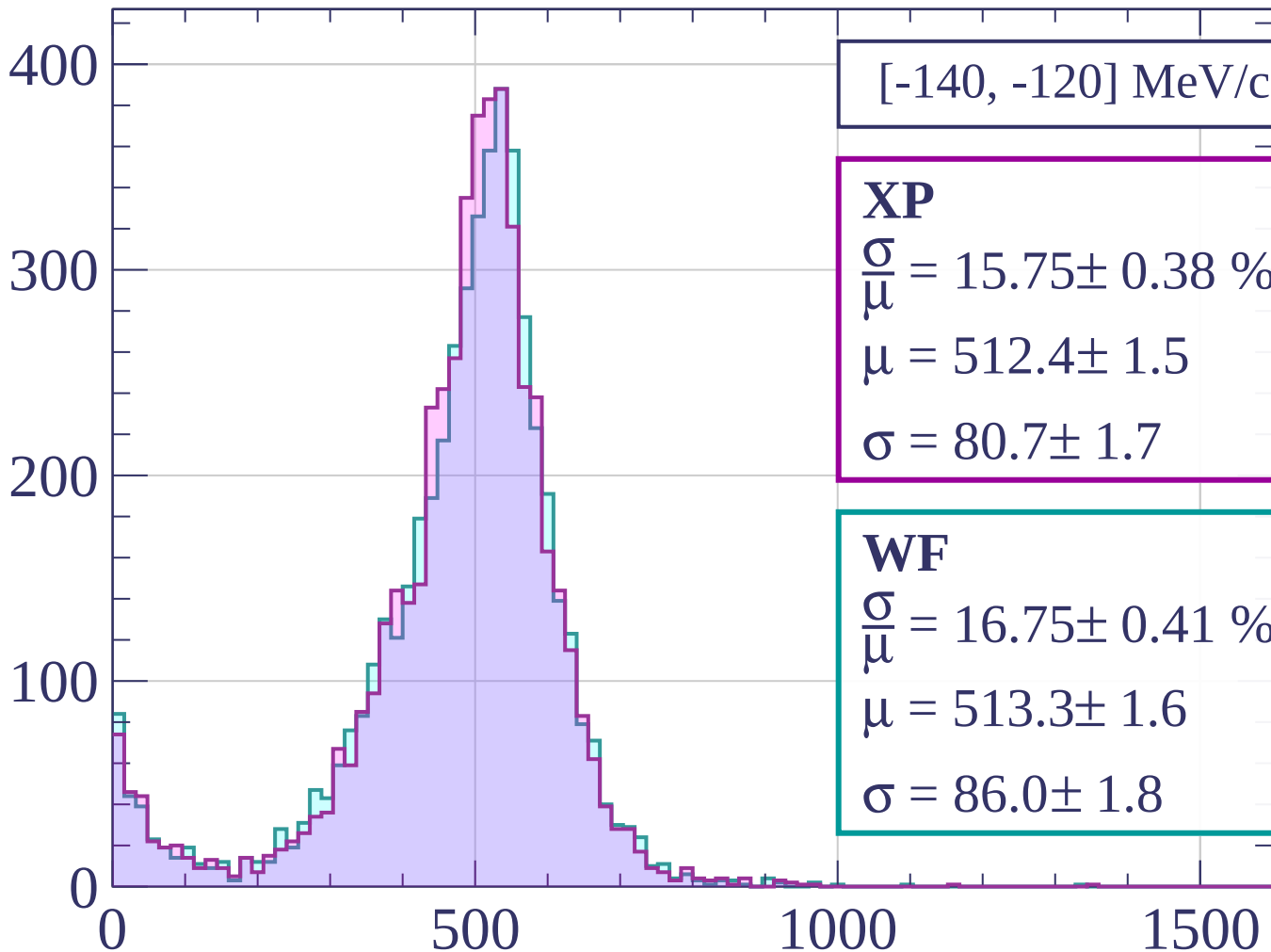
$$\sigma = 80.7 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 16.75 \pm 0.41 \%$$

$$\mu = 513.3 \pm 1.6$$

$$\sigma = 86.0 \pm 1.8$$



dE/dx [ADC counts/cm]

Count

[-120, -100] MeV/c

XP

$$\frac{\sigma}{\mu} = 20.84 \pm 0.49 \%$$

$$\mu = 525.9 \pm 2.3$$

$$\sigma = 109.6 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 23.22 \pm 0.52 \%$$

$$\mu = 521.8 \pm 2.4$$

$$\sigma = 121.1 \pm 2.1$$

300

200

100

0

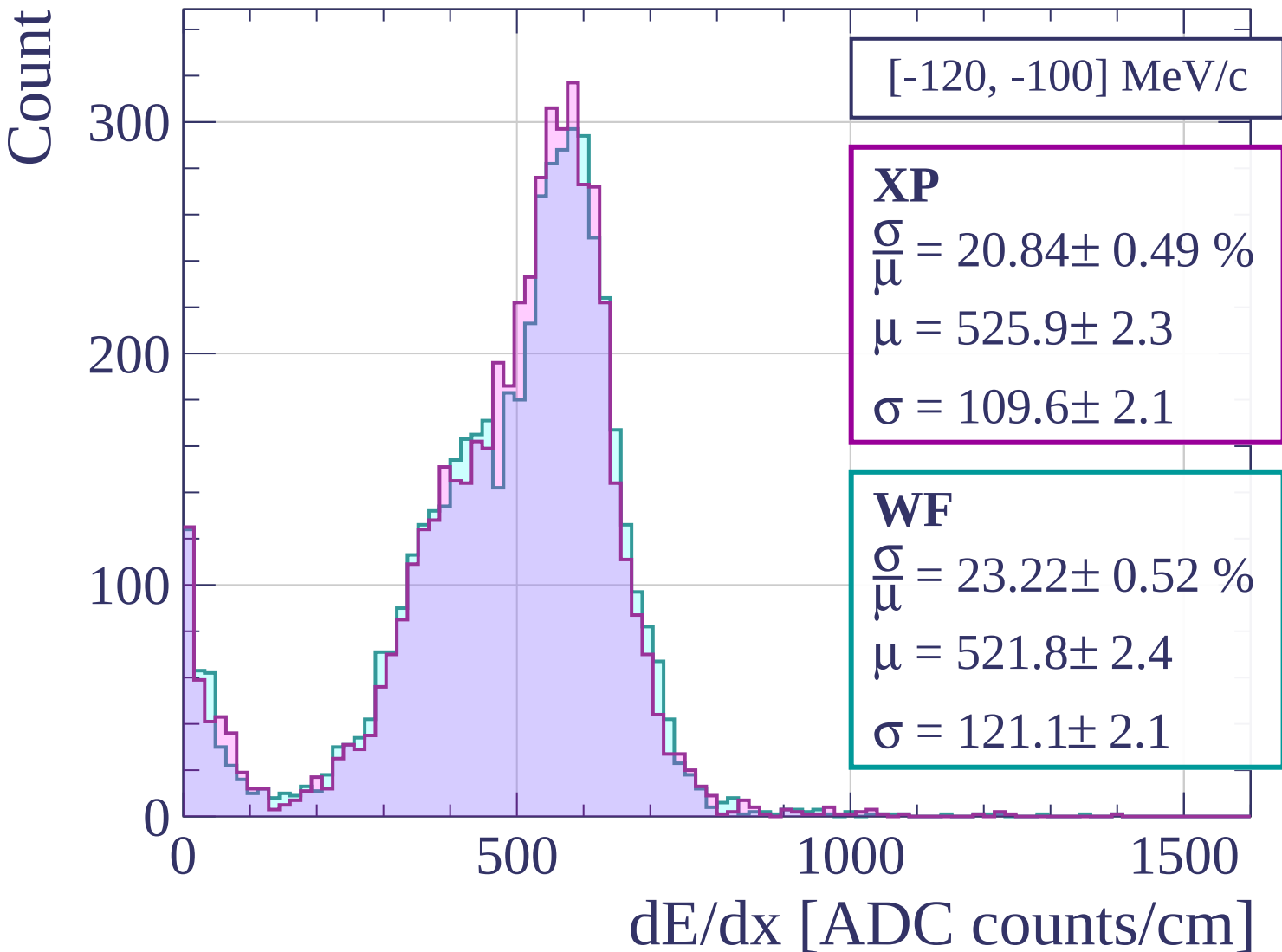
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-100, -80] MeV/c

XP

$$\frac{\sigma}{\mu} = 27.41 \pm 0.49 \%$$

$$\mu = 535.5 \pm 2.3$$

$$\sigma = 146.8 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 28.92 \pm 0.53 \%$$

$$\mu = 532.6 \pm 2.4$$

$$\sigma = 154.0 \pm 2.1$$

200

100

0

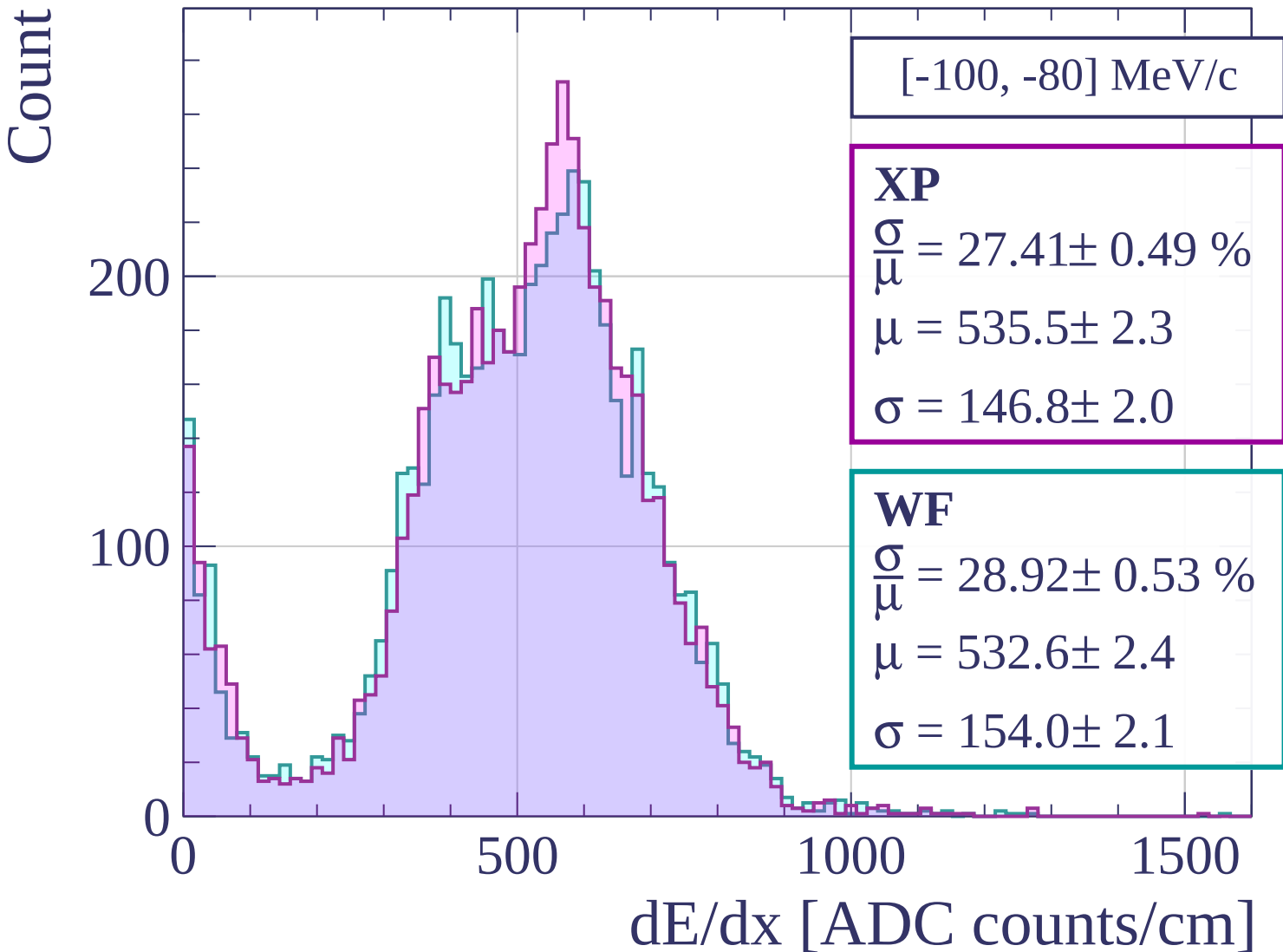
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-80, -60] MeV/c

XP

$$\frac{\sigma}{\mu} = 24.19 \pm 0.53 \%$$

$$\mu = 498.1 \pm 2.3$$

$$\sigma = 120.5 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 26.04 \pm 0.53 \%$$

$$\mu = 490.3 \pm 2.3$$

$$\sigma = 127.7 \pm 2.0$$

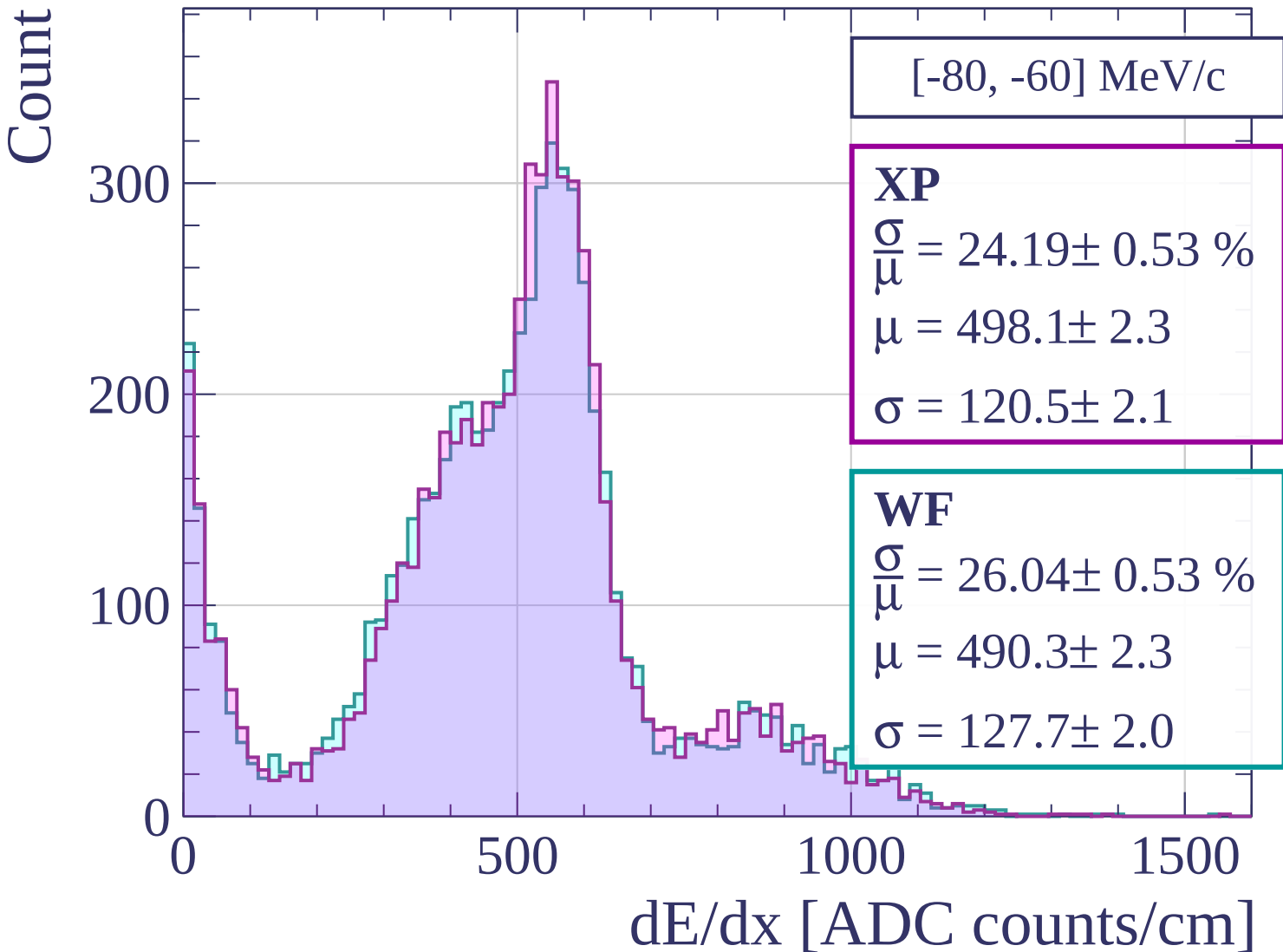
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-60, -40] MeV/c

XP

$$\frac{\sigma}{\mu} = 20.00 \pm 0.38 \%$$

$$\mu = 507.7 \pm 1.8$$

$$\sigma = 101.5 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 21.82 \pm 0.42 \%$$

$$\mu = 499.6 \pm 2.0$$

$$\sigma = 109.0 \pm 1.6$$

400

200

0

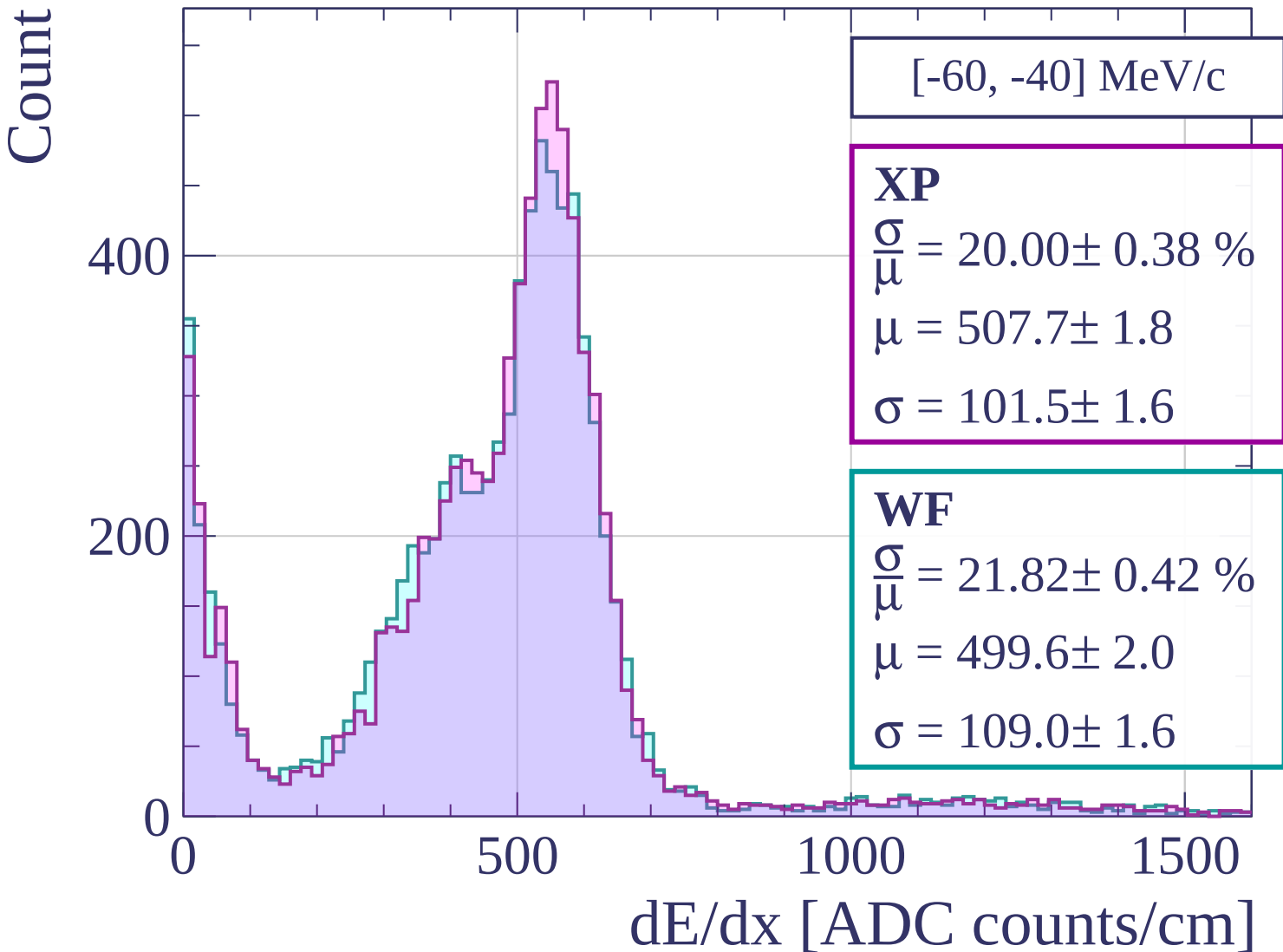
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-40, -20] MeV/c

XP

$$\frac{\sigma}{\mu} = 13.86 \pm 0.21 \%$$

$$\mu = 526.2 \pm 0.9$$

$$\sigma = 72.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 15.99 \pm 0.25 \%$$

$$\mu = 519.2 \pm 1.1$$

$$\sigma = 83.0 \pm 1.1$$

800

600

400

200

0

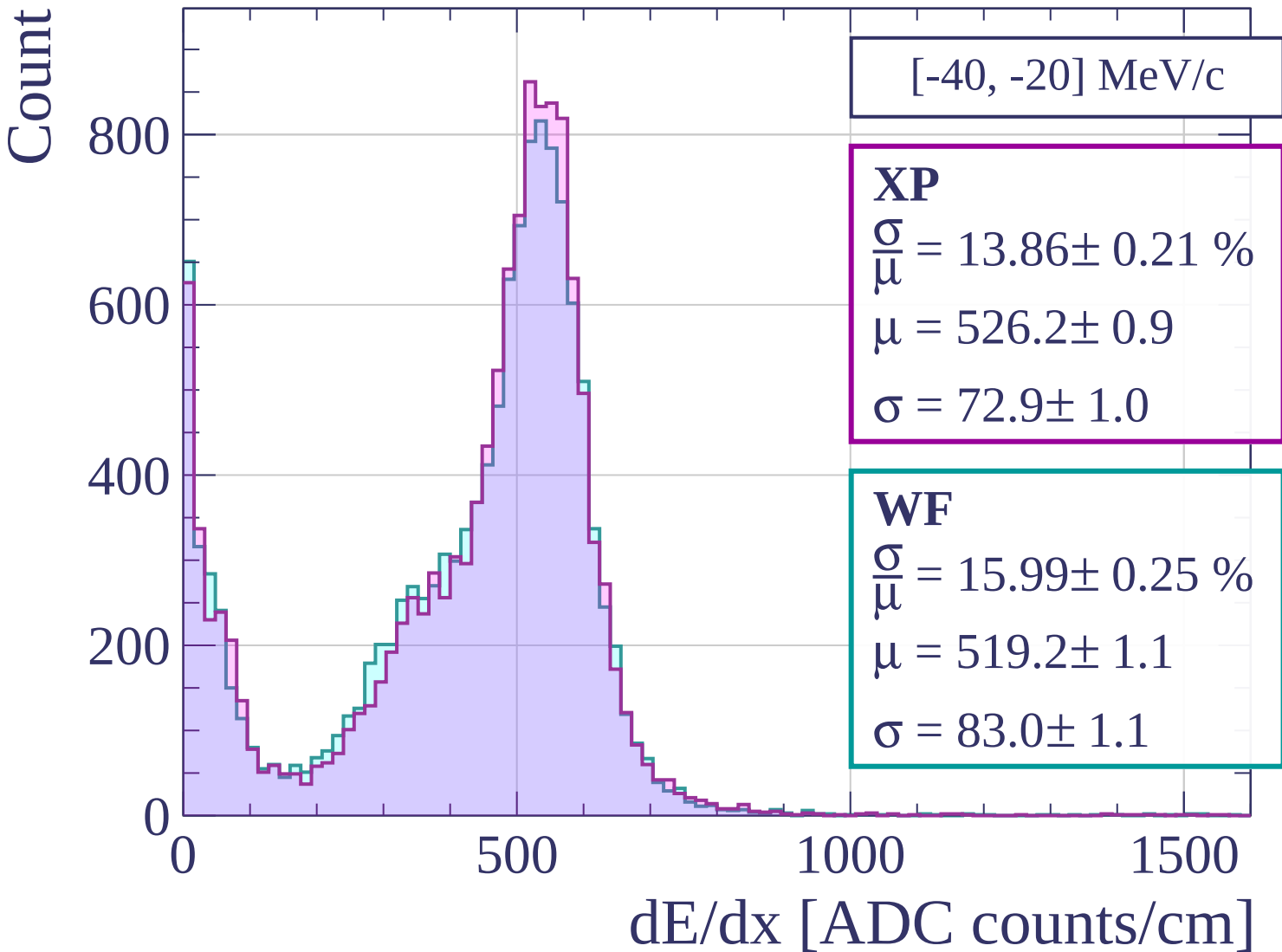
0

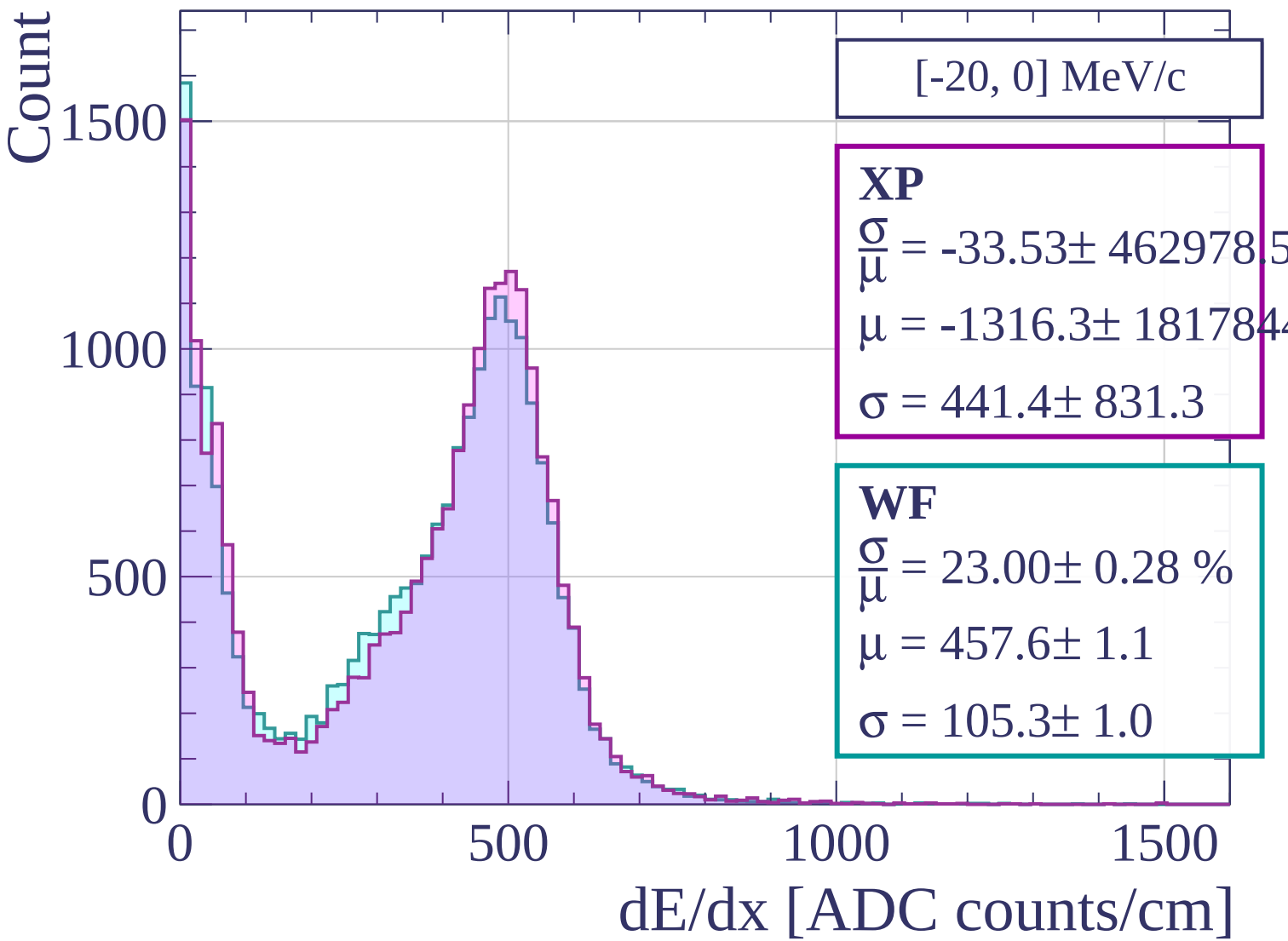
500

1000

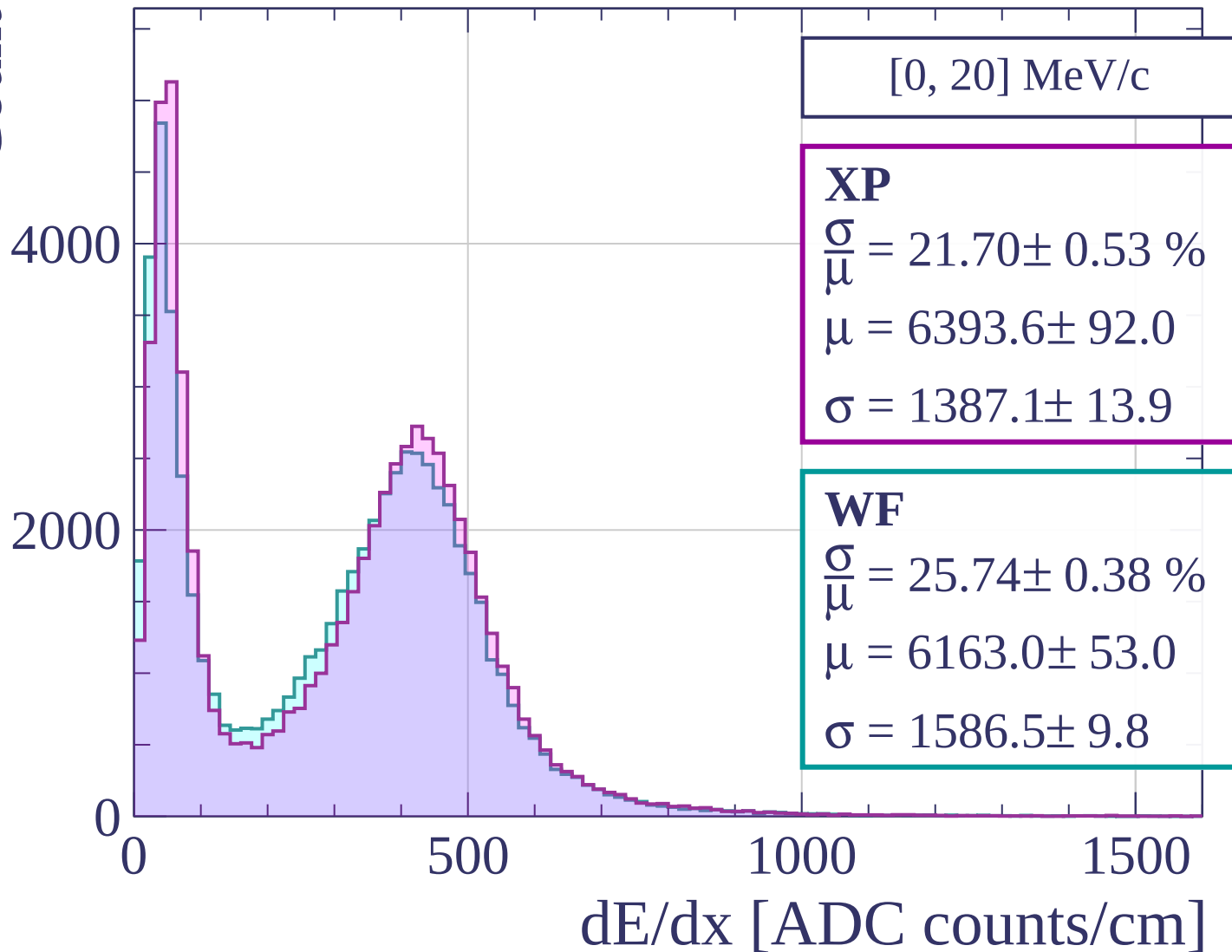
1500

dE/dx [ADC counts/cm]

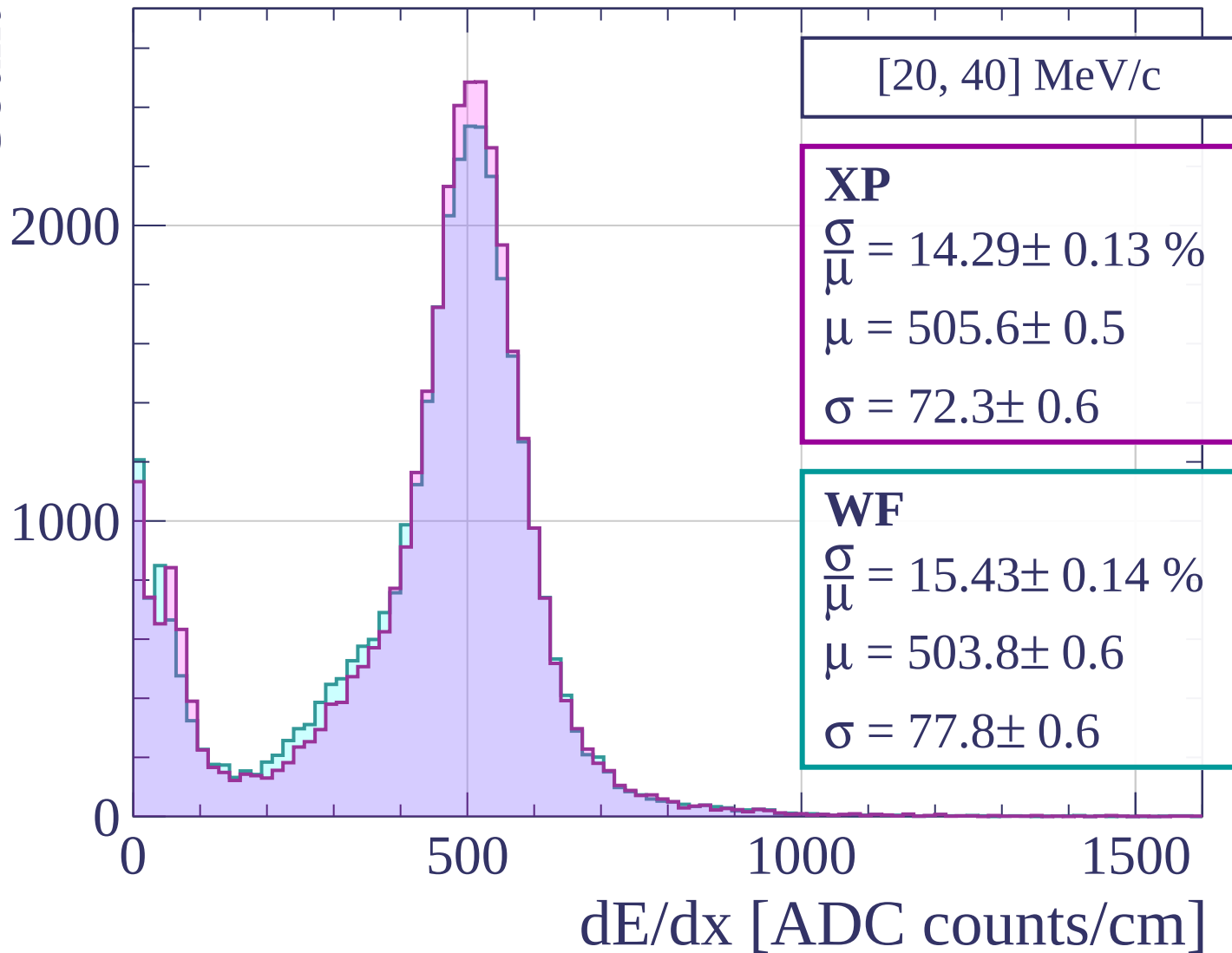


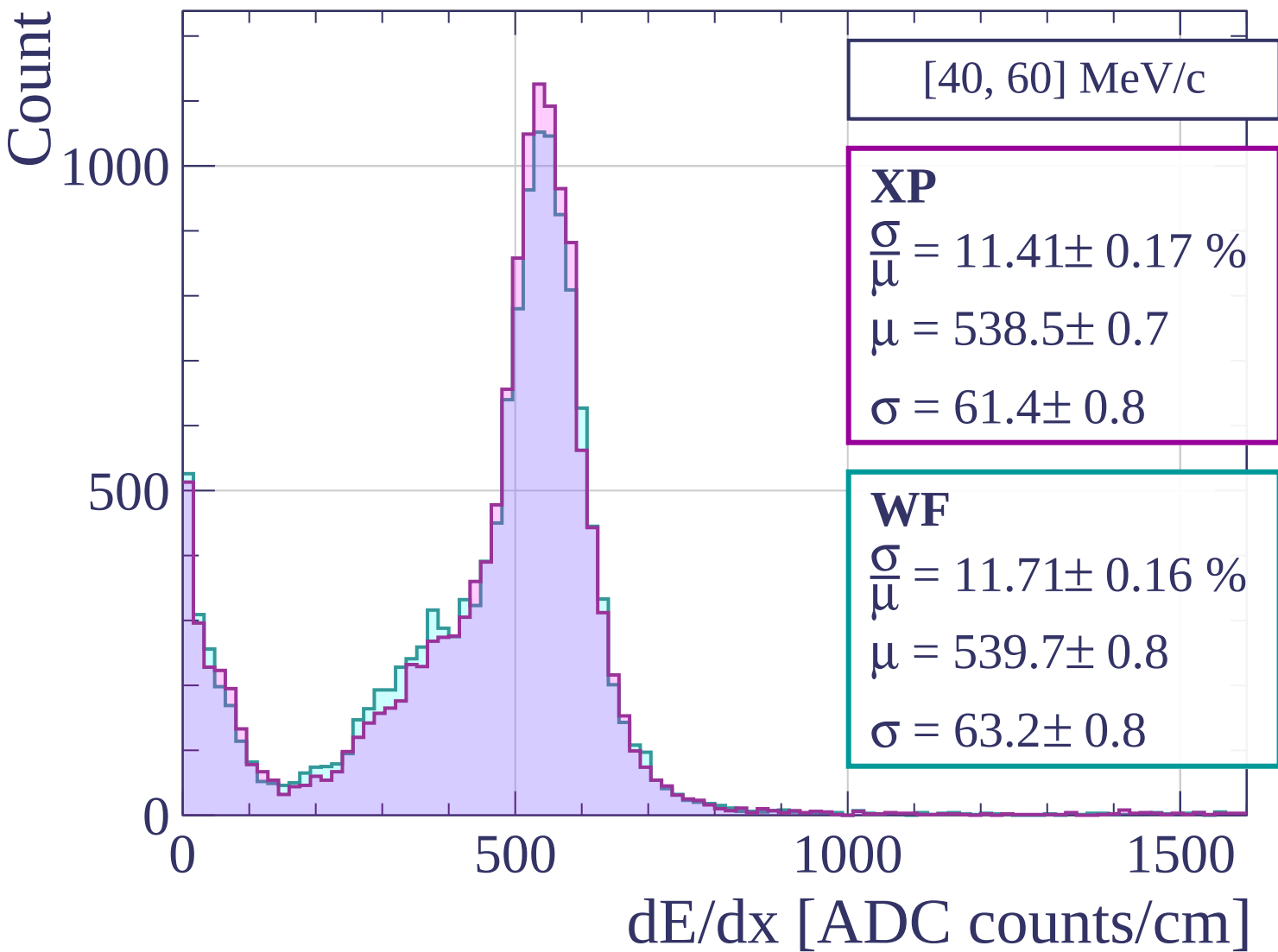


Count



Count





Count

[60, 80] MeV/c

XP

$$\frac{\sigma}{\mu} = 22.83 \pm 0.44 \%$$

$$\mu = 494.3 \pm 2.0$$

$$\sigma = 112.9 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 25.58 \pm 0.49 \%$$

$$\mu = 482.9 \pm 2.2$$

$$\sigma = 123.5 \pm 1.8$$

400

300

200

100

0

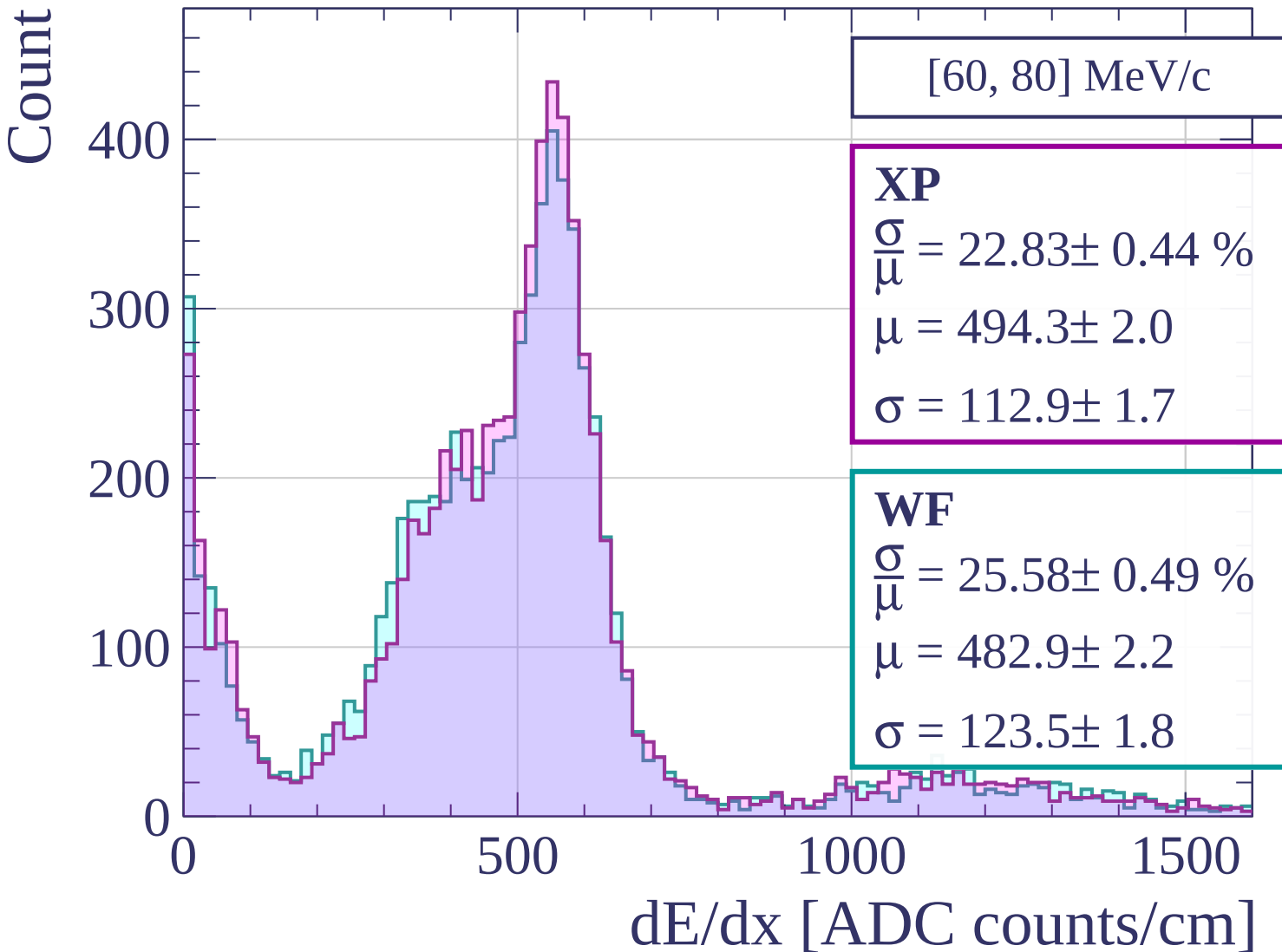
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[80, 100] MeV/c

XP

$$\frac{\sigma}{\mu} = 27.58 \pm 0.61 \%$$

$$\mu = 484.3 \pm 2.4$$

$$\sigma = 133.5 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 28.60 \pm 0.59 \%$$

$$\mu = 473.1 \pm 2.4$$

$$\sigma = 135.3 \pm 2.1$$

0

100

200

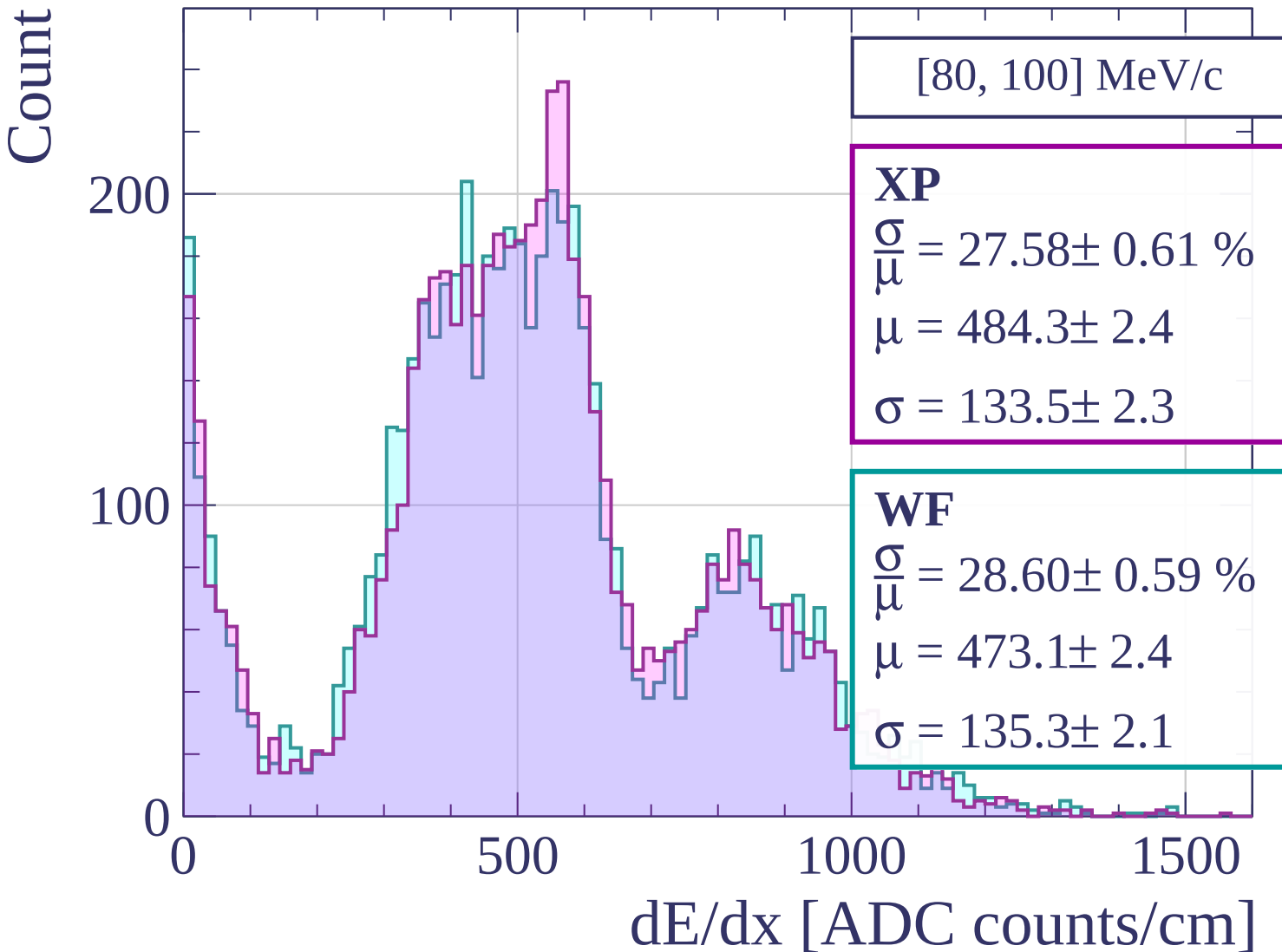
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[100, 120] MeV/c

XP

$$\frac{\sigma}{\mu} = 28.32 \pm 0.46 \%$$

$$\mu = 557.6 \pm 2.4$$

$$\sigma = 157.9 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 30.62 \pm 0.54 \%$$

$$\mu = 553.5 \pm 2.7$$

$$\sigma = 169.5 \pm 2.1$$

200

150

100

50

0

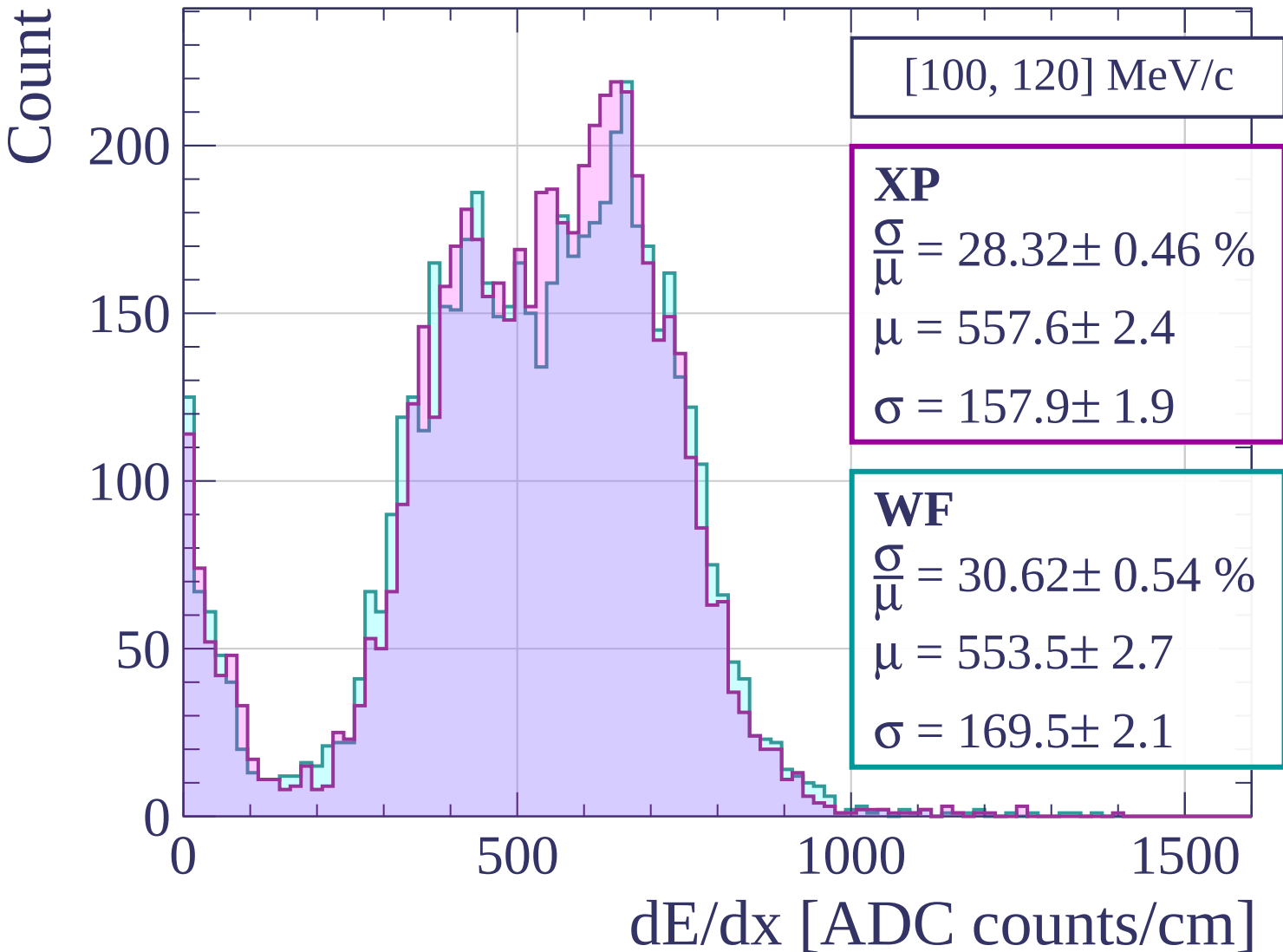
0

500

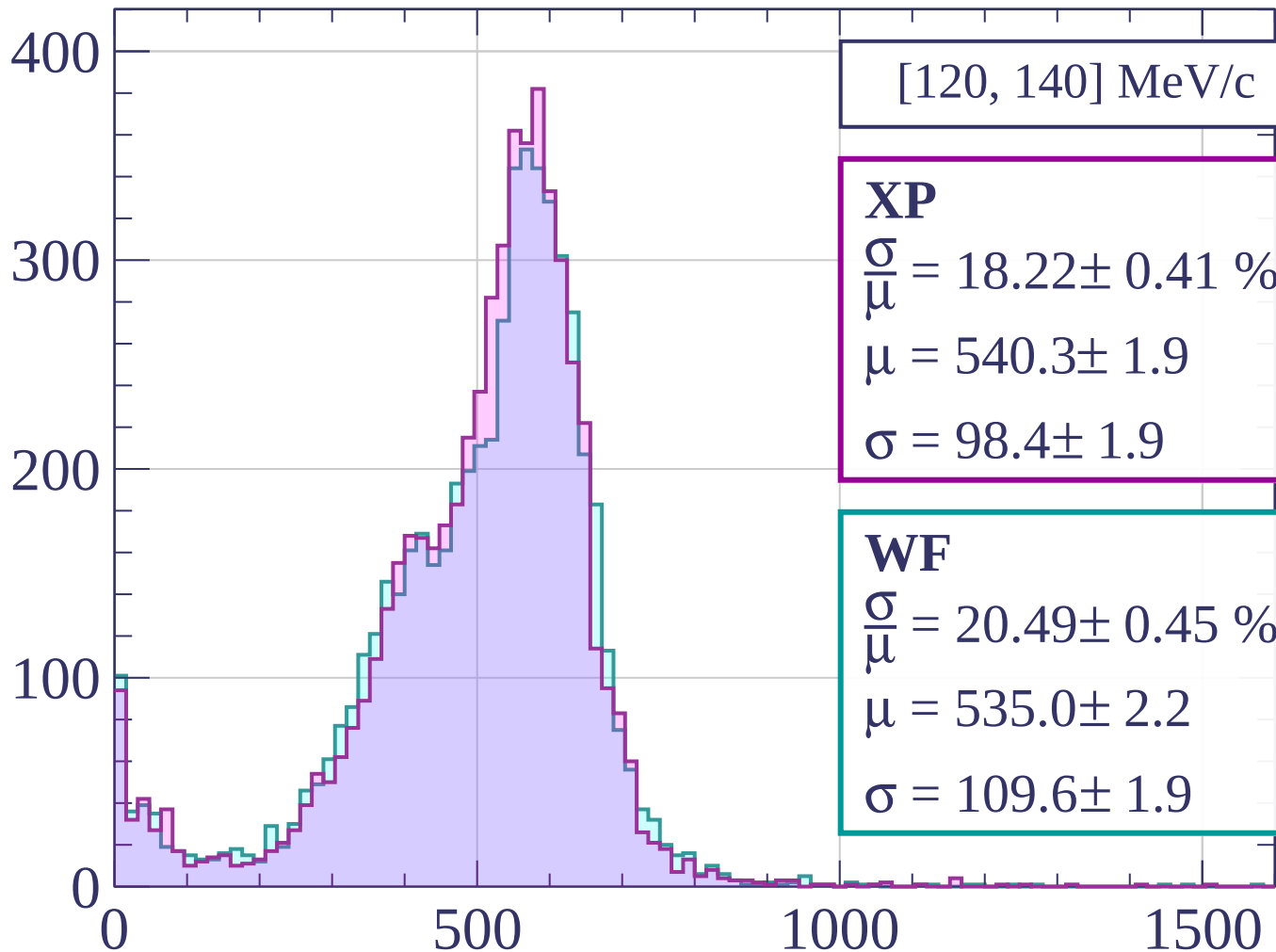
1000

1500

dE/dx [ADC counts/cm]

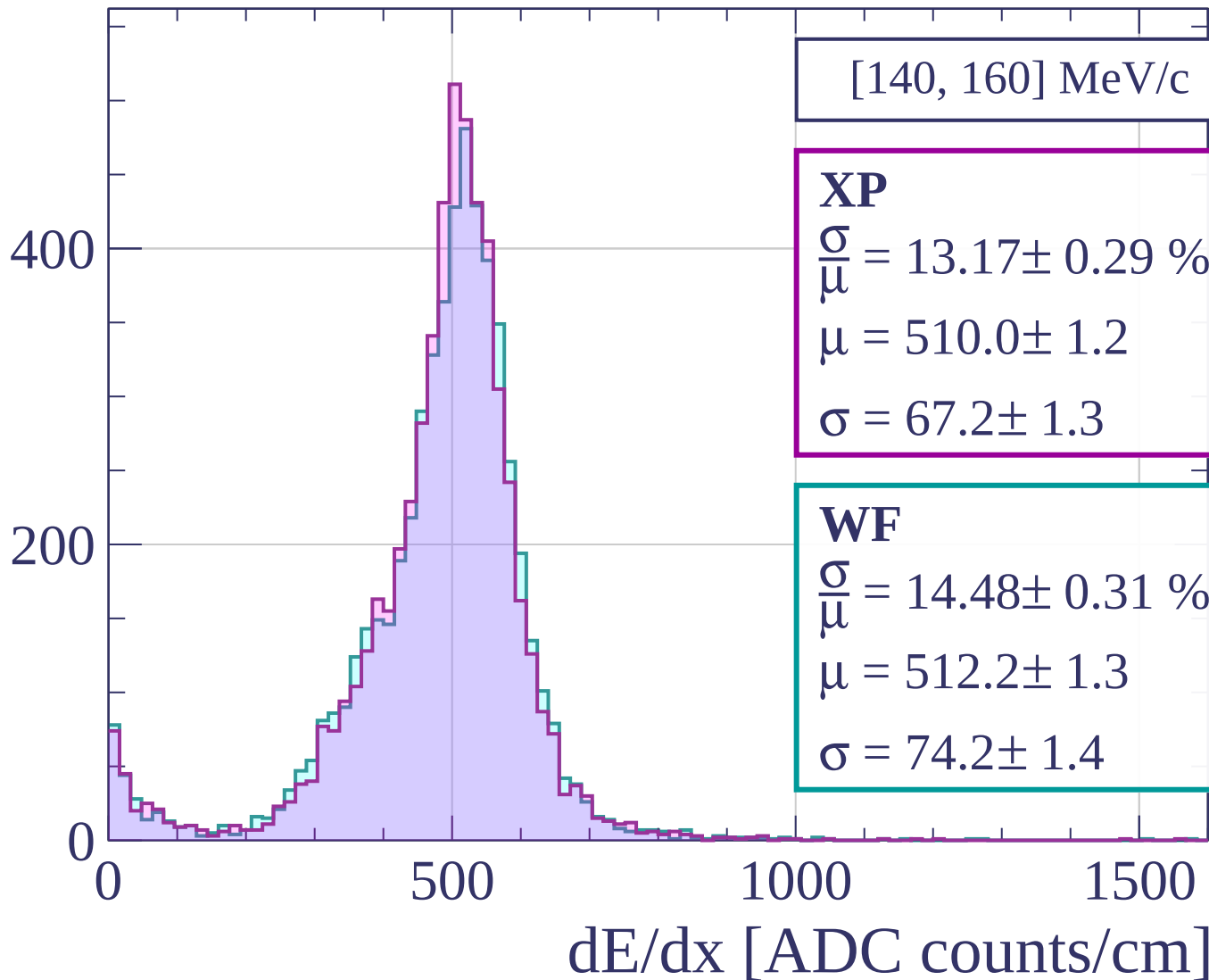


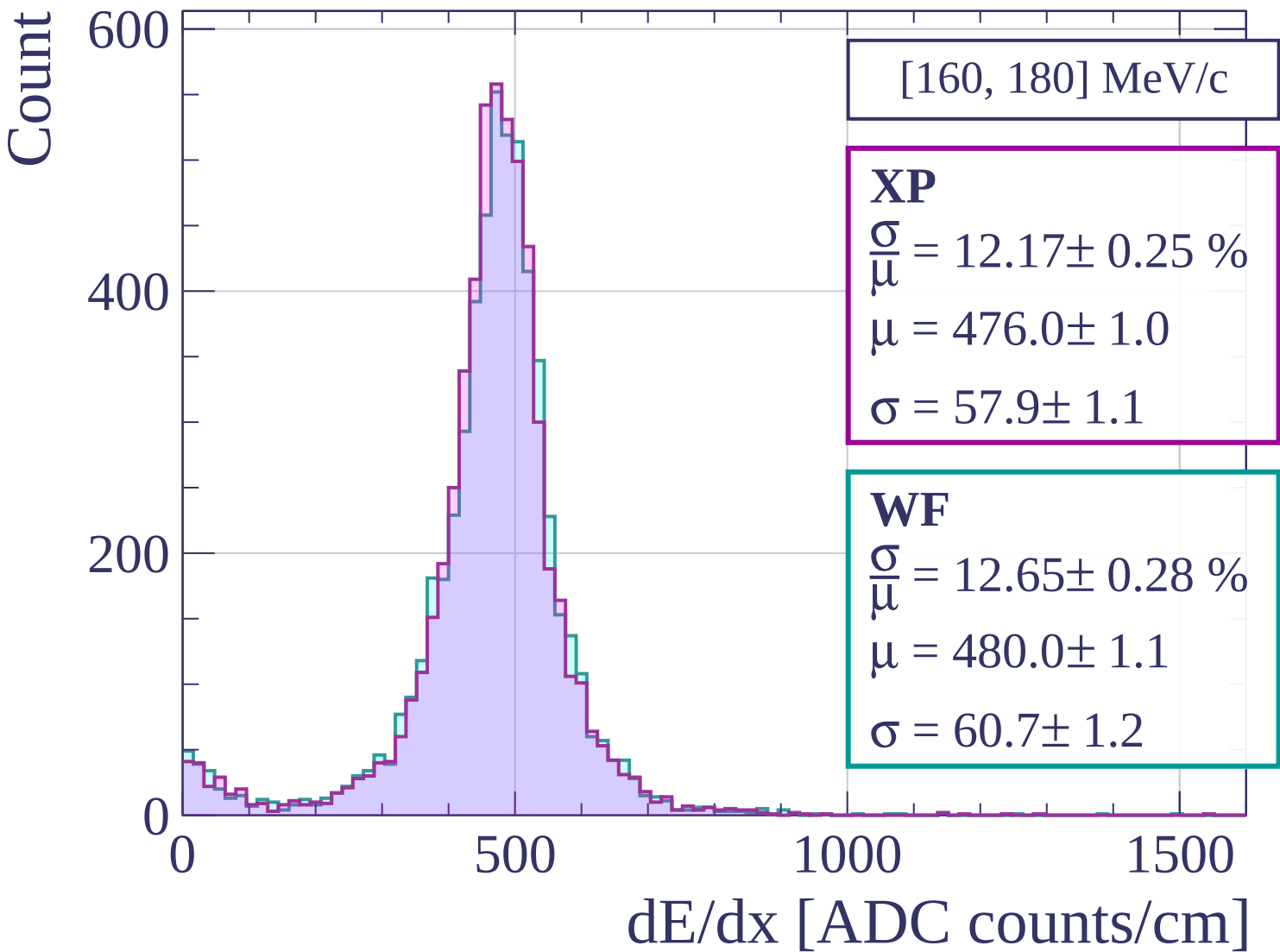
Count



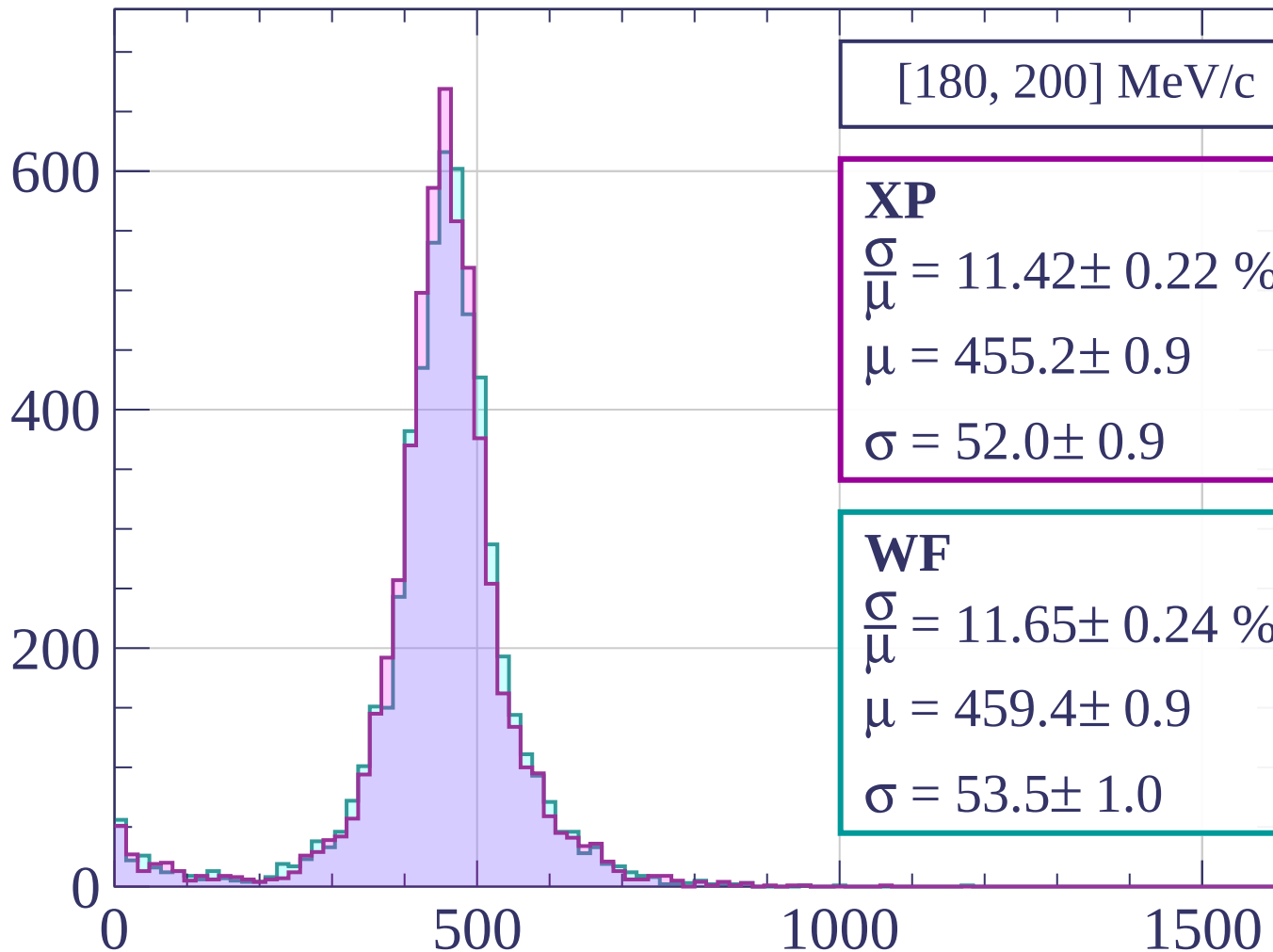
dE/dx [ADC counts/cm]

Count





Count



$[180, 200] \text{ MeV/c}$

XP

$$\frac{\sigma}{\mu} = 11.42 \pm 0.22 \%$$

$$\mu = 455.2 \pm 0.9$$

$$\sigma = 52.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.65 \pm 0.24 \%$$

$$\mu = 459.4 \pm 0.9$$

$$\sigma = 53.5 \pm 1.0$$

dE/dx [ADC counts/cm]

Count

[200, 220] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.93 \pm 0.21 \%$$

$$\mu = 437.5 \pm 0.8$$

$$\sigma = 47.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 11.54 \pm 0.22 \%$$

$$\mu = 442.5 \pm 0.8$$

$$\sigma = 51.1 \pm 0.9$$

600

400

200

0

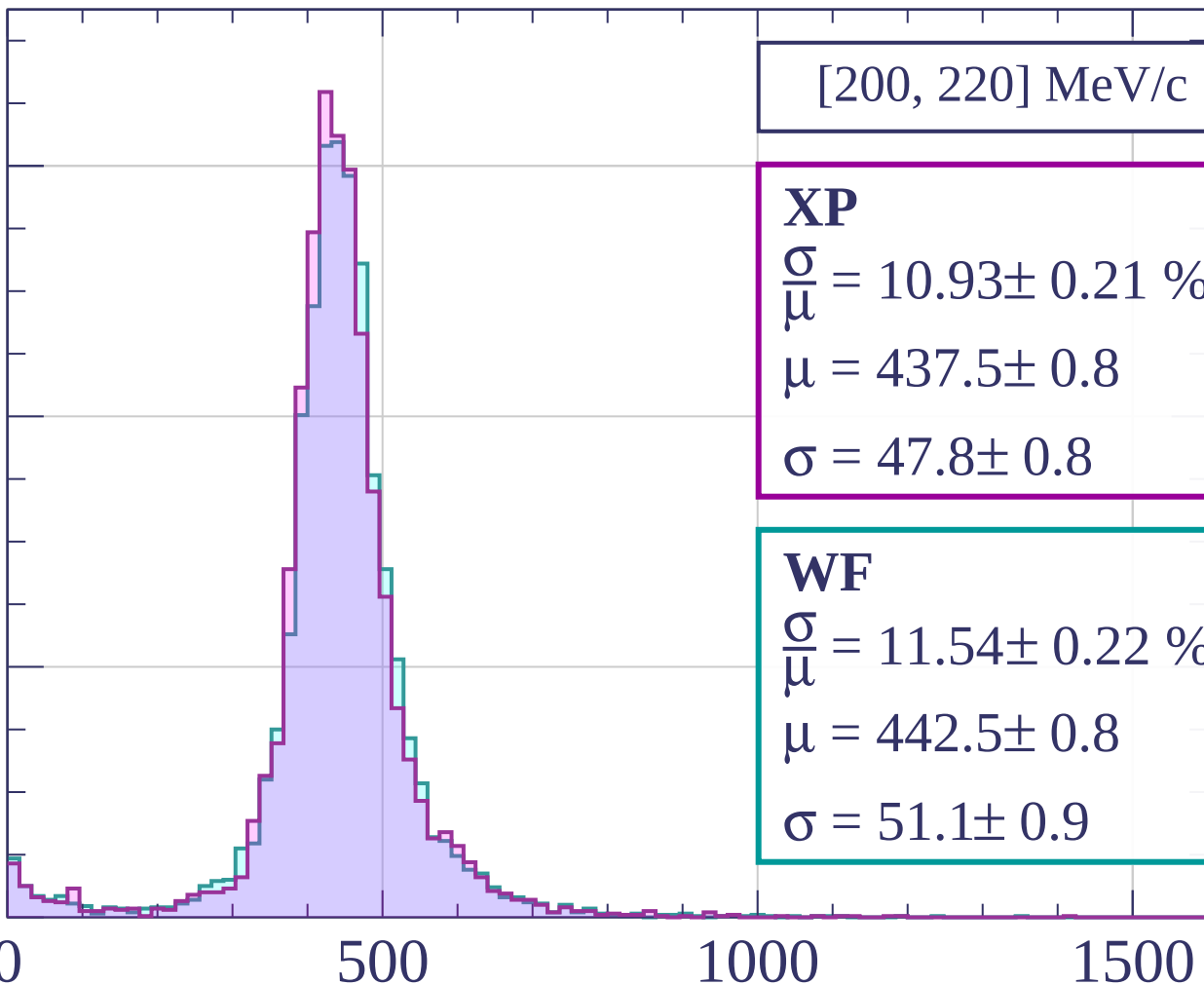
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[220, 240] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.56 \pm 0.19 \%$$

$$\mu = 423.2 \pm 0.7$$

$$\sigma = 40.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.24 \pm 0.20 \%$$

$$\mu = 428.3 \pm 0.8$$

$$\sigma = 43.9 \pm 0.8$$

600

400

200

0

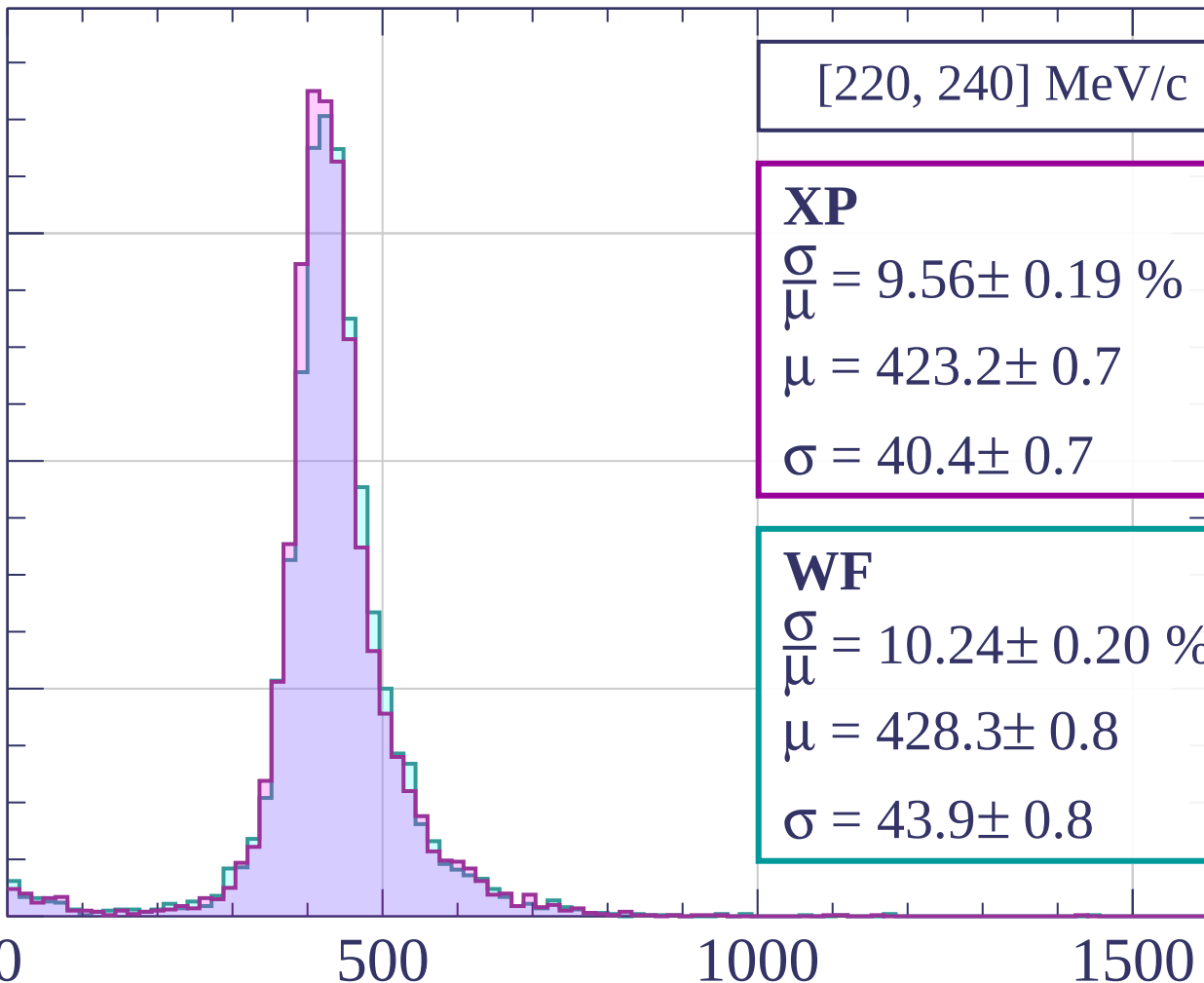
0

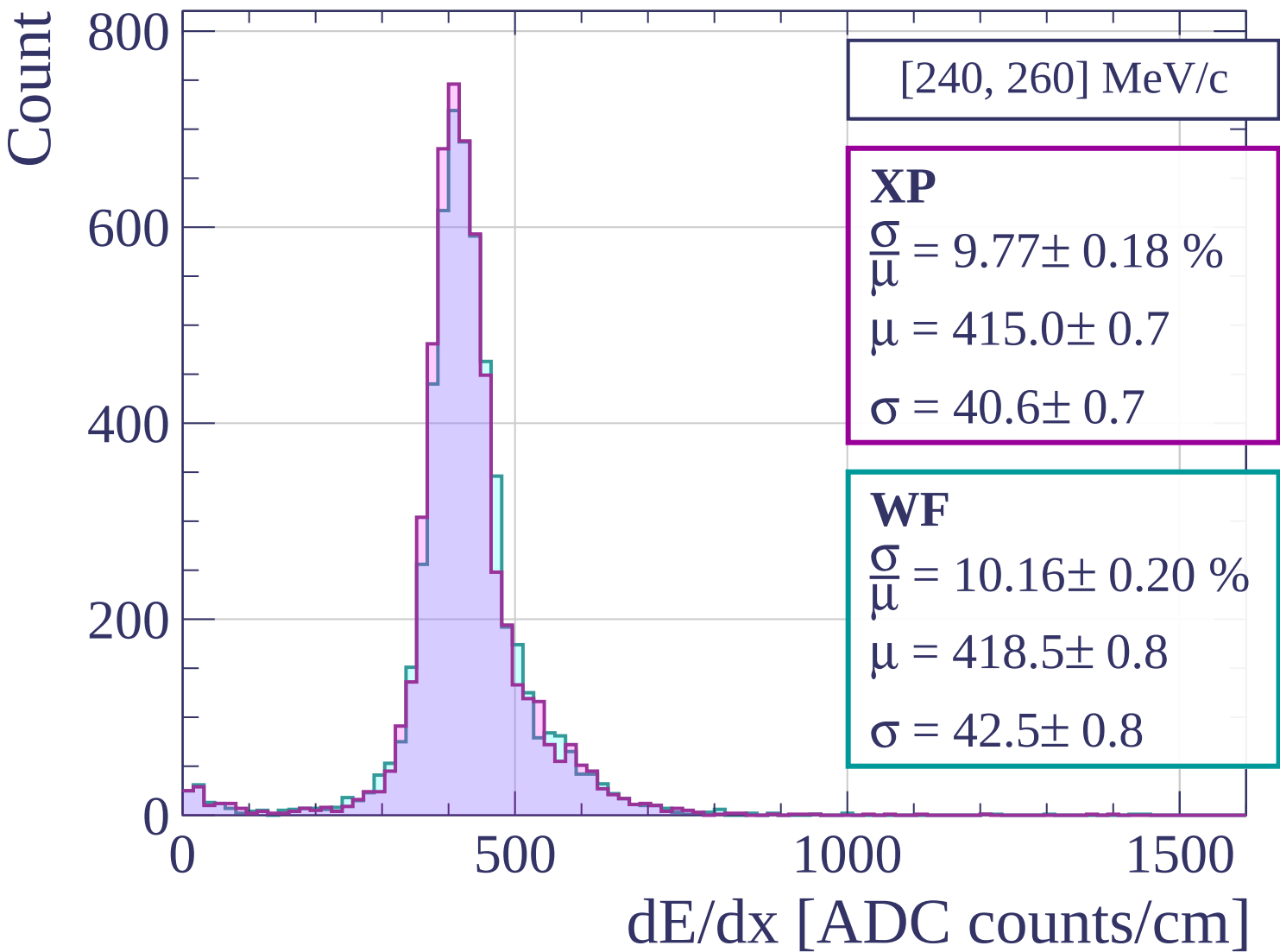
500

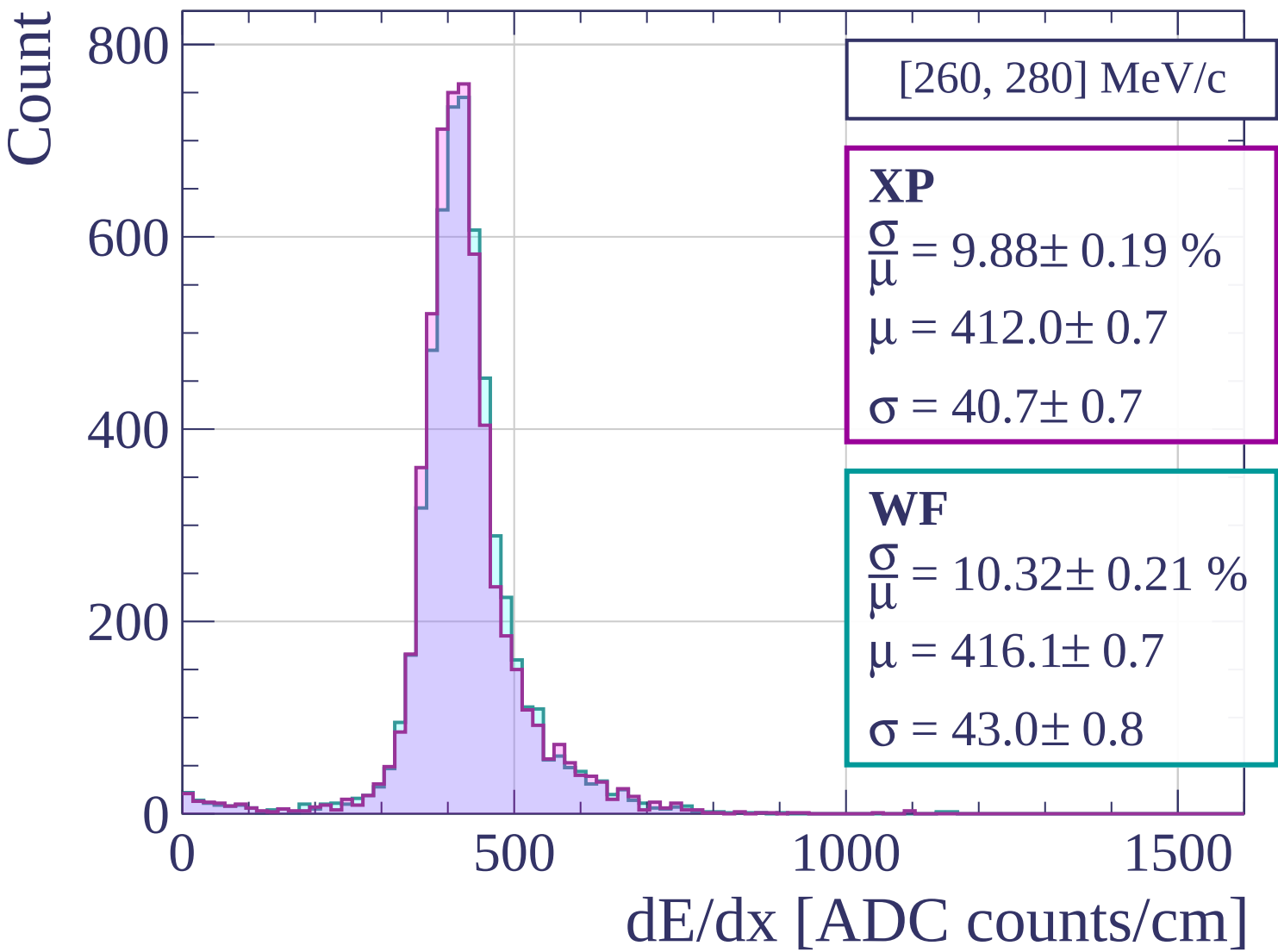
1000

1500

dE/dx [ADC counts/cm]







Count

[280, 300] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.78 \pm 0.20 \%$$

$$\mu = 405.9 \pm 0.7$$

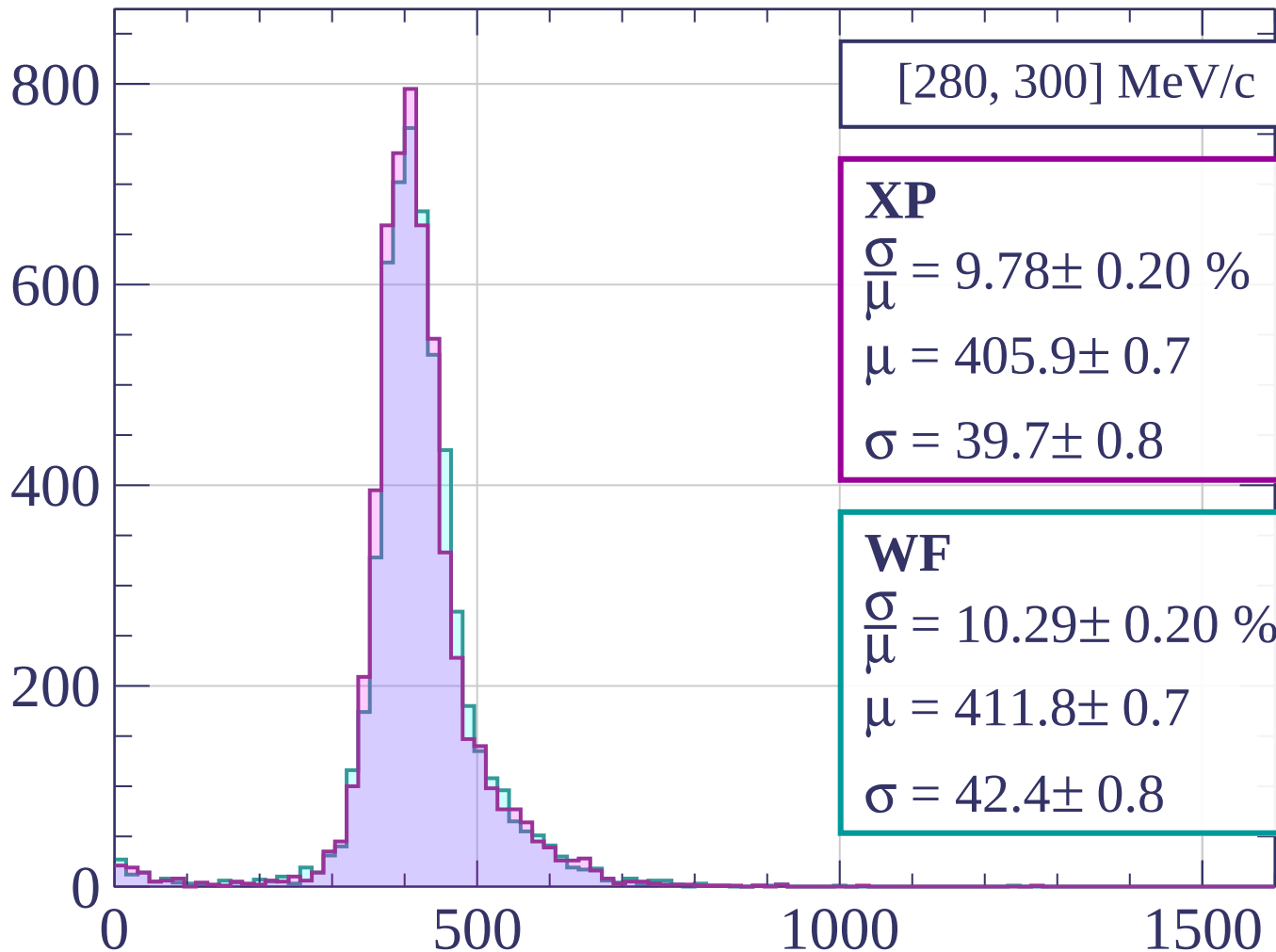
$$\sigma = 39.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.29 \pm 0.20 \%$$

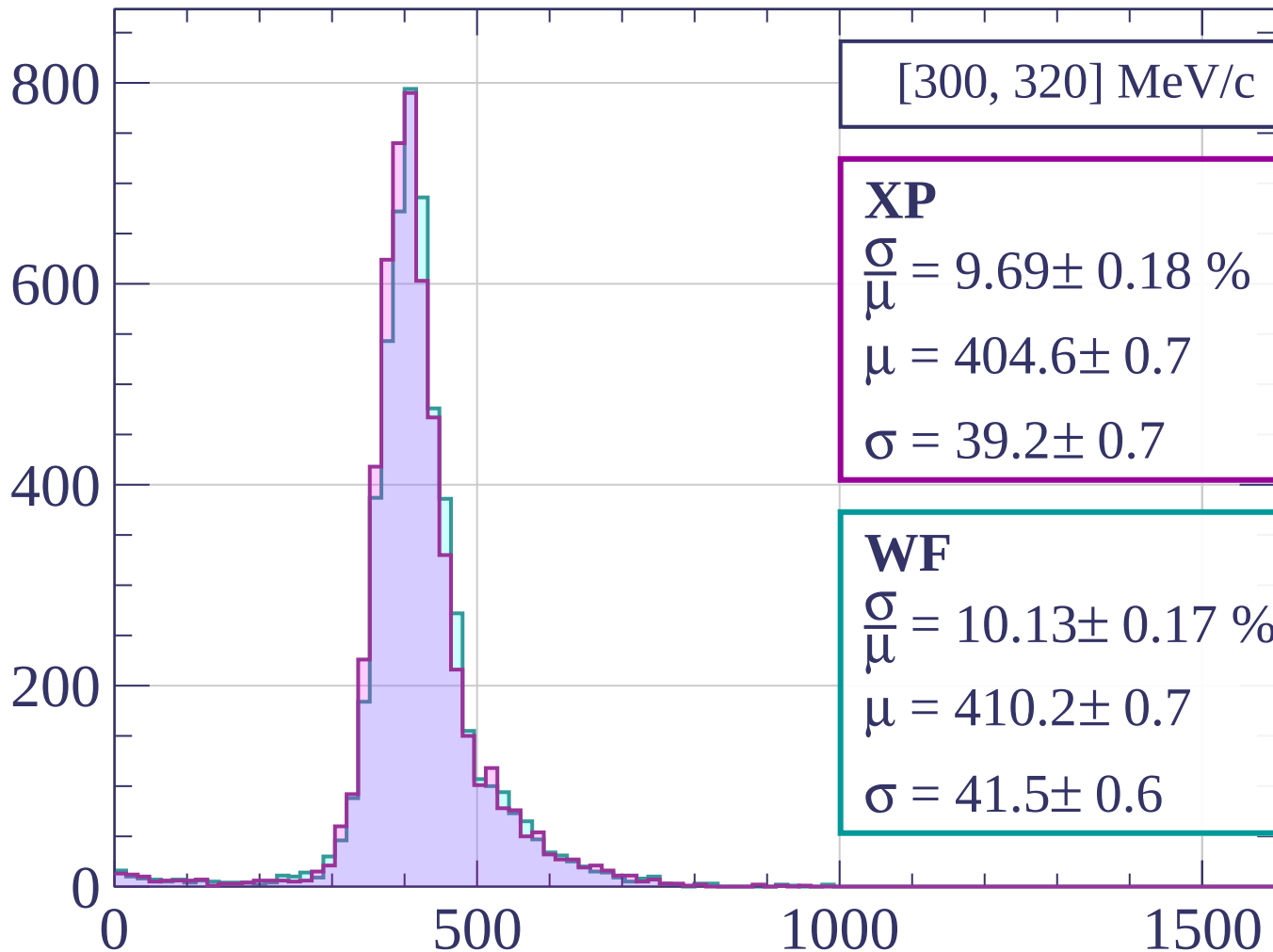
$$\mu = 411.8 \pm 0.7$$

$$\sigma = 42.4 \pm 0.8$$



dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

[320, 340] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.46 \pm 0.20 \%$$

$$\mu = 402.3 \pm 0.7$$

$$\sigma = 38.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.56 \pm 0.21 \%$$

$$\mu = 407.0 \pm 0.7$$

$$\sigma = 38.9 \pm 0.8$$

800

600

400

200

0

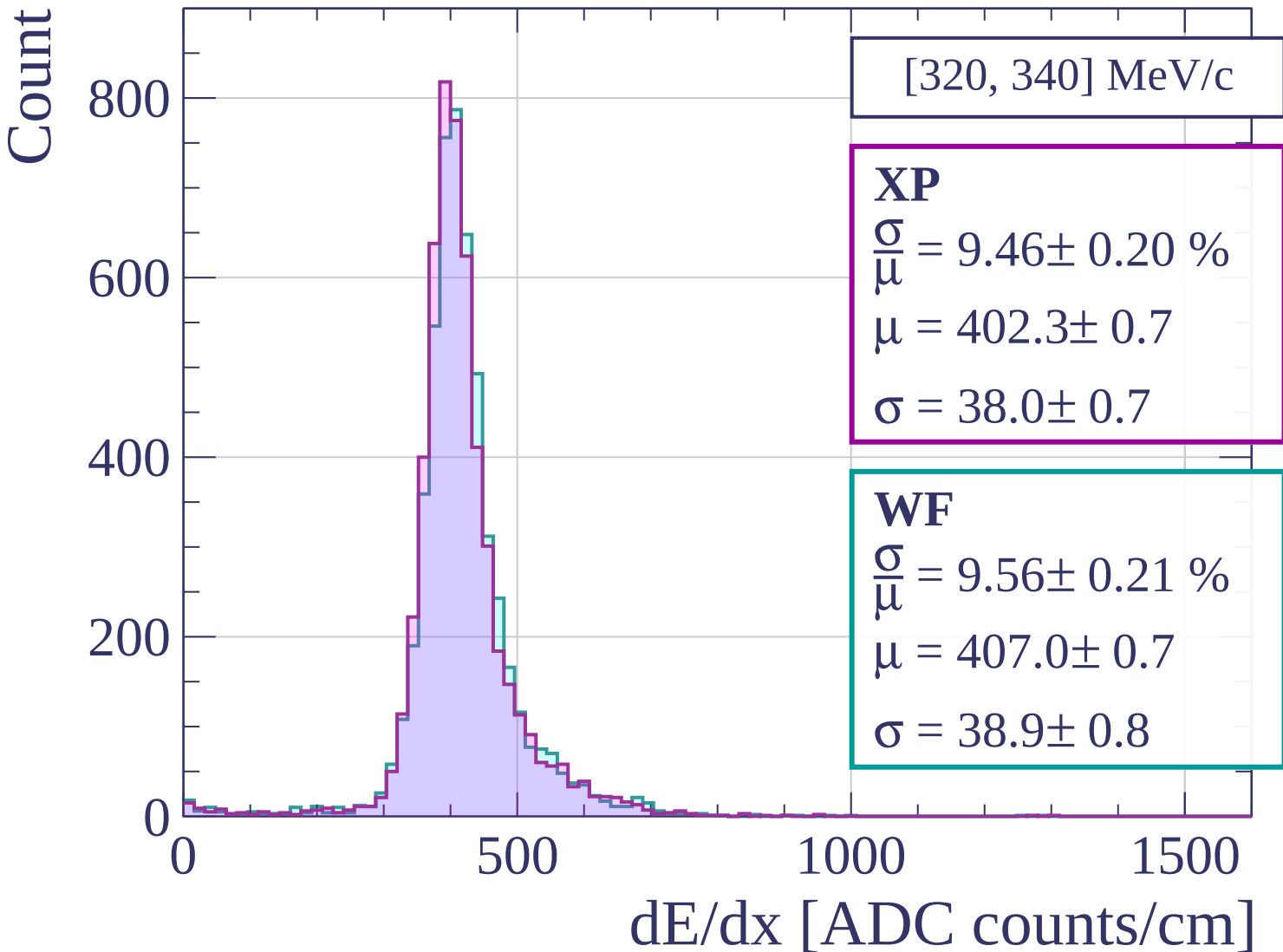
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[340, 360] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.46 \pm 0.17 \%$$

$$\mu = 401.0 \pm 0.6$$

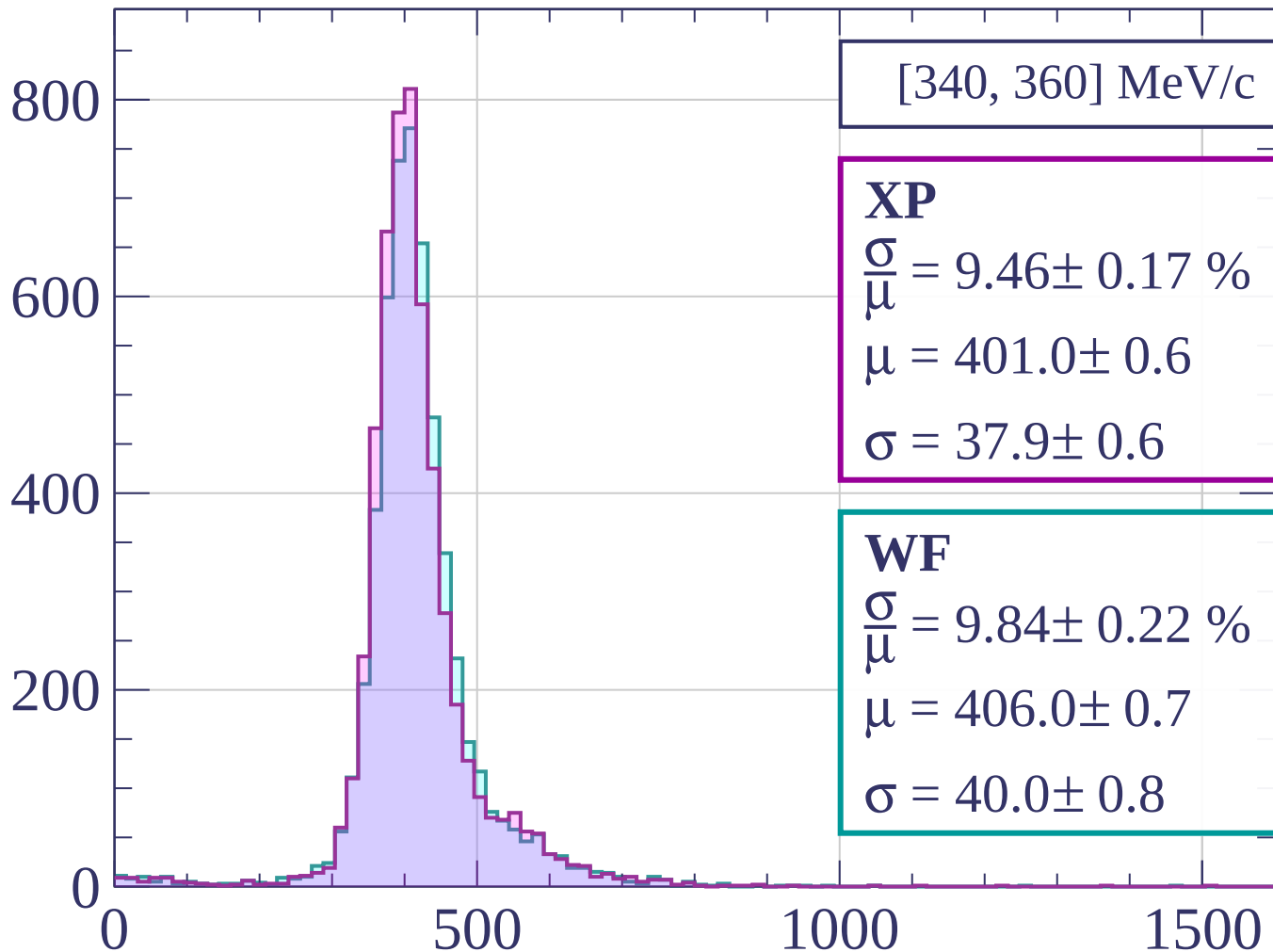
$$\sigma = 37.9 \pm 0.6$$

WF

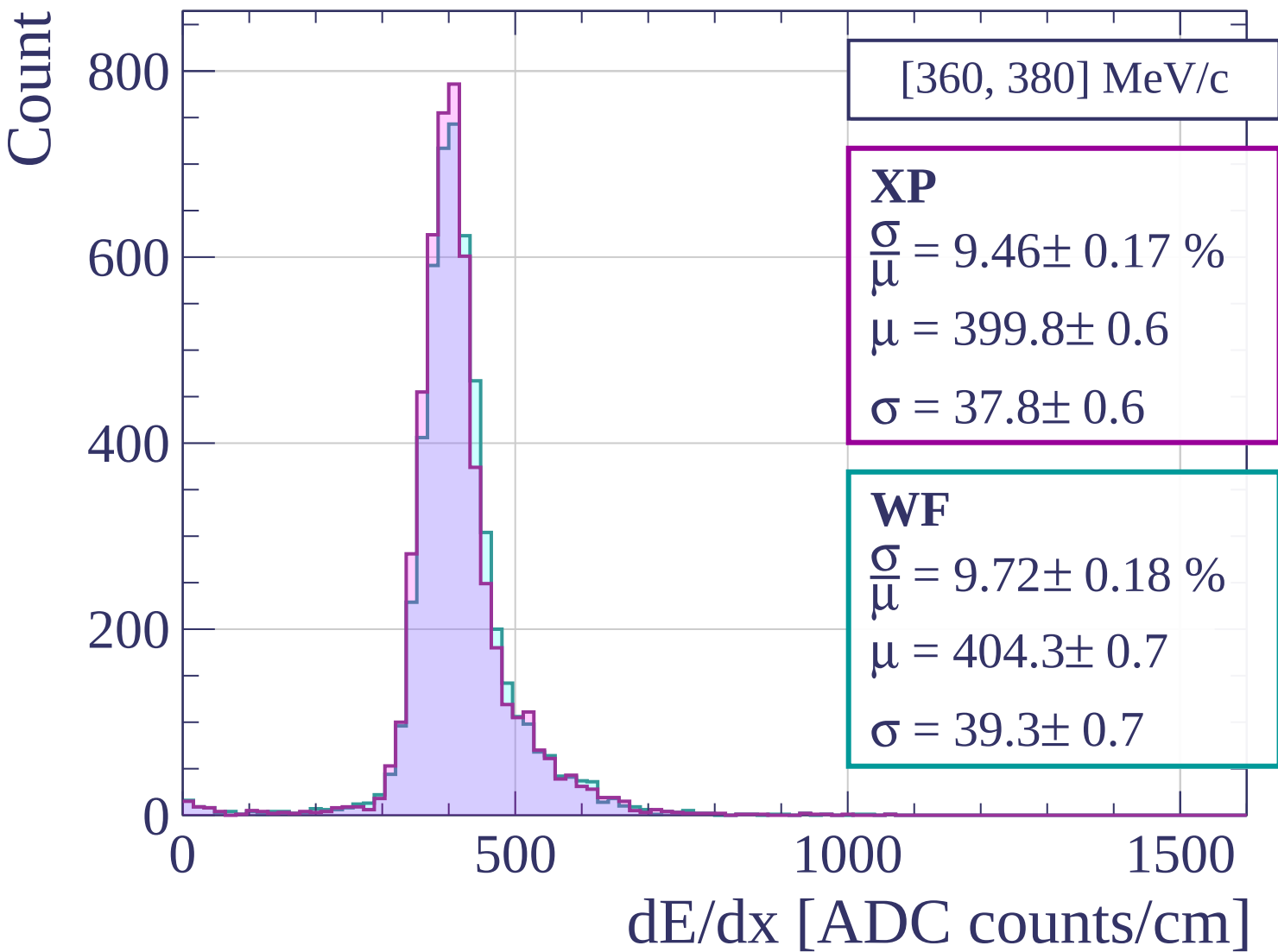
$$\frac{\sigma}{\mu} = 9.84 \pm 0.22 \%$$

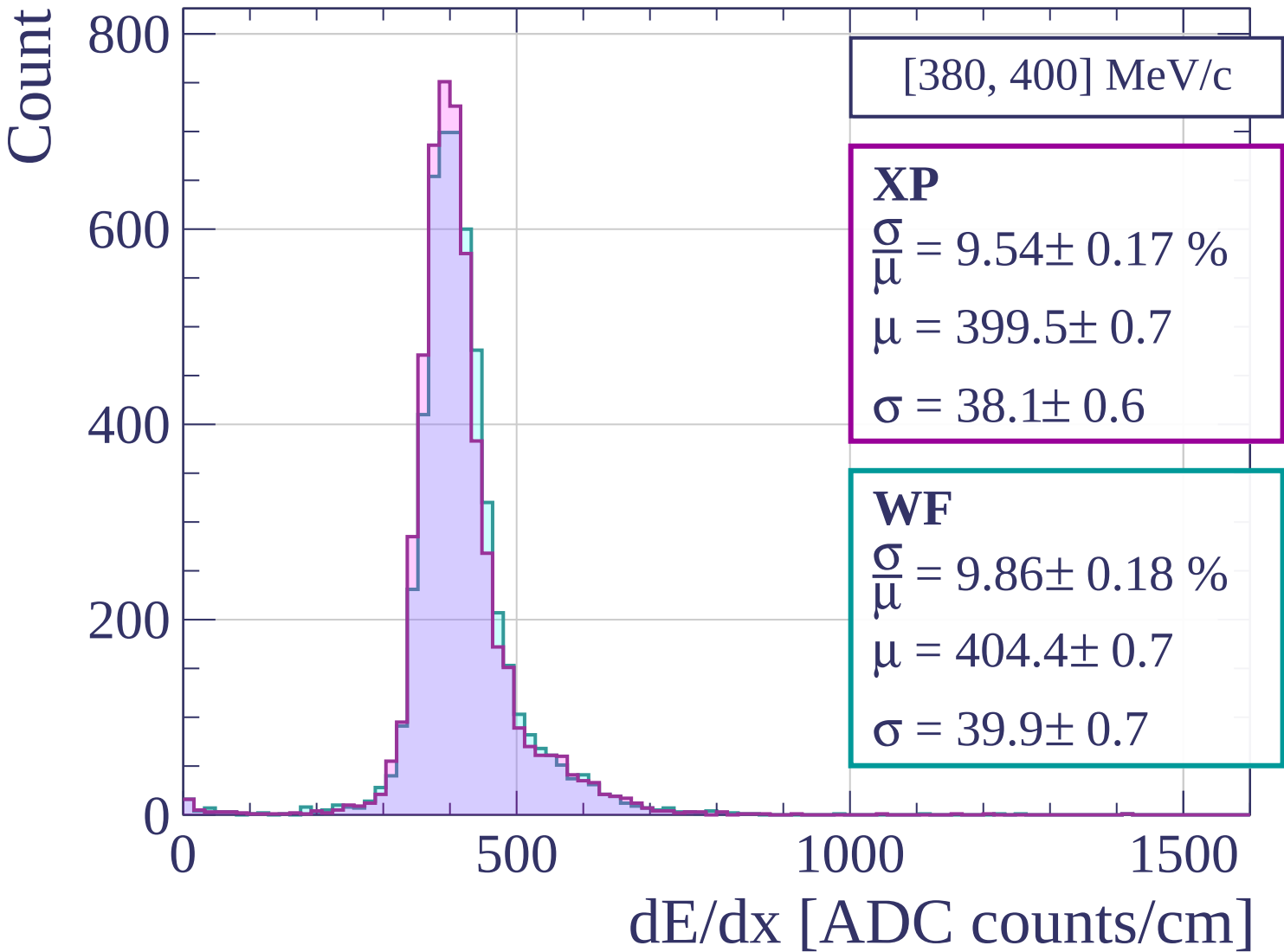
$$\mu = 406.0 \pm 0.7$$

$$\sigma = 40.0 \pm 0.8$$



dE/dx [ADC counts/cm]





Count

[400, 420] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.42 \pm 0.16 \%$$

$$\mu = 399.5 \pm 0.6$$

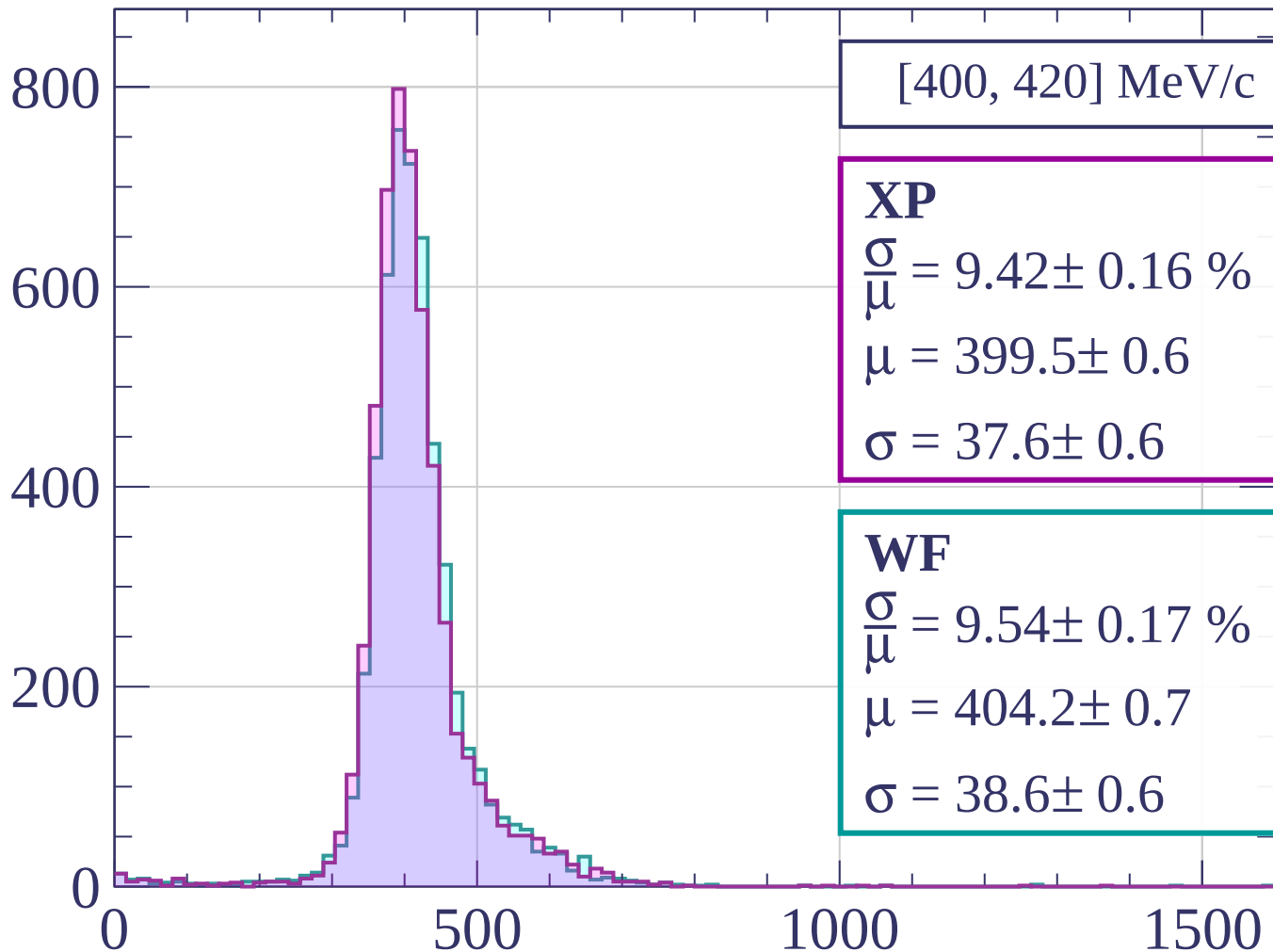
$$\sigma = 37.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.54 \pm 0.17 \%$$

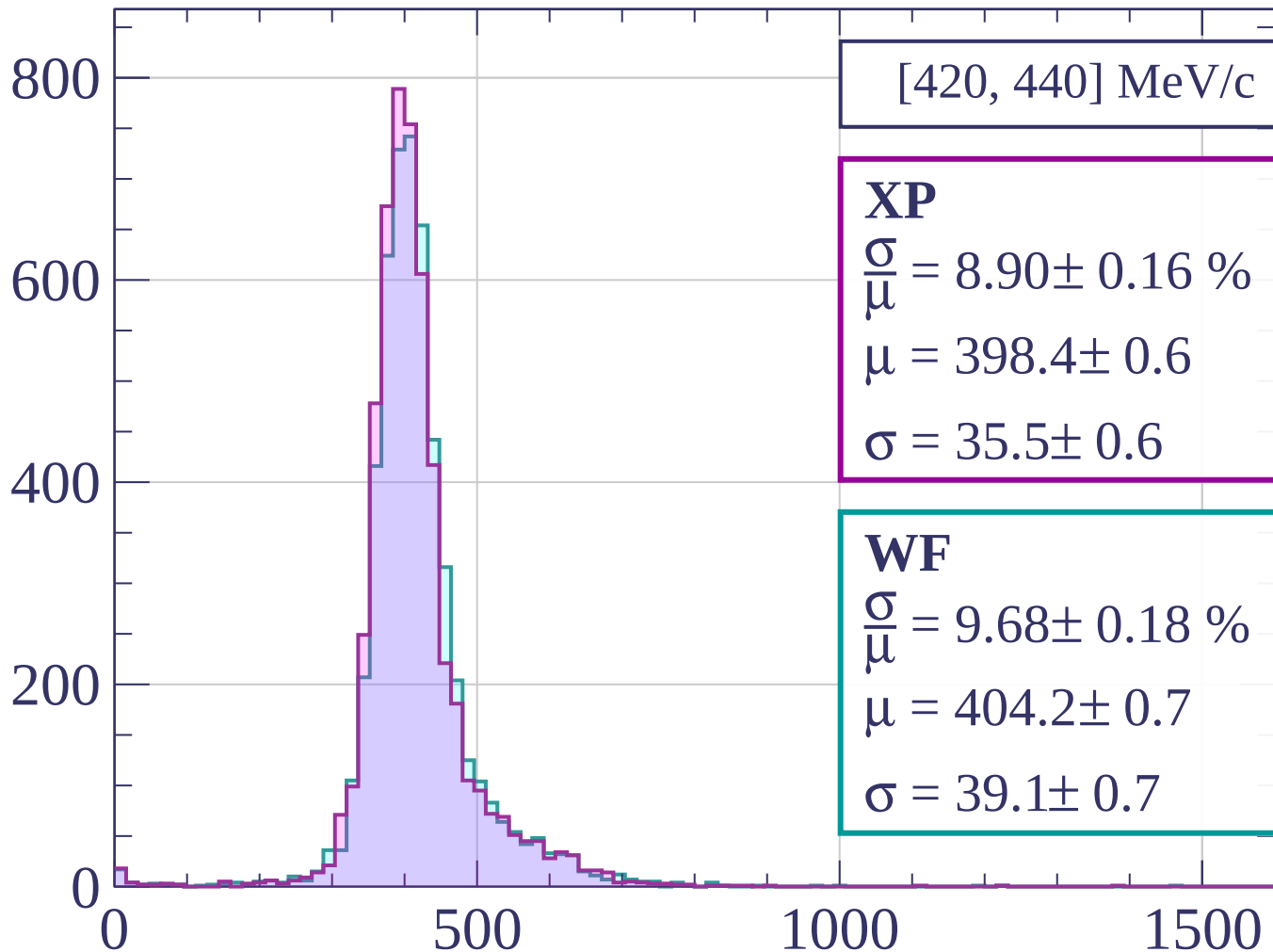
$$\mu = 404.2 \pm 0.7$$

$$\sigma = 38.6 \pm 0.6$$



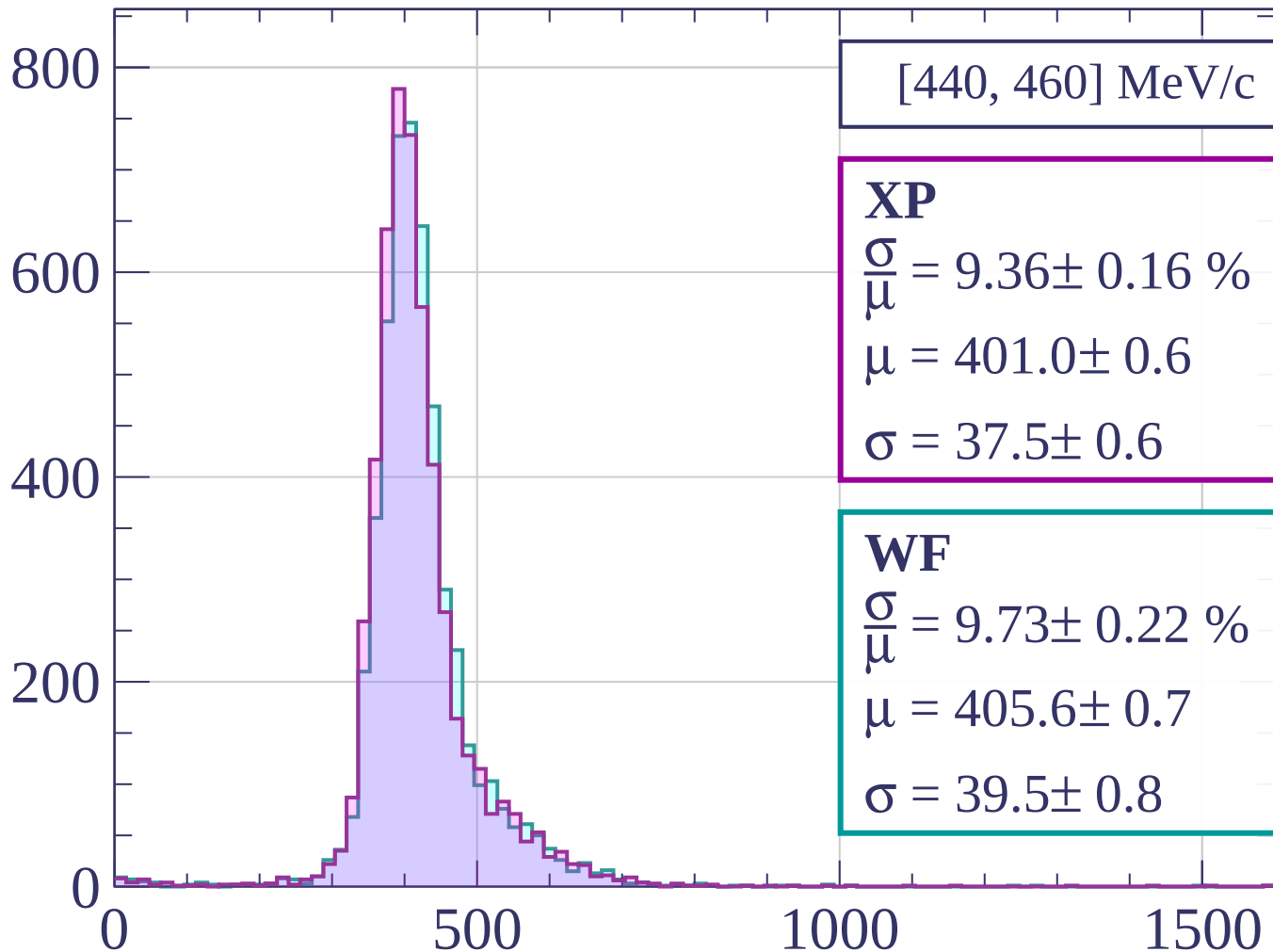
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

[460, 480] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.16 \pm 0.16 \%$$

$$\mu = 401.9 \pm 0.6$$

$$\sigma = 36.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.66 \pm 0.21 \%$$

$$\mu = 406.9 \pm 0.7$$

$$\sigma = 39.3 \pm 0.8$$

800

600

400

200

0

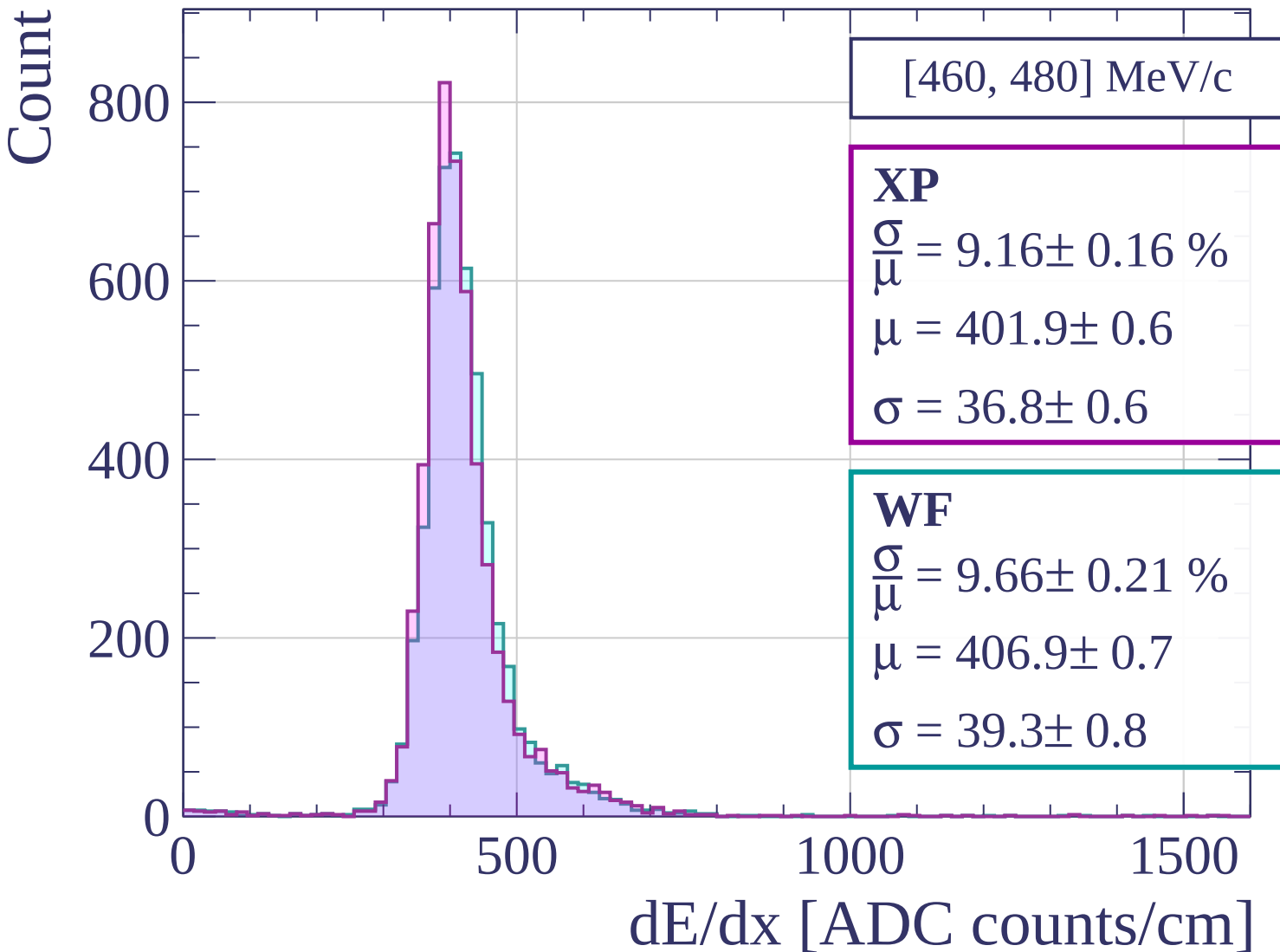
0

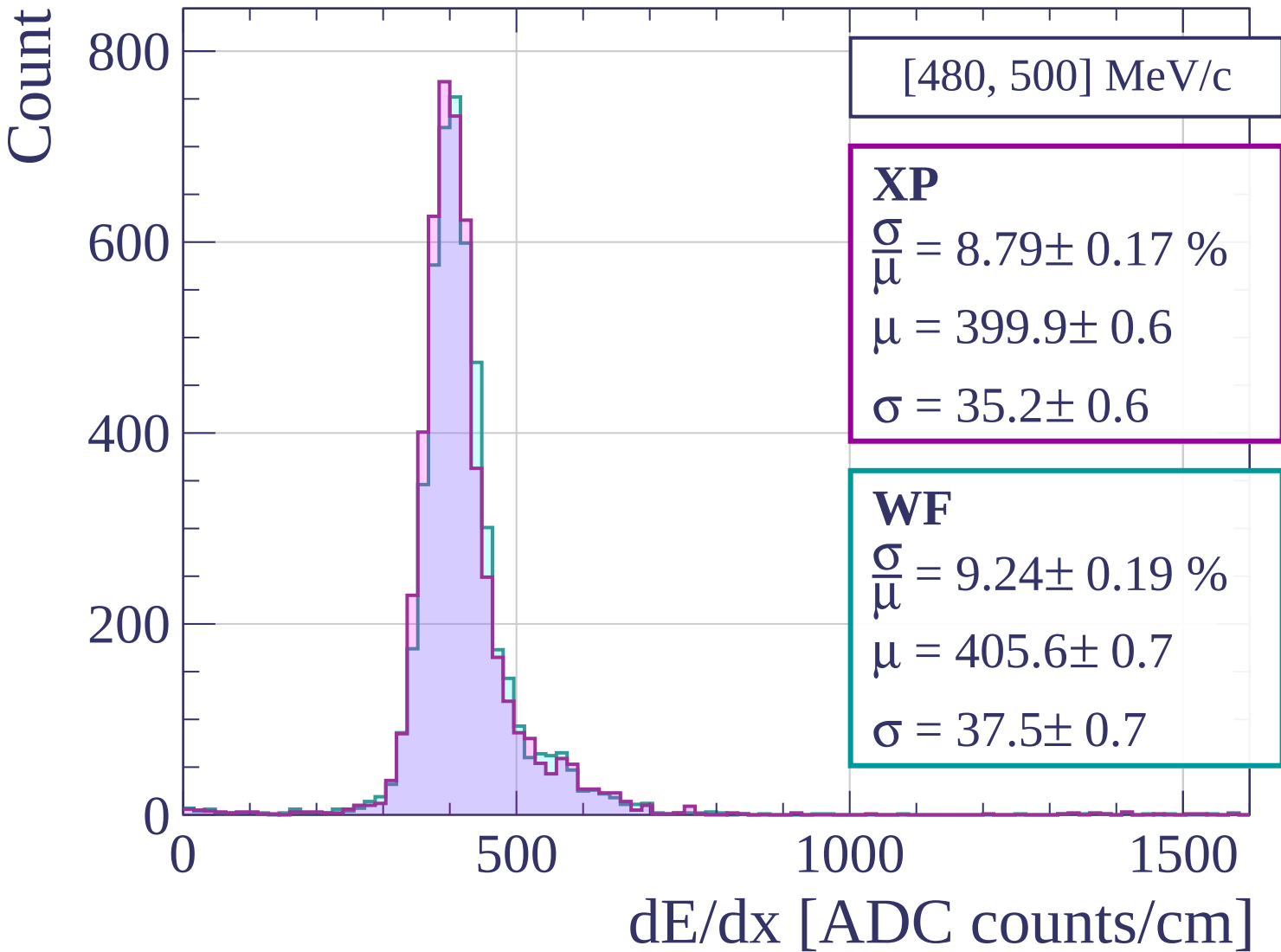
500

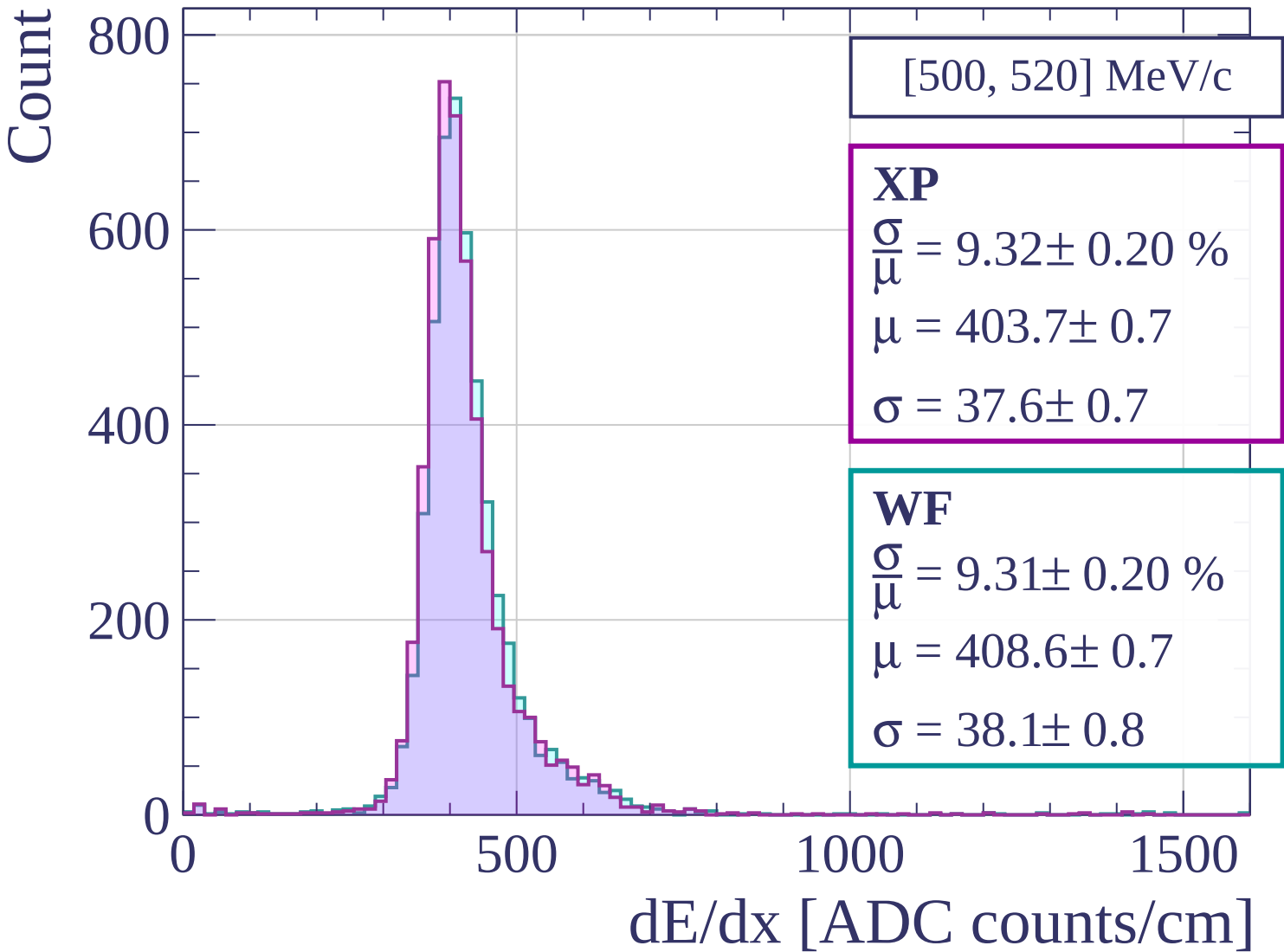
1000

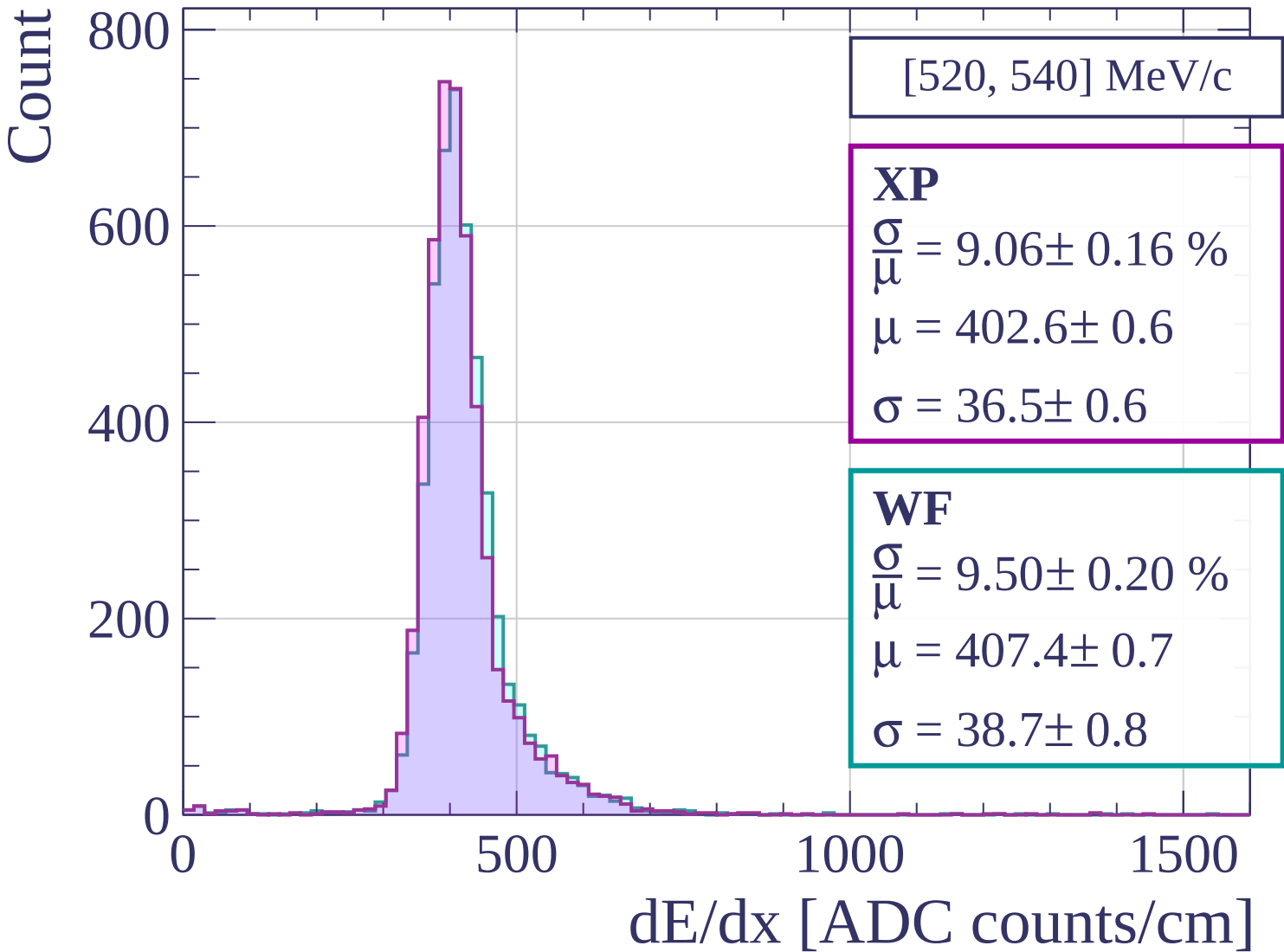
1500

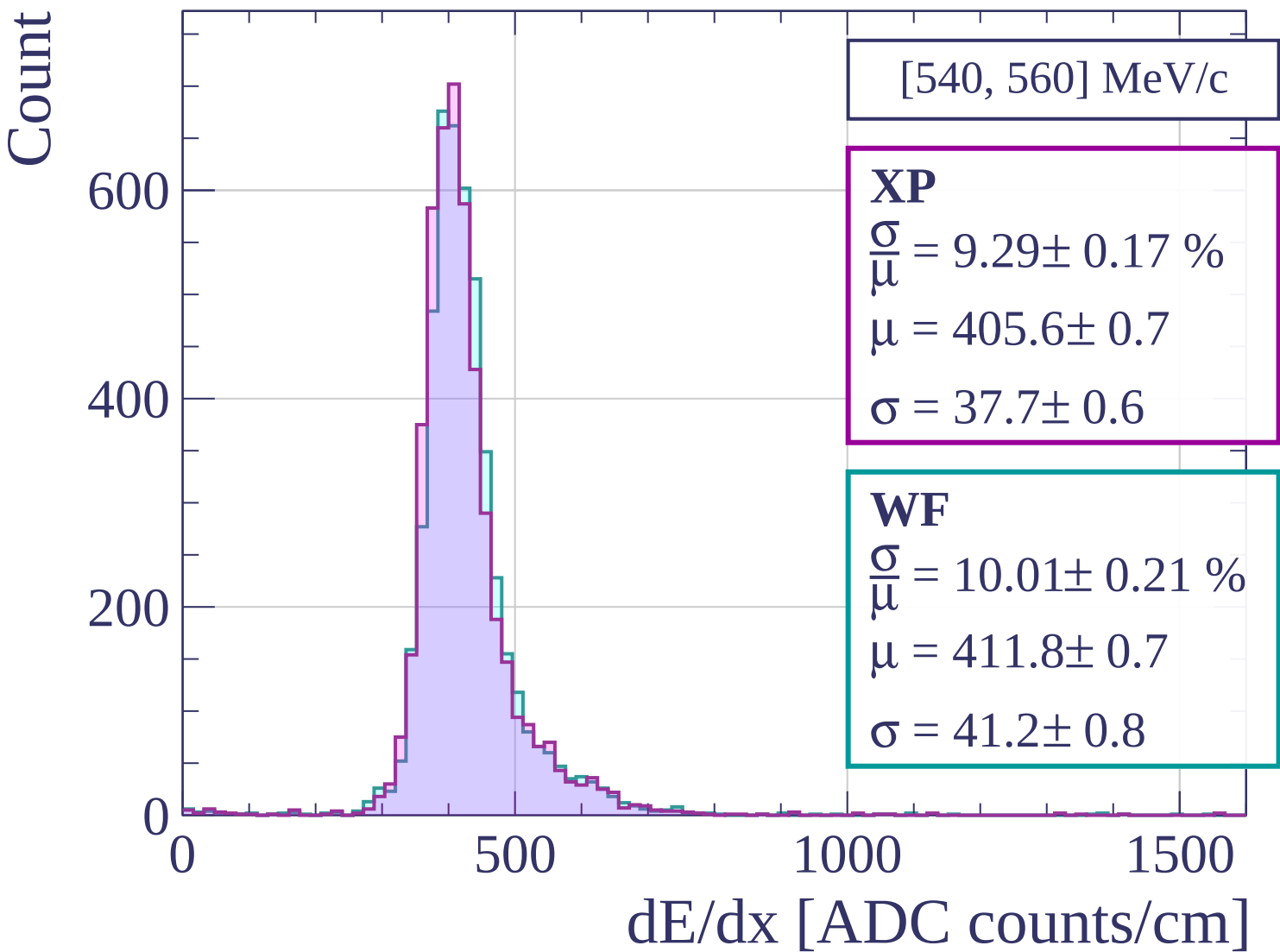
dE/dx [ADC counts/cm]











Count

[560, 580] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.53 \pm 0.21 \%$$

$$\mu = 405.6 \pm 0.7$$

$$\sigma = 38.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.92 \pm 0.21 \%$$

$$\mu = 412.0 \pm 0.7$$

$$\sigma = 40.9 \pm 0.8$$

600

400

200

0

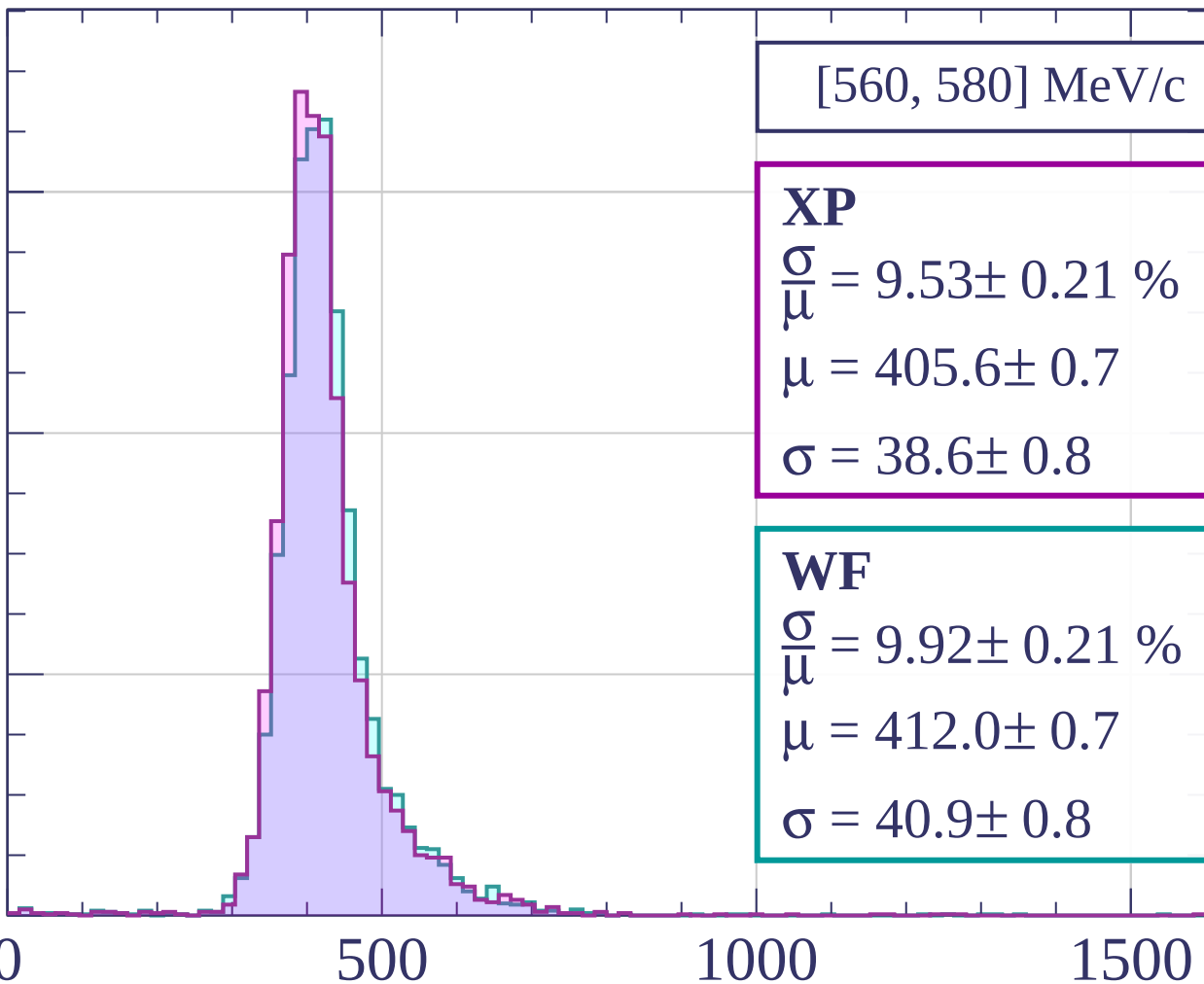
0

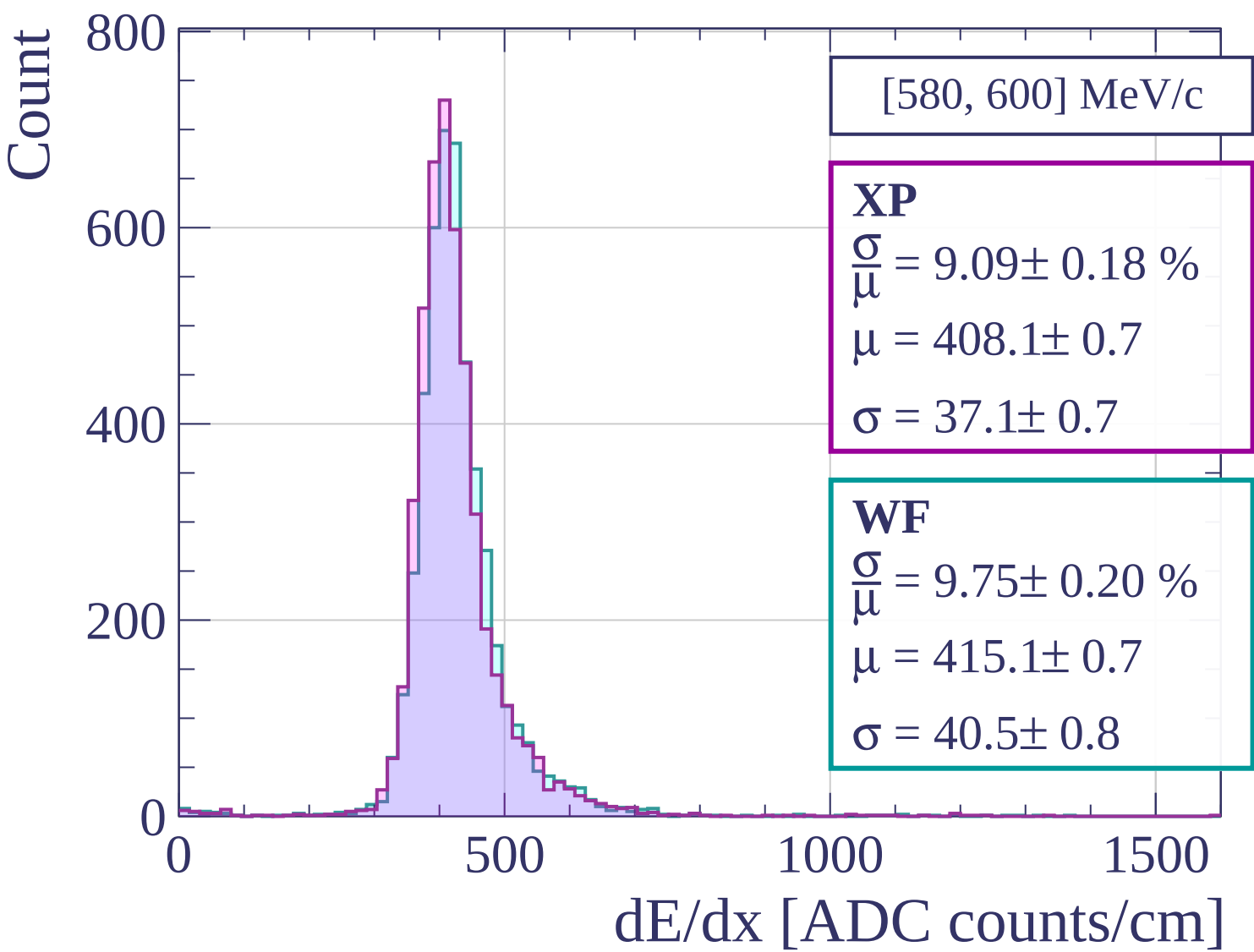
500

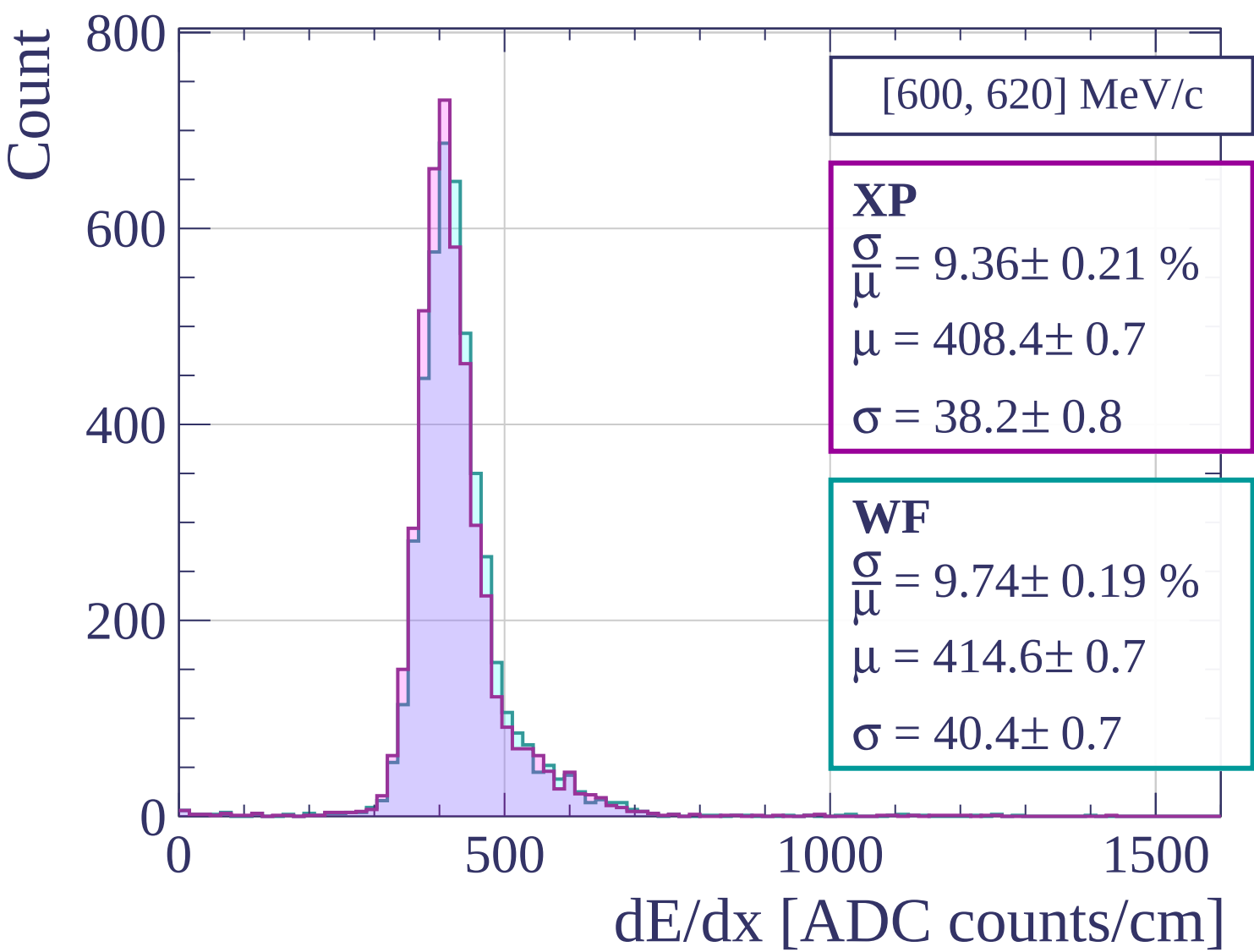
1000

1500

dE/dx [ADC counts/cm]







Count

[620, 640] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.33 \pm 0.20 \%$$

$$\mu = 409.1 \pm 0.7$$

$$\sigma = 38.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.71 \pm 0.19 \%$$

$$\mu = 415.5 \pm 0.7$$

$$\sigma = 40.3 \pm 0.7$$

600

400

200

0

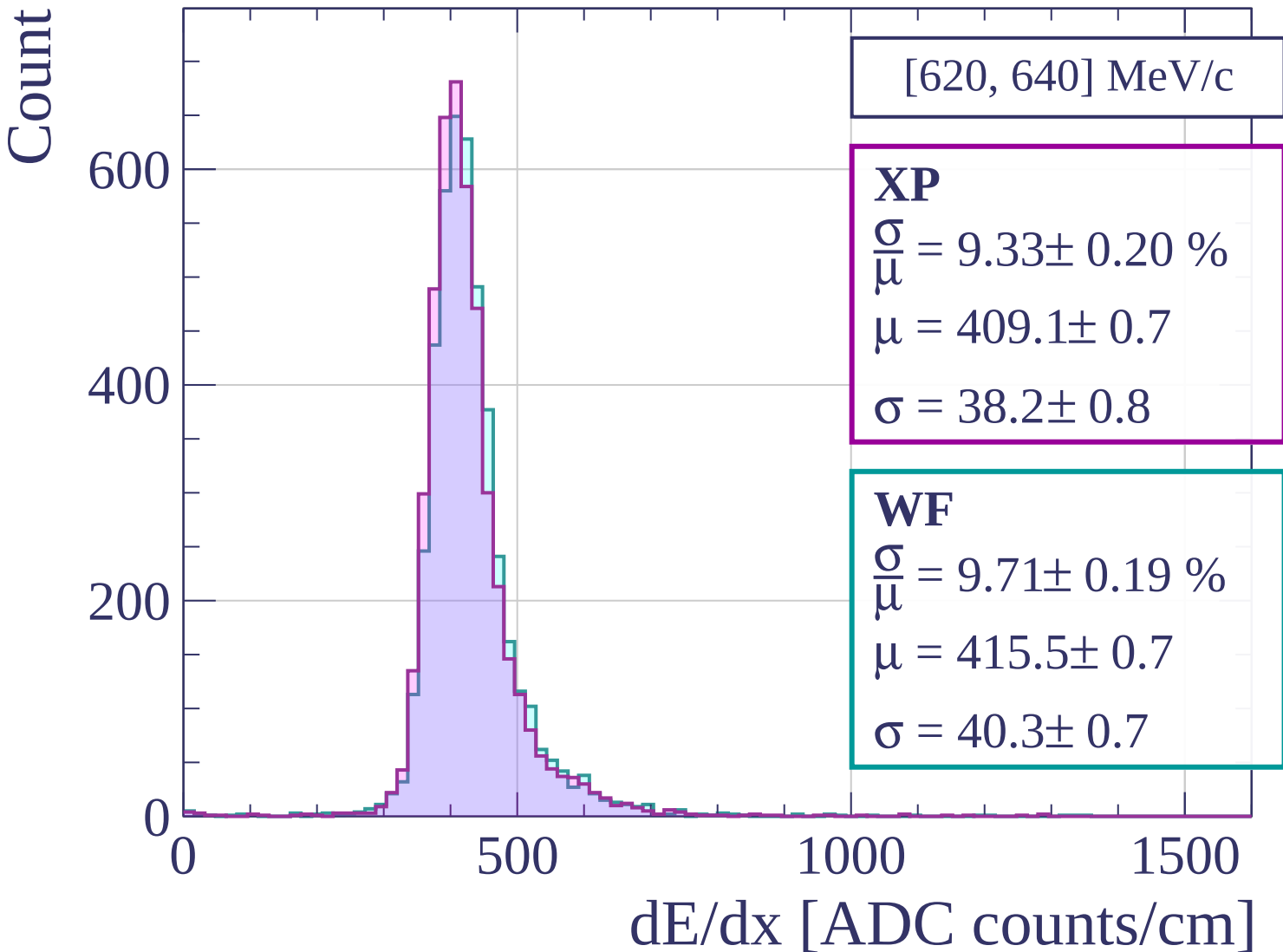
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[640, 660] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.18 \pm 0.19 \%$$

$$\mu = 410.4 \pm 0.7$$

$$\sigma = 37.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.42 \pm 0.18 \%$$

$$\mu = 417.0 \pm 0.7$$

$$\sigma = 39.3 \pm 0.7$$

600

400

200

0

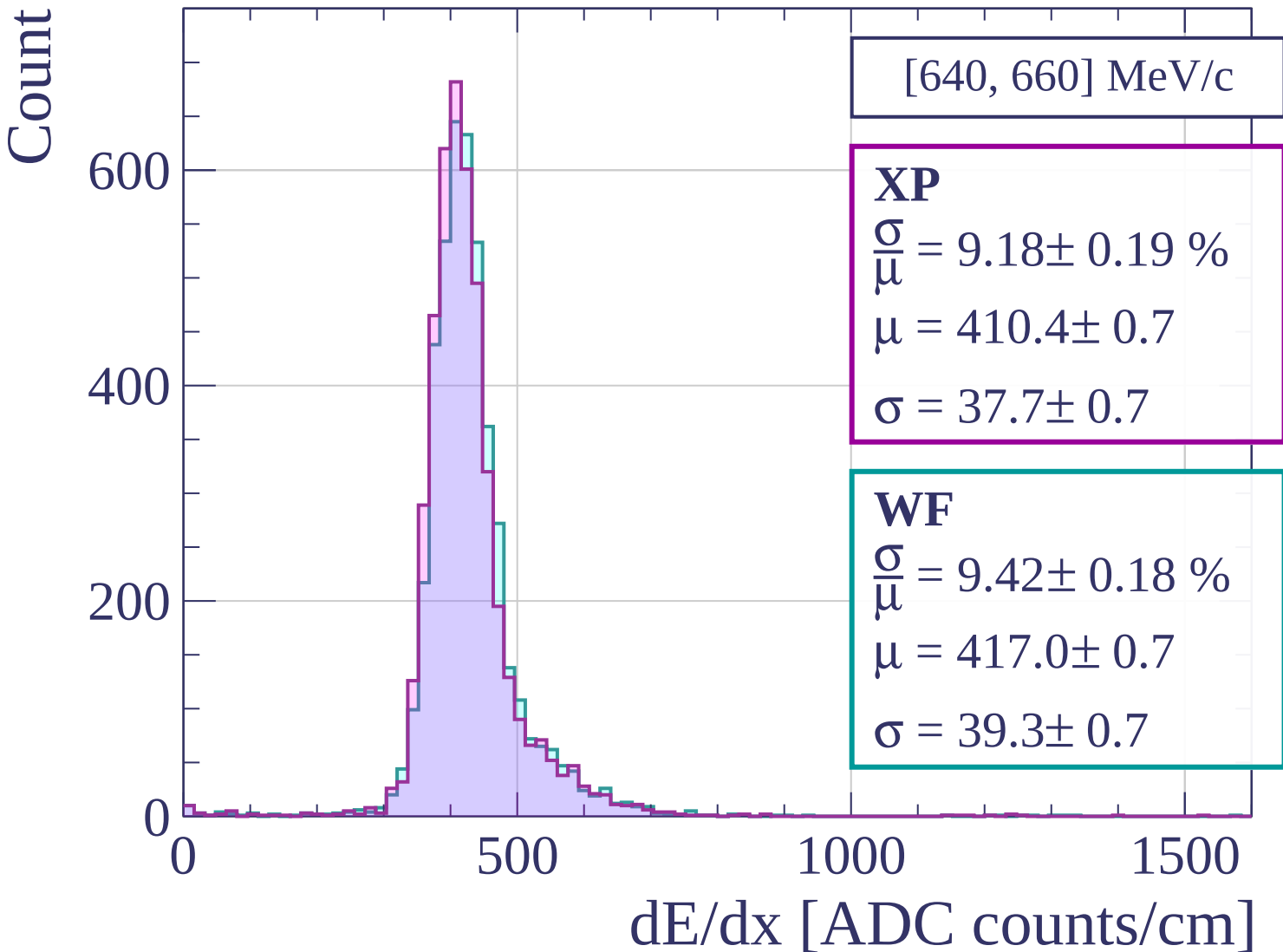
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[660, 680] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.99 \pm 0.19 \%$$

$$\mu = 412.2 \pm 0.7$$

$$\sigma = 37.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.61 \pm 0.20 \%$$

$$\mu = 418.5 \pm 0.8$$

$$\sigma = 40.2 \pm 0.7$$

600

400

200

0

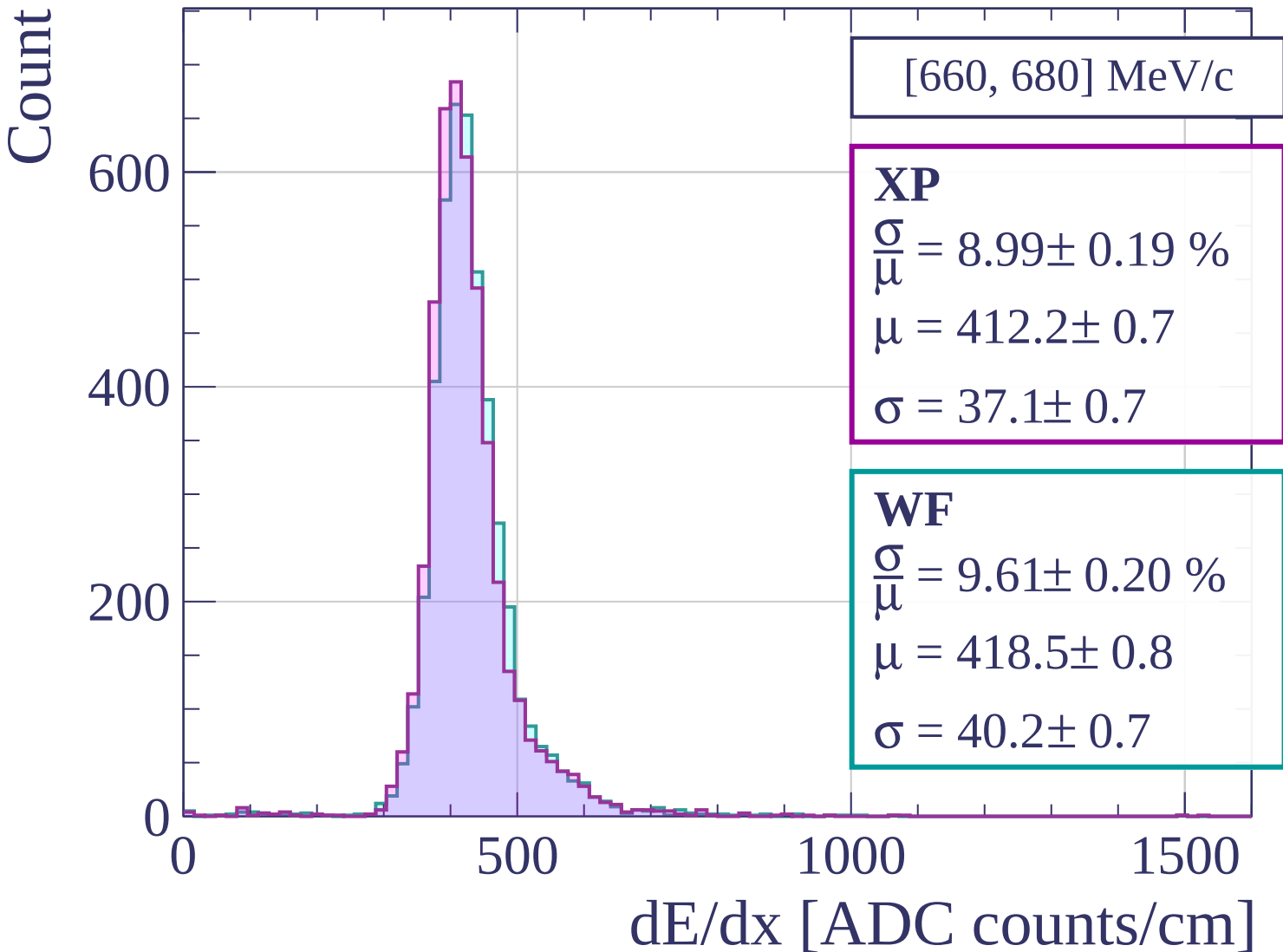
0

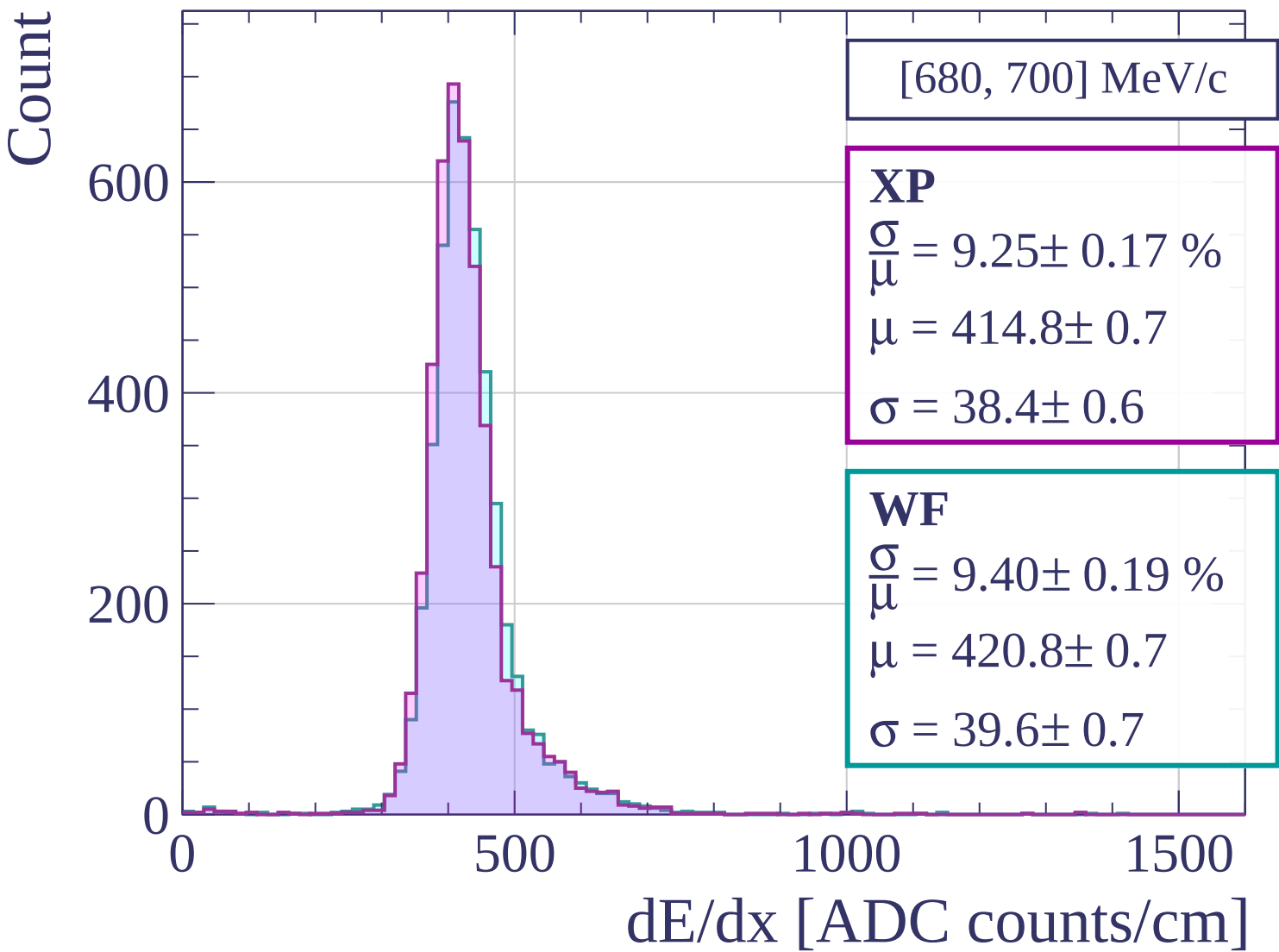
500

1000

1500

dE/dx [ADC counts/cm]





Count

[700, 720] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.74 \pm 0.18 \%$$

$$\mu = 414.6 \pm 0.7$$

$$\sigma = 36.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.20 \pm 0.21 \%$$

$$\mu = 421.3 \pm 0.7$$

$$\sigma = 38.8 \pm 0.8$$

600

400

200

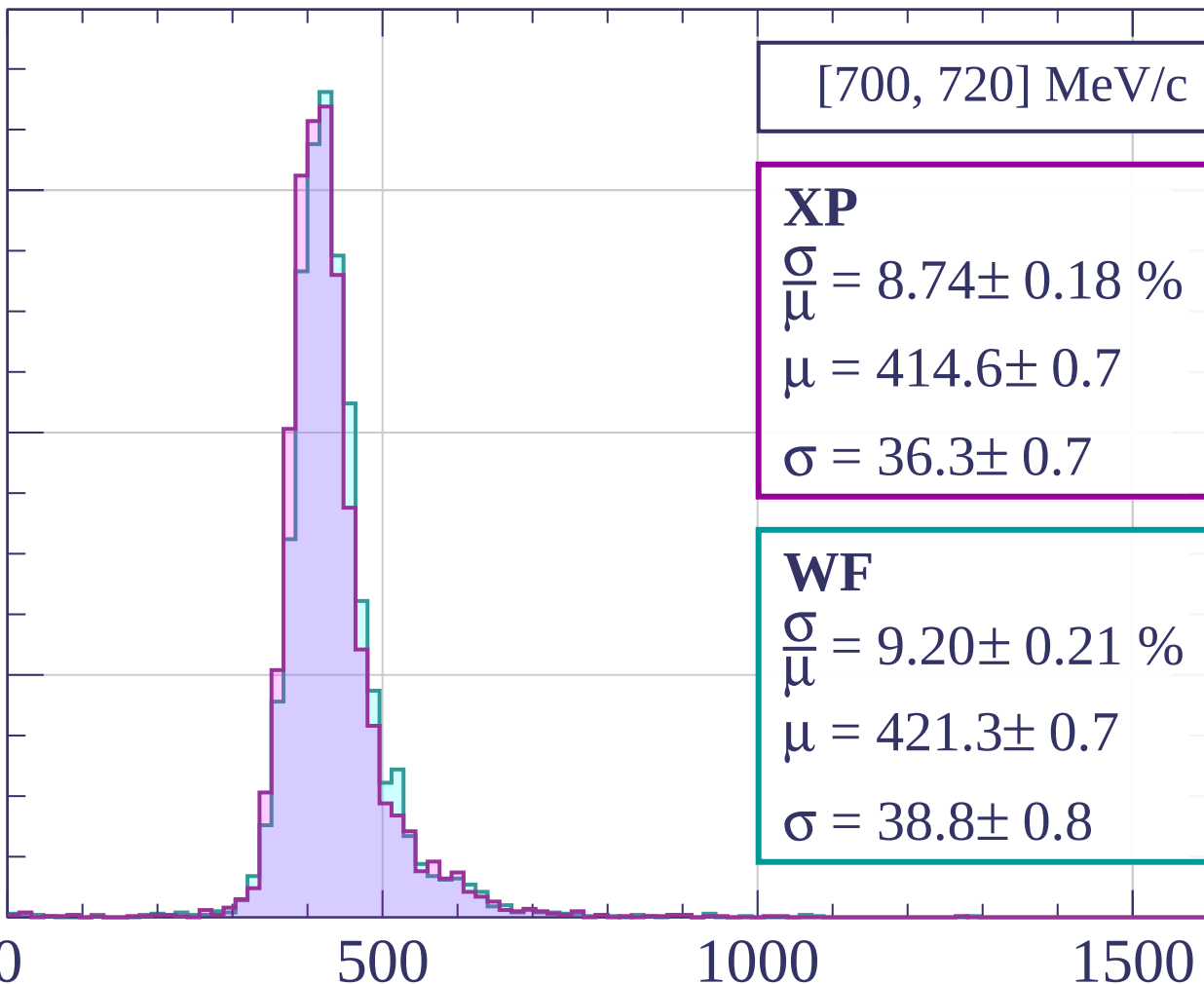
0

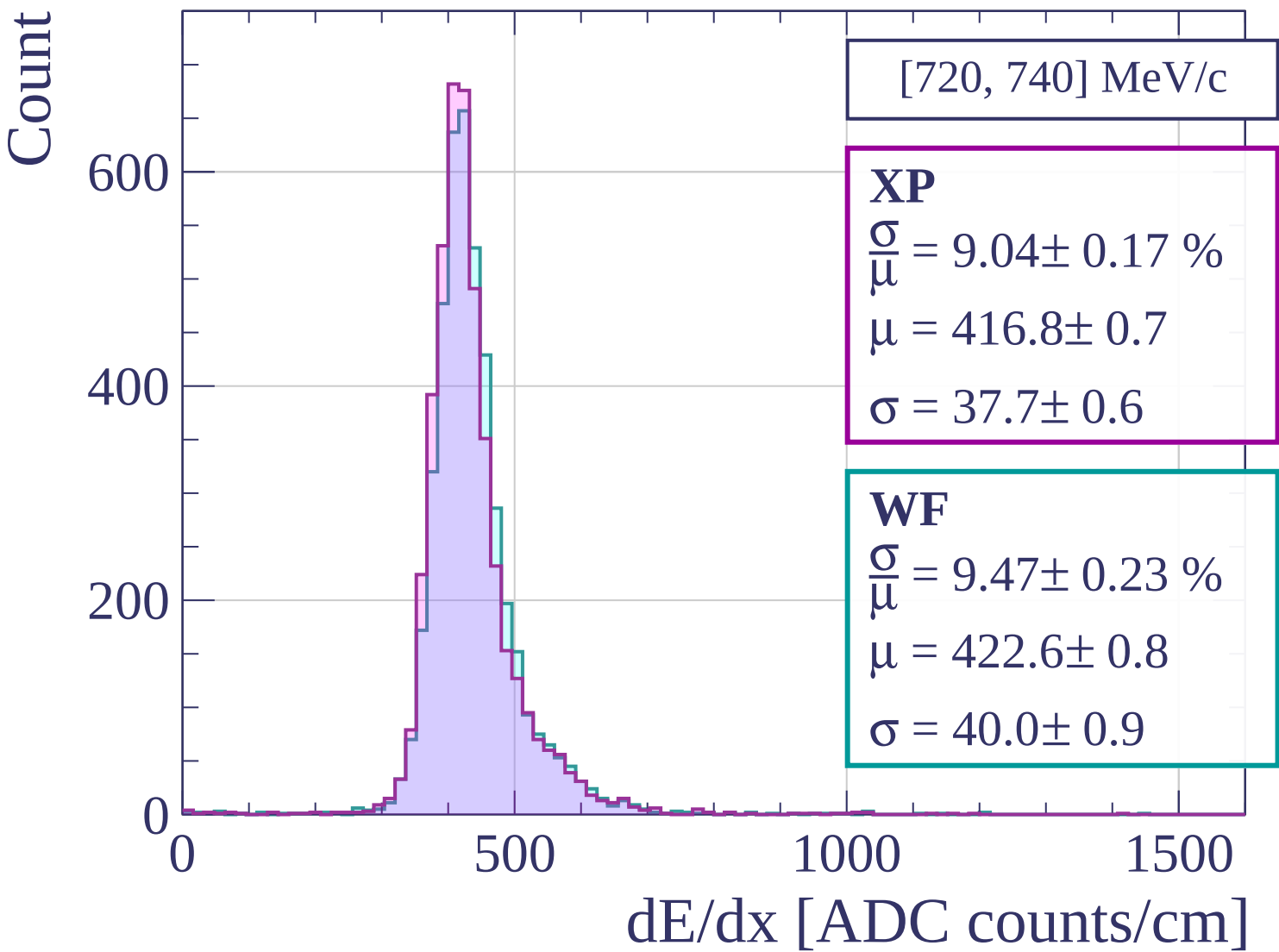
500

1000

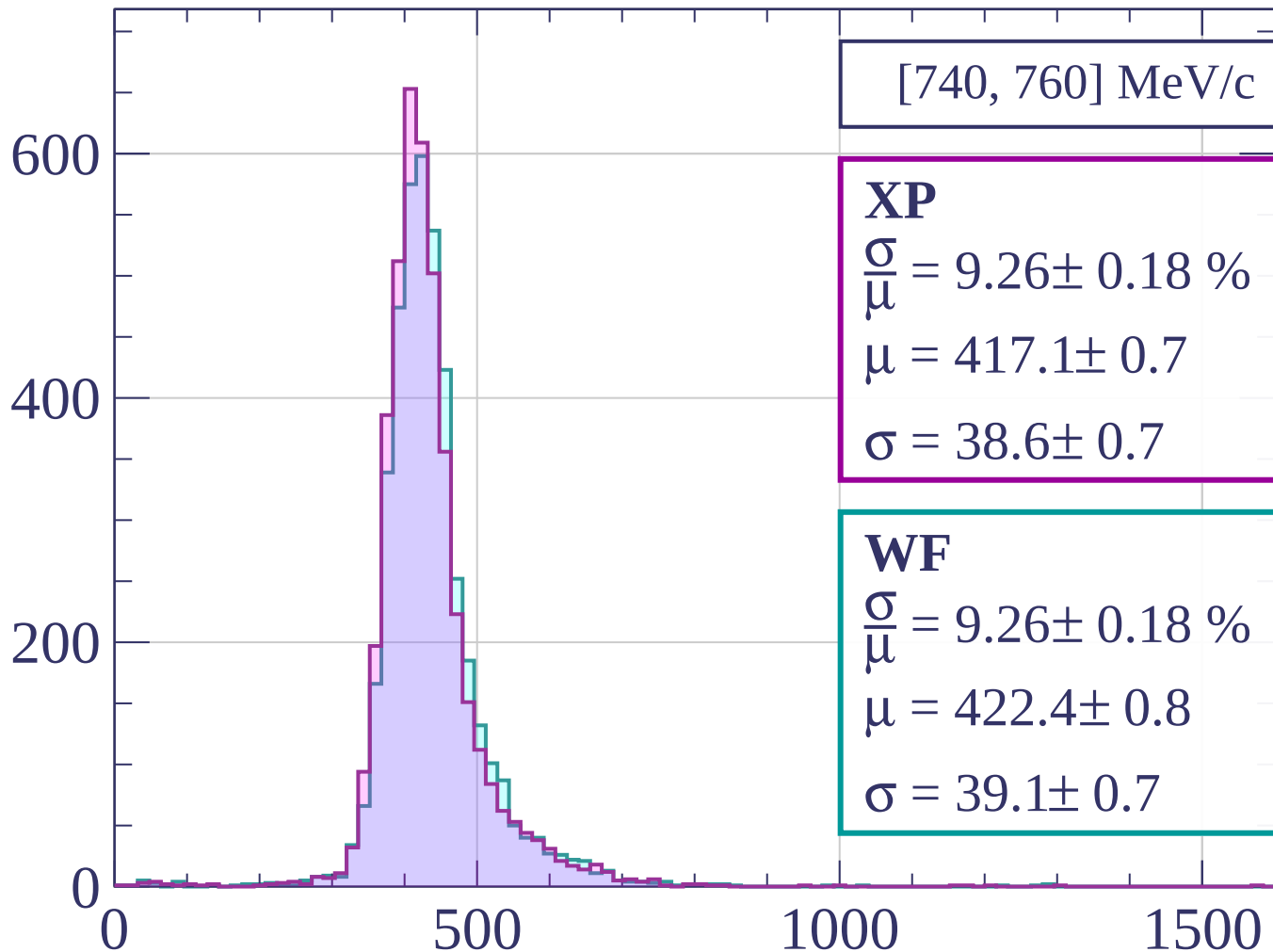
1500

dE/dx [ADC counts/cm]





Count



XP

$$\frac{\sigma}{\mu} = 9.26 \pm 0.18 \%$$

$$\mu = 417.1 \pm 0.7$$

$$\sigma = 38.6 \pm 0.7$$

WF

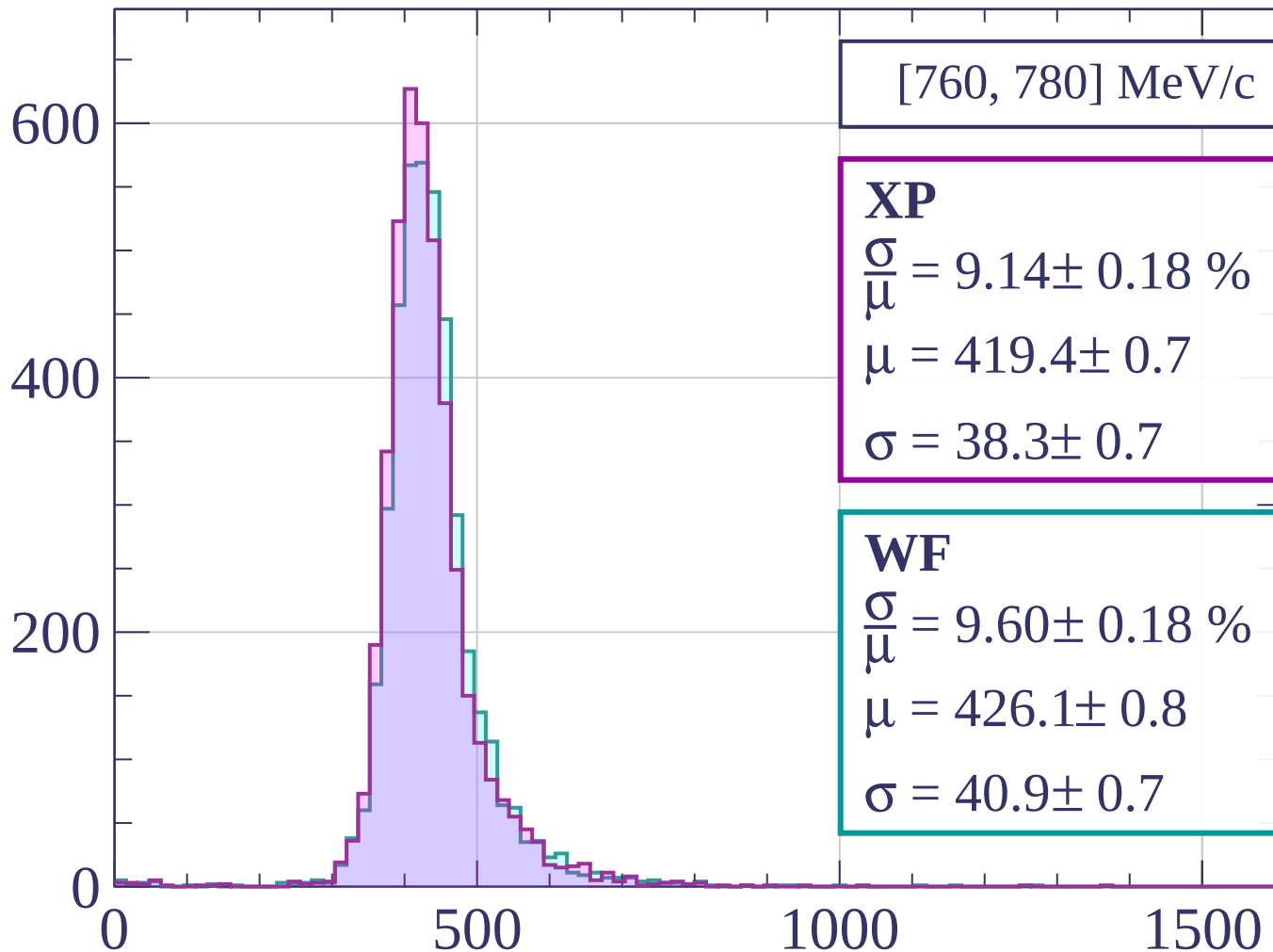
$$\frac{\sigma}{\mu} = 9.26 \pm 0.18 \%$$

$$\mu = 422.4 \pm 0.8$$

$$\sigma = 39.1 \pm 0.7$$

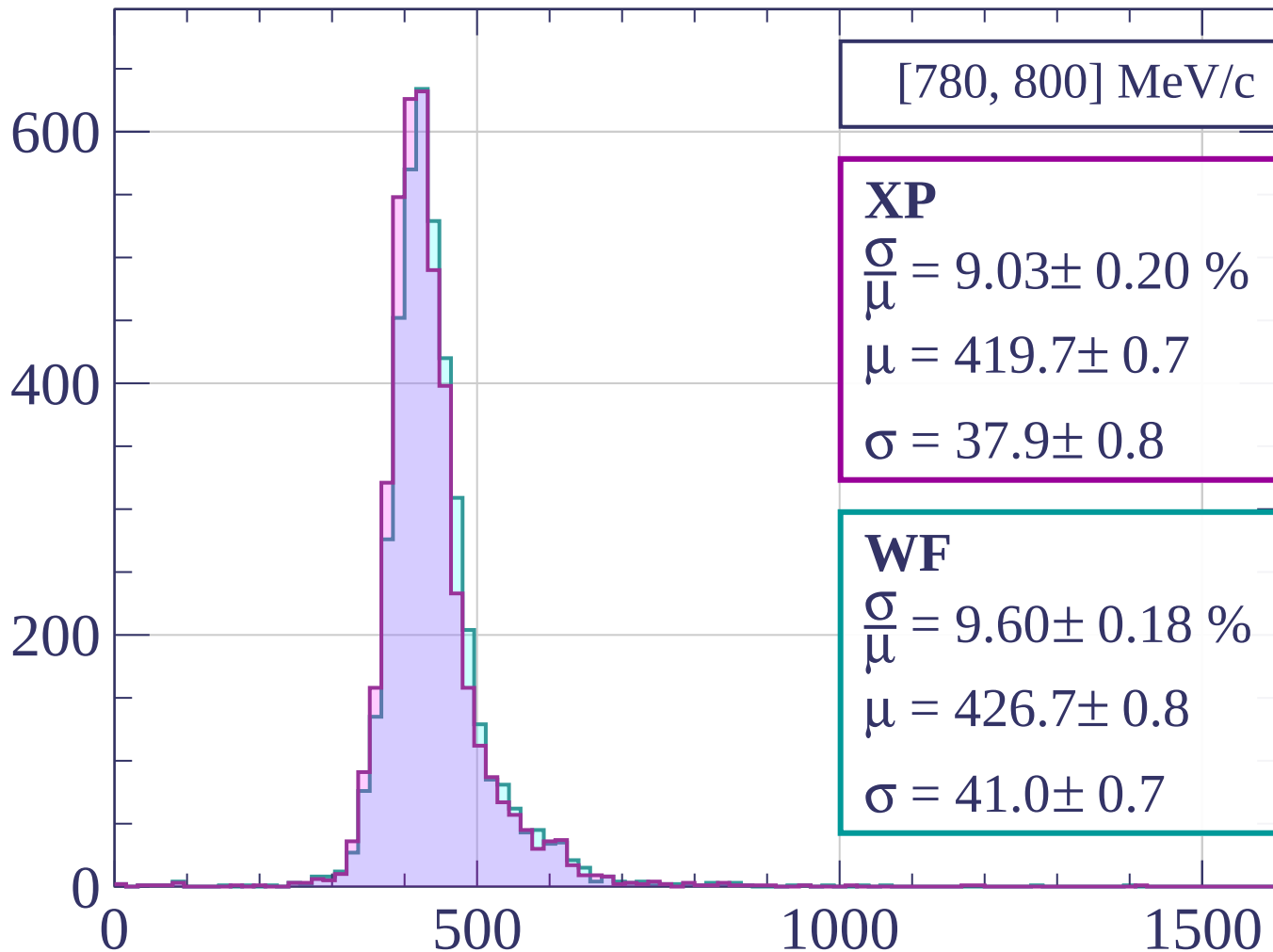
dE/dx [ADC counts/cm]

Count



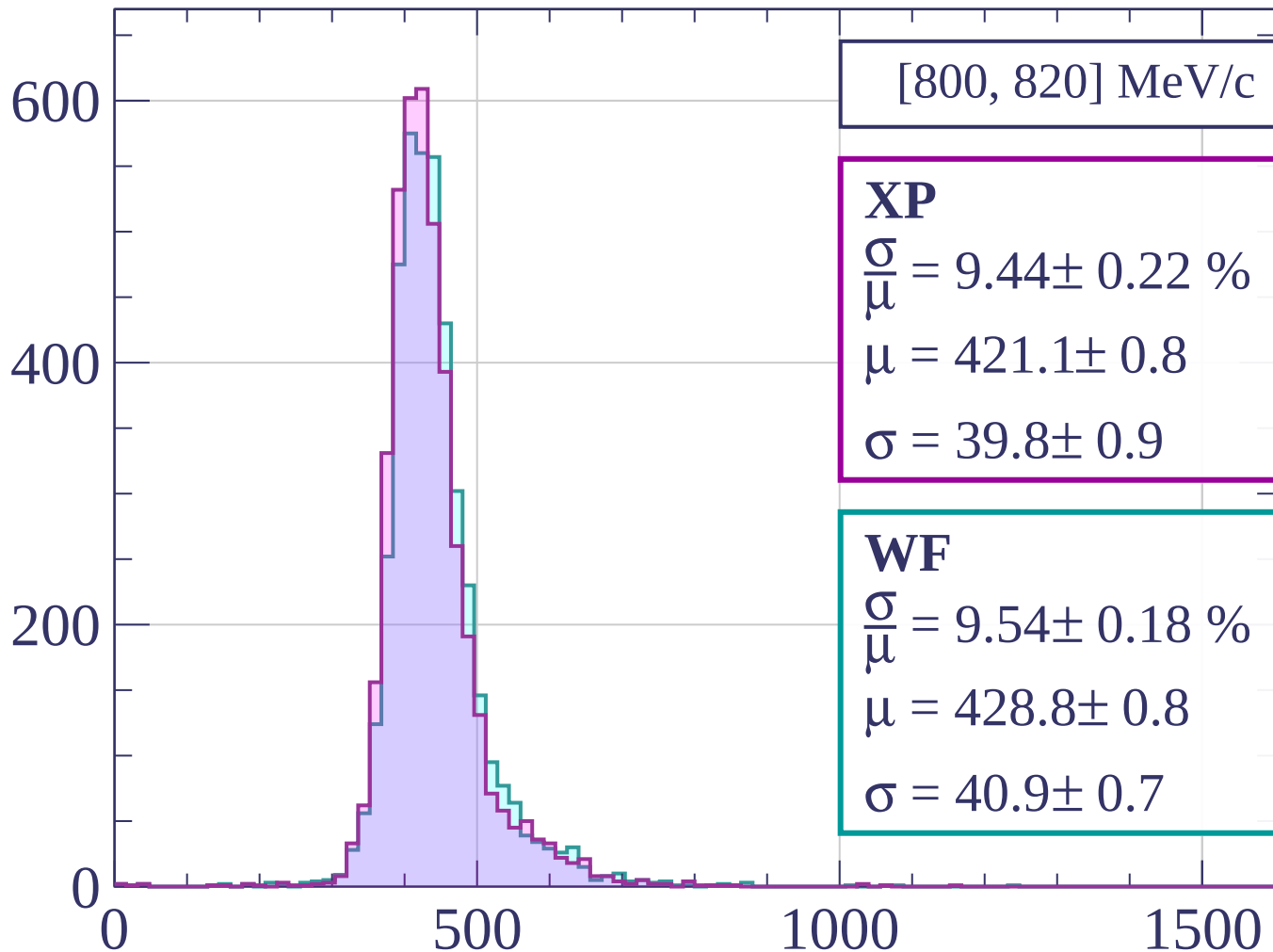
dE/dx [ADC counts/cm]

Count



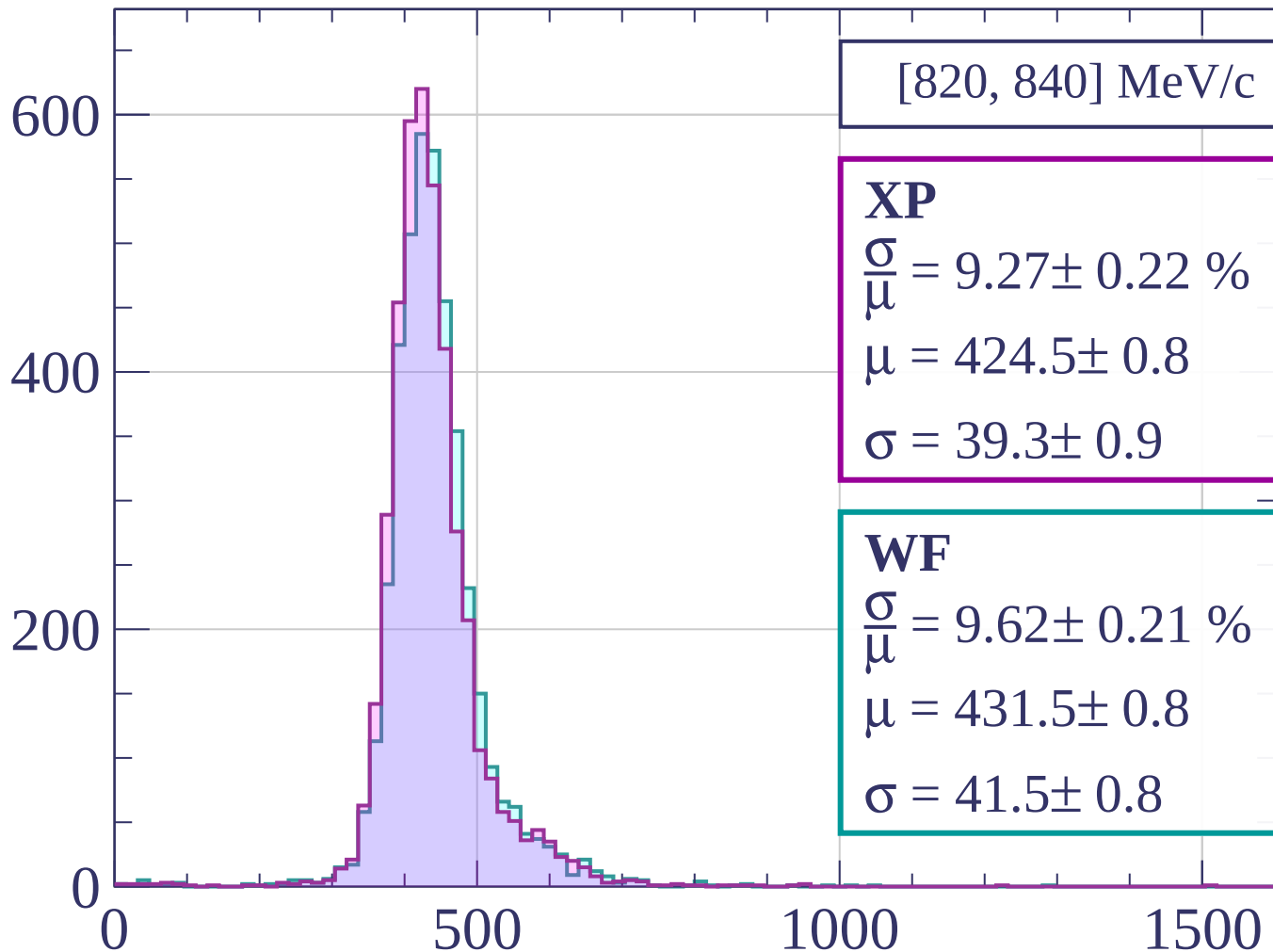
dE/dx [ADC counts/cm]

Count



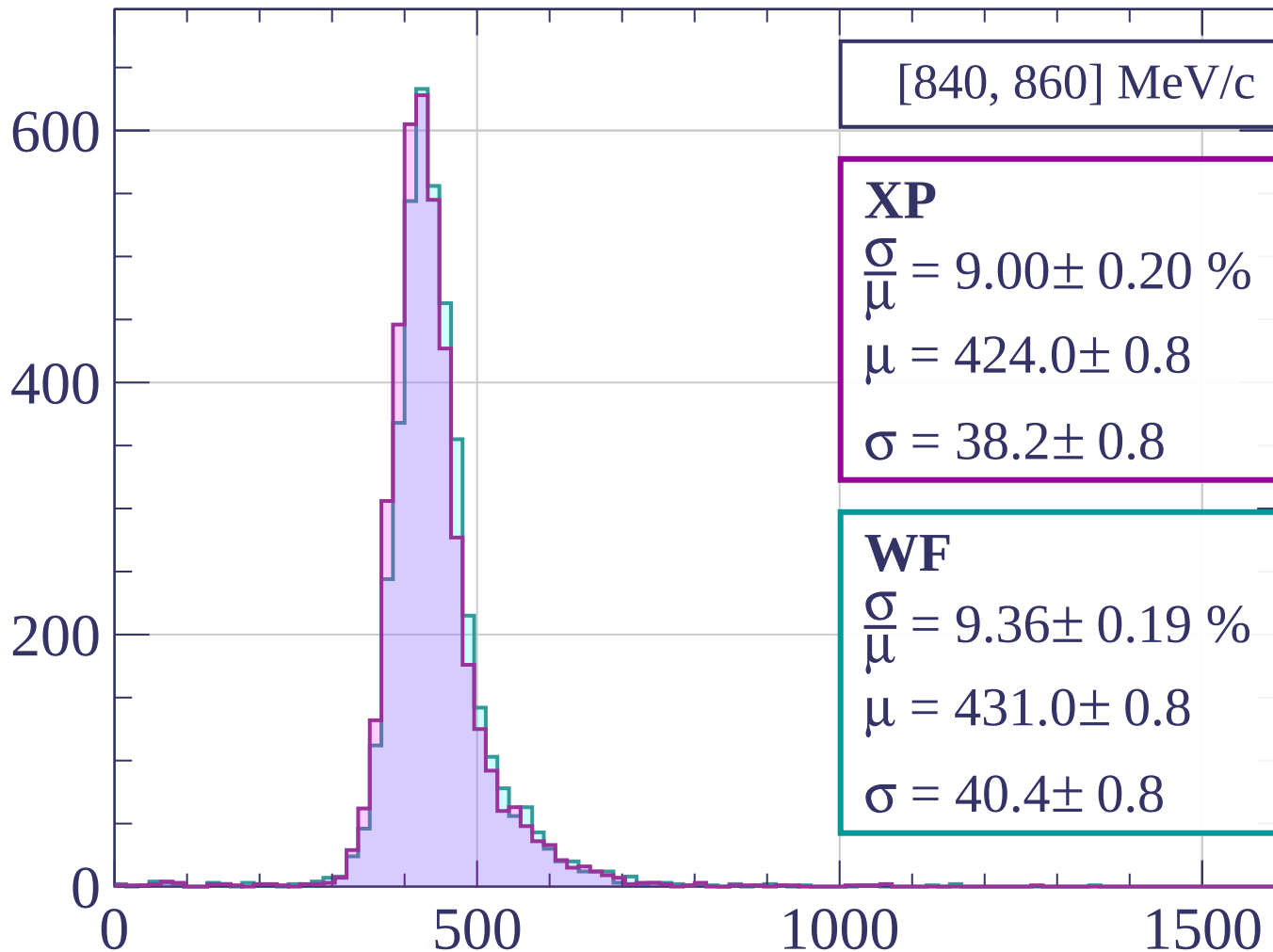
dE/dx [ADC counts/cm]

Count



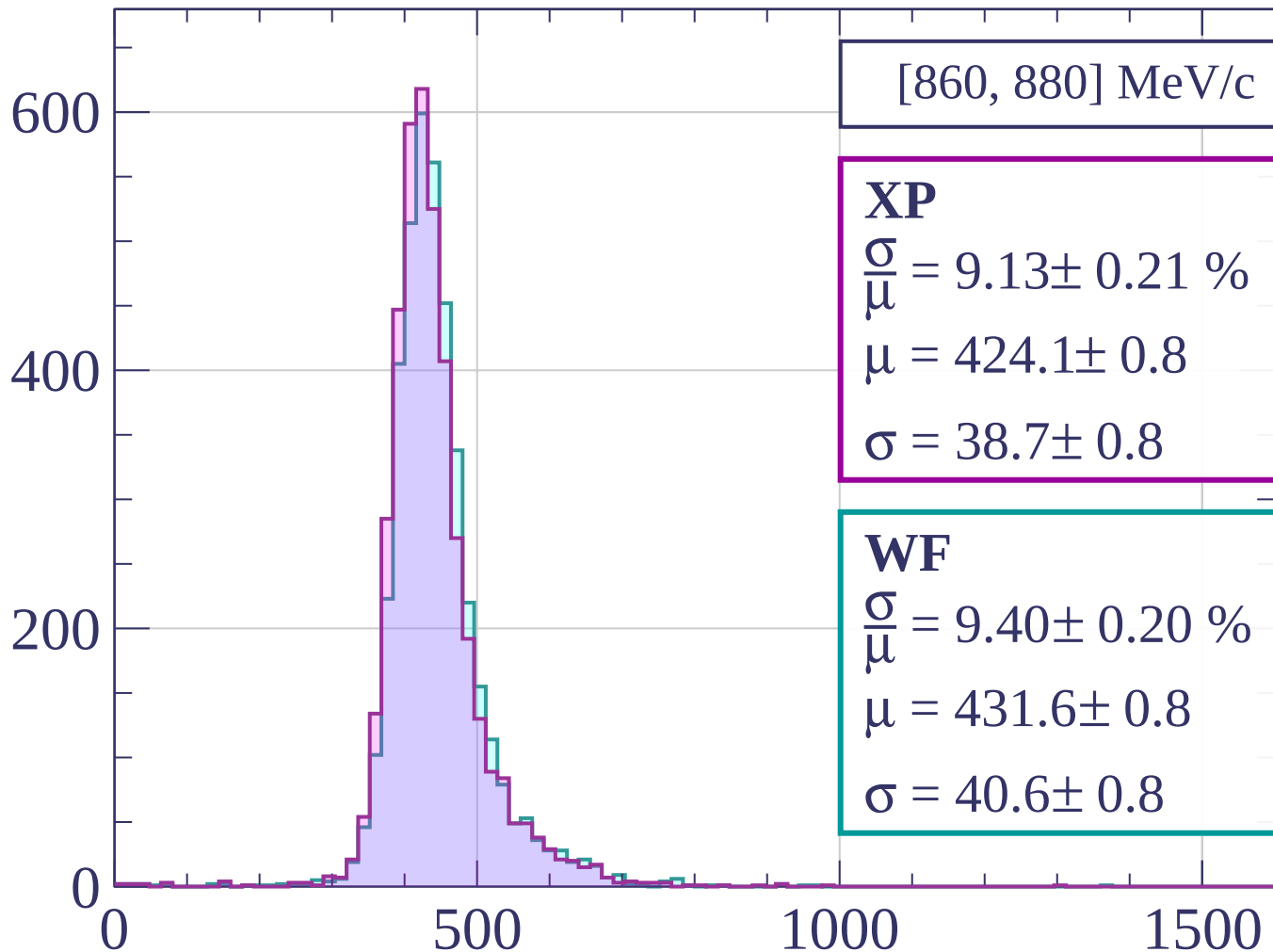
dE/dx [ADC counts/cm]

Count



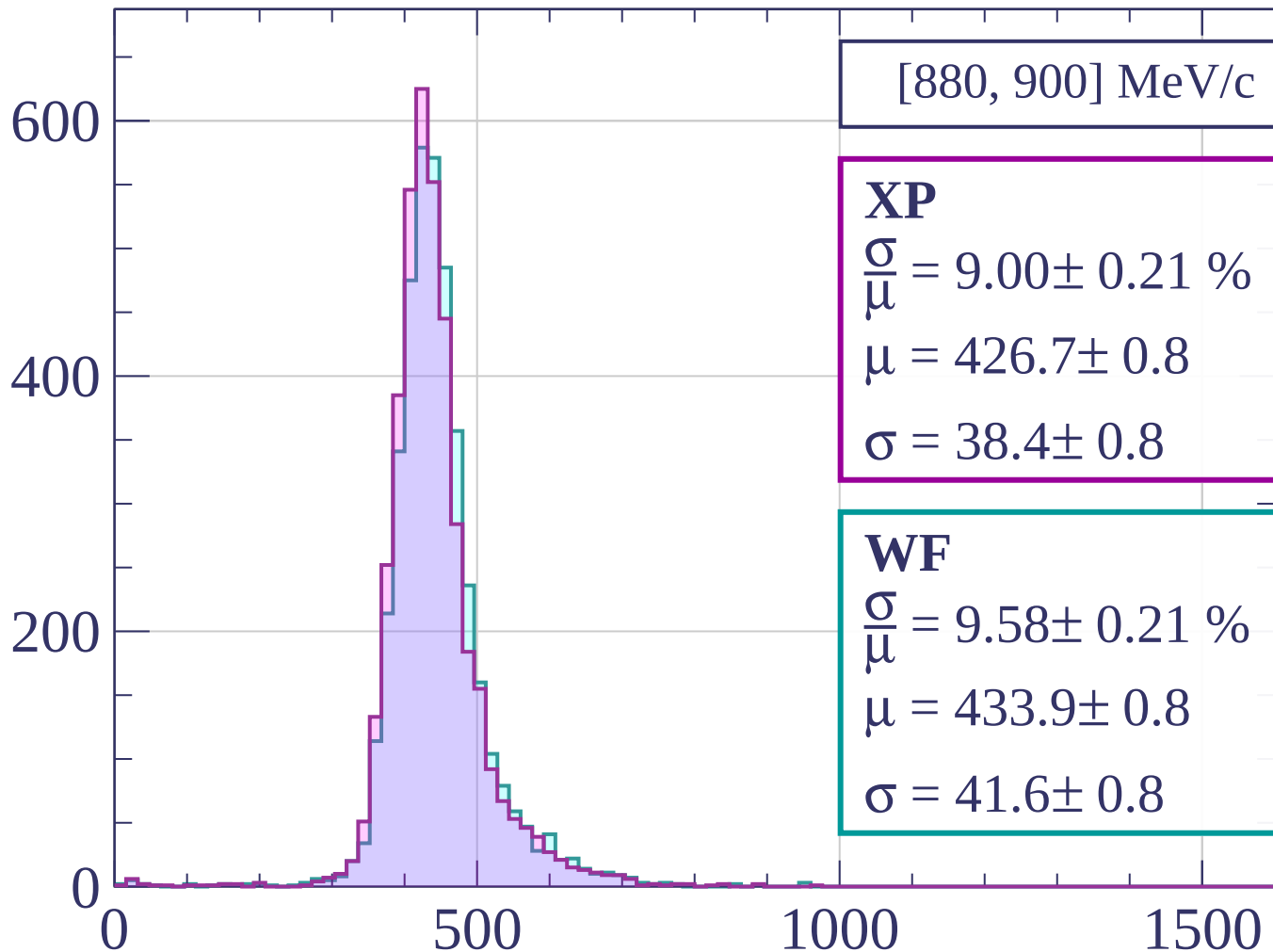
dE/dx [ADC counts/cm]

Count

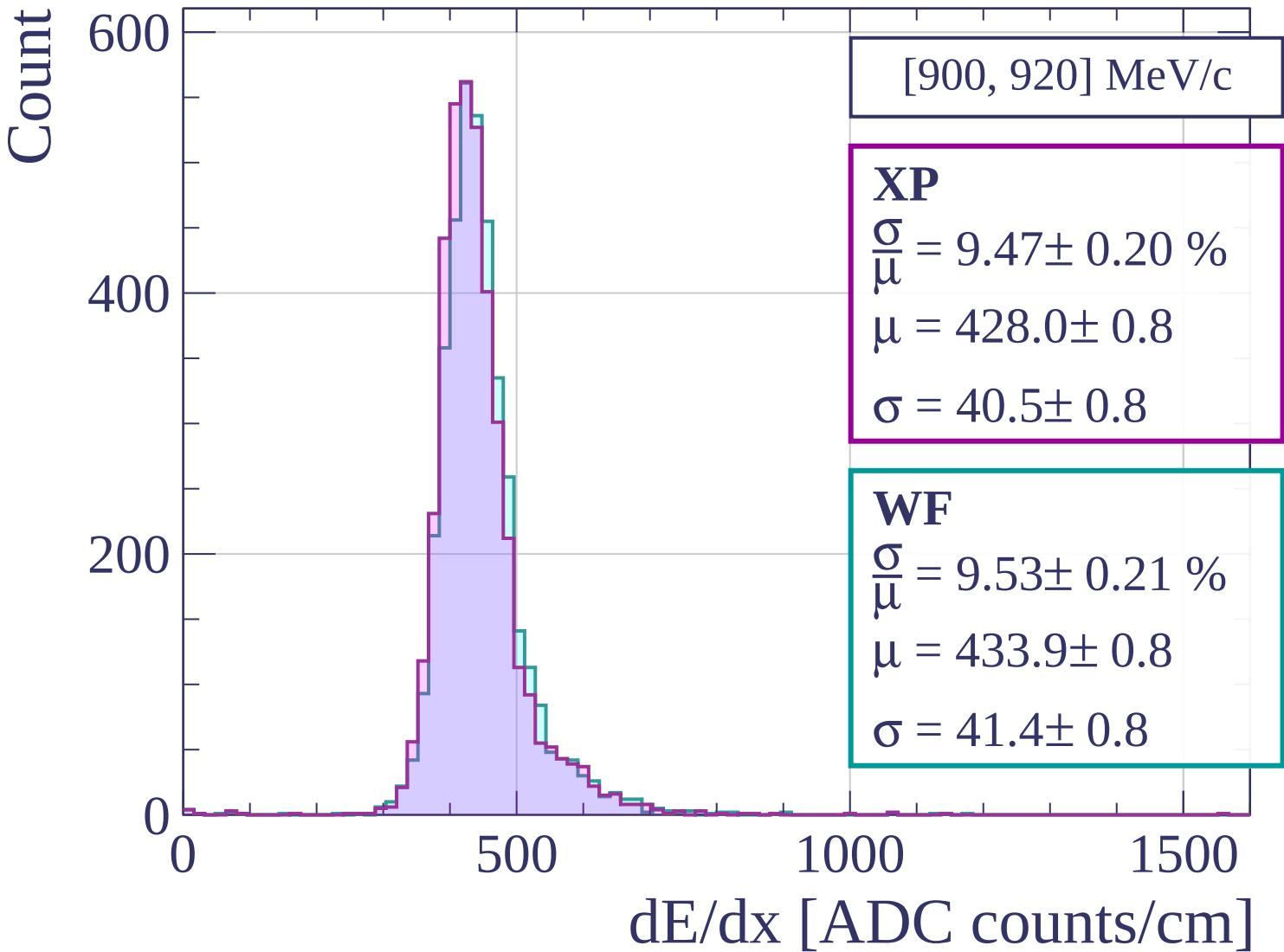


dE/dx [ADC counts/cm]

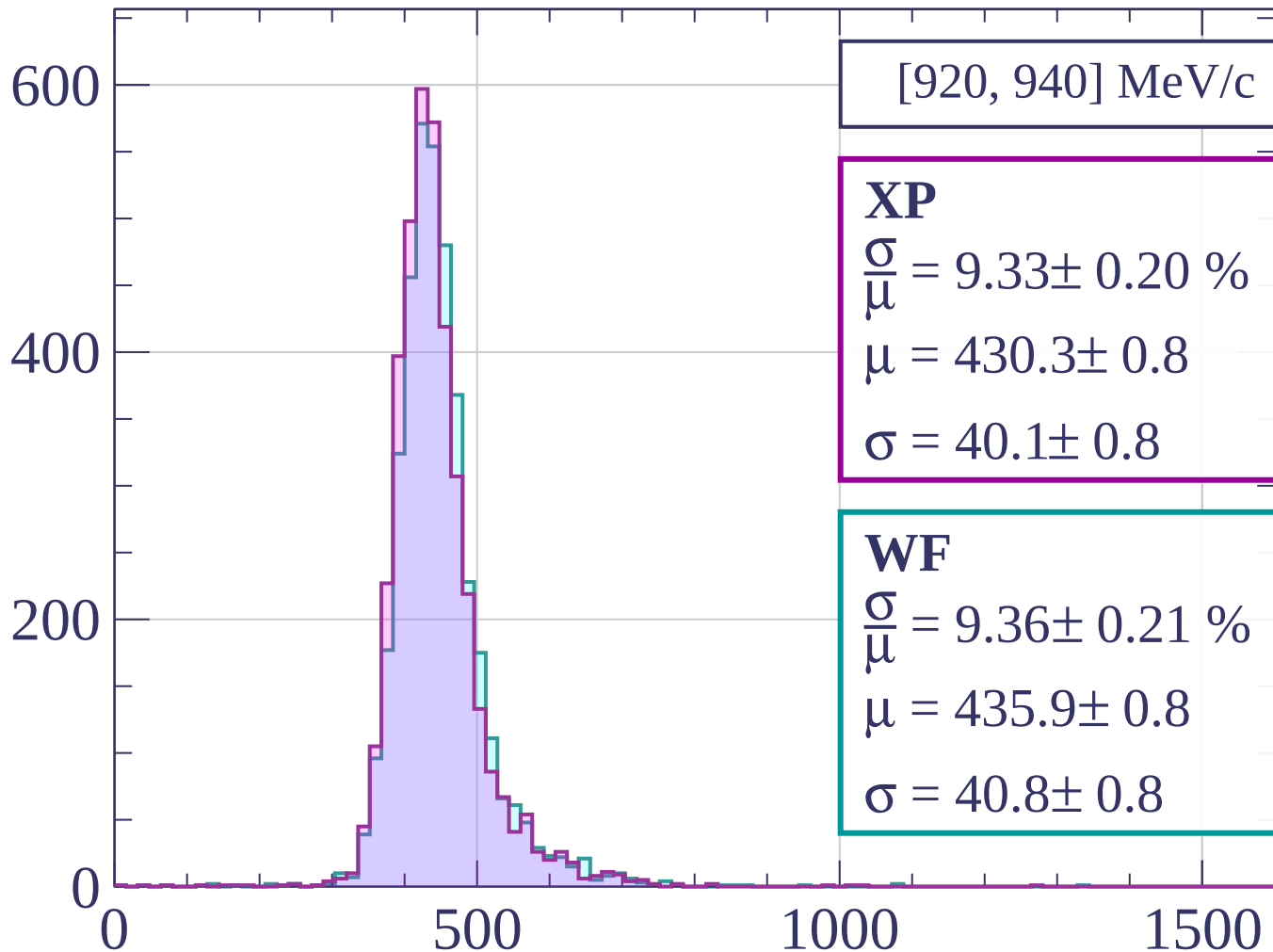
Count



dE/dx [ADC counts/cm]

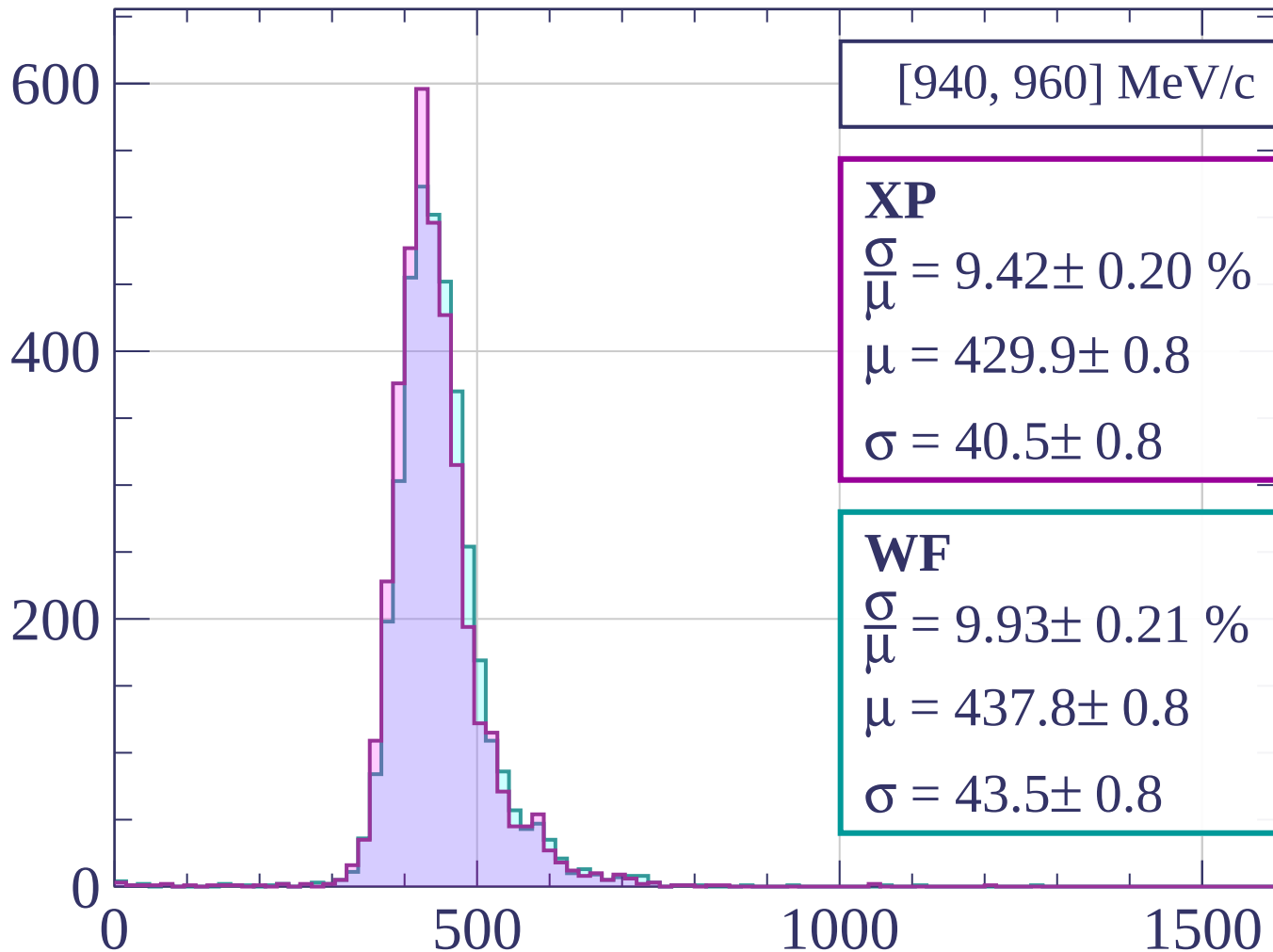


Count

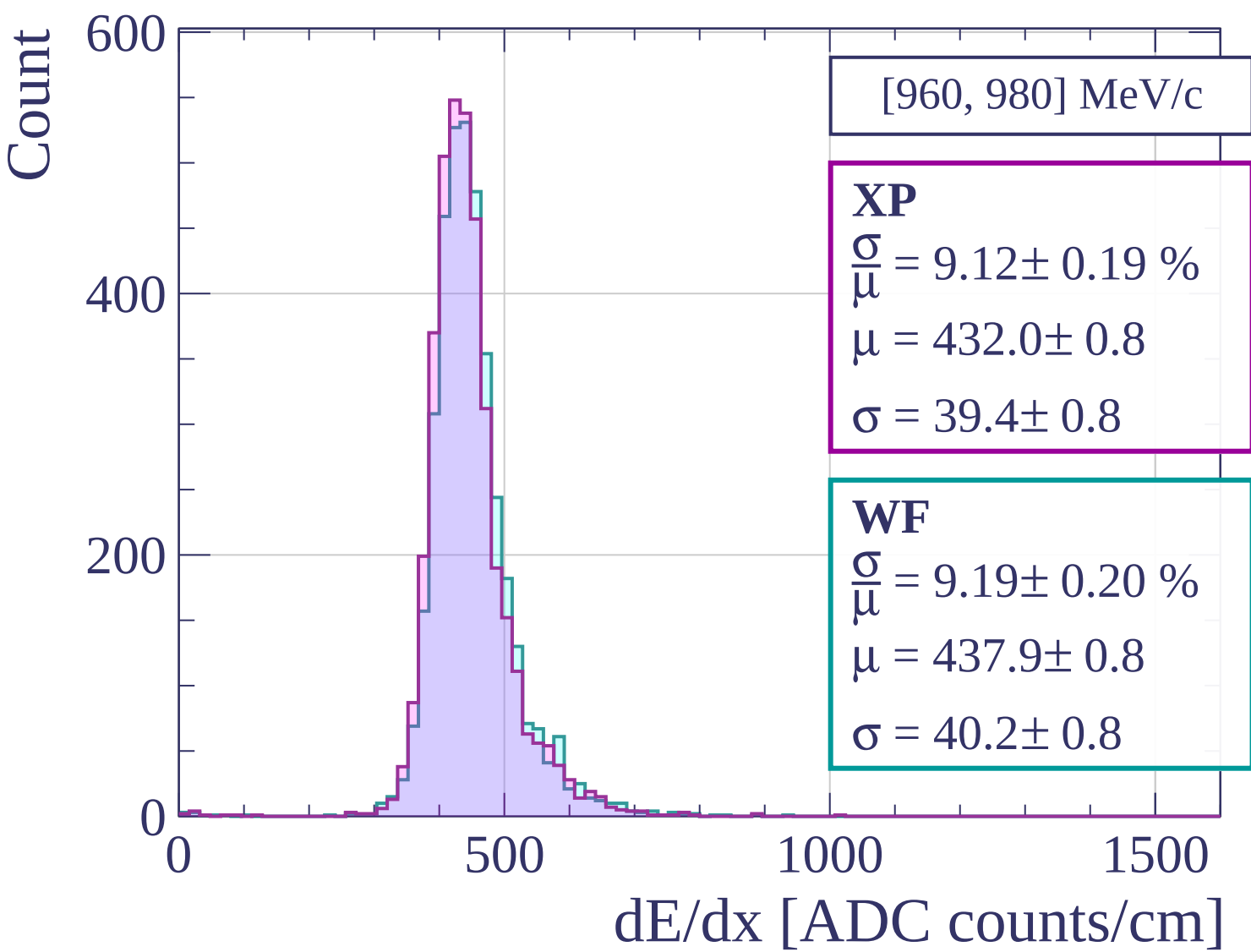


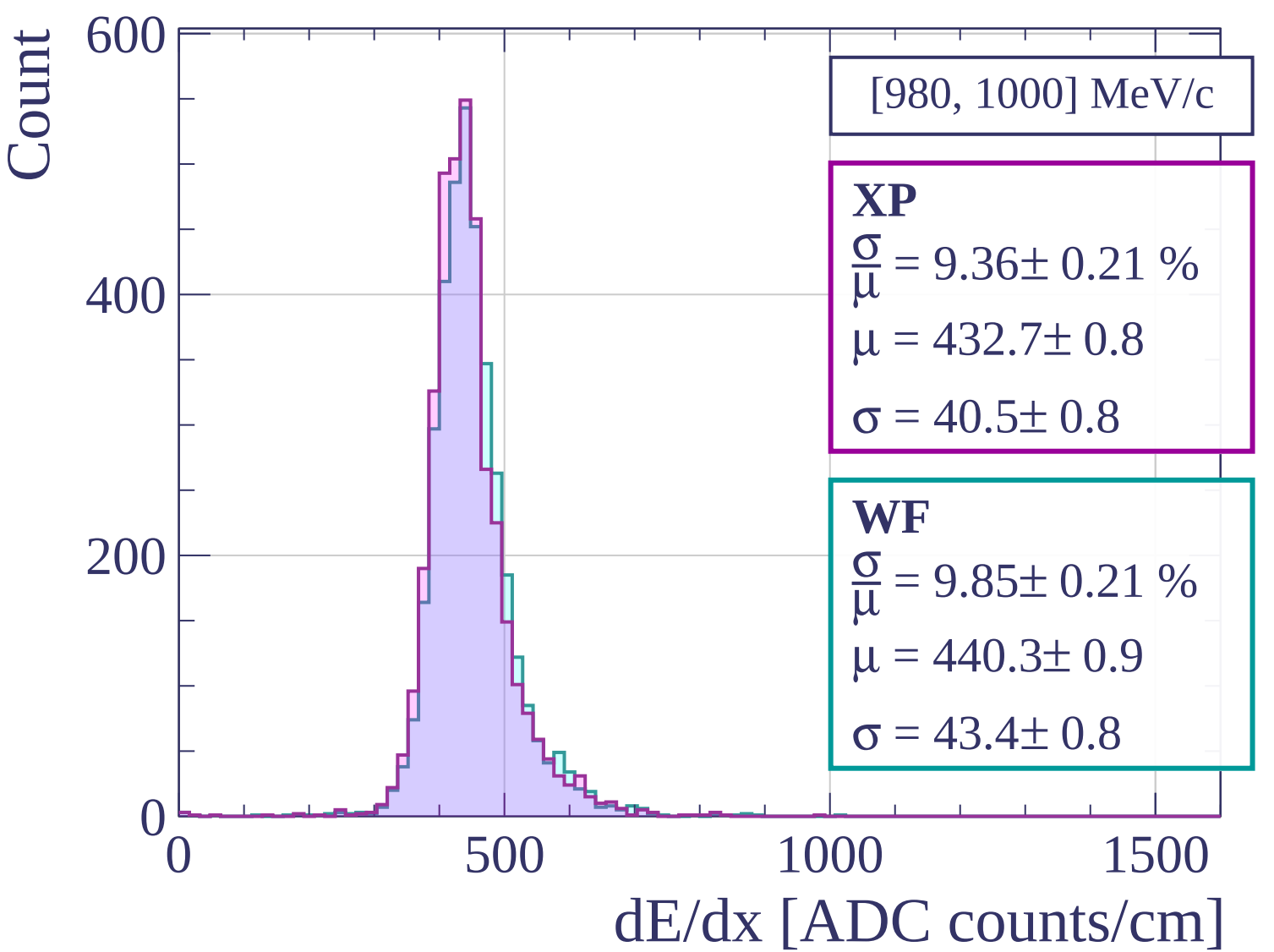
dE/dx [ADC counts/cm]

Count

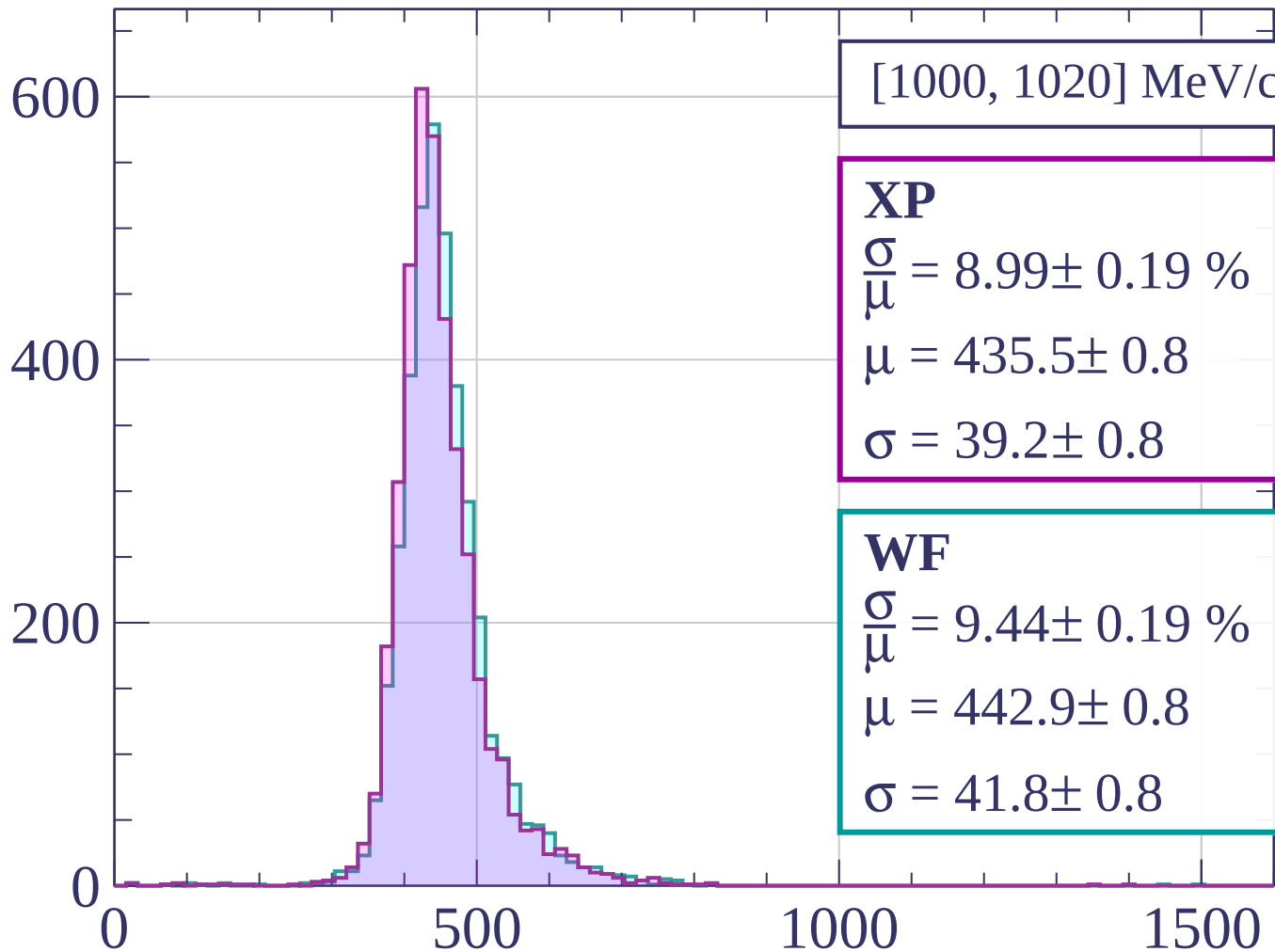


dE/dx [ADC counts/cm]





Count



[1000, 1020] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.99 \pm 0.19 \%$$

$$\mu = 435.5 \pm 0.8$$

$$\sigma = 39.2 \pm 0.8$$

WF

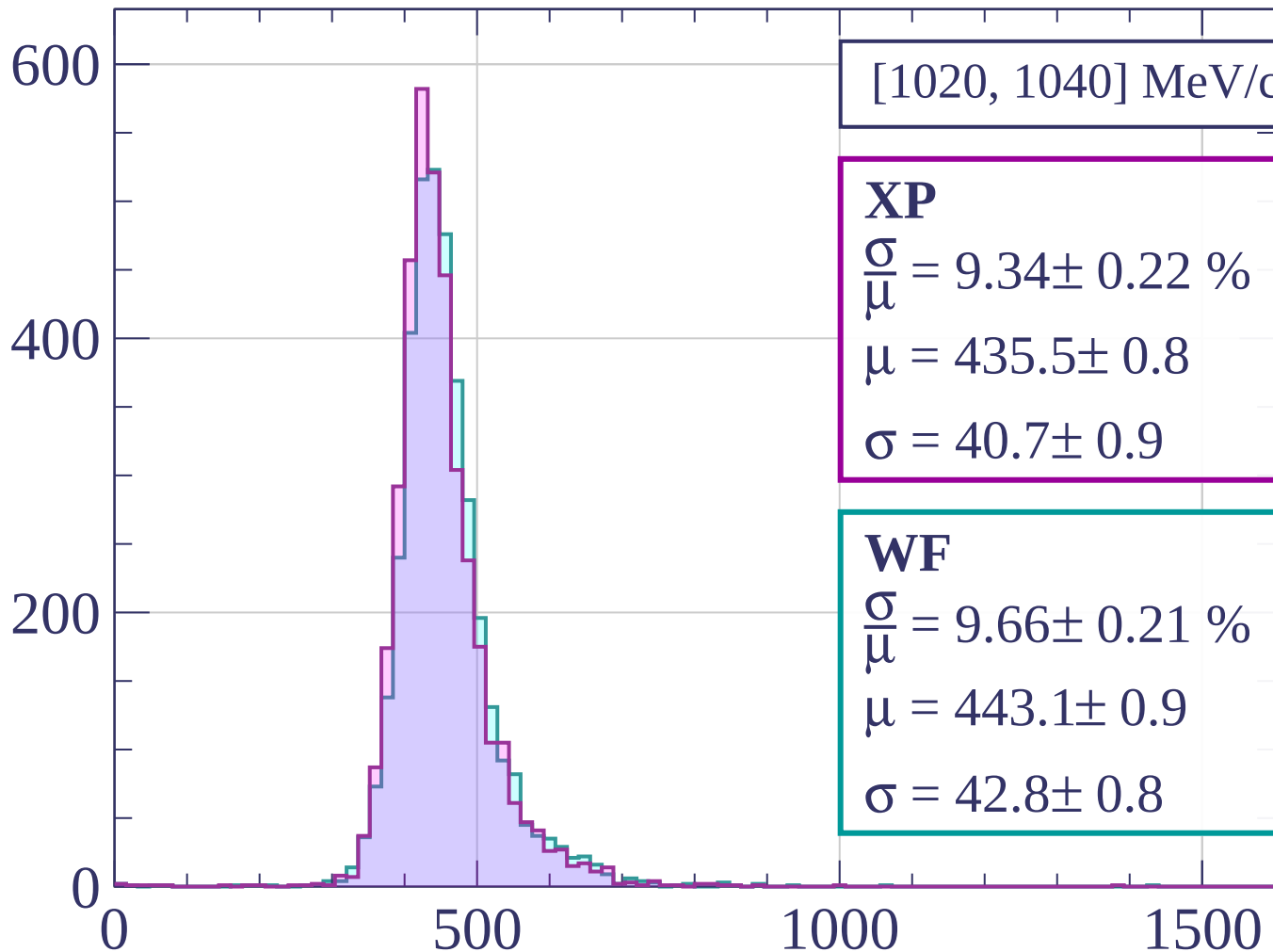
$$\frac{\sigma}{\mu} = 9.44 \pm 0.19 \%$$

$$\mu = 442.9 \pm 0.8$$

$$\sigma = 41.8 \pm 0.8$$

dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

[1040, 1060] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.79 \pm 0.19 \%$$

$$\mu = 436.0 \pm 0.8$$

$$\sigma = 38.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.27 \pm 0.19 \%$$

$$\mu = 443.9 \pm 0.8$$

$$\sigma = 41.1 \pm 0.7$$

400

200

0

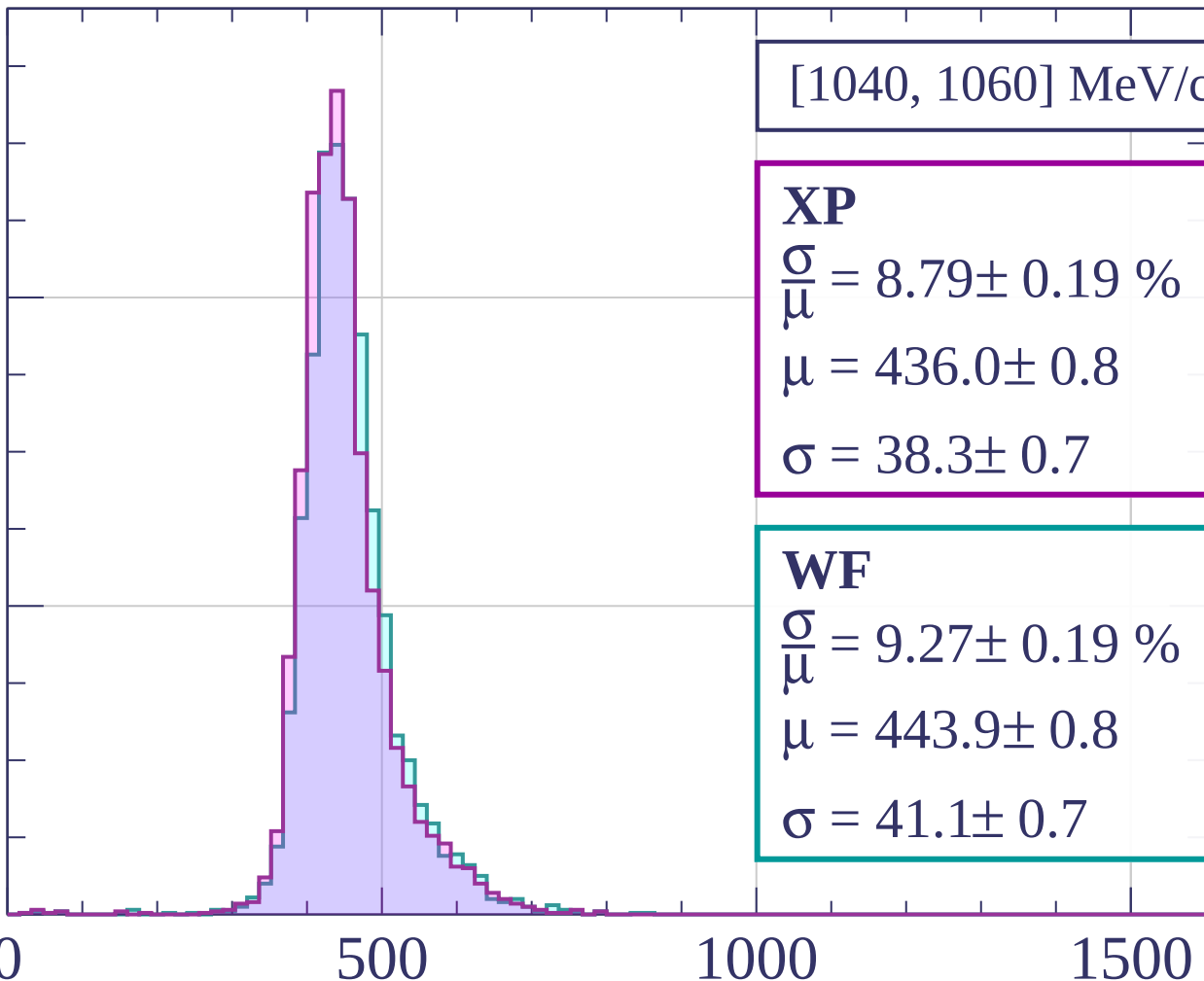
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1060, 1080] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.04 \pm 0.19 \%$$

$$\mu = 437.8 \pm 0.8$$

$$\sigma = 39.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.37 \pm 0.20 \%$$

$$\mu = 445.6 \pm 0.9$$

$$\sigma = 41.8 \pm 0.8$$

400

200

0

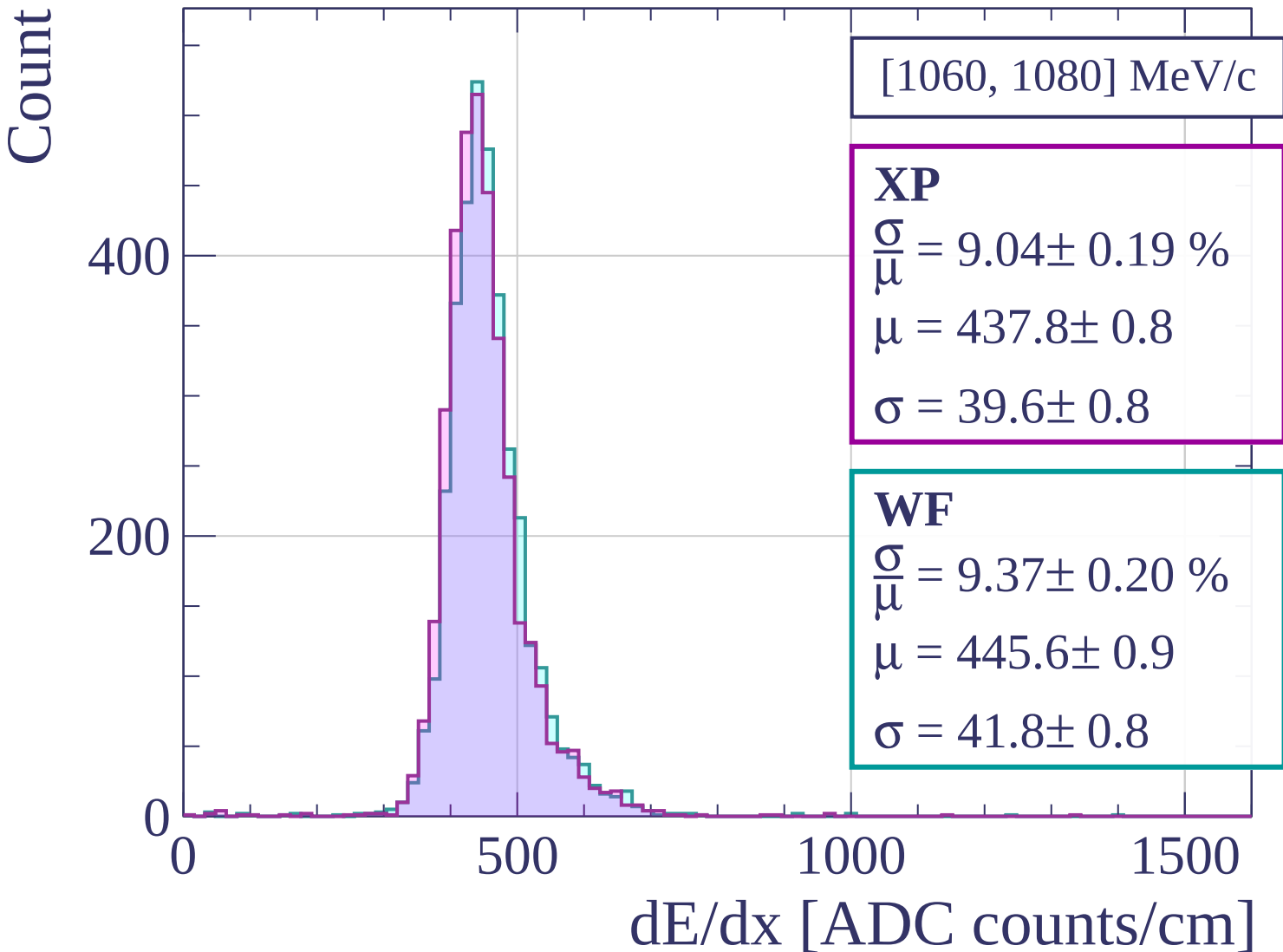
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1080, 1100] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.20 \%$$

$$\mu = 440.6 \pm 0.8$$

$$\sigma = 41.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.56 \pm 0.20 \%$$

$$\mu = 446.9 \pm 0.9$$

$$\sigma = 42.7 \pm 0.8$$

400

200

0

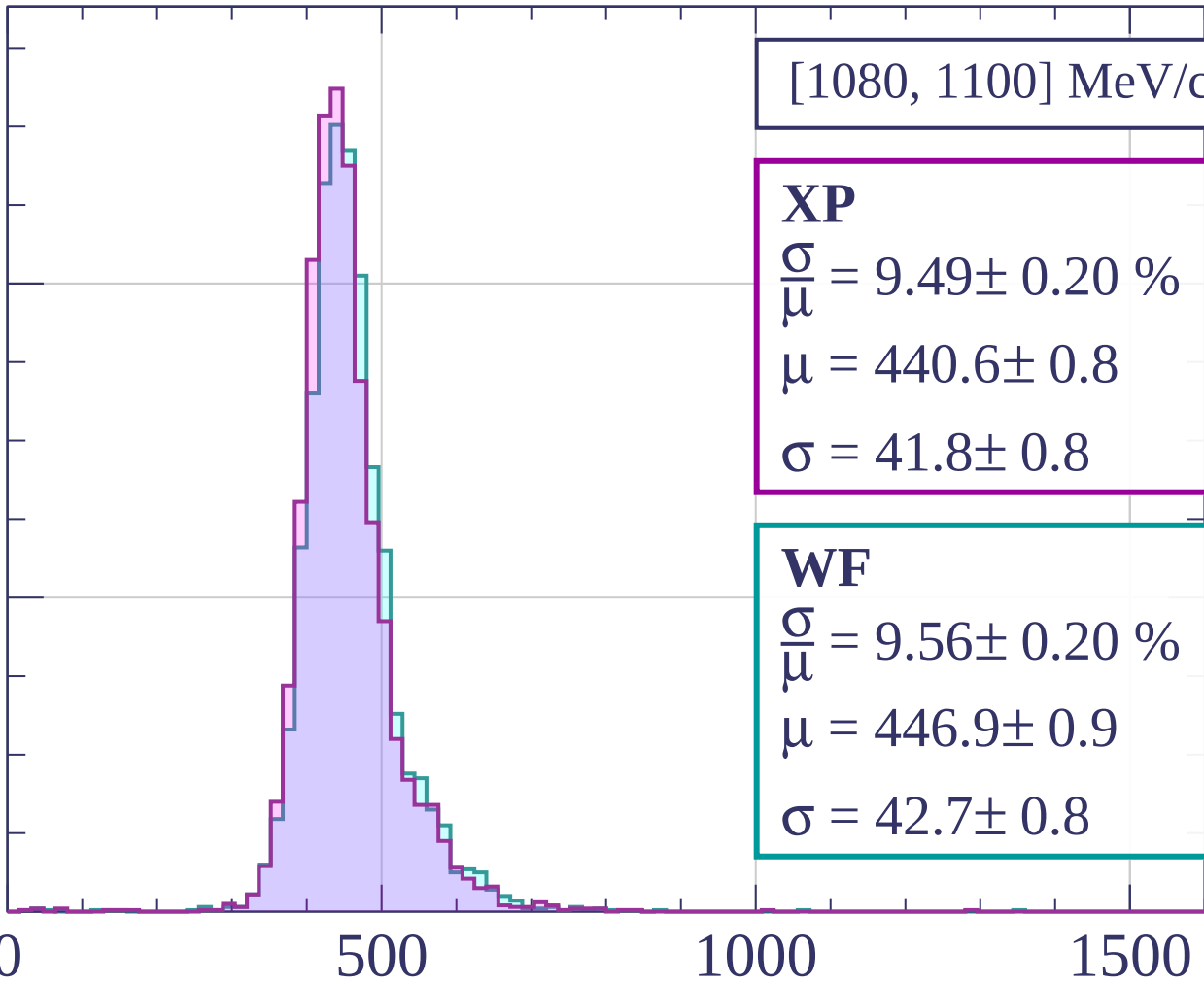
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1100, 1120] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.18 \pm 0.25 \%$$

$$\mu = 439.1 \pm 0.9$$

$$\sigma = 40.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.40 \pm 0.23 \%$$

$$\mu = 447.7 \pm 0.9$$

$$\sigma = 42.1 \pm 0.9$$

400

200

0

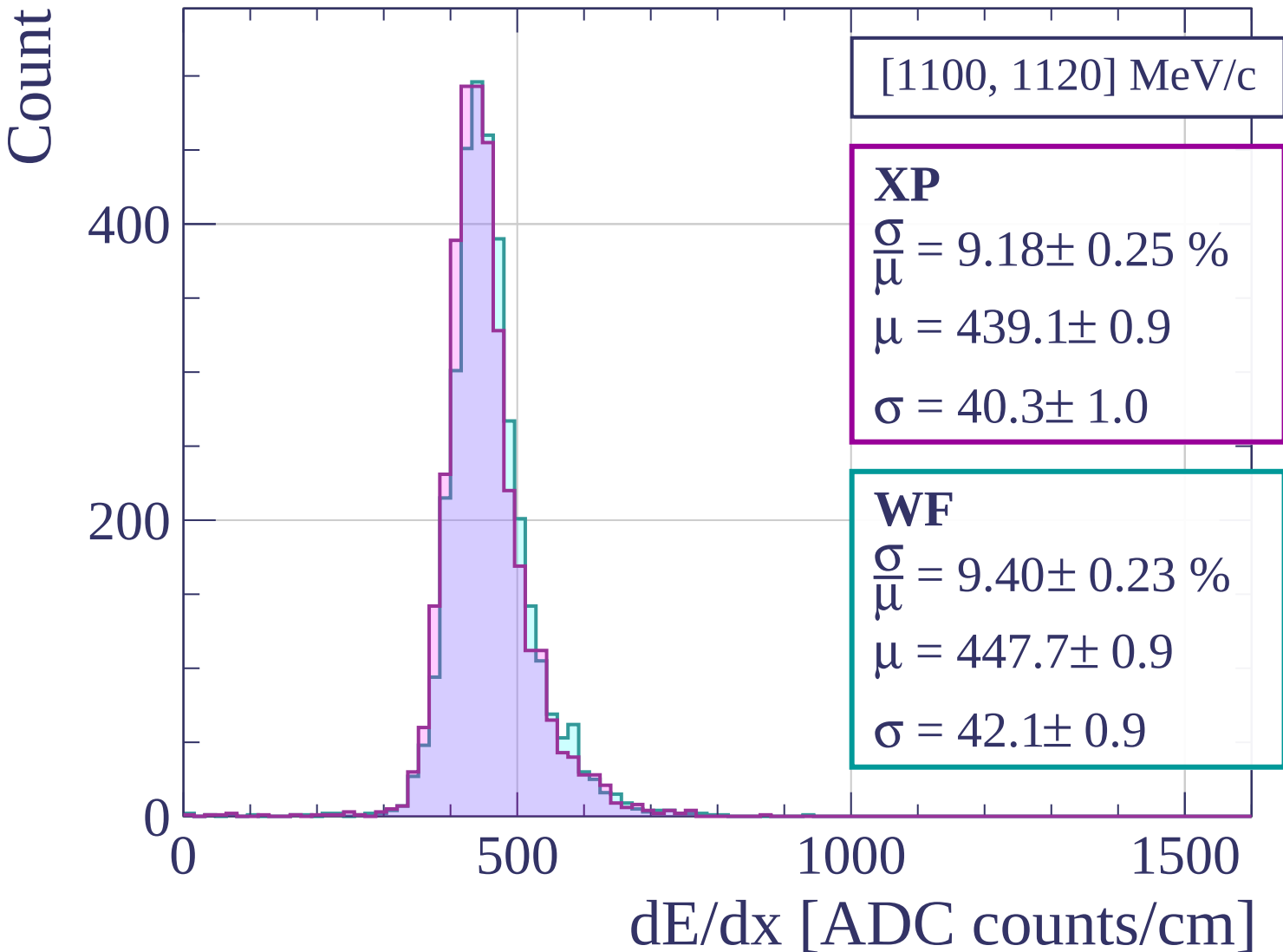
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1120, 1140] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.20 \%$$

$$\mu = 442.1 \pm 0.9$$

$$\sigma = 41.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.65 \pm 0.19 \%$$

$$\mu = 450.8 \pm 0.9$$

$$\sigma = 43.5 \pm 0.8$$

400

200

0

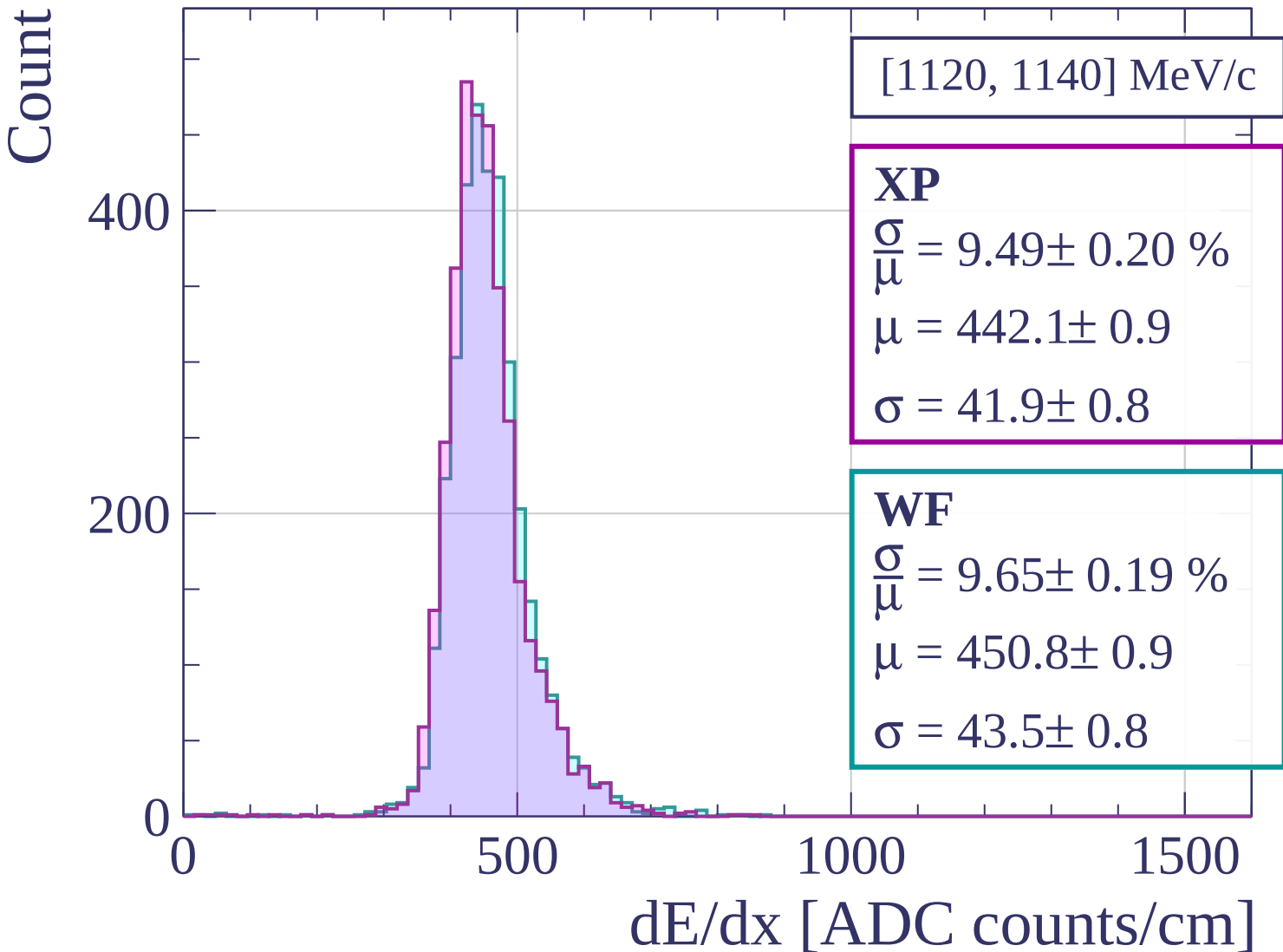
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1140, 1160] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.21 \pm 0.18 \%$$

$$\mu = 444.0 \pm 0.8$$

$$\sigma = 40.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.31 \pm 0.22 \%$$

$$\mu = 449.8 \pm 0.9$$

$$\sigma = 41.9 \pm 0.9$$

400

200

0

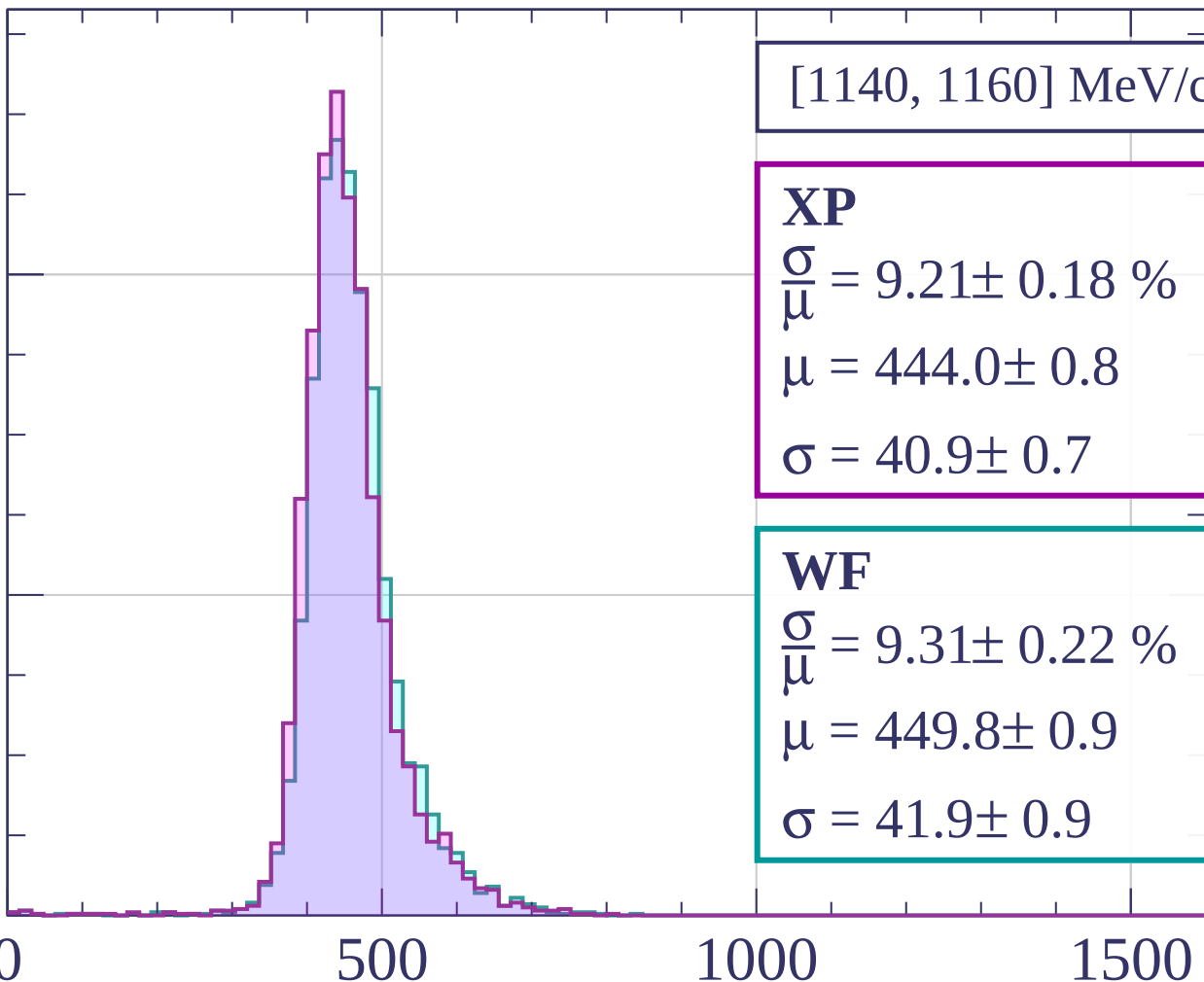
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1160, 1180] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.32 \pm 0.19 \%$$

$$\mu = 445.0 \pm 0.8$$

$$\sigma = 41.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.42 \pm 0.23 \%$$

$$\mu = 450.7 \pm 0.9$$

$$\sigma = 42.5 \pm 1.0$$

400

200

0

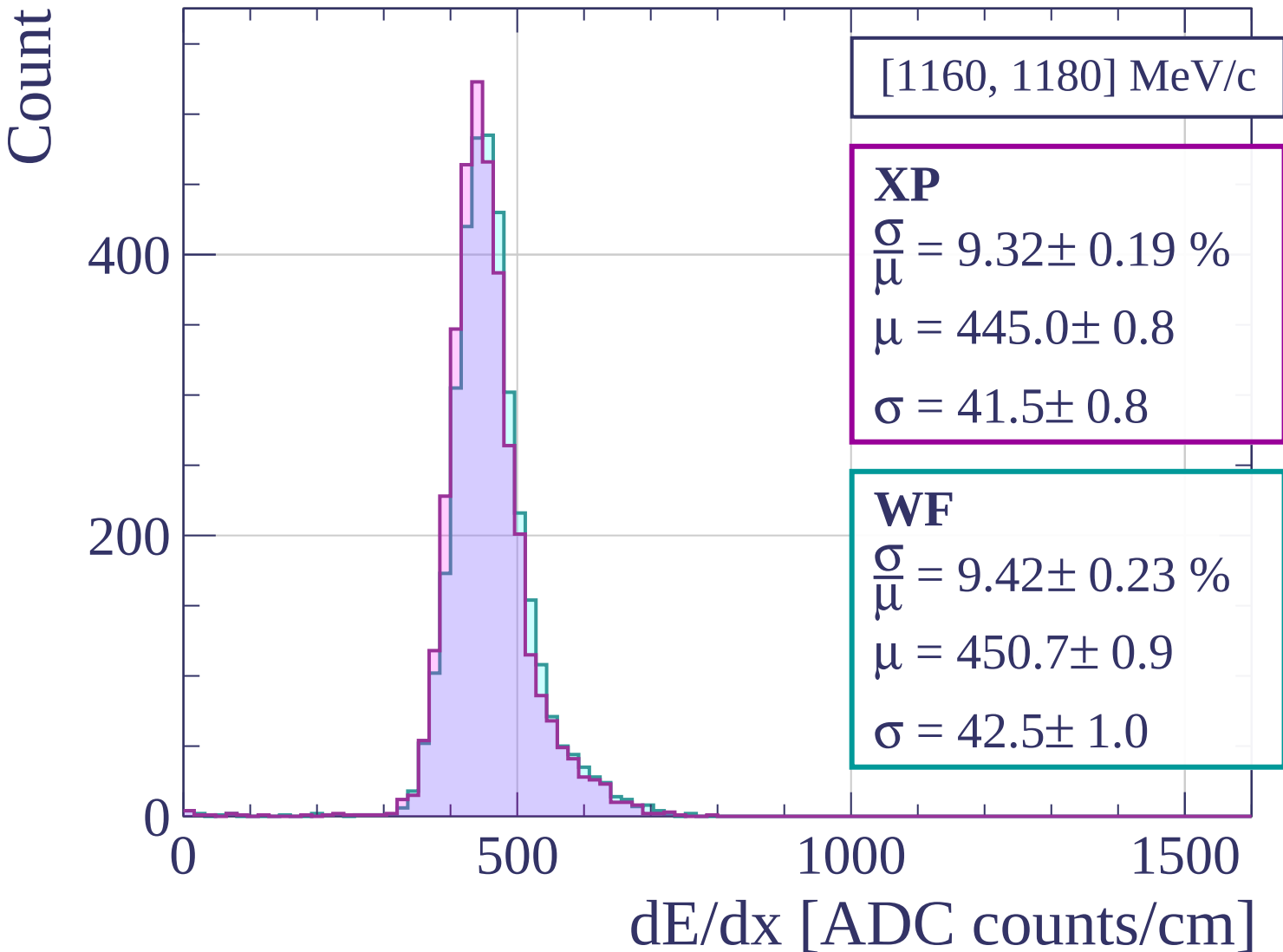
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1180, 1200] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.43 \pm 0.23 \%$$

$$\mu = 445.9 \pm 0.9$$

$$\sigma = 42.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.70 \pm 0.22 \%$$

$$\mu = 453.3 \pm 0.9$$

$$\sigma = 44.0 \pm 0.9$$

400

200

0

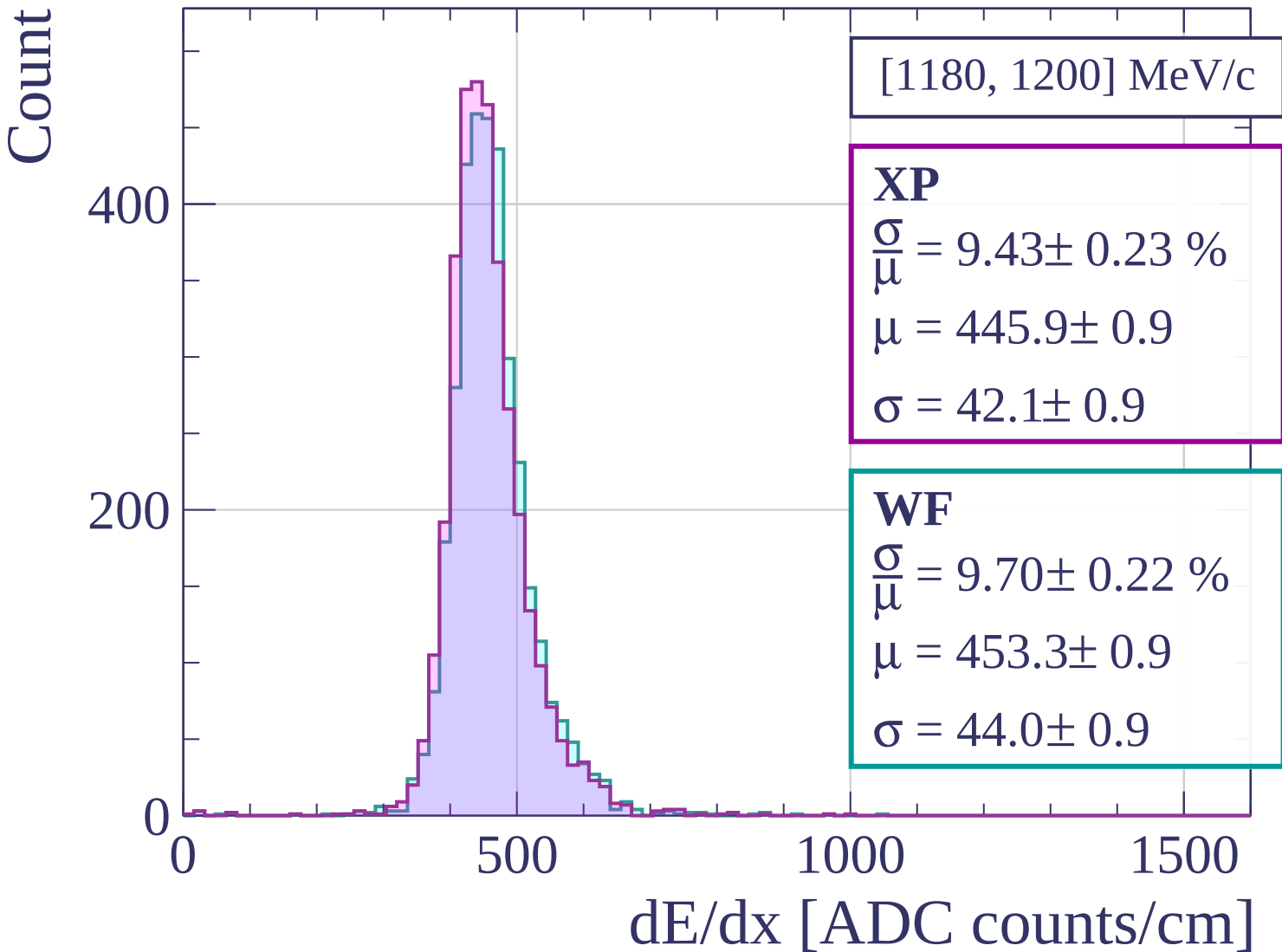
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1200, 1220] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.10 \pm 0.22 \%$$

$$\mu = 446.4 \pm 0.9$$

$$\sigma = 40.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.38 \pm 0.21 \%$$

$$\mu = 454.2 \pm 0.9$$

$$\sigma = 42.6 \pm 0.8$$

400

200

0

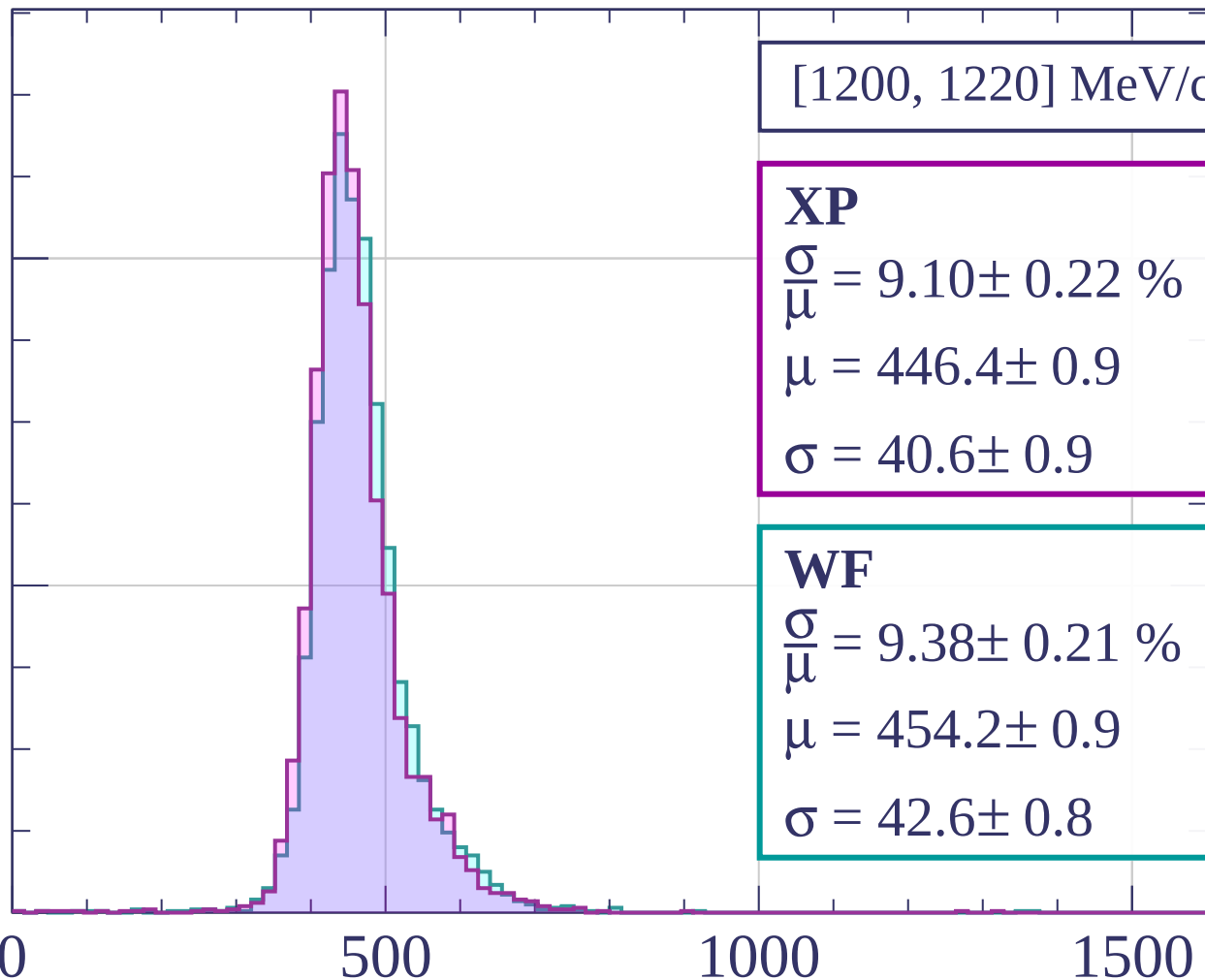
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1220, 1240] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.54 \pm 0.24 \%$$

$$\mu = 448.1 \pm 0.9$$

$$\sigma = 42.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.79 \pm 0.23 \%$$

$$\mu = 455.5 \pm 0.9$$

$$\sigma = 44.6 \pm 0.9$$

400

200

0

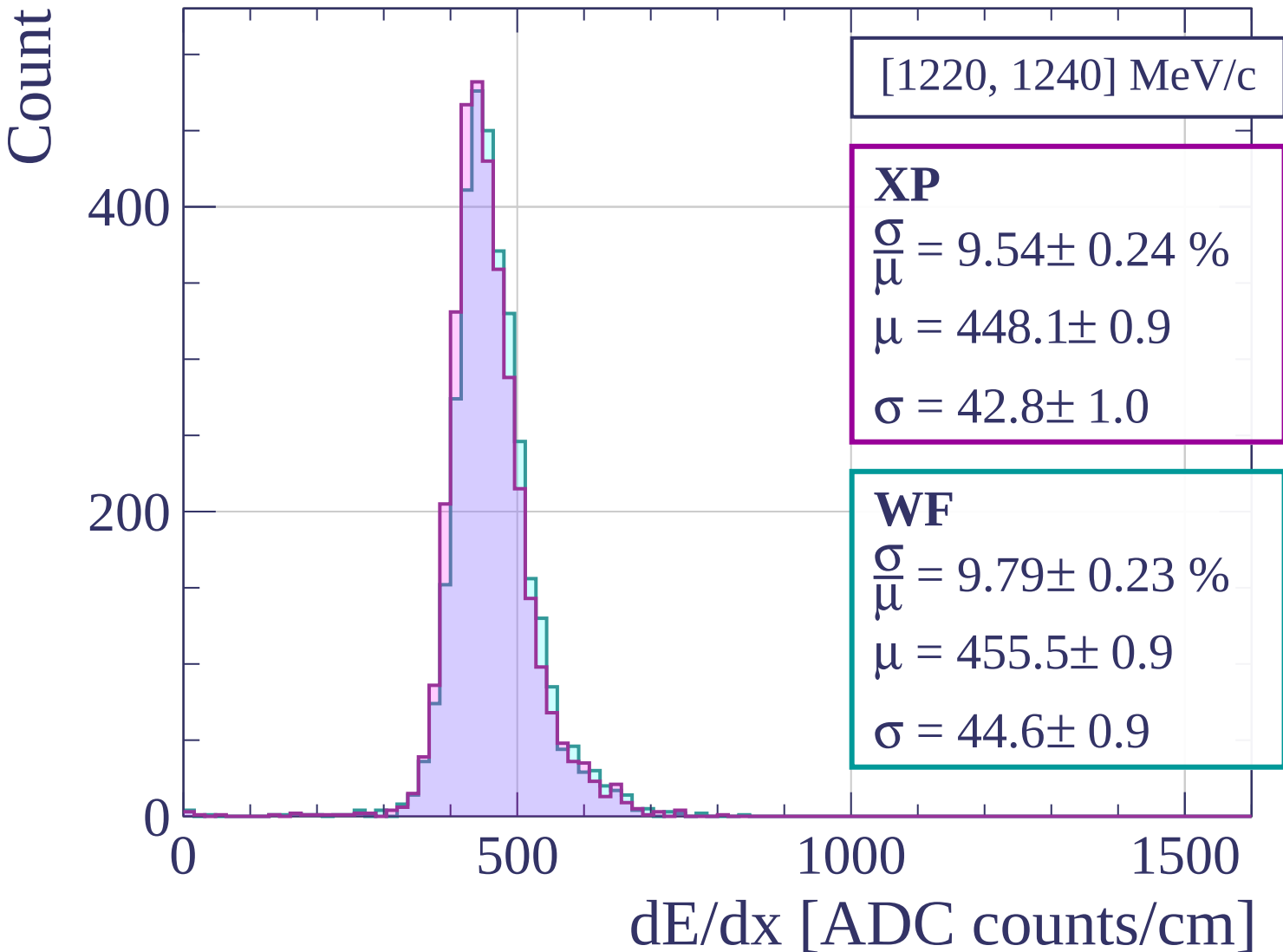
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1240, 1260] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.56 \pm 0.24 \%$$

$$\mu = 448.1 \pm 1.0$$

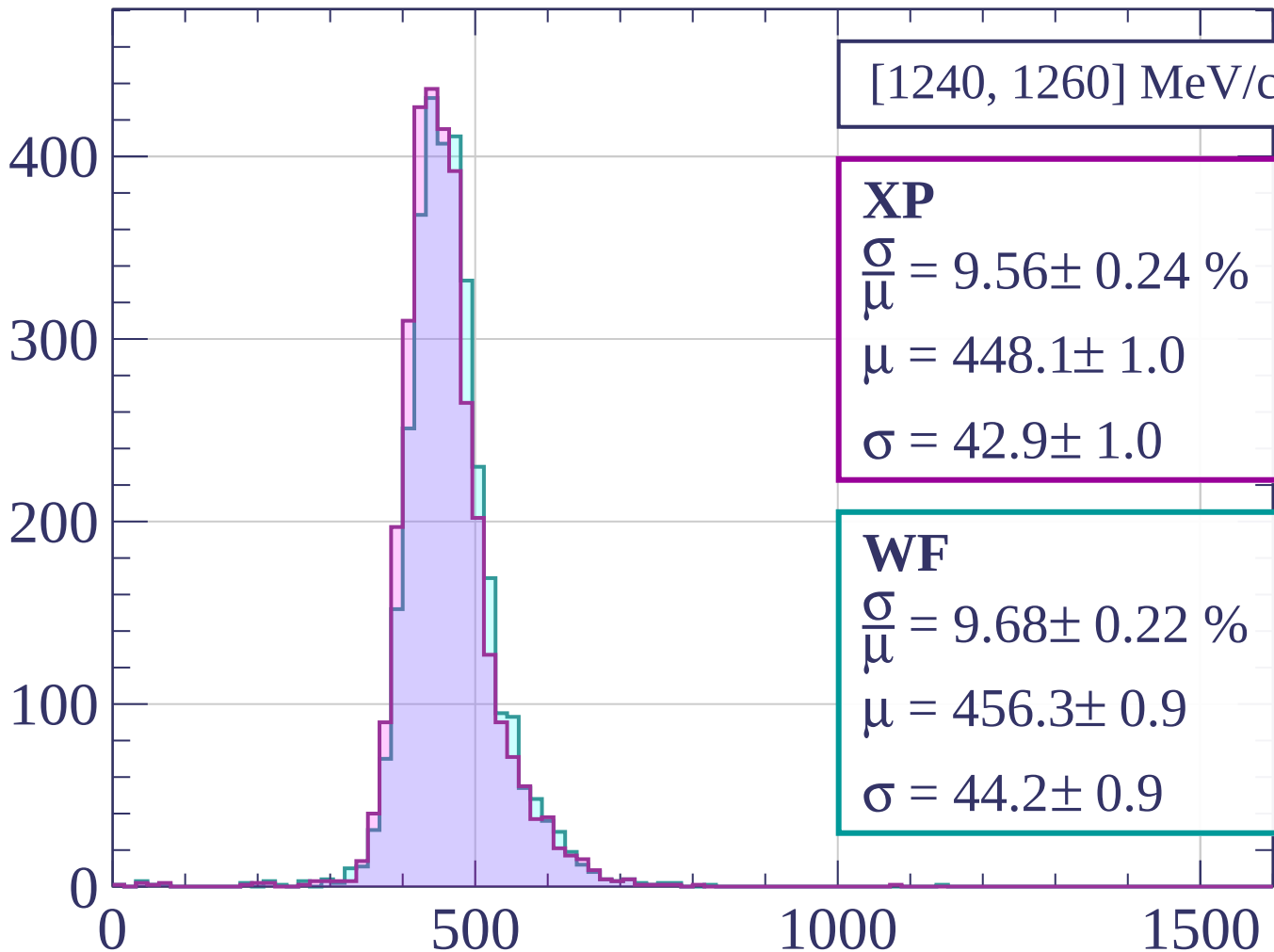
$$\sigma = 42.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.68 \pm 0.22 \%$$

$$\mu = 456.3 \pm 0.9$$

$$\sigma = 44.2 \pm 0.9$$



dE/dx [ADC counts/cm]

Count

[1260, 1280] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.23 \pm 0.25 \%$$

$$\mu = 450.2 \pm 1.0$$

$$\sigma = 46.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.04 \pm 0.24 \%$$

$$\mu = 457.1 \pm 1.0$$

$$\sigma = 45.9 \pm 1.0$$

400

200

0

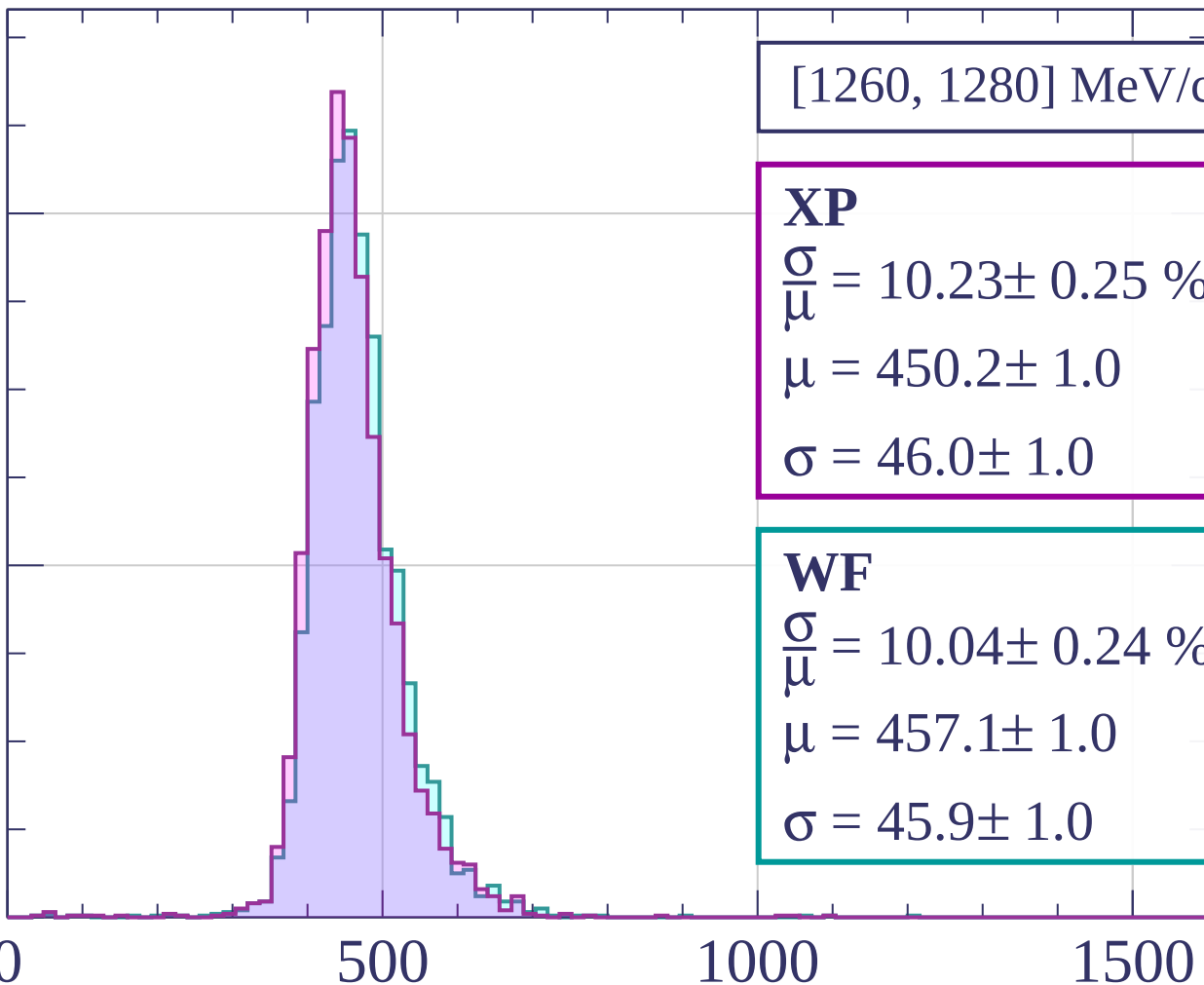
0

500

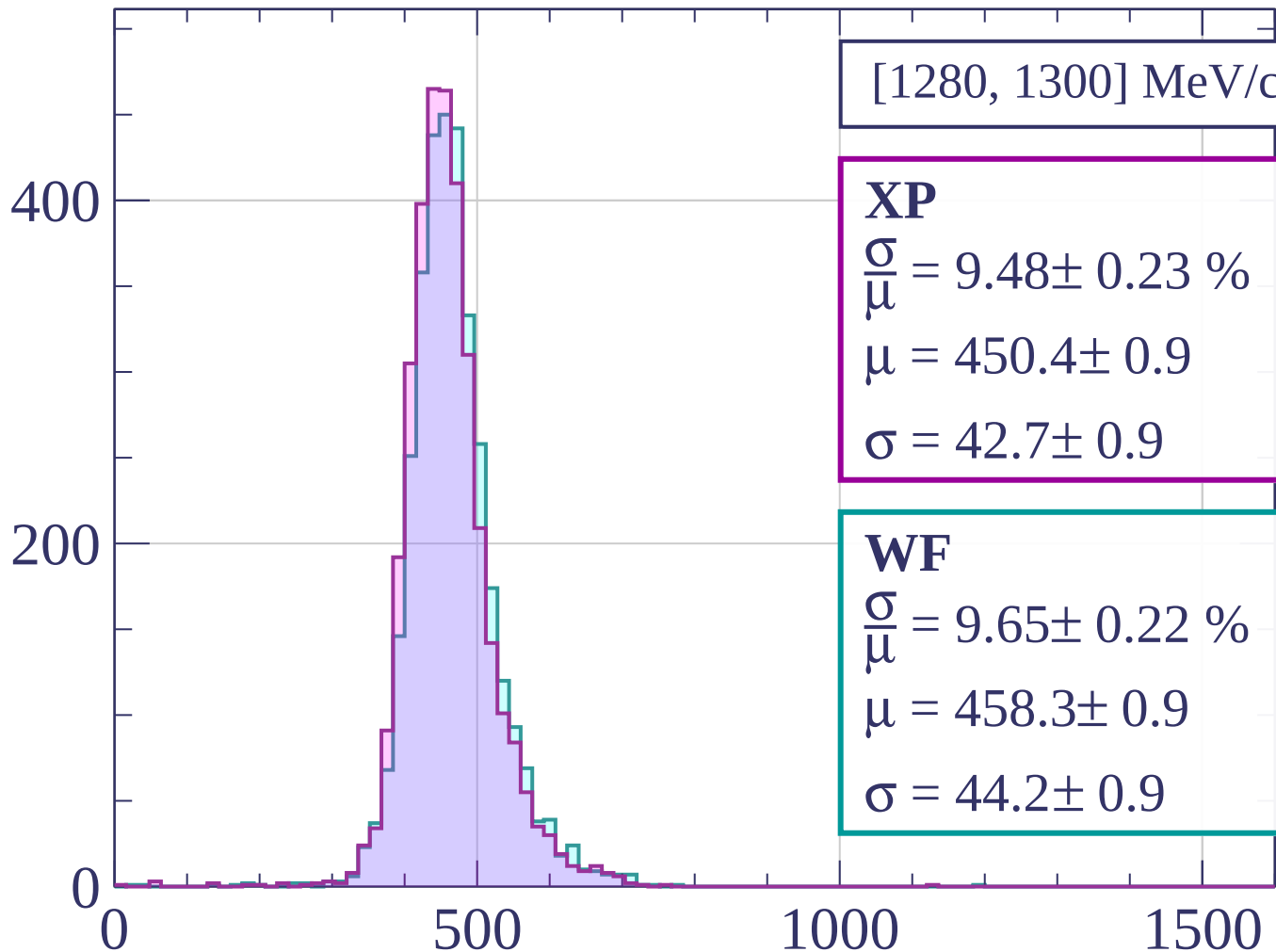
1000

1500

dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]

Count

[1300, 1320] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.46 \pm 0.22 \%$$

$$\mu = 452.5 \pm 0.9$$

$$\sigma = 42.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.48 \pm 0.23 \%$$

$$\mu = 459.4 \pm 1.0$$

$$\sigma = 43.5 \pm 1.0$$

400

200

0

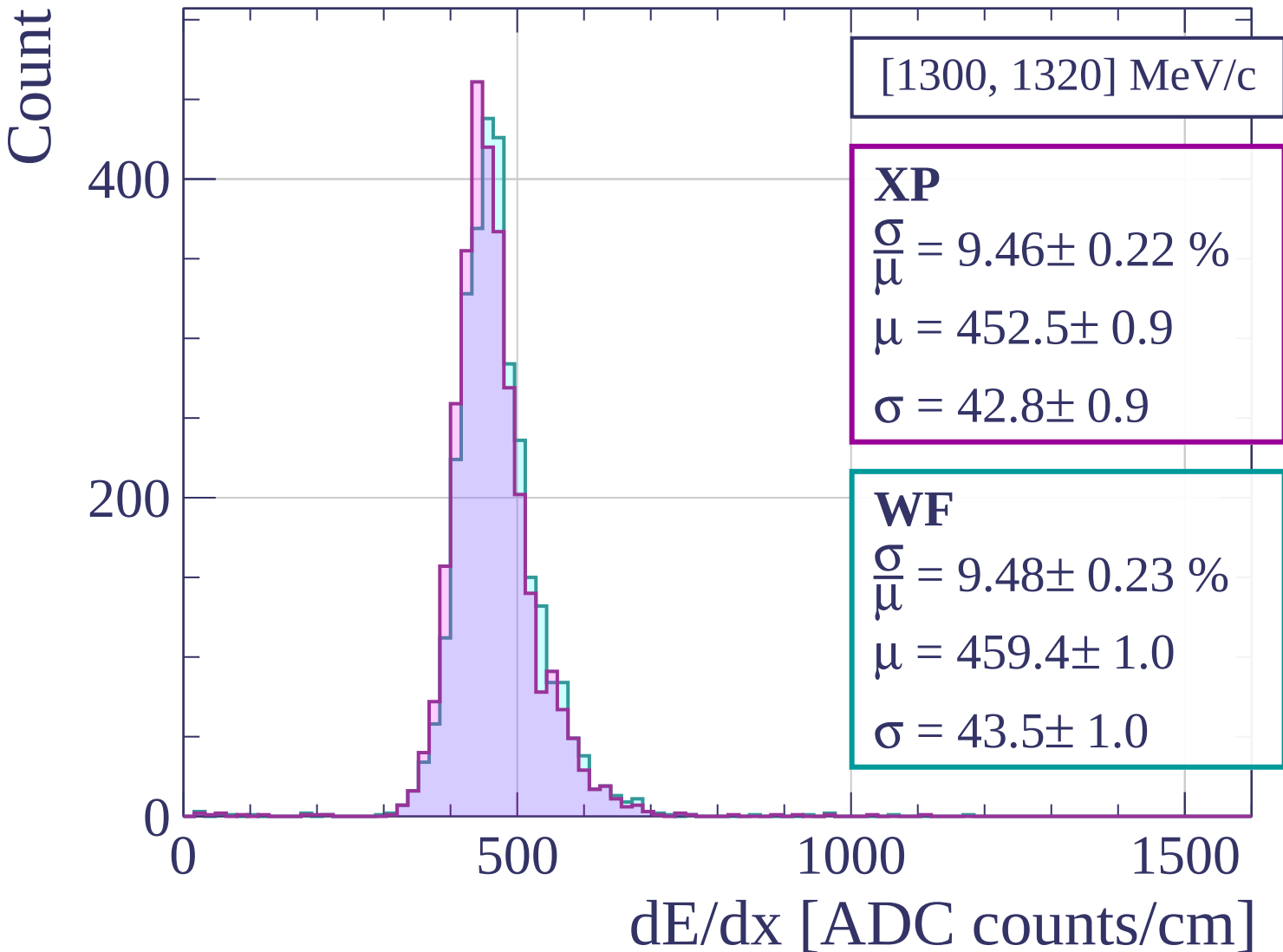
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1320, 1340] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.64 \pm 0.23 \%$$

$$\mu = 453.1 \pm 0.9$$

$$\sigma = 43.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.66 \pm 0.22 \%$$

$$\mu = 459.3 \pm 1.0$$

$$\sigma = 44.3 \pm 0.9$$

400

200

0

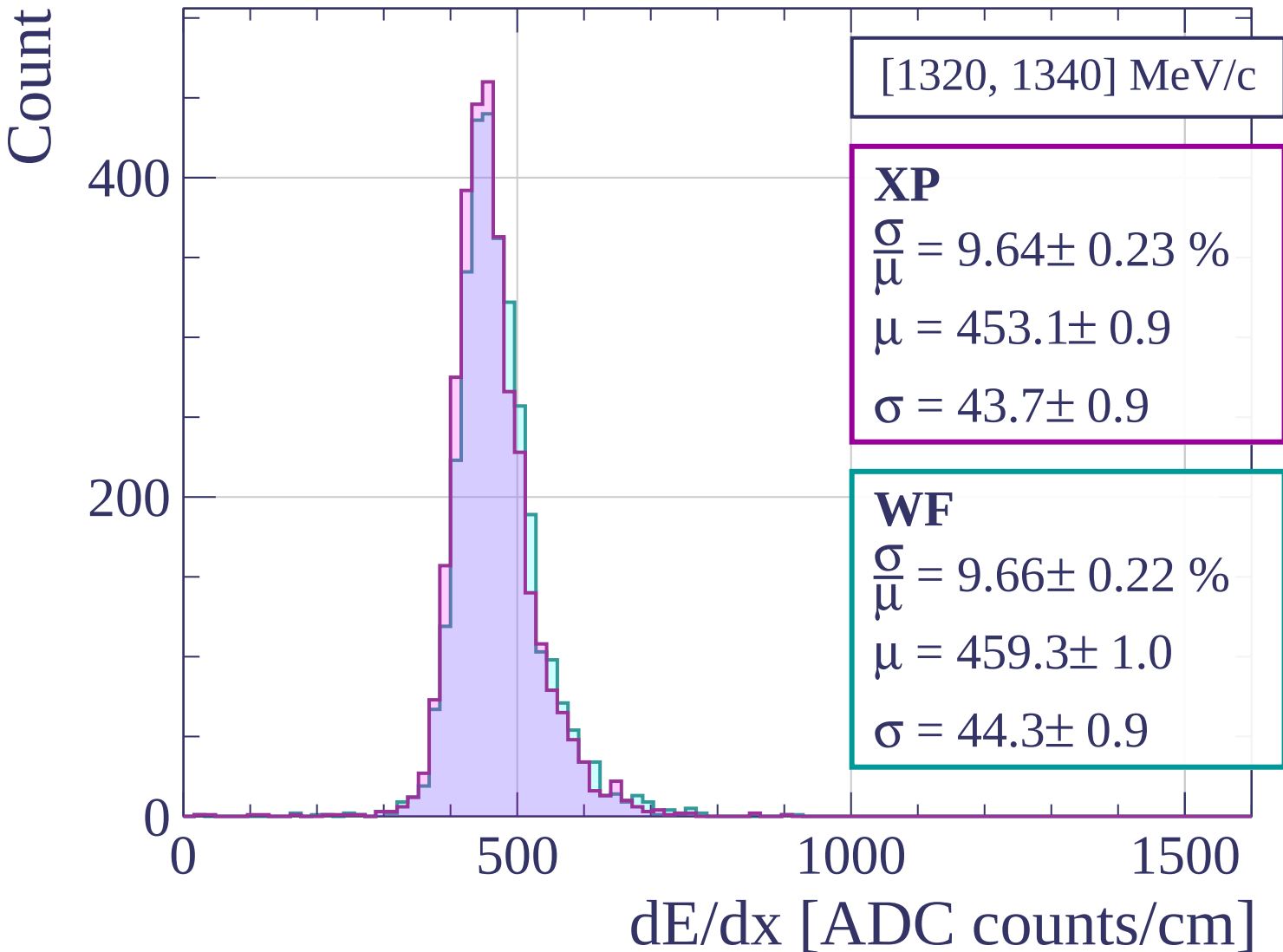
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1340, 1360] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.99 \pm 0.25 \%$$

$$\mu = 453.1 \pm 1.0$$

$$\sigma = 45.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.97 \pm 0.24 \%$$

$$\mu = 459.0 \pm 1.0$$

$$\sigma = 45.8 \pm 1.0$$

400

300

200

100

0

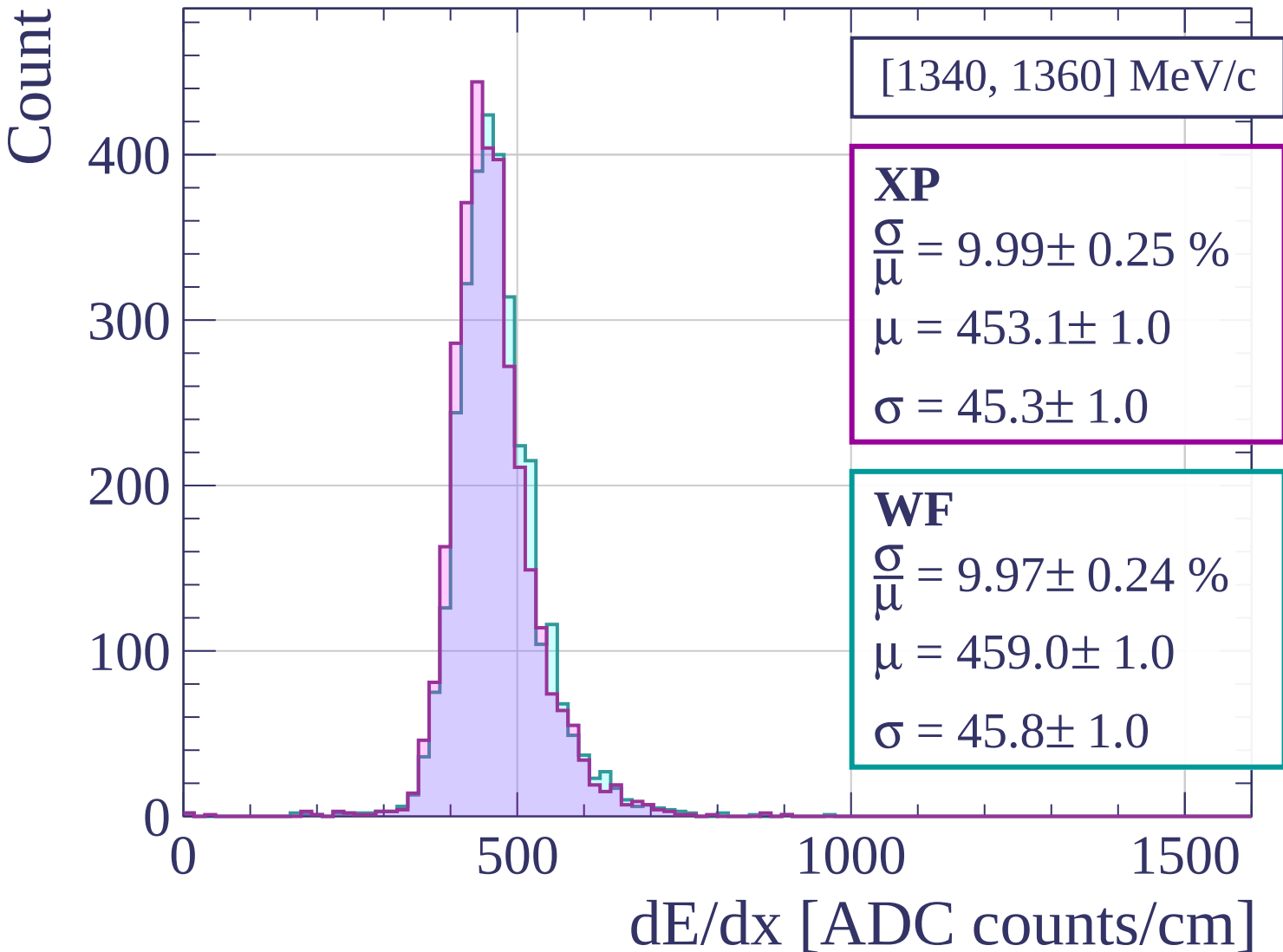
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1360, 1380] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.56 \pm 0.23 \%$$

$$\mu = 455.5 \pm 1.0$$

$$\sigma = 43.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.33 \pm 0.22 \%$$

$$\mu = 461.3 \pm 1.0$$

$$\sigma = 43.1 \pm 0.9$$

400

200

0

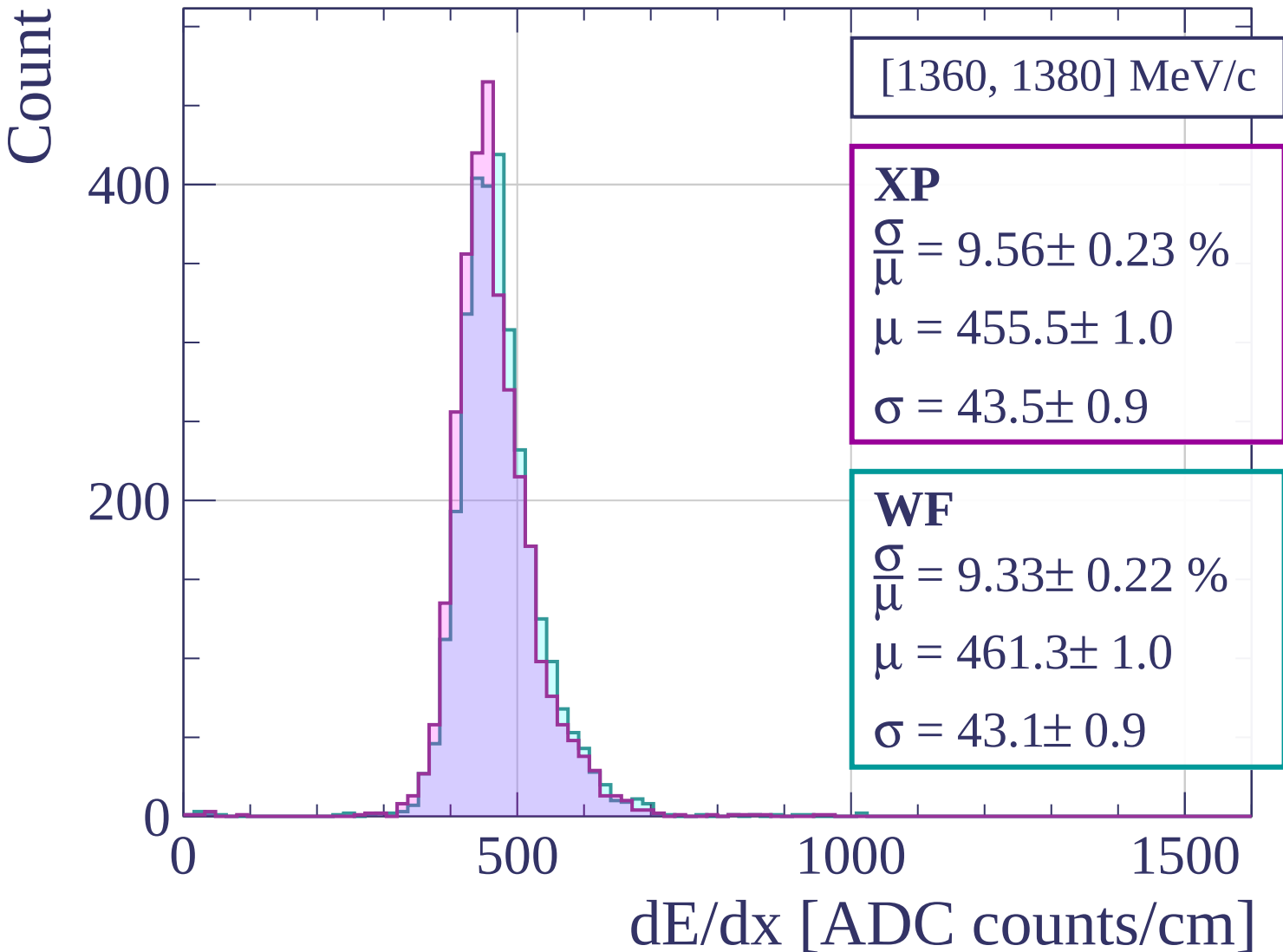
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1380, 1400] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.78 \pm 0.24 \%$$

$$\mu = 454.3 \pm 1.0$$

$$\sigma = 44.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.59 \pm 0.24 \%$$

$$\mu = 461.1 \pm 1.0$$

$$\sigma = 44.2 \pm 1.0$$

400

200

0

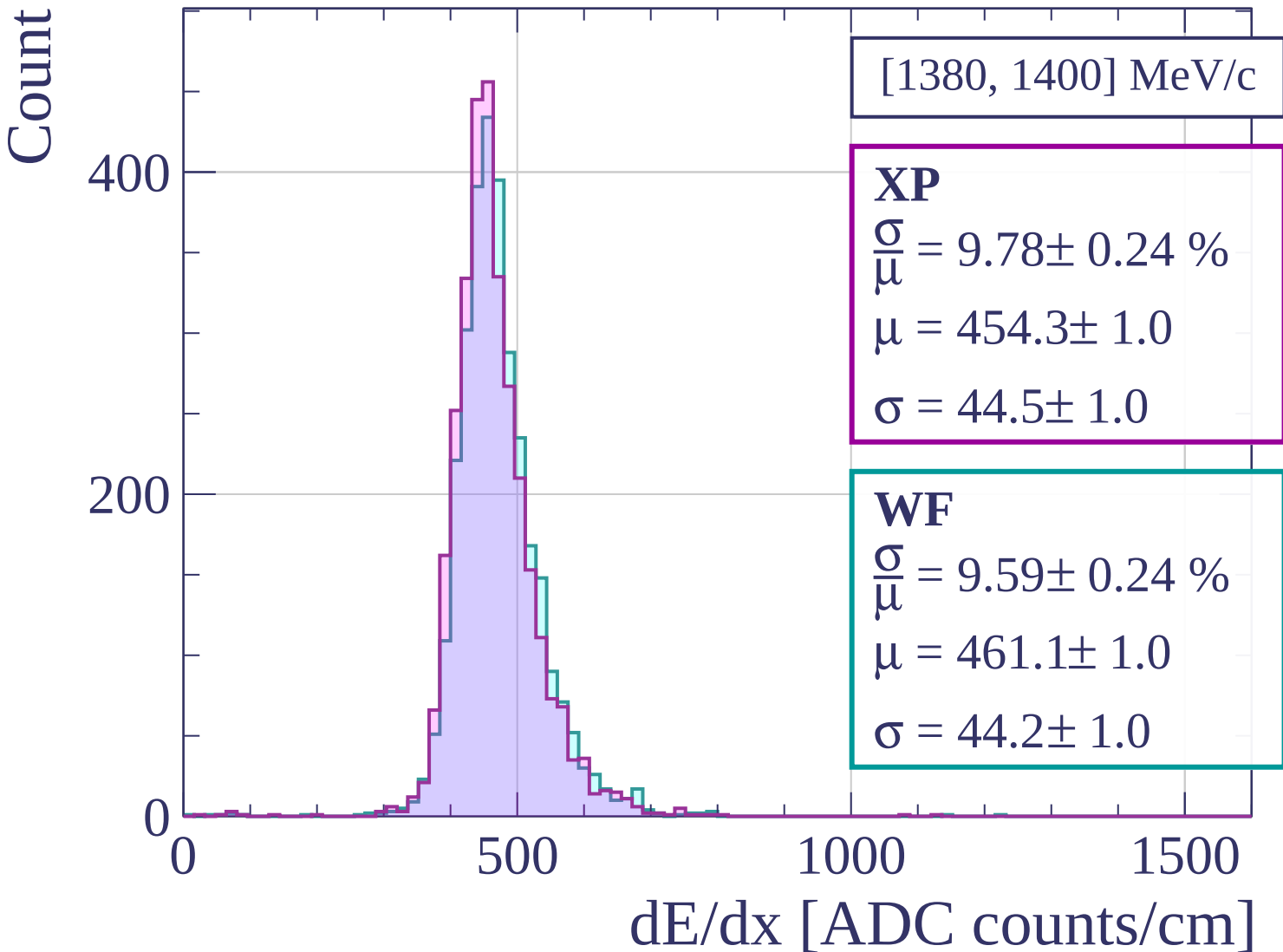
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1400, 1420] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.77 \pm 0.23 \%$$

$$\mu = 458.6 \pm 1.0$$

$$\sigma = 44.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.89 \pm 0.22 \%$$

$$\mu = 467.0 \pm 1.0$$

$$\sigma = 46.2 \pm 0.9$$

400

300

200

100

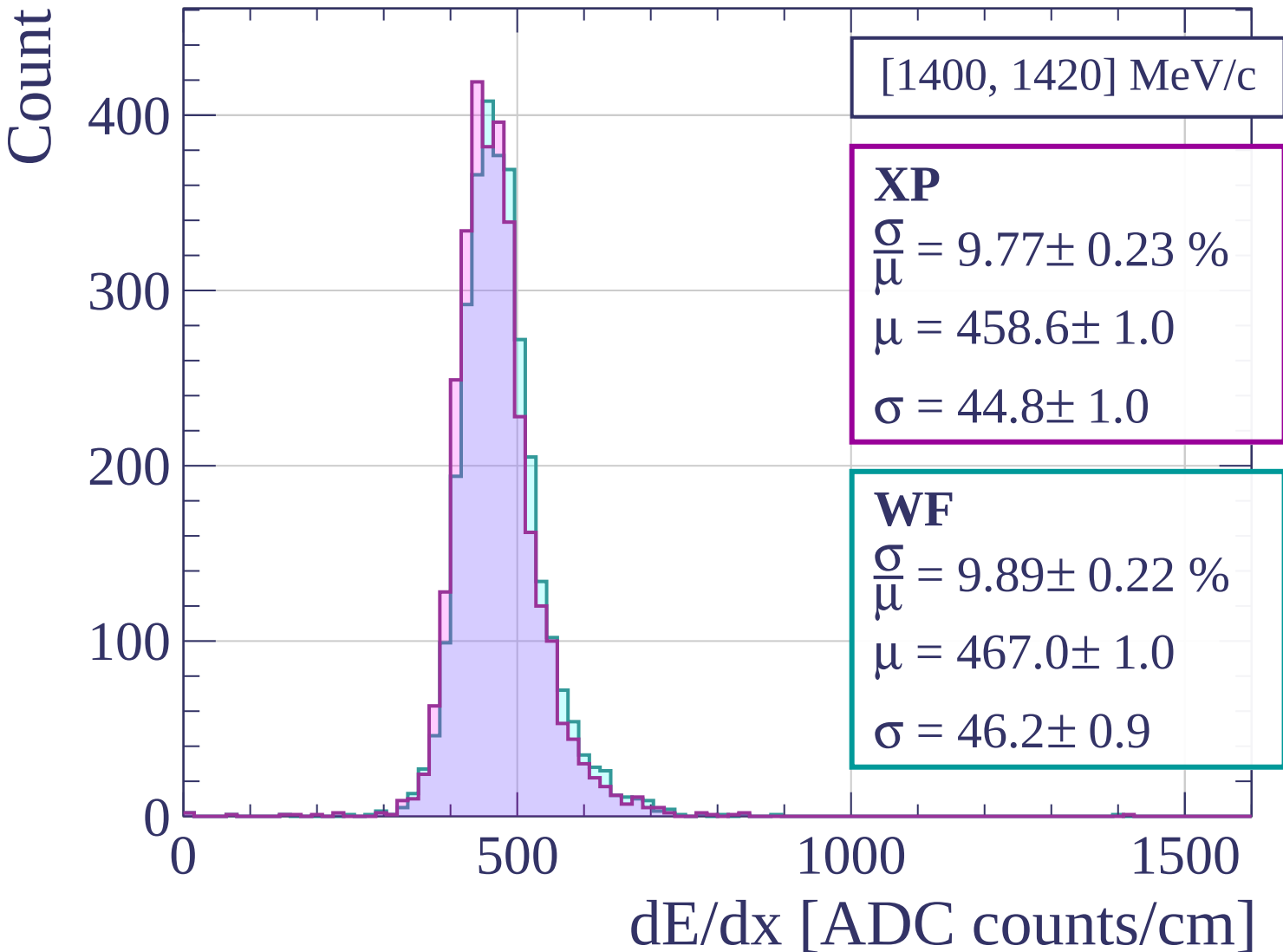
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1420, 1440] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.23 \pm 0.27 \%$$

$$\mu = 456.9 \pm 1.1$$

$$\sigma = 46.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.66 \pm 0.27 \%$$

$$\mu = 464.9 \pm 1.1$$

$$\sigma = 49.5 \pm 1.1$$

400

300

200

100

0

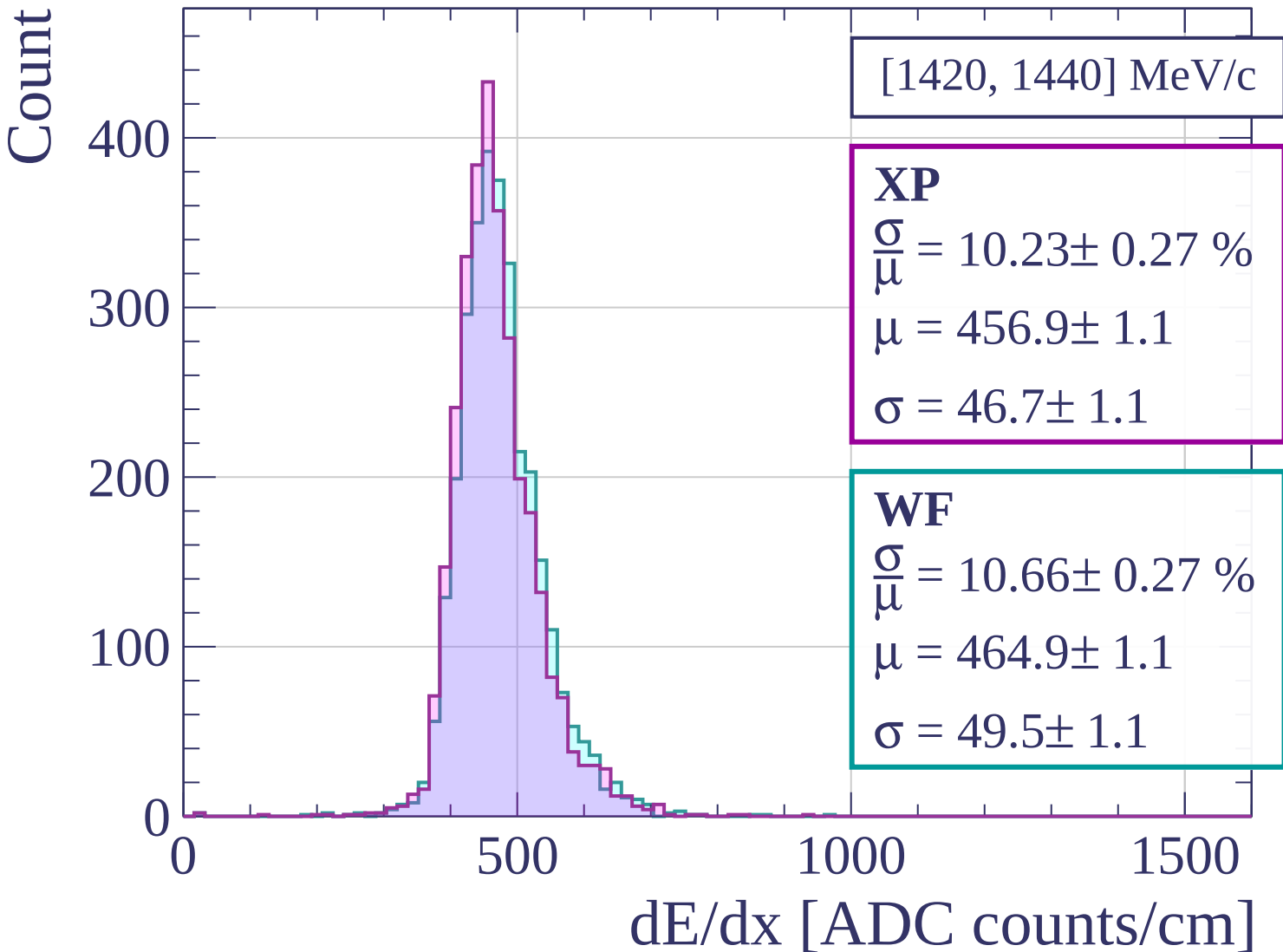
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1440, 1460] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.04 \pm 0.25 \%$$

$$\mu = 459.0 \pm 1.1$$

$$\sigma = 46.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.25 \pm 0.24 \%$$

$$\mu = 466.5 \pm 1.1$$

$$\sigma = 47.8 \pm 1.0$$

400

300

200

100

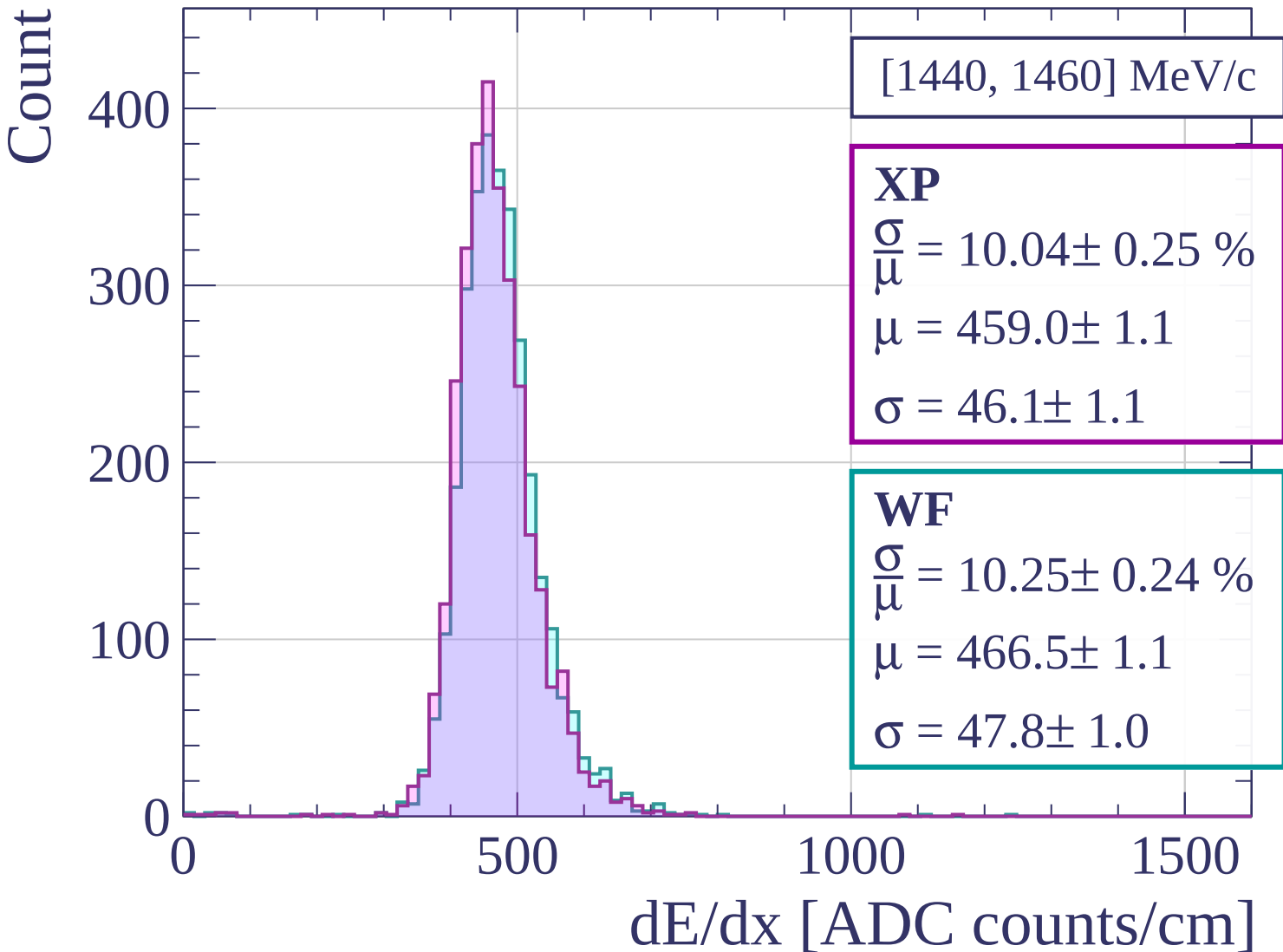
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1460, 1480] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.23 \%$$

$$\mu = 459.1 \pm 1.0$$

$$\sigma = 43.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.71 \pm 0.21 \%$$

$$\mu = 467.7 \pm 1.0$$

$$\sigma = 45.4 \pm 0.9$$

400

300

200

100

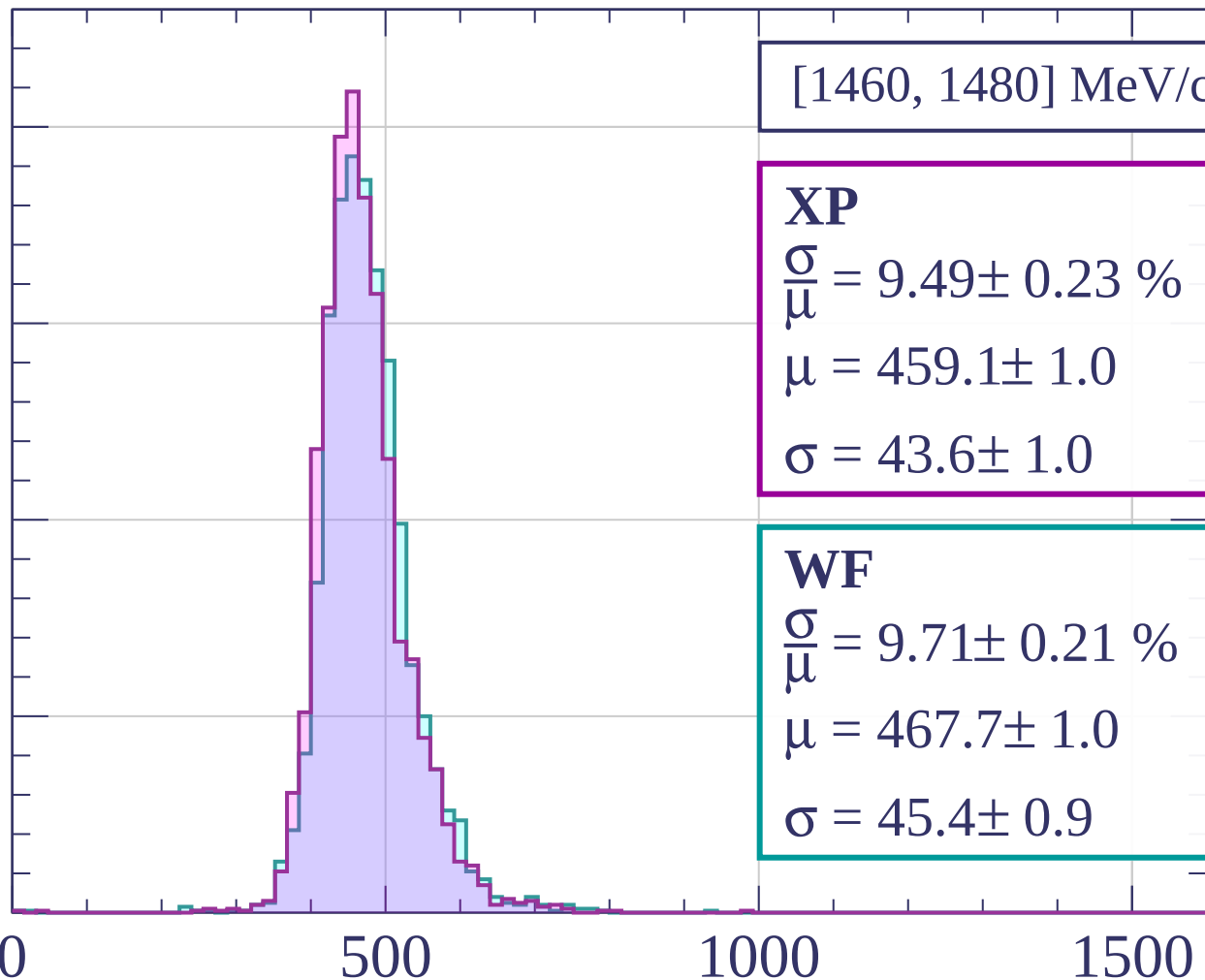
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1480, 1500] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.67 \pm 0.24 \%$$

$$\mu = 460.2 \pm 1.0$$

$$\sigma = 44.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.78 \pm 0.22 \%$$

$$\mu = 469.1 \pm 1.0$$

$$\sigma = 45.9 \pm 1.0$$

400

300

200

100

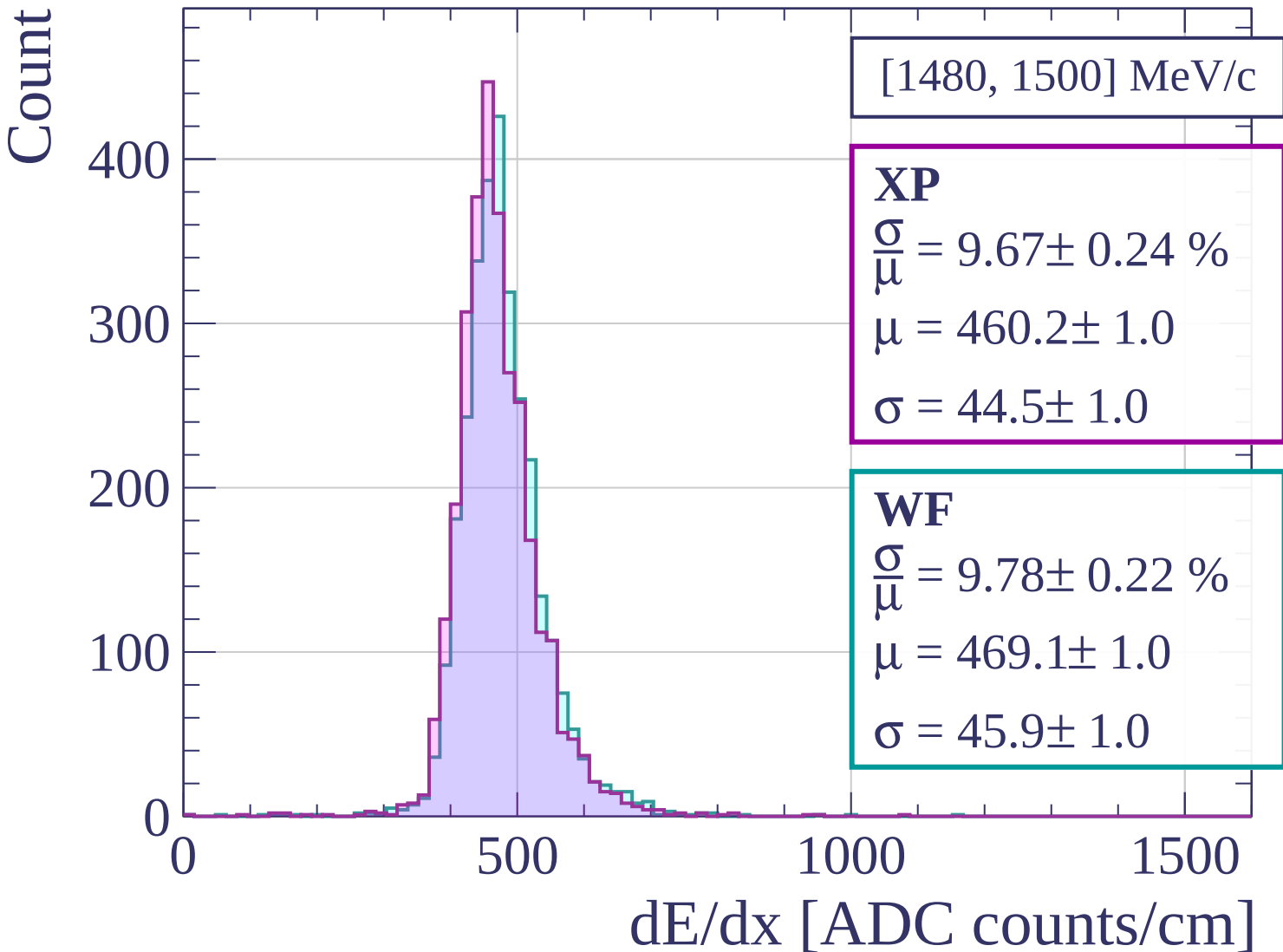
0

500

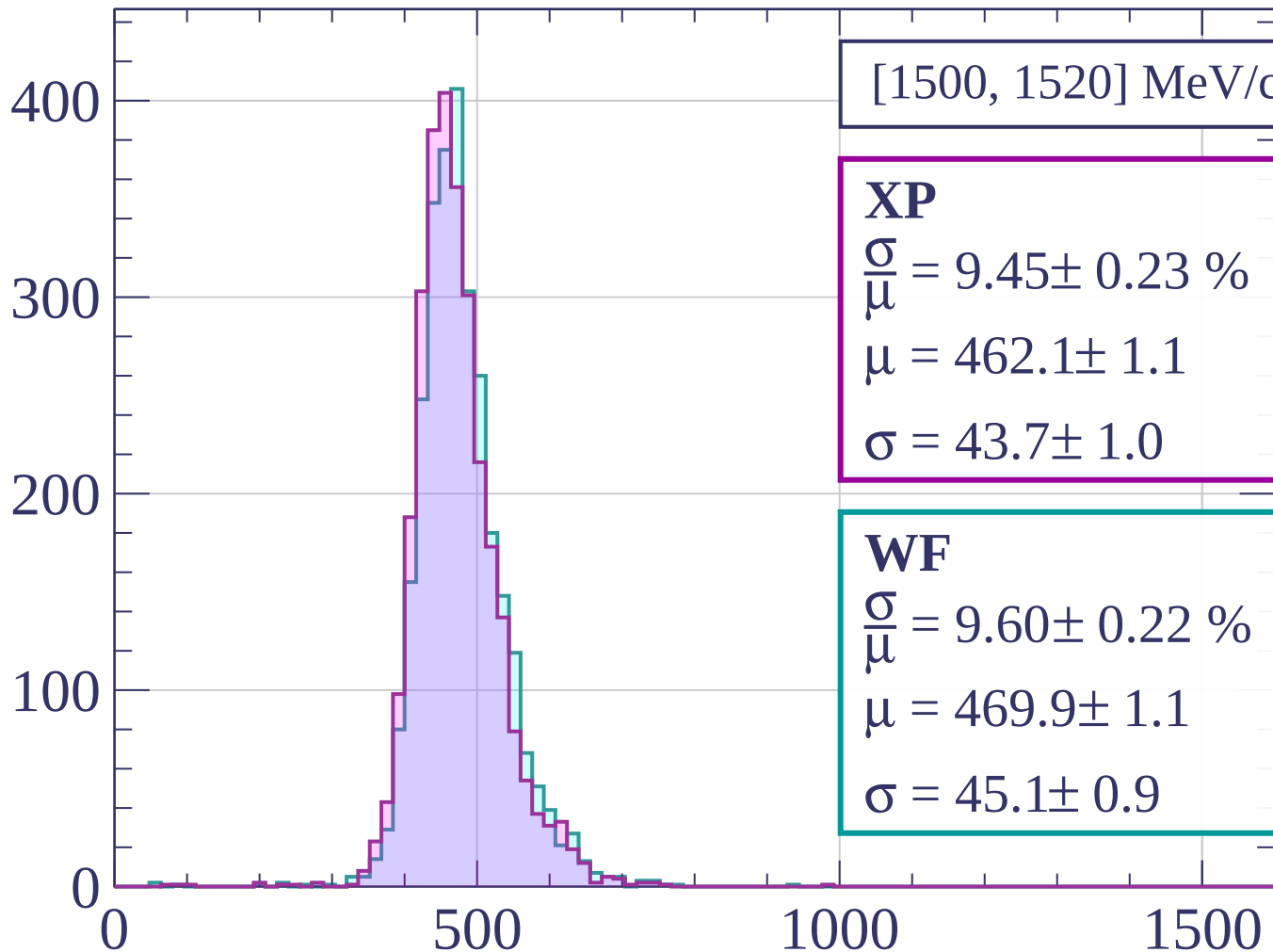
1000

1500

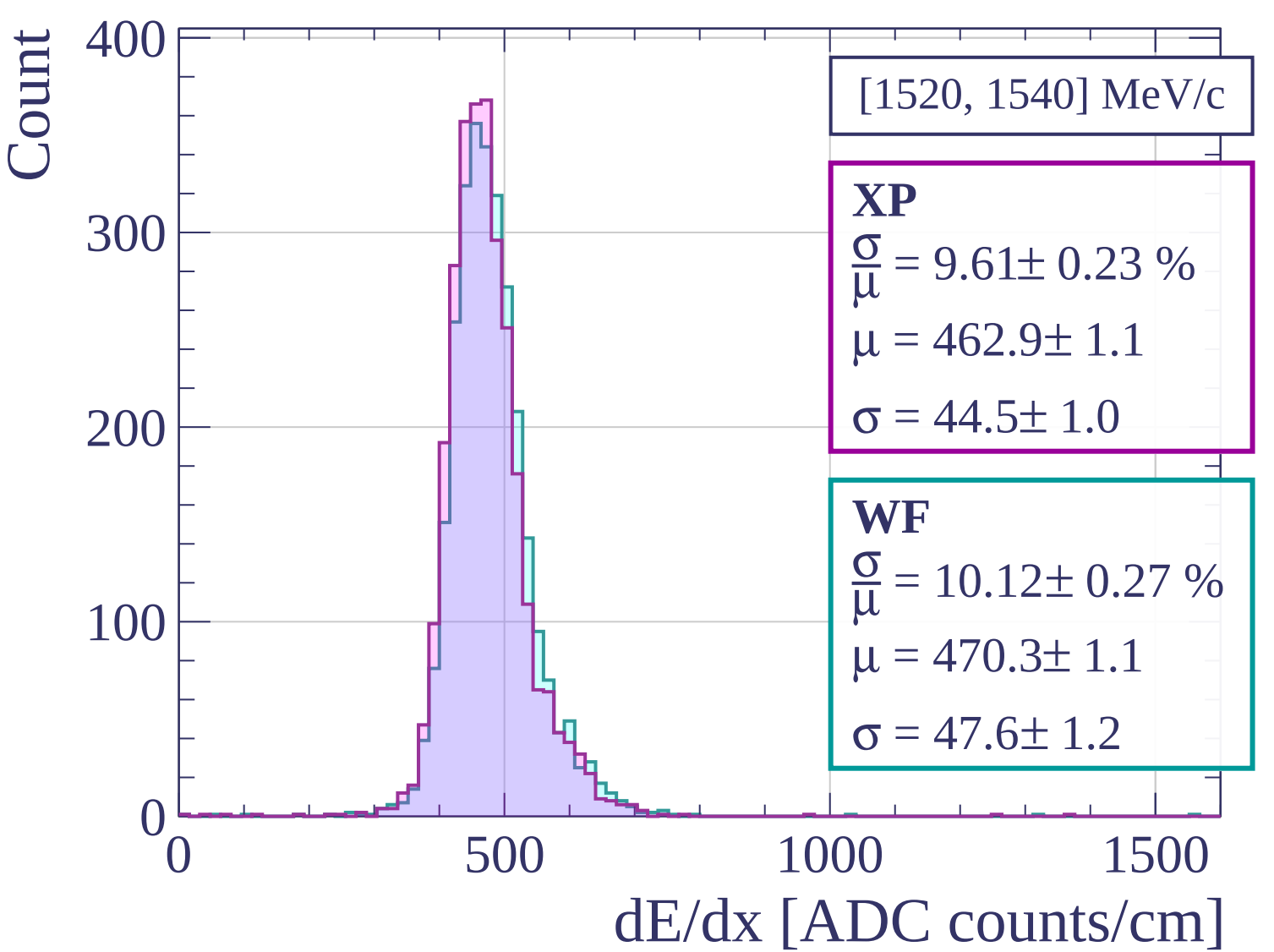
dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]



Count

[1540, 1560] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.43 \pm 0.24 \%$$

$$\mu = 462.6 \pm 1.1$$

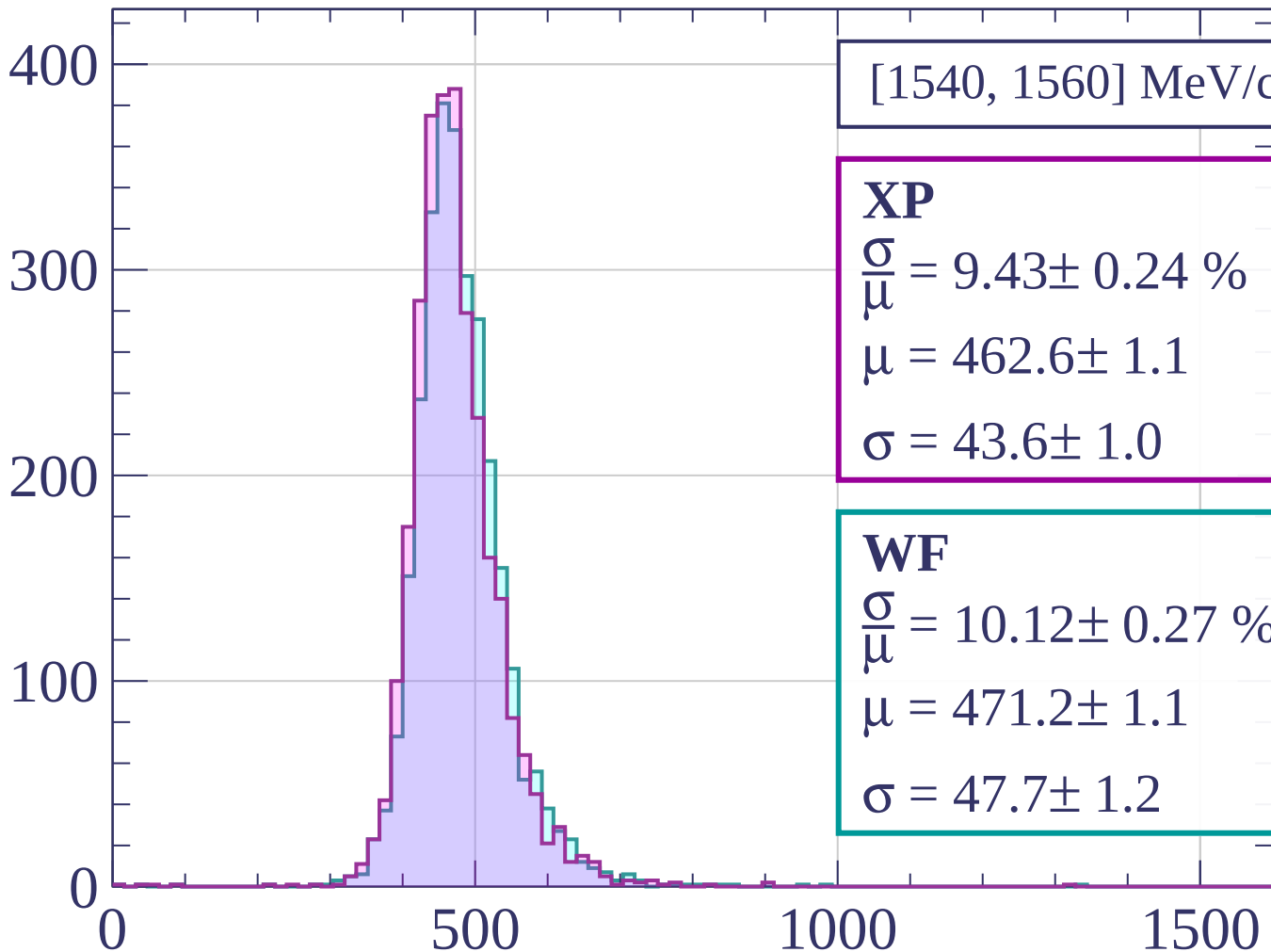
$$\sigma = 43.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.12 \pm 0.27 \%$$

$$\mu = 471.2 \pm 1.1$$

$$\sigma = 47.7 \pm 1.2$$



dE/dx [ADC counts/cm]

Count

[1560, 1580] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.88 \pm 0.23 \%$$

$$\mu = 465.7 \pm 1.1$$

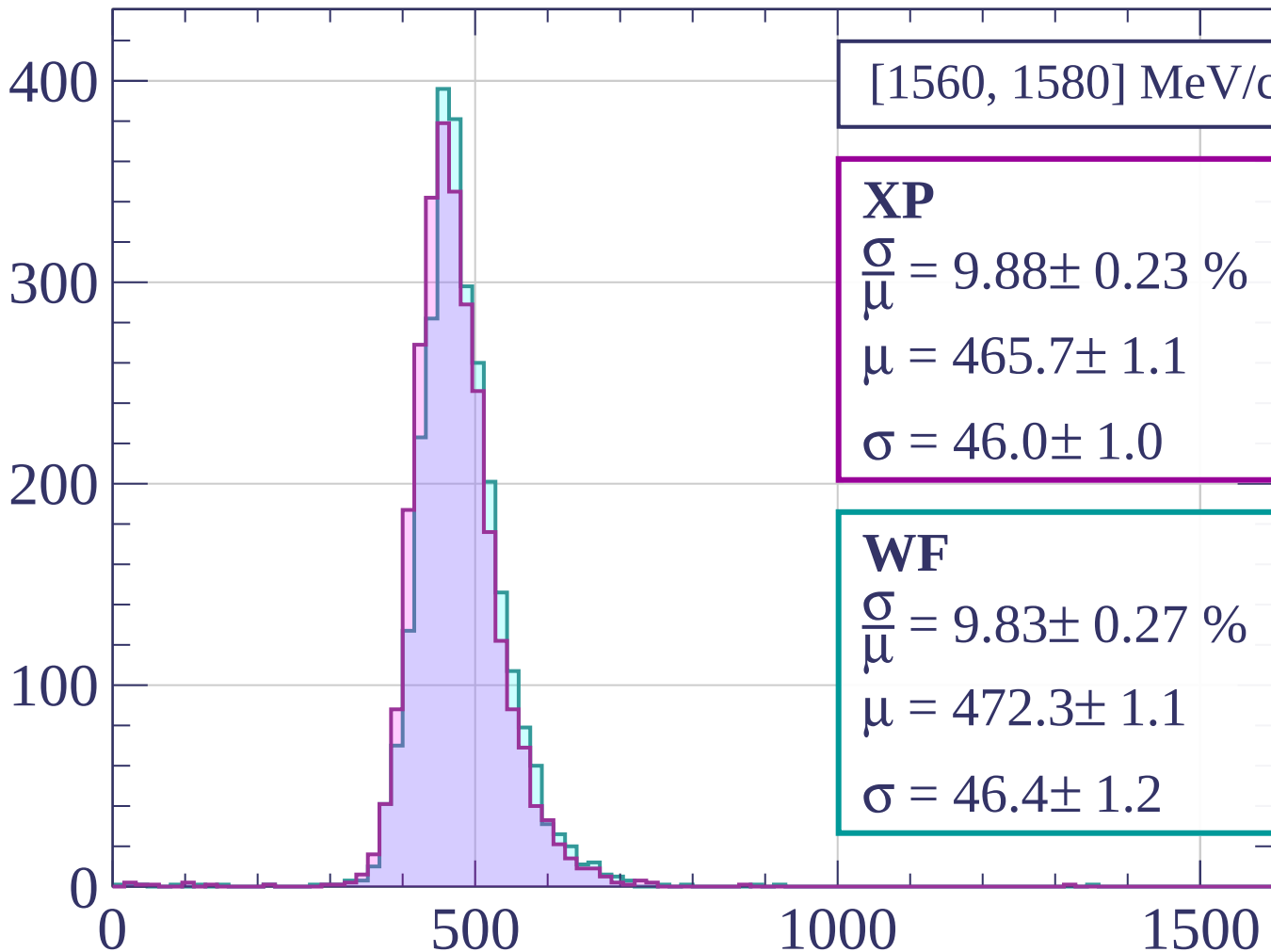
$$\sigma = 46.0 \pm 1.0$$

WF

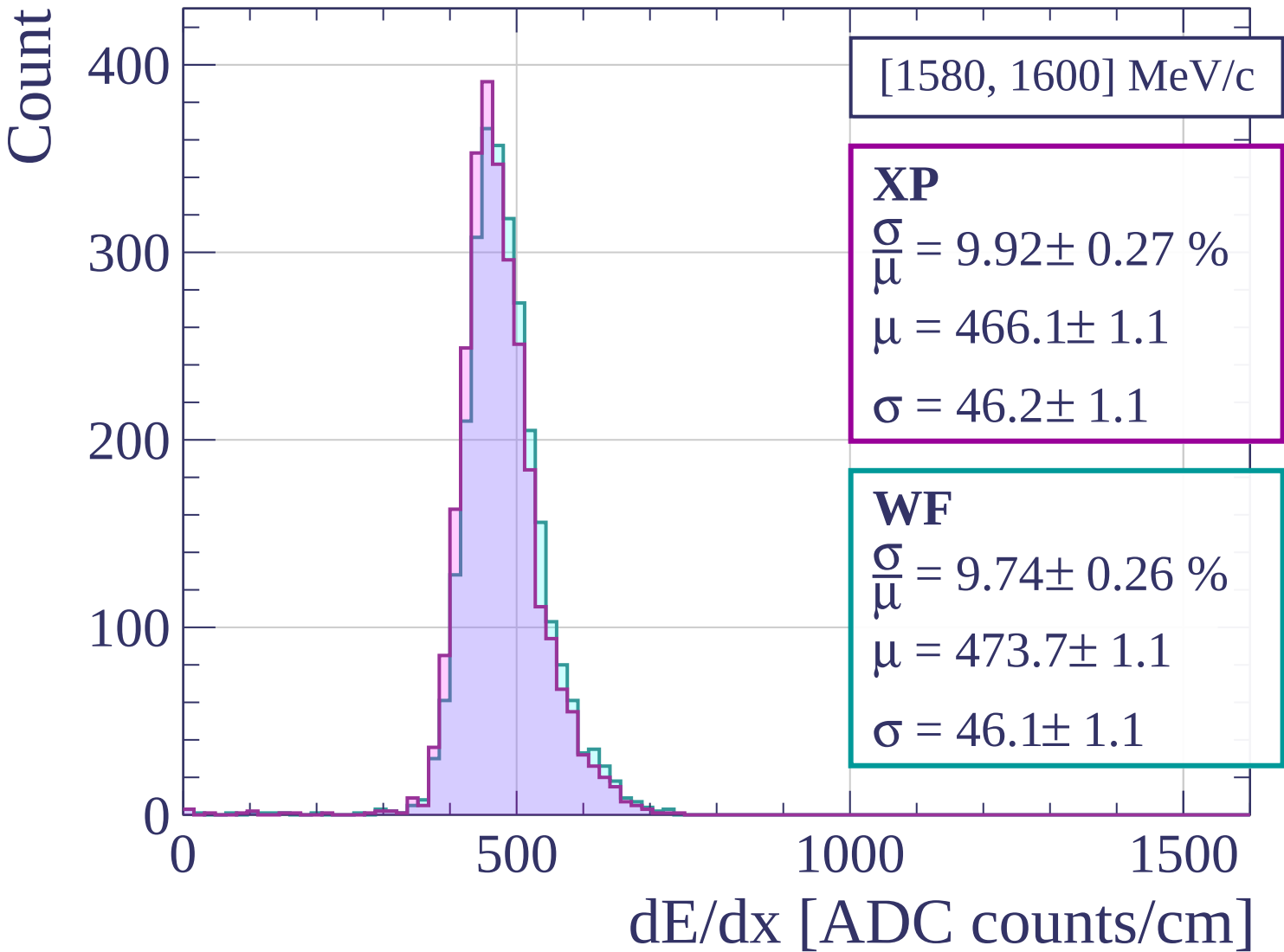
$$\frac{\sigma}{\mu} = 9.83 \pm 0.27 \%$$

$$\mu = 472.3 \pm 1.1$$

$$\sigma = 46.4 \pm 1.2$$



dE/dx [ADC counts/cm]



Count

[1600, 1620] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.70 \pm 0.23 \%$$

$$\mu = 467.4 \pm 1.0$$

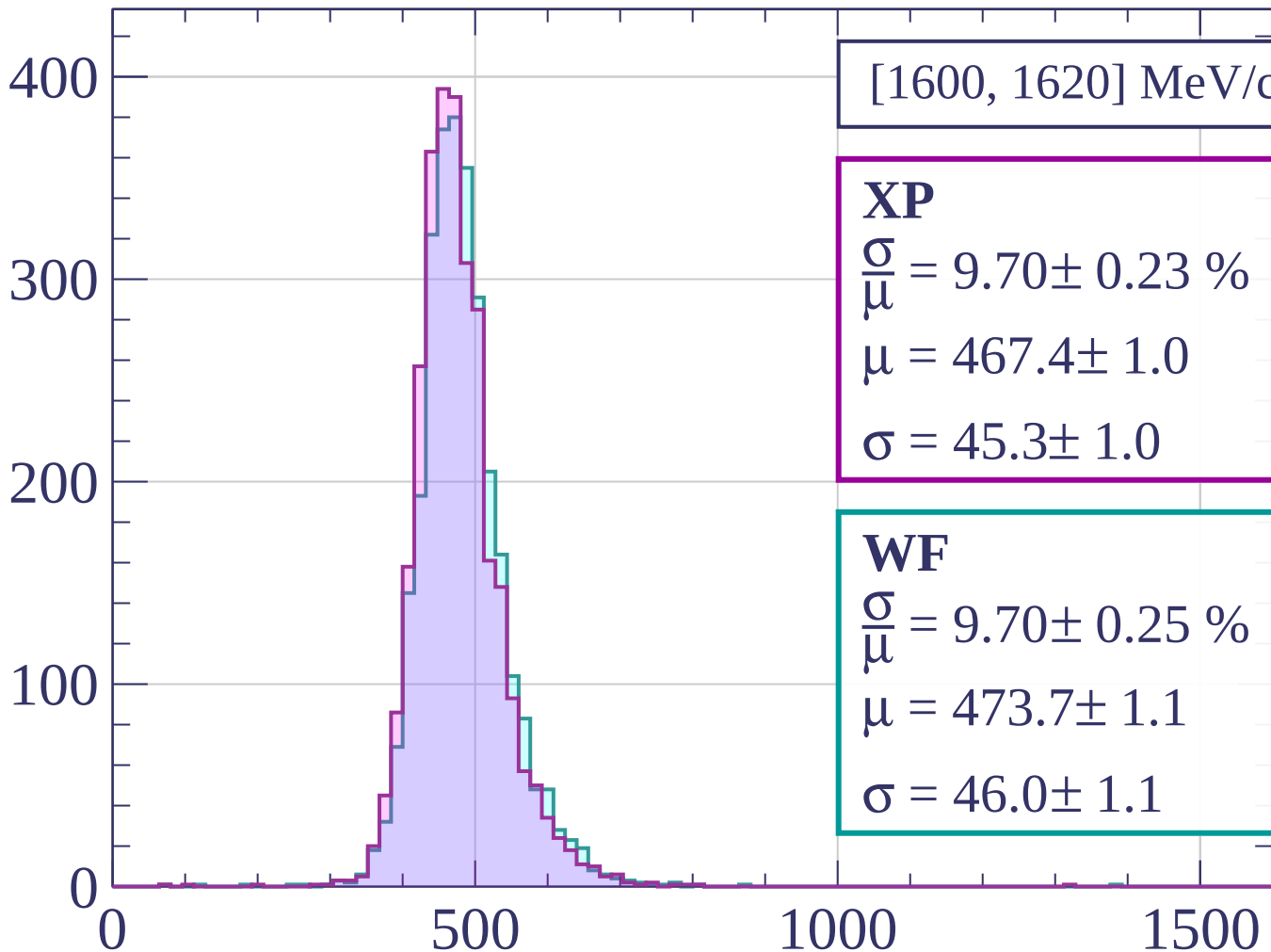
$$\sigma = 45.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.70 \pm 0.25 \%$$

$$\mu = 473.7 \pm 1.1$$

$$\sigma = 46.0 \pm 1.1$$



dE/dx [ADC counts/cm]

Count

[1620, 1640] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.92 \pm 0.24 \%$$

$$\mu = 468.7 \pm 1.1$$

$$\sigma = 46.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.23 \pm 0.26 \%$$

$$\mu = 476.5 \pm 1.1$$

$$\sigma = 48.7 \pm 1.1$$

300

200

100

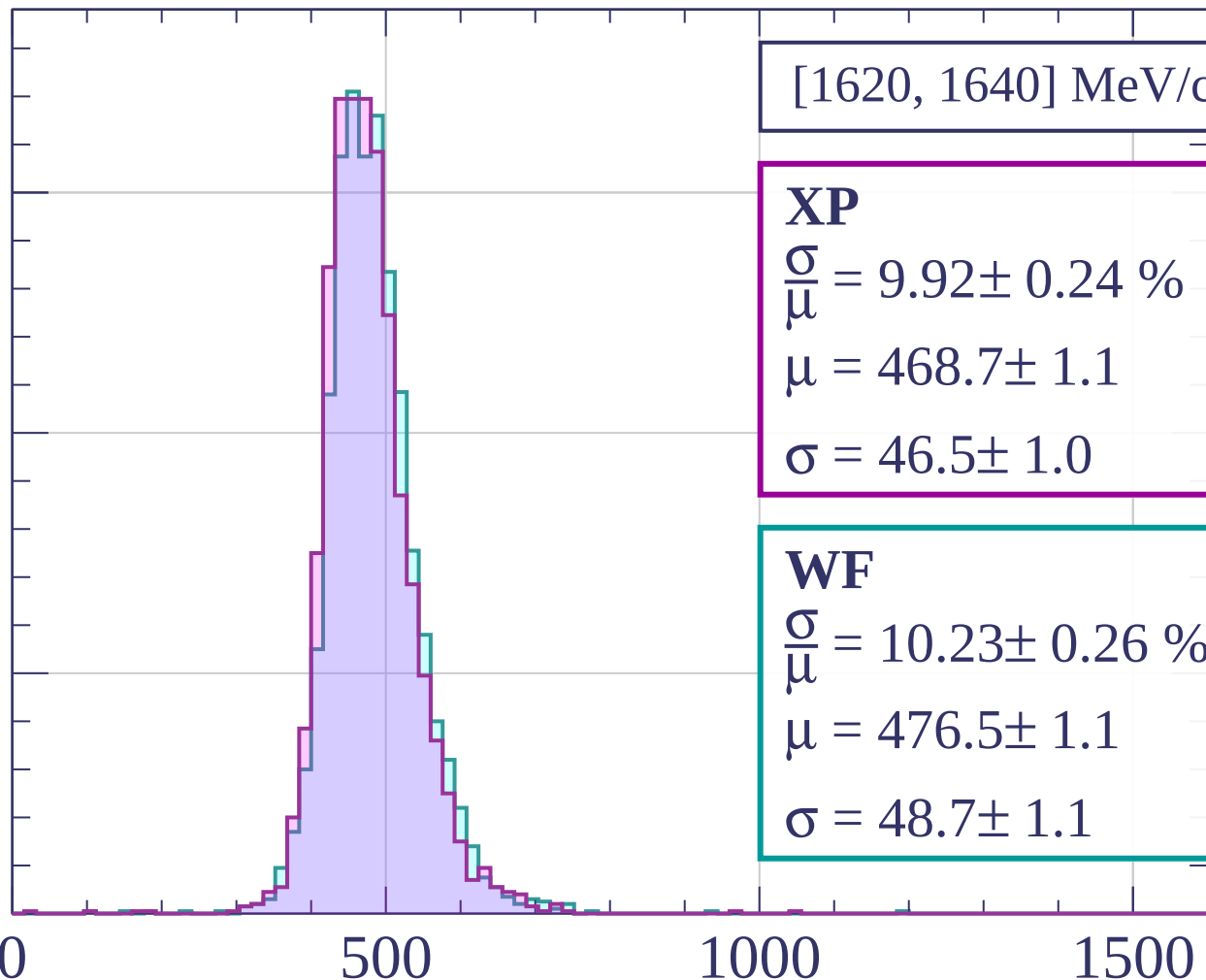
0

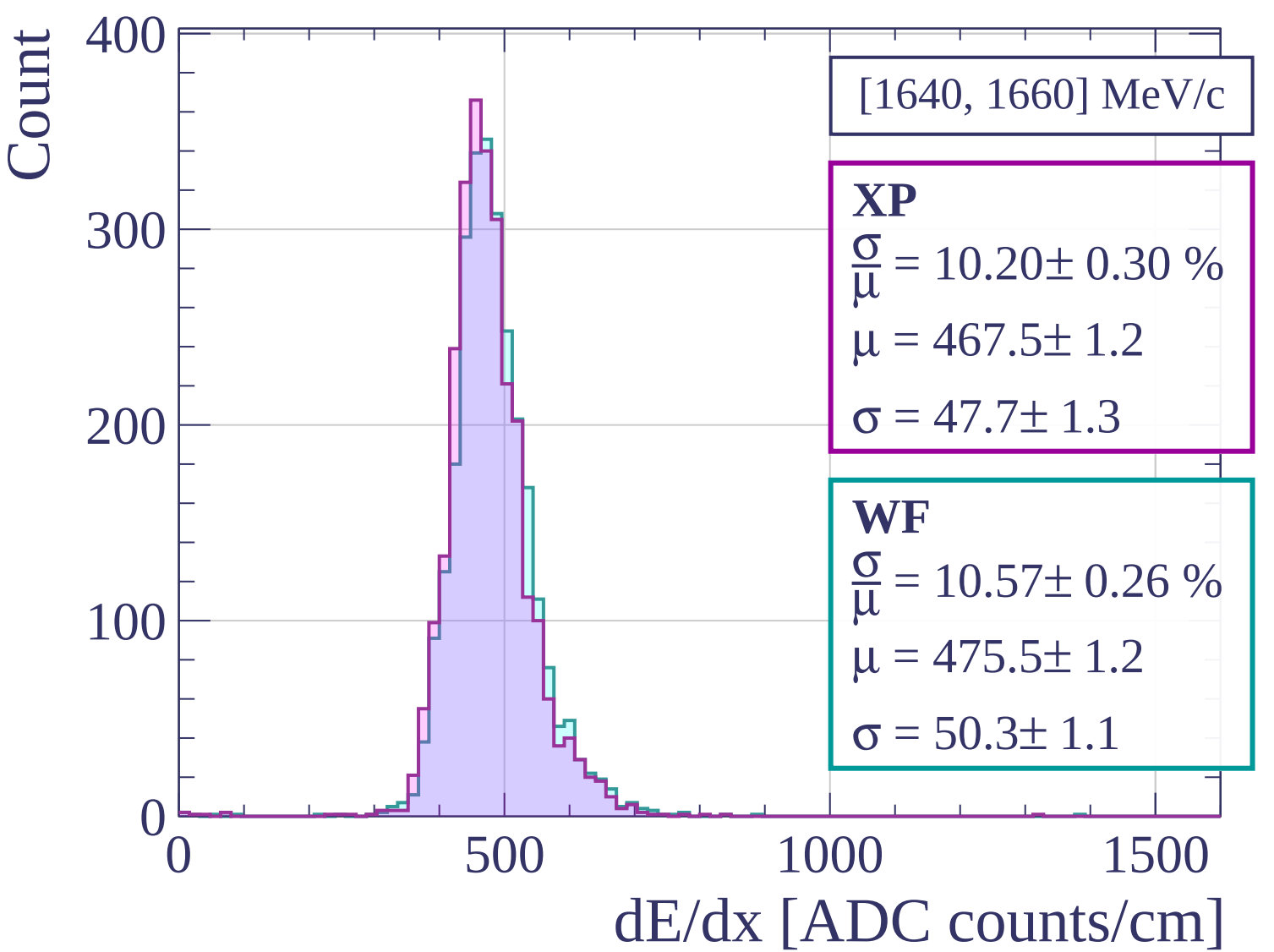
500

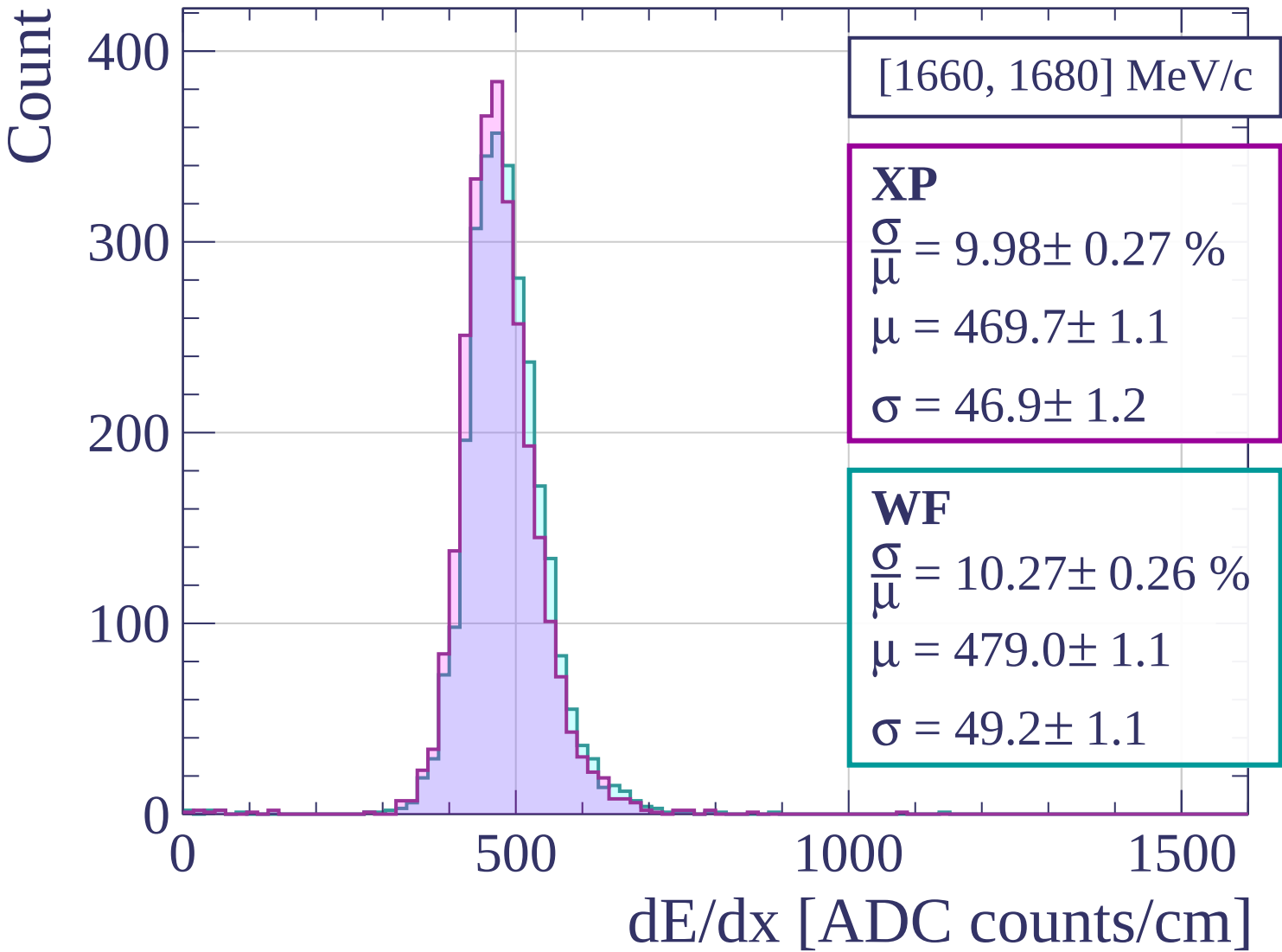
1000

1500

dE/dx [ADC counts/cm]







Count

[1680, 1700] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.57 \pm 0.26 \%$$

$$\mu = 467.4 \pm 1.1$$

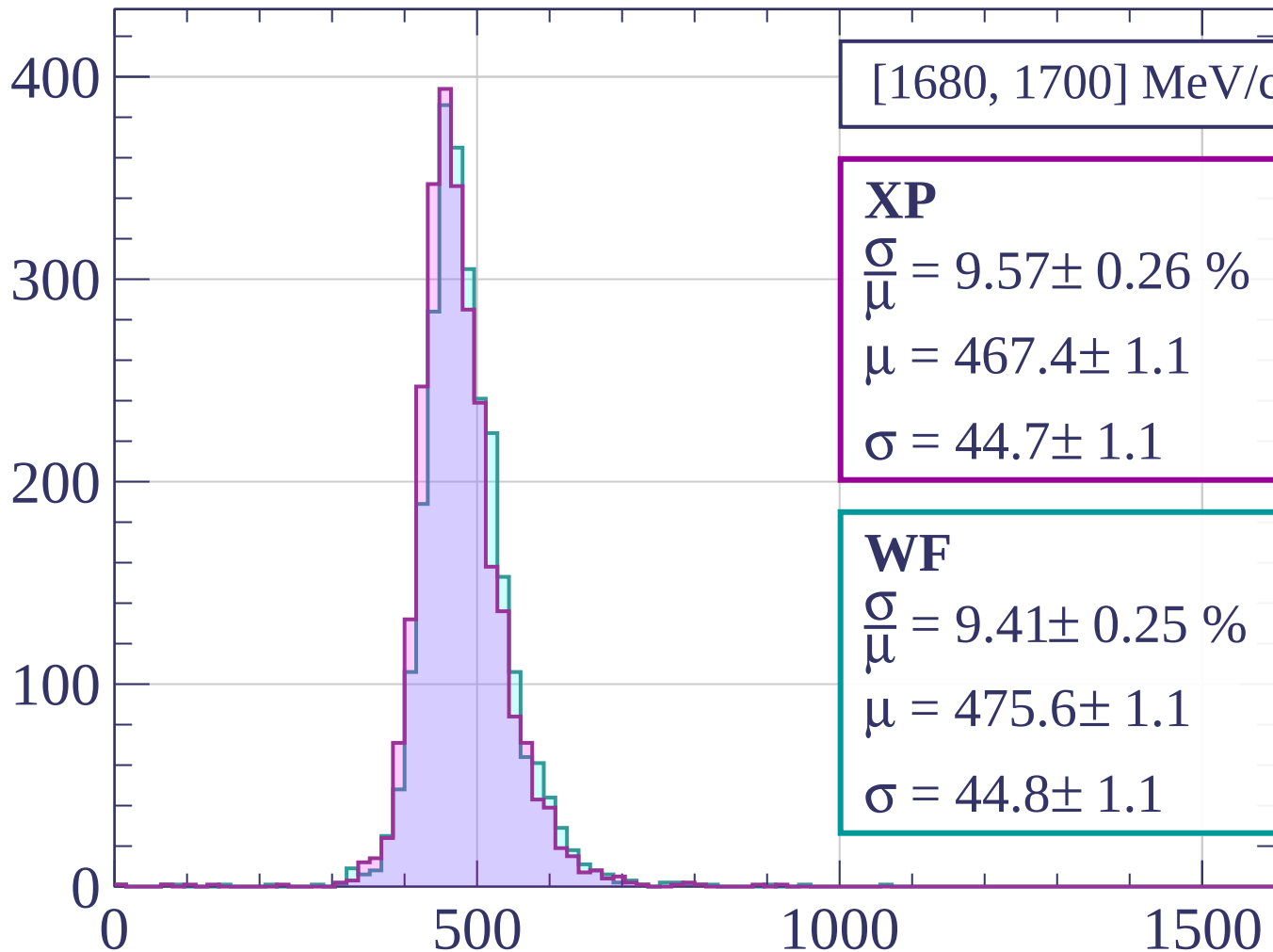
$$\sigma = 44.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.41 \pm 0.25 \%$$

$$\mu = 475.6 \pm 1.1$$

$$\sigma = 44.8 \pm 1.1$$



dE/dx [ADC counts/cm]

Count

[1700, 1720] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.87 \pm 0.26 \%$$

$$\mu = 471.3 \pm 1.1$$

$$\sigma = 46.5 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.13 \pm 0.26 \%$$

$$\mu = 480.9 \pm 1.2$$

$$\sigma = 48.7 \pm 1.1$$

300

200

100

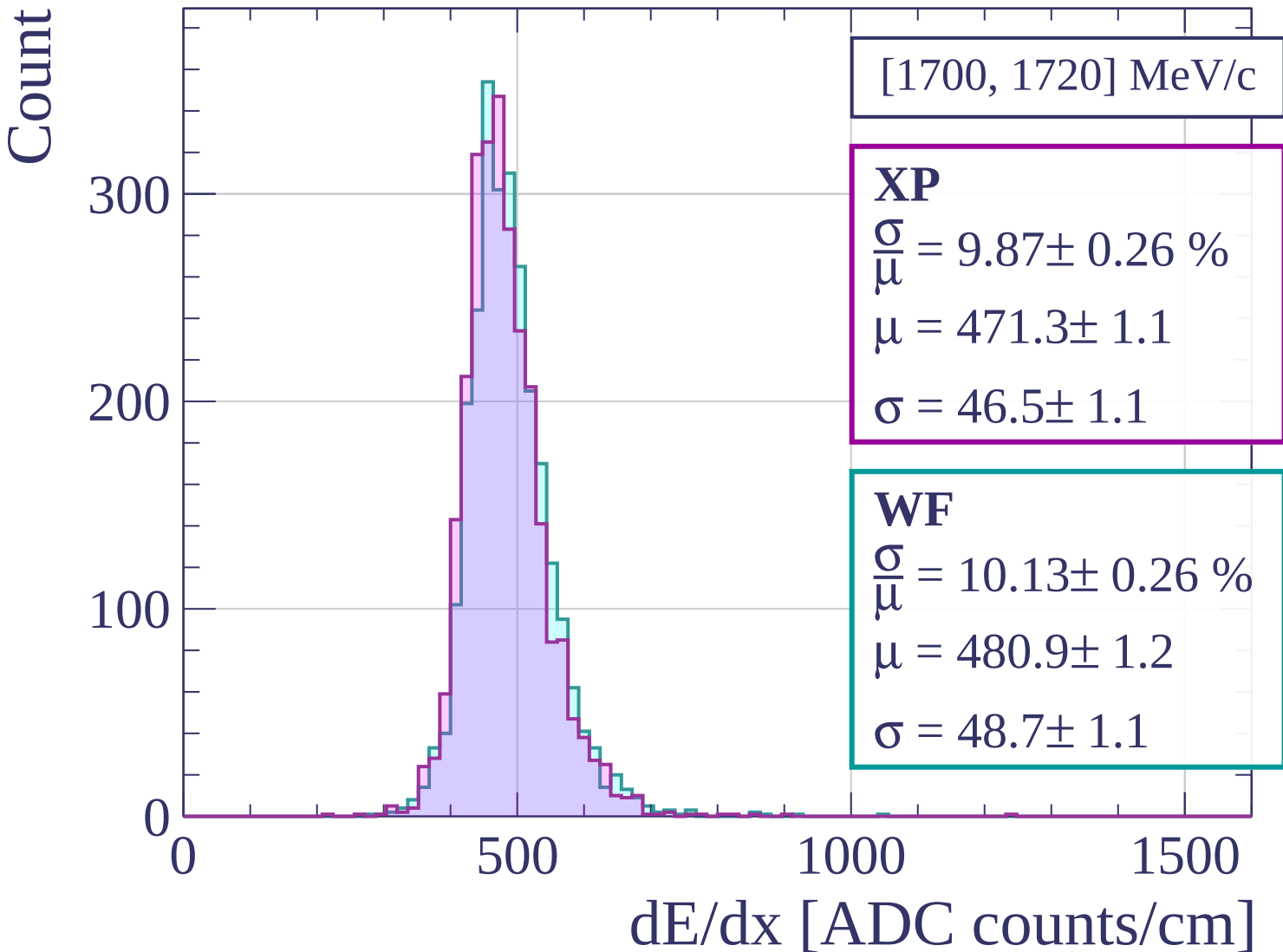
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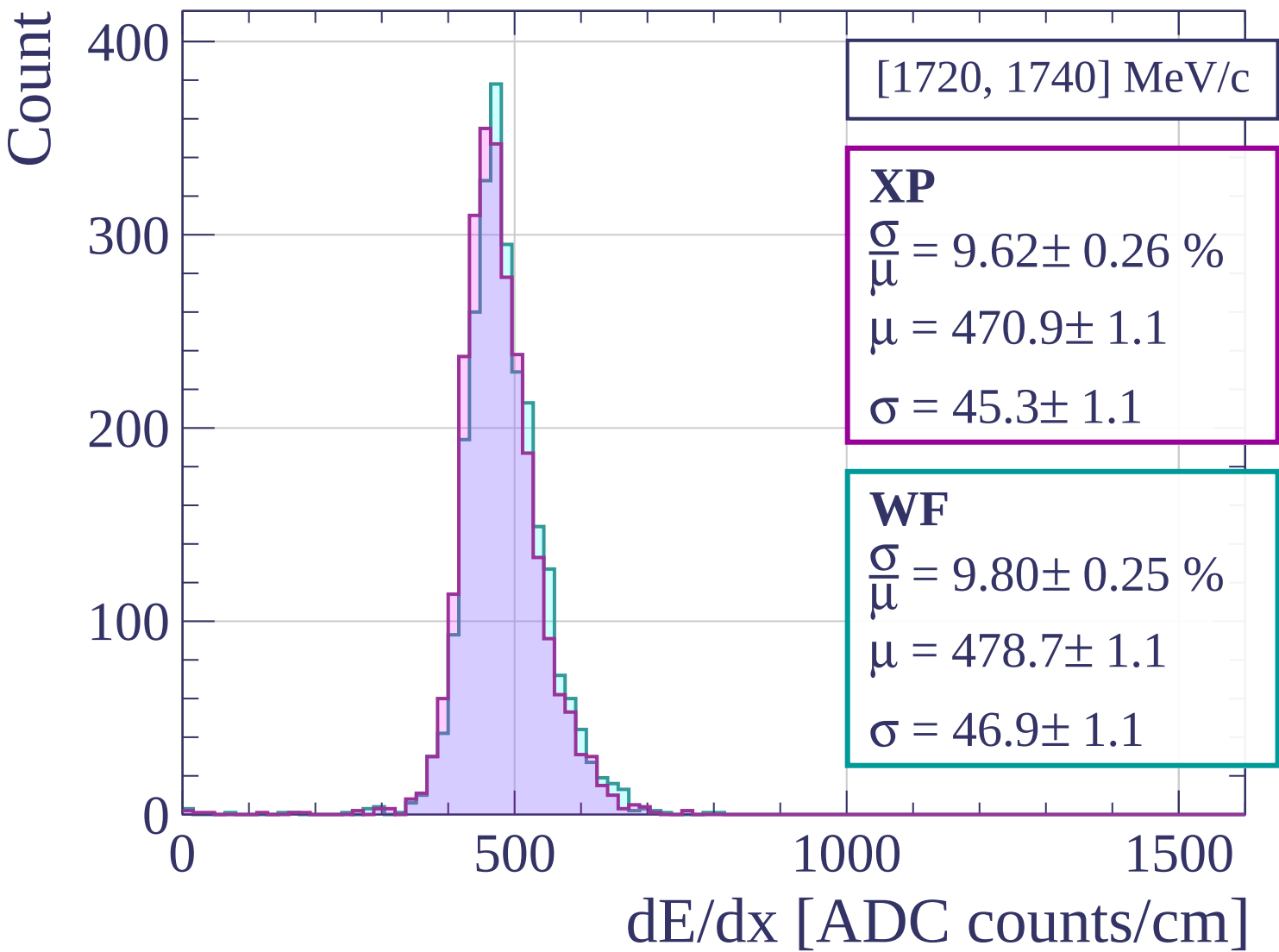
500

1000

1500

dE/dx [ADC counts/cm]





Count

[1740, 1760] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.93 \pm 0.27 \%$$

$$\mu = 472.7 \pm 1.2$$

$$\sigma = 46.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.40 \pm 0.27 \%$$

$$\mu = 481.6 \pm 1.2$$

$$\sigma = 50.1 \pm 1.2$$

300

200

100

0

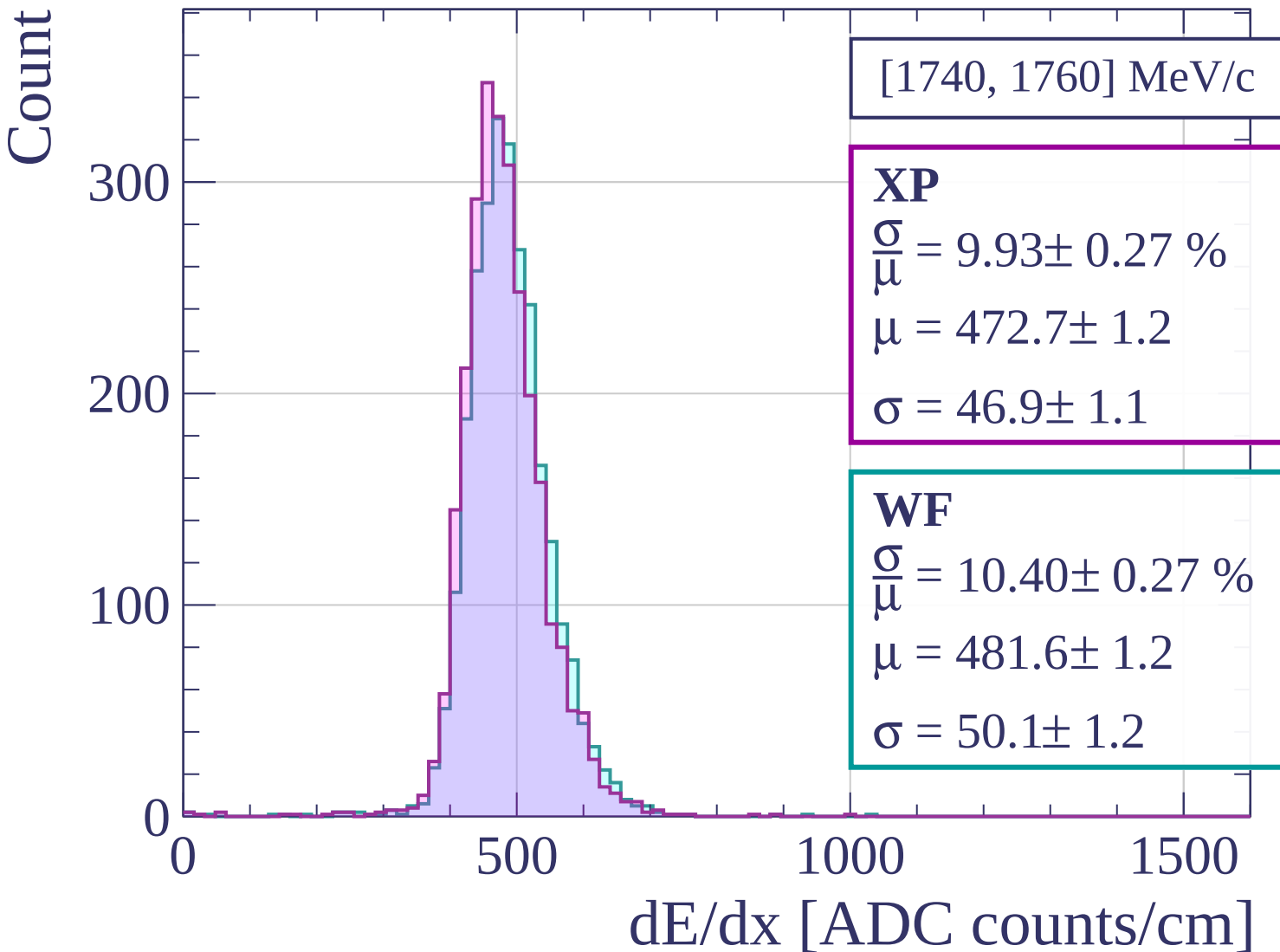
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1760, 1780] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.08 \pm 0.30 \%$$

$$\mu = 471.9 \pm 1.2$$

$$\sigma = 47.6 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.23 \pm 0.28 \%$$

$$\mu = 481.6 \pm 1.2$$

$$\sigma = 49.2 \pm 1.2$$

300

200

100

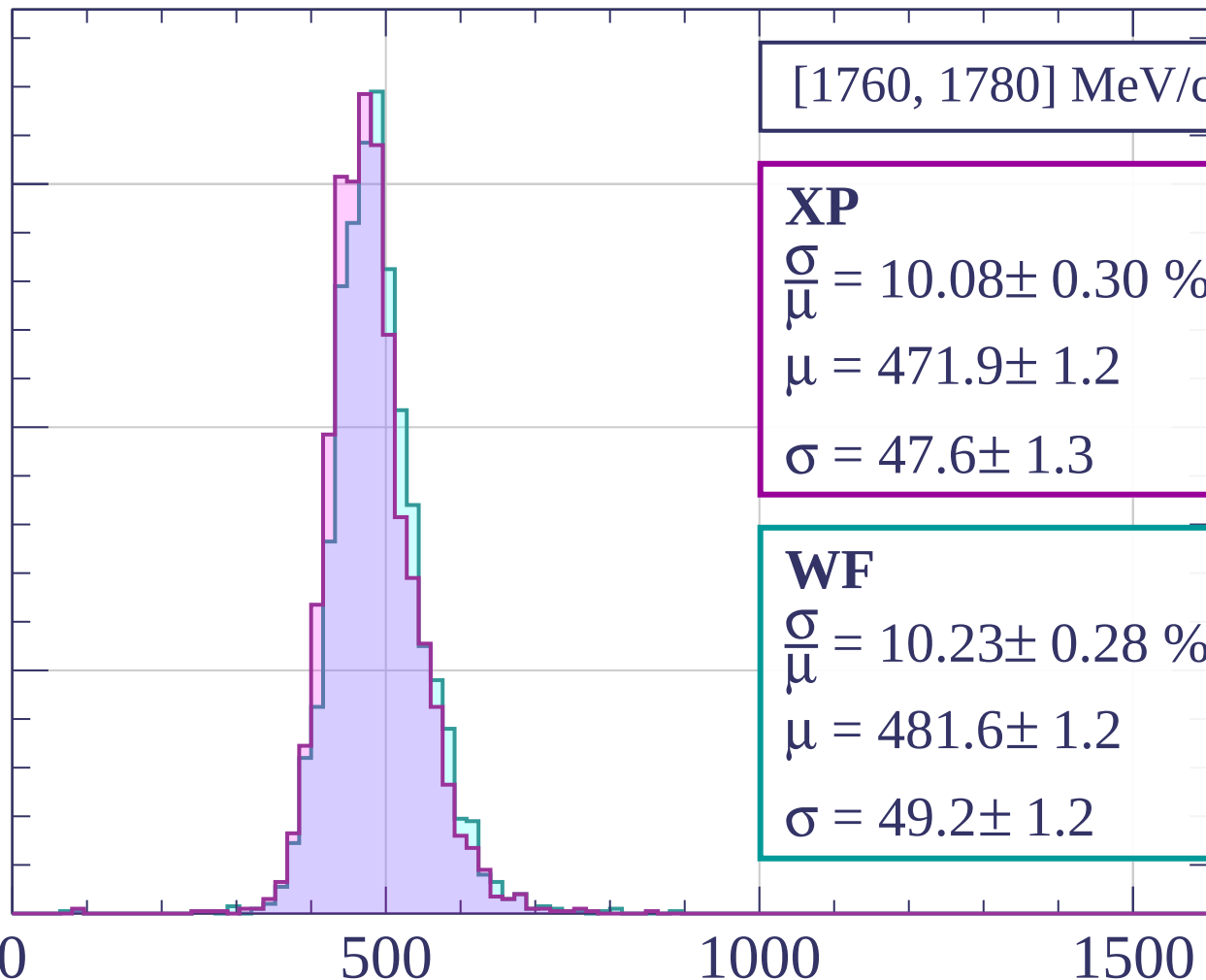
0

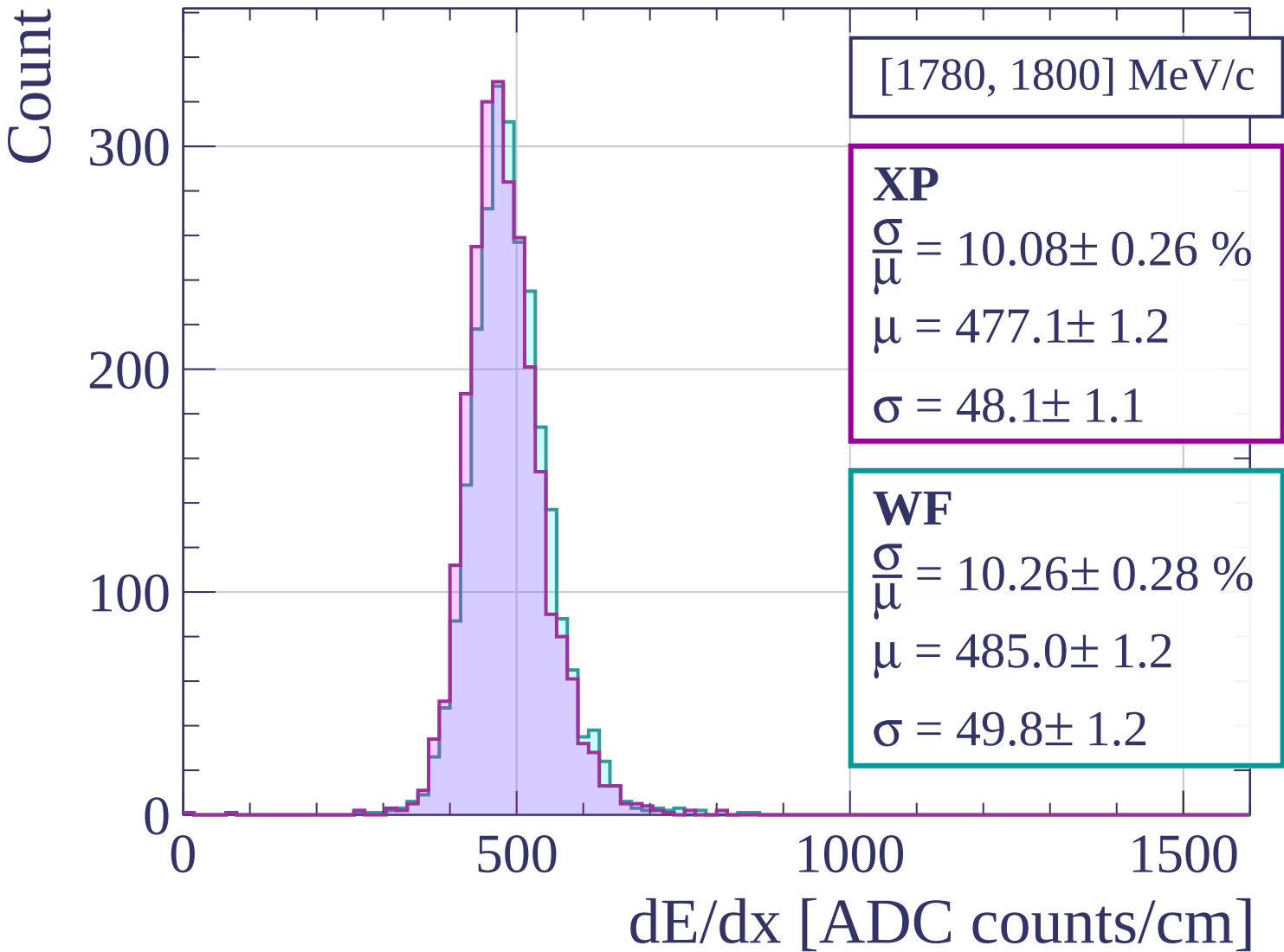
500

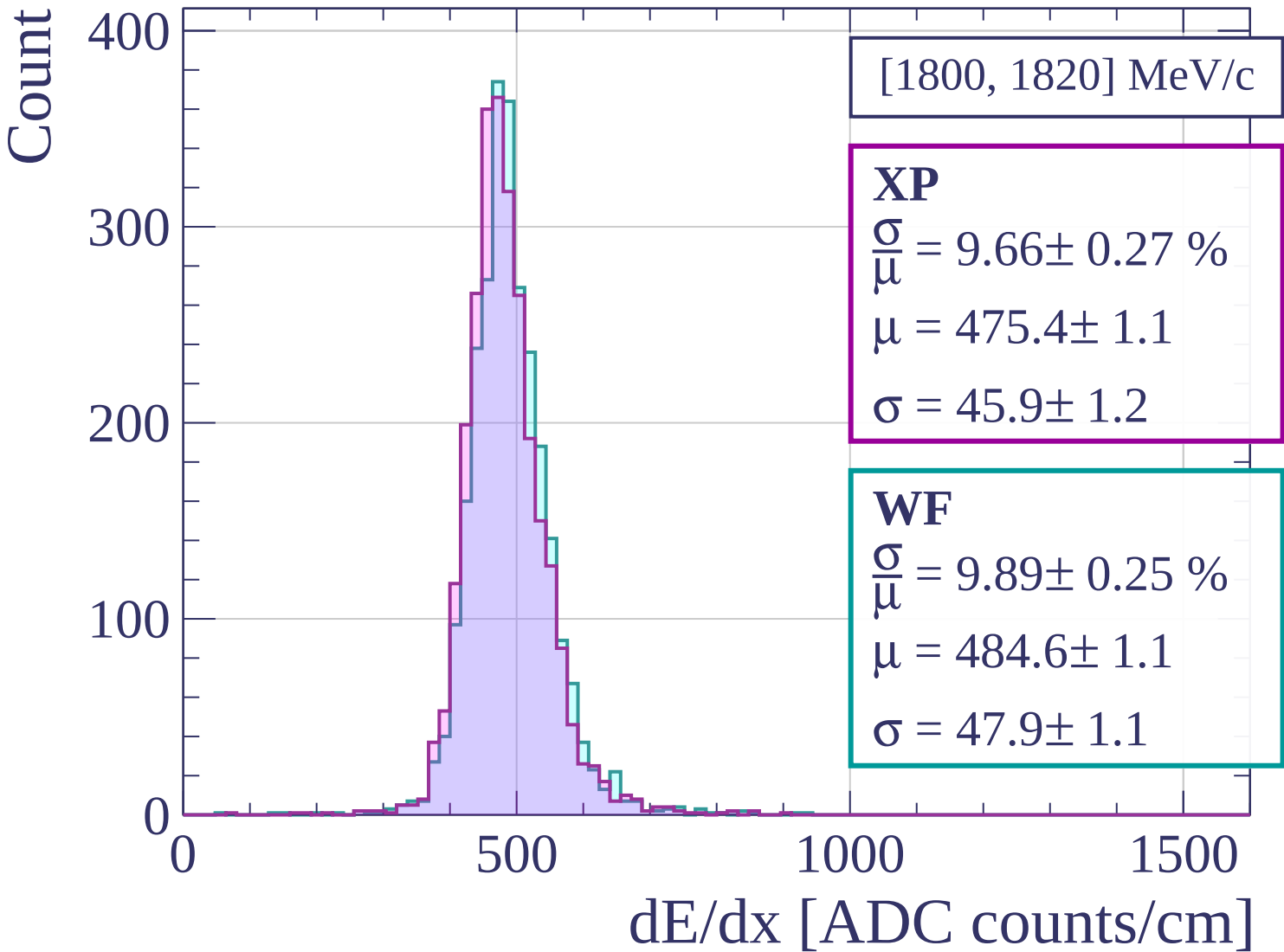
1000

1500

dE/dx [ADC counts/cm]







Count

[1820, 1840] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.95 \pm 0.29 \%$$

$$\mu = 472.4 \pm 1.2$$

$$\sigma = 47.0 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.95 \pm 0.25 \%$$

$$\mu = 482.2 \pm 1.2$$

$$\sigma = 48.0 \pm 1.1$$

300

200

100

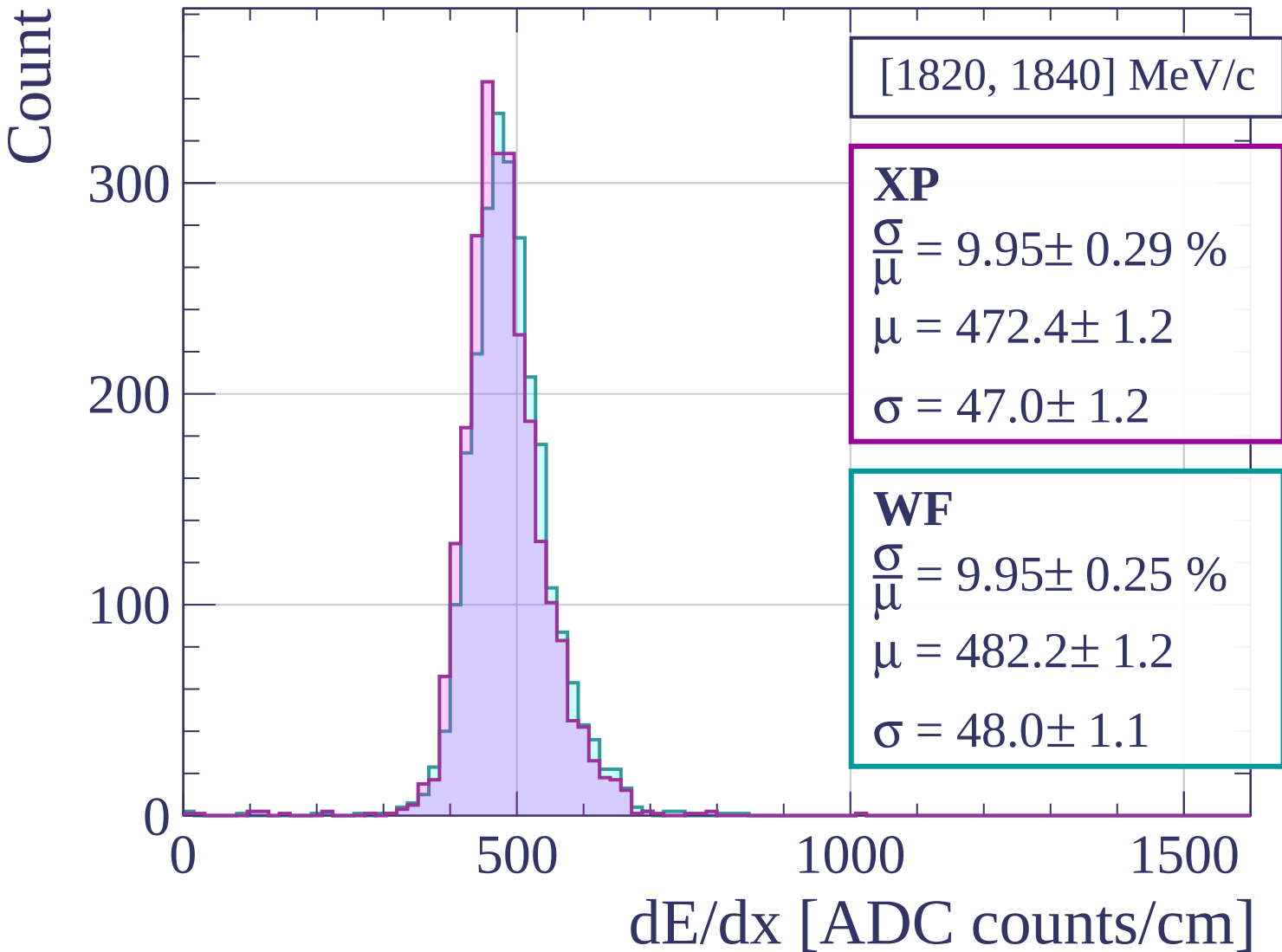
0

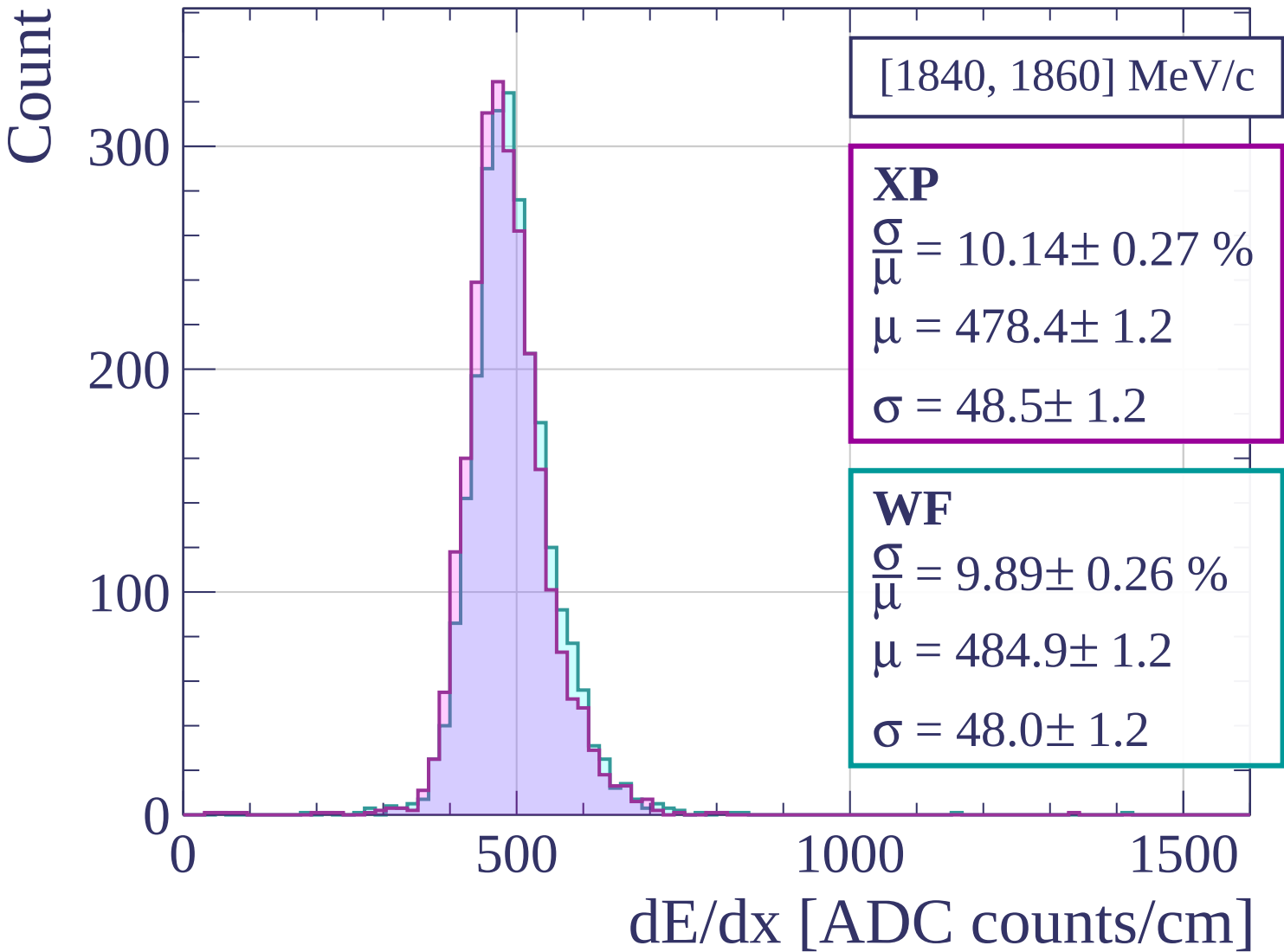
500

1000

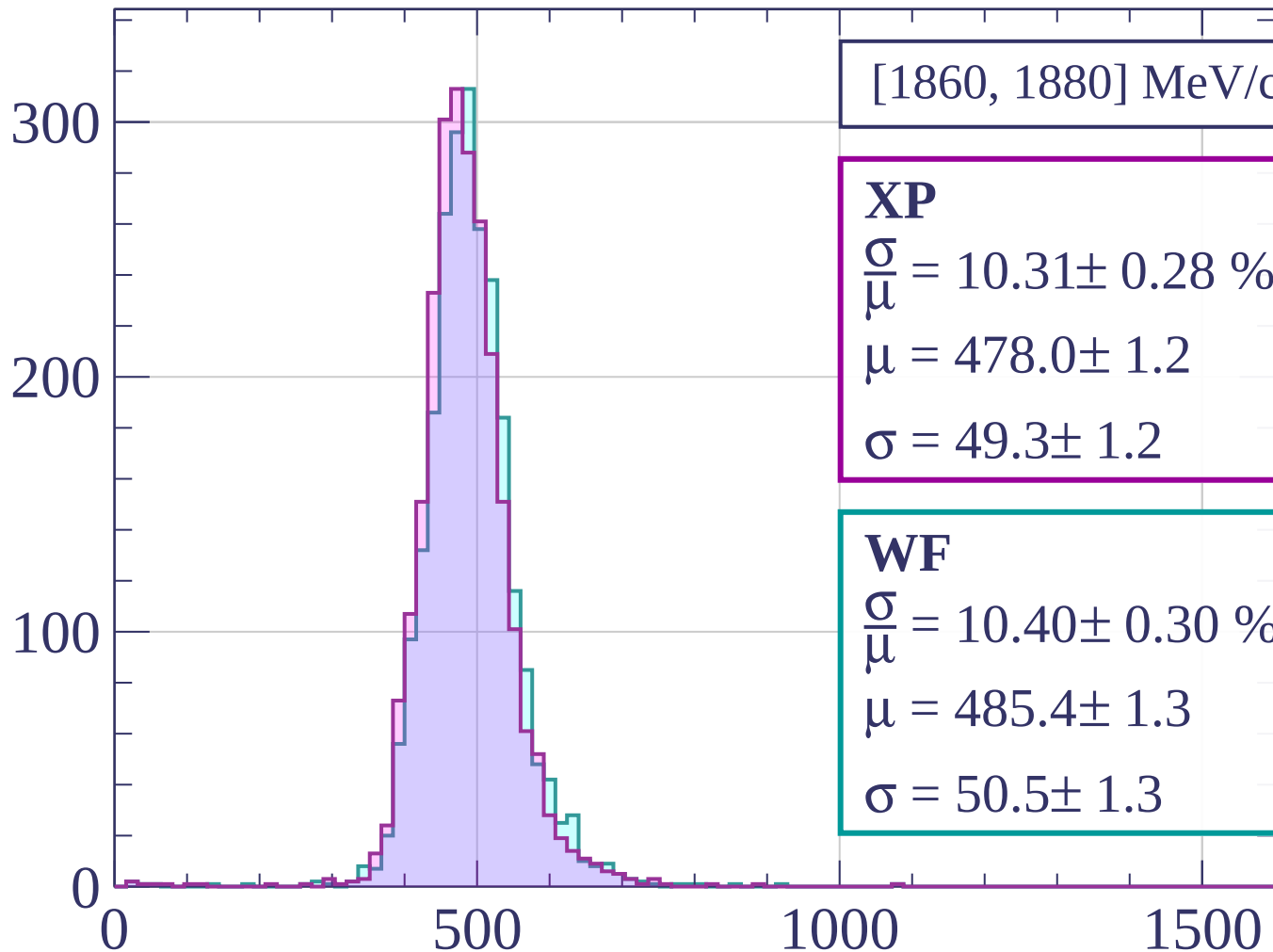
1500

dE/dx [ADC counts/cm]





Count



[1860, 1880] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.31 \pm 0.28 \%$$

$$\mu = 478.0 \pm 1.2$$

$$\sigma = 49.3 \pm 1.2$$

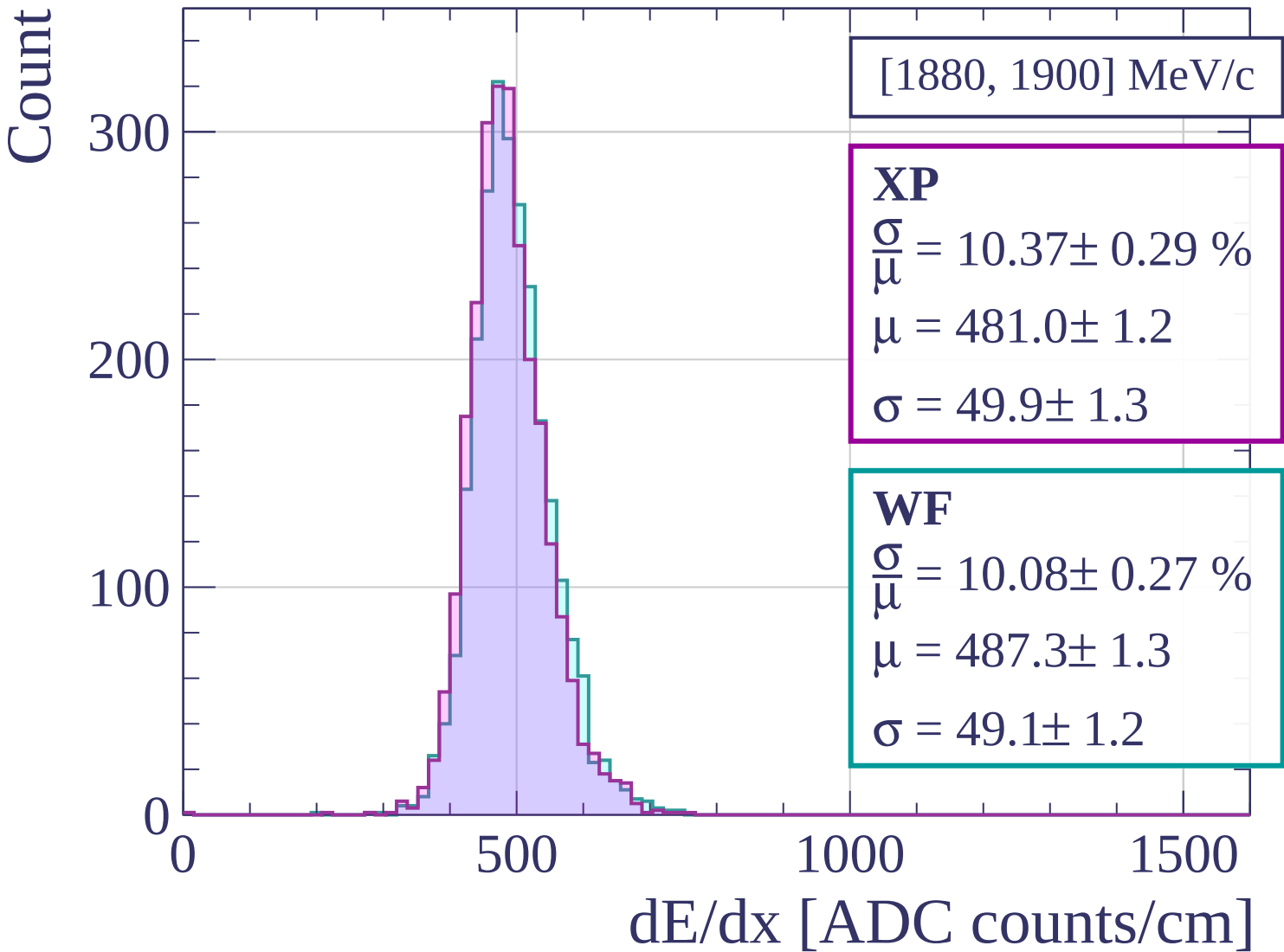
WF

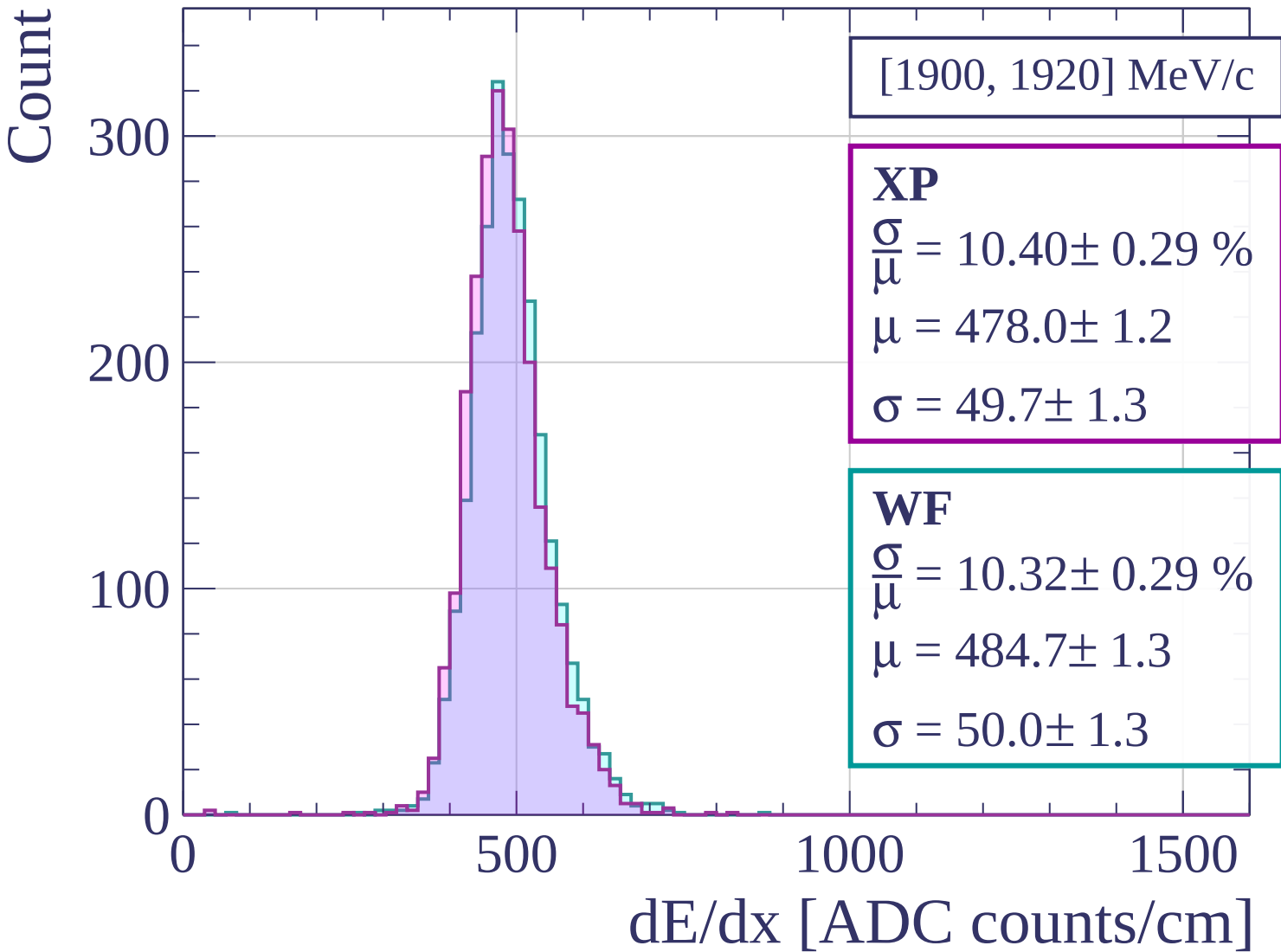
$$\frac{\sigma}{\mu} = 10.40 \pm 0.30 \%$$

$$\mu = 485.4 \pm 1.3$$

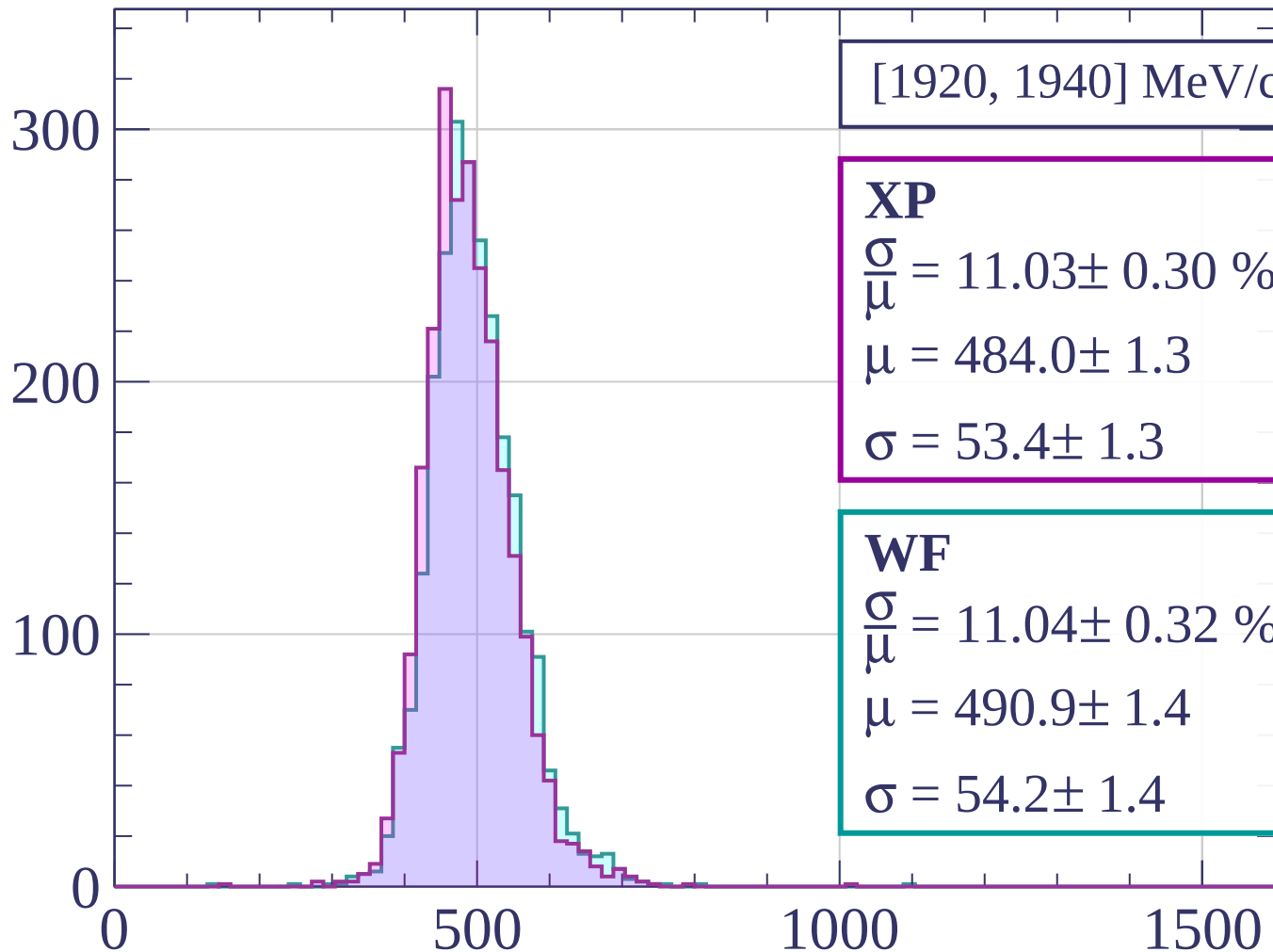
$$\sigma = 50.5 \pm 1.3$$

dE/dx [ADC counts/cm]



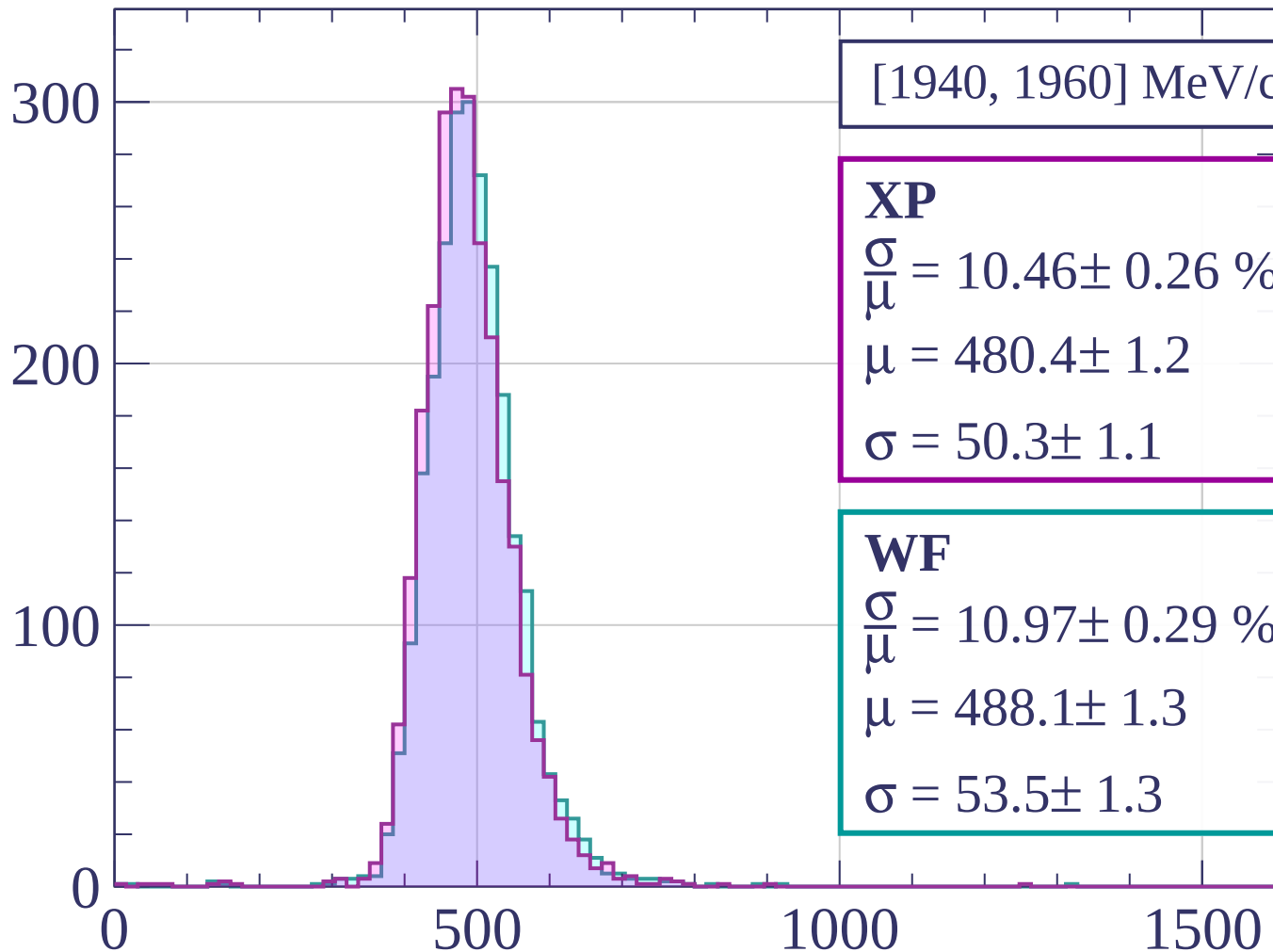


Count



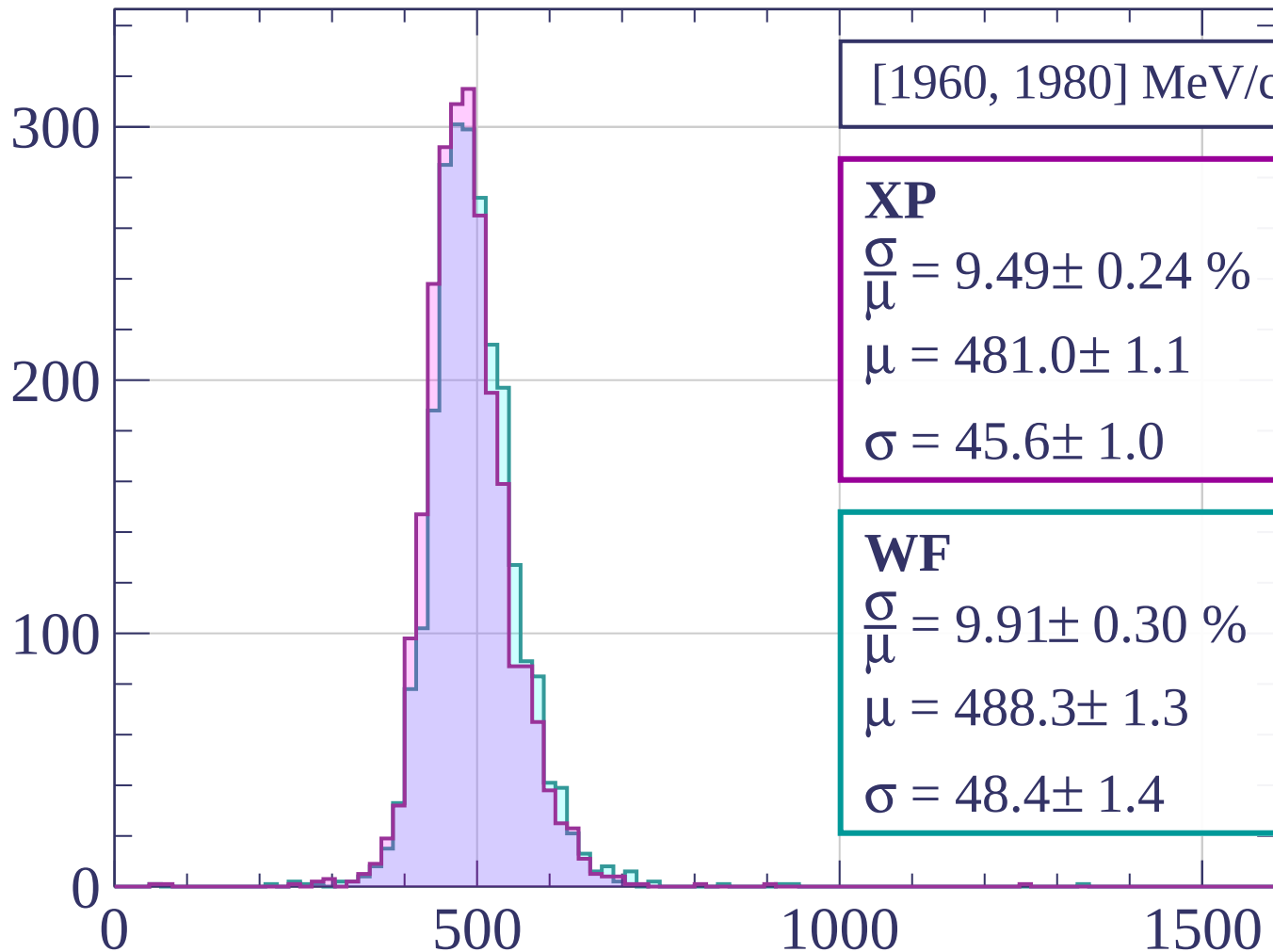
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

[1980, 2000] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.86 \pm 0.26 \%$$

$$\mu = 482.6 \pm 1.2$$

$$\sigma = 47.6 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.09 \pm 0.25 \%$$

$$\mu = 491.2 \pm 1.2$$

$$\sigma = 49.6 \pm 1.1$$

300

200

100

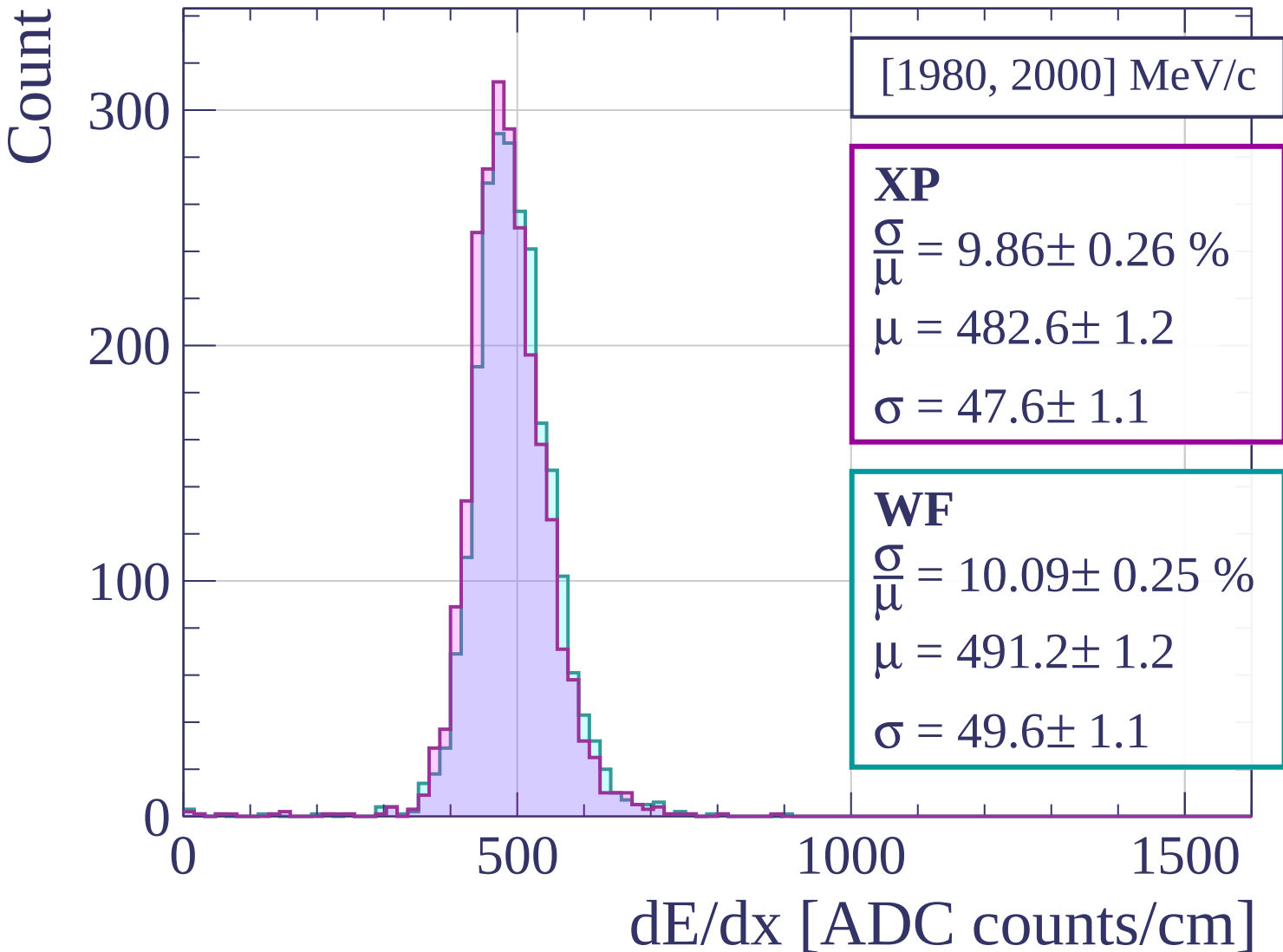
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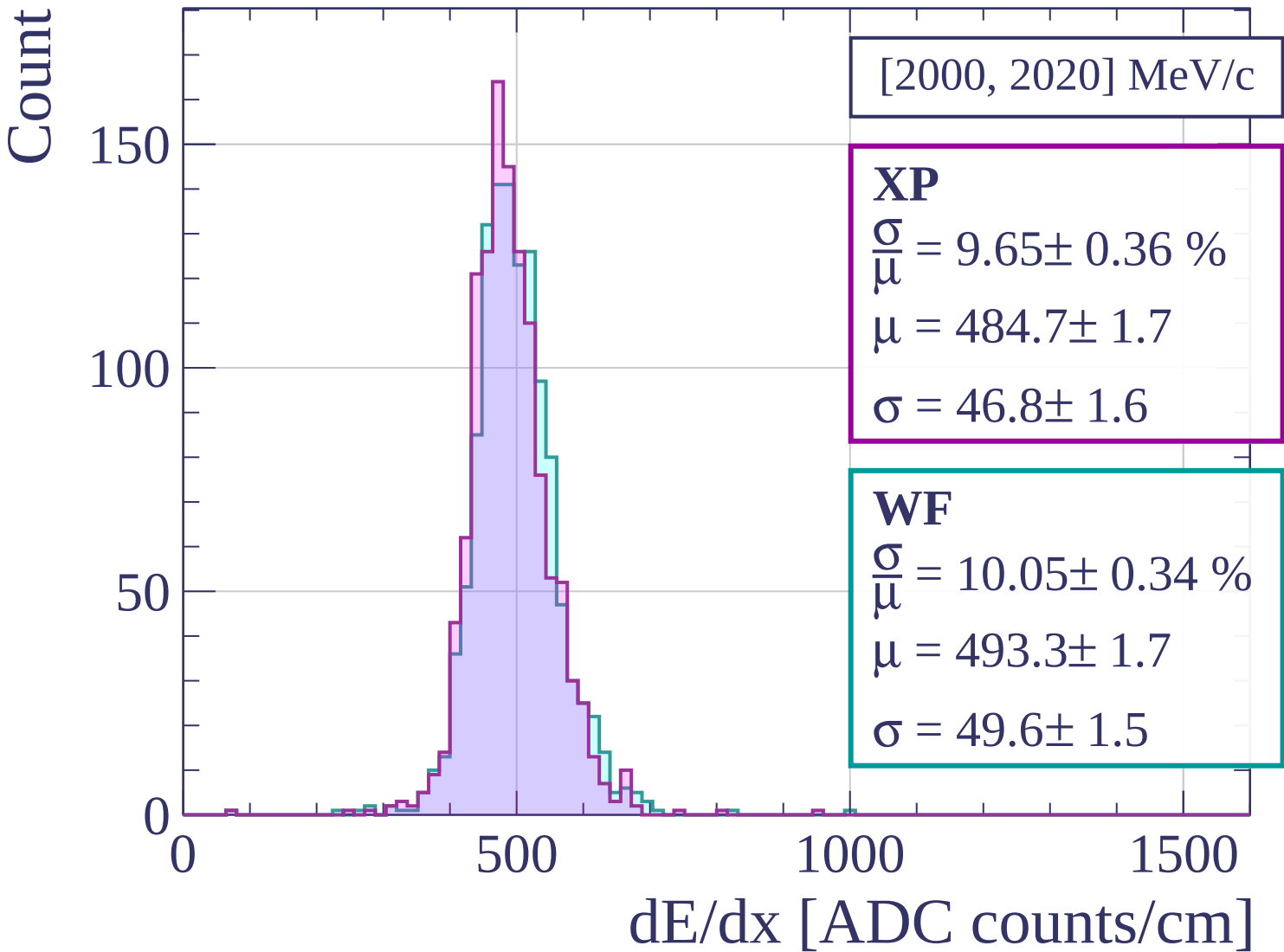
500

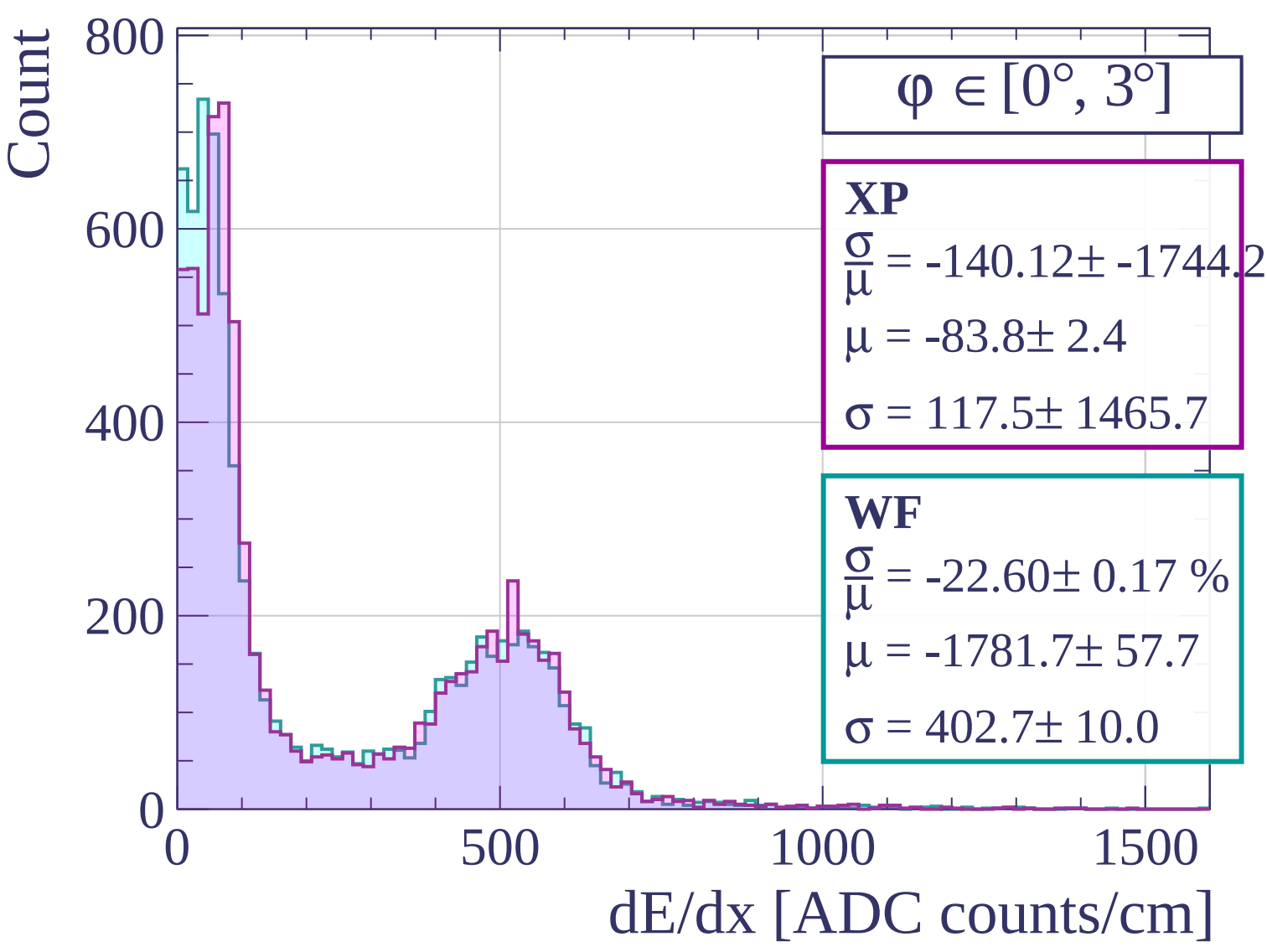
1000

1500

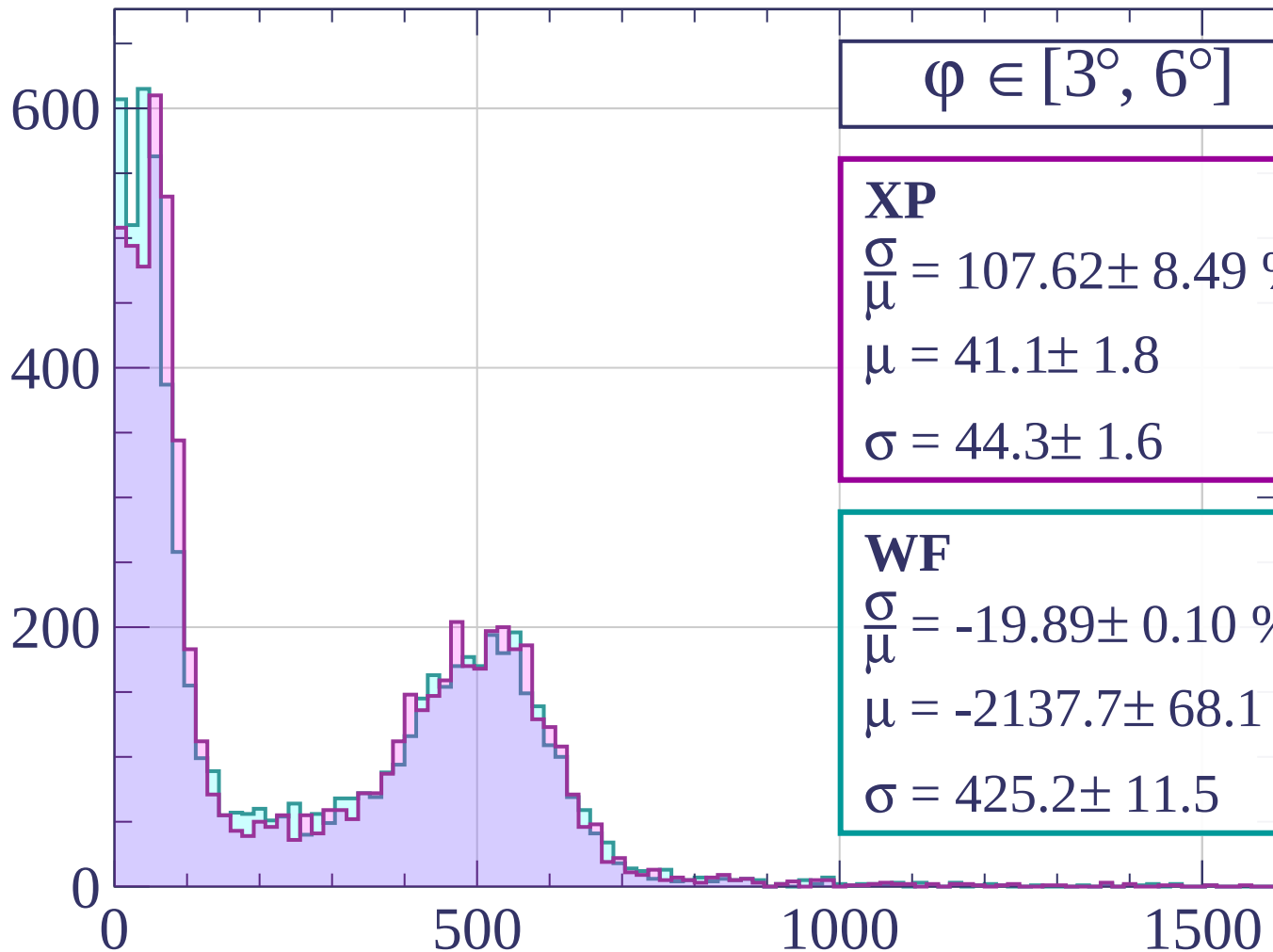
dE/dx [ADC counts/cm]



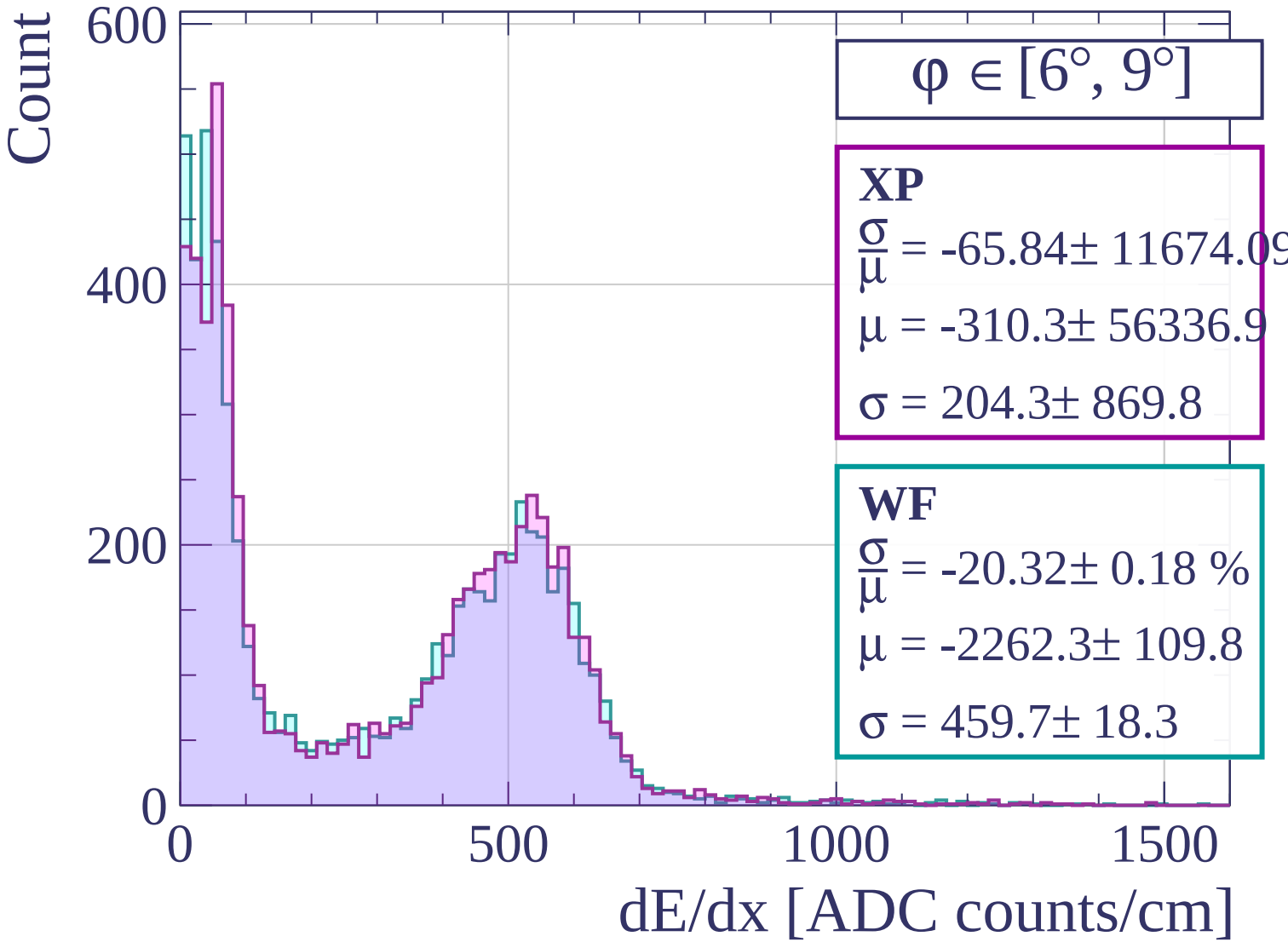




Count



dE/dx [ADC counts/cm]



Count

$$\varphi \in [9^\circ, 12^\circ]$$

XP

$$\frac{\sigma}{\mu} = 23.17 \pm 4.38 \%$$

$$\mu = 4443.5 \pm 404.9$$

$$\sigma = 1029.5 \pm 100.8$$

WF

$$\frac{\sigma}{\mu} = -19.79 \pm 0.13 \%$$

$$\mu = -2221.8 \pm 107.5$$

$$\sigma = 439.6 \pm 18.5$$

400

300

200

100

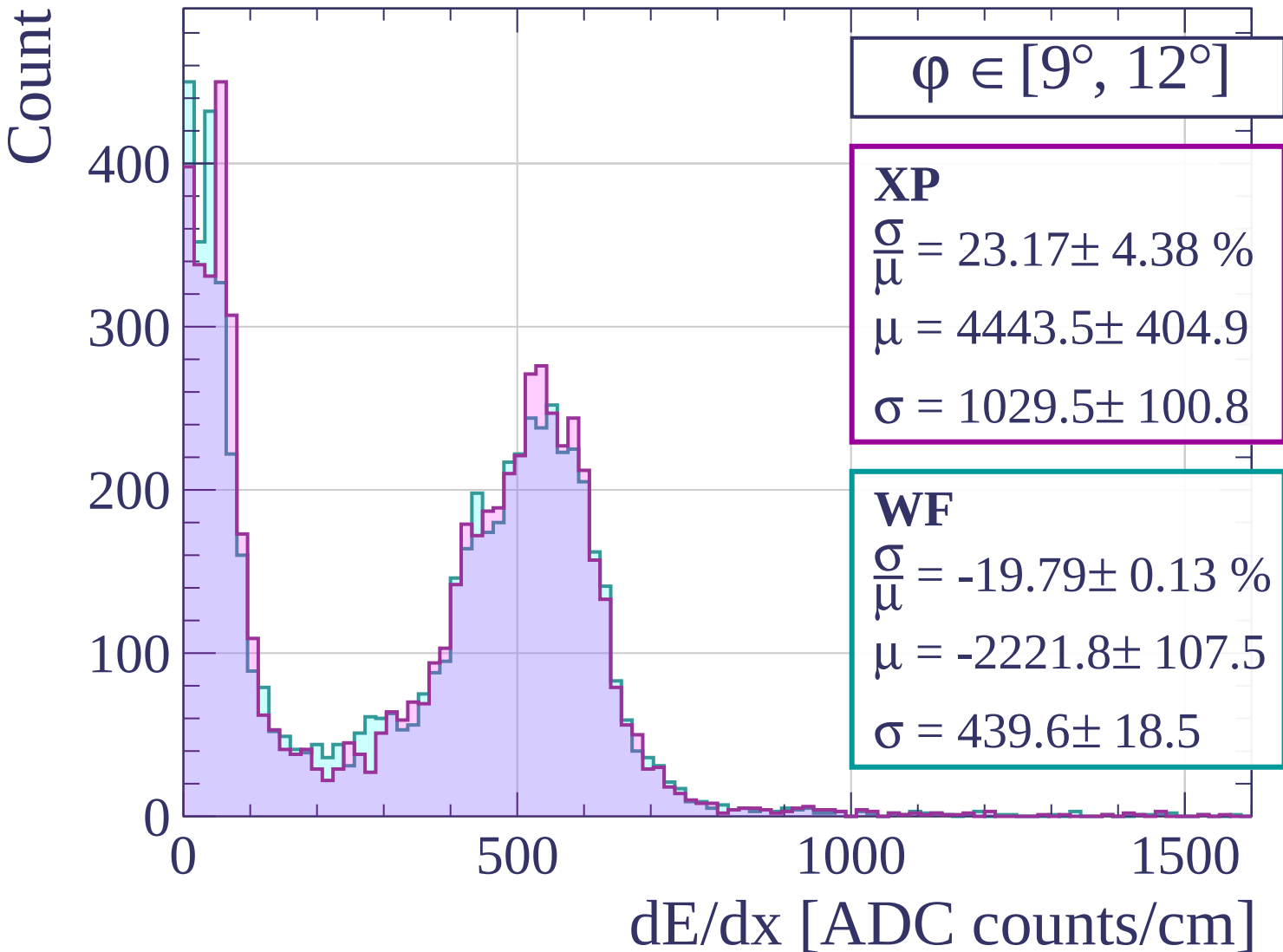
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [12^\circ, 15^\circ]$

XP

$$\frac{\sigma}{\mu} = 22.52 \pm 4.55 \%$$

$$\mu = 4187.7 \pm 423.9$$

$$\sigma = 943.2 \pm 95.2$$

WF

$$\frac{\sigma}{\mu} = -100.99 \pm 5206.17 \%$$

$$\mu = -110.5 \pm 6531.4$$

$$\sigma = 111.6 \pm 842.4$$

400

200

0

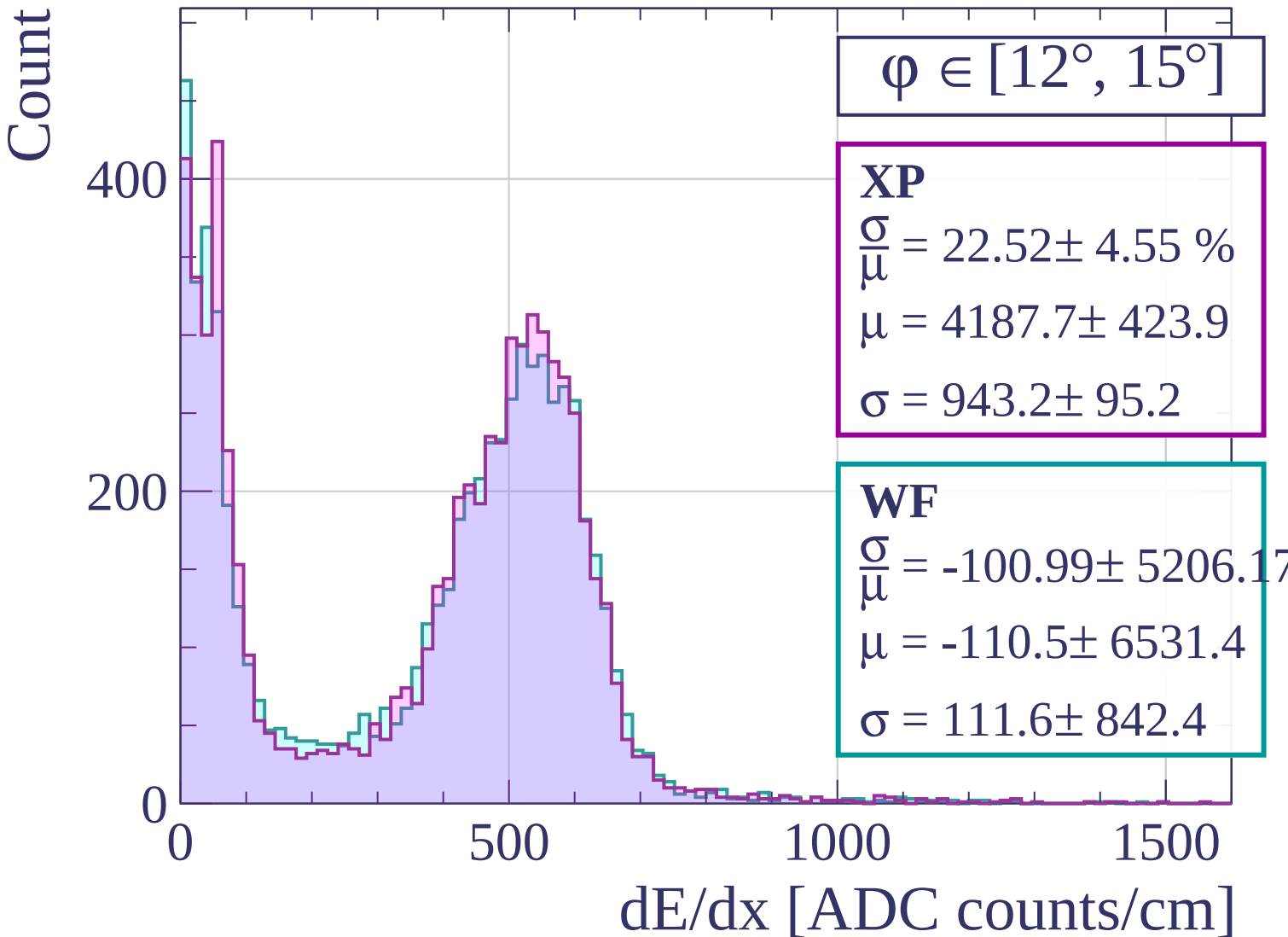
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\varphi \in [15^\circ, 18^\circ]$

XP

$$\frac{\sigma}{\mu} = 17.45 \pm 0.30 \%$$

$$\mu = 524.2 \pm 1.5$$

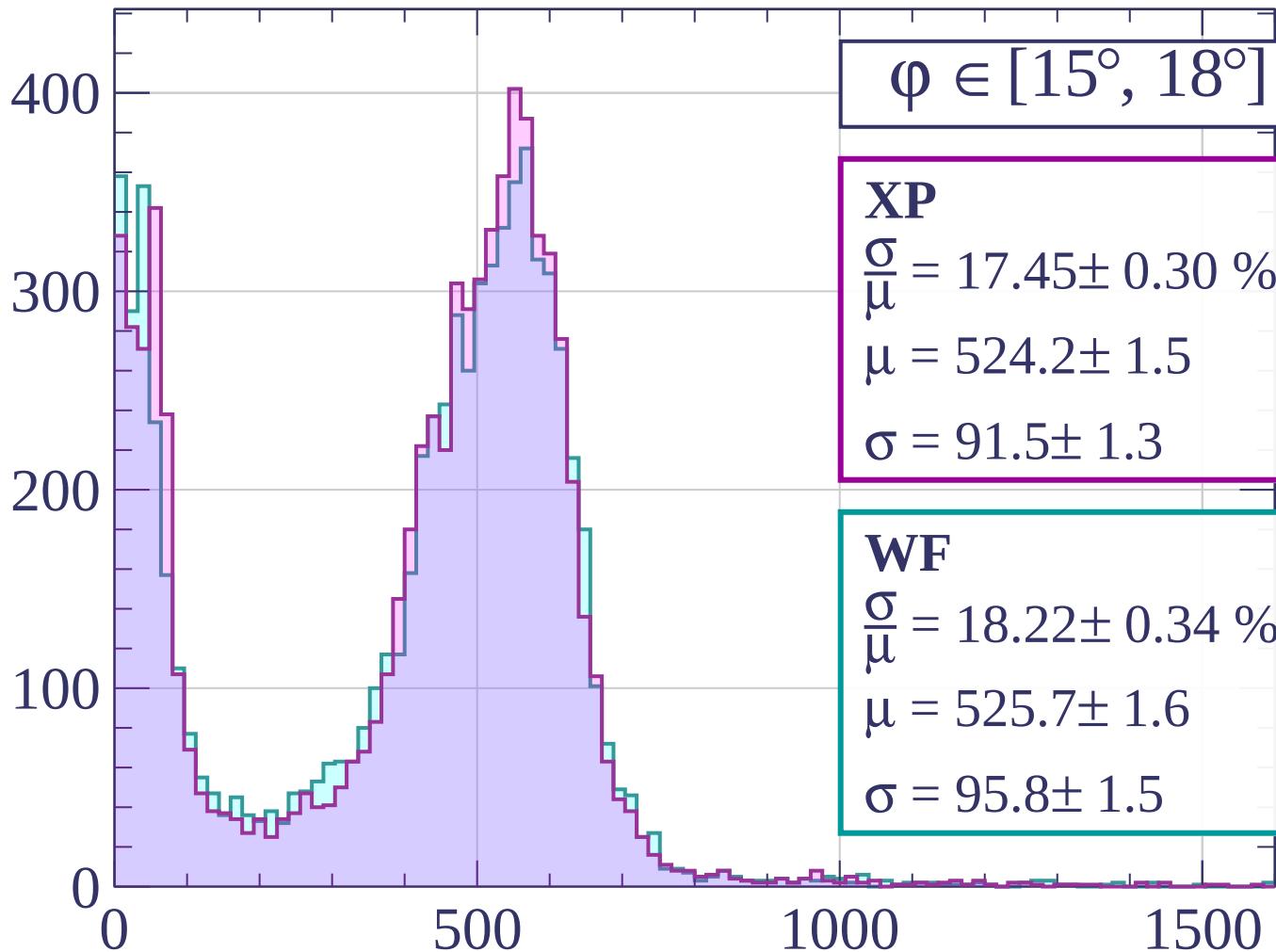
$$\sigma = 91.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 18.22 \pm 0.34 \%$$

$$\mu = 525.7 \pm 1.6$$

$$\sigma = 95.8 \pm 1.5$$



dE/dx [ADC counts/cm]

Count

$\varphi \in [18^\circ, 21^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.64 \pm 0.28 \%$$

$$\mu = 523.7 \pm 1.4$$

$$\sigma = 87.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 17.35 \pm 0.27 \%$$

$$\mu = 525.0 \pm 1.4$$

$$\sigma = 91.1 \pm 1.2$$

400

200

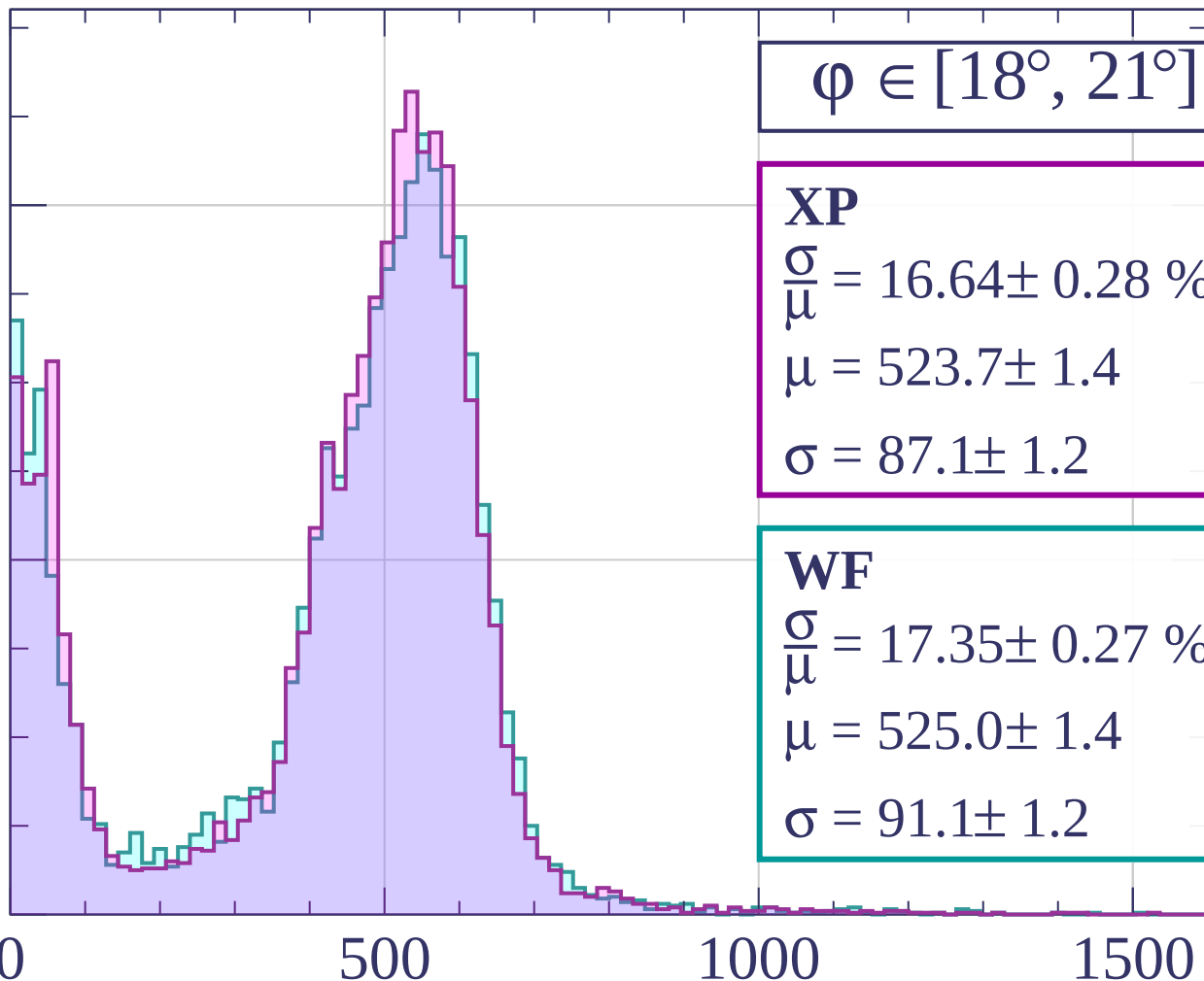
0

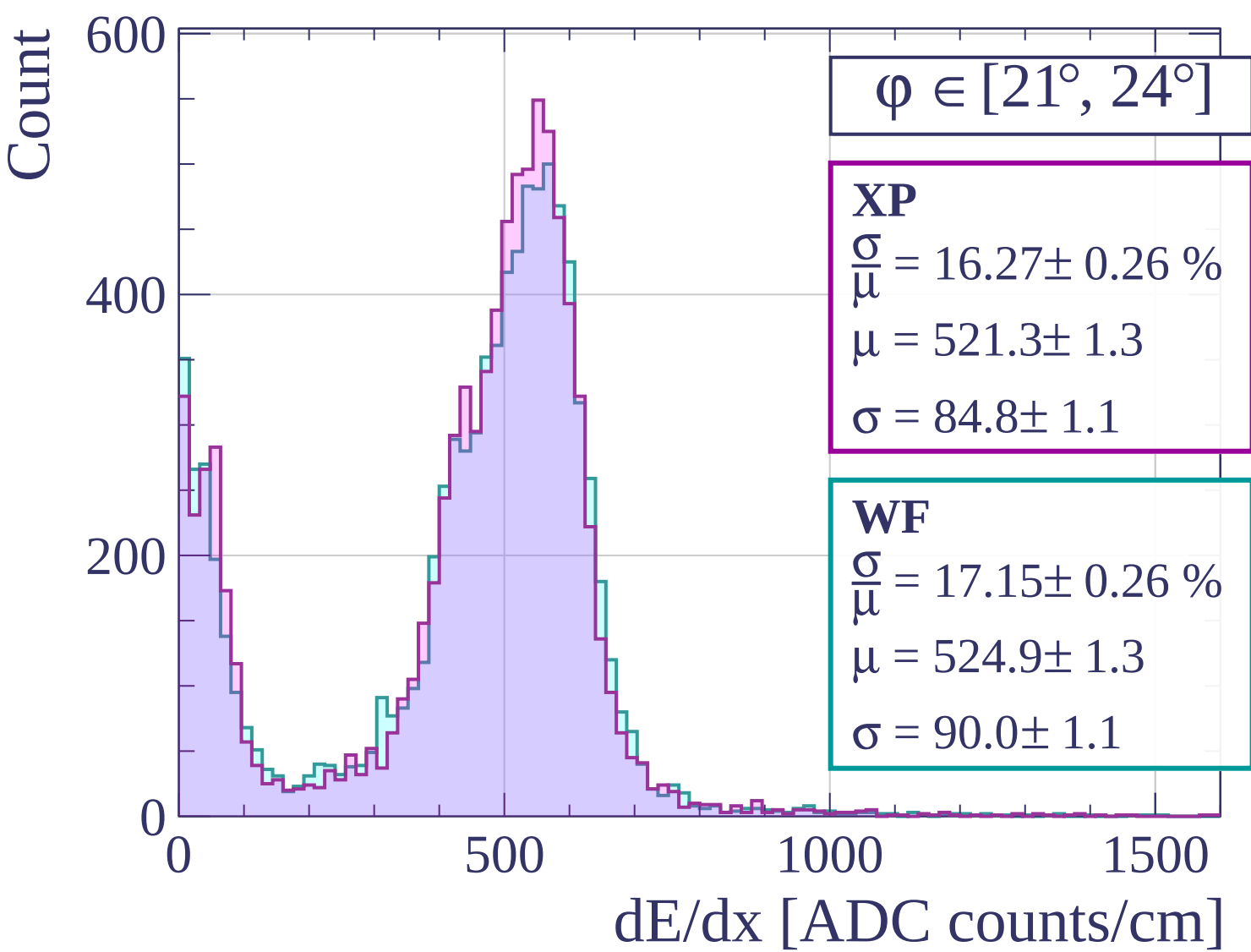
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\phi \in [24^\circ, 27^\circ]$

XP

$$\frac{\sigma}{\mu} = 17.06 \pm 0.23 \%$$

$$\mu = 511.1 \pm 1.2$$

$$\sigma = 87.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 18.07 \pm 0.26 \%$$

$$\mu = 514.3 \pm 1.3$$

$$\sigma = 93.0 \pm 1.1$$

400

200

0

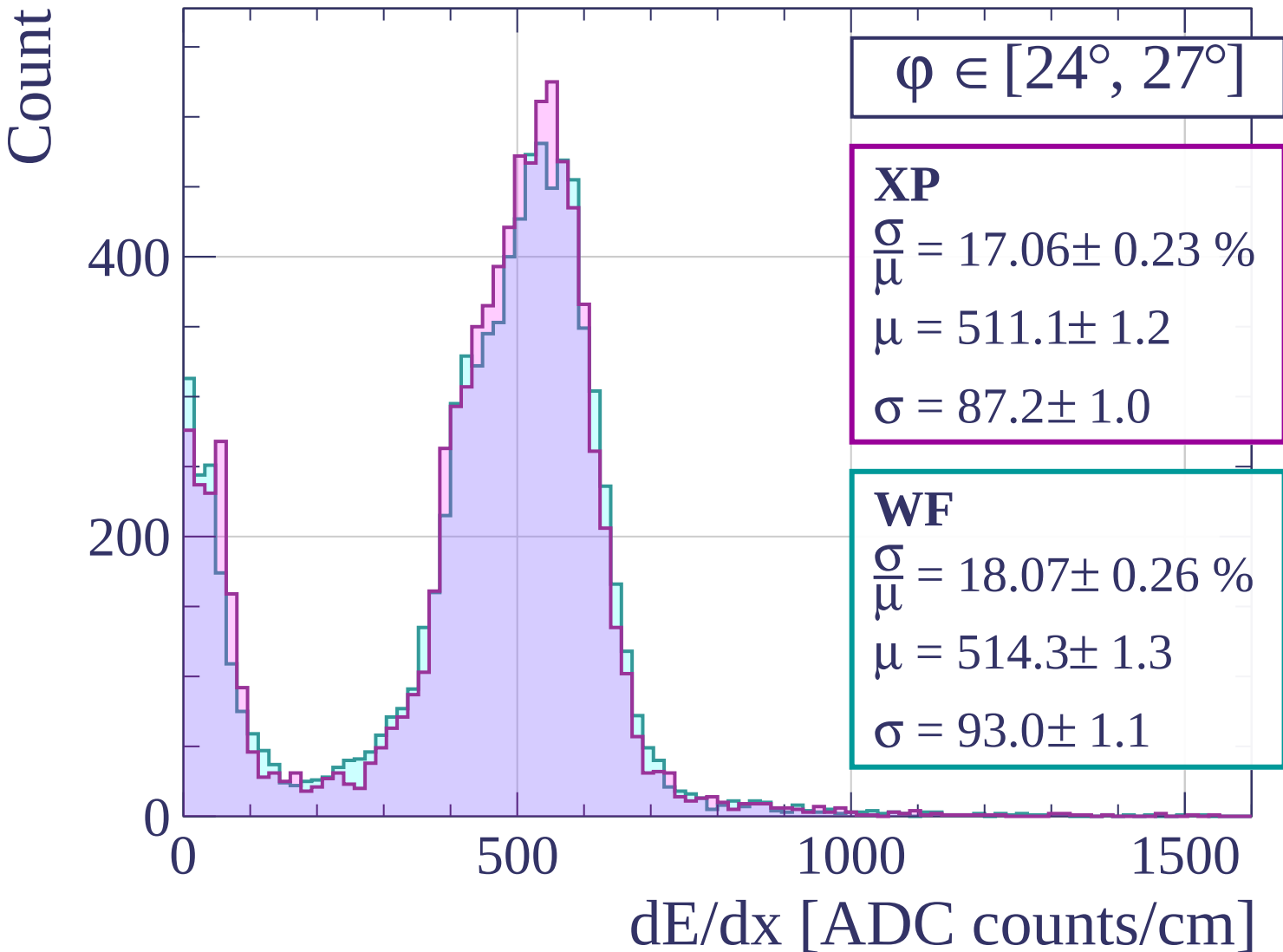
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [27^\circ, 30^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.59 \pm 0.23 \%$$

$$\mu = 519.8 \pm 1.1$$

$$\sigma = 86.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 16.67 \pm 0.22 \%$$

$$\mu = 518.2 \pm 1.1$$

$$\sigma = 86.4 \pm 0.9$$

600

400

200

0

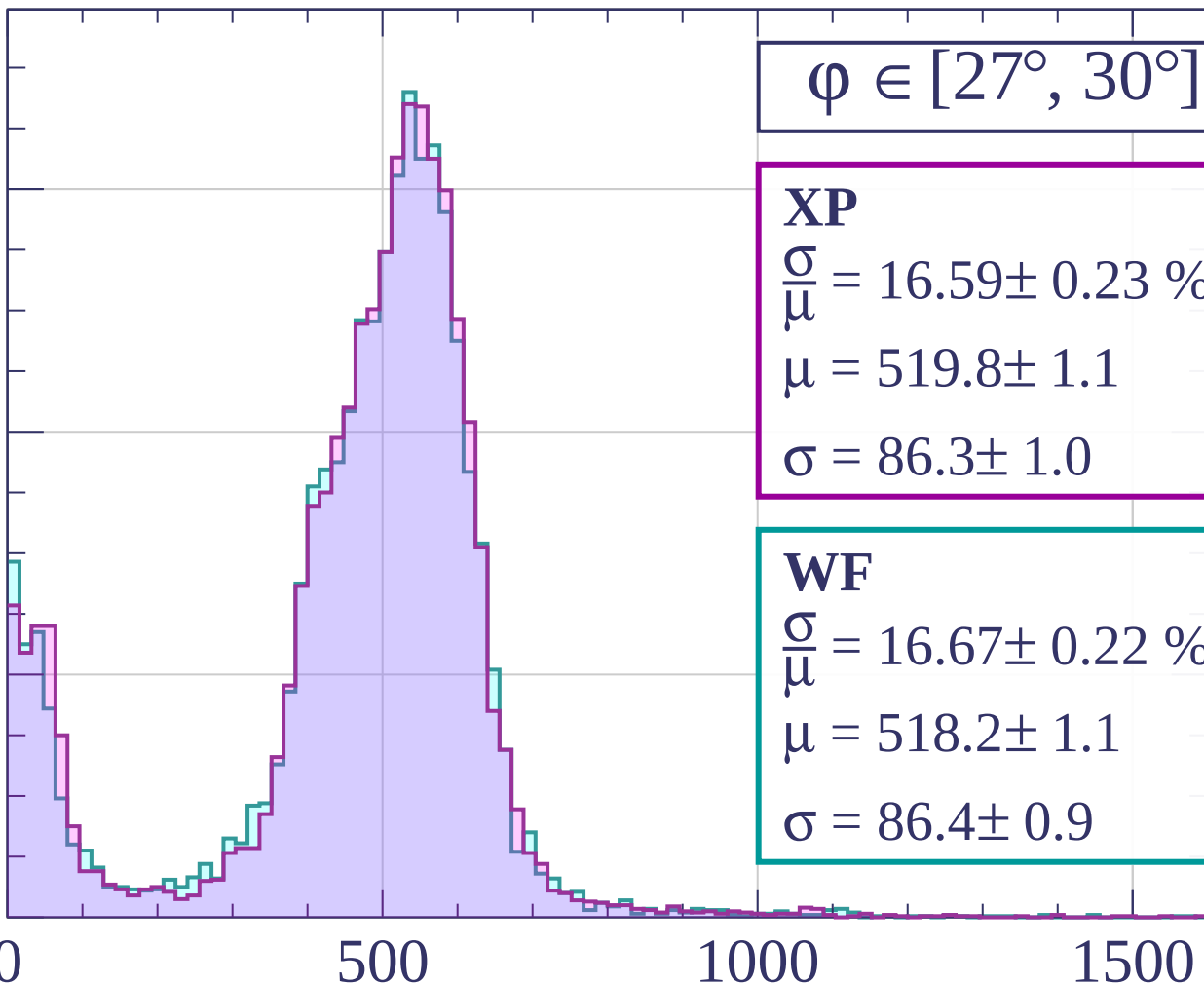
0

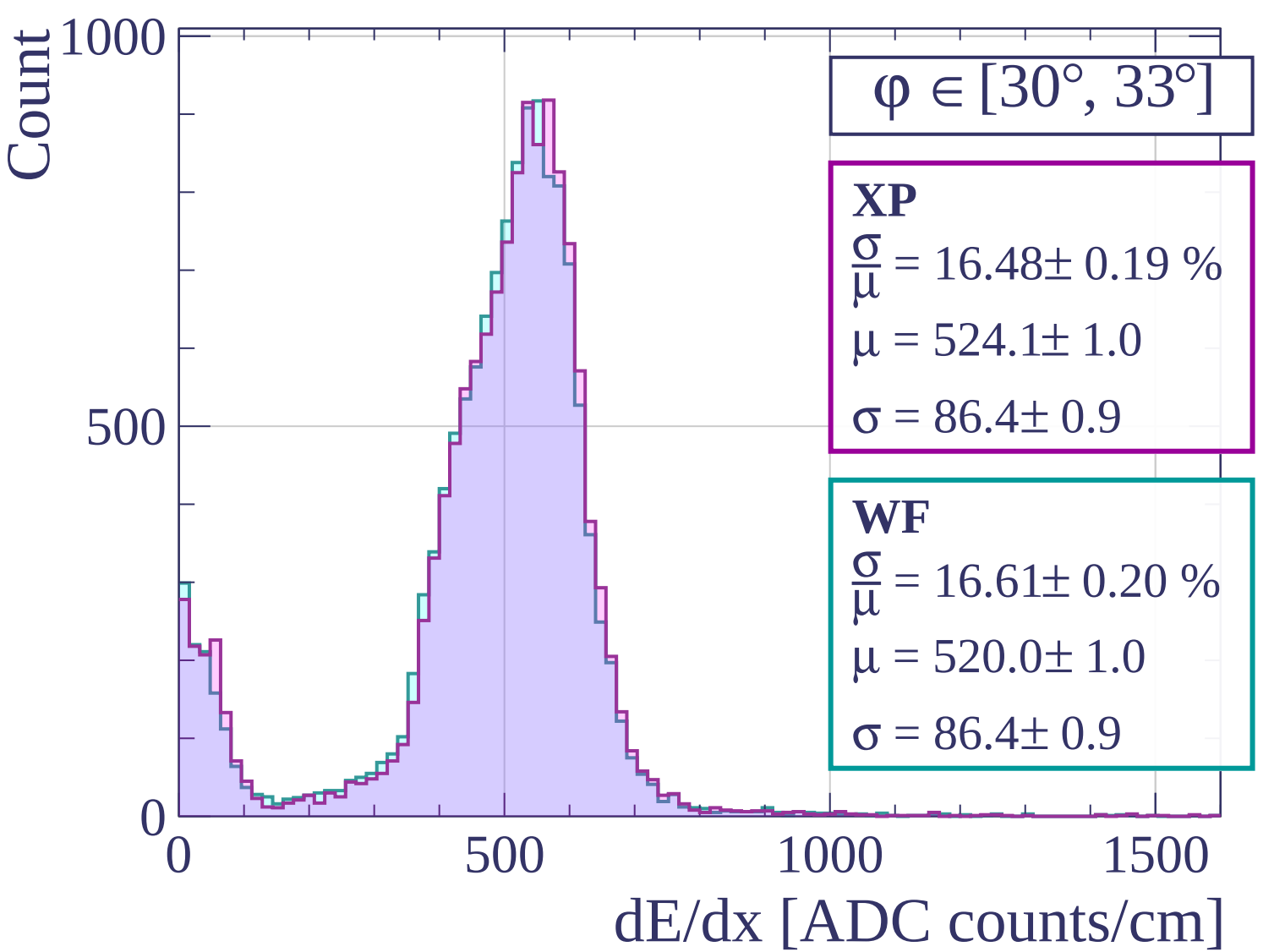
500

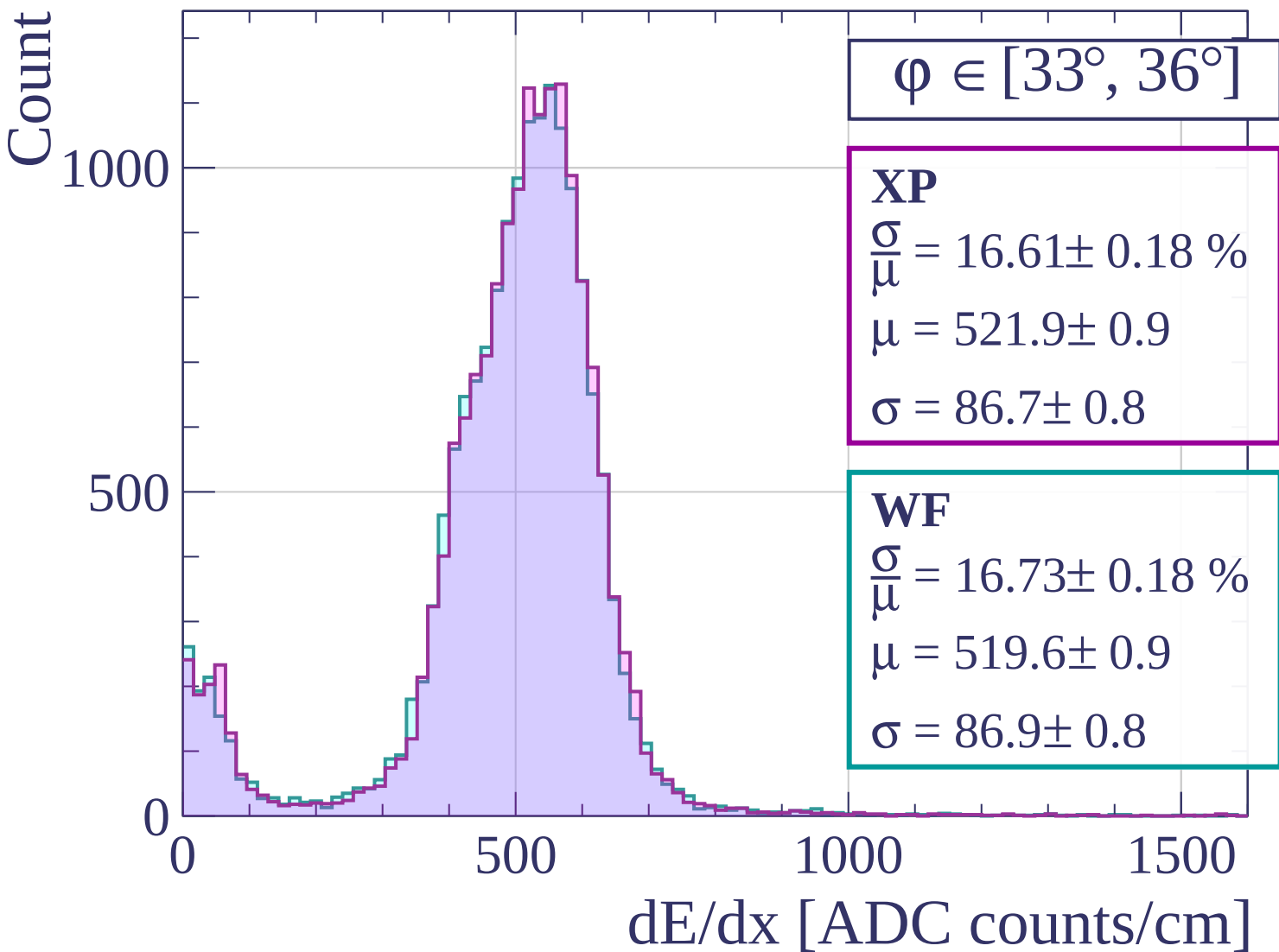
1000

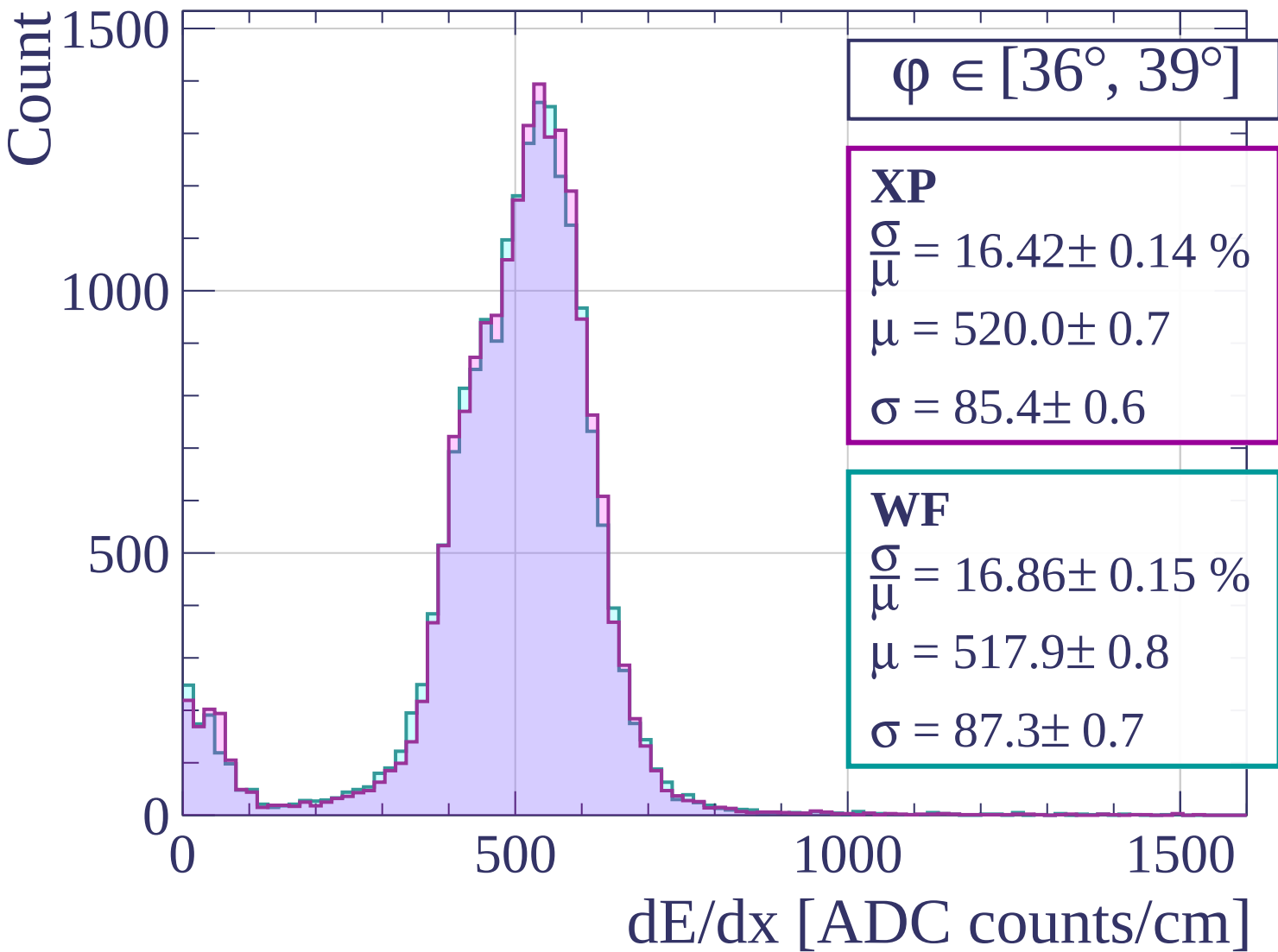
1500

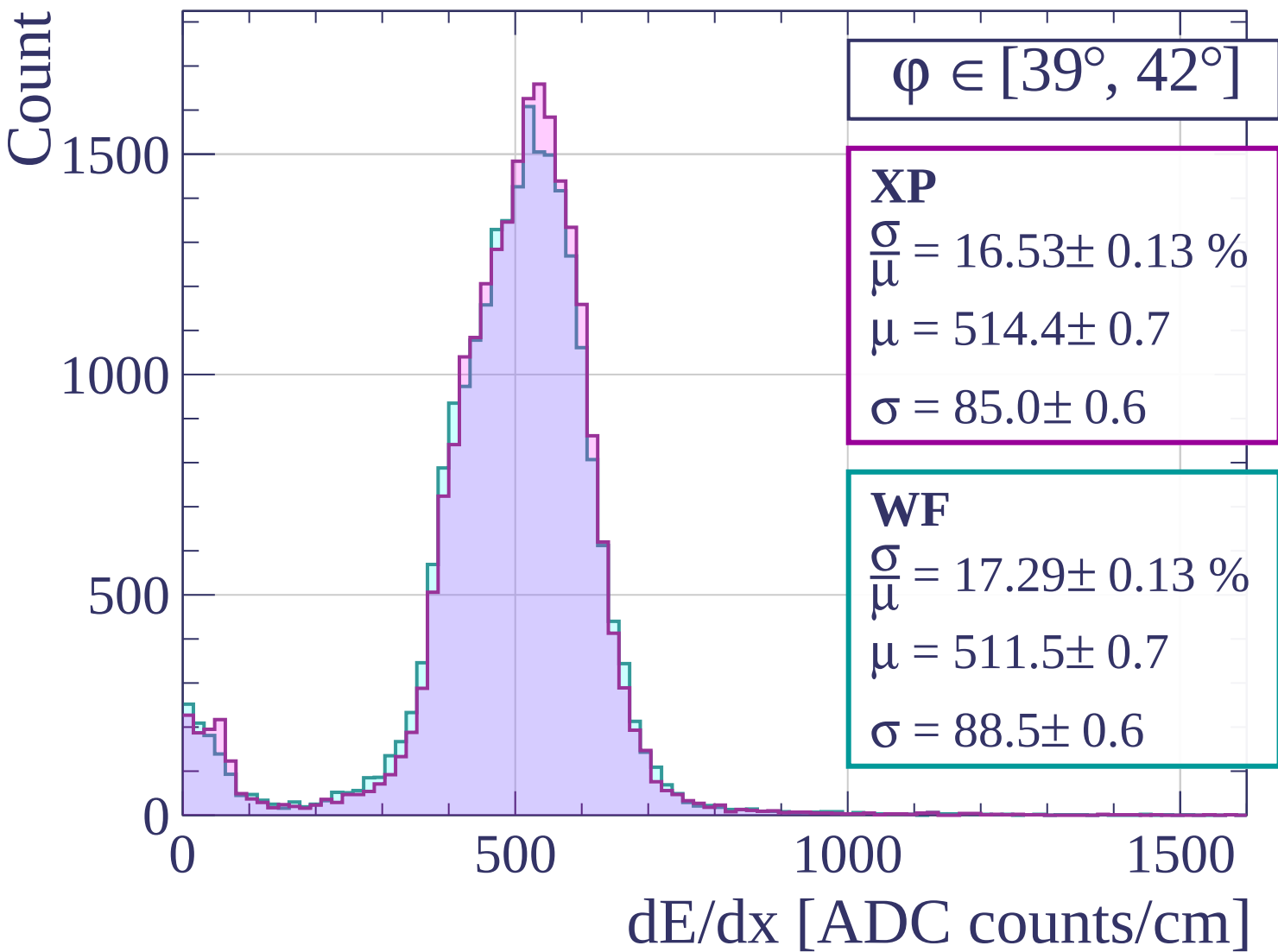
dE/dx [ADC counts/cm]

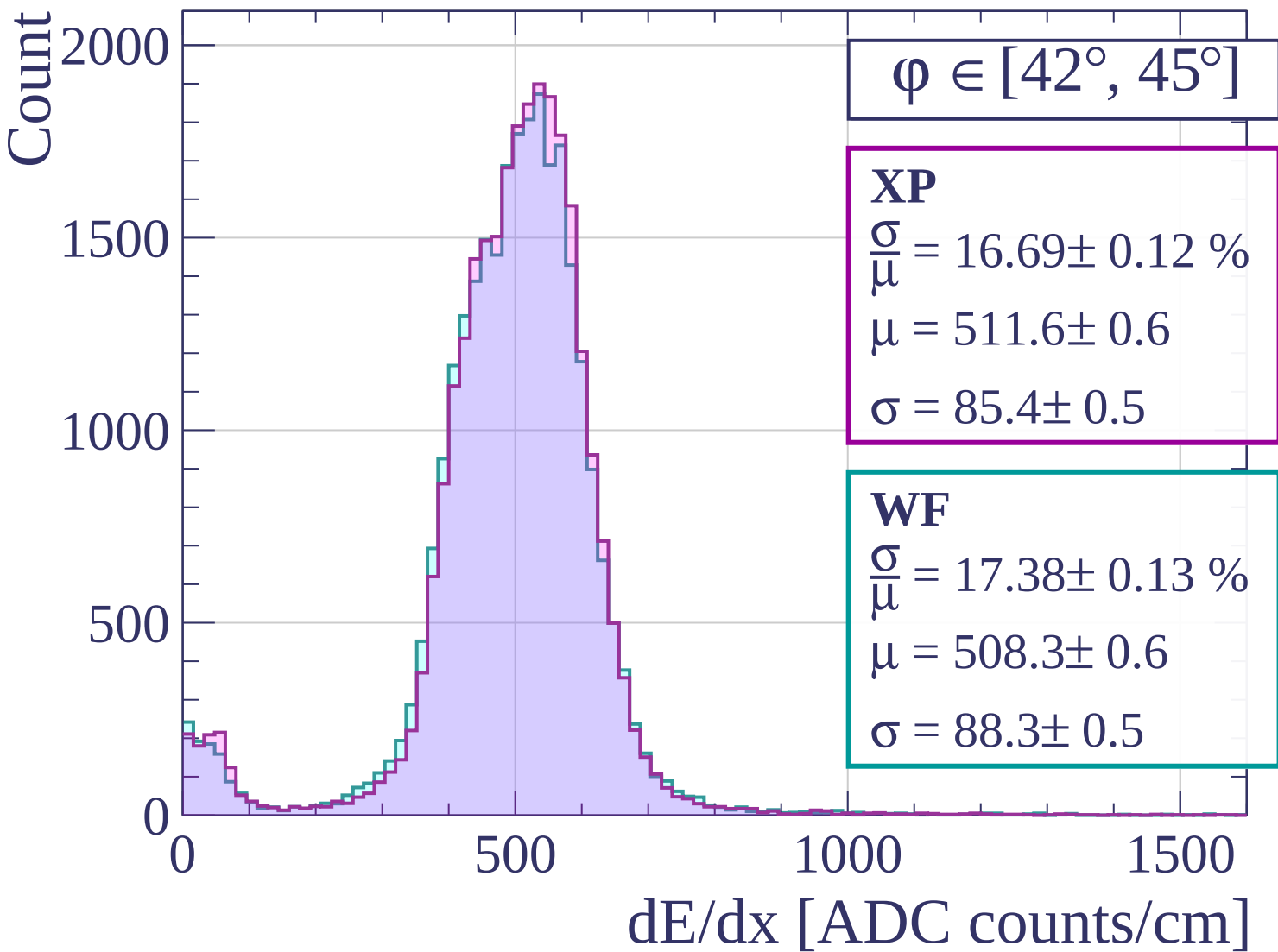


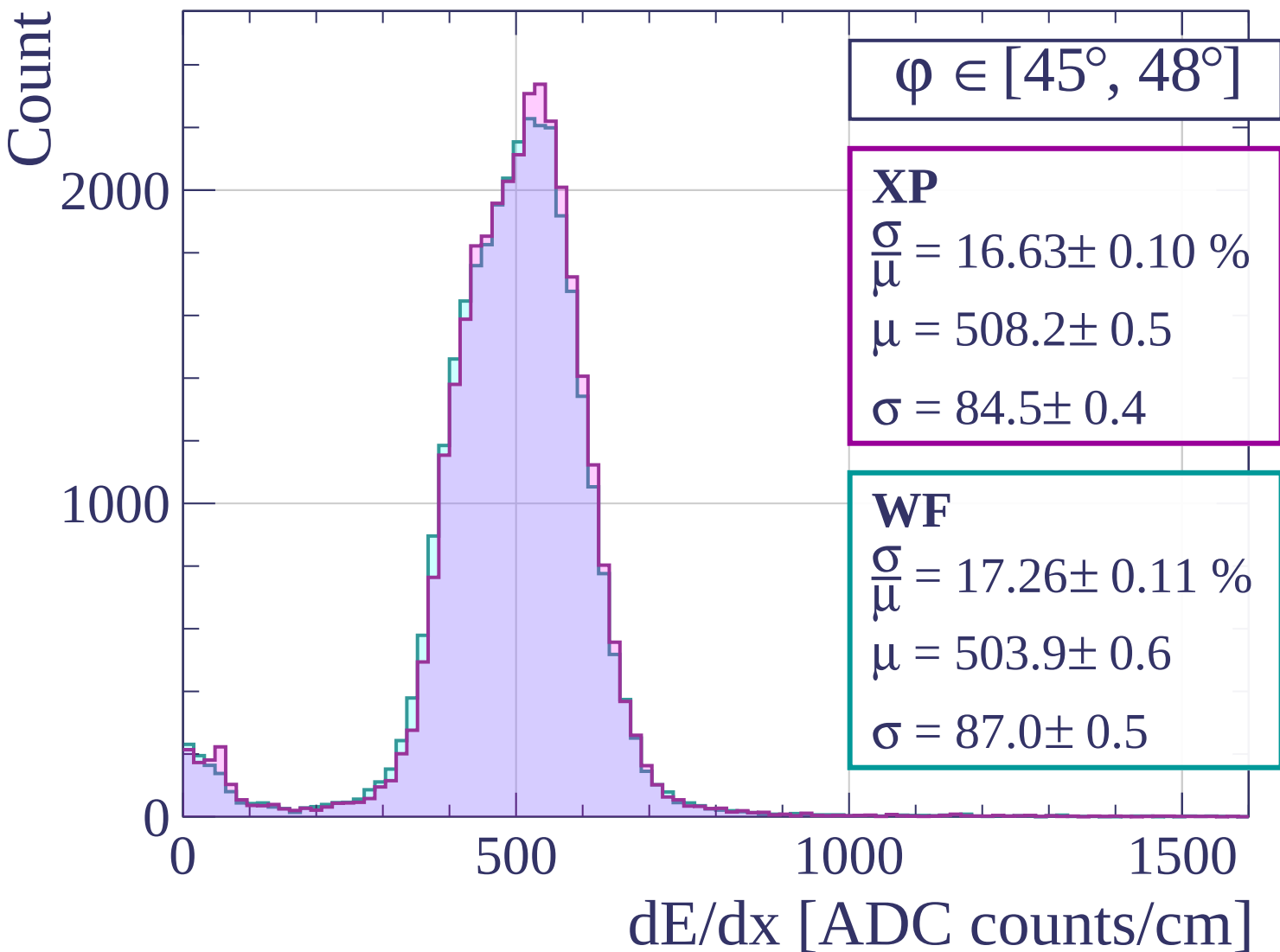




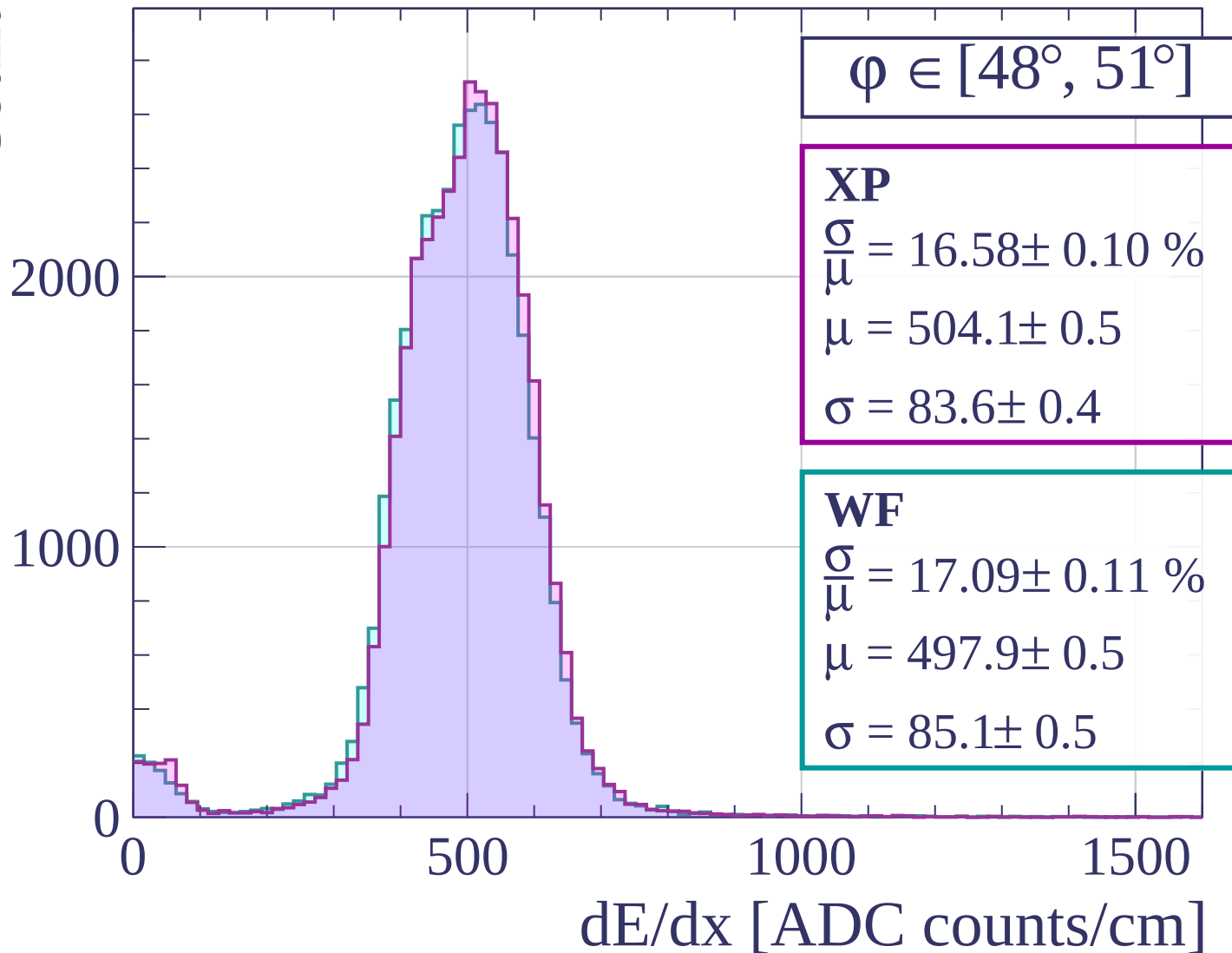


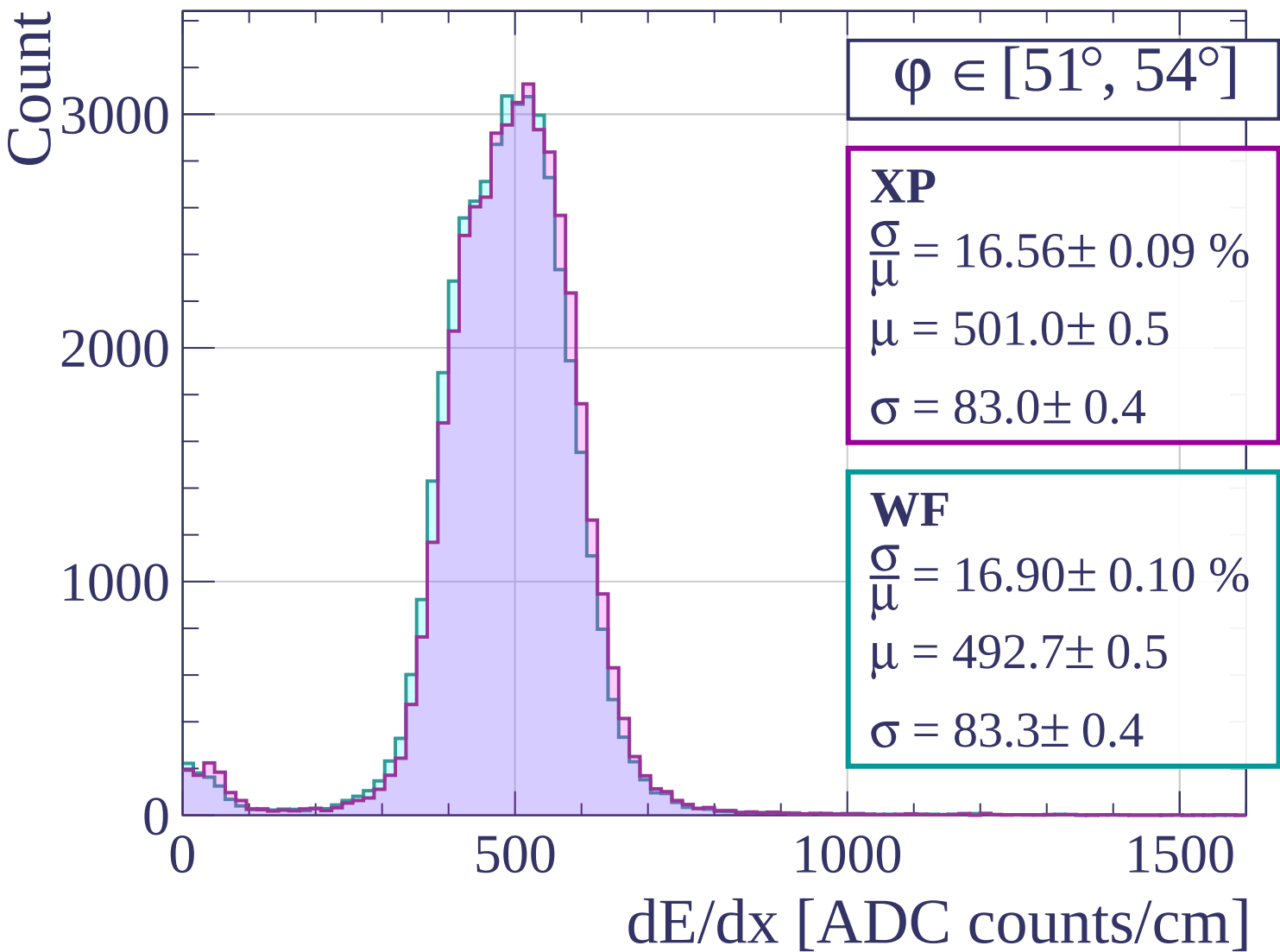


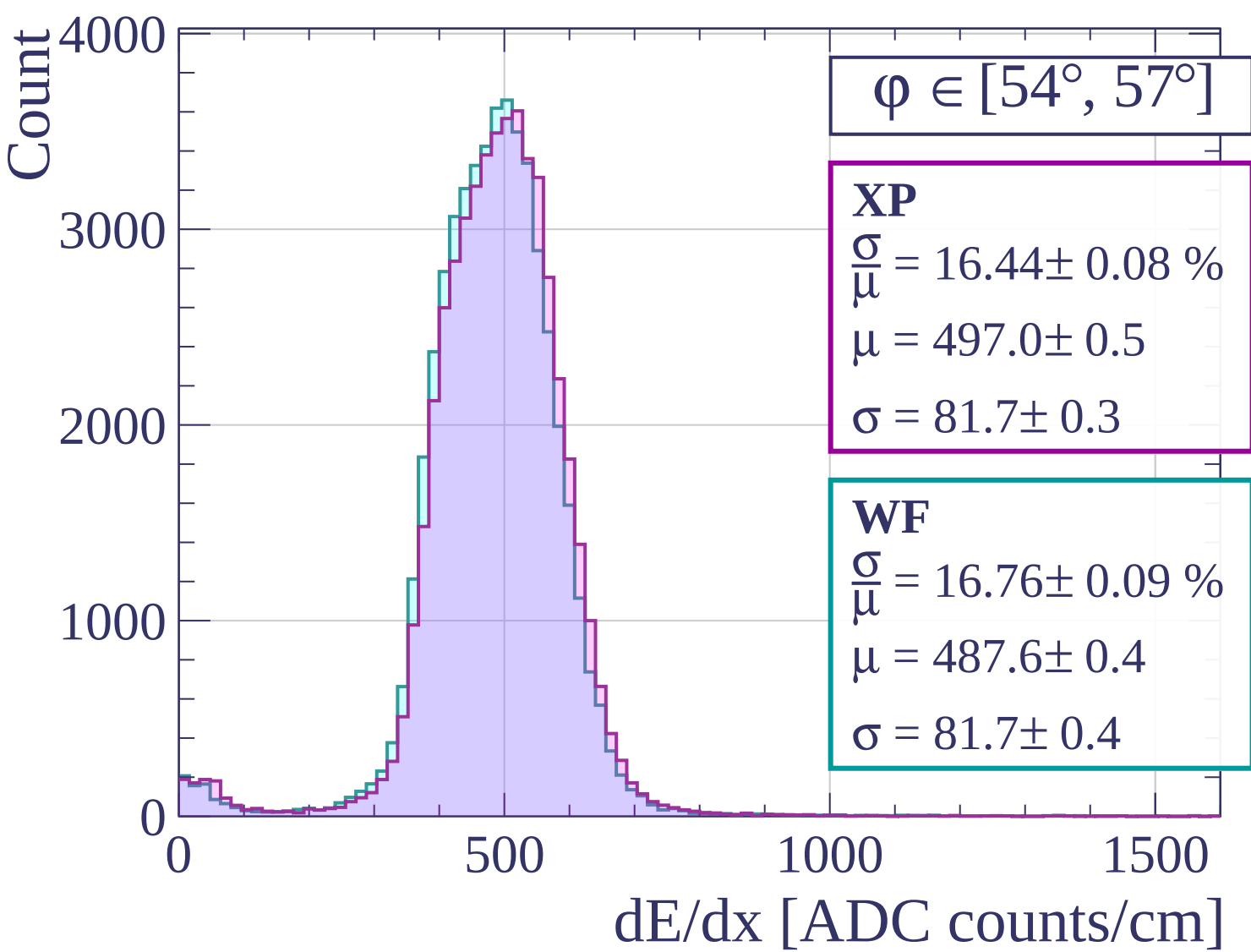


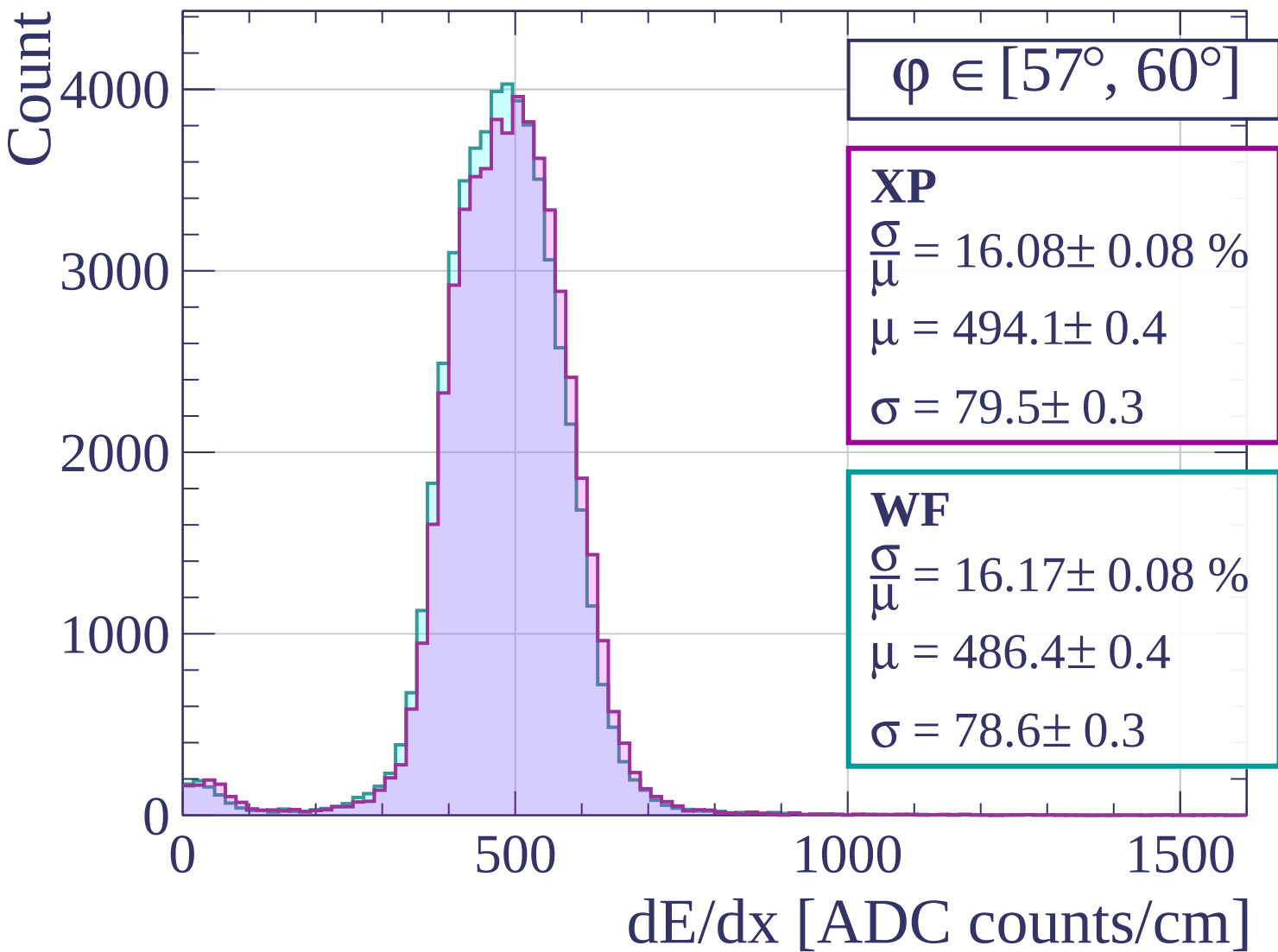


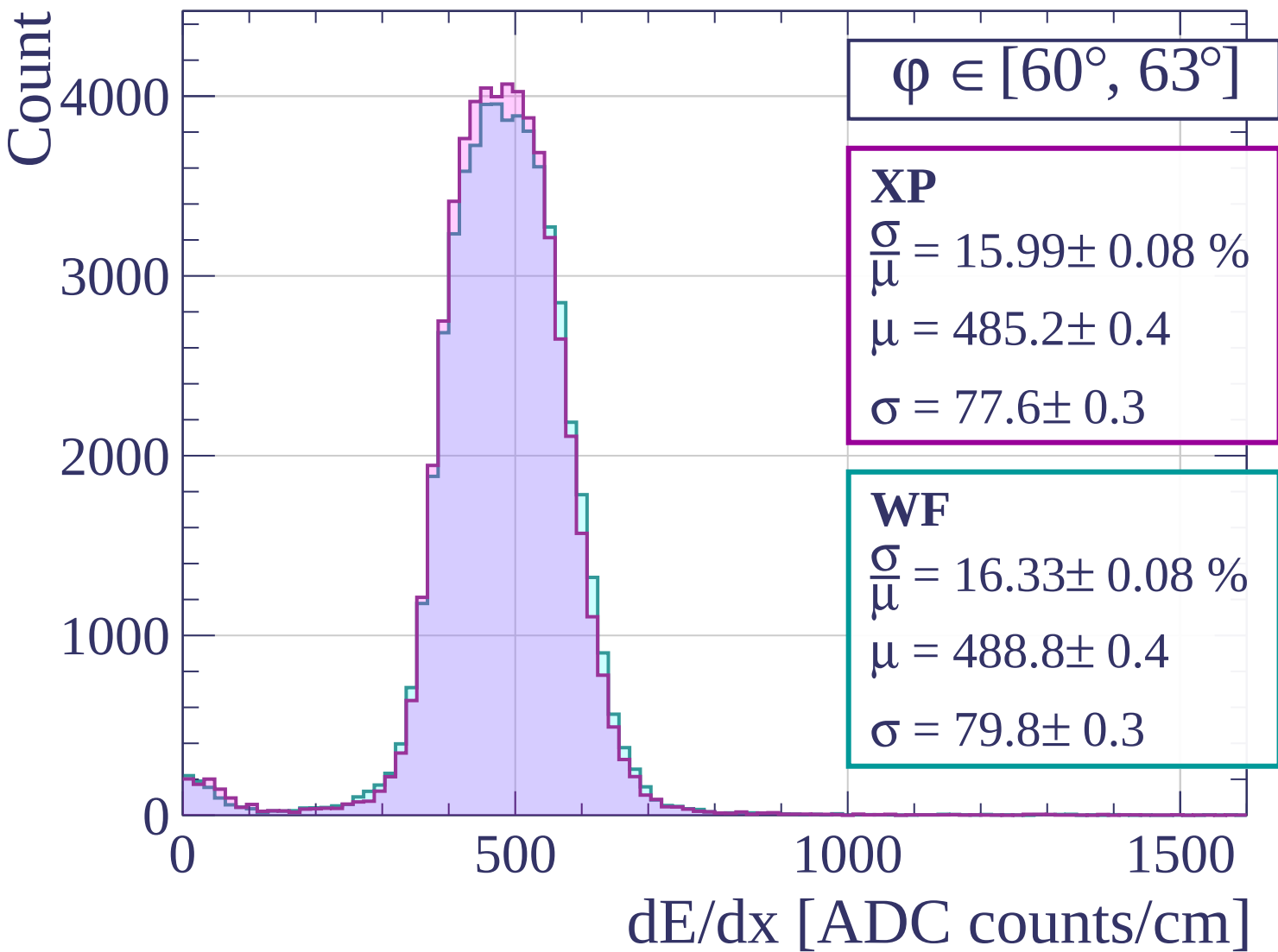
Count



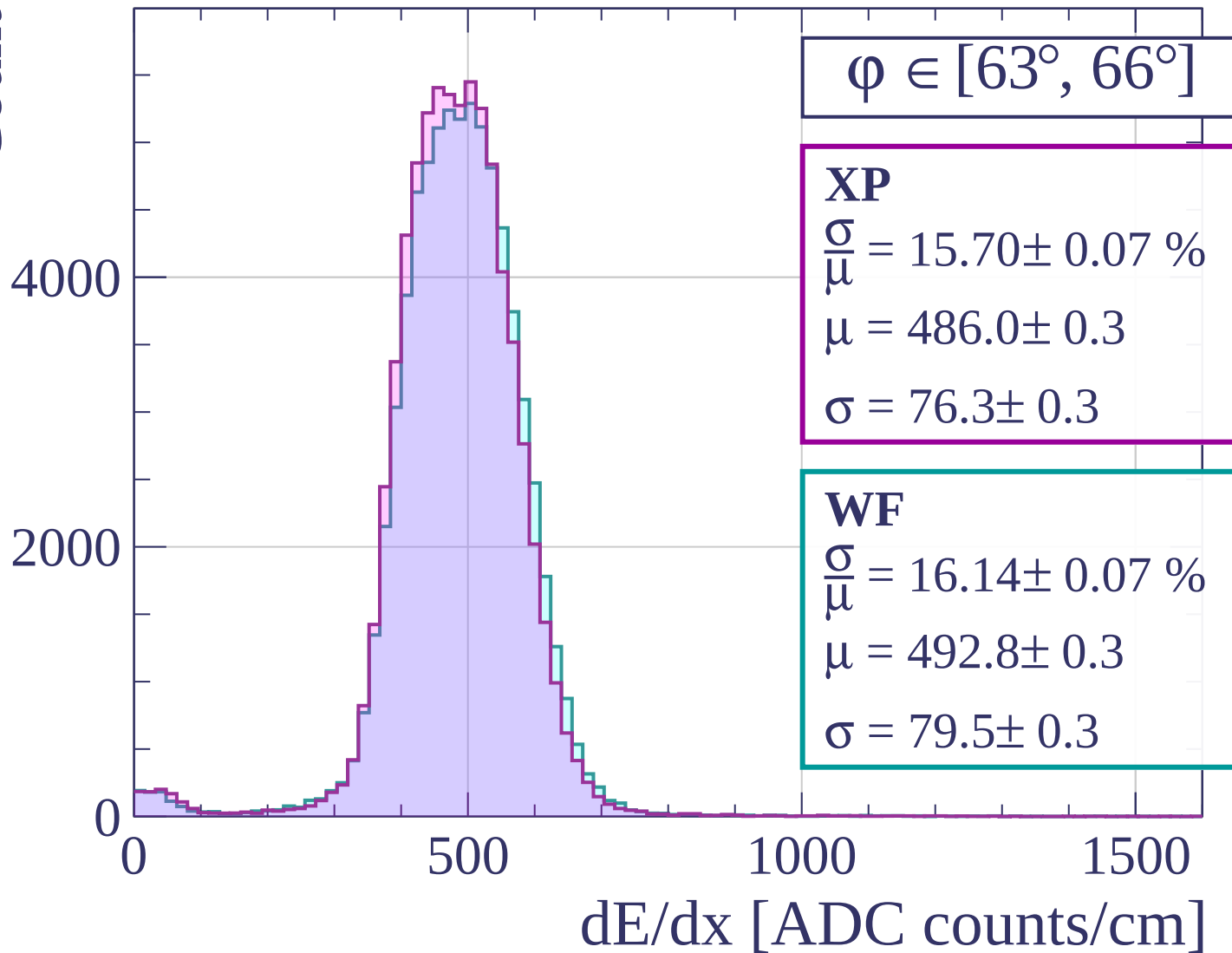


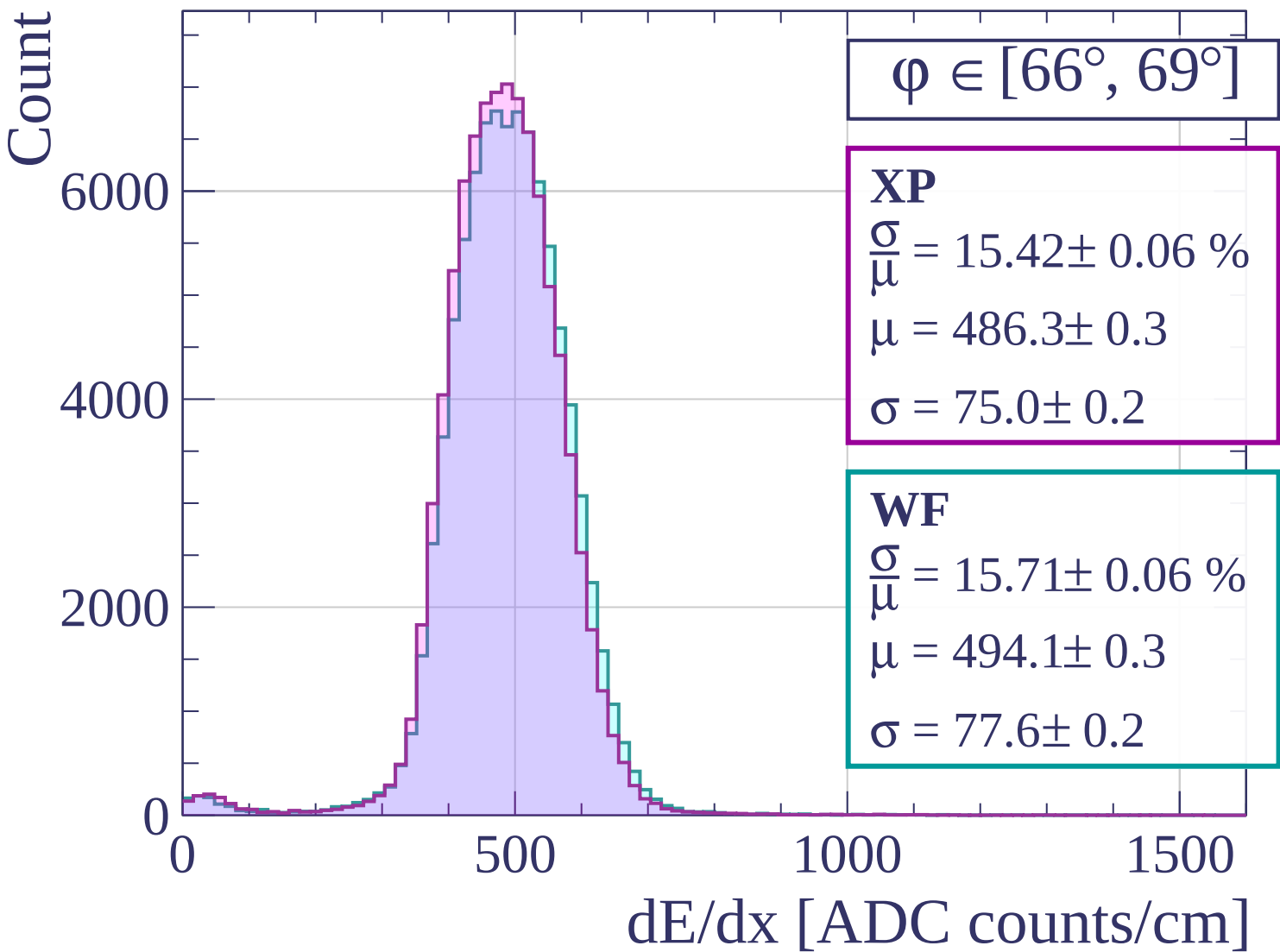


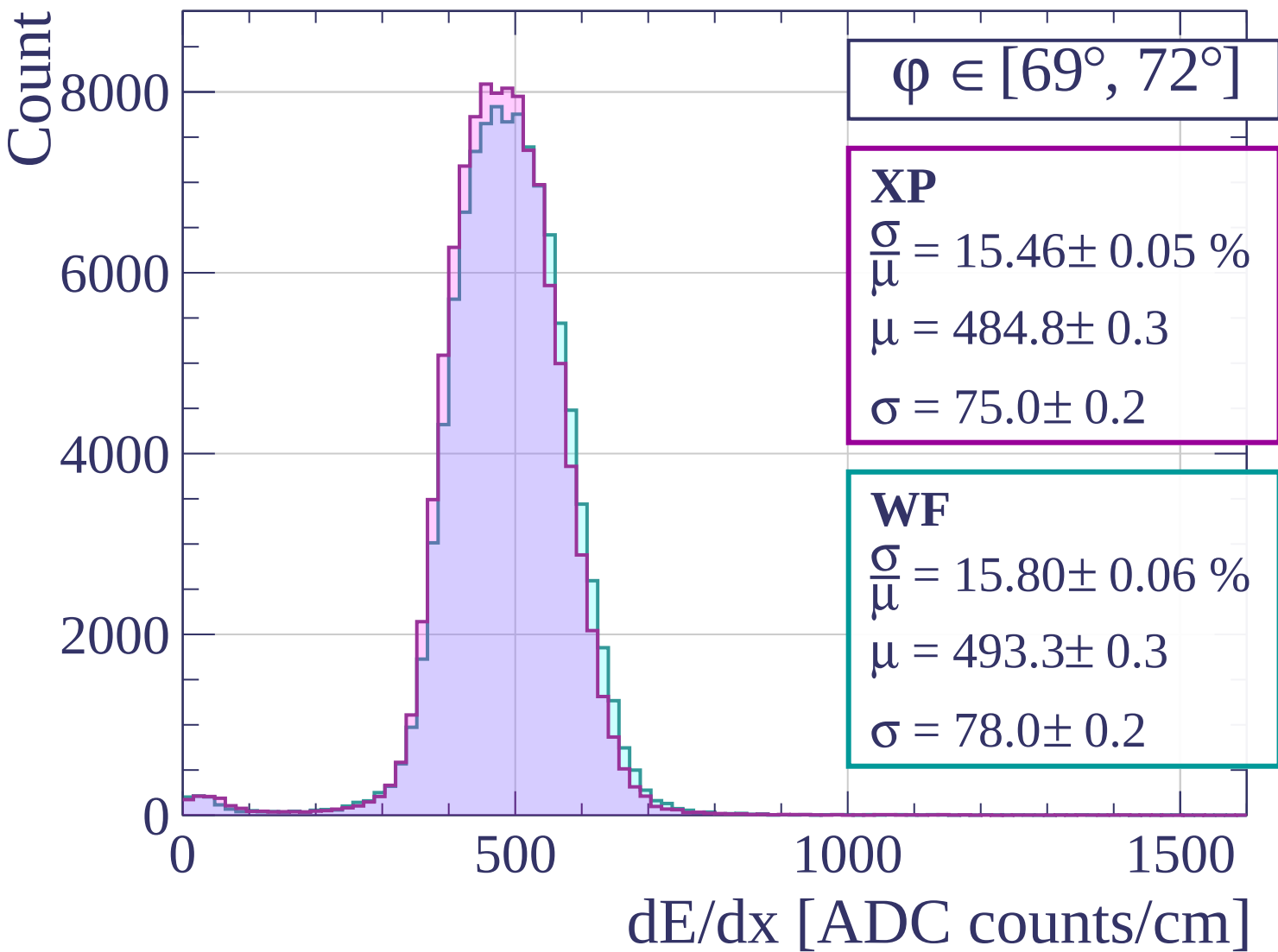


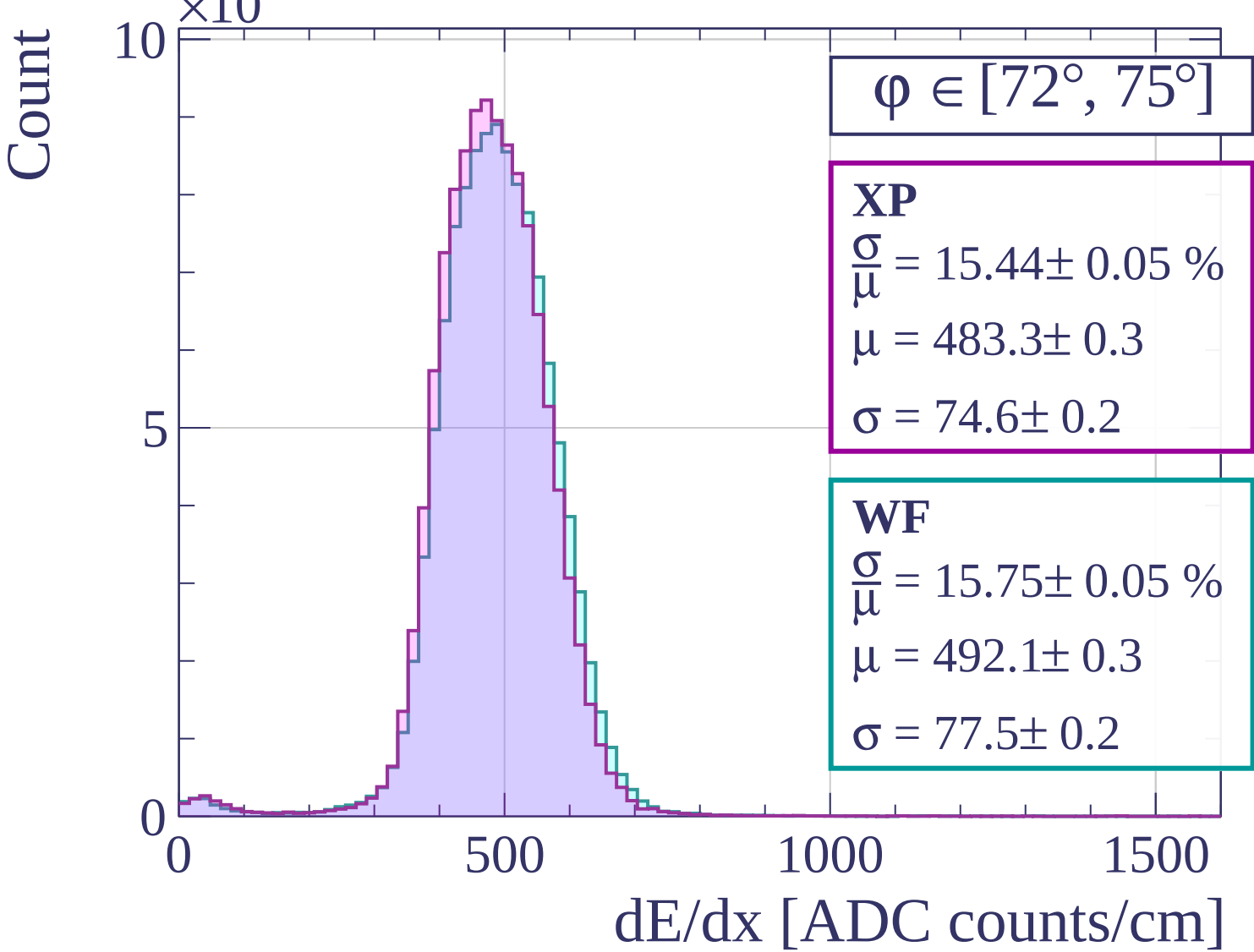


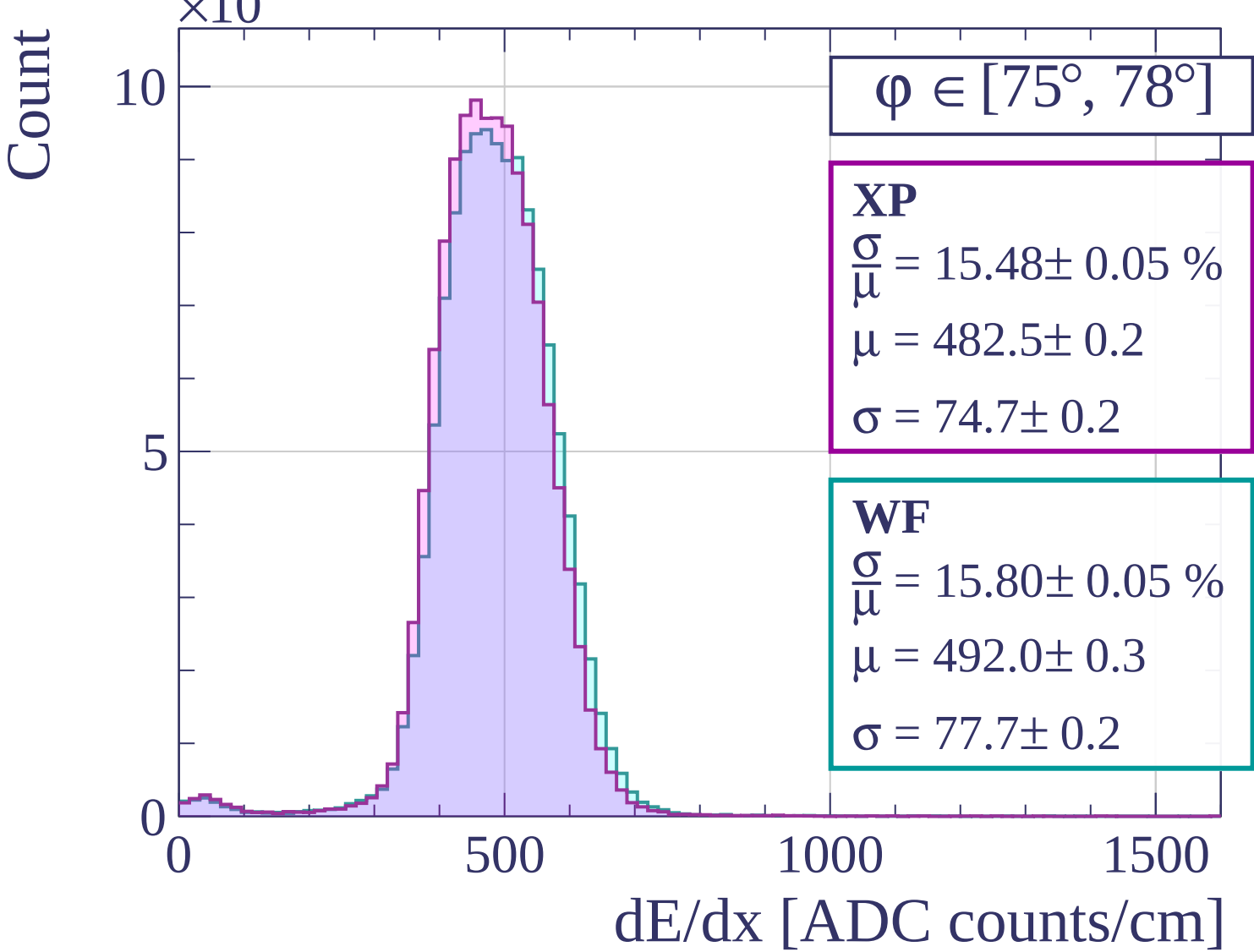
Count

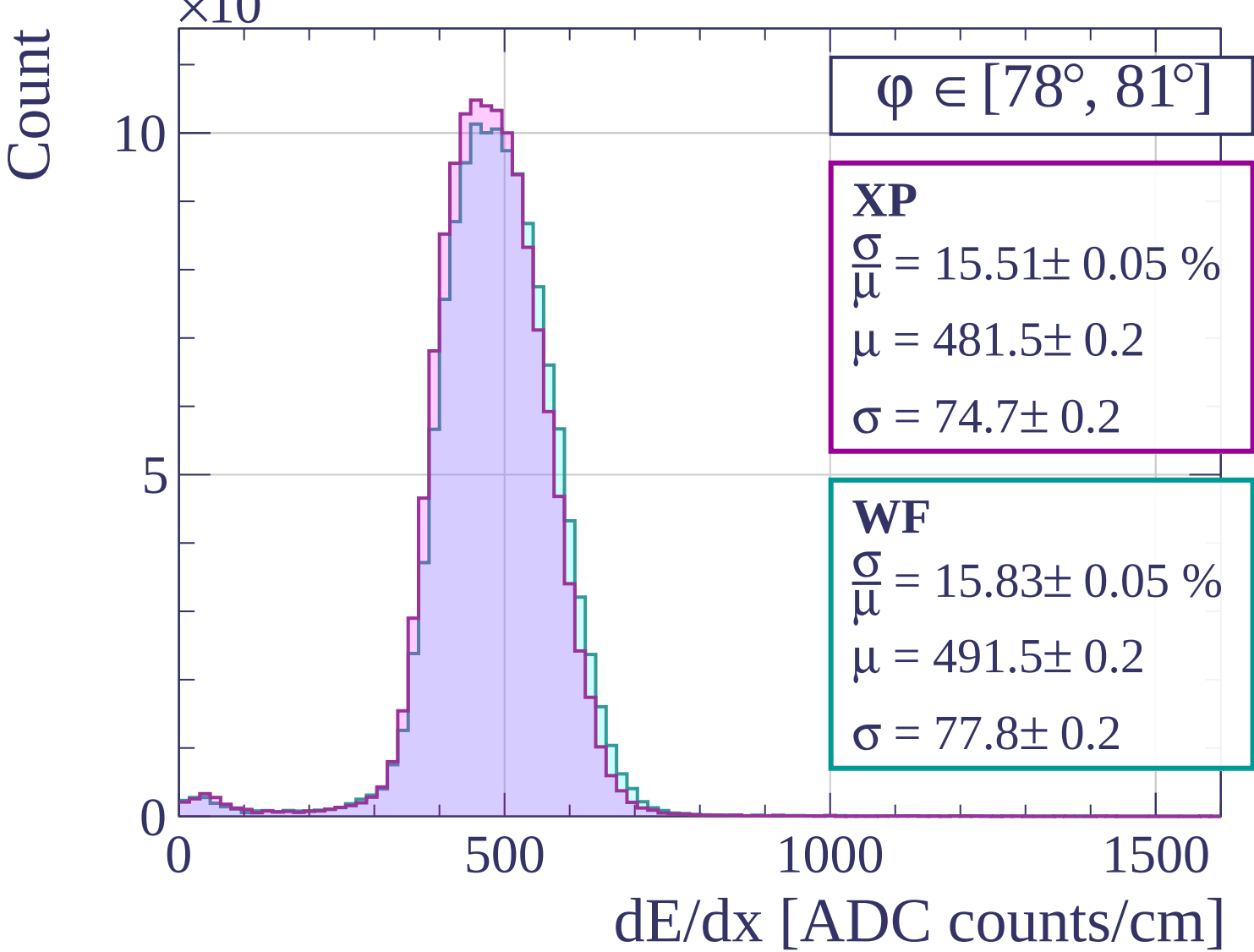


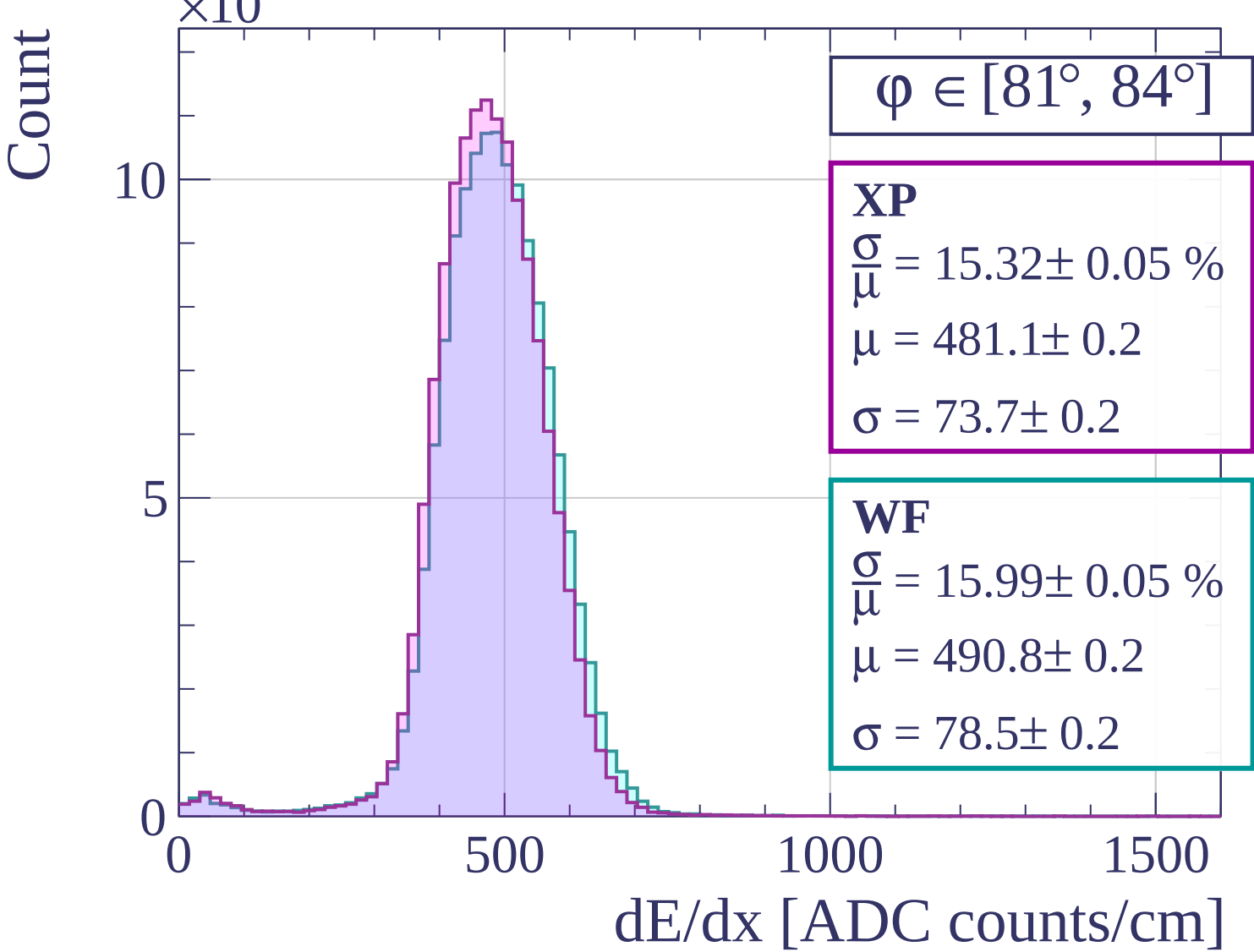


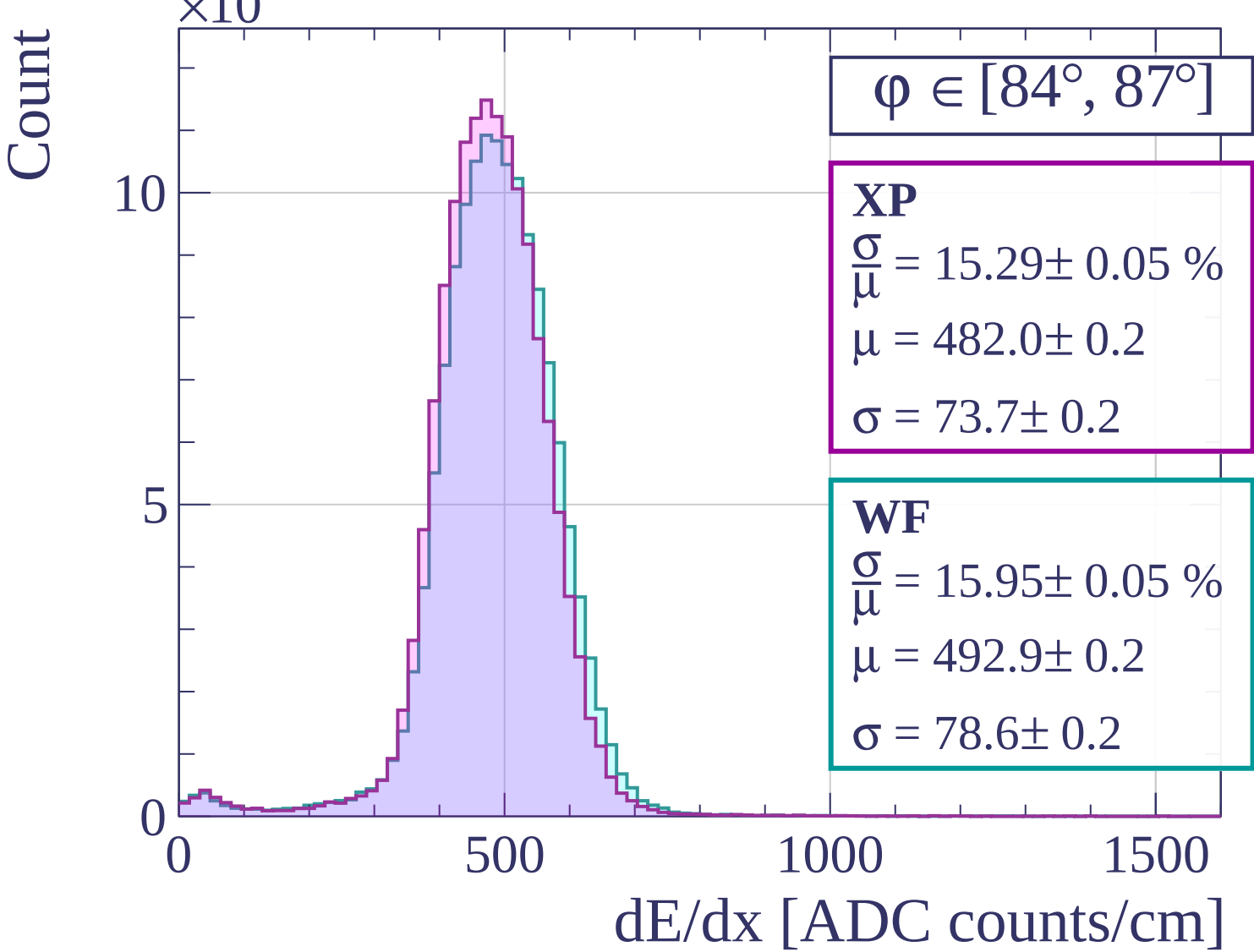


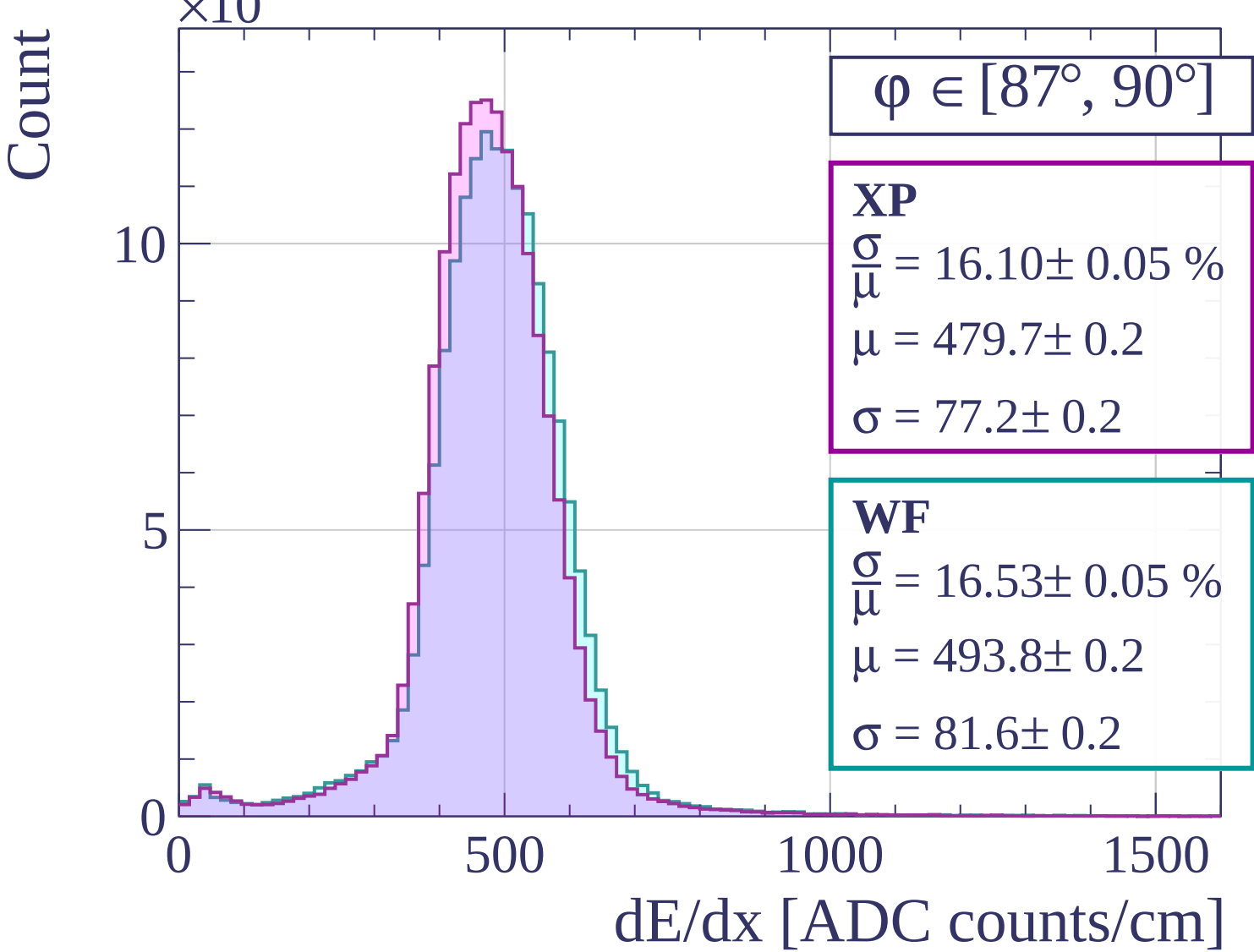


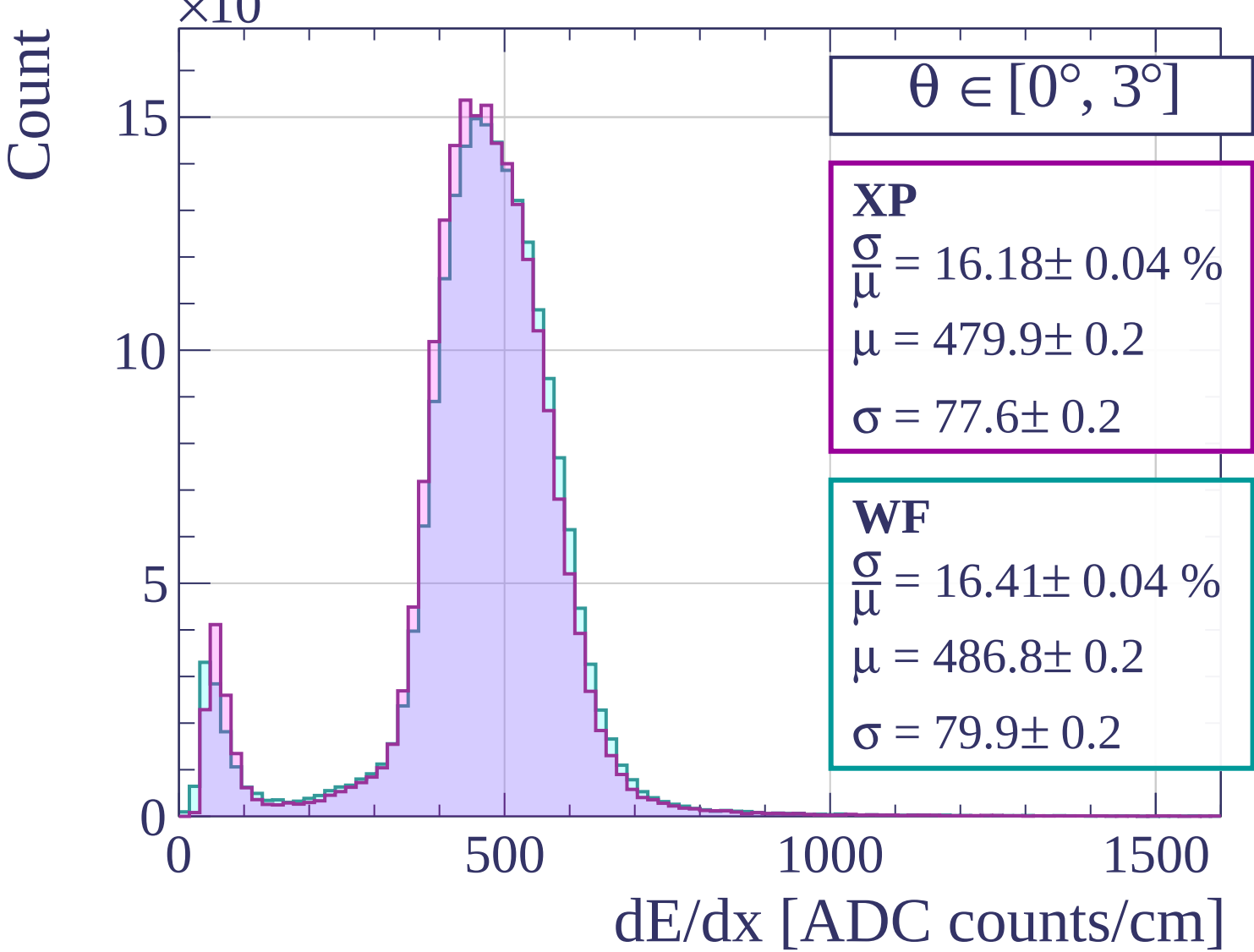


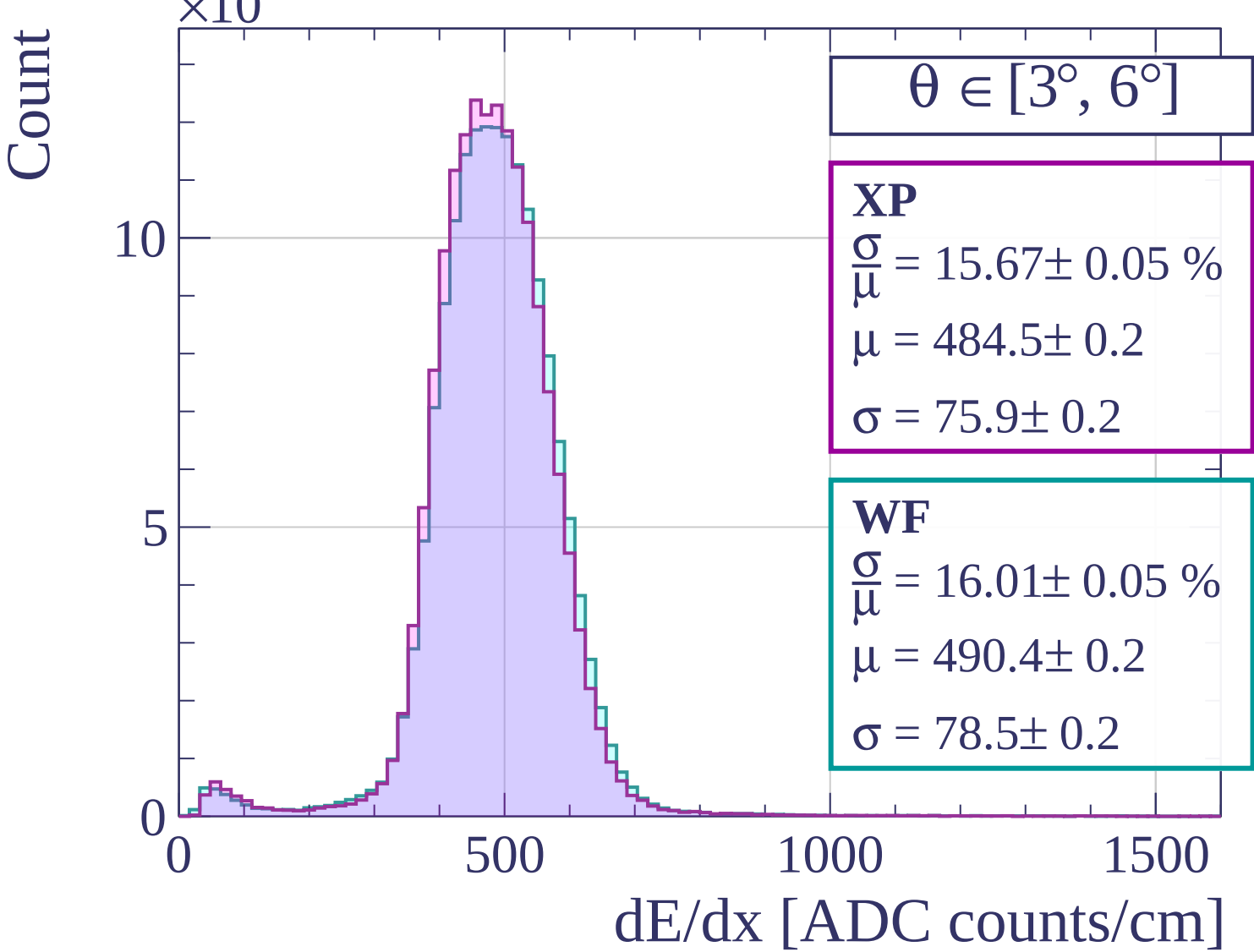


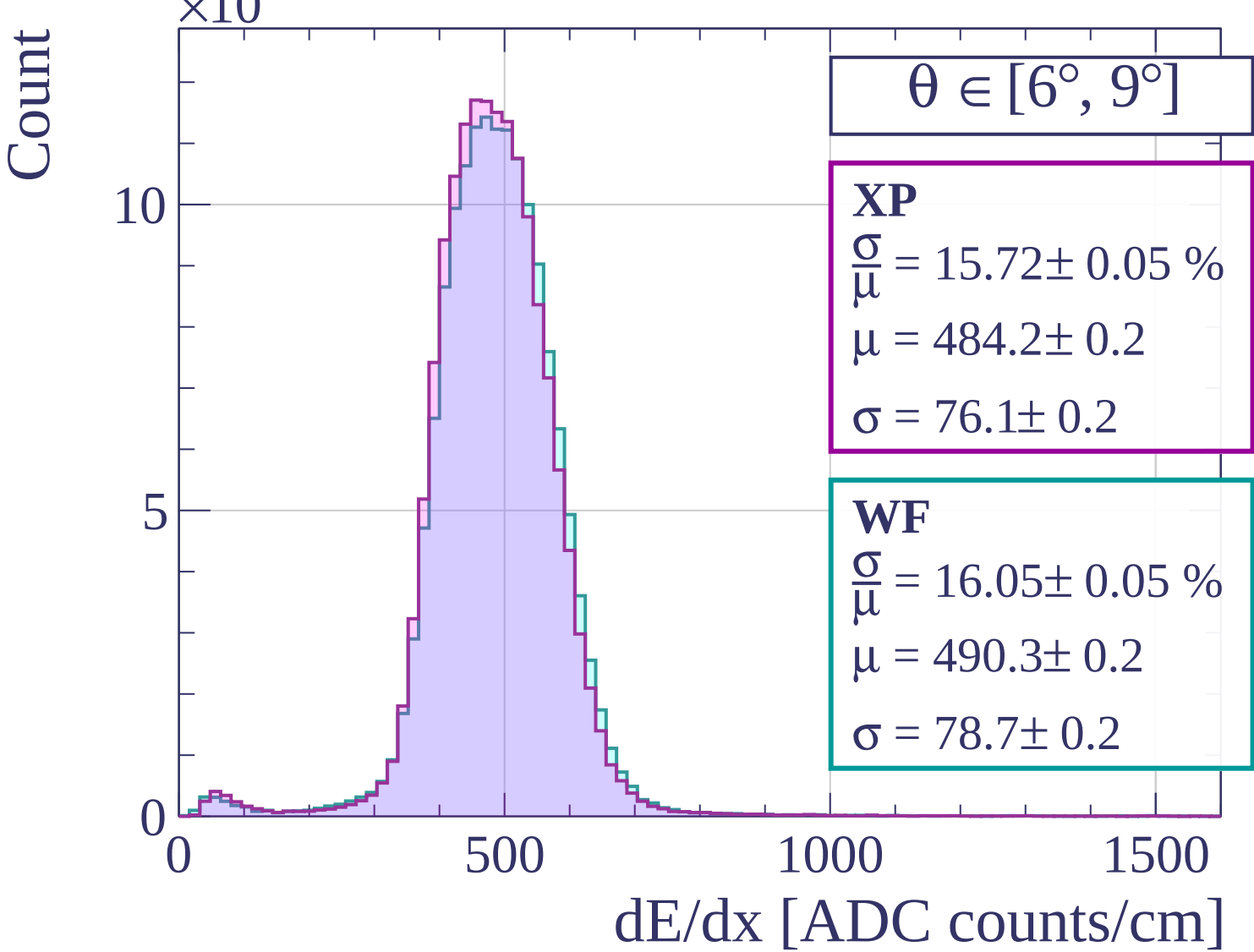


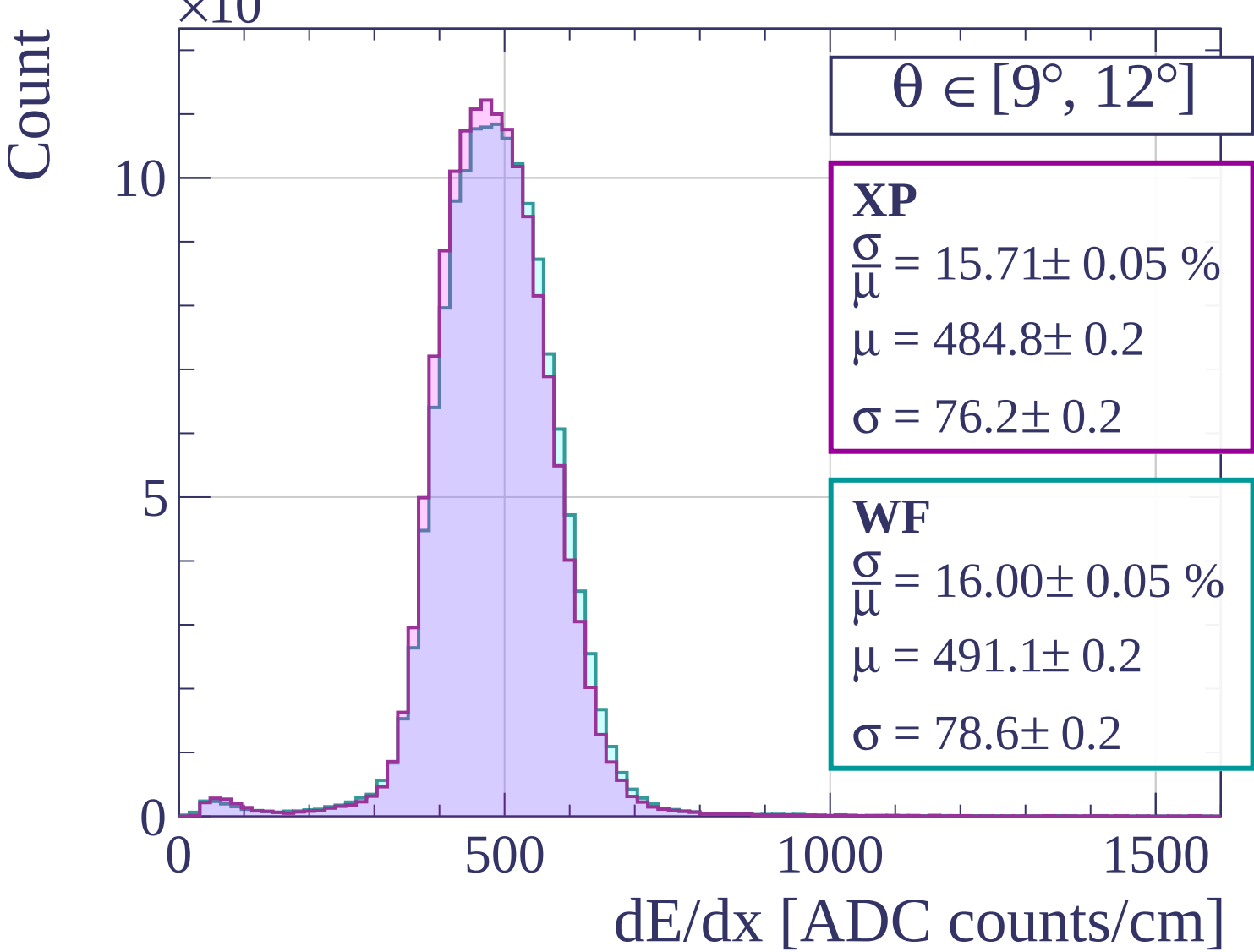


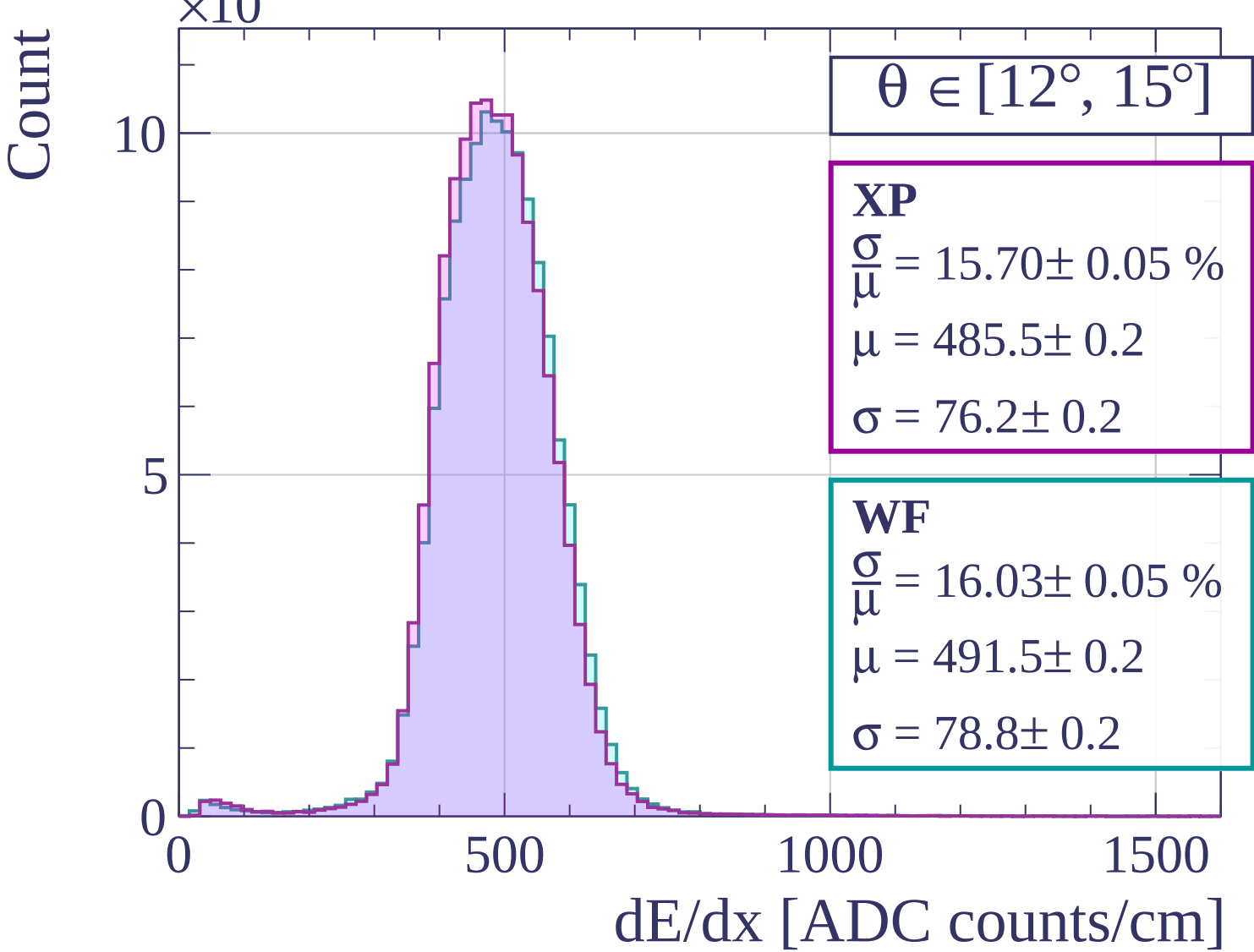


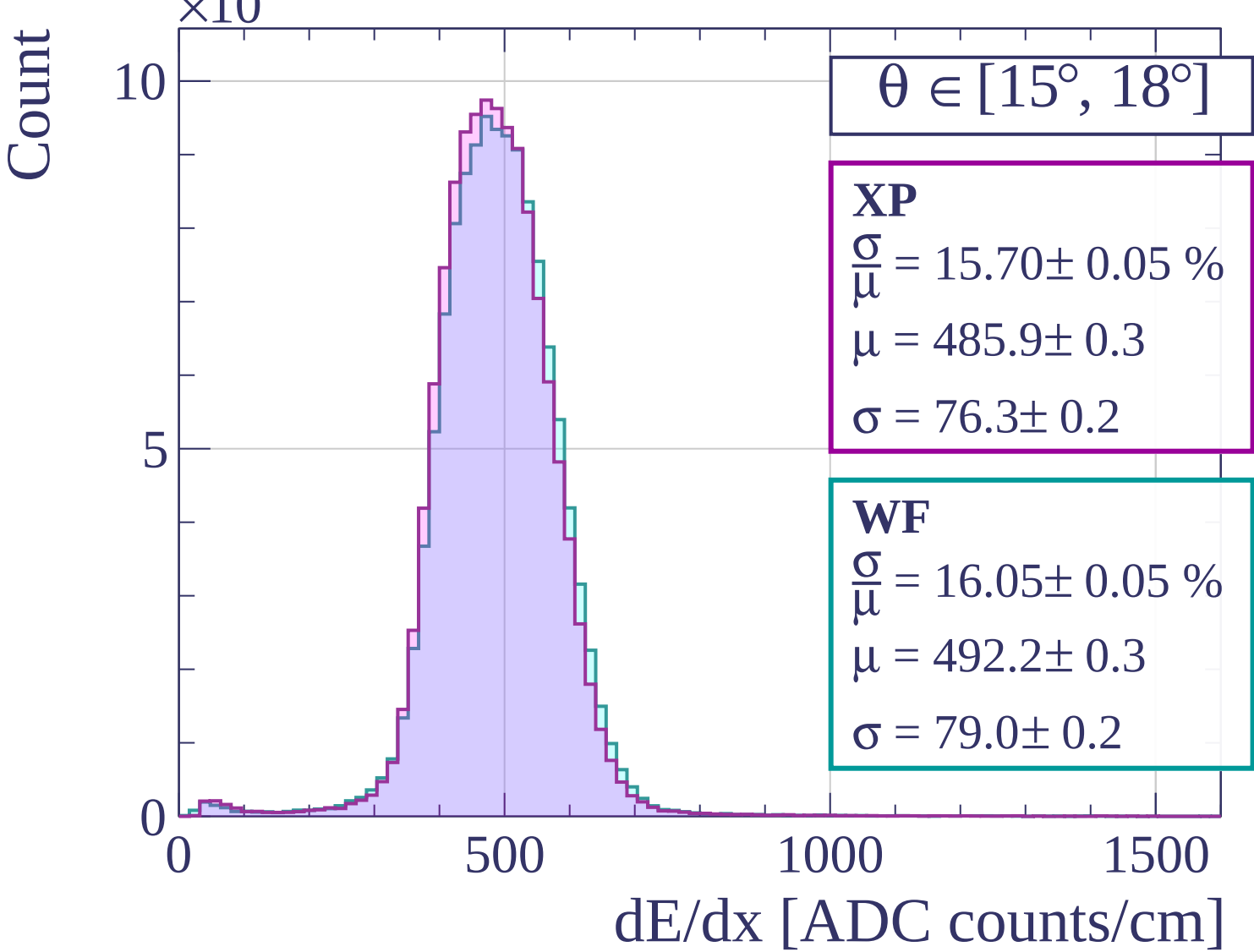


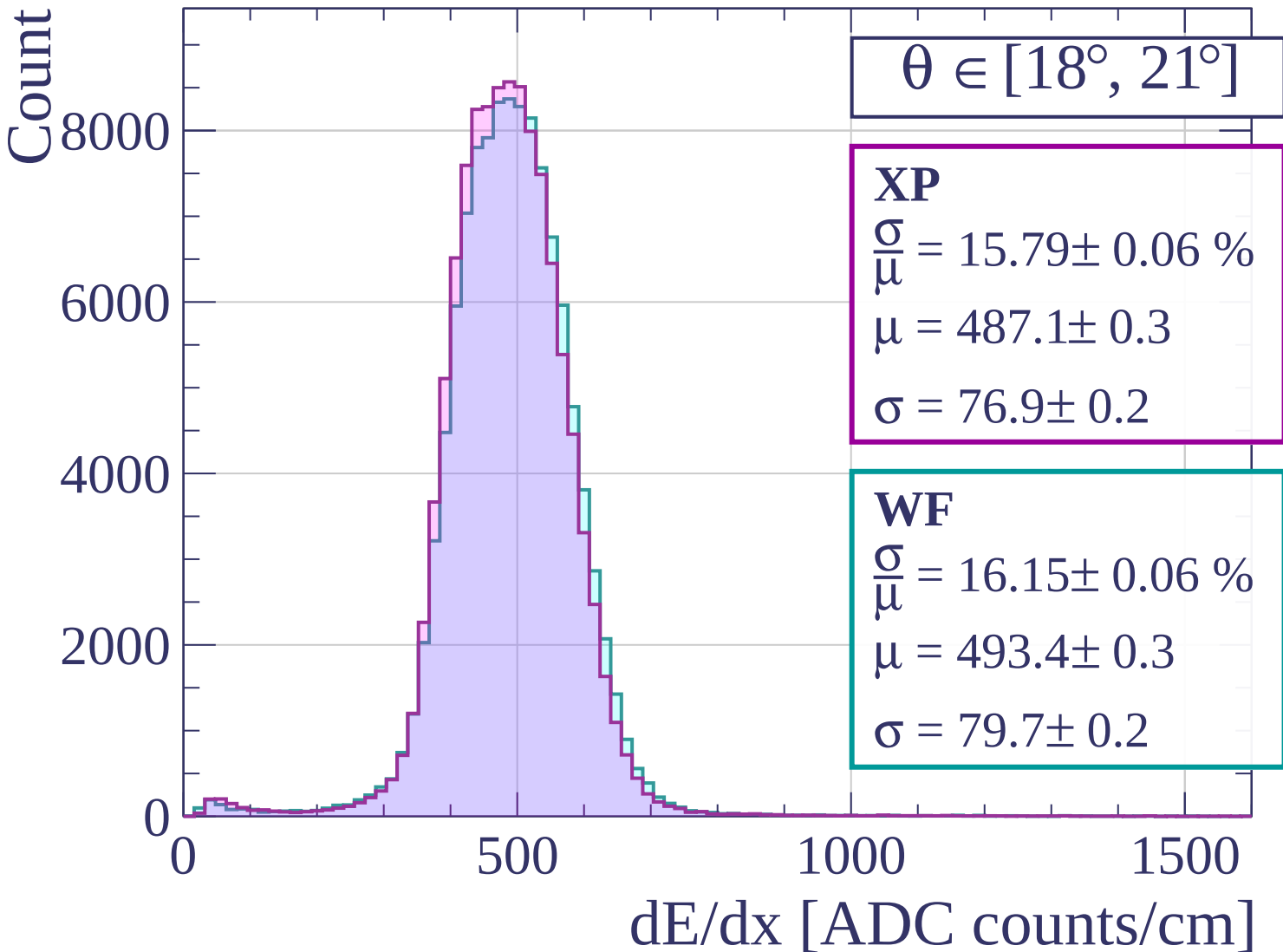


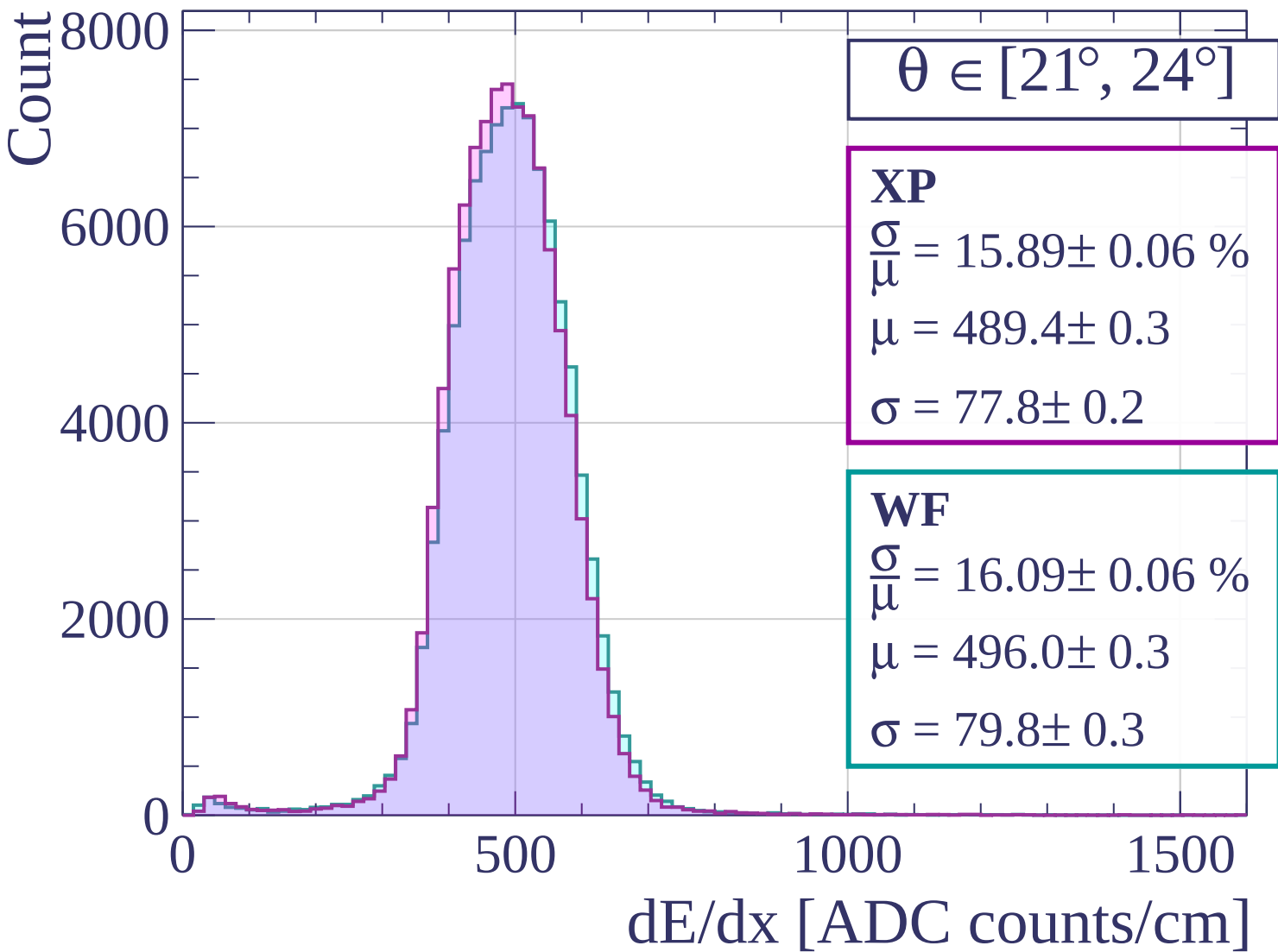


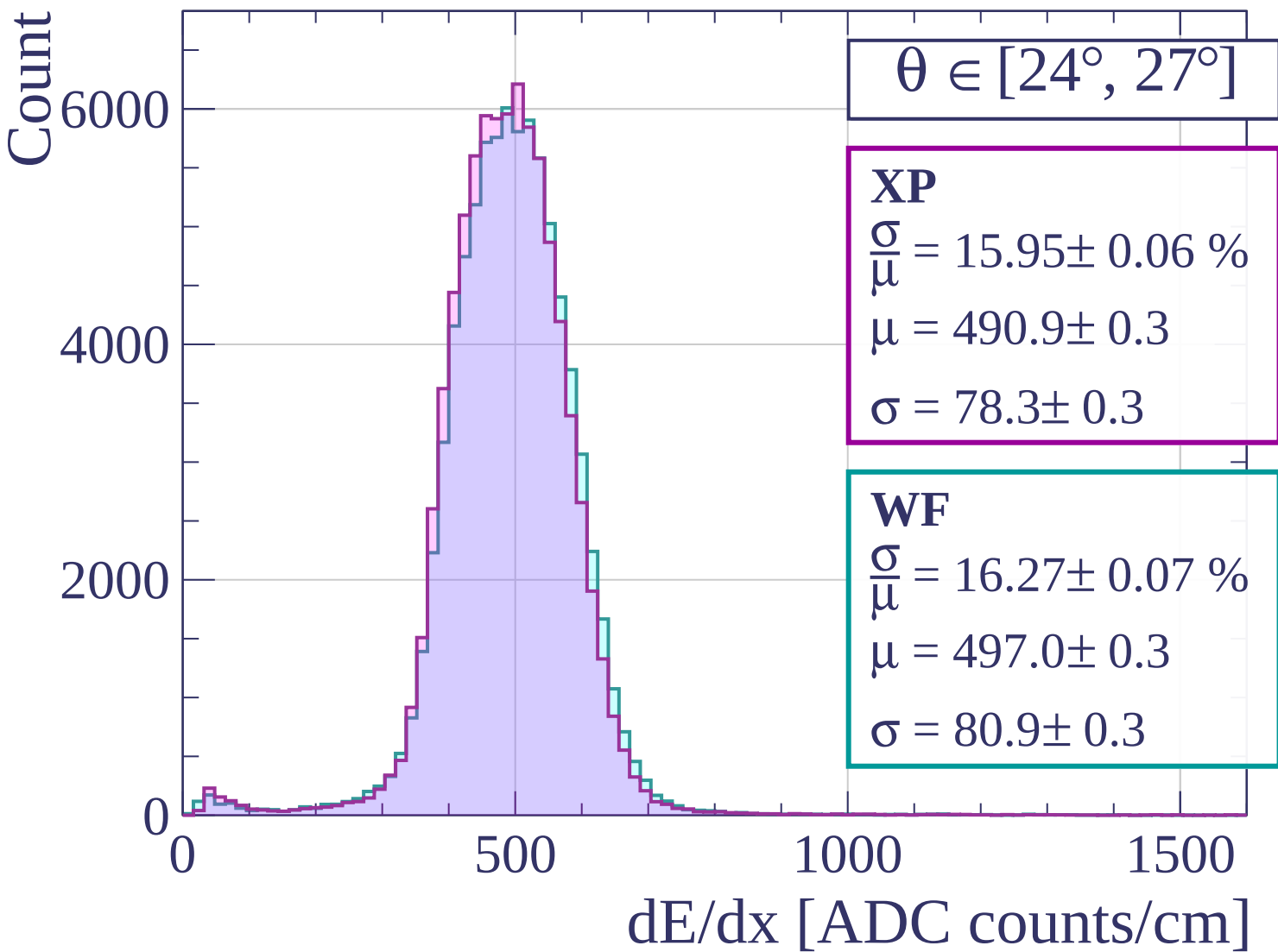


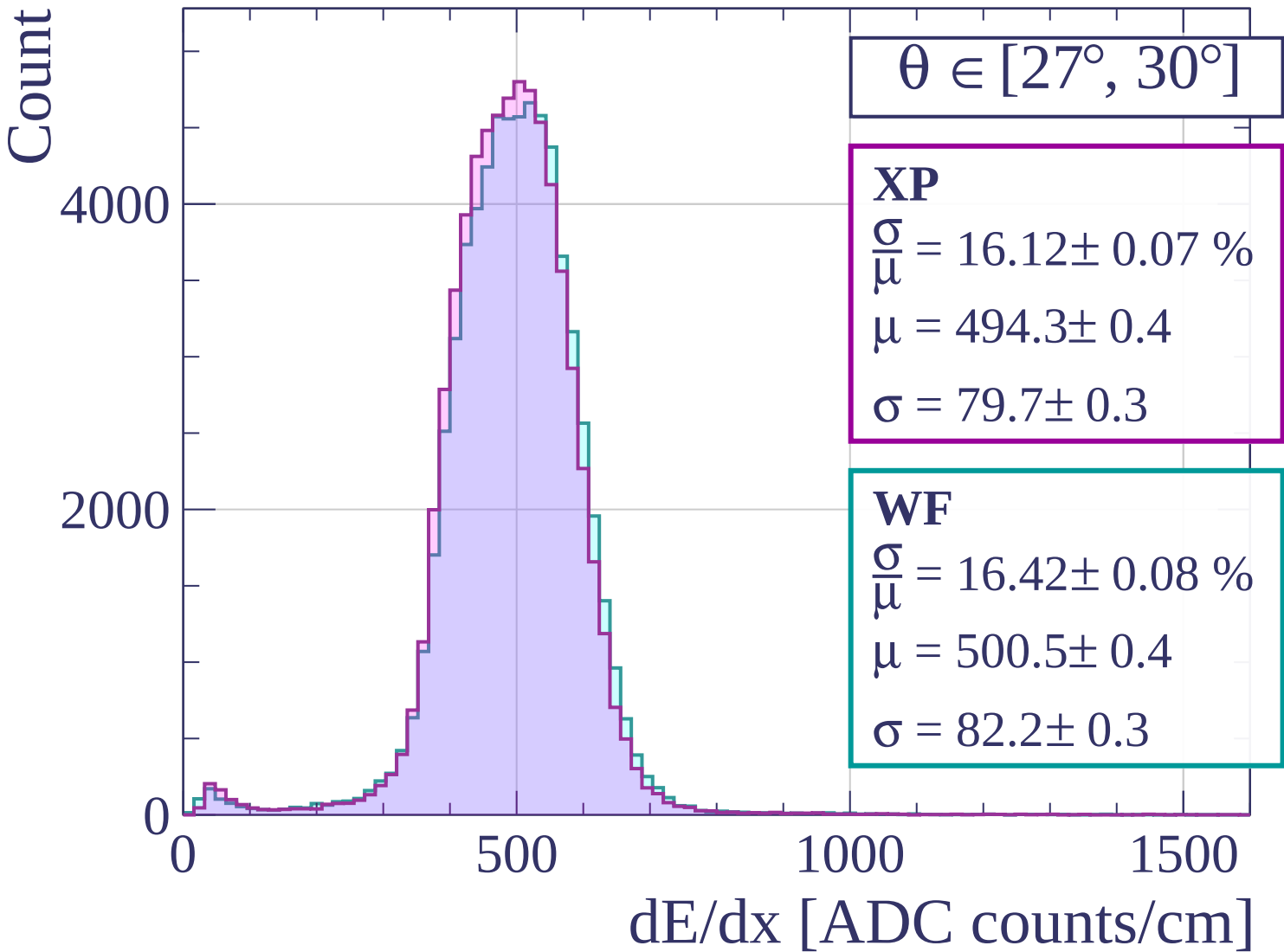


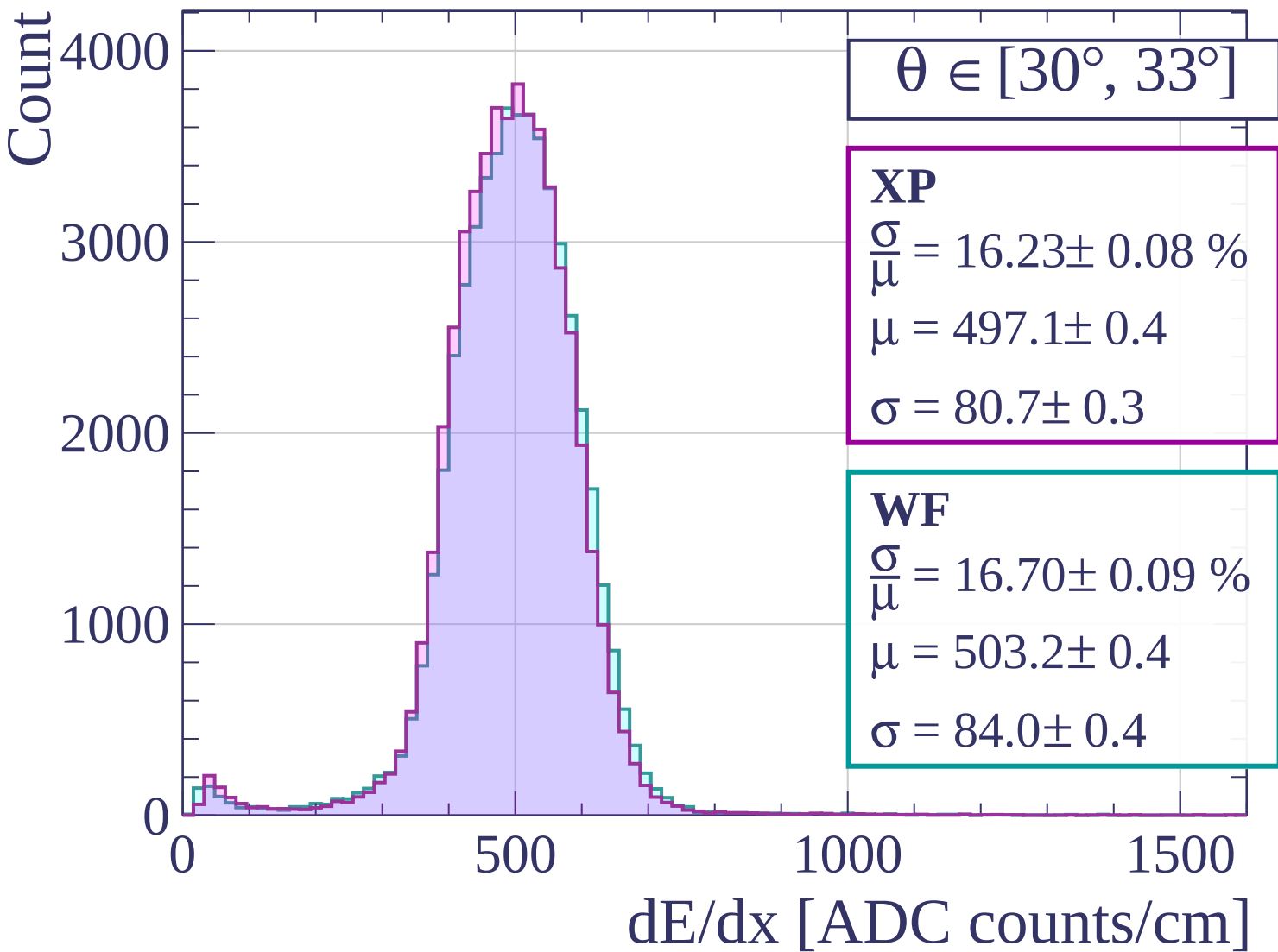


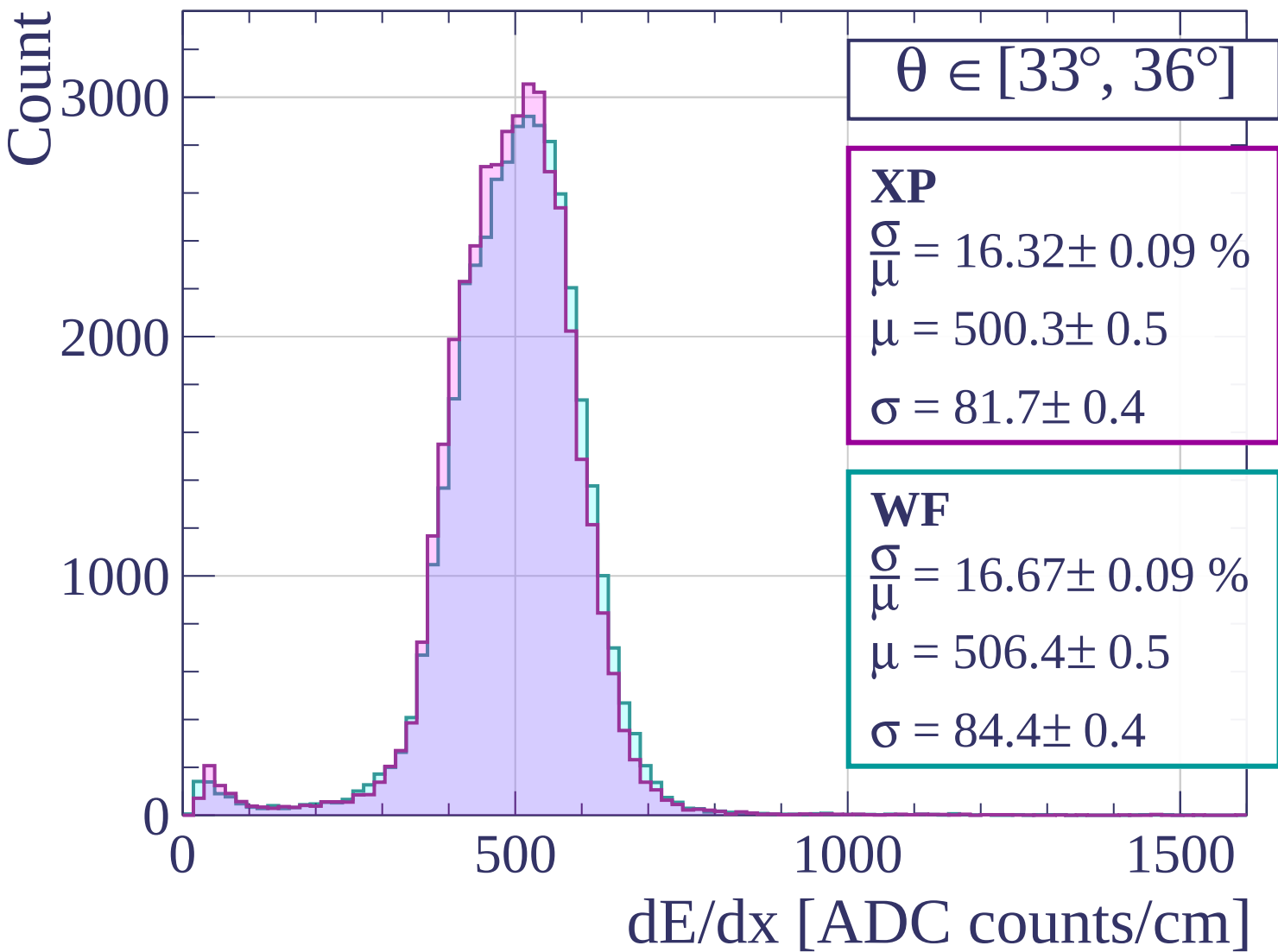


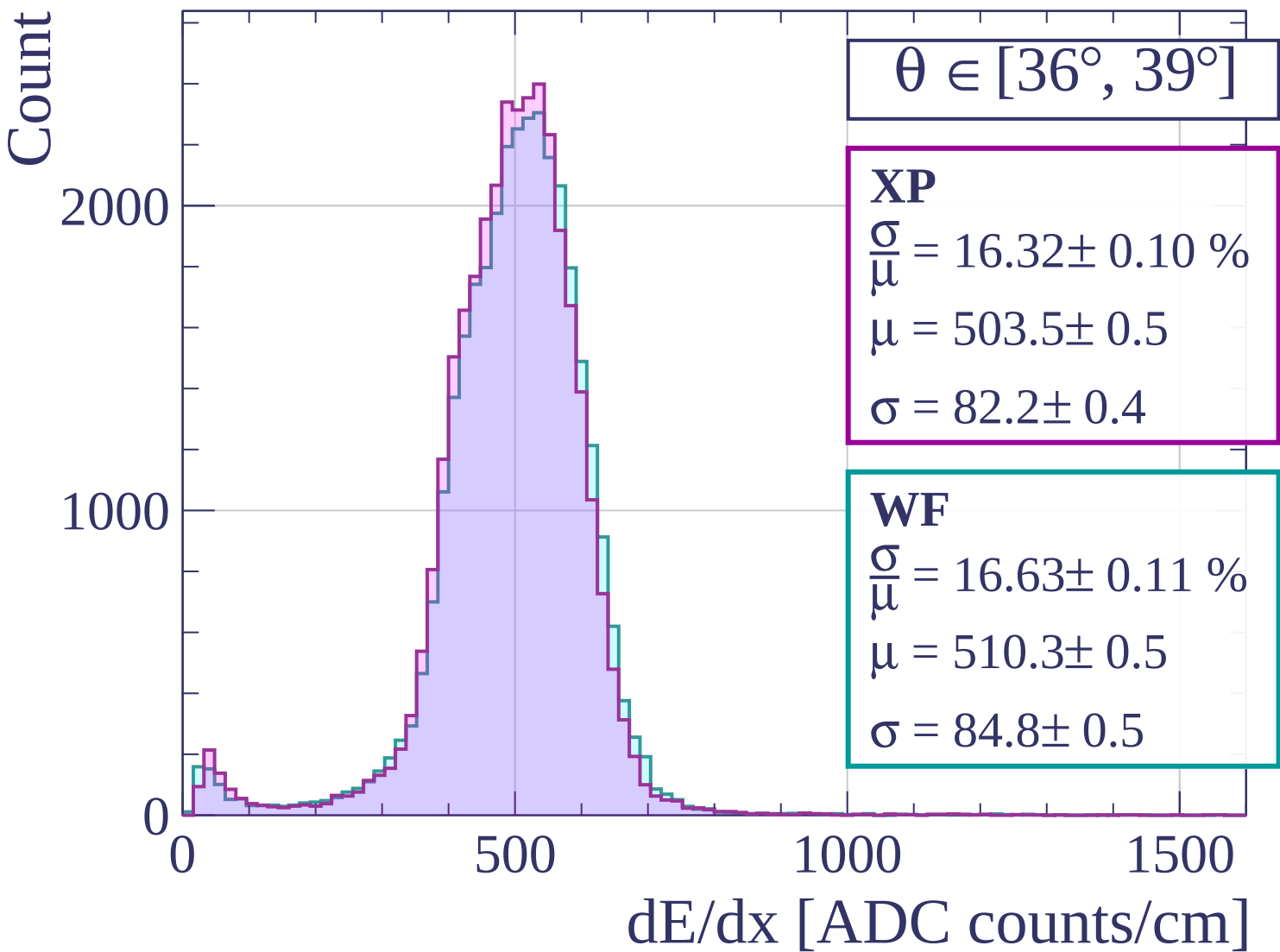


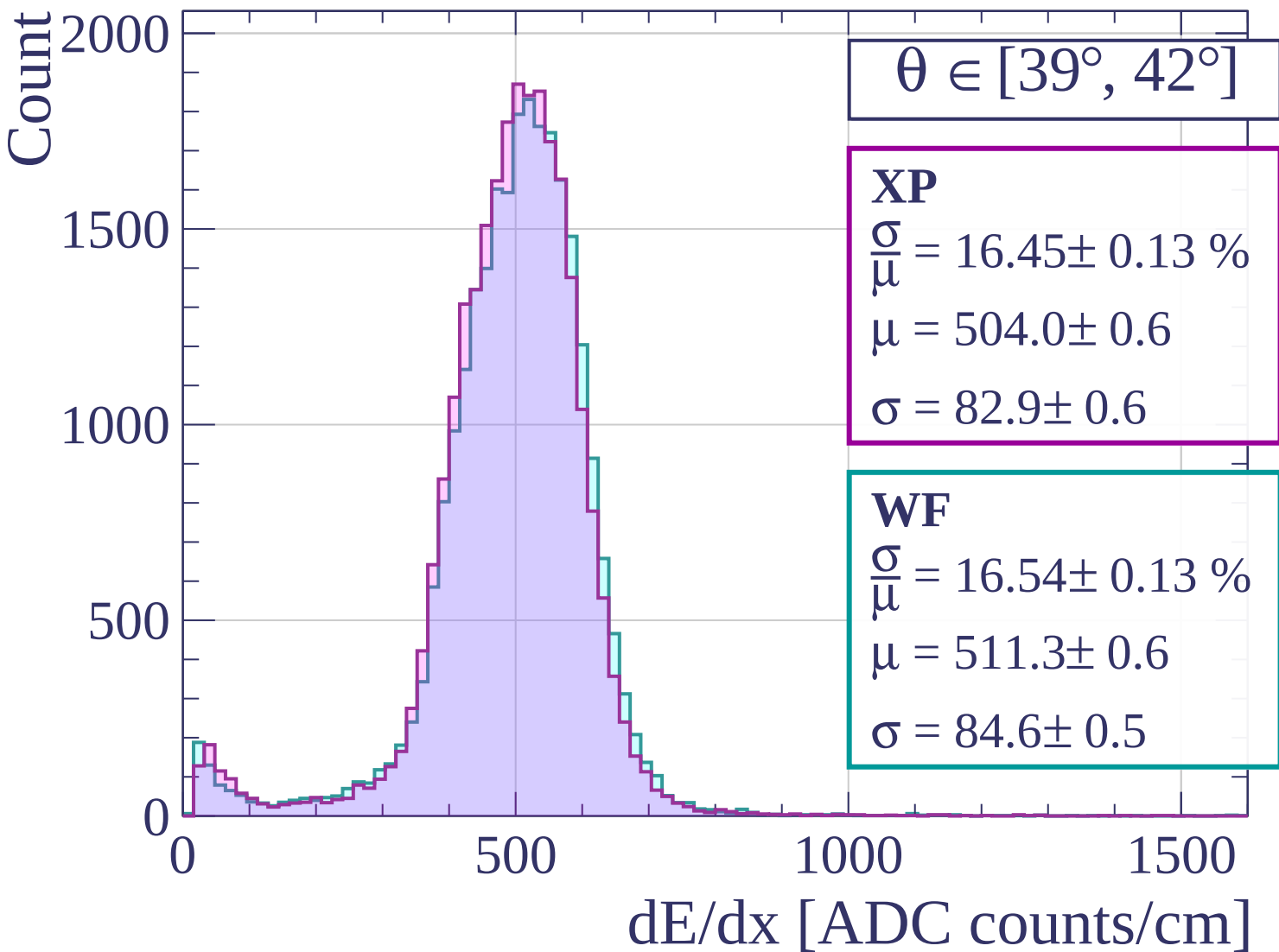


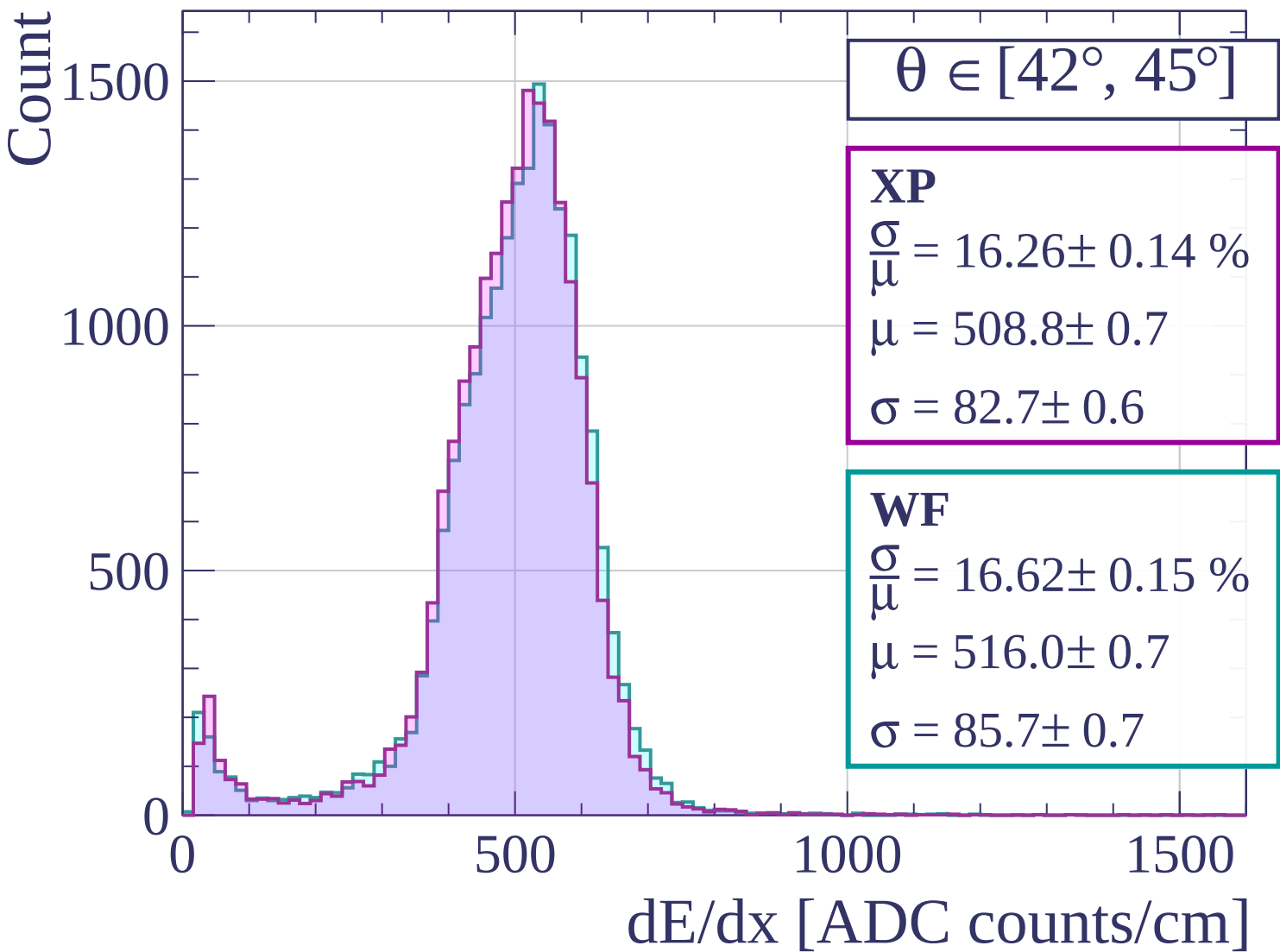


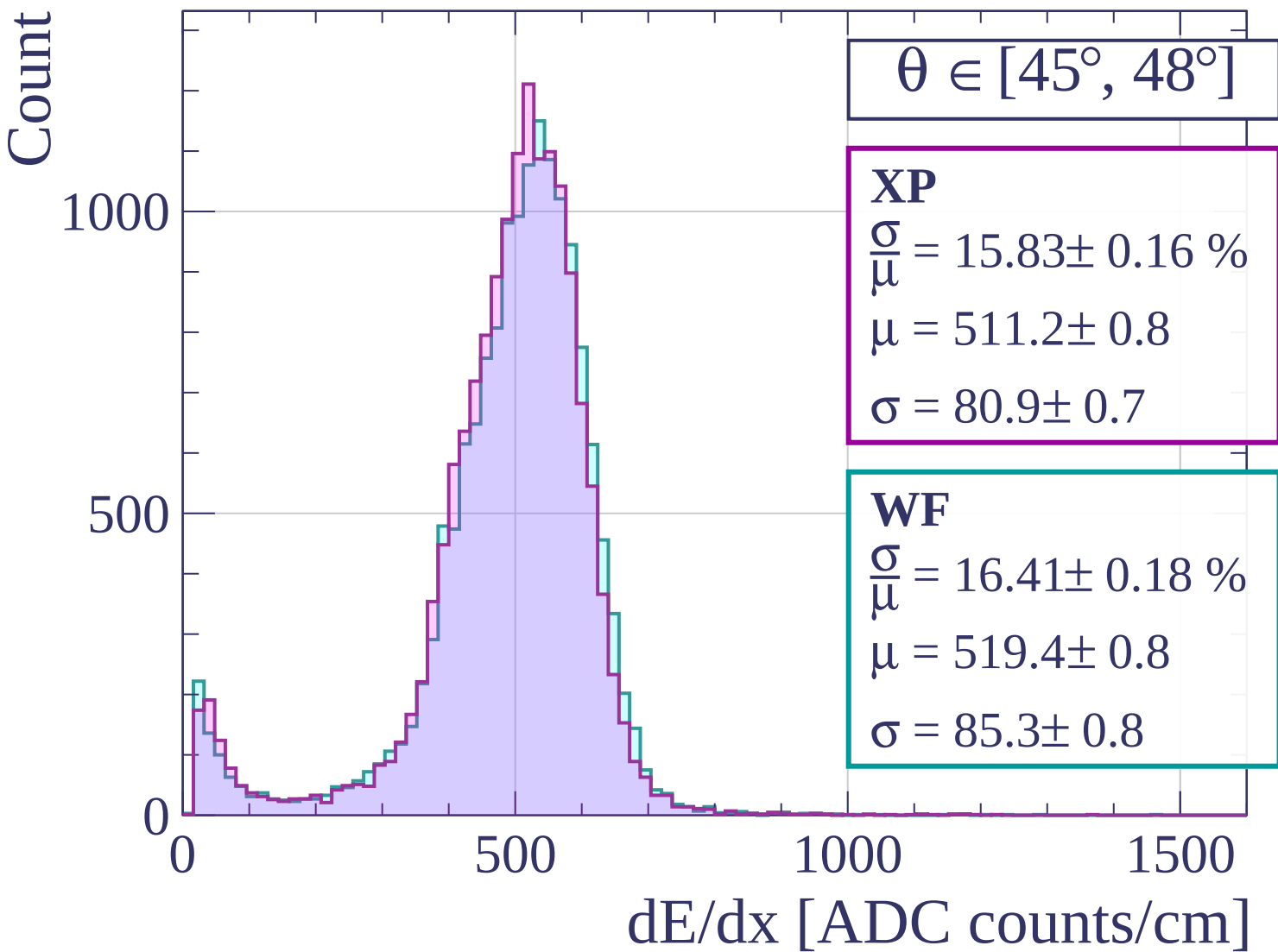












Count

$\theta \in [48^\circ, 51^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.21 \pm 0.20 \%$$

$$\mu = 512.9 \pm 0.9$$

$$\sigma = 83.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 16.45 \pm 0.20 \%$$

$$\mu = 520.6 \pm 1.0$$

$$\sigma = 85.7 \pm 0.9$$

800

600

400

200

0

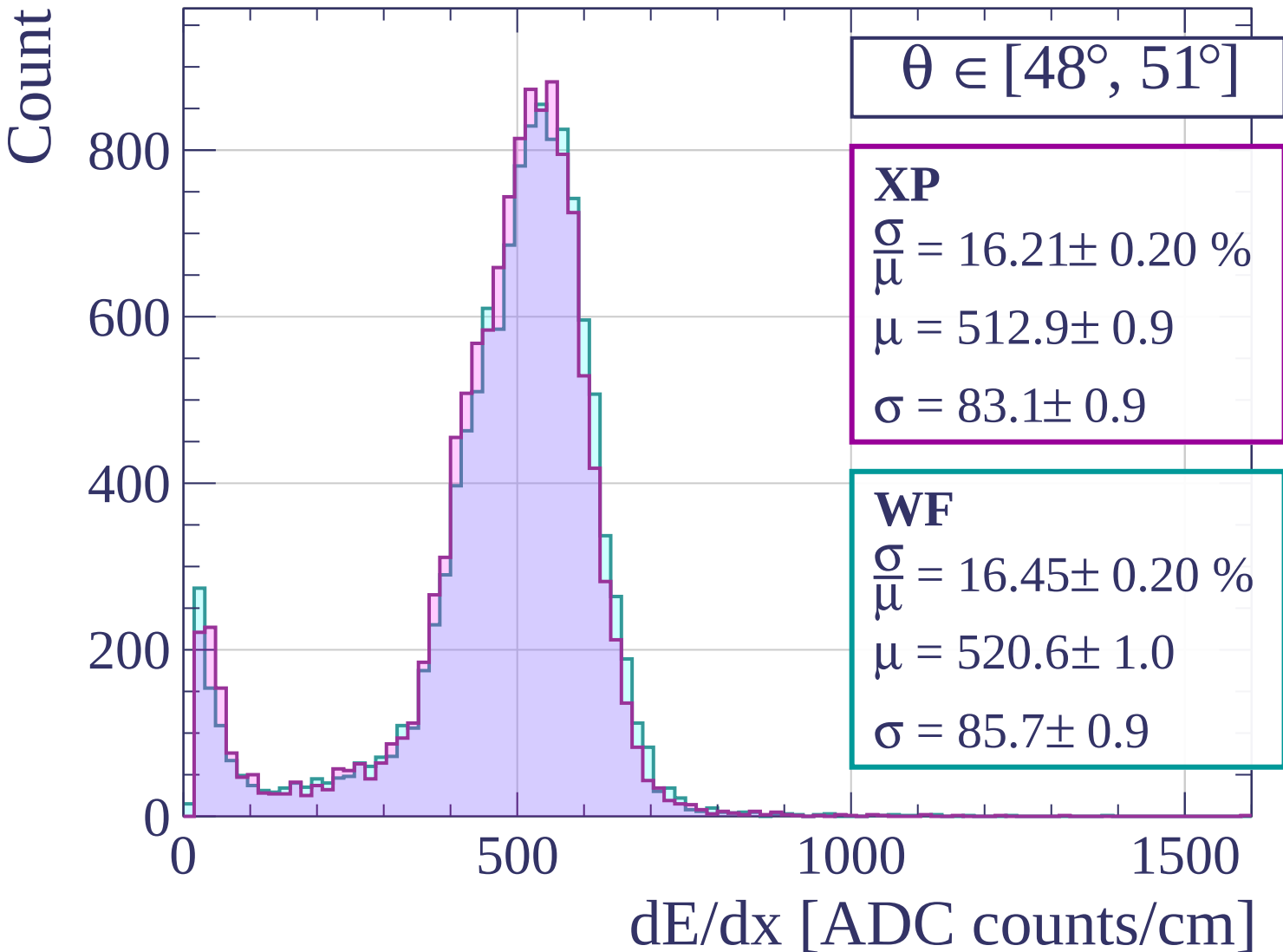
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [51^\circ, 54^\circ]$

XP

$$\frac{\sigma}{\mu} = 15.81 \pm 0.23 \%$$

$$\mu = 515.0 \pm 1.1$$

$$\sigma = 81.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 16.07 \pm 0.22 \%$$

$$\mu = 522.8 \pm 1.1$$

$$\sigma = 84.0 \pm 1.0$$

600

400

200

0

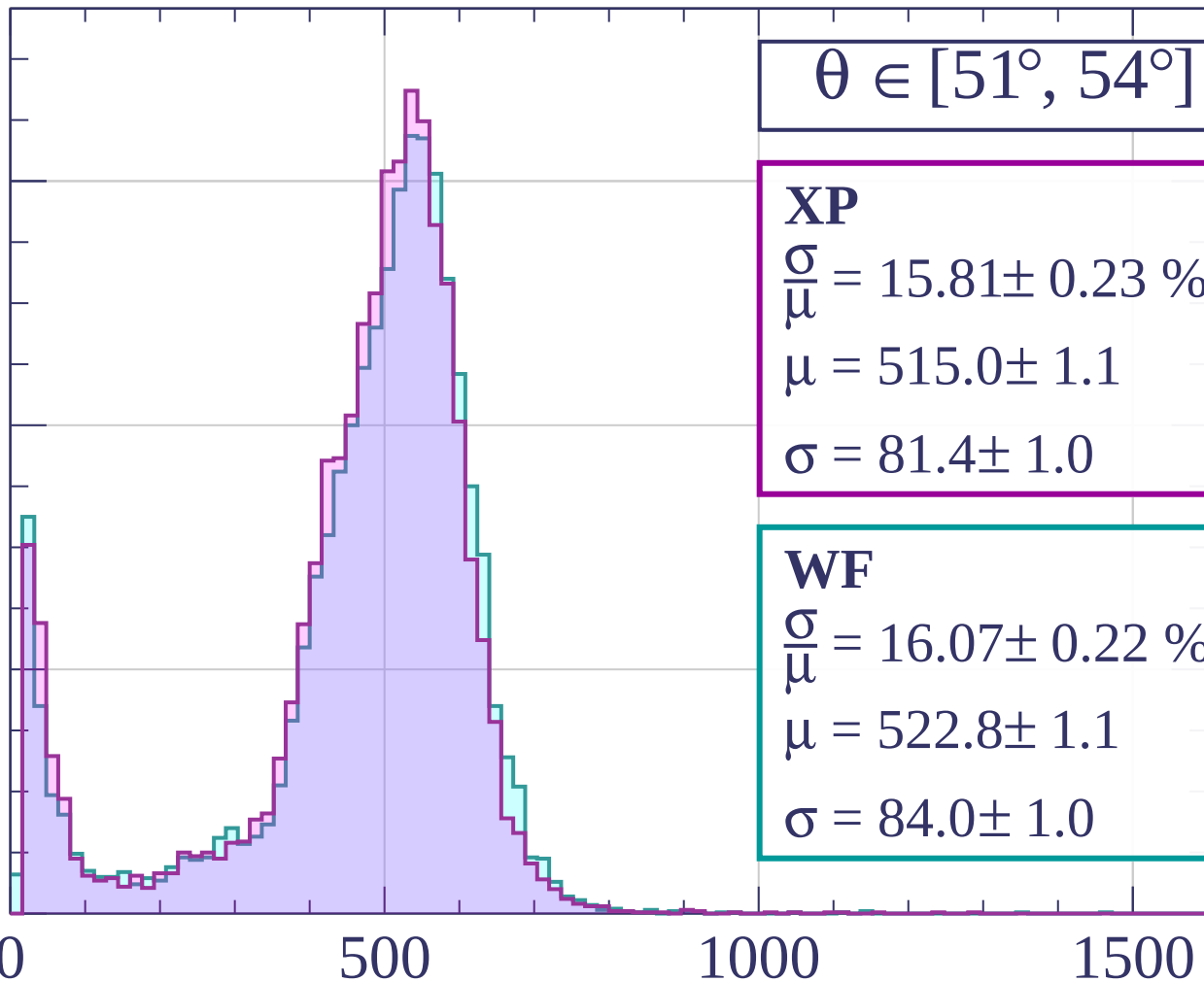
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$$\theta \in [54^\circ, 57^\circ]$$

XP

$$\frac{\sigma}{\mu} = 15.39 \pm 0.25 \%$$

$$\mu = 517.0 \pm 1.3$$

$$\sigma = 79.5 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 15.80 \pm 0.29 \%$$

$$\mu = 526.2 \pm 1.4$$

$$\sigma = 83.1 \pm 1.3$$

400

200

0

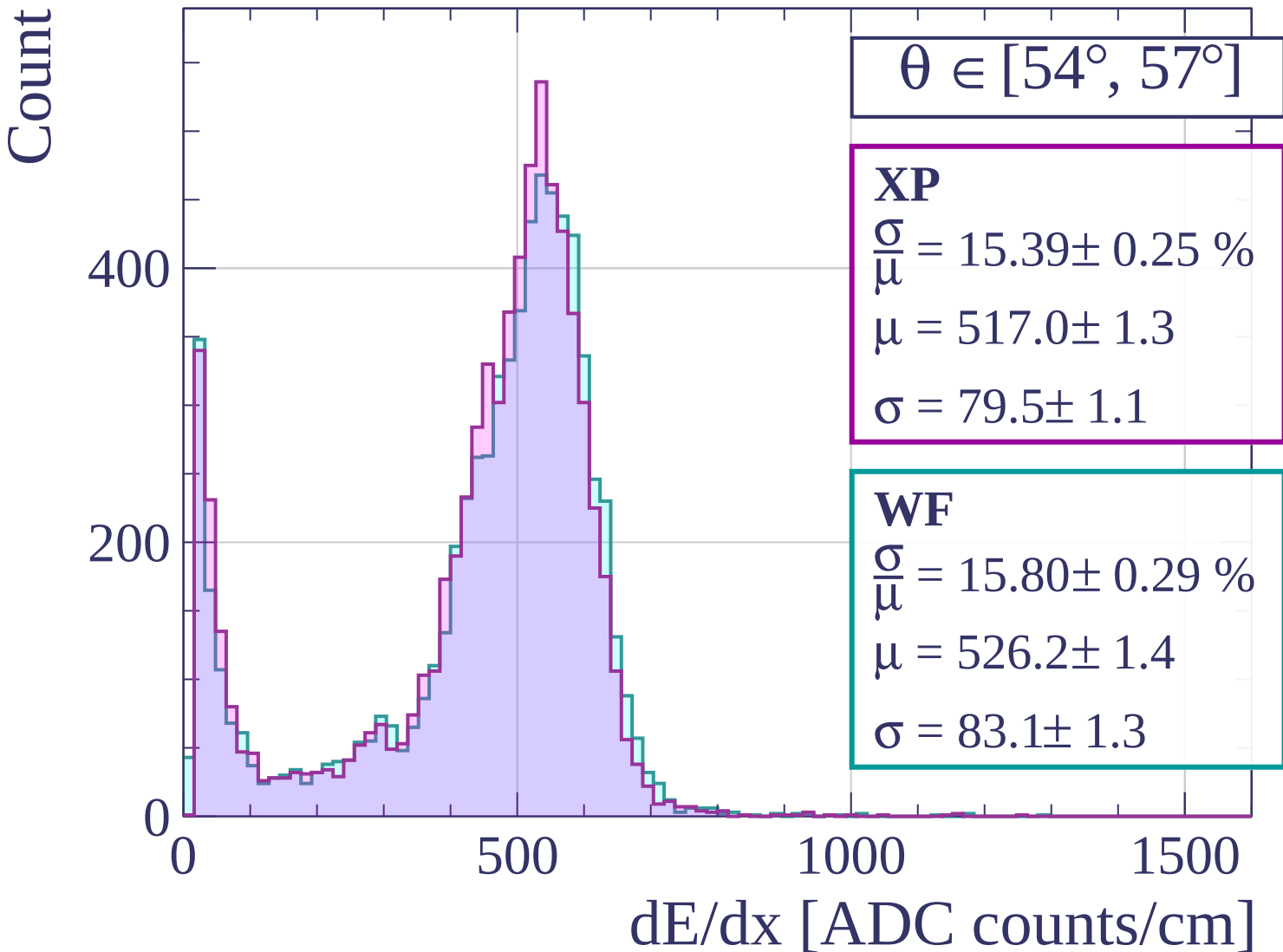
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [57^\circ, 60^\circ]$

XP

$$\frac{\sigma}{\mu} = 15.00 \pm 0.28 \%$$

$$\mu = 513.0 \pm 1.5$$

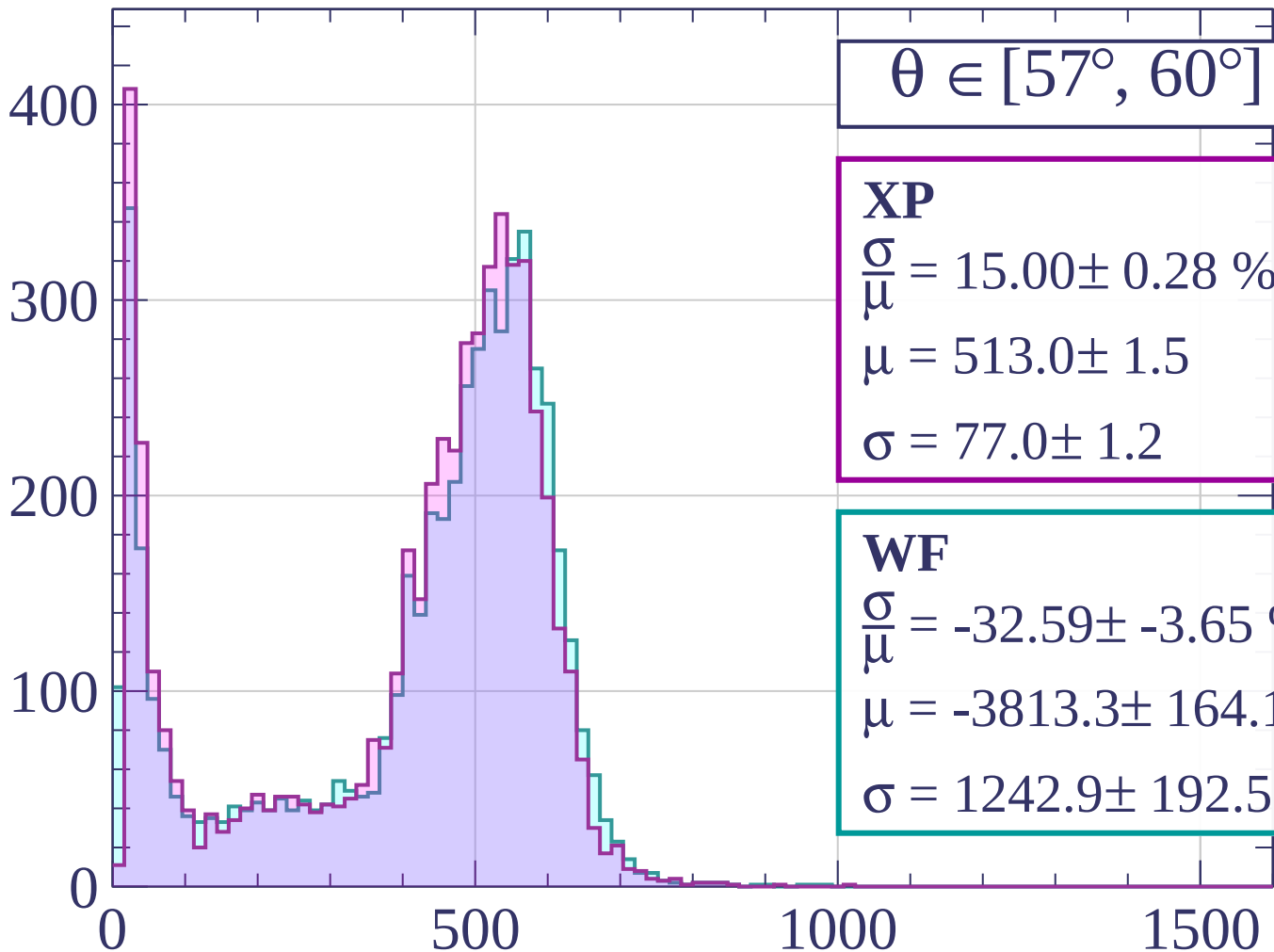
$$\sigma = 77.0 \pm 1.2$$

WF

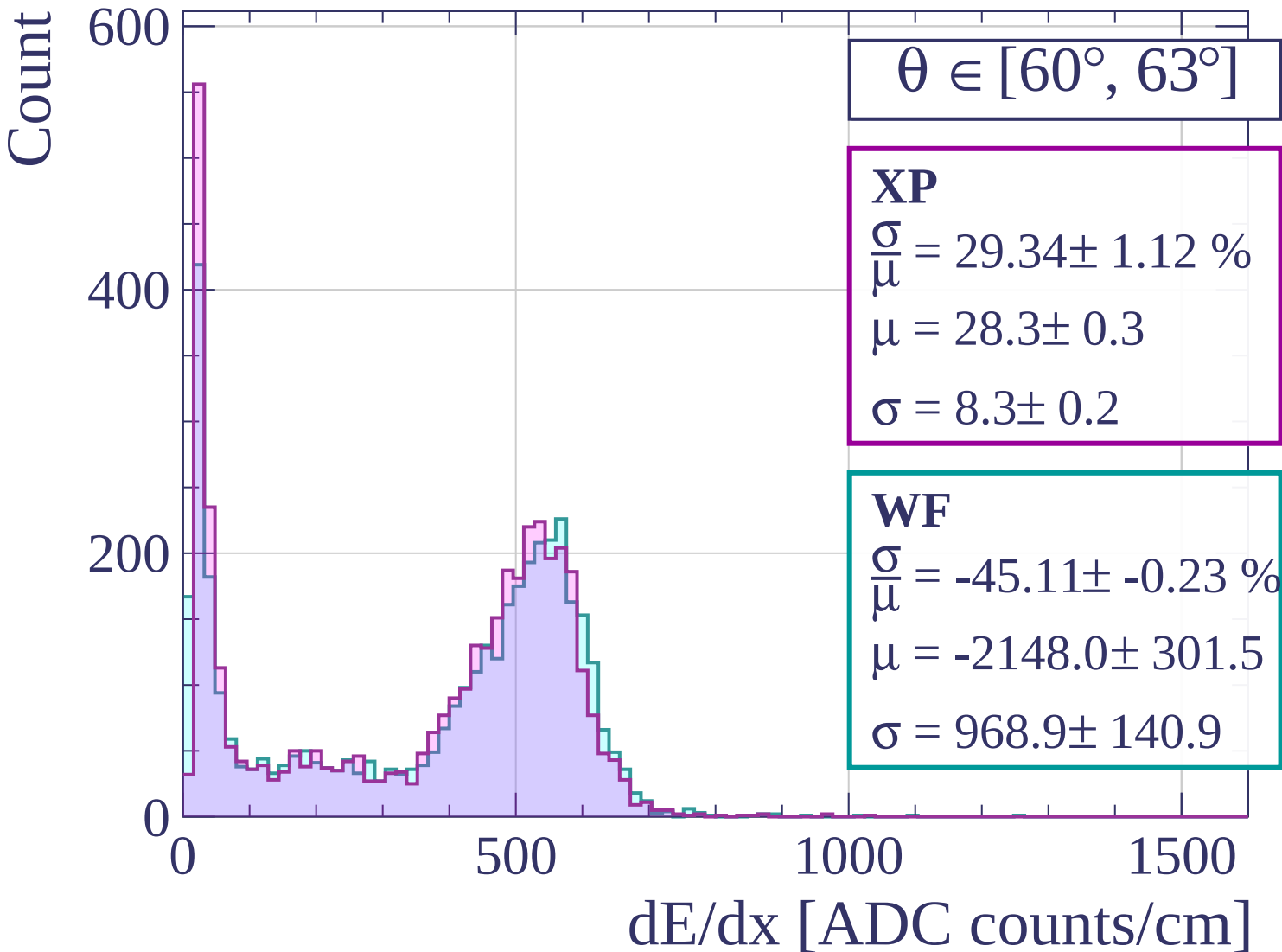
$$\frac{\sigma}{\mu} = -32.59 \pm -3.65 \%$$

$$\mu = -3813.3 \pm 164.1$$

$$\sigma = 1242.9 \pm 192.5$$



dE/dx [ADC counts/cm]



Count

$\theta \in [63^\circ, 66^\circ]$

XP

$$\frac{\sigma}{\mu} = -23.50 \pm 0.21 \%$$

$$\mu = -3942.2 \pm 322.0$$

$$\sigma = 926.3 \pm 67.6$$

WF

$$\frac{\sigma}{\mu} = -22.05 \pm -0.13 \%$$

$$\mu = -2303.0 \pm 252.8$$

$$\sigma = 507.7 \pm 58.7$$

600

400

200

0

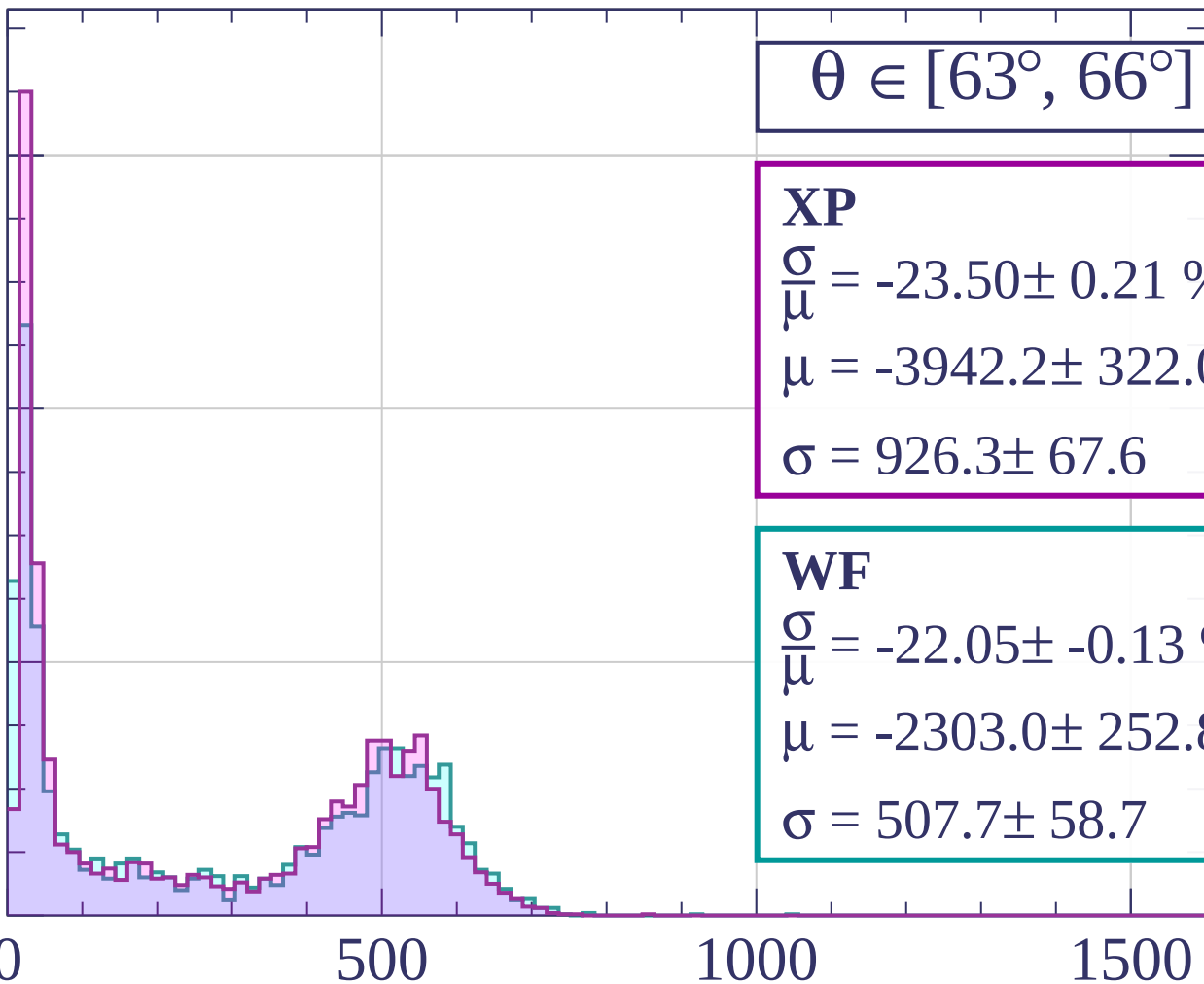
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [66^\circ, 69^\circ]$

XP

$$\frac{\sigma}{\mu} = -20.98 \pm 0.09 \%$$

$$\mu = -1692.7 \pm 119.5$$

$$\sigma = 355.2 \pm 23.5$$

WF

$$\frac{\sigma}{\mu} = -21.22 \pm 0.20 \%$$

$$\mu = -1065.1 \pm 78.1$$

$$\sigma = 226.0 \pm 14.5$$

600

400

200

0

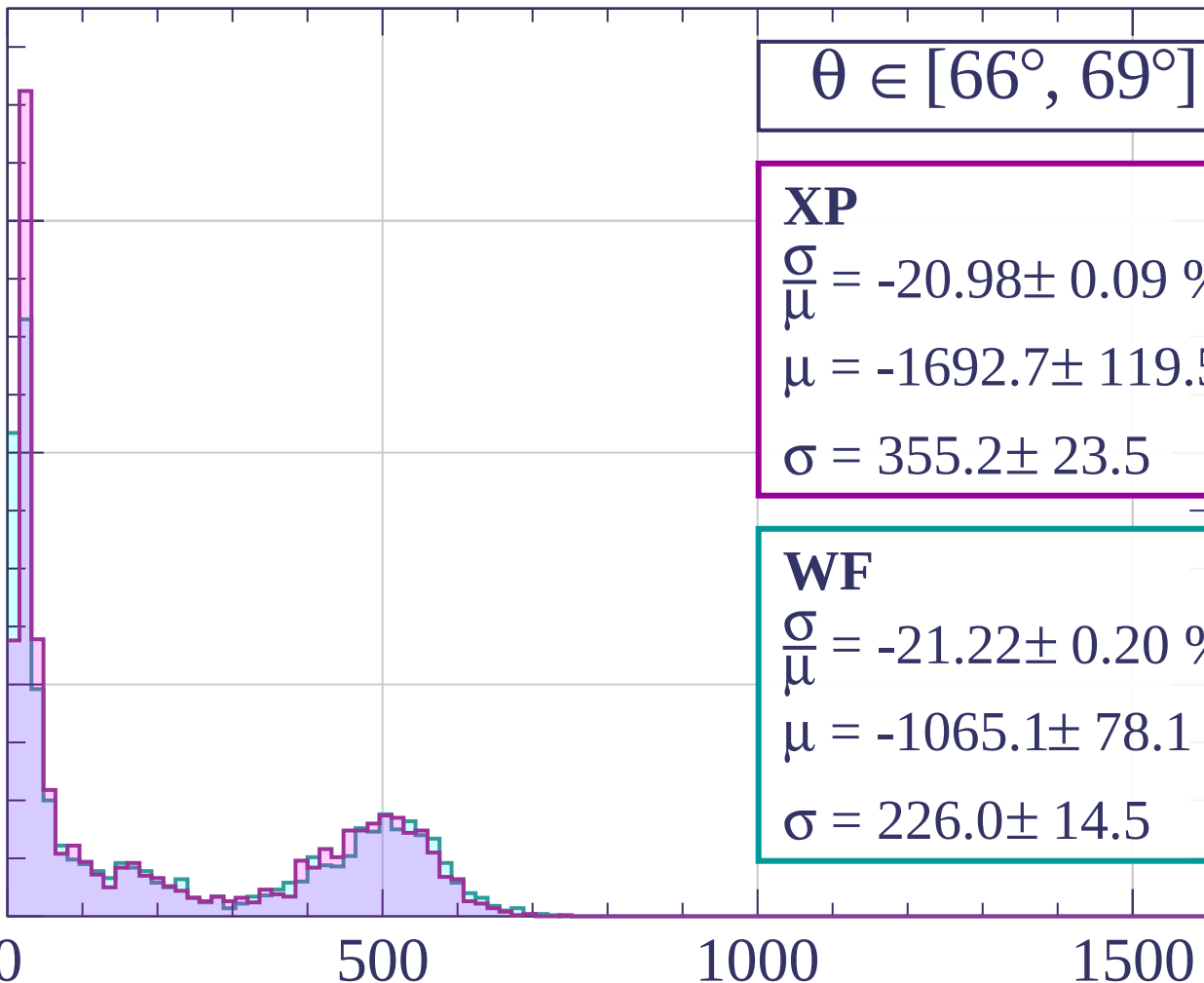
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [69^\circ, 72^\circ]$

XP

$$\frac{\sigma}{\mu} = 53.98 \pm 2.44 \%$$

$$\mu = 22.2 \pm 0.4$$

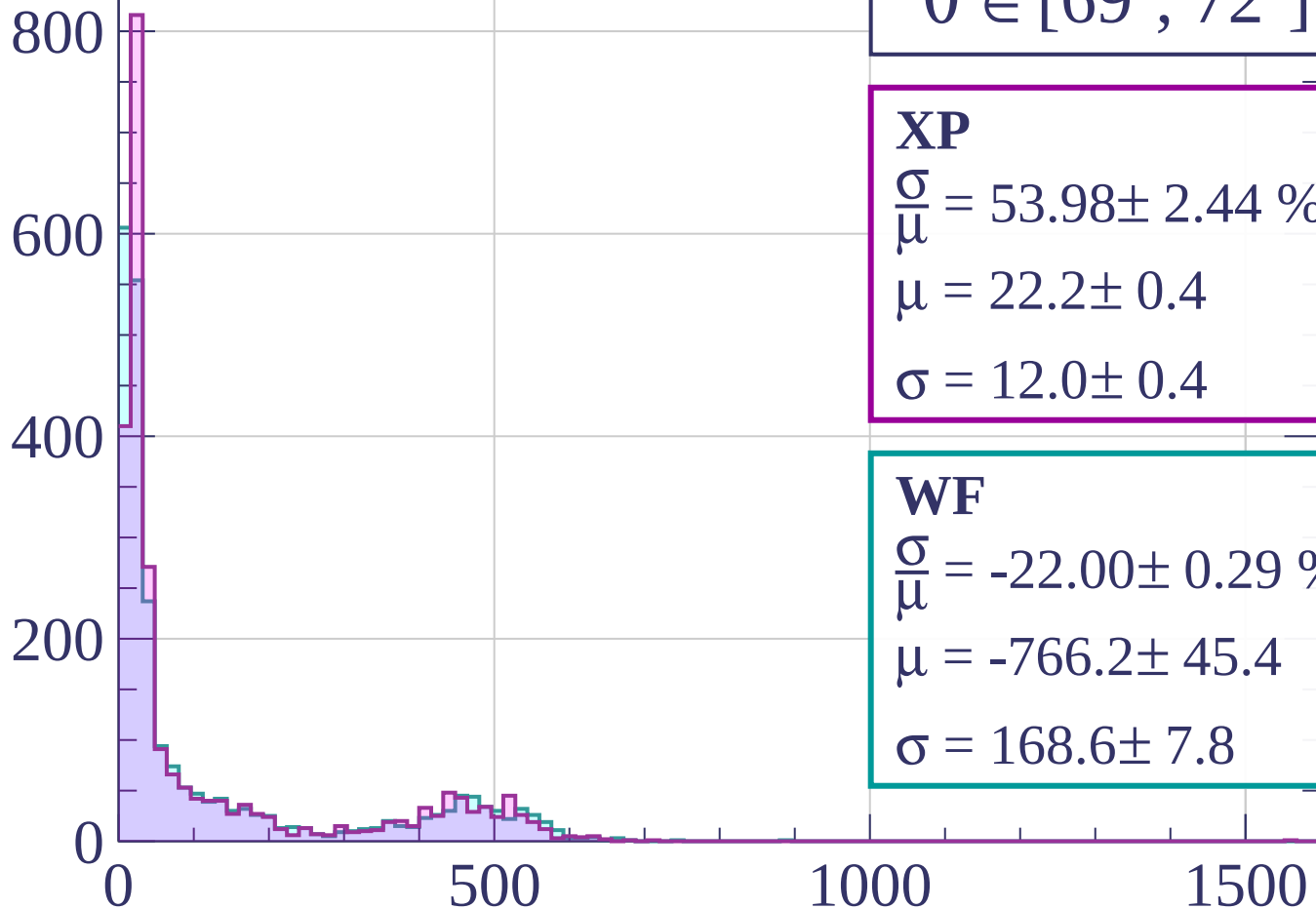
$$\sigma = 12.0 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = -22.00 \pm 0.29 \%$$

$$\mu = -766.2 \pm 45.4$$

$$\sigma = 168.6 \pm 7.8$$



dE/dx [ADC counts/cm]

Count

$\theta \in [72^\circ, 75^\circ]$

XP

$$\frac{\sigma}{\mu} = 2.28 \pm 77.25 \%$$

$$\mu = 39.2 \pm 0.6$$

$$\sigma = 0.9 \pm 30.2$$

WF

$$\frac{\sigma}{\mu} = -21.70 \pm -0.05 \%$$

$$\mu = -620.6 \pm 17.4$$

$$\sigma = 134.7 \pm 4.1$$

800

600

400

200

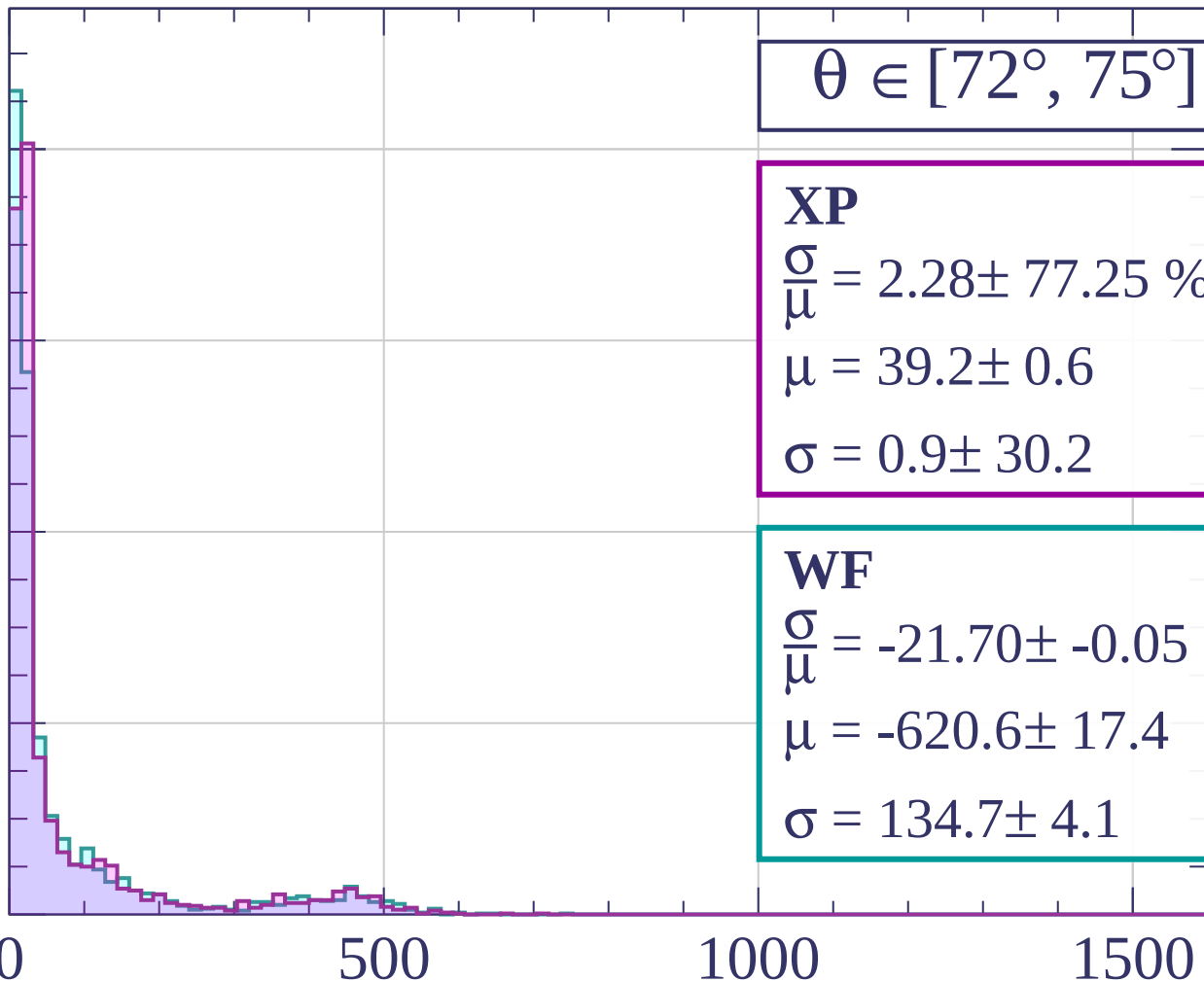
0

500

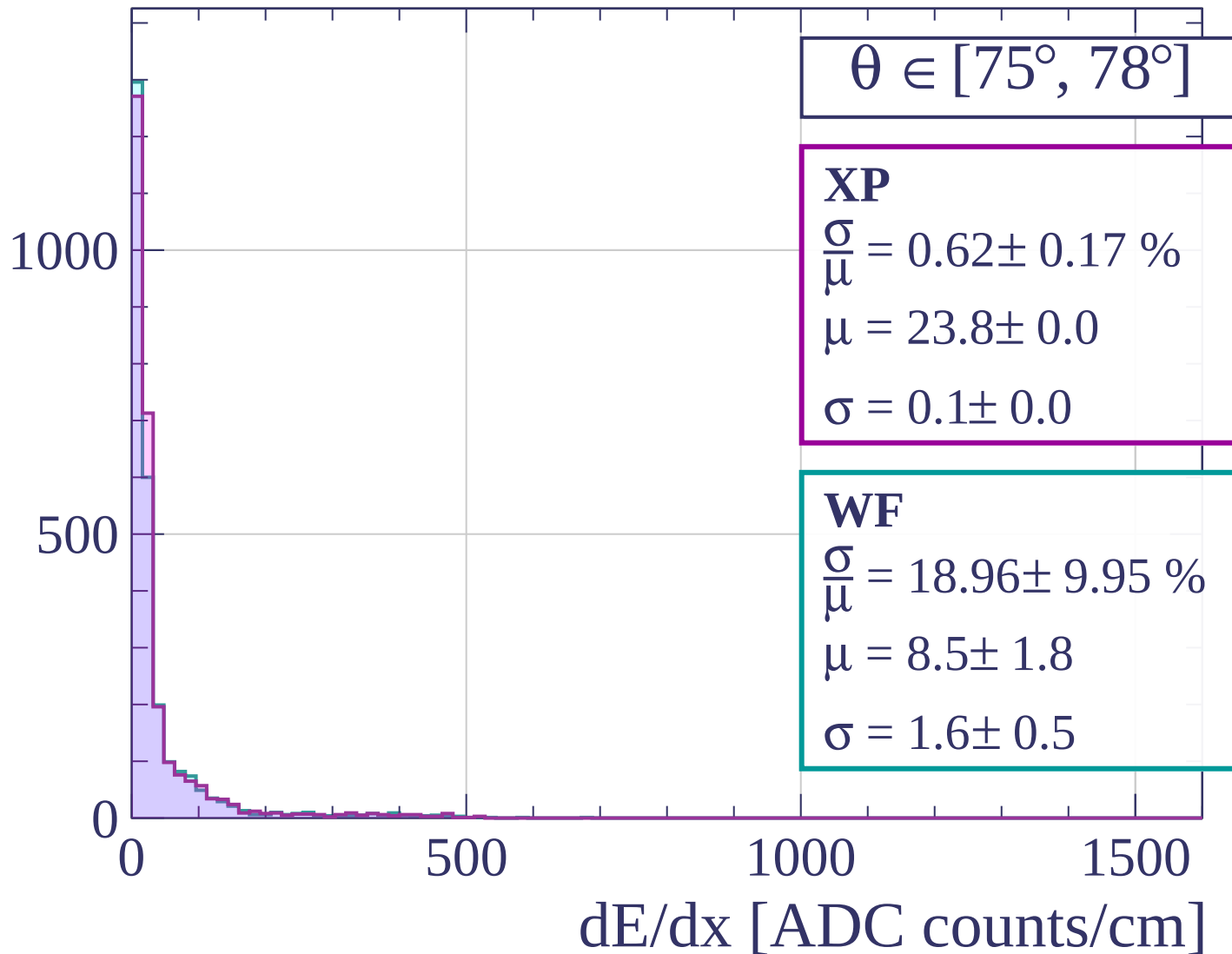
1000

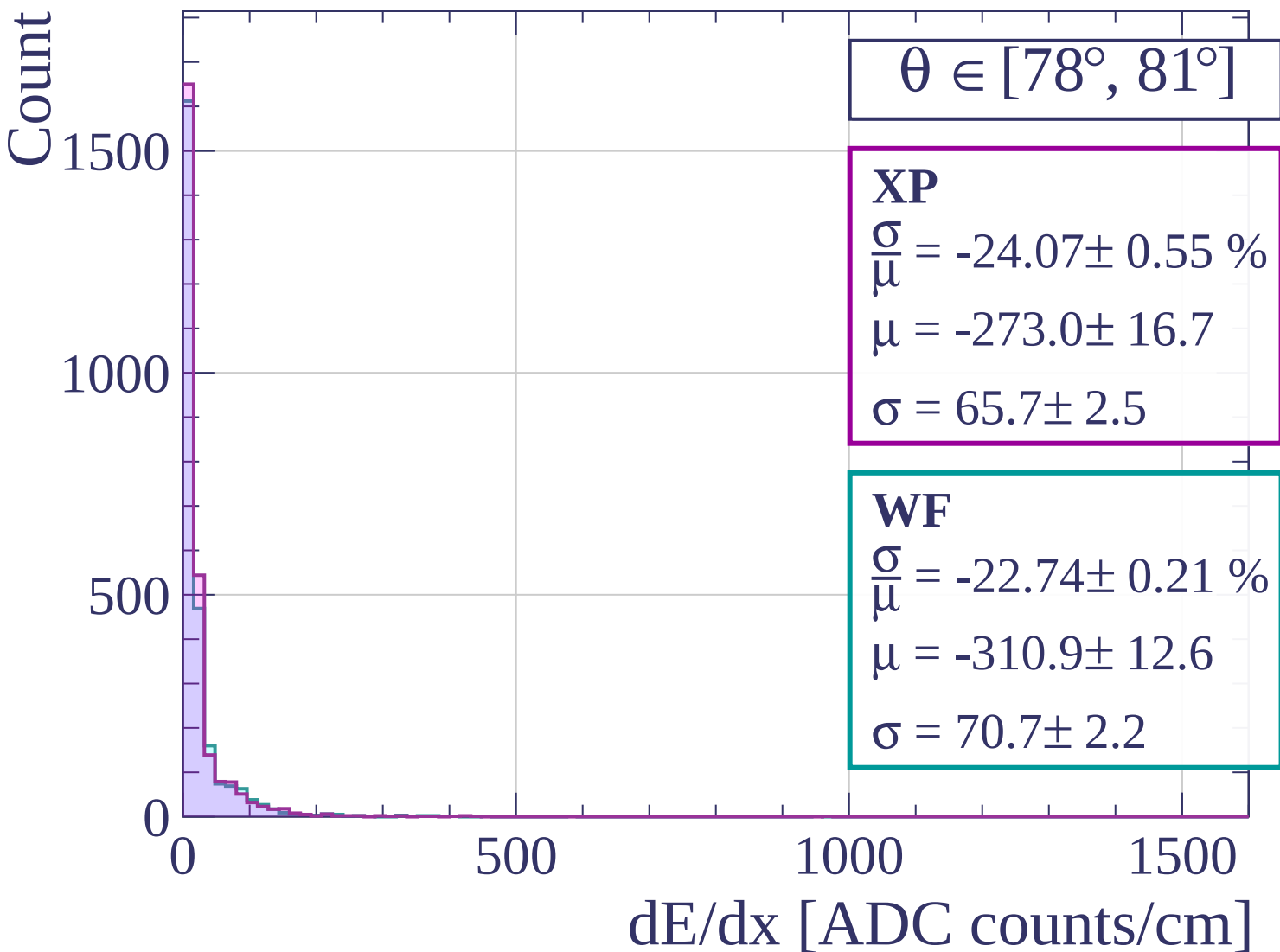
1500

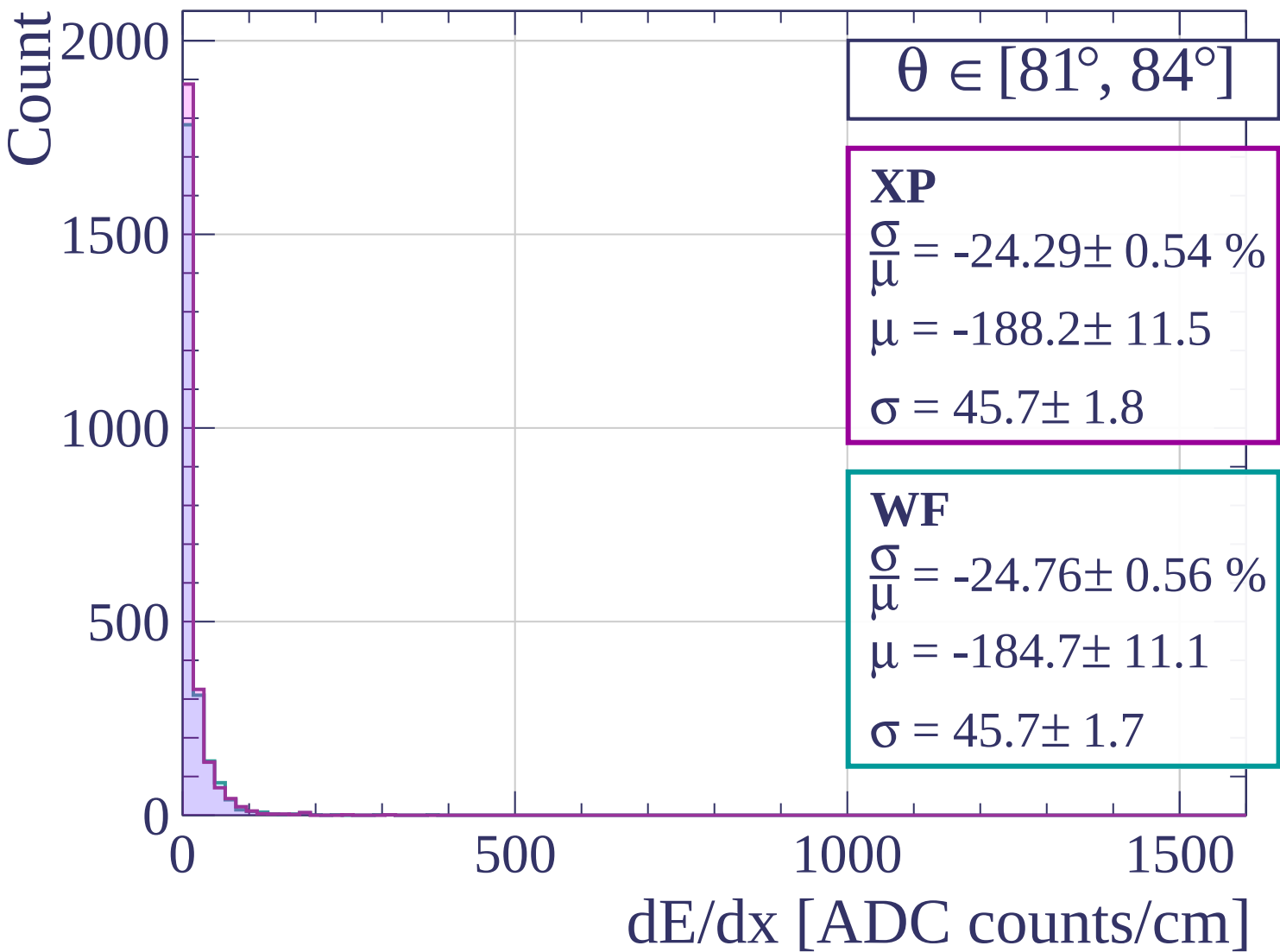
dE/dx [ADC counts/cm]



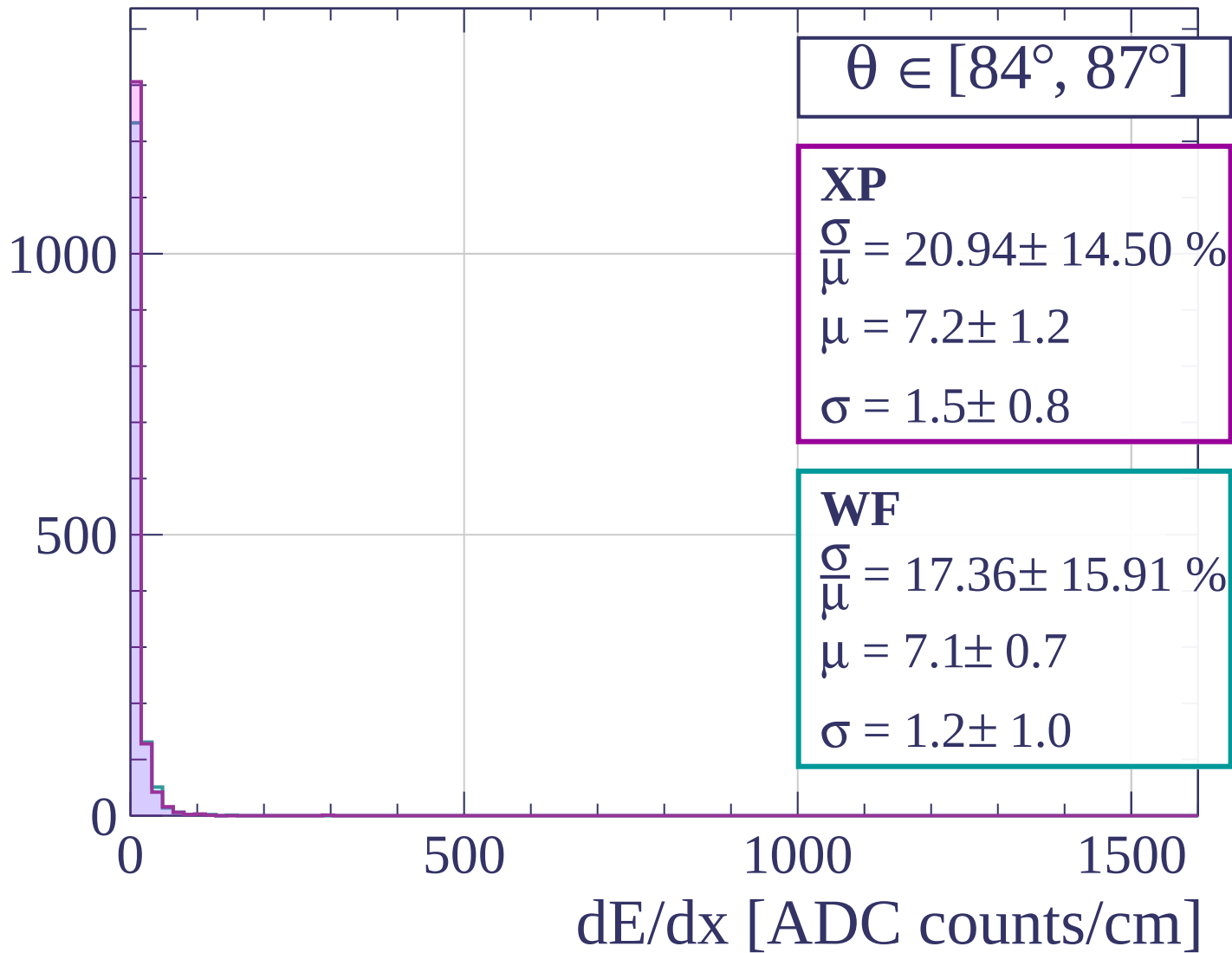
Count







Count



Count

$\theta \in [87^\circ, 90^\circ]$

XP

$$\frac{\sigma}{\mu} = 3.42 \pm 4.87 \%$$

$$\mu = 7.7 \pm 0.2$$

$$\sigma = 0.3 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 3.42 \pm 5.14 \%$$

$$\mu = 7.7 \pm 0.2$$

$$\sigma = 0.3 \pm 0.4$$

150

100

50

0

0

500

1000

1500

dE/dx [ADC counts/cm]

